

Prepladder Version X Qbank

Subject: Ophthalmology

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Introduction to Ophthalmology

1. A 25-year-old patient presents with redness and discomfort in the eyes. On examination, the physician notes inflammation confined to the conjunctiva attached to the posterior tarsal plate. Which part of the conjunctiva is primarily affected in this patient?

- A. Bulbar conjunctiva
 - B. Forniceal conjunctiva
 - C. Palpebral conjunctiva
 - D. Limbal conjunctiva
-

2. Which of the following structures do not contain any ganglion cells?

- A. Macula
 - B. Fovea
 - C. Foveola
 - D. Periphery of the retina
-

3. Which of the following is the anteroposterior diameter of the adult human eyeball?

- A. 23 mm
 - B. 23.5 mm
 - C. 24 mm
 - D. 75 mm
-

4. Which structure primarily offers resistance to aqueous humor outflow within the iridocorneal angle?

- A. Iris
 - B. Ciliary Body
 - C. Descemet's Membrane
 - D. Trabecular Meshwork
-

5. Which of the following statements accurately describes the role of retinal ganglion cells in the visual process?

- A. They are the primary photoreceptors responsible for converting light into electrical signals.
 - B. They are concentrated in the fovea and are responsible for colour vision and high visual acuity.
 - C. Their axons form the optic nerve, which transmits visual information to the brain.
 - D. They contain pigments that undergo structural changes when exposed to light, triggering signal transduction.
-

6. What is the primary use of the instrument shown in the image given below?



- A. To measure intraocular pressure.
 - B. To examine the anterior segment of the eye.
 - C. To provide a wide-field view of the retina.
 - D. To capture high-resolution images of the retina.
-

7. Which of the following arteries is not a branch of the ophthalmic artery?

- A. Supratrochlear Artery
 - B. Dorsal Nasal Artery
 - C. Transverse Facial Artery
 - D. Short Posterior Ciliary Artery
-

8. A 32-year-old patient presents with an orbital infection characterized by severe pain, swelling, and redness around the eye. Despite treatment, the patient develops a headache, fever, and neurological deficits, leading to a diagnosis of a brain abscess. What is the most likely route of spread of the infection?

- A. Ethmoidal veins to the cavernous sinus
- B. Facial vein to the cavernous sinus
- C. Superior ophthalmic vein to the cavernous sinus

D. Angular vein to the internal jugular vein

9. A patient presents to the ophthalmology OPD with the following features: Which of the following conditions is most likely responsible for these symptoms?



- A. Abducens nerve palsy
 - B. Optic neuritis
 - C. Oculomotor nerve palsy
 - D. Trochlear nerve palsy
-

10. During an anatomy lecture, a student learns about the innervation of the eye. The pupil dilates in response to low light conditions. Which nerve fibers are primarily responsible for the dilation of the pupil?

- A. Oculomotor nerve fibers
 - B. Trigeminal nerve fibers
 - C. Sympathetic nerve fibers
 - D. Facial nerve fibers
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
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Solution for Question 1:

Correct Answer: C) Palpebral conjunctiva

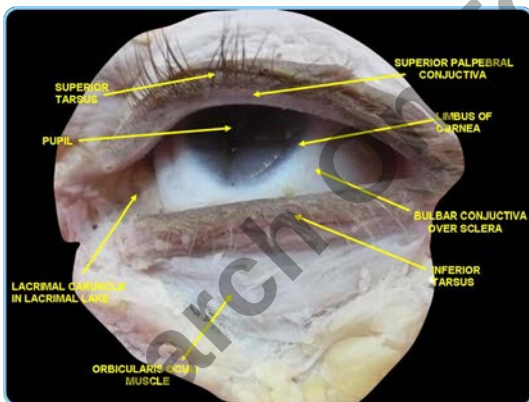
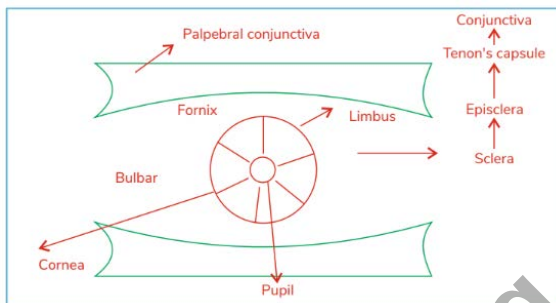
Explanation:

The palpebral conjunctiva lines the inner surface of the eyelids, which start at lid margins and are attached to posterior tarsal plates; while the bulbar conjunctiva covers the anterior sclera, which is continuous with corneal epithelium at limbus.

- The conjunctiva is a transparent mucous membrane lining the inner eyelid surface and anterior globe, terminating at the corneoscleral limbus.
- Blood supply: Anterior ciliary and palpebral arteries
- Lymphatic drainage to the preauricular and submandibular nodes (similar to the eyelids).
- Function: Provides both passive and active immunity.

- Anatomical divisions include: Palpebral conjunctiva: Starts at the mucocutaneous junction of the lid margins and attaches firmly to the posterior tarsal plates, with vertically oriented blood vessels along the tarsus. Forniceal conjunctiva: Loose and redundant, allowing for eye movement. Bulbar conjunctiva: Covers the anterior sclera, merging with the corneal epithelium at the limbus. The radial ridges at the limbus form the palisades of Vogt, the likely reservoirs for corneal stem cells. The bulbar conjunctival stroma is loosely attached to the underlying Tenon's capsule, except at the limbus where both layers fuse. Additional structures include the plica semilunaris (a semilunar fold on the nasal side) and the caruncle, a fleshy nodule containing modified cutaneous tissue, situated medially to the plica.
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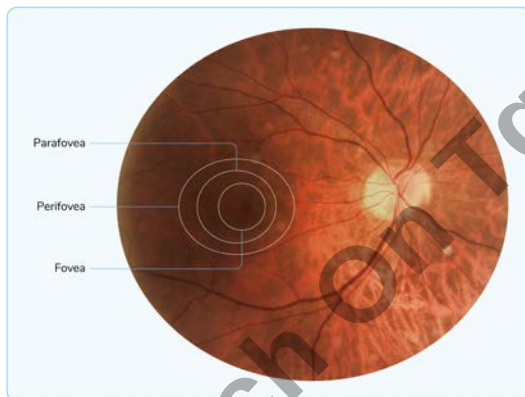
Solution for Question 2:

Correct Answer: C) Foveola

Explanation:

The foveola is the central part of the fovea and is unique in that it lacks ganglion cells, which allows for optimal light entry, contributing to high visual acuity.

- The macula is a circular area measuring 5 to 5.5 mm in diameter, covering the central 15–20° of the visual field.
- It contains high concentrations of yellow carotenoid pigments, lutein and zeaxanthin, hence the name "macula lutea".
- The fovea, a depression located at the center of the macula, has a diameter of about 1.5 mm, similar to that of the optic disc.
- The foveola, forming the central floor of the fovea, measures 0.35 mm in diameter and is the thinnest part of the retina, containing only cone photoreceptors and Müller cells, with no ganglion cells.
- The umbo is the central depression of the foveola and is associated with the foveolar light reflex; its loss may signal early damage.
- The foveal avascular zone is a blood vessel-free area within the fovea but surrounded by capillaries, with an average diameter of 0.6 mm, which varies with age and disease and is best assessed via fluorescein angiography.



Solution for Question 3:

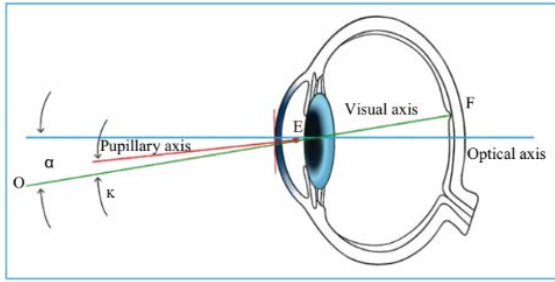
Correct Answer: C) 24 mm

Explanation:

Anteroposterior diameter of the adult human eyeball is 24 mm.

Axes and angle kappa:

- The anatomical (pupillary) axis runs from the posterior pole of the eye through the center of the cornea, but it usually doesn't align with the visual axis because the fovea is slightly temporal.
- The visual axis runs from the fovea through the eye's nodal point to the fixation point.
- In normal binocular single vision (BSV), the visual axes of both eyes meet at the fixation point, where images are aligned by the fusion reflex and combined by binocular cells in the visual cortex to produce BSV.
- Alpha angle (■) is the angle formed between the visual and optic axes.
- Angle kappa (κ) is the angle between the visual and anatomical axes. The fovea is normally located temporal to the anatomical center of the posterior pole, causing the eyes to slightly abduct for bifoveal fixation. A light on the cornea creates a reflex just nasal to the corneal center, known as a positive angle kappa. A large positive angle kappa (e.g. temporally displaced macula) may give a pseudoexotropia. A negative angle kappa occurs when the fovea is nasal to the posterior pole (e.g., high myopia), causing a temporal corneal reflex that can simulate esotropia.
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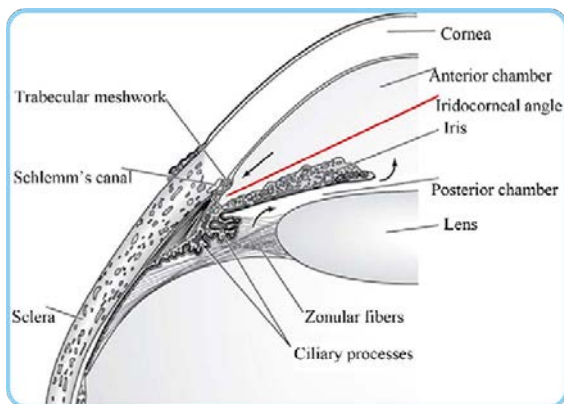
Solution for Question 4:

Correct Answer: D) Trabecular Meshwork

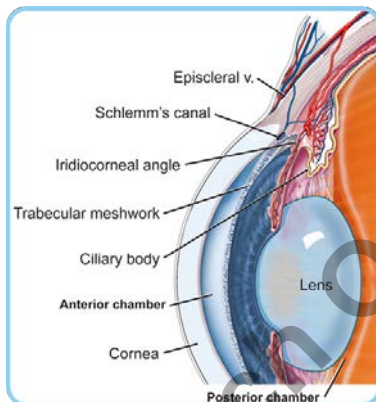
Explanation:

Angle of Anterior Chamber

- Angle of Anterior Chamber, found at the periphery of the anterior chamber, is bordered by the corneo sclera anteriorly and the iris root and ciliary body posteriorly.
- It houses the trabecular meshwork and the canal of Schlemm, both crucial for draining aqueous humor.
- Trabecular Meshwork: A triangular, loosely arranged tissue network between the anterior chamber and the canal of Schlemm. Its coarse (collagen and elastin) and fine (mucopolysaccharides) frameworks create resistance to aqueous flow. (Option D)
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- Its coarse (collagen and elastin) and fine (mucopolysaccharides) frameworks create resistance to aqueous flow. (Option D)
- Canal of Schlemm: A circular venous sinus within the sclera, crucial for draining aqueous humour
- A circular venous sinus within the sclera, crucial for draining aqueous humour
- A circular venous sinus within the sclera, crucial for draining aqueous humour



- The diagram above shows the outflow.
- The aqueous that is formed in the posterior chamber goes through the pupil into the anterior chamber, and through the trabecular pathway.
- When it passes through the uveosclera, it is called the uveoscleral pathway.

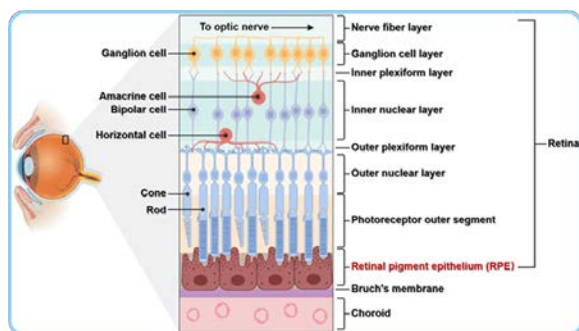
Iris (Option A) does not play a significant role in regulating the aqueous humour outflow or offering resistance to its drainage.

Ciliary body (Option B) produces aqueous humour but does not resist its outflow.

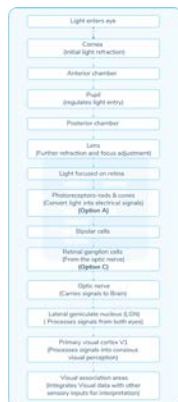
Descemet's membrane (Option C) contributes to structural support, not resistance.

Solution for Question 5:

Correct Answer: C) Their axons form the optic nerve, which transmits visual information to the brain.



Explanation:



Cones, not retinal ganglion cells, are concentrated in the fovea and responsible for colour vision and high visual acuity. (Option B) Photopigments like rhodopsin in rods undergo structural changes when exposed to light, not retinal ganglion cells. (Option D)

- Cones, not retinal ganglion cells, are concentrated in the fovea and responsible for colour vision and high visual acuity. (Option B)
- Photopigments like rhodopsin in rods undergo structural changes when exposed to light, not retinal ganglion cells. (Option D)

Solution for Question 6:

Correct Answer: C) To provide a wide-field view of the retina.

Explanation:

The given image is an Indirect Ophthalmoscope, device used to examine the retina and provide wide-field view of the retina.



- Wide Field of View: Provides a broad, panoramic view of the retina.
- Magnification: Offers magnified images of the retinal structures.
- Lens and Light Source: Uses a special lens and light source to visualize the retina.
- Peripheral Retina Examination: Allows for detailed examination of the peripheral retina.
- Diagnosis and Monitoring: Useful for detecting and monitoring conditions like retinal detachments, retinal degenerations, etc.
- Enhanced Detail: Enables a more comprehensive view compared to direct ophthalmoscopy.

Solution for Question 7:

Correct Answer: C) Transverse Facial Artery

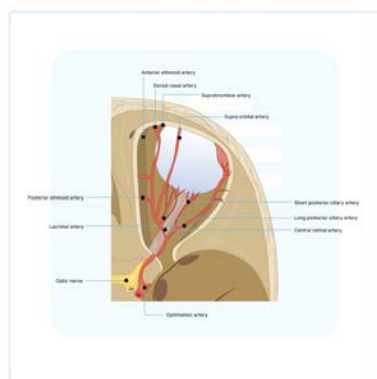
Explanation:

Transverse Facial Artery: This artery is a branch of the superficial temporal artery and not ophthalmic artery, which is a branch of the external carotid artery. It supplies the skin of the face.

Arterial Supply of eyeball and orbital structures

Ophthalmic artery: Branch of internal carotid artery constitutes the main source of blood supply for the eyeball and orbital structures.

Branches of Ophthalmic artery:



Ocular Branches	Orbital Branches
Central retinal artery (supplies inner 6 layers of retina) Long posterior ciliary arteries Short posterior ciliary arteries Anterior ciliary arteries - Muscular branch	Lacrimal artery - Lateral palpebral artery Recurrent meningeal artery Supraorbital artery Medial palpebral artery Posterior ethmoidal artery Anterior ethmoidal artery Dorsal nasal artery (terminal branch) Supratrochlear artery (terminal branch)

Solution for Question 8:

Correct Answer: C) Superior ophthalmic vein to the cavernous sinus

Explanation:

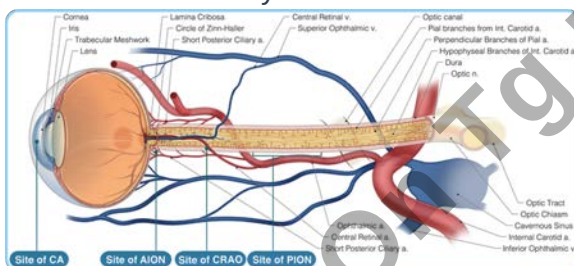
- The superior ophthalmic vein directly drains blood from the orbit into the cavernous sinus, providing a pathway for infections to enter the cranial cavity.
- This route is significant in cases of orbital infections, as it can lead to serious complications such as brain abscesses and cavernous sinus thrombosis.

The venous drainage of the eye primarily occurs through:

- Superior ophthalmic vein: It receives drainage from the angular, supratrochlear, and supraorbital veins, superior vortex veins, anterior ethmoidal vein, and receives collateral

drainage from the inferior ophthalmic vein. It continues posteriorly to drain into the cavernoussinus.

- It receives drainage from the angular, supratrochlear, and supraorbital veins, superior vortex veins, anterior ethmoidal vein, and receives collateral drainage from the inferior ophthalmic vein.
- It continues posteriorly to drain into the cavernous sinus.
- Inferior ophthalmic vein: It originates from a plexus in the inferior orbit and collects blood from the inferior vortex veins. It eventually drains into the cavernous sinus.
- It originates from a plexus in the inferior orbit and collects blood from the inferior vortex veins.
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- It continues posteriorly to drain into the cavernous sinus.
- It originates from a plexus in the inferior orbit and collects blood from the inferior vortex veins.
- It eventually drains into the cavernous sinus.



Solution for Question 9:

Correct Answer: C) Oculomotor nerve palsy

Explanation:

- The eye appears downward and abducted, commonly known as down and out pupil due to paralysis of the inferior oblique, and medial rectus muscles.

- There is ptosis of right eye (upper eyelid droop, due to denervation to the levator palpebrae superioris), and the pupil is dilated in the right eye (as the constrictor pupillae is denervated).
- These signs are suggestive of oculomotor nerve palsy.

Nerves of the eye:

Optic nerve (CN II)	Purely sensory Senses the incoming light and image displayed on retina
Oculomotor nerve (CN III)	Innervates majority of the extraocular muscles Motor innervation to the superior rectus muscle, medial rectus muscle, inferior rectus muscle, inferior oblique muscle, levator palpebrae superioris muscle, ciliary muscle, and the sphincter muscle.
Trochlear nerve (CN IV)	Motor innervation to the superior oblique muscle
Trigeminal Nerve (CN V)	Ophthalmic branch of the trigeminal nerve provides sensory innervation to the eye Afferent part of corneal and lacrimation reflex
Abducens Nerve (CN VI)	Motor innervation to the lateral rectus muscle
Facial Nerve (CN VII)	Motor innervation of the orbicularis oculi muscle Efferent part of corneal and lacrimation reflex
Sympathetic Nervous System	Long ciliary nerves: Innervates the pupillary dilator muscles causing mydriasis
Parasympathetic nervous system	Short ciliary nerves: Innervates the sphincter pupillae muscles causing miosis

Solution for Question 10:

Correct Answer: C) Sympathetic nerve fibers

Explanation:

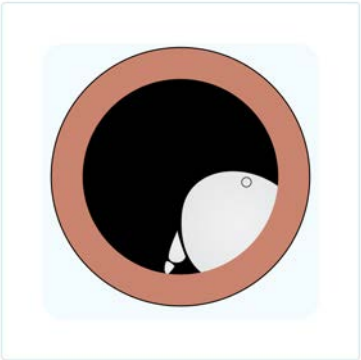
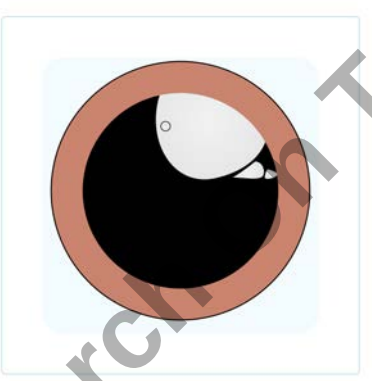
- Sympathetic nerve fibers are responsible for the dilation of the pupil (mydriasis).
- These fibers originate from the sympathetic chain and supply the dilator pupillae muscle, allowing the pupil to expand in response to low light conditions, thus enabling more light to enter the eye.

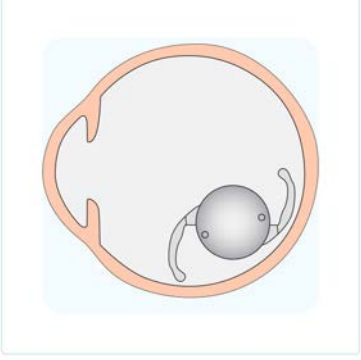
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Lens and Disorders

1. Match the images of IOL malpositions with their names: Image Syndrome a. 1. Sunrise syndrome b. 2. Lost lens syndrome c. 3. Sunset syndrome

Image	Syndrome
 ... (content truncated)	1. Sunrise syndrome
 ... (content truncated)	2. Lost lens syndrome

 ... (content truncated)	3. Sunset syndrome
--	---------------------------

- A. a-1,b-2,c-3
- B. a-3,b-1,c-2
- C. a-2,b-1,c-3
- D. a-1,b-3,c-2

2. Toric IOLs are specifically designed to correct which type of refractive error?

- A. Myopia
- B. Hypermetropia
- C. Astigmatism
- D. Presbyopia

3. Which of the following is considered the best formula for intraocular lens (IOL) power calculation after LASIK surgery?

- A. SRK
- B. Hoffer Q
- C. Haigis

D. Holladay I

4. In which scenario is a posterior capsule tear more commonly observed during cataract surgery?

- A. During phacoemulsification with high energy settings
 - B. During cortical clean-up with a manual irrigation-aspiration technique
 - C. During hydrodelineation
 - D. During anterior chamber maintenance with a viscoelastic substance
-

5. A 70-year-old woman with a history of hypertension and high myopia undergoes cataract surgery. During the procedure, she suddenly experiences severe ocular pain and a rapid rise in intraocular pressure. Fundoscopic examination reveals a dark red elevation in the choroidal region. Which of the following is the most appropriate immediate management step?

- A. Intraoperative posterior sclerostomy
 - B. Increase the infusion pressure in the anterior chamber
 - C. Perform immediate vitrectomy to drain the hemorrhage
 - D. Close the surgical wound and monitor intraocular pressure
-

6. Which of the following is not a condition associated with a shallow anterior chamber?

- A. Wound leak with hypotony
- B. Ciliochoroidal detachment
- C. Vitreous bulge
- D. Iridodialysis

7. A 67-year-old man had undergone cataract surgery. He presents with endophthalmitis after 3 days. His infection is severe with vision reduced to hand movements close to the face. What is the most appropriate management?

- A. Intravitreal antibiotics
 - B. Systemic antibiotics
 - C. Steroid therapy
 - D. Vitrectomy
-

8. A 65-year-old man has undergone cataract surgery. He has now returned after 6 weeks with endophthalmitis. What is the most likely organism that caused the infection?

- A. Staphylococcus aureus
 - B. Streptococcus pyogenes
 - C. Propionibacterium acnes
 - D. Staphylococcus epidermidis
-

9. Soemmering's Ring and Elschnig's Pearls are examples of which of the following?

- A. After cataracts
 - B. Senile cataracts
 - C. Congenital cataracts
 - D. Toxic cataracts
-

10. A key advantage of peribulbar anesthesia over retrobulbar anesthesia is:

- A. Faster onset of anesthesia
 - B. Less risk of globe perforation
 - C. Better akinesia of the eye
 - D. Reduced risk of systemic toxicity
-

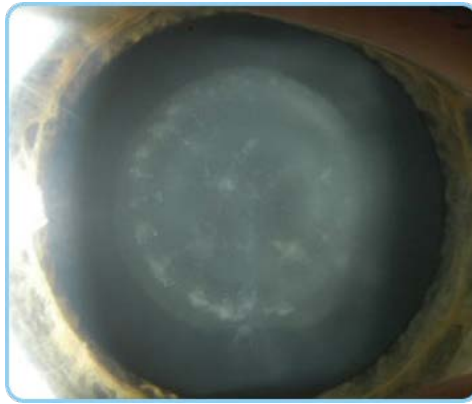
11. Which of the following statements is true regarding Cystoid Macular Edema?

- A. Collection of fluid in Henle's layer
 - B. Honeycomb appearance in fundoscopy
 - C. Antiprostaglandin is both treatment and prevention
 - D. All of the above
-

12. What is the common inheritance pattern of hereditary cataracts?

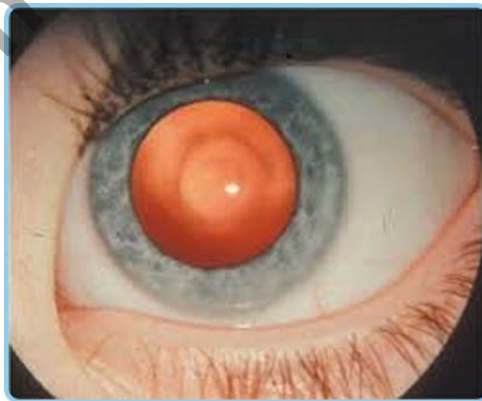
- A. Autosomal dominant
 - B. Autosomal recessive
 - C. X linked recessive
 - D. X linked dominant
-

13. Doctor noticed leukocoria in a 3-month-old baby during a routine examination and the eye appeared as shown below. Choose the least appropriate statement regarding the most likely diagnosis.



- A. Most common type of congenital cataract
- B. Commonly inherited as autosomal dominant trait
- C. Usually zone of embryonic nucleus is involved
- D. Associated with vitamin-D deficiency

14. An infant with leukocoria presents with recurrent gastrointestinal infections and developmental regression. Bilateral eye examination revealed the following. Choose the most appropriate next step in evaluation.



- A. Serum estimation of glucose, calcium, and phosphorous levels
- B. Estimation of antibody titers for infections
- C. Urine chromatography for amino acids

D. Urine assay for the presence of reducing substances

15. A 1-month-old infant is diagnosed with a dense unilateral cataract. Which of the following is the most appropriate management strategy?

- A. Delay surgery until the child is 1–2 years old to reduce surgical risks
 - B. Perform immediate cataract surgery
 - C. Prescribe glasses and monitor the cataract progression
 - D. Wait until the child develops strabismus before considering surgery
-

16. A newborn presented with the following findings. There is a positive maternal history for fever during the first trimester of pregnancy. Which among the following is least likely to be seen in this newborn?





- A. Patent ductus arteriosus
- B. Limb hypoplasia
- C. Microphthalmia
- D. Sensorineural hearing loss

17. A school-going child is diagnosed with a connective tissue disorder characterized by abnormalities of the eye, short stature, brachydactyly, joint stiffness, and cardiovascular defects. Which among the following is incorrect regarding the likely diagnosis?

- A. Follows autosomal recessive mode of inheritance
- B. Associated with supero-temporal lens dislocation
- C. Molecular genetic analysis is confirmatory
- D. Usual age of presentation is 5-10 years

18. A 30-year-old male with retinitis pigmentosa has developed a cataract as a complication. Which of the following is a feature of his cataract? Posterior subcapsular cortical cataract Irregular in outline Variable density Breadcrumb appearance Polychromatic luster Diffuse yellow haze in the adjoining cortex

- A. 1,6,3,6
 - B. 2,4,5,6
 - C. 1,2,3,4
 - D. 1,2,3,4,5,6
-

19. What is the consequence of age-related decline in the Na⁺-K⁺ ATPase pump in the lens?

- A. Increased intralenticular K⁺ levels and decreased Na⁺ levels.
 - B. Increased intralenticular Na⁺ levels and decreased K⁺ levels.
 - C. Enhanced ion exchange efficiency.
 - D. Reduced intralenticular Ca²⁺ levels.
-

20. Which among the following is/are true statements regarding lens proteins? $\alpha/\beta/\gamma$ -Crystallins are water-soluble proteins Major intrinsic proteins contribute to lens transparency An increase in crystallins can lead to cataract β/γ -Crystallin family are the largest crystallins

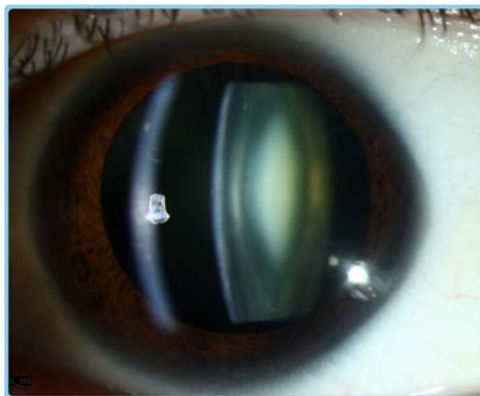
- A. Only 1
 - B. 1 & 2
 - C. 2, 3 & 4
 - D. 1, 2, 3 & 4
-

21. A 70-year-old male patient comes to your clinic with complaints of glare and diminution of vision for 3 months. On examination, you notice the following appearance Choose the correct statement regarding the correct diagnosis.



- A. The Iris shadow test becomes positive only when the whole cortex is involved.
- B. Patients see better in dark surroundings.
- C. Characterized by saucer-shaped opacities in the center of the posterior cortex.
- D. Vision loss occurs at a late stage.

22. An 80-year-old female patient came with complaints of gradual progressive diminution of vision for 2 years in both her eyes. She reports that over the last year, she noticed an improvement in her ability to read without glasses. The following finding is seen on slit lamp examination over the right eye and the left eye has similar opacity. Choose the most appropriate course of management for this patient.



- A. Advise lifestyle modification and review after 3 months
- B. Recommend corticosteroids eye drops and reassess in 6 months.

- C. Plan for Phacoemulsification.
 - D. Prescribe her rigid gas-permeable contact lenses.
-

23. An elderly male diabetic patient came with loss of vision in the left eye associated with pain in the left eye. On examination, the left eye appears red, and visual acuity- Perception of light+ and IOP is raised in the same eye. The patient gives history of blurring and diminution of vision for the past 1 year. The slit lamp examination is given below. Choose the correct diagnosis and appropriate pathophysiology for this patient.



- A. Phacomorphic glaucoma - Secondary open-angle glaucoma
 - B. Phacomorphic glaucoma - Secondary closed-angle glaucoma
 - C. Phacolytic glaucoma - Secondary open-angle glaucoma
 - D. Phacolytic glaucoma - Secondary closed-angle glaucoma
-

24. A 45-year-old man is having difficulty in near vision. The ophthalmologist has prescribed him a convex lens and informed him that it is an age-related physiological condition. Which of the following is an age-related change that would have caused his condition?

- A. Decrease in elasticity of lens capsule
- B. Sclerosis of lens substance

- C. Reduced ciliary muscle power
 - D. All of the above
-

25. A 6-month-old girl was diagnosed with galactosemia. An ocular examination has revealed a cataract which developed as a complication. Which of the following statements regarding her cataract is false?

- A. Bilateral
 - B. Oil droplet appearance
 - C. Periphery of the lens
 - D. May be reversed
-

26. Which of the following is not a feature of chalcosis?

- A. Kayser-Fleischer ring
 - B. Sunflower cataract
 - C. Retinal golden plaques
 - D. Oil droplet cataract
-

27. Which of the following is the thinnest part of the lens capsule?

- A. Anterior pole
 - B. Posterior pole
 - C. Equator
 - D. Pre-equatorial region
-

28. Match the type of cataract with the etiology

Type of cataract	Etiology
a. Posterior subcapsular	1. Glassblowers
b. Anterior subcapsular-granular	2. Miotics
c. Posterior cortical	3. Steroids

- A. a1,b2,c3
- B. a3,b2,c1
- C. a2,b1,c3
- D. a2,b3,c1

29. A 35-year-old man presents with decreased vision in his right eye following a blunt trauma a few weeks ago. On examination, feathery opacities are noted. What is the most likely type of cataract this patient has developed?

- A. Subepithelial opacities
- B. Diffuse concussion cataract
- C. Rosette cataract
- D. Morgagnian cataract

30. A 38-year-old patient with diabetes presents with rapidly worsening vision. Upon examination, the ophthalmologist notes the presence of bilateral, snowflake-like white opacities in the cortex of the lens. Which of the following is the most likely diagnosis?

- A. Senile cataract
 - B. Christmas tree cataract
 - C. Diabetic cataract
 - D. Posterior subcapsular cataract
-

31. Which surgical approach is the most appropriate for a 65 year old male with a visually significant cataract and a history of narrow-angle glaucoma?

- A. Intracapsular Cataract Extraction
 - B. Extracapsular Cataract Extraction
 - C. Phacoemulsification
 - D. Small Incision Cataract Surgery
-

32. Which of the following steps is used to maintain the depth of the anterior chamber and protect the corneal endothelium during Conventional ECCE?

- A. Anterior capsulotomy
 - B. Viscoelastic injection
 - C. Corneoscleral section completion
 - D. Hydrodissection
-

33. A 70-year-old woman presents for cataract surgery in her right eye. She has moderate cataract density with some nuclear sclerosis and a clear cornea. She expresses a strong desire for a procedure that minimizes recovery time and provides optimal visual outcomes. Which of the following statements accurately reflects the advantage of Phacoemulsification Over Manual SICS?

- A. Manual SICS typically requires a smaller incision compared to Phacoemulsification.

- B. Manual SICS allows for faster visual recovery compared to Phacoemulsification.
 - C. Manual SICS is less suitable for very hard cataracts compared to Phacoemulsification.
 - D. Manual SICS usually results in a higher level of postoperative astigmatism compared to Phacoemulsification.
-

34. What are the steps involved in FLACS?

- A. Capsulorrhexis → Corneal Incision → Lens Fragmentation → IOL implantation
 - B. Corneal Incision → Capsulorrhexis → Lens Fragmentation → IOL implantation
 - C. Capsulorrhexis → Lens Fragmentation → Corneal Incision → IOL implantation
 - D. Corneal Incision → Lens Fragmentation → Capsulorrhexis → IOL implantation
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	3
Question 4	2
Question 5	4
Question 6	4
Question 7	4
Question 8	3
Question 9	1
Question 10	2
Question 11	4
Question 12	1
Question 13	1
Question 14	4
Question 15	2
Question 16	2
Question 17	2
Question 18	4
Question 19	2
Question 20	2
Question 21	4
Question 22	3

Question 23	3
Question 24	4
Question 25	3
Question 26	4
Question 27	2
Question 28	2
Question 29	3
Question 30	3
Question 31	1
Question 32	2
Question 33	4
Question 34	3

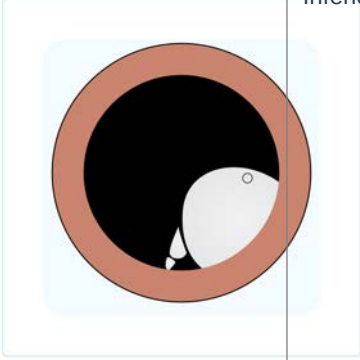
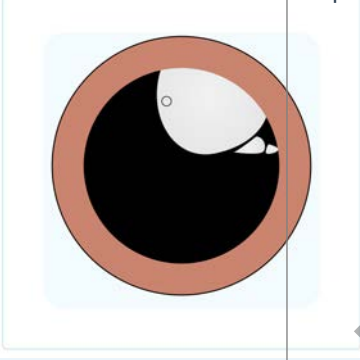
Solution for Question 1:

Correct Answer: B) a-3,b-1,c-2

Explanation:

IOL malpositions post cataract surgery:

Name	Malposition
------	-------------

 ... (content truncated)	Inferior subluxation of IOL
 ... (content truncated)	Superior subluxation of IOL
 ... (content truncated)	Complete dislocation of an IOL into the vitreous cavity
Windshield wiper syndrome	When a very small IOL is placed vertically in the sulcus the superior loop moves to the left and right, with movements of the head

Solution for Question 2:

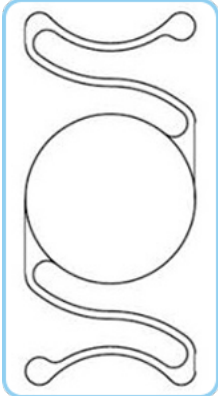
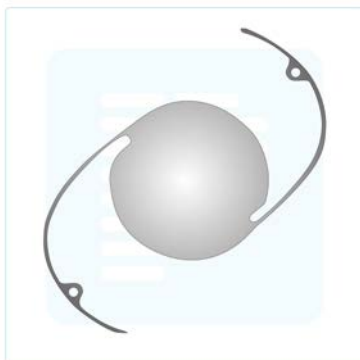
Correct Answer: C) Astigmatism

Explanation:

Astigmatism: Common refractive error where refraction changes in different meridians of the eye. Toric IOLs have an integral cylindrical refractive component to compensate for pre-existing corneal astigmatism and improve vision clarity for patients with astigmatism.

Types of Intraocular Lenses (IOLs):

Based on method of fixation in the eye:

Anterior chamber IOL	Iris supported lens
 ... (content truncated)	 Lenses are fixed on the Iris, with the help of suture, claws or loops. They have high incidents of post-operative complications. Eg - Singh and Worst's Iris claw lens
Posterior chamber IOL	Scleral fixated IOL (SFIOL)
 ... (content truncated)	 ... (content truncated)
Special function IOL	
Toric IOL: Designed to correct astigmatism (Option C) Aspheric IOL: Designed to reduce spherical aberrations	

Aniridia IOL: Designed to cover defects of aniridia or partial iris loss.

Based on the focussing ability the IOLs are of following types:

Unifocal IOLs are the most commonly used ones having unifocal power. Depending upon the power of IOLs implanted, these can make the patient emmetropic, myopic or hypermetropic



Multifocal IOLs have separate optics to focus for distance and near, so also known as simultaneous vision lenses. These are of 2 types, either refractive or diffractive optics types. These are also called pseudo accommodative IOLs.



Accommodative IOLs exhibit some anterior movement of optic to improve the near vision. Available models of accommodative IOLs are: Crystalens and Synchrony lens



Extended-depth-of-Focus IOL- e.g. Technis Symphony, provides uncorrected distance, intermediate as well as near vision to some extent due to its elongated focus.

Myopia (Option A): Not specifically corrected by Toric IOLs. Standard monofocal IOLs are typically used for correcting myopia.

Hypermetropia (Option B): Can be corrected with standard IOLs, but Toric IOLs are not designed specifically for this condition.

Presbyopia (Option D): Age-related condition where the eye loses its ability to focus on near objects. Multifocal or EDOF IOLs are typically used to address presbyopia, not Toric IOLs.

Solution for Question 3:

Correct Answer: C) Haigis

Explanation:

Haigis formula is particularly well-regarded for post-LASIK eyes.

- It uses two variables: axial length & anterior chamber depth, making it more adaptable to eyes that have undergone refractive surgery.
- This adaptability generally provides more accurate IOL power calculations in post-LASIK cases.

Biometry: Method of calculating IOL power by 2 methods - Keratometry and Axial length

- Equipments used: A scan ultrasound for axial length

Formulas used:

SRK (Option A)	Hoffer Q (Option B)	Holladay I (Option D)
$P = A - 2.5L - 0.9K$ (L= axial length) (K= constant) Normal axial length (24 mm)	Short axial length (<22mm)	Long axial length (>24.5 mm)

Solution for Question 4:

Correct Answer: B) During cortical clean-up with a manual irrigation-aspiration technique

Explanation:

Posterior capsule tears are more common during the cortical clean-up phase of cataract surgery.

- This is due to the mechanical stress and manipulation involved in removing residual cortex, which can sometimes lead to inadvertent damage to the posterior capsule.
 - More common in SICS and Phacoemulsification procedures.
 - Can occur in following situations
- During forceful hydrodissection.
 - Direct injury with Sinskey's hook, chopper.
 - During cortex aspiration.
 - Manual irrigation-aspiration technique.

Operative complications of Cataract surgery:

- Superior rectus muscle laceration
- Button holding of anterior wall of tunnel
- Scleral disinsertion
- Descemet's detachment
- Iridodialysis
- Zonular dehiscence
- Vitreous prolapse and loss
- Nucleus drop into vitreous cavity
- Posterior loss of lens fragments
- Expulsive choroidal haemorrhage

During phacoemulsification with high energy settings (Option A): They are more likely to cause issues like corneal edema or endothelial damage rather than directly causing a posterior capsule tear.

During hydrodelineation (Option C): This technique generally helps separate the lens from the capsule and is less likely to cause a tear if performed correctly.

During anterior chamber maintenance with a viscoelastic substance (Option D): Maintaining the anterior chamber with viscoelastic substances is designed to protect intraocular structures and reduce complications.

Solution for Question 5:

Correct Answer: D) Close the surgical wound and monitor intraocular pressure

Explanation:

The given clinical scenario suggests suprachoroidal hemorrhage. The immediate steps involve closing the wound to prevent further prolapse of intraocular contents and monitoring intraocular pressure.

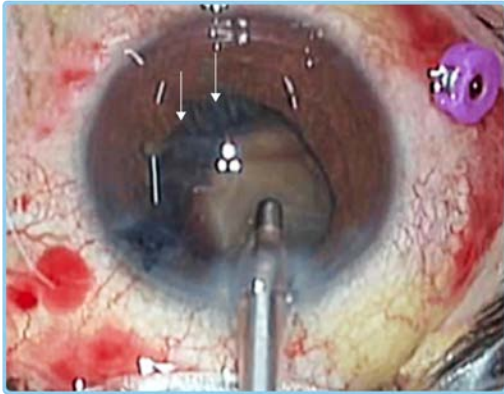
Suprachoroidal hemorrhage:

- A suprachoroidal haemorrhage involves a bleed into the suprachoroidal space from a ruptured short posterior ciliary artery.
- If sufficiently severe, it may result in extrusion of intraocular contents (expulsive haemorrhage).
- It is a dreaded complication, but extremely rare (0.04%) with phacoemulsification.
- Signs: Progressive shallowing of the AC, increased IOP, and iris prolapse. Vitreous extrusion, loss or partial obscuration of the red reflex and the appearance of a dark mound behind the pupil. In severe cases, posterior segment contents may be extruded into the AC and through the incision.
- Progressive shallowing of the AC, increased IOP, and iris prolapse.
- Vitreous extrusion, loss or partial obscuration of the red reflex and the appearance of a dark mound behind the pupil.
- In severe cases, posterior segment contents may be extruded into the AC and through the incision.
- Management:
- Progressive shallowing of the AC, increased IOP, and iris prolapse.
- Vitreous extrusion, loss or partial obscuration of the red reflex and the appearance of a dark mound behind the pupil.
- In severe cases, posterior segment contents may be extruded into the AC and through the incision.

Immediate	Subsequent
Closure of the incision with a suture to retain residual viscoelastic in the eye. Diagnosis should be confirmed at the slit-lamp as soon as possible. Acetazolamide to lower intraocular pressure. Intensive topical and systemic steroids to reduce inflammation, with standard postoperative antibiotic treatment and IOP management.	Drainage, if subsequent hemorrhage fails to resolve (after 7-14 days, by which liquefaction of blood clot has taken place). Pars plana vitrectomy may be considered when the retina appears adherent or detached.

Intraoperative posterior sclerostomy (Option A) should not be performed during the management of suprachoroidal hemorrhage because it increases the risk of complications such as infection and retinal detachment. The immediate focus should be on stabilizing the eye by controlling intraocular pressure and closing the incision. Once stabilized, further surgical interventions can be evaluated in a safer environment.

Increasing infusion pressure (Option B) may worsen the bleeding, and immediate vitrectomy (Option C) is generally avoided initially as it could exacerbate the hemorrhage.



Solution for Question 6:

Correct answer: D) Iridodialysis

Explanation:

Iridodialysis, which is a separation of the iris from its attachment to the ciliary body, typically does not cause a shallow anterior chamber.

Flat/Shallow anterior chamber

- Early Postoperative complications of cataract surgery.
- It may occur due to wound leak, ciliochoroidal detachment and pupil block.

Flat anterior chamber with wound leak: (Option A)

- Associated with hypotony, diagnosed by Seidel's test. Fluorescein diluted by aqueous at site of leakage.
- Treatment: Most cases cured within 4 days with pressure bandage and oral acetazolamide. If persistent: injection of air in anterior chamber and suturing of leaking wound.

Ciliochoroidal detachment: (Option B)

- Presents as convex brownish mass in an involved quadrant with shallow anterior chamber.
- Treatment: Most cases cured within 4 days with pressure bandage and oral acetazolamide. If persistent, suprachoroidal drainage with injection of air in anterior chamber.

Pupil block due to vitreous bulge: (Option C)

- Occurs after ICCE and in ECCE technique due to adhesions with IOL (more common with ACIOLs).
- Leads to formation of iris bombe and shallowing of anterior chamber.
- If persistent for 5-7 days, it may form permanent peripheral anterior synechiae (PAS) leading to secondary angle closure glaucoma.
- Treatment: Initially treated with mydriatic, hyperosmotic agents (e.g. 20% mannitol) and acetazolamide. If not relieved, laser peripheral iridotomy should be performed to bypass the pupillary block.

Solution for Question 7:

Correct Answer: D) Vitrectomy

Explanation:

The given scenario is of post-operative acute bacterial endophthalmitis.



A vitrectomy is done if no improvement in 48-72 hours or if severe infection with vision reduced to hand movements close to the face

Acute endophthalmitis:

Definition	Inflammation of inner structures of the eye (uveal tissue and retina) with exudates in the vitreous, anterior, and posterior chambers. It spares sclera. Acute bacterial endophthalmitis happens 1-7 days post intraocular surgery
M/C causative organism	Staphylococcus epidermidis
Presentation	Severe ocular pain Redness Lacrimation Photophobia Loss of vision Red and swollen lids Conjunctival chemosis Circumcorneal congestion Cloudy cornea Ring infiltration Yellow necrotic wound edges Hypopyon Oedematous and muddy iris ... (content truncated)
Treatment	Intravitreal antibiotics: First choice: Vancomycin 1 mg + Ceftazidime 2.25 mg Second choice: Vancomycin 1 mg + Amikacin 0.4 mg Topical antibiotics Systemic antibiotics Steroid Therapy Supportive Therapy Vitrectomy - if no improvement in 48-72 hours or if severe infection with vision reduced to hand movements close to the face If blind - no perception of light - Evisceration.

Solution for Question 8:

Correct Answer: C) Propionibacterium acnes

Explanation:

The scenario is that of late-onset postoperative endophthalmitis

- Endophthalmitis is inflammation of the inner structures of the eye (uveal tissue and retina) with exudates in the vitreous, anterior, and posterior chambers

Delayed onset endophthalmitis:

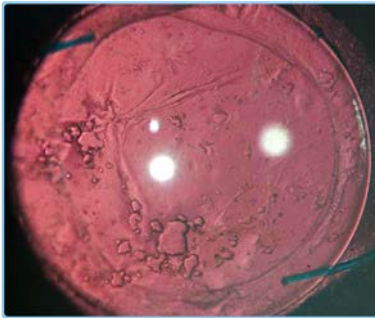
Onset	After 6-weeks of intraocular surgery
Course	Slow, with a panuveitis-like picture
Clinical Features	Granulomatous keratic precipitates Inflammatory deposits over intraocular lens (IOL) White capsular plaque Anterior chamber reaction Vitritis Cystoid macular edema (CME)
Response to Treatment	May respond to periocular or systemic steroids but usually recurs
Common Causative Organisms	Bacterial: <i>Propionibacterium acnes</i> (most common), Coagulase-negative <i>Staphylococcus</i> Fungal: <i>Aspergillus</i> , <i>Candida</i> , and others

Solution for Question 9:

Correct Answer: A) After cataracts

Explanation:

After cataract = Secondary cataract:

Definition	M/C postoperative complication following extracapsular lens extraction.
Clinical Types	 ... (content truncated)
Treatment	ND YAG laser posterior capsulotomy

Solution for Question 10:

Correct Answer: B) Less risk of globe perforation

Explanation:

Peribulbar anesthesia is associated with a lower risk of globe perforation because the anesthetic is injected outside the muscle cone, reducing the likelihood of accidentally penetrating the eye.

Features	Retrobulbar block	Peribulbar block
Injection Site	Inside the muscle cone (intraconal)	Outside the muscle cone (extraconal)
Onset of Anesthesia	Faster onset (Option A)	Slower onset
Akinesia (Eye Movement)	Better akinesia due to direct nerve blockade (Option C)	Variable akinesia, may require multiple injections
Risk of Globe Perforation and Optic nerve injury	Higher risk due to proximity to the globe and optic nerve	Lower risk
Hemorrhage Risk	Higher risk of retrobulbar hemorrhage	Lower risk of severe hemorrhage

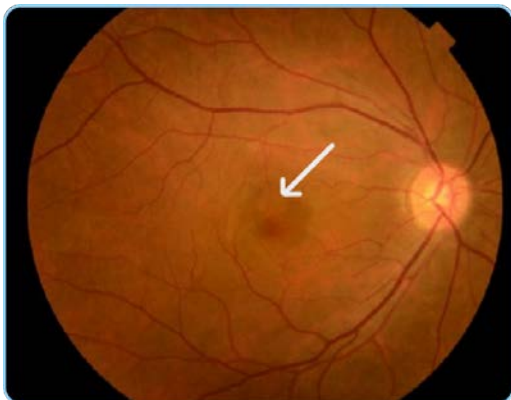
Reduced risk of systemic toxicity (Option D) is generally low in both techniques, but peribulbar anesthesia does not significantly reduce this risk compared to retrobulbar anesthesia.

Solution for Question 11:

Correct Answer: D) All of the above

Explanation:

Cystoid macular edema:



Definition	Collection of fluid in cystic loculi in the Henle's layer of the macula Defect in OPL layer of retina(inner retinal blood barrier break defect) Frequent complication of cataract surgery
Fundoscopy	Honeycomb appearance
Fluorescein Angiography	Flower petal pattern
Prevention	Preoperative and postoperative use of anti prostaglandin eye drops (indomethacin, flurbiprofen, or ketorolac)
Treatment	Anterior vitrectomy Steroids Antiprostaglandins

Solution for Question 12:

Correct Answer: A) Autosomal dominant

Explanation:

Cataracts can be inherited without systemic disorders in the autosomal dominant pattern

Etiology of congenital cataract:

Idiopathic	Sporadic cases with unknown etiology.
------------	---------------------------------------

Hereditary	Inherited without Systemic Disorders: Usually autosomal dominant. Genes: Cry-G (crystallins). Connexins. MIP (Major intrinsic protein). Inherited with Systemic Disorders: Chromosomal disorders (e.g., trisomy 21) Skeletal disorders (e.g., Stickler syndrome) CNS disorders (e.g., cerebro-oculo-facial syndrome) Renal system disorders (e.g., Lowe's syndrome)
Maternal Factors	1. Malnutrition during Pregnancy: Associated with nonfamilial zonular cataract. 2. Infections: Rubella (50% of cases), toxoplasmosis, cytomegalic inclusion disease. 3. Drug Ingestion: Certain drugs (e.g., thalidomide, corticosteroids) during pregnancy. 4. Radiation Exposure: Maternal exposure to radiation during pregnancy.
Foetal or Infantile Factors	1. Deficient Oxygenation (Anoxia): Due to placental hemorrhage. 2. Birth Trauma 3. Metabolic Disorders: Galactosemia, galactokinase deficiency, neonatal hypoglycemia. 4. Associated with Congenital Anomalies: Down's syndrome, Lowe's syndrome, myotonia dystrophica, congenital ichthyosis. 5. Ocular Diseases: Persistent hyperplastic primary vitreous (PHPV), aniridia, anterior chamber cleavage syndrome, retinopathy of prematurity, lenticonus posterior, microphthalmos. 6. Malnutrition in Early Infancy

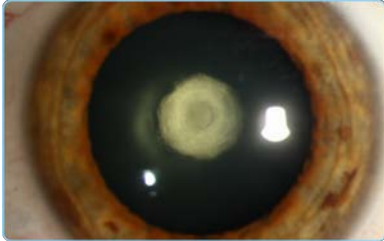
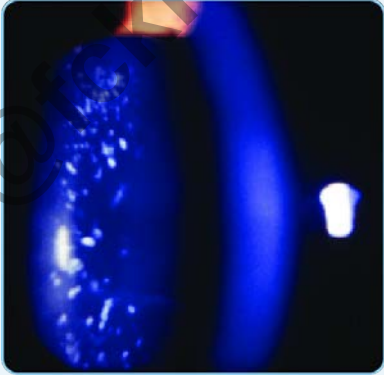
Solution for Question 13:

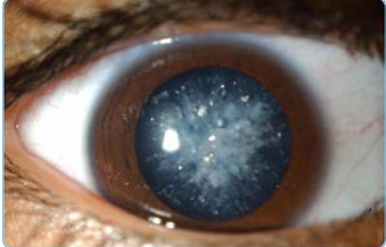
Correct Answer: A) Most common type of congenital cataract

Explanation:

The image is of lamellar cataract

Types of Congenital cataract:

Polar cataracts	Anterior polar cataract: Usually associated with persistent pupillary membrane. Non-axial, stationary, asymptomatic.
 Posterior polar cataract: Onion ring / whorled appearance Associated with : mittendorf dots PHPV ... (content truncated)	
Congenital nuclear cataracts	 ... (content truncated)
 Lamellar cataract: (zonular cataract) Most common congenital cataract affecting vision ... (content truncated)	

 Sutural cataract: Punctate opacities scattered around Y-sutures Patterns: Floriform, Coralliform, Dendritic sutural	
Generalized cataracts	Coronary cataract: Adolescent onset cataract Numerous club-shaped opacities in the periphery
 Blue dot cataract (cataracta-punctata-caerulea): Most common type of congenital cataract (Option A) Developmental cataract (seen > 1 year child) Does not cause vision problems Associated with Down's syndrome	
Total congenital cataract (TORCH infections)	

Solution for Question 14:

Correct Answer: D) Urine assay for the presence of reducing substances

Explanation:

Given clinical scenario is indicative of Classical galactosemia.

Description	Inborn error of galactose metabolism.
-------------	---------------------------------------

Description and types	Classical Galactosaemia: deficiency of galactose-1-phosphate uridylyltransferase (GPUT). Non-classical type: Galactokinase deficiency
Characteristic Features	Frequently leads to bilateral cataract formation. Cataracts often present as oil droplet central lens opacities. Recurrent infections with Escherichia coli. Mental retardation/developmental delay.
Management and prognosis	Avoidance of milk and milk-based products. Cataract changes are reversible.

Biochemical analysis:

Blood analysis	Levels of blood glucose, calcium and phosphorus to rule out Hyperglycemia and Hypocalcemia. (Option A)
Red blood cell transferase and galactokinase levels - for Galactosemia.	
Antibody titres for infections such as toxoplasmosis, rubella, cytomegalovirus and herpes simplex virus (TORCH) and hepatitis B virus. (Option B)	
Urine analysis	Urine assay for reducing substances - for Galactosaemia. (Option D) Screening for amino acids in the urine - if Lowe syndrome is suspected. (Option C) Blood in urine - Alport syndrome Estimation of copper in urine - Wilson disease

Solution for Question 15:

Correct Answer: B) Perform immediate cataract surgery

Explanation:

Management of congenital cataract:

Timing of Surgery	Unilateral cataracts: as early as possible. (Option B) After 1 month of age for bilateral cataracts. (Option A)
Surgical Technique	Small, ideally 1–1.5 mm limbal incisions. Posterior capsulorhexis with pars plana vitrectomy is done additionally to reduce the risk of After-cataract.

Optical Rehabilitation	Contact lenses/glasses: for children below 2 years age IOLs: Generally preferred in children above 2 years age Foldable hydrophobic acrylic IOLs are preferred over PMMA. Always placed with undercorrection.
------------------------	--

Prescribe glasses and monitor the cataract progression (Option C): Glasses are used post cataract surgery and are not the mainstay of treatment.

Wait until the child develops strabismus before considering surgery (Option D): Waiting until complications is not advised in dense unilateral cataracts as it can lead to permanent vision impairment.

Solution for Question 16:

Correct Answer: B) Limb hypoplasia

Explanation:

Pearly white cataract and salt & pepper fundus with antenatal history of fever in early pregnancy is classical of Congenital Rubella syndrome.

Cause	Maternal rubella infection acquired in 1st trimester
Clinical Features	a. Ocular Defects: Congenital cataract (2nd most common manifestation) Salt and pepper chorioretinopathy (most common feature) Microphthalmia (Option C) Cloudy cornea Poorly dilating pupil b. Ear Defects: Sensorineural hearing loss (Option D) c. Cardiovascular Defects: Patent ductus arteriosus (Option A) Pulmonary stenosis Ventricular septal defects
Diagnosis	RT-PCR/IgM antibody titres
Prevention	Administration of rubella vaccine at least three months before pregnancy.

Differential diagnoses for Salt and pepper fundus:



Congenital Rubella syndrome
 Congenital Syphilis
 Myotonic dystrophy
 Retinitis pigmentosa sine pigmento
 Leber's amaurosis

Solution for Question 17:

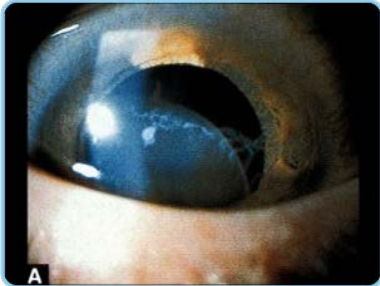
Correct Answer: B) Associated with supero-temporal lens dislocation

Explanation:

The given clinical scenario is suggestive of Weill-Marchesani syndrome, in which there is infero-nasal subluxation of lens not supero-temporal displacement

Ectopia Lentis - Systemic associations:

Marfan's Syndrome	Autosomal dominant
 ... (content truncated)	
Arachnodactyly, long limbs, hypermobile joints, cardiovascular defects	
Homocystinuria	Autosomal recessive

 ... (content truncated)	
Detection of homocysteine in urine using the sodium nitroprusside test.	
Weill-Marchesani Syndrome	Autosomal recessive (Option A)
 Microspherophakia, forward subluxation of lens	
Short stature, brachydactyly, joint stiffness, cardiovascular defects.	
Ehlers-Danlos Syndrome	Lens subluxation, blue sclera, Hyperextensible joints
Stickler Syndrome	Liquified vitreous, retinal detachment, Pre-senile cataract
Oro-facial abnormalities, arthropathy	
Sulphite Oxidase Deficiency	Autosomal recessive
Ectopia lentis, Progressive muscular rigidity	
Death usually by 5 years of age	

Molecular genetic analysis is confirmatory (Option C): Genetic testing for ADAMTS10/ADAMTS17 (Autosomal Recessive) or FBN1 (Autosomal dominant) variants is diagnostic of Weill-Marchesani syndrome.

The usual age of presentation is 5-10 years (Option D): The mean age of presentation is 7.5 years, the majority of cases present in the first decade of life.

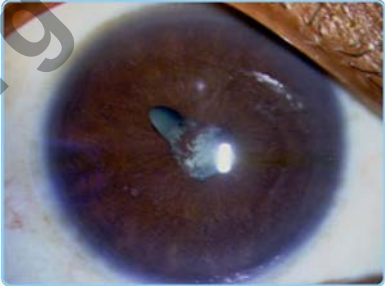
Solution for Question 18:

Correct Answer: D) 1,2,3,4,5,6

Explanation:

All of the listed features are that of complicated cataracts

Complicated cataract:

Definition	Cataracts which develop as a complication of some other intraocular disease
Causes (mnemonic: UMAR: -Uveitis -Myopia -Angle-closure glaucoma -Retinitis pigmentosa)	Inflammations (Anterior uveitis is the most common cause of complicated cataracts) Degenerative conditions such as retinitis pigmentosa and myopic chorioretinal degeneration Long-standing retinal detachment Glaucoma (primary or secondary) Intraocular tumors in late stages (retinoblastoma or melanoma)
Features	 Starts as a posterior subcapsular cortical cataract (PSC) Irregular outline Variable in density ... (content truncated)

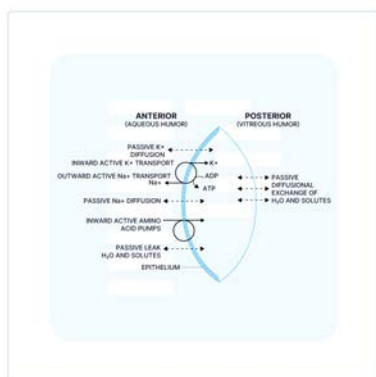
Solution for Question 19:

Correct answer: B) Increased intralenticular Na⁺ levels and decreased K⁺ levels.

Explanation:

Pump leak theory:

- The mechanism that prevents water from getting into the lens is known as the pump-leak theory.
- Over the Anterior surface of the lens, Na⁺-K⁺ ATPase pumps out Na⁺ and pumps in K⁺ into the lens.
- The leak happens in the posterior surface i.e. ions (K⁺) and molecules passively diffuse across a concentration gradient.



Enhanced ion exchange efficiency (Option C): Dysfunction of the sodium-potassium ATPase pump leads to decreased ion exchange efficiency, not enhanced efficiency.

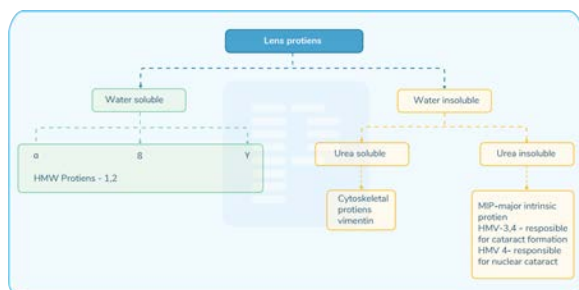
Solution for Question 20:

Correct Answer: B) 1 & 2

Explanation:

- Lens proteins are broadly divided into soluble fractions and insoluble fractions.
- The water-soluble fraction consists of crystallins $\alpha/\beta/\gamma$ (Statement 1), which make up about 90% of the lens's total protein content.
- The α crystallins are the largest crystallins (Statement 4) followed by β and γ respectively.
- The insoluble fraction of the lens consists of albuminoids - mainly major intrinsic proteins (MIP26/Aquaporin-0) and other cytoskeletal proteins. (Statement 2)

- A decrease in crystallins and an increase in albuminoids can lead to cataract formation. (Statement 3)
- Other proteins: HMV-3,4; accumulation of HMV-4 can give rise to nuclear cataract.



Solution for Question 21:

Correct Answer: D) Vision loss occurs at a late stage.

Explanation:

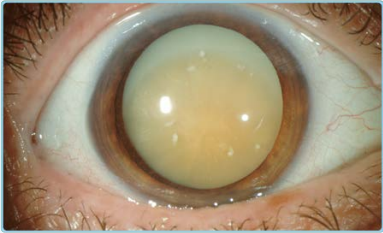
An elderly patient with clinical features (glare/ diminution of vision/ blurred vision) suggestive of cataract and characteristic wedge-shaped opacities starting from periphery to center are noted, which points to a diagnosis of Cuneiform Senile Cortical Cataract.



Type of Cataract	Characteristics	Visual Impact
Cuneiform Senile Cortical Cataract	Wedge-shaped opacities with clear areas in between. Extends from periphery to center. First appears in the lower nasal quadrant. Present in both anterior and posterior cortex.	Visual disturbances are noted at a comparatively late stage (since it starts at the periphery). (Option D) Vision improves in bright light with contracted pupils. (Option B)

	The radial spoke-like pattern of opacities is seen.	
Cupuliform Senile Cortical Cataract	Saucer-shaped opacities are typically in the central part of the posterior cortex (posterior subcapsular cataract). (Option C) Extends from center to periphery.	Causes an early loss of visual acuity.- Due to the involvement of the nodal point Glare seen in PSC Better vision in dim light with dilated pupils.

Stages of Cortical cataract:

Stage	Description	Characteristics
1. Stage of Lamellar Separation	Earliest stage of senile cataract	Formation of vacuoles and water clefts within the cortex Reversible change Can only be demonstrated by slit lamp examination
2. Stage of Incipient Cataract	Detectable opacities with clear areas present between. Causes visual disturbances.	Cuneiform type: Wedge-shaped opacities, extending from the equator appear as radial spoke-like opacities.
Cupuliform type: Saucer-shaped opacity in the central posterior cortex.		
3. Immature Senile Cataract	Progression of opacification	Iris shadow visible (clear cortex visible) -Iris Shadow present here due to opacities + clear areas (Option A) Can progress to Intumescent Cataract leading to the shallow anterior chamber.
4. Mature Senile Cataract (aka intumescent cataract)	Complete opacification of cortex	The lens appears pearly white Referred to as a "ripe cataract" No iris shadow as the whole lens is opacified with a chalky white color
5. Hypermature Senile Cataract (HMSC)	Advanced stage of cataract beyond maturity	 Morgagnian Hypermature Cataract: Cortex liquefies Nucleus sinks to the bottom Bag of milky fluid appearance

Sclerotic Hypermature Cataract: The cortex disintegrates, lens and capsule shrink. Deep anterior chamber Iridodonesis may be present.		
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Solution for Question 22:

Correct Answer: C) Plan for Phacoemulsification.

Explanation:

Gradual progressive loss of vision and improvement of near vision ("Second sight") with the slit lamp finding points towards diagnosis of Nuclear Cataract.



Description and Pathophysiology	Nuclear cataract/sclerosis- The etiology is nuclear sclerosis. Results from the gradual loss of lens elasticity and transparency due to dehydration, and increase in water-insoluble proteins.
Clinical Features	Glare Diplopia Colored Halos Blurred Vision/ Gradual Vision Loss Black spots in front of eyes. Deterioration of vision: In some patients with nuclear sclerosis, near vision improves due to progressive index myopia. Such patients may have improvement in their near-vision glasses. This improvement in near vision is

	referred to as 'second sight.'
Grading	Nuclear Opacity (NO): Ranges from 1 (slight opacification) to 5 (severe opacity). Nuclear Color (NC): Ranges from 1 (amber) to 5 (black). Cataracta brunescens (amber-brown), Cataracta rubra (red), Cataracta nigra (black)
Management	Surgical measures: Mainstay of treatment. Cataract surgeries: ICCE (not preferred now) ECCE SICS (Small Incision Cataract Surgery) Phacoemulsification (most preferred) Followed by IOL implantation: PC-IOL (Posterior chamber-intraocular lens) is preferred.

Advise lifestyle modification and review after 3 months (Option A): Would not address the cataract directly and would not improve the visual acuity.

Recommend corticosteroids eye drops and reassess in 6 months (Option B): The use of topical corticosteroids after cataract surgery reduces postoperative noninfectious inflammation. But it is not used as the treatment for nuclear cataract.

Prescribe her rigid gas-permeable contact lenses (Option D): Rigid gas permeable contact lenses are indicated for early-stage keratoconus with astigmatism.

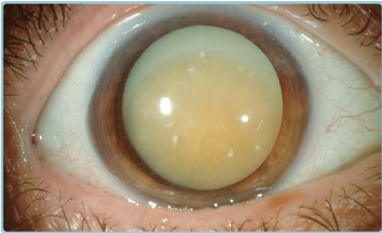
Solution for Question 23:

Correct answer: C) Phacolytic glaucoma - Secondary open-angle glaucoma

Explanation:

Given clinical scenario is suggestive of Glaucoma and the slit lamp image of a milky white cortex with a sunken nucleus corresponds to the Morgagnian type of hyper mature cataract - leading to secondary open-angle glaucoma called Phacolytic glaucoma. (Option C) (Option D)

Feature	Phacomorphic Glaucoma	Phacolytic Glaucoma
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Pathogenesis	Intumescent cataract blocks the pupil. this pushes the iris forward, blocking the anterior chamber angle, leading to angle-closure glaucoma.	 Leakage of lens proteins from a hypermature cataract clogs the trabecular meshwork.
Causes	Intumescent cataract (mature cataract) Anterior subluxation of lens Spherophakia	Hypermature (Morgagnian) cataract Obstruction of the trabecular meshwork by lens proteins.
Clinical Features	Symptoms are similar to acute primary angle-closure glaucoma. Shallow anterior chamber. Ciliary congestion, dilated pupil, and swollen cataract is visible. Closed anterior chamber angle on gonioscopy.	Features of acute congestive glaucoma due to high IOP. Deep anterior chamber. Open anterior chamber angle on gonioscopy.
Management	Medical Treatment: IV mannitol, systemic acetazolamide, topical beta-blockers. Laser Iridotomy: To relieve angle closure. Surgical Treatment: Cataract extraction with PCIOL implantation.	Medical Therapy: To lower IOP. Surgical Treatment: Extraction of the hypermature cataract and PCIOL implantation.

Phacomorphic glaucoma - Secondary open-angle glaucoma (Option A), Phacomorphic glaucoma - Secondary closed-angle glaucoma (Option B): Phacomorphic glaucoma is seen in the intumescent stage of cataract - a type of secondary closed-angle glaucoma.

Solution for Question 24:

Correct Answer: D) All of the above

Explanation:

The scenario is that of presbyopia

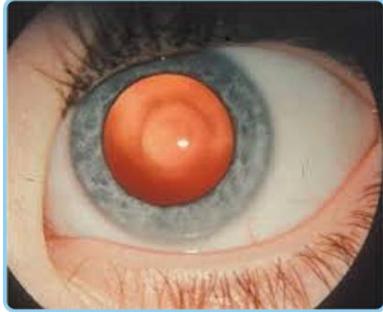
Accommodation	The mechanism by which the eyes focus diverging rays from a near object on the retina
Mechanism	When looking at a near object from a far object: Ciliary muscles contract → Zonules relax → Lens becomes more convex anteriorly
Presbyopia	Physiological insufficiency of accommodation, causing worsening near vision after the age of 40
Causes	Decrease in elasticity of lens capsule (Option A) Sclerosis of lens substance (Option B) Reduced ciliary muscle power (Option C)
Treatment	Convex glasses for near-work A guide for powers required at various ages: 45 years: +1 to +1.25D 50 years: +1.5 to +1.75D 55 years: +2 to +2.25D 60 years: +2.5 to +3.0D

Solution for Question 25:

Correct Answer: C) Periphery of the lens

Explanation:

Galactosemia cataracts are central lens cataracts, not peripheral lens cataracts

Galactosemia	Inborn error of galactose metabolism
Enzyme deficiency	Galactose-1-phosphate uridylyltransferase (GPUT) Galactokinase (GK)
Type of cataract	 Bilateral (Option A) "Oil droplet" (Option B) ... (content truncated)

Features	May be reversed (Option D) May be prevented if milk and milk products are eliminated at an early stage
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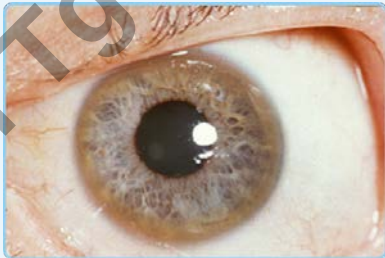
Solution for Question 26:

Correct Answer: D) Oil droplet cataract

Explanation:

Oil droplet cataract is seen in galactosemia, not chalcosis.

Chalcosis:

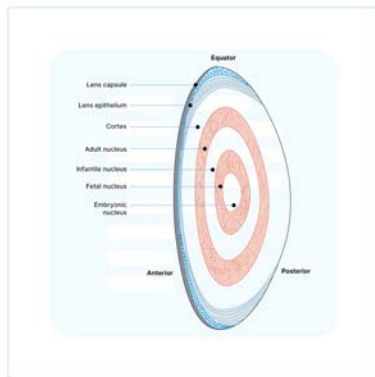
Definition	Specific changes produced by the alloy of copper in the eye
Mechanism	Copper ions from the alloy are dissociated electrolytically and deposited under the membranous structures of the eye
Clinical features	 ... (content truncated)

Solution for Question 27:

Correct Answer: B) Posterior pole

Explanation:

The Capsule at the posterior pole (4 microns) is the thinnest part.



Lens capsule: Thickest basement membrane in the body.

Posterior pole: Thinnest part (Capsule), measuring around 4 micrometers, due to fewer cells and less collagen.

Lens Nucleus: The central part contains the oldest fibers in different zones, which are laid down successively as the development proceeds.

- Embryonic nucleus: Innermost part of nucleus corresponding to the lens up to the first 3 months of gestation. Consists of primary lens fibers.
- Fetal nucleus: Surrounding embryonic nucleus which corresponds to the lens from 3 months of gestation till birth. Fibers are anteriorly Y-shaped and posteriorly inverted Y-shaped.
- Infantile nucleus: Corresponds to the lens from birth to puberty and surrounds the fetal nucleus.
- Adult nucleus: Corresponds to lens fibers formed after puberty to rest of life.

Anterior pole (Option A): Thicker part of the lens capsule, providing support and maintaining curvature.

Equator (Option C): Moderately thick, where zonular fibers attach for lens accommodation.

Pre-equatorial region (Option D): The thickest portion of the lens capsule measuring 12-14 micrometers supports zonular attachments.

Solution for Question 28:

Correct Answer: B) a3,b2,c1

Explanation:

Toxic cataract:

- Steroids - Prolonged use in high doses may cause posterior subcapsular cataracts. Children are more susceptible
- Other causes of posterior subcapsular cataract: Galactosemia, Myotonic dystrophy, BLUNT trauma, complicated cataract, chloroquine, busulfan, NF2,
- Glaucoma by TOPICAL steroids and cataract by SYSTEMIC STEROIDS (mnemonic: GTCS)
- Miotics - Especially long-acting cholinesterase inhibitors such as echothiophate, demecarium bromide, and diisopropyl fluorophosphate (DFP) may cause anterior subcapsular-granular cataracts
- Other causes of anterior subcapsular cataract - Amiodarone, chlorpromazine, busulfan, gold, and allopurinol
- AACG (Mnemonic): Atopic dermatitis- shield cataract, Amiodarone, Chlorpromazine, Gold deposit, AACG(acute angle closure glaucoma)

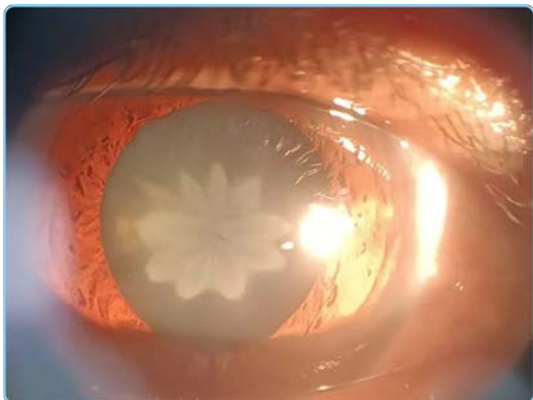
Radiation cataract: Infrared (heat) cataract-



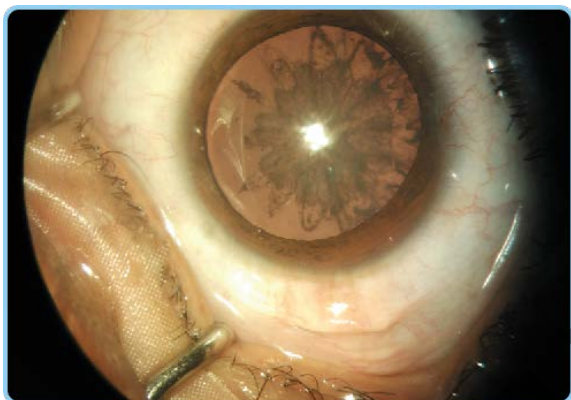
- Infrared aka glassblower cataract
- Lens capsule exfoliation with discoid posterior opacities
- X-ray, gamma rays, neutron exposure: Inadequately protected technicians, patients treated for malignant tumors, and workers of atomic energy plants are at risk.

Traumatic cataract: blunt trauma: Rosette cataract

- It is a cortical cataract that starts from the posterior cortex.



Electric Shock cataract/ Lightning cataract: It leads to Stellate ASC



Solution for Question 29:

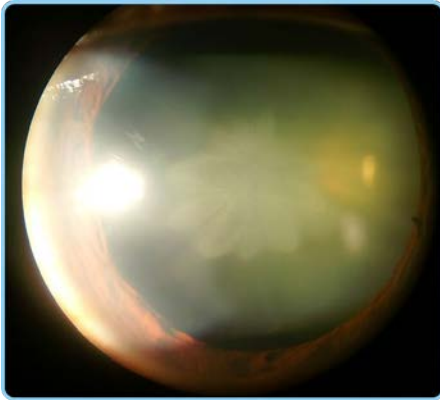
Correct Answer: C) Rosette cataract

Explanation:

Early Rosette Cataract is characterized by feathery opacities along suture lines, as seen in this patient.

- A traumatic cataract is an acquired cataract resulting from ocular trauma, such as penetrating or blunt injuries.
- It forms due to degeneration in previously normal lens fibers.
- Morphological Forms Subepithelial Opacities: Most common presentation. Rosette Cataracts:

- Subepithelial Opacities: Most common presentation.
- Rosette Cataracts:
- Subepithelial Opacities: Most common presentation.
- Rosette Cataracts:



- Early Rosette: Feathery opacities along posterior cortex suture lines; typical of concussion cataracts. (Option C)
- Late Rosette: Develops 1-2 years post-injury, with shorter, compact sutural extensions.
- Diffuse concussion cataract: Common presentation of total cataract following blunt trauma, which may accelerate senile cataract development.

Subepithelial opacities (Option A) are common but do not typically present with feathery opacities along suture lines.

Diffuse concussion cataract (Option B) involves widespread lens opacity, not localized feathery patterns.

Solution for Question 30:

Correct Answer: C) Diabetic cataract

Explanation:

Diabetic cataract (aka Presenile cataract) is characterized by bilateral, snowflake-like white opacities in the lens cortex, typically seen in young adults with diabetes. Opacities may either resolve spontaneously or progress to maturity within days.

- Senile cataract (Option A) typically affects older individuals, although it can occur earlier in diabetics; however, it does not present with snowflake-like opacities.
- Christmas tree cataract (Option B) is associated with myotonic dystrophy, not diabetes.
- Posterior subcapsular cataract (Option D) can be seen in various conditions, including myotonic dystrophy, but does not present with the snowflake-like opacity as seen in true diabetic cataract.

Solution for Question 31:

Correct Answer: A) Intracapsular Cataract Extraction (ICCE)

Explanation:

ICCE may be more suitable due to the patient's narrow-angle glaucoma history. ICCE allows for the complete removal of the lens along with the capsule, which may help prevent potential complications associated with retained lens material. However, it is worth noting that ECCE and phacoemulsification are more common and may be preferred in most scenarios.

- Effective when zonules are weak or damaged.
- Indication: Primarily used for significantly subluxated or dislocated lenses.

Lens Extraction: Methods include Indian Smith, cryoextraction, or capsule forceps.

- Indian Smith Method: Pressure applied at 6 and 12 o'clock to express the lens.
- Cryoextraction
- Forceps Method: Hold the lens with Arrugas forceps.

Solution for Question 32:

Correct Answer: B) Viscoelastic injection

Explanation:

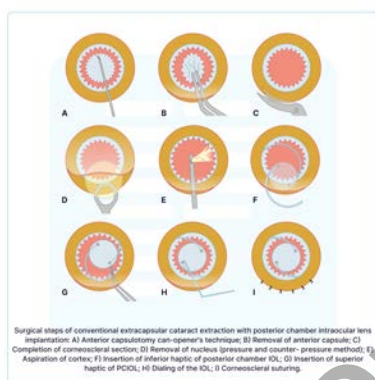
This step involves injecting a viscoelastic substance into the anterior chamber to maintain its depth and protect the corneal endothelium during the surgery.

- Extracapsular Cataract Extraction (ECCE): A type of cataract surgery where the lens is removed but the posterior capsule remains intact, allowing for various techniques to be applied based on patient age and lens stability.

ECCE Techniques

Surgery	Site of incision	Size of incision
Conventional ECCE	At the limbus	8-10mm
SICS	Sclera	5-7mm
Phacoemulsification	Cornea	2.2-3.2mm
Microincision (MICS)	Cornea	<1.8mm

Surgical Steps of Conventional ECCE



Advantages of ECCE over ICCE

- ECCE is suitable for all ages with intact zonules.
- ECCE allows for posterior chamber IOL implantation, unlike ICCE.
- Lower risk of vitreous-related complications with ECCE.
- Lower rates of endophthalmitis, cystoid macular edema, and retinal detachment with ECCE.
- ECCE reduces the risk of secondary rubeosis in diabetics.
- ECCE carries a lower risk of Postoperative Astigmatism.
- Better prognosis for later glaucoma surgery or corneal transplantation with ECCE.

Solution for Question 33:

Correct Answer: D) Manual SICS usually results in a higher level of postoperative astigmatism compared to Phacoemulsification.

Explanation

Manual SICS usually results in a higher level of postoperative astigmatism compared to Phacoemulsification as they use larger incisions.

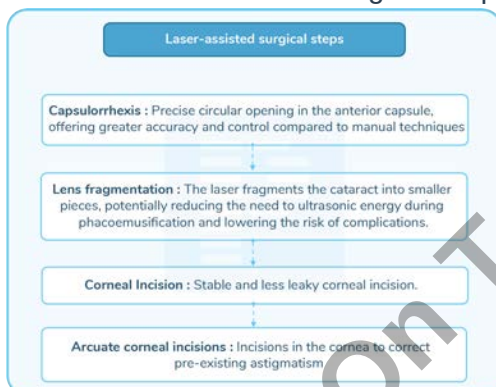
Aspect	Manual SICS	Phacoemulsification
Procedure Type	Extracapsular Cataract Extraction (ECCE)	Extracapsular Cataract Extraction (ECCE)
Incision Type	Self-sealing, valvular, sclerocorneal tunnel incision	Self-sealing, valvular, sclerocorneal tunnel incision or clear corneal incision.
Anesthesia	Local anesthesia (can be performed under)	Local anesthesia (can also be performed under topical anesthesia)
Nucleus Removal Method	Manually extracted through the incision after being maneuvered out of the capsular bag	Fragmented using an ultrasonic probe and then aspirated
Suitability (Option C)	Suitable for all cataract densities, including very hard cataracts	Typically requires a smaller incision, less suitable for very hard cataracts
Visual Recovery (Option B)	Slower visual recovery, delayed return to optimal vision	Faster visual rehabilitation due to smaller incision and reduced tissue manipulation
Astigmatism (Option A)	Higher levels of surgically induced astigmatism, larger incision	Less postoperative astigmatism due to smaller incision size
Complication Risks	Lower risk of nuclear drop Higher risk of postoperative pain and inflammation, hyphema and conjunctival congestion	Lower risk of postoperative pain, conjunctival congestion, hyphema and inflammation
Overall Accessibility	More accessible due to cost-effectiveness and versatility	Preferred in settings with better resources and where faster recovery is prioritized

Solution for Question 34:

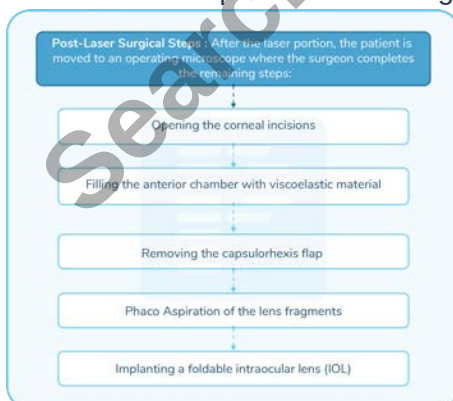
Correct Answer: C) Capsulorrhexis → Lens Fragmentation → Corneal Incision → IOL implantation

Explanation:

- Femtosecond laser-assisted cataract surgery (FLACS) represents an advanced method in cataract treatment, classified as a microincision cataract surgery (MICS) technique due to its use of a very small incision, typically less than 2 mm.
- FLACS employs a femtosecond laser (FSL) to perform specific surgical steps with enhanced precision and safety. This precision can reduce the amount of ultrasound energy needed during phacoemulsification, which is advantageous for patients with preexisting retinal conditions, such as diabetic retinopathy.
- FLACS is suitable for patients with pre-existing astigmatism, as the femtosecond laser can create precise incisions that help correct astigmatism while offering enhanced precision in other surgical steps.
- Laser-Assisted Surgical Steps:



- Post-Laser Surgical Steps: After the laser portion, the patient is moved to an operating microscope where the surgeon completes the remaining steps:



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Previous Year Questions

1. A patient with a long-standing history of diabetes presents with cataracts. Which of the following metabolic products is primarily responsible for cataract formation in this condition?

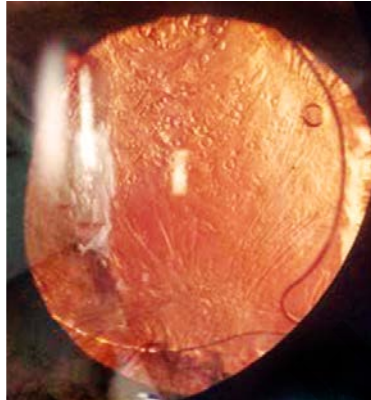
- A. Mannitol
- B. Galactitol
- C. Sorbitol
- D. Fructose

2. A 2-month-old child is brought to the ophthalmology OPD with the following presentation. What is the ideal management?



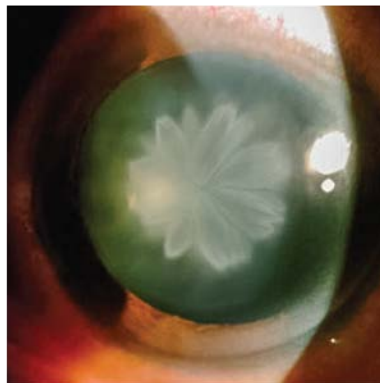
- A. Medical management
- B. Operate immediately
- C. Surgery after 6 months
- D. Surgery after 2 years

3. A patient presents with decreased vision three years after surgery and exhibits this condition. What is the most likely diagnosis?



- A. Pseudo-bullous keratopathy
- B. Corneal dystrophy
- C. Posterior subcapsular cataract
- D. Posterior capsular opacification (PCO)

4. Which is the type of cataract seen following a blunt trauma?



- A. Rosette cataract

- B. Christmas tree
 - C. Snowflake
 - D. Sunflower cataract
-

5. Oil drop cataract is seen in?

- A. Galactosemia
 - B. Wilson's disease
 - C. Gaucher's disease
 - D. Tay Sachs disease
-

6. What dye is used in staining for the anterior capsule of lens?

- A. Trypan blue
 - B. Brilliant blue
 - C. Fluorescent green
 - D. Lissamine green
-

7. In which type of cataract surgery is suturing with 10-0 nylon commonly performed?

- A. Phacoemulsification
 - B. Extracapsular cataract extraction (ECCE)
 - C. Intracapsular cataract extraction (ICCE)
 - D. Manual small incision cataract surgery (MSICS)
-

8. Which muscle is paralyzed in a patient who has left-sided head tilt resulting in right hypotropia that worsens with right head tilt and dextroversion?

- A. Right superior oblique
 - B. Right superior rectus
 - C. Right inferior oblique
 - D. Left inferior oblique
-

9. Ectopia lentis is seen in?

- A. Homocystinuria
 - B. Refsum's disease
 - C. Wilson's disease
 - D. Sarcoidosis
-

10. Which laser is considered the most appropriate for cataract surgery?

- A. Nd-YAG
 - B. Diode
 - C. Excimer
 - D. Femtosecond laser
-

11. What could be the most probable cause for the progressive vision loss in both eyes of a 25-year-old male who has been undergoing low-dose oral steroid therapy for the past decade?

- A. Cataract
 - B. Glaucoma
 - C. Cystoid macular edema
 - D. Retinal detachment
-

12. Which of the following does not characterize complicated cataracts?

- A. Krukenberg spindle
 - B. Polychromatic lustre
 - C. Occurs after uveitis
 - D. Breadcrumb appearance
-

13. After cataract surgery, endophthalmitis has developed in a patient. On examination, the vision is better in hand movements. What is the treatment?

- A. Intravitreal antibiotic injections
 - B. Pars planar vitrectomy
 - C. Redo surgery
 - D. Topical antibiotics
-

14. Which of the following cataracts is seen in blunt trauma?

- A. Rosette cataract
- B. Toxic cataract
- C. Metabolic cataract
- D. Nuclear cataract

15. What kind of laser can be utilized to treat posterior capsular opacification, which is observed during an examination of a patient who has reported decreased visual acuity following cataract surgery?

- A. Femto laser
 - B. Nd YAG laser
 - C. Argon fluoride
 - D. Argon
-

16. Rosette cataract is seen after?

- A. Blunt trauma to eye
 - B. Copper foreign body in eye
 - C. Penetrating injury to eye
 - D. Infection
-

17. What could be the likely diagnosis for a 55-year-old male patient who visited the OPD with complaints of experiencing glare while driving at night, despite having cataract surgery a year ago? The patient's best corrected visual acuity is 6/12 in the right eye and 6/9 in the left eye, with no improvement observed when using a pinhole.

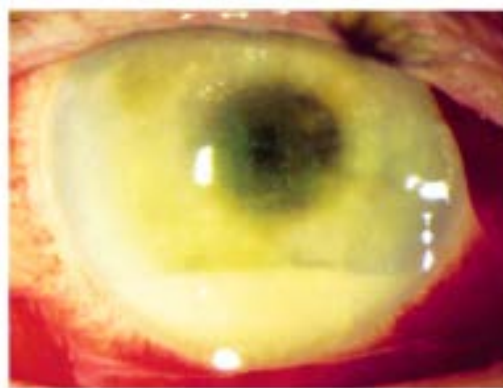
- A. Cystoid macular edema
- B. Pseudophakic bullous keratopathy
- C. ARMD
- D. Posterior capsular opacification

18. The following image depicts a particular type of cataract. What is the most common cause of this cataract?



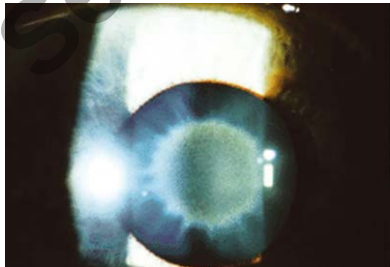
- A. Penetrating trauma
- B. Steroid use
- C. Diabetes
- D. Blunt trauma

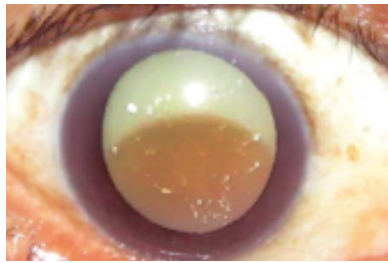
19. A patient, 36 hours after cataract surgery, presents with complaints of pain and visual disturbance. His eye is as shown below. What is the management of this condition?



- A. Intravenous steroids
- B. Intravenous antibiotics
- C. Bandage and padding of the eye
- D. Intravitreal antibiotics

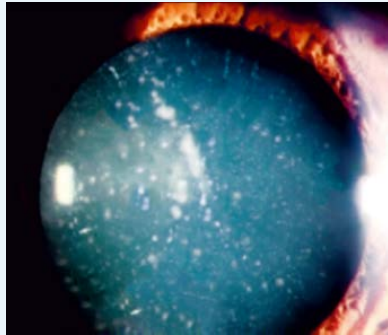
20. Match the given images to their descriptions: 1. a) Sunflower cataract 2. b) Morgagnian cataract 3. c) Snowflake cataract 4. d) Posterior subcapsular cataract

 1.	a) Sunflower cataract
 2.	b) Morgagnian cataract



3.

c) Snowflake cataract

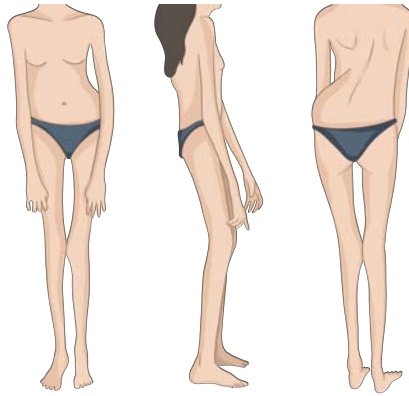


4.

d) Posterior subcapsular cataract

- A. 1-a, 2-d, 3-b, 4-c.
- B. 1-d, 2-b, 3-a, 4-c.
- C. 1-d, 2-a, 3-b, 4-c.
- D. 1-c, 2-b, 3-a, 4-d.

21. Riya was found to have a height of more than 95 percentile for her age and an arm-span to height ratio of more than 1.05. On further examination, she had flat feet, mild scoliosis, and long slender fingers. An image of her showing characteristic signs is provided below. What is the most common ocular complication expected in this child?



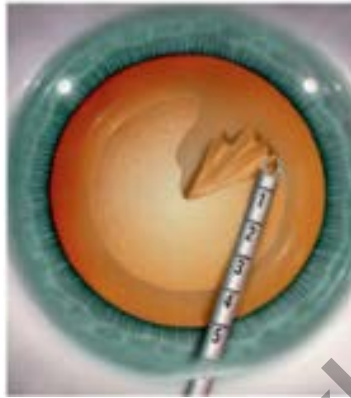
- A. Hypertensive retinopathy
- B. Rosette cataract
- C. Retinitis pigmentosa
- D. Subluxation of lens

22. Can you determine the observation indicated by the arrow in the image provided?



- A. Vossius ring
- B. Weiss ring
- C. Fleischer's ring
- D. Kayser Fleischer ring

23. Identify the surgical step shown in the image given below:



- A. Capsulorrhexis
- B. Hydrodissection
- C. Intraocular lens implantation
- D. Lens aspiration

24. A 65-year-old diabetic woman presents with painless loss of vision. The ocular findings are shown in the image below. What is the likely diagnosis and management for this patient?



- A. Zonular cataract and lensectomy

- B. Rosette cataract and sugar control
- C. Sunflower cataract and lesion extraction
- D. Immature senile cataract and phacoemulsification with IOL

25. A child with elevated liver enzymes presented to the OPD with the following eye finding on gonioscopy. It is associated with?



- A. Copper
- B. Zinc
- C. Selenium
- D. Iron

26. What is the probable medical condition for a 70-year-old individual who has relied on presbyopia glasses but can now comfortably read the newspaper without them?

- A. Index myopia
- B. Index hypermetropia
- C. presbyopia
- D. None of the above

27. What substance is responsible for the development of a diabetic cataract in a 68-year-old female patient, which has been present for 3 months?

- A. Glucose
 - B. Fructose
 - C. Lactose + Glucose
 - D. Sorbitol
-

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Correct Answers

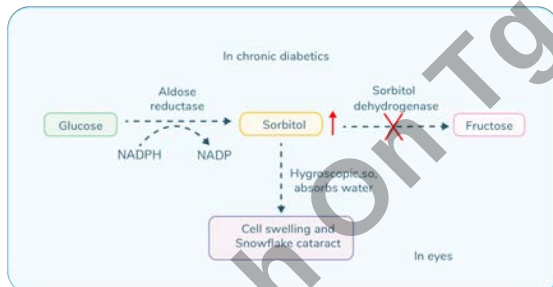
Question	Correct Answer
Question 1	3
Question 2	2
Question 3	4
Question 4	1
Question 5	1
Question 6	1
Question 7	2
Question 8	2
Question 9	1
Question 10	4
Question 11	1
Question 12	1
Question 13	1
Question 14	1
Question 15	2
Question 16	1
Question 17	4
Question 18	4
Question 19	4
Question 20	3
Question 21	4
Question 22	1

Question 23	1
Question 24	4
Question 25	1
Question 26	1
Question 27	4

Solution for Question 1:

- In individuals with diabetes, high blood glucose levels can lead to increased activity of the enzyme aldose reductase. This enzyme converts glucose to sorbitol.
- Sorbitol is poorly metabolized and can accumulate within the lens of the eye.
- The accumulation of sorbitol leads to osmotic changes, drawing water into the lens and causing it to swell. This swelling and disruption of lens fibers leads to cataract formation.

Pathophysiology of Diabetic Cataract Formation:



Mannitol (Option A): Mannitol is an osmotic diuretic and is not involved in cataract formation.

Galactitol (Option B): Galactitol (dulcitol) accumulates in the lens in galactosemia, leading to early-onset cataracts, but not in diabetic cataracts.

Fructose (Option D): While sorbitol is eventually converted to fructose by sorbitol dehydrogenase, the rate is slow in the lens, making sorbitol the primary culprit in diabetic cataracts.

Solution for Question 2:

Correct Option B - Operate immediately:

- In this situation, the image reveals bilateral congenital cataracts, meaning both eyes are affected. If not treated promptly, this condition can result in irreversible vision loss due to amblyopia. For bilateral cataracts, surgery is typically performed after the child reaches one month of age, so immediate surgery is necessary in this case.

Incorrect Options:

Options A, C, and D are incorrect, as explained above.

Solution for Question 3:

Correct Option D - Posterior capsular opacification (PCO):

- Posterior Capsular Opacification can occur after cataract surgery when residual lens epithelial cells migrate and proliferate on the posterior capsule. Elschnig Pearls are clusters of proliferating lens epithelial cells that form a circular, pearl-like opacity on the posterior capsule, contributing to the visual disturbances(as seen in the given image).

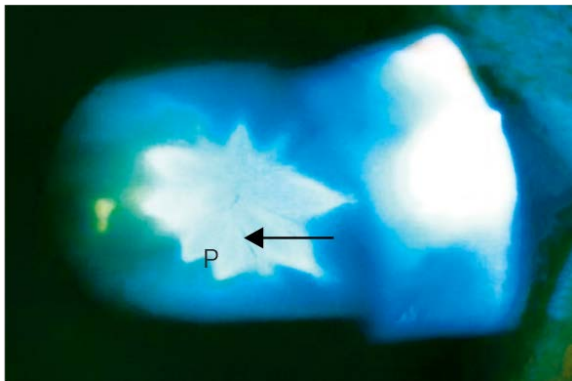
Incorrect Options:

Options A, B, and C do not exhibit Elschnig pearls.

Solution for Question 4:

Correct Option A - Rosette cataract:

- Rosette cataract is usually associated with Blunt trauma
- It is a cortical cataract that starts from the posterior cortex.



Incorrect Options:

Option B - Christmas tree: Christmas-Tree Cataract is associated with Myotonic Dystrophy.

Option C - Snowflake cataract: Diabetes mellitus is associated with Snowflake cataract

Option D - Sunflower cataract: Sunflower cataract is associated with Wilson's disease

Solution for Question 5:

Correct Option A - Galactosemia:

- Galactosemia is associated with Oil drop cataract

Incorrect Options:

- Options B, C & D: Wilson's disease, Gaucher's disease and Tay Sachs disease are not associated with Oil drop cataract

Solution for Question 6:

Correct Option A - Trypan Blue

- Trypan blue is a dye that can be used to stain the anterior capsule of the lens during certain ophthalmic surgeries, such as cataract surgery. It helps to increase the visibility of the

anterior capsule of the lens, making the surgical procedure easier and safer.

Incorrect Options:

Option B - Brilliant Blue: This dye is sometimes used in eye surgery, specifically for staining the internal limiting membrane (ILM) during vitrectomy procedures.

Option C - Fluorescent green (Fluorescein): This dye is used in diagnostic procedures to highlight damage to the cornea.

Option D - Lissamine Green: This dye is used in eye exams to detect damage to the eye's surface, particularly the cornea and conjunctiva.

Solution for Question 7:

Correct Option B - Extracapsular cataract extraction (ECCE)

- In extracapsular cataract extraction (ECCE), the surgeon makes a larger incision on the eye's surface, typically around 10-12 mm in size. This incision allows the surgeon to remove the cataractous lens while leaving the posterior lens capsule intact. After removing the lens, the incision is usually closed with sutures, often using 10-0 nylon. ECCE requires suturing because the incision size is relatively large, and proper wound closure is necessary to ensure the eye heals correctly and maintains its structural integrity.

Incorrect Options:

Option A - Phacoemulsification: Phacoemulsification is a modern and minimally invasive cataract surgery technique. In this procedure, a small incision is made through which an ultrasonic probe (phacoemulsifier) is inserted to break up the cataract into tiny fragments that are then suctioned out. Suturing with 10-0 nylon is not commonly performed in phacoemulsification surgery because the incision size is typically very small and may not require sutures to close.

Option C - Intracapsular cataract extraction (ICCE): Intracapsular cataract extraction involves removing the entire lens, including the lens capsule, in one piece. This method is now less commonly used due to advances in surgical techniques. Suturing with 10-0 nylon is not commonly performed in ICCE surgery because the procedure involves removing the entire lens and capsule, making the need for suturing less relevant.

Option D - Manual small incision cataract surgery (MSICS): Manual small incision cataract surgery is a technique similar to ECCE but uses a smaller incision (around 6-7 mm) compared to

traditional ECCE. While suturing may still be performed in MSICS, the use of 10-0 nylon is more commonly associated with traditional ECCE due to the larger incision size.

Solution for Question 8:

Correct Option B:

Paralysis of the right superior rectus muscle is involved in the described clinical presentation.

- The right superior rectus muscle is responsible for elevating and adducting the eye. When this muscle is paralyzed, it leads to a weakness or paresis of the muscle, resulting in an inability to fully elevate and adduct the affected eye. As a result, the patient presents with a left-sided head tilt to compensate for the vertical misalignment.
- When the patient tries to straighten their head, the misalignment becomes more prominent, leading to right hypotropia (downward deviation of the affected eye). This occurs because the unopposed action of the intact inferior rectus muscle causes the eye to be pulled downward.
- On dextroversion (looking to the right) and right head tilt, the imbalance between the paralyzed right superior rectus muscle and the intact muscles on the opposite side becomes more pronounced, exacerbating the hypotropia.

Incorrect Options:

Option A: Paralysis of the right superior oblique muscle

- Paralysis of the right superior oblique muscle would result in a different pattern of eye movement abnormalities known as trochlear nerve palsy.
- Trochlear nerve palsy typically presents with vertical diplopia (double vision), which worsens on looking downward and toward the affected side.
- It is not associated with head tilt or hypotropia on straightening or dextroversion.

Option C: Paralysis of the right inferior oblique muscle

- Paralysis of the right inferior oblique muscle would lead to a different pattern of eye movement abnormalities.
- Paralysis of the inferior oblique muscle typically causes hypertropia (upward deviation) of the affected eye, not hypotropia (downward deviation) as described in the patient's presentation.

- It is not associated with head tilt or worsening of symptoms on dextroversion or right head tilt.

Option D: Paralysis of the left inferior oblique muscle

- Paralysis of the left inferior oblique muscle would not explain the specific clinical findings mentioned in the patient's presentation.
- The left inferior oblique muscle is not directly involved in the described pattern of head tilt, hypotropia, and worsening on dextroversion or right head tilt.
- It does not account for the specific muscle weakness leading to the observed eye movement abnormalities.

Solution for Question 9:

Correct Option A -Homocystinuria:

- Ectopia lentis refers to the abnormal displacement or dislocation of the lens of the eye. It can occur as a result of various underlying conditions. Among the given options condition associated with ectopia lentis is homocystinuria.
- Other diseases associated with ectopia lentis are: Marfan syndrome. Weill Marchesani syndrome. Ehler's Danlos syndrome. Hyperlysinemia. Sulphite oxidase deficiency. Stickler syndrome.
- Marfan syndrome.
- Weill Marchesani syndrome.
- Ehler's Danlos syndrome.
- Hyperlysinemia.
- Sulphite oxidase deficiency.
- Stickler syndrome.
- Marfan syndrome.
- Weill Marchesani syndrome.
- Ehler's Danlos syndrome.
- Hyperlysinemia.
- Sulphite oxidase deficiency.
- Stickler syndrome.

Incorrect Options:

Option B - Refsum's disease: Refsum's disease, also known as hereditary ataxia polyneuritis, is a rare genetic disorder characterized by the accumulation of phytanic acid in the body. It primarily affects the nervous system and other organs but is not associated with ectopia lentis.

Option C - Wilson's disease: Wilson's disease is an inherited disorder that leads to abnormal copper metabolism in the body, resulting in copper accumulation in various organs, particularly the liver and brain. It manifests as Kayser-Fleischer ring in the cornea.

Option D - Sarcoidosis: Sarcoidosis is a systemic inflammatory disease that can affect multiple organs, including the eyes. Ocular manifestations include band-shaped keratopathy, and uveitis.

Solution for Question 10:

Correct Option D - Femtosecond laser:

- The Femtosecond laser is the most suitable laser for cataract surgery presently. It is used in a procedure called Femtosecond Laser-Assisted Cataract Surgery (FLACS).
- It replaces several manual steps of phacoemulsification with an automated process. The corneal incisions, the capsulorhexis, and initial fragmentation of the crystalline lens, as well as astigmatism-relieving incisions.

Incorrect Options:

Option A - Nd-YAG laser: Nd-YAG laser is primarily used for posterior capsulotomy for the management of posterior capsular opacification.. Nd-YAG laser is not the primary laser used for cataract removal.

Option B - Diode laser: Diode lasers are used in iridotomy for the management of glaucoma.

Option C - Excimer laser: Excimer lasers are primarily used in refractive surgeries such as LASIK (laser-assisted in situ keratomileusis) to reshape the cornea and correct refractive errors.

Solution for Question 11:

Correct Option A - Cataract:

- The most likely reason for the progressive loss of vision in a 25-year-old male who has been on low-dose oral steroid therapy for the last 10 years is Cataract.
- Toxic cataract is caused by prolonged use of steroids that causes disturbance in the osmotic balance in the lens.
- Other drugs causing toxic cataract includes Phenothiazines, Busulfan, Gold, Chloroquine, Amiodarone, Long-acting miotics.

Incorrect Options:

Option B - Glaucoma is the most common complication of topical steroids.

Option C - Cystoid macular edema may be associated with steroid use, but it typically presents with symptoms such as blurry vision, central vision loss, and distortion.

Option D - Retinal detachment causes progressive loss of vision which is not consistent with the above described case.

Solution for Question 12:

Correct Option A - Krukenberg spindle:

- A complicated cataract is a type of cataract that develops secondary to other ocular conditions, such as uveitis, or systemic diseases.
- Krukenberg Spindle refers to a vertical deposition of pigmented material on the posterior surface of the cornea's central endothelium. It's commonly associated with pigment dispersion syndrome and pigmentary glaucoma rather than cataracts.

Incorrect Options:

Option B - Polychromatic Lustre: Aka 'oil droplet' or 'iridescent' cataract is characterized by the presence of multicolored reflections from the lens due to light scattering. This phenomenon is frequently observed in complicated cataracts.

Option C - Occurs after Uveitis: Chronic anterior uveitis is the most common cause of complicated cataract.

Option D - Breadcrumb Appearance: The lens opacities in complicated cataracts resemble breadcrumbs appearance.

Solution for Question 13:

Correct Option A - Intravitreal antibiotic injections:

- Within a week after surgery one can get endophthalmitis
- Endophthalmitis is a purulent inflammation of intraocular fluids
- Treatment: it will depend upon whether the patient is able to perceive the hand movement or doesn't have hand movement. When there is hand movement and above, the treatment of choice would be intravitreal antibiotic injections. In case of only perception of light and no hand movement, Pars plana vitrectomy (option B) is done.
- When there is hand movement and above, the treatment of choice would be intravitreal antibiotic injections.
- In case of only perception of light and no hand movement, Pars plana vitrectomy (option B) is done.
- When there is hand movement and above, the treatment of choice would be intravitreal antibiotic injections.
- In case of only perception of light and no hand movement, Pars plana vitrectomy (option B) is done.

Incorrect Options:

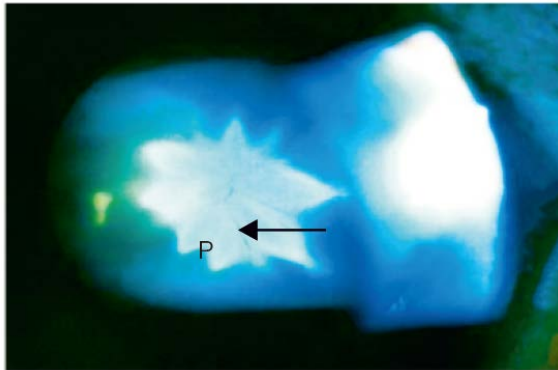
Options C, and D are incorrect.

Solution for Question 14:

Correct Option A - Rosette cataract:

Traumatic cataract:

- Perforating/ penetrating: cataract if it hits the lens
- Blunt trauma: concussion injury
- Features of blunt trauma Rosette-shaped cataract: it is a cortical cataract that starts from the posterior cortex.
- Rosette-shaped cataract: it is a cortical cataract that starts from the posterior cortex.
- Rosette-shaped cataract: it is a cortical cataract that starts from the posterior cortex.



Incorrect Options:

Option B - Toxic cataract:

- It can be caused by steroids, phenothiazines, long acting miotics, amiodarone, busulfan, gold, chloroquine

Option C - Metabolic cataract:

- Due to Diabetes mellitus
- It can occur due to 2 enzyme deficiencies Galactokinase - causes lamellar cataract GPUT - causes oil drop cataract
- Galactokinase - causes lamellar cataract
- GPUT - causes oil drop cataract
- Galactokinase - causes lamellar cataract
- GPUT - causes oil drop cataract

Option D - Nuclear cataract:

- It is due to nuclear sclerosis
- Sclerosis can be caused due to denaturation of the protein resulting in an increase in insoluble protein and a decrease in soluble protein, and deposition of pigments such as melanin and urochrome.

Solution for Question 15:

Correct Option B - Nd YAG laser:

- In the clinical scenario mentioned above decreased visual acuity after the cataract surgery with the presence of posterior capsular opacification is suggestive of After cataract, most common late complication of uncomplicated cataract surgery.
- It is caused by the proliferation of lens epithelial cells that have remained within the capsular bag following cataract extraction.
- Nd YAG laser is the preferred technique used to create an opening in the posterior capsule termed as posterior capsulotomy to treat patients with after cataract.
- Indication for Nd YAG laser posterior capsulotomy are: Reduced visual acuity. Glare.
- Reduced visual acuity.
- Glare.
- Reduced visual acuity.
- Glare.

Incorrect Options:

Option A - Femto laser: FLACS - Femtosecond laser-assisted cataract surgery is used in refractive surgery. The laser used is a neodymium glass (Nd: glass) laser with a wavelength of 1053 nanometers.

Option C - Argon fluoride: It is a deep ultraviolet laser with a wavelength of 193 nanometers. It is an excimer laser that is used in refractive surgeries, trabeculoplasty, and glaucoma. The mechanism of action is photoablation which is the flattening of the cornea.

Option D - Argon: Argon lasers are photocoagulative lasers with wavelengths of 488 nm and 514 nm used for trabeculoplasty, lipid keratopathy, and glaucoma.

Solution for Question 16:

Correct Option A - Blunt trauma to eye:

- Rosette-shaped cataract a cortical cataract that starts from the posterior cortex, also known as traumatic cataract, is a type of cataract that occurs as a result of blunt trauma to the eye. The impact from the trauma can cause disruption and damage to the lens, leading to the formation of a characteristic rosette-like pattern in the lens fibers.

Incorrect Options:

Option B - Copper foreign body in eye: It can lead to Sunflower cataract.

Option C - Penetrating injury to the eye: A penetrating injury can result in sympathetic ophthalmitis.

Option D - Infection: Infections of the eye, such as endophthalmitis, can lead to severe inflammation and damage to various eye structures, including the lens.

Solution for Question 17:

Correct Option D - Posterior capsular opacification:

- Based on the patient's history of cataract surgery and the symptoms of glare during night drives, posterior capsular opacification is the most likely diagnosis.
- During cataract surgery, the posterior capsule of the lens is left intact to support the intraocular lens (IOL) implant. It is caused by the proliferation of lens epithelial cells that have remained within the capsular bag following cataract extraction. . This condition is known as posterior capsular opacification (PCO) or secondary cataract or after cataract.

Incorrect Options:

Option A - Cystoid macular edema: Cystoid macular edema (CME) is a condition characterized by the accumulation of fluid in the macula, which is the central part of the retina responsible for sharp central vision. It is characterized by blurring, especially for near tasks, and sometimes distortion.

Option B - Pseudophakic bullous keratopathy: Pseudophakic bullous keratopathy is a condition that can occur after cataract surgery, characterized by corneal edema and swelling.

Option C - ARMD (Age-Related Macular Degeneration): Age-related macular degeneration is a progressive degenerative disease affecting the macula, which leads to central vision loss. It commonly presents in individuals over the age of 50 and can cause symptoms such as blurred or distorted central vision.

Solution for Question 18:

Correct Option D - Blunt trauma:

- Blunt trauma refers to injuries caused by a direct blow or impact to the eye without penetration. Blunt trauma can result in damage to the lens, leading to the development of rosette-shaped cataract. It is a cortical cataract that starts from the posterior cortex.

Incorrect Options:

Option A - Penetrating Trauma: Penetrating trauma refers to injuries that result in the direct penetration of an object into the eye, causing damage to the lens. Penetrating trauma can result in sympathetic ophthalmitis.

Option B - Steroid Use: Prolonged or excessive use of corticosteroid medications, either in the form of eye drops or systemic administration, can increase the risk of developing toxic cataracts.

Option C - Diabetes: Diabetes is associated with snowflake cataract.

Solution for Question 19:

Correct Option D - Intravitreal antibiotics:

- Given image of eye shows inflammation, congestion, corneal edema and hypopyon after 36 hours of undergoing cataract surgery is suggestive of acute onset endophthalmitis
- Treatment of acute onset endophthalmitis will depend upon whether the patient is able to perceive the hand movement or doesn't have hand movement. In the case of only perception of light and no hand movement, Pars Plana Vitrectomy (PPV) helps to quickly reduce the incidence of vision loss by 50%. When there is hand movement and above, the treatment of choice would be intravitreal antibiotic injections (ceftazidime (gram -ve) and vancomycin (gram +ve)). Systemic antibiotics, mainly Quinolones, are preferred. Topical steroids and cycloplegics are used in the case of uveitis.

- In the case of only perception of light and no hand movement, Pars Plana Vitrectomy (PPV) helps to quickly reduce the incidence of vision loss by 50%.
- When there is hand movement and above, the treatment of choice would be intravitreal antibiotic injections (ceftazidime (gram -ve) and vancomycin (gram +ve)).
- Systemic antibiotics, mainly Quinolones, are preferred. Topical steroids and cycloplegics are used in the case of uveitis.
- In the case of only perception of light and no hand movement, Pars Plana Vitrectomy (PPV) helps to quickly reduce the incidence of vision loss by 50%.
- When there is hand movement and above, the treatment of choice would be intravitreal antibiotic injections (ceftazidime (gram -ve) and vancomycin (gram +ve)).
- Systemic antibiotics, mainly Quinolones, are preferred. Topical steroids and cycloplegics are used in the case of uveitis.

Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 20:

Correct Option C- 1-d, 2-a, 3-b, 4-c:

Posterior Subcapsular Cataract:

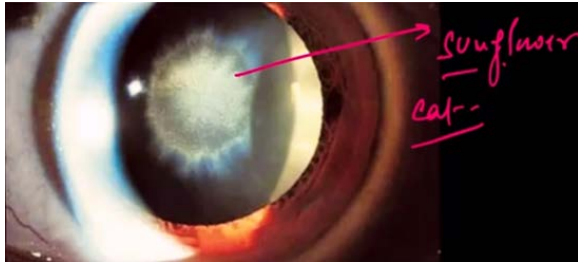
- PSC is the lenticular opacity that causes maximum diminution of vision because this opacity is in proximity to the nodal point of the eye i.e., the first focal point/optical center of the eye where light rays are focused. It is just behind the lens.



- PSC forms due to dysplasia of E cells that migrate posteriorly. Wedl cells/Bladder cells are E cells that are responsible for formation of PSC.

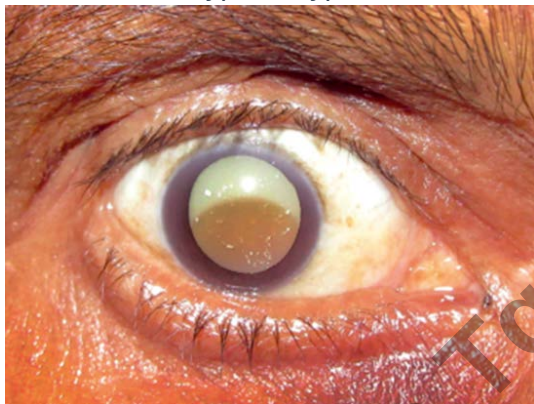
Sunflower Cataract:

- The image below depicts a Sunflower cataract: It is associated with chalcosis condition characterized by accumulation of copper or copper alloy in the eye.
- It is caused by foreign copper body or Wilson's disease.



Morgagnian Cataract:

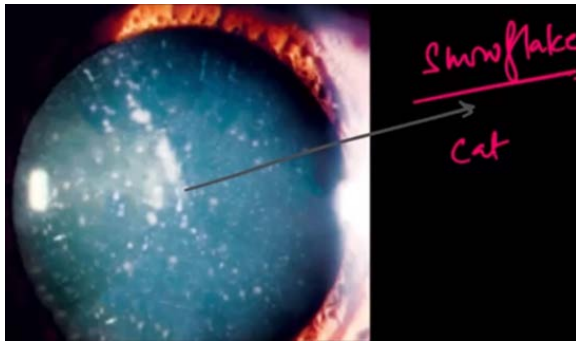
- It is a type of hyper mature cataract



- Liquefaction of the cortex causes the nucleus to float, Wrinkling of the capsule are observed.

Snowflake/ Snowstorm cataract:

- It is associated with diabetes mellitus
- It occurs due to sorbitol accumulation. Sorbitol is hyperosmotic, and any hydration will lead to cataracts.
- The enzyme responsible for the Sorbitol pathway is NADPH-dependent Aldose Reductase.



Solution for Question 21:

Correct Option D - Subluxation of lens:

- The physical presentation of tall, thin stature with disproportionately long limbs (arm span > height), arachnodactyly, scoliosis is suggestive of musculoskeletal disorder i.e., Marfan's syndrome.
- Marfan's syndrome is associated with superolateral ectopia lentis (subluxation of lens).
- Other ocular features of Marfan's syndrome Long eyeball results in Axial myopia. Megalocornea (>13mm diameter). Blue sclera. Cornea plana (lesser curvature). Hypoplasia of dilator papillae. Lattice degeneration. Rhegmatogenous Retinal detachment.
- Long eyeball results in Axial myopia.
- Megalocornea (>13mm diameter).
- Blue sclera.
- Cornea plana (lesser curvature).
- Hypoplasia of dilator papillae.
- Lattice degeneration.
- Rhegmatogenous Retinal detachment.
- Long eyeball results in Axial myopia.
- Megalocornea (>13mm diameter).
- Blue sclera.
- Cornea plana (lesser curvature).
- Hypoplasia of dilator papillae.
- Lattice degeneration.

- Rhegmatogenous Retinal detachment.

Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 22:

Correct Option A - Vossius ring:

- The given image is suggestive of Vossius ring, a condition that is typically detected during an ophthalmological examination following a blunt trauma to the eye.
- The Vossius ring itself is identified by the presence of pigment dispersion in a circular shape on the anterior surface of the lens. This dispersion or imprint of pigmented cells from the posterior surface of the iris occurs during the incident of trauma when the iris is pushed against the lens. The resulting ring is usually complete and prominent and is often centrally located on the lens.

Incorrect Options:

Option B - Weiss Ring: This is an opacified ring-shaped structure that can be seen floating within the vitreous humor of the eye, often signifying a posterior vitreous detachment, a condition in which the vitreous gel separates from the retina.

Option C - Fleischer's Ring: This refers to a pigmented, circular line found at the edge of the cornea in keratoconus patients. It's composed of iron deposits and signifies corneal thinning and protrusion.

Option D - Kayser-Fleischer Ring: This is a brown-colored ring found at the periphery of the cornea, most commonly seen in patients with Wilson's disease, a rare genetic disorder leading to excessive copper deposition in the body.

Solution for Question 23:

Correct Option A - Capsulorrhexis:

- The surgical procedure being performed in the image is Continuous curvilinear capsulorhexis. It refers to the creation of a circular tear in the anterior capsule of the lens. This procedure allows the surgeon to remove the cloudy lens and insert an artificial lens.
- In a successful capsulorhexis, the tear is continuous and circular, providing a clear opening through which the lens material can be removed. This is ideally achieved using a special instrument like cystotome or a bent hypodermic needle, or using laser like femtolas.

Incorrect Options:

Option B - Hydrodissection: This is a surgical technique commonly used during cataract surgery, where balanced salt solution is injected between the lens and anterior capsule, separating the lens from capsule, thereby facilitating lens removal.

Option C - Intraocular Lens Implantation: This procedure involves the surgical placement of an artificial lens, typically following cataract removal, to restore vision by replicating the focusing power of the eye's natural lens.

Option D - Lens Aspiration: This surgical method involves the removal of the eye's natural lens by pressure and counterpressure technique using corneal forceps, lens spatula, and lens hook or by irrigating wire vectus technique.

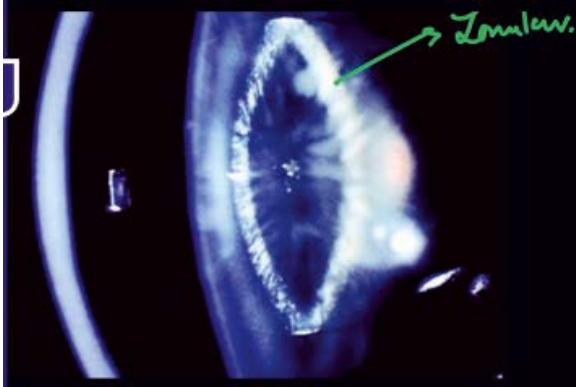
Solution for Question 24:

Correct Option D - Immature senile cataract and phacoemulsification with IOL:

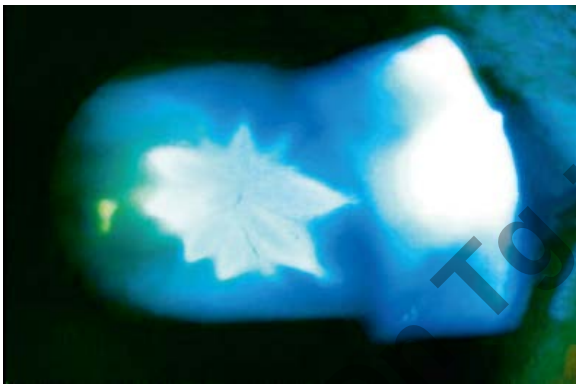
- The patient's painless loss of vision, in conjunction with the findings on the ocular examination i.e., clouding of lens in cuneiform pattern is consistent with an immature senile cataract.
- Diabetic patients are at an increased risk of developing cataracts, and the patient's age of 65 is also a common time for cataracts to develop.
- The management of immature senile cataracts involves surgical intervention, with phacoemulsification and intraocular lens (IOL) implantation being the preferred method. Phacoemulsification is a minimally invasive procedure in which the cloudy lens is broken into small pieces using ultrasound waves and then removed from the eye through a small incision. An IOL is then inserted to replace the removed lens and restore the patient's vision.

Incorrect Options:

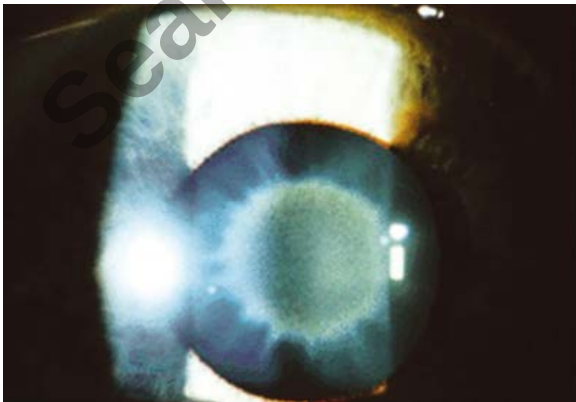
Option A - Zonular cataract and lensectomy: A zonular cataract is a type of congenital cataract that affects the lens zonules, which are fibers that hold the lens in place. This type of cataract requires a lensectomy, which is the surgical removal of the entire lens.



Option B - Rosette cataract and sugar control: Rosette cataracts are rare and occur after blunt trauma to the eye. This type of cataract can be managed by phacoemulsification.



Option C - Sunflower cataract and lesion extraction: Sunflower cataracts are caused due to any copper-containing foreign body or Wilson's disease. Lesion extraction is not an appropriate treatment for this type of cataract. Treatment of the underlying cause is preferred (copper chelating agents).



Solution for Question 25:

Correct Options A - Copper:

- The ring-like structure noted in the periphery on ocular examination is the Kayser-Fleischer ring characterized by brownish-yellow zone of fine copper dusting in peripheral Descemet membrane. This is pathognomonic of Wilson's disease.
- Wilson disease (hepatolenticular degeneration) is a condition involving the widespread abnormal deposition of copper in tissues. It is caused by a deficiency of ceruloplasmin, the major copper-carrying blood protein and presents with liver disease, psychiatric disturbance, and KF ring with neurological symptoms..

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 26:

Correct Option A - Index myopia:

- Index myopia is a term used in the context of contact lenses and eyeglasses prescription. It refers to a situation where refractive index of the lens is higher than the contact lens or eyeglass prescription.
- When someone has index myopia, it means that their eye's natural lens, which changes shape to focus on objects at various distances (accommodation), is overcorrecting or causing their vision to be more myopic (nearsighted) than what is indicated by the prescription in their eyeglasses or contact lenses.
- In patients with nuclear sclerosis, distant vision deteriorates due to progressive index myopia. Such patients may be able to read without presbyopic glasses. This improvement in near vision is referred to as 'second sight':

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Incorrect Options:

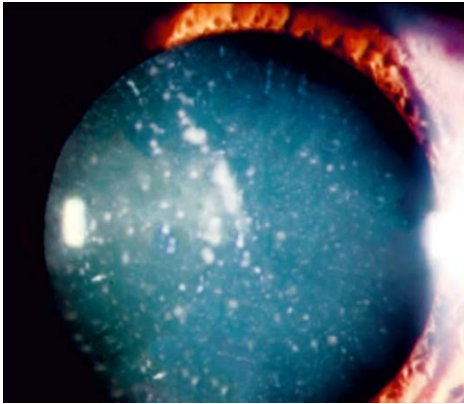
Option B - Index hypermetropia: It is due to the change in the refractive index of the crystalline lens, which occurs in old age or diabetics. The refractive index gradually increases from the center to the periphery.

Option C - Presbyopia: It is an error of accommodation in which with advancing age the accommodative power of crystalline lens decreases with increasing age due to decrease in elasticity of lens and sclerosis of lens.

Solution for Question 27:

Correct Option D - Sorbitol:

- The accumulation of sorbitol is responsible for developing a diabetic cataract.
- In diabetes, excess glucose in the blood accumulates in the lens and is converted to sorbitol by the enzyme NADPH-dependent aldose reductase.
- Sorbitol is a hyperosmotic agent leading to oxidative damage, changes in refractive index of the eye (myopia), and formation of opacities (snowflake) leading to formation of diabetic cataract aka snowflake cataract.



Incorrect Options:

- Option A, B, and C are incorrect. Refer to Option D for an explanation.

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Glaucoma-PACG, Treatment, Congenital Glaucoma

1. A 65-year-old female presents with acute onset pain, redness, and blurring of vision in the right eye. On examination, her intraocular pressure (IOP) is 48 mmHg, and there is a shallow anterior chamber with a hazy cornea. Slit-lamp examination reveals an intumescent, swollen lens with the iris pushed forward. Gonioscopy shows a closed angle. What is the most likely underlying mechanism for her glaucoma?

- A. Obstruction of the trabecular meshwork by inflammatory cells
- B. Forward displacement of the lens due to an abnormal zonular attachment
- C. Disruption of aqueous humor flow due to posterior synechiae
- D. Pupillary block caused by the swollen lens

2. Which of the following conditions is not typically associated with secondary congenital glaucoma?

- A. Aniridia
- B. Marfan Syndrome
- C. Turner Syndrome
- D. Angel man syndrome

3. A 6-month-old infant with congenital glaucoma has corneal enlargement, clouding, and Haab's striae. Medical treatment has failed, and the angle is difficult to visualize due to corneal clouding. Which surgical procedure is most appropriate for this infant?

- A. Goniotomy

- B. Trabeculotomy with Trabeculectomy
 - C. Trabeculectomy with antimetabolites
 - D. Glaucoma drainage devices
-

4. A 1-month-old child presents with persistent eye watering, light sensitivity, and frequent eye rubbing. Examination shows enlarged corneas with Haab's striae, cloudiness, a deep anterior chamber, and thin, blue sclera. The child has had ocular enlargement since birth. The ophthalmologist suspects congenital glaucoma. Which type of glaucoma best fits this presentation?

- A. Juvenile Glaucoma
 - B. Infantile Glaucoma
 - C. True Congenital Glaucoma
 - D. Secondary Congenital Glaucoma
-

5. A 50-year-old female presents with gradually progressive vision loss. On examination, her intraocular pressure (IOP) is elevated, and a cup-to-disc ratio of 0.7 is noted in both eyes. She is prescribed a prostaglandin analog as first-line therapy. Which of the following is the most likely mechanism of action and a common side effect of this medication?

- A. Reduces aqueous humor production; Lid retraction
 - B. Increases trabecular outflow; Brow ache
 - C. Increases uveoscleral outflow; Iris pigmentation
 - D. Decreases aqueous humor secretion; Allergic blepharoconjunctivitis
-

6. All of the following are penetrating glaucoma surgeries except:

- A. Trabeculectomy
 - B. Deep sclerectomy
 - C. Minimum incision glaucoma surgery
 - D. Glaucoma drainage device implantation
-

7. Which of the following laser procedures are used for Open angle glaucoma (OAG) and angle closure glaucoma (ACG) respectively?

- A. Argon laser iridotomy for open-angle glaucoma and selective laser trabeculoplasty (SLT) for angle closure glaucoma
 - B. Selective laser trabeculoplasty for open-angle glaucoma and laser iridotomy for angle closure glaucoma
 - C. Laser iridoplasty for open-angle glaucoma and argon laser iridotomy for angle closure glaucoma
 - D. Laser iridotomy for open-angle glaucoma and laser iridoplasty for angle closure glaucoma
-

8. A 65-year-old male presents to the ophthalmology clinic for a routine eye examination. During the examination, his intraocular pressure (IOP) is measured at 28 mmHg in both eyes. The ophthalmologist diagnoses him with primary open-angle glaucoma. Which of the following medications is primarily designed to lower intraocular pressure by reducing the production of aqueous humor?

- A. Travoprost
 - B. Pilocarpine
 - C. Acetazolamide
 - D. Unoprostone
-

9. Which ICE syndrome is characterized by the presence of pigmented nodular lesions on the iris?

- A. Progressive iris atrophy
 - B. Chandler's syndrome
 - C. Cogan-Reese syndrome
 - D. Both A & C
-

10. Which of the following statements is false about traumatic glaucoma?

- A. Traumatic glaucoma can develop from both perforating and blunt injuries.
 - B. Angle recession glaucoma typically presents with elevated IOP shortly after the trauma.
 - C. Lens-induced glaucoma can occur due to a ruptured, swollen, or dislocated lens following trauma.
 - D. Peripheral anterior synechiae from corneal injury can lead to angle-closure glaucoma.
-

11. A 30-year-old male presents with unilateral eye discomfort, blurred vision, and seeing halos around lights. On examination, his intraocular pressure (IOP) is measured at 48 mm Hg. The anterior segment shows a few small, flat, nonpigmented keratic precipitates located inferiorly, mild redness, and an aqueous flare. The pupil is dilated, but there are no signs of a posterior synechiae. The patient reports recurrent episodes that resolve within days to weeks. Given these findings, what is the most likely diagnosis?

- A. Secondary Angle-Closure Glaucoma with Pupillary Block
 - B. Secondary Angle-Closure Glaucoma without Pupillary Block
 - C. Open-Angle Glaucoma Due to Uveitis
 - D. Glaucomatocyclitic Crisis
-

12. A 70-year-old patient with a history of primary angle closure glaucoma undergoes a trabeculectomy to control intraocular pressure. Postoperatively, the patient presents with severe eye pain, blurred vision, a significantly elevated intraocular pressure (IOP), and a shallow anterior chamber. Seidel's test is negative. The condition does not improve with miotic agents. What is the most appropriate initial treatment for this condition?

- A. Oral corticosteroids
- B. Pars plana Vitrectomy
- C. Atropine drops and oral acetazolamide
- D. Peripheral iridectomy

13. A 58-y/o diabetic patient presents with decreased vision, eye pain, and redness in the right eye. Examination reveals iris neovascularization, elevated intraocular pressure, and proliferative diabetic retinopathy. What is the most appropriate management strategy to prevent further neovascularization?

- A. Topical beta-blockers and oral carbonic anhydrase inhibitors
- B. Panretinal photocoagulation
- C. Systemic corticosteroids
- D. VEGF inhibitors

14. A 55-year-old Asian woman presents with acute eye pain and blurred vision. Her ophthalmologist suspects primary angle closure glaucoma (PACG). Which of the following statements about the etiopathogenesis of PACG is incorrect?

- A. Pupillary block is the most common mechanism in PACG
- B. Phacomorphic mechanism involves an abnormally large lens pushing the iris forward

- C. Plateau iris configuration is more common in younger patients with PACG
 - D. Caucasians have a higher prevalence of PACG compared to other ethnicities
-

15. A 72-year-old male presents with gradual vision loss in the right eye. On slit-lamp examination, dandruff-like material is noted on the anterior lens capsule and iris. Intraocular pressure is elevated, and the patient is diagnosed with pseudoexfoliative glaucoma. Which of the following is most likely to be observed in this patient's eye?

- A. Posterior synechiae
 - B. Zonular instability and phacodonesis
 - C. Corneal guttata
 - D. Iris heterochromia
-

16. A 35-year-old myopic male presents with intermittent blurring of vision and mild ocular pain after exercise. On examination, IOP is 28 mmHg, vertical line of pigment on corneal endothelium, heavily pigmented trabecular meshwork on gonioscopy, transillumination defects in the mid-peripheral iris. Which of the following is the first line treatment?

- A. Nd:YAG laser peripheral iridotomy
 - B. Selective laser trabeculoplasty
 - C. Topical prostaglandin analogue
 - D. Argon laser peripheral iridoplasty
-

17. A 45-year-old male presents with unilateral eye pain, redness, and moderately elevated intraocular pressure (IOP) of 28 mmHg. On slit-lamp examination, you observe fine, pigmented deposits on the corneal endothelium in a vertical spindle

pattern. The patient O/E has symmetric transillumination defects in the mid-peripheral iris. Based on these findings, which of the following is the most likely cause of this patient's secondary glaucoma?

- A. Pseudoexfoliation syndrome
- B. Pigment dispersion syndrome
- C. Neovascular glaucoma
- D. Lens-particle glaucoma

18. A 65-year-old male presents with a painful, blind right eye. On examination, the eye is stony hard with no light perception, 100% cupping of the optic disc, and a fixed, mid-dilated pupil. Which of the following is the correct initial management?

- A. Topical timolol and brimonidine
- B. Trabeculectomy
- C. Cyclodestructive procedure
- D. Evisceration

19. All of the following components of Vogt's triad are typically observed in a patient who has experienced and resolved an acute primary angle-closure glaucoma (APAC) attack except?

- A. Glaucomflecken
- B. Iris atrophy
- C. Slightly dilated non-reactive pupil
- D. Progressive retinal damage

20. A 55-year-old female presents for a routine eye examination. During gonioscopy, the examination reveals an occludable anterior chamber angle bilaterally. The patient reports no visual symptoms, and intraocular pressure (IOP) measurements are within normal limits. There is no evidence of glaucomatous damage on optic nerve examination. What is the most appropriate next step in management for this patient?

- A. I.V Mannitol/Acetazolamide
 - B. Prophylactic peripheral iridotomy
 - C. Filtration surgery
 - D. Topical beta-blocker therapy
-

21. What aspect of the anterior chamber does the Van Herick grading system evaluate during slit lamp examination?

- A. The curvature of the cornea
 - B. The width of the anterior chamber angle
 - C. The condition of the retina
 - D. The intraocular pressure
-

22. In the Shaffer classification system for grading the anterior chamber angle, which grade indicates that the angle is fully open with visible structures such as the scleral spur and trabecular meshwork?

- A. Grade 0
 - B. Grade 1
 - C. Grade 2
 - D. Grade 3
-

23. Which of the following best describes the principle of gonioscopy?

- A. It involves measuring intraocular pressure.
 - B. It is a technique for visualizing the anterior chamber angle of the eye using a special lens
 - C. It is a technique for visualizing the posterior chamber angle of the eye using a special lens
 - D. It evaluates the retina.
-

24. A 35-year-old woman presents with intermittent eye pain and blurred vision. Gonioscopy reveals narrow angles, and the angle remains narrow after laser peripheral iridotomy. Which of the following statements about plateau iris configuration is incorrect ?

- A. It is characterized by an anteriorly positioned ciliary body
 - B. It is more commonly seen in older patients with primary angle closure glaucoma
 - C. Plateau iris configuration can persist after iridotomy
 - D. Ultrasound biomicroscopy (UBM) can be useful in diagnosing this condition
-

25. A 50-year-old patient presents with symptoms suggesting possible angle-closure glaucoma. You want to perform a comprehensive assessment of the anterior chamber angle. Which of the following statements is incorrect regarding anterior chamber angle assessment and anatomy?

- A. Gonioscopy is the definitive diagnostic tool for primary angle closure.
- B. Normal angle structures from anterior to posterior include: Schwalbe's line, non-pigmented trabecular meshwork, pigmented trabecular meshwork, scleral spur, and ciliary body band.

C. An angle is considered occludable if the pigmented trabecular meshwork is not visible in less than 180 degrees without indentation.

D. Indirect gonioscopes used to measure angle of AC are goldmann, posner-ritch, zeiss

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Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	2
Question 4	3
Question 5	3
Question 6	2
Question 7	2
Question 8	3
Question 9	3
Question 10	2
Question 11	4
Question 12	3
Question 13	2
Question 14	4
Question 15	2
Question 16	3
Question 17	2
Question 18	3
Question 19	4
Question 20	2
Question 21	2
Question 22	4

Question 23	2
Question 24	2
Question 25	3

Solution for Question 1:

Correct Answer: D) Pupillary block caused by the swollen lens

Explanation:

- Lens-induced glaucoma is a type of secondary angle-closure glaucoma associated with a disorder of the lens.
- In this case, the enlarged lens prevents aqueous humor from flowing through the pupil into the anterior chamber, leading to a pupillary block.
- The iris is pushed forward, resulting in angle closure and increased intraocular pressure.

Lens-induced glaucoma can be categorized into the following types:

- Lens-induced secondary angle-closure glaucoma: Phacomorphic glaucoma: Caused by a swollen or intumescent lens. Phacotopic glaucoma: Due to anterior displacement of the lens.
- Phacomorphic glaucoma: Caused by a swollen or intumescent lens.
- Phacotopic glaucoma: Due to anterior displacement of the lens.
- Lens-induced secondary open-angle glaucoma: Phacolytic glaucoma: Results from leakage of lens proteins. Lens particle glaucoma: Triggered by lens fragments in the anterior chamber. Phacoanaphylactic glaucoma: Caused by an immune response to lens proteins.
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- Phacoanaphylactic glaucoma: Caused by an immune response to lens proteins.

Other options describe mechanisms for other forms of secondary glaucoma, such as uveitic glaucoma (Option A), phacodonesis (Option B), or chronic iritis (Option C), but are not

characteristic of lens-induced glaucoma.

Solution for Question 2:

Correct Answer: D) Angel man syndrome

Explanation:

- Angelman Syndrome (AS) is a rare neurogenetic disorder caused by loss of function of the UBE3A gene on chromosome 15. It primarily affects the nervous system, leading to severe developmental delays, intellectual disability, speech impairment, and movement disorders.
- Ophthalmic findings that can be seen in Angelman syndrome include Strabismus, Refractive errors, Oculocutaneous albinism, It is not associated with secondary glaucoma.

Secondary Developmental/ Congenital Glaucoma	
Associated Ocular Anomalies	Associated Systemic Anomalies
Glaucoma Associated with Iridodysgenesis: Aniridia (50% of cases) (Option A ruled out) Familial Iris Hypoplasia Congenital Ectropion Uvea Congenital Microcornea Congenital Nanophthalmos	Chromosomal Disorders: Trisomy 13 (Patau Syndrome) Trisomy 18 (Edward's Syndrome) Trisomy 21 (Down Syndrome) Turner Syndrome (Option C ruled out)
Ectopia Lentis Syndromes: Marfan Syndrome (Option B ruled out) Weill-Marchesani Syndrome Homocystinuria	
Glaucoma Associated with Iridocorneal Dysgenesis: Posterior Embryotoxon: Prominent Schwalbe's ring. Axenfeld-Rieger Syndrome: Axenfeld Anomaly: Posterior embryotoxon with peripheral iris tissue strands. Rieger Anomaly: Posterior embryotoxon, iris hypoplasia, ectropion, iris defects. Rieger Syndrome: Rieger anomalies with dental, facial, and other systemic anomalies. Peter's Anomaly: Central corneal opacity with or without iris or lens adhesions. Combined Rieger's Syndrome and Peter's Anomaly: Features of both conditions.	Phakomatosis : Sturge-Weber Syndrome (50% of cases) Von Recklinghausen's Neurofibromatosis (25% of cases)

Metabolic Syndromes: Lowe's Syndrome: Oculocerebrorenal syndrome. Hurler's Syndrome: Mucopolysaccharidosis. Zellweger Syndrome: Hepato-cerebral renal syndrome.	
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Solution for Question 3:

Correct Answer: B) Trabeculotomy with Trabeculectomy

Explanation:

- Congenital glaucoma is diagnosed in this infant based on corneal enlargement, clouding, Haab's striae, and elevated intraocular pressure.
- When corneal clouding prevents proper angle visualization, and medical treatment fails, Trabeculotomy with Trabeculectomy becomes the procedure of choice.
- Trabeculotomy helps expose Schlemm's canal to improve aqueous outflow, and trabeculectomy provides additional filtration, offering better intraocular pressure control.
- This combined approach is effective when angle visualization is not possible and helps manage congenital glaucoma in infants.

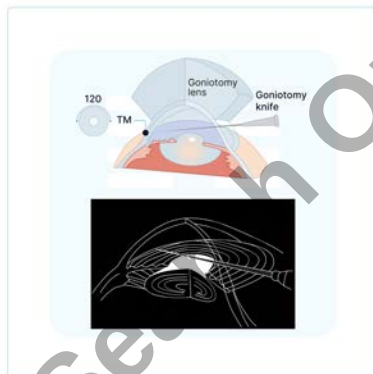
Medical Treatment:

- Purpose: Used to lower intraocular pressure (IOP) until surgery can be performed.
- Medications: Hyperosmotic agents: Reduce IOP by drawing fluid out of the eye. Carbonic anhydrase inhibitors: Decrease fluid production. Beta-blockers (e.g., timolol 0.5%): Often the first choice for IOP reduction.
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- Carbonic anhydrase inhibitors: Decrease fluid production.
- Beta-blockers (e.g., timolol 0.5%): Often the first choice for IOP reduction.
- Avoid: Miotics: Can paradoxically increase IOP. Alpha-2 agonists (e.g., brimonidine): Contraindicated in children due to CNS depression.
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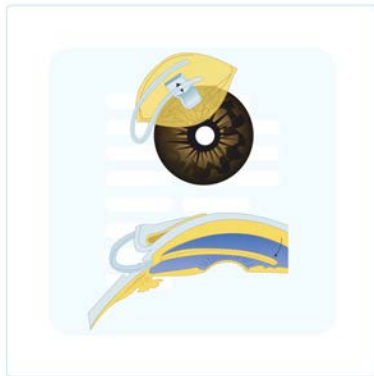
- Miotics: Can paradoxically increase IOP.
- Alpha-2 agonists (e.g., brimonidine): Contraindicated in children due to CNS depression.

Surgical Procedures:

- Incisional Angle Surgery: Goniotomy: Involves an incision in the angle of the eye using a Barkan's goniotomy knife, performed under gonioscopic control. Success rate is around 85%. Interestingly, angle surgery is most successful in patients presenting between 2 months and 1 year of age (infantile-onset PCG) with 90% success, and less in newborn or late-recognized PCG with <50% success. In general, IOP control is obtained in greater than 80% of patients.
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- Trabeculotomy: Used when corneal clouding prevents visualization or if goniotomy fails. Involves exposing Schlemm's canal and breaking the inner wall with a trabeculotome.
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- Filtration Surgery: Trabeculectomy with antimetabolites: Often provides good results. Combined trabeculotomy and trabeculectomy with antimetabolites: Considered a standard procedure for better outcomes.
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- Combined trabeculotomy and trabeculectomy with antimetabolites: Considered a standard procedure for better outcomes.
- Glaucoma Drainage Devices (GDD): Used in severe, resistant cases where other methods have failed.
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- Trabeculectomy with antimetabolites: Often provides good results.
- Combined trabeculotomy and trabeculectomy with antimetabolites: Considered a standard procedure for better outcomes.
- Used in severe, resistant cases where other methods have failed.

Goniotomy (Option A): This procedure involves making an incision in the angle of the eye and is best suited when the angle can be visualized clearly.

Trabeculectomy with antimetabolites (Option C): While this is a common and effective procedure for many glaucoma cases, it might not be the first choice in this scenario due to the presence of corneal clouding and difficulty in initial angle procedures.

Glaucoma Drainage Devices (Option D): These are typically reserved for severe, resistant cases when other surgical methods have failed. They are not the first-line approach for the scenario described but might be considered if other surgical options are unsuccessful.

Solution for Question 4:

Correct Answer: C) True Congenital Glaucoma

Explanation:

- True Congenital Glaucoma is characterized by elevated intraocular pressure present during intrauterine life, resulting in ocular enlargement at birth.
- The presence of Haab's striae, early onset, enlarged corneas, and the classic triad of symptoms (watering, photophobia, and blepharospasm) are all consistent with newborn glaucoma.

Primary Congenital Glaucoma	
Types	1. Newborn glaucoma/ True congenital glaucoma: Elevated intraocular pressure is present during intrauterine life, resulting in ocular enlargement at birth. (40% cases) (Option C) Typically presents within the first month of life. Often associated with developmental anomalies of the anterior segment, such as primary congenital glaucoma, which can result from genetic mutations or other congenital conditions. 2. Infantile: ≤ 3 yrs of age- (Option B ruled out) Infantile glaucoma is labelled when the disease manifests prior to the child's third birthday Affects infants older than 1 month and children up to 3 years. Can also result from genetic factors or be secondary to other conditions like Sturge-Weber syndrome or other syndromic associations. 3. Juvenile: > 3 yrs of age (Option A ruled out) who develop pressure rise after 3 years but before adulthood. Also known as Juvenile primary open angle glaucoma(POAG) usually occurs between 10 to 35 years of age. Buphthalmos, meaning "bull-like eyes," is a condition characterized by the enlargement of the eyeball. It typically manifests before the age of 3 years and results from the accumulation and retention of aqueous humor.
Pathogenesis	Trabeculodysgenesis: Maldevelopment of the trabeculum and iridotrabecular junction from neural crest cells impairs aqueous outflow, causing elevated IOP. Angle Recess Absence: The angle recess is absent, with the iris inserting either flatly or concavely into the trabeculum. Iris Insertion Types: Flat Insertion: Reveals the ciliary body and scleral spur. Concave Insertion: Obscures the scleral spur and ciliary body.

Clinical Features	 Classic Triad: Lacrimation, photophobia, and blepharospasm; often accompanied by eye rubbing. Corneal Signs: Oedema: Early sign, may progress to permanent opacity. Enlargement: Corneal diameter >13 mm confirms enlargement; >16 mm indicates poor prognosis. Haab's Striae: Tears in Descemet's membrane, visible as opaque lines. Additional Features: Thin Sclera: Appears blue due to underlying uveal tissue. Deep Anterior Chamber: Indicates elevated IOP. ... (content truncated)
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Secondary Congenital Glaucoma (Option D):

- Usually associated with other ocular or systemic anomalies, which is not indicated in this case.
- The clinical features align more with true congenital glaucoma.

Solution for Question 5:

Correct Answer: C) Increases uveoscleral outflow; Iris pigmentation

Explanation:

Anti-Glaucoma Drugs:

1. Prostaglandin F2 α Agonists (Option C)

- Examples: Latanoprost, Bimatoprost, Travoprost, Tafluprost
- Mechanism: Increases uveoscleral outflow
- Uses: First-line for primary open-angle and angle-closure glaucoma
- Side Effects: Sandy eyes, orbital fat atrophy, anterior uveitis, cystoid macular edema, heterochromia iridis, hypertrichosis
- Contraindications: Cystoid macular edema, diabetes mellitus

2. Beta Blockers (Option D)

- Non-Selective: Timolol, Levobunolol, Carteolol, Metipranolol More effective but contraindicated in asthma
- More effective but contraindicated in asthma
- Selective: Betaxolol (retinoprotective), Levobetaxolol Safer in asthma
- Safer in asthma
- Mechanism: Reduces aqueous secretion from ciliary epithelium
- Side Effects: Allergic blepharoconjunctivitis, dry eyes, corneal hypoesthesia
- More effective but contraindicated in asthma
- Safer in asthma

3. Alpha-2 Agonists (Option A)

- Examples: Brimonidine (long-term use), Apraclonidine (IOP spikes)
- Mechanism: Reduces aqueous production, increases uveoscleral outflow
- Side Effects: Lid retraction, respiratory depression (Brimonidine), allergic blepharoconjunctivitis
- Contraindications: Children <2 years
- Brimonidine: Safest during pregnancy (Category B)

4. Miotics (Option B)

- Examples: Pilocarpine, Physostigmine, Echothiophate
- Mechanism: Stimulates M3 receptors, increasing trabecular outflow
- Side Effects: Brow pain, miosis, cataract (Echothiophate), retinal tears

5. Carbonic Anhydrase Inhibitors

- Examples: Acetazolamide (systemic), Brinzolamide, Dorzolamide (topical)
- Mechanism: Inhibits aqueous secretion from ciliary epithelium
- Uses: Acute congestive glaucoma (with IV mannitol)
- Side Effects: Ciliary edema, secondary angle-closure glaucoma (Topiramate, Zonisamide)

6. Newer Drugs

- Netarsudil (Rho kinase inhibitor): Increases trabecular outflow; Side effect - vortex keratopathy
- Omidenepag (PGE2 agonist): Increases both trabecular and uveoscleral outflow

Solution for Question 6:

Correct Answer: B) Deep Sclerectomy

Explanation:

- In penetrating surgeries an incision is made to create a new drainage pathway, while non-penetrating surgeries aim to enhance the eye's natural drainage mechanism without incisions.

Penetrating surgeries

- Involves making incisions or creating openings that penetrate the eye's interior to establish a new drainage pathway for aqueous humor.
- Trabeculectomy is a classic example of a penetrating surgery; Involves resecting the trabecular meshwork and creating a fistula between the anterior chamber and the sub-conjunctival space. A partial-thickness V-shaped scleral incision is made to expose the trabecular meshwork, which is removed using Kelly's punch. An iridectomy is performed to prevent pupillary block. The scleral flap is repositioned and loosely sutured to allow controlled leakage of aqueous humor. A conjunctival flap covers the area, with the presence of a bleb indicating a successful procedure.
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- The scleral flap is repositioned and loosely sutured to allow controlled leakage of aqueous humor.
- A conjunctival flap covers the area, with the presence of a bleb indicating a successful procedure.

Non-penetrating surgeries

- Aim to improve the eye's natural drainage system without making a fistula.
- Examples include procedures like deep sclerectomy or viscocanalostomy.
- They enhance the outflow of aqueous humor by modifying or removing parts of the trabecular meshwork and adjacent structures but do not involve creating a direct drainage pathway through the eye wall.

Penetrating surgeries:

Trabeculectomy: (Option A)

- A fistula is created between the anterior chamber and sub tenon's space
- MCC of failure: fibrosis of fistula
- To prevent fibrosis , use antimetabolites(mitomycin, 5-FU)
- The trabecular meshwork is bypassed
- They have less chance of postoperative complications

Glaucoma drainage device / seton surgery: (Option D)

- It is done in refractory cases , uncontrolled glaucoma and glaucoma with aniridia.
- Express Implant / Express mini shunt is a modification of trabeculectomy where a needle is used (made up of titanium) instead of a punch. No peripheral iridotomy is done.
- Ahmed Glaucoma Valve (AGV): Silicon tube and valve with polypropylene body
- Molteno implant: Silicone tubes with polypropylene plates
- Baerveldt: Both silicon tube and plate

drainage device / seton surgery" data-author="" data-hash="1827" data-license="" data-source="" data-tags="March2025" src="https://image.prepladder.com/notes/6fINf1LTfBr2EJPi31KA1742979359.png" />

Minimum incision glaucoma surgery: (MIGS): (Option C)

- In mild to moderate glaucoma with modest target pressure goal.
- Commonly done with cataract surgery.
- This is mainly done in combination with phacoemulsification

Non penetrating surgeries:

- Deep sclerectomy
- Viscocanalostomy

Solution for Question 7:

Correct Answer: B) Selective laser trabeculoplasty for open-angle glaucoma and laser iridotomy for angle closure glaucoma

Explanation:

Selective Laser Trabeculoplasty (SLT)

- It is commonly used for open-angle glaucoma.
- SLT improves aqueous humor outflow by targeting the trabecular meshwork, which helps lower intraocular pressure over time.

Laser trabeculoplasty:

- There are 2 types:

a) Argon Laser trabeculoplasty

b) Selective Laser trabeculoplasty- Laser applied selectively to pigmented trabecular meshwork

- Mechanism of action of argon and Nd: YAG laser is photocoagulation
- It increases fibrosis in one part, leading to contraction of that area, which then pulls the mesh, leading to the opening of the mesh to the other side.
- Selective Laser trabeculoplasty is the treatment of choice for prophylaxis in the other eye

Laser Iridotomy

- It is used for angle closure glaucoma.
- This procedure creates a small hole in the peripheral iris to allow better fluid drainage and relieve the blockage of the anterior chamber angle.
- This helps lower intraocular pressure and prevent further episodes of angle closure.

Argon laser iridotomy (Option A) is used for angle closure glaucoma, not open-angle glaucoma.

- Also, SLT is not typically used for angle closure glaucoma.

Laser iridoplasty (Option C) is used for some cases of angle closure glaucoma, particularly when the angle is very narrow or if iridotomy is not feasible.

- However, argon laser iridotomy is the more common procedure for angle closure glaucoma. Laser iridoplasty is not used for open-angle glaucoma.

Laser iridotomy (Option D) is not used for open-angle glaucoma; it is used for angle-closure glaucoma.

- Similarly, laser iridoplasty is a treatment for angle closure glaucoma, not open-angle glaucoma.

Solution for Question 8:

Correct Answer: C) Acetazolamide

Explanation:

Treatment of glaucoma:

- There are 3 types of treatment modalities:

1. Medical:

The drugs either decrease the formation of aqueous or increase the drainage to decrease IOP. This is of 2 types:

Topical:

- Beta-blockers These decrease the formation of aqueous Eg: Timolol, betaxolol C/I in asthma S/E: corneal anesthesia and blepharoconjunctivitis
- These decrease the formation of aqueous
- Eg: Timolol, betaxolol
- These decrease the formation of aqueous
- Eg: Timolol, betaxolol
- Alpha agonists: They have dual action i.e. they decrease the formation of aqueous and increase the trabecular outflow. Eg: Adrenaline/epinephrine, dipivefrin, brimonidine, and apraclonidine Brimonidine is safe in pregnancy Adrenaline is C/I in aphakic glaucoma
- They have dual action i.e. they decrease the formation of aqueous and increase the trabecular outflow.
- Eg: Adrenaline/epinephrine, dipivefrin, brimonidine, and apraclonidine
- Brimonidine is safe in pregnancy
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- They have dual action i.e. they decrease the formation of aqueous and increase the trabecular outflow.
- Eg: Adrenaline/epinephrine, dipivefrin, brimonidine, and apraclonidine

- Brimonidine is safe in pregnancy
- Adrenaline is C/I in aphakic glaucoma
- Miotics: These increase the trabecular outflow and open the angle blockage Eg: Pilocarpine (Option B) S/E of Pilocarpine: Pseudo myopia, retinal detachment, shallow anterior chamber, iris cysts
- These increase the trabecular outflow and open the angle blockage
- Eg: Pilocarpine (Option B)
- These increase the trabecular outflow and open the angle blockage
- Eg: Pilocarpine (Option B)
- Topical carbonic anhydrase inhibitors: These decrease the formation of aqueous Eg: dorzolamide and brinzolamide These are C/I in sulpha allergies
- These decrease the formation of aqueous
- Eg: dorzolamide and brinzolamide
- These are C/I in sulpha allergies
- These decrease the formation of aqueous
- Eg: dorzolamide and brinzolamide
- These are C/I in sulpha allergies
- Newer anti-glaucoma drugs: Netarsudil: this increases the contractile property of the trabecular meshwork, thus increasing the trabecular outflow and it also decreases the aqueous production. It is a Rho-kinase inhibitor S/E: vortex keratopathy
- Netarsudil: this increases the contractile property of the trabecular meshwork, thus increasing the trabecular outflow and it also decreases the aqueous production.
- It is a Rho-kinase inhibitor
- Netarsudil: this increases the contractile property of the trabecular meshwork, thus increasing the trabecular outflow and it also decreases the aqueous production.
- It is a Rho-kinase inhibitor
- PGF2 alpha agonist: It increases the uveoscleral outflow These are also contraindicated in uveitis and asthma For eg- Latanoprost, bimatoprost, travoprost (Option A), tafluprost, and unoprostone isopropyl (Option D) Latanoprost causes heterochromia iridis Bimatoprost increases both uveoscleral and trabecular outflow Side effects: Hypertrichosis
- It increases the uveoscleral outflow
- These are also contraindicated in uveitis and asthma
- For eg- Latanoprost, bimatoprost, travoprost (Option A), tafluprost, and unoprostone isopropyl (Option D)
- Latanoprost causes heterochromia iridis

- Bimatoprost increases both uveoscleral and trabecular outflow
- Side effects: Hypertrichosis
- It increases the uveoscleral outflow
- These are also contraindicated in uveitis and asthma
- For eg- Latanoprost, bimatoprost, travoprost (Option A), tafluprost, and unoprostone isopropyl (Option D)
- Latanoprost causes heterochromia iridis
- Bimatoprost increases both uveoscleral and trabecular outflow
- Side effects: Hypertrichosis

Systemic:

- Carbonic anhydrase inhibitor For eg: acetazolamide(Option C), methazolamide S/E: Tingling and numbness, constipation, renal stones, and a bad metallic taste
- For eg: acetazolamide(Option C), methazolamide
- For eg: acetazolamide(Option C), methazolamide
- Hyperosmotic agents: These draw aqueous out of the eye For eg: I/V mannitol (C/I in heart disease), glycerol (C/I in diabetes), isosorbide, and urea
- These draw aqueous out of the eye
- For eg: I/V mannitol (C/I in heart disease), glycerol (C/I in diabetes), isosorbide, and urea
- These draw aqueous out of the eye
- For eg: I/V mannitol (C/I in heart disease), glycerol (C/I in diabetes), isosorbide, and urea

2. Laser :

- Laser trabeculoplasty can be done using argon laser (ALT), or Micropulse diode (MLT) and selective laser trabeculoplasty (SLT).

3. Surgery:

- Surgical treatment of POAG primarily consists of a fistulizing (filtration) surgery which provides a new channel for aqueous outflow and successfully controls the IOP .
- Trabeculectomy is the most frequently performed filtration surgery nowadays.

Solution for Question 9:

Correct Answer: C) Cogan-Reese syndrome

Explanation:

Cogan-Reese Syndrome is known for its distinctive iris lesions.

- It is characterized by the presence of pigmented nodular lesions on the iris, which can be diffuse or localized.
- These pigmented lesions are a key feature that helps distinguish it from other ICE syndromes.

Glaucomas Associated with Iridocorneal Endothelial (ICE) Syndromes:

- ICE Syndromes Include: Progressive iris atrophy Chandler's syndrome Cogan-Reese syndrome
- Progressive iris atrophy
- Chandler's syndrome
- Cogan-Reese syndrome
- Pathogenesis: Abnormal corneal endothelial cells proliferate to form a membrane at the angle of the anterior chamber. Glaucoma is caused by secondary synechial angle closure due to contraction of this endothelial membrane.
- Abnormal corneal endothelial cells proliferate to form a membrane at the angle of the anterior chamber.
- Glaucoma is caused by secondary synechial angle closure due to contraction of this endothelial membrane.
- Clinical Features: Typically affects middle-aged women. Raised intraocular pressure (IOP) associated with specific features of the causative condition: Progressive iris atrophy: Iris shows corectopia, atrophy, and hole formation. (Options A & D ruled out) Chandler's syndrome: Mild or no iris changes, corneal edema predominates, even at normal IOP. (Option B ruled out) Cogan-Reese syndrome: Iris has pigmented nodular or diffuse lesions (also called iris nevus syndrome), and may have corneal changes. (Option C)
- Typically affects middle-aged women.
- Raised intraocular pressure (IOP) associated with specific features of the causative condition: Progressive iris atrophy: Iris shows corectopia, atrophy, and hole formation. (Options A & D ruled out) Chandler's syndrome: Mild or no iris changes, corneal edema predominates, even at normal IOP. (Option B ruled out) Cogan-Reese syndrome: Iris has pigmented nodular or diffuse lesions (also called iris nevus syndrome), and may have corneal changes. (Option C)
- Progressive iris atrophy: Iris shows corectopia, atrophy, and hole formation. (Options A & D ruled out)

- Chandler's syndrome: Mild or no iris changes, corneal edema predominates, even at normal IOP. (Option B ruled out)
- Cogan-Reese syndrome: Iris has pigmented nodular or diffuse lesions (also called iris nevus syndrome), and may have corneal changes. (Option C)
- Treatment: Glaucoma drainage device (artificial filtration shunt) may help control IOP. Medical treatment is often ineffective. Trabeculectomy surgery usually fails.
- Glaucoma drainage device (artificial filtration shunt) may help control IOP.
- Medical treatment is often ineffective.
- Trabeculectomy surgery usually fails.
- Progressive iris atrophy
- Chandler's syndrome
- Cogan-Reese syndrome
- Abnormal corneal endothelial cells proliferate to form a membrane at the angle of the anterior chamber.
- Glaucoma is caused by secondary synechial angle closure due to contraction of this endothelial membrane.
- Typically affects middle-aged women.
- Raised intraocular pressure (IOP) associated with specific features of the causative condition: Progressive iris atrophy: Iris shows corectopia, atrophy, and hole formation. (Options A & D ruled out) Chandler's syndrome: Mild or no iris changes, corneal edema predominates, even at normal IOP. (Option B ruled out) Cogan-Reese syndrome: Iris has pigmented nodular or diffuse lesions (also called iris nevus syndrome), and may have corneal changes. (Option C)
- Progressive iris atrophy: Iris shows corectopia, atrophy, and hole formation. (Options A & D ruled out)
- Chandler's syndrome: Mild or no iris changes, corneal edema predominates, even at normal IOP. (Option B ruled out)
- Cogan-Reese syndrome: Iris has pigmented nodular or diffuse lesions (also called iris nevus syndrome), and may have corneal changes. (Option C)
- Progressive iris atrophy: Iris shows corectopia, atrophy, and hole formation. (Options A & D ruled out)
- Chandler's syndrome: Mild or no iris changes, corneal edema predominates, even at normal IOP. (Option B ruled out)
- Cogan-Reese syndrome: Iris has pigmented nodular or diffuse lesions (also called iris nevus syndrome), and may have corneal changes. (Option C)
- Glaucoma drainage device (artificial filtration shunt) may help control IOP.

- Medical treatment is often ineffective.
- Trabeculectomy surgery usually fails.

Solution for Question 10:

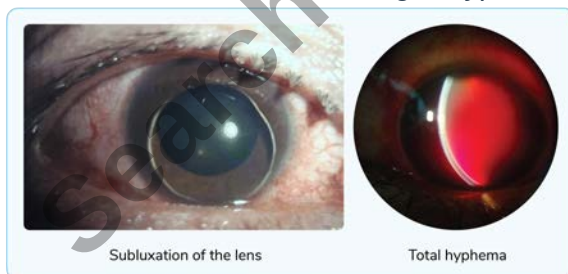
Correct Answer: B) Angle recession glaucoma typically presents with elevated IOP shortly after the trauma.

Explanation:

- Angle recession glaucoma often develops months to years after the initial blunt trauma due to long-term fibrosis of the trabecular meshwork.
- It is not typically associated with immediate elevated IOP.

Traumatic Glaucoma

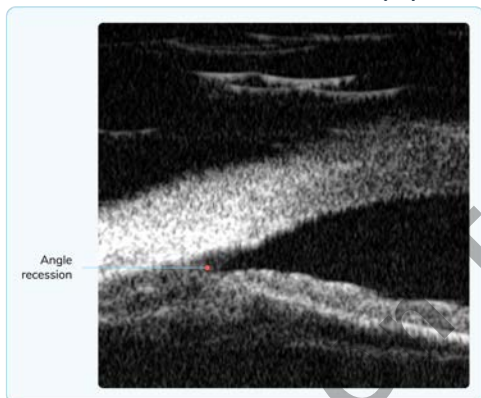
- A secondary glaucoma may complicate perforating as well as blunt injuries. (Option A ruled out)
- Causes: Inflammatory Glaucoma: Results from iridocyclitis. Intraocular Hemorrhage: Hyphaema inside the eye leading to glaucoma.
- Inflammatory Glaucoma: Results from iridocyclitis.
- Intraocular Hemorrhage: Hyphaema inside the eye leading to glaucoma.
- Inflammatory Glaucoma: Results from iridocyclitis.
- Intraocular Hemorrhage: Hyphaema inside the eye leading to glaucoma.



- Lens-Induced Glaucoma: Due to a ruptured, swollen, or dislocated lens. (Option C ruled out)
- Angle-Closure: Caused by peripheral anterior synechiae formation after corneal injury. (Option D ruled out)
- Epithelial or Fibrous Infiltration: May involve the trabecular meshwork.
- Angle Recession Glaucoma: Due to angle recession (The recession is due to a tear in the ciliary body) from blunt trauma after years (10yrs), leading to fibrosis of the trabecular

meshwork over time. (Option B)

- Lens-Induced Glaucoma: Due to a ruptured, swollen, or dislocated lens. (Option C ruled out)
- Angle-Closure: Caused by peripheral anterior synechiae formation after corneal injury. (Option D ruled out)
- Epithelial or Fibrous Infiltration: May involve the trabecular meshwork.
- Angle Recession Glaucoma: Due to angle recession (The recession is due to a tear in the ciliary body) from blunt trauma after years (10yrs), leading to fibrosis of the trabecular meshwork over time. (Option B)
- Lens-Induced Glaucoma: Due to a ruptured, swollen, or dislocated lens. (Option C ruled out)
- Angle-Closure: Caused by peripheral anterior synechiae formation after corneal injury. (Option D ruled out)
- Epithelial or Fibrous Infiltration: May involve the trabecular meshwork.
- Angle Recession Glaucoma: Due to angle recession (The recession is due to a tear in the ciliary body) from blunt trauma after years (10yrs), leading to fibrosis of the trabecular meshwork over time. (Option B)



- Management: Medical Therapy: Includes topical 0.5% timolol and oral acetazolamide. Treatment of Underlying Causes: For instance, using atropine and steroids to control inflammation. Surgical Intervention: Based on the specific cause and situation.
- Medical Therapy: Includes topical 0.5% timolol and oral acetazolamide.
- Treatment of Underlying Causes: For instance, using atropine and steroids to control inflammation.
- Surgical Intervention: Based on the specific cause and situation.
- Medical Therapy: Includes topical 0.5% timolol and oral acetazolamide.
- Treatment of Underlying Causes: For instance, using atropine and steroids to control inflammation.
- Surgical Intervention: Based on the specific cause and situation.

Solution for Question 11:

Correct Answer: D) Glaucomatocyclitic Crisis

Explanation:

Glaucomatocyclitic Crisis (Posner-Schlossman Syndrome) is characterized by very high IOP (often between 40 and 50 mm Hg), minimal inflammation, a few small keratic precipitates, no posterior synechiae, and recurrent episodes that resolve within days to weeks.

- The treatment involves controlling IOP with anti-glaucoma medications, followed by steroid therapy to manage the uveitis.

Glaucoma Associated with Uveitis

Types:

1. Secondary Angle-closure glaucoma with a pupillary block (Option A): Caused by pupillary block due to annular synechiae, leading to iris bombe and subsequent angle closure.
2. Secondary Angle-closure glaucoma without a pupillary block (Option B) : Results from inflammatory debris organizing in the angle, which pulls the iris over the trabecular meshwork, causing progressive angle closure and elevated IOP.
3. Open-angle glaucoma (Option C) : This is caused by the blocking/inflammation of the trabecular meshwork by aqueous cells.
4. Specific Hypertensive Uveitis Syndromes: Glaucomatocyclitic crisis

Uveitis with Open-Angle Glaucoma (Glaucomatocyclitic Crisis)

- It is called glaucomatocyclitic crisis or PSS (Posner-Schlossman Syndrome) or hypertensive uveitis (There is high intraocular pressure).
- The glaucoma is out of proportion to the minimal inflammation present.
- The condition is characterized by acute, recurrent episodes that resolve within days to weeks. Recurrence is common.
- Etiology: It is commonly seen in young males. This can be due to HLA-BW54 association has been documented. CMV or H. Pylori infection.
- HLA-BW54 association has been documented.
- CMV or H. Pylori infection.

- Pathogenesis: It is caused due to trabeculitis and blocking of trabecular meshwork happening due to uveitis.
- HLA-BW54 association has been documented.
- CMV or H. Pylori infection.



Posner-Schlossman Syndrome

- On examination Keratic Precipitates: Few small, flat, nonpigmented keratic precipitates, typically located inferiorly. Dilated pupil. Aqueous Flare is present. No posterior synechiae. Intraocular pressure: Very high, often between 40 and 50 mm Hg. White eye (no congestion)
- Keratic Precipitates: Few small, flat, nonpigmented keratic precipitates, typically located inferiorly.
- Dilated pupil.
- Aqueous Flare is present.
- No posterior synechiae
- Intraocular pressure: Very high, often between 40 and 50 mm Hg.
- White eye (no congestion)
- Keratic Precipitates: Few small, flat, nonpigmented keratic precipitates, typically located inferiorly.
- Dilated pupil.
- Aqueous Flare is present.
- No posterior synechiae
- Intraocular pressure: Very high, often between 40 and 50 mm Hg.
- White eye (no congestion)
- Treatment The treatment of choice is anti-glaucoma drugs and once IOP is under control, steroids are given. Thus, steroids are given under anti-glaucoma cover.
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- The treatment of choice is anti-glaucoma drugs and once IOP is under control, steroids are given. Thus, steroids are given under anti-glaucoma cover.

Solution for Question 12:

Correct Answer: C) Atropine drops and oral acetazolamide

Explanation:

- The clinical scenario describes a 70-year-old patient who, following trabeculectomy, experiences severe eye pain and blurred vision, elevated intraocular pressure (IOP), a shallow anterior chamber, a negative Seidel's test, and no improvement with miotic agents which are characteristic of ciliary block glaucoma (malignant glaucoma).
- This condition occurs due to a mechanical blockage of aqueous humor flow, often after intraocular surgery.

Initial Treatment for Malignant Glaucoma:

- Atropine Drops: These are used to dilate the ciliary ring and alleviate the ciliary block causing the increased IOP and shallow anterior chamber.
- Oral Acetazolamide: This carbonic anhydrase inhibitor helps reduce aqueous humor production, which is crucial for managing the elevated IOP in ciliary block glaucoma.

Ciliary Block Glaucoma (Malignant Glaucoma/ Aqueous misdirection syndrome)

- A rare form of glaucoma that may occur as a complication of intraocular surgery.
- Characterized by markedly raised intraocular pressure (IOP) and a shallow or absent anterior chamber.
- Post-surgical Complication: Common after peripheral iridectomy (Option D ruled out), trabeculectomy, cataract surgery, capsulotomy, and laser iridotomy.
- Cilio-Lenticular Block: In phakic eyes, the ciliary processes rotate forward and press against the lens.
- Cilio-IOL Block: In pseudophakic eyes, the ciliary processes press against the intraocular lens.
- Cilio-Vitreous Block: In aphakic eyes, the ciliary processes press against the anterior hyaloid face of the vitreous.
- Outcome: Blocks forward flow of aqueous humor, causing aqueous pockets to form in the vitreous, pushing the iris-lens diaphragm forward, increasing IOP, and flattening the anterior chamber.

- Symptoms: Severe pain and blurred vision post-intraocular surgery.
- Signs: Persistent shallow anterior chamber following surgery. Markedly raised IOP in the early postoperative period. Negative Seidel's test (Seidel's test: Tear film is stained to detect leakage of aqueous from the anterior chamber) Poor response or aggravation by miotics.
- Persistent shallow anterior chamber following surgery.
- Markedly raised IOP in the early postoperative period.
- Negative Seidel's test (Seidel's test: Tear film is stained to detect leakage of aqueous from the anterior chamber)
- Poor response or aggravation by miotics.
- Persistent shallow anterior chamber following surgery.
- Markedly raised IOP in the early postoperative period.
- Negative Seidel's test (Seidel's test: Tear film is stained to detect leakage of aqueous from the anterior chamber)
- Poor response or aggravation by miotics.
- Types: Can be phakic, aphakic, or pseudophakic.
- Medical Therapy (Initial): Atropine Drops/Ointment (1%): To dilate the ciliary ring and relieve ciliary block. Acetazolamide (250 mg QID): To reduce aqueous humor production. Timolol Maleate (0.5%): To decrease aqueous production. Intravenous Mannitol: To reduce vitreous gel volume and decrease IOP.
- Atropine Drops/Ointment (1%): To dilate the ciliary ring and relieve ciliary block.
- Acetazolamide (250 mg QID): To reduce aqueous humor production.
- Timolol Maleate (0.5%): To decrease aqueous production.
- Intravenous Mannitol: To reduce vitreous gel volume and decrease IOP.
- Laser Therapy: YAG laser hyaloidotomy for aphakic and pseudophakic patients.
- Surgical Therapy: Pars plana vitrectomy with or without lensectomy if medical therapy fails within 4-5 days. (Option B ruled out)
- Atropine Drops/Ointment (1%): To dilate the ciliary ring and relieve ciliary block.
- Acetazolamide (250 mg QID): To reduce aqueous humor production.
- Timolol Maleate (0.5%): To decrease aqueous production.
- Intravenous Mannitol: To reduce vitreous gel volume and decrease IOP.
- Surgical intervention is often effective but may need to be repeated if the condition recurs.
- The fellow eye is also at risk of developing similar issues.

Oral Corticosteroids (Option A):

- While helpful in some types of inflammation, they are not the primary treatment for ciliary block glaucoma, which is caused by mechanical blockage rather than inflammation.

Solution for Question 13:

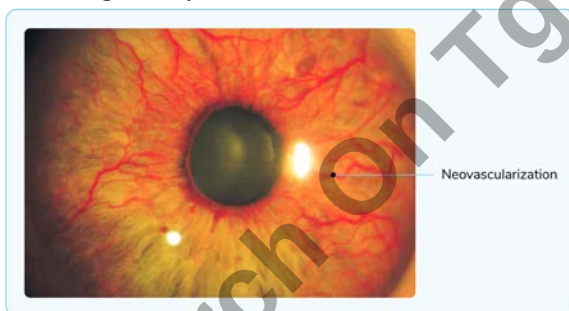
Correct Answer: B) Panretinal photocoagulation

Explanation:

- Based on these findings, the diagnosis of neovascular glaucoma is supported by the presence of iris neovascularization and elevated IOP in the context of proliferative diabetic retinopathy.
- Panretinal photocoagulation is the most appropriate treatment as it targets the underlying cause of neovascular glaucoma.
- This treats the retinal ischemia, reducing the stimulus for abnormal blood vessel growth and neovascularization.

Neovascular Glaucoma (NVG)

- NVG is a severe type of glaucoma caused by neovascularization (abnormal blood vessel growth) in the iris and anterior chamber angle of the eye.



- Causes: Common: Proliferative diabetic retinopathy Central retinal vein occlusion Sick cell retinopathy Eales' disease Rare: Chronic intraocular inflammation Intraocular tumors Long-standing retinal detachment Central retinal artery occlusion
- Common: Proliferative diabetic retinopathy Central retinal vein occlusion Sick cell retinopathy Eales' disease
- Proliferative diabetic retinopathy
- Central retinal vein occlusion
- Sick cell retinopathy
- Eales' disease

- Rare: Chronic intraocular inflammation Intraocular tumors Long-standing retinal detachment Central retinal artery occlusion
- Chronic intraocular inflammation
- Intraocular tumors
- Long-standing retinal detachment
- Central retinal artery occlusion
- Mechanism: Neovascularization (rubeosis iridis) on the iris leads to fibrosis. Results in zip-like adhesions of the iris to the cornea at the angle.
- Neovascularization (rubeosis iridis) on the iris leads to fibrosis.
- Results in zip-like adhesions of the iris to the cornea at the angle.
- Stages: Pre-glaucomatous Stage: Presence of rubeosis iridis (neovascularization of the iris). Open-Angle Glaucoma Stage: Formation of a pretrabecular neovascular membrane, leading to increased intraocular pressure (IOP). Secondary Angle Closure Glaucoma Stage: Neovascular membrane causes goniosynechiae (adhesions) leading to angle closure.
- Pre-glaucomatous Stage: Presence of rubeosis iridis (neovascularization of the iris).
- Open-Angle Glaucoma Stage: Formation of a pretrabecular neovascular membrane, leading to increased intraocular pressure (IOP).
- Secondary Angle Closure Glaucoma Stage: Neovascular membrane causes goniosynechiae (adhesions) leading to angle closure.
- Treatment: Panretinal Photocoagulation or Cryotherapy: Aimed at preventing further neovascularization. Anti-VEGF Agents: Used as an adjunctive treatment . Help reduce neovascularization.(Option D) Medical Therapy: Often ineffective for controlling IOP. Trabeculectomy: A surgical procedure to lower intraocular pressure, often combined with antifibroblastic agents. Glaucoma Drainage Device: Artificial filtration shunt (Seton operation) can help manage IOP.
- Panretinal Photocoagulation or Cryotherapy: Aimed at preventing further neovascularization.
- Anti-VEGF Agents: Used as an adjunctive treatment . Help reduce neovascularization.(Option D)
- Medical Therapy: Often ineffective for controlling IOP.
- Trabeculectomy: A surgical procedure to lower intraocular pressure, often combined with antifibroblastic agents.
- Glaucoma Drainage Device: Artificial filtration shunt (Seton operation) can help manage IOP.
- Common: Proliferative diabetic retinopathy Central retinal vein occlusion Sickle-cell retinopathy Eales' disease

- Proliferative diabetic retinopathy
- Central retinal vein occlusion
- Sickle-cell retinopathy
- Eales' disease
- Rare: Chronic intraocular inflammation Intraocular tumors Long-standing retinal detachment
- Central retinal artery occlusion
- Chronic intraocular inflammation
- Intraocular tumors
- Long-standing retinal detachment
- Central retinal artery occlusion
- Proliferative diabetic retinopathy
- Central retinal vein occlusion
- Sickle-cell retinopathy
- Eales' disease
- Chronic intraocular inflammation
- Intraocular tumors
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- Neovascularization (rubeosis iridis) on the iris leads to fibrosis.
- Results in zip-like adhesions of the iris to the cornea at the angle.
- Pre-glaucomatous Stage: Presence of rubeosis iridis (neovascularization of the iris).
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- Panretinal Photocoagulation or Cryotherapy: Aimed at preventing further neovascularization.
- Anti-VEGF Agents: Used as an adjunctive treatment . Help reduce neovascularization.(Option D)
- Medical Therapy: Often ineffective for controlling IOP.
- Trabeculectomy: A surgical procedure to lower intraocular pressure, often combined with antifibroblastic agents.
- Glaucoma Drainage Device: Artificial filtration shunt (Seton operation) can help manage IOP.

Topical beta-blockers and oral carbonic anhydrase inhibitors (Option A):

- These medications can help manage intraocular pressure (IOP) but do not address the underlying cause of neovascular glaucoma, which is the retinal ischemia and resulting neovascularization.

Systemic corticosteroids (Option C):

- While corticosteroids can reduce inflammation, they do not address neovascularization or the elevated IOP associated with neovascular glaucoma.
- They are not the primary treatment for this condition.

Solution for Question 14:

- PACG is more common in Asians compared to all other ethnicities.
- The risk factors and etiopathogenesis mechanisms for primary angle closure glaucoma (PACG) are given in the table below:

Risk Factors	Mechanisms
Demographic factors: Age: 6th and 7th decades of life Gender: Female to male ratio 1:3 Ethnicity: More common in Asians	1. Pupillary block (70% of cases): (Option A ruled out) Most common mechanism (70% of cases) Increased contact between iris and lens Precipitating factors: Physiological mydriasis (e.g., dim illumination, emotional stress) Pharmacological mydriasis (e.g., tropicamide, cyclopentolate) Results in relative pupil block
Anatomical factors: Hypermetropic eyes Shallow anterior chamber Short axial length Thick, anteriorly positioned lens Small corneal diameter Plateau iris configuration	2. Plateau iris configuration: Anteriorly positioned ciliary processes More common in younger patients (Option C ruled out) Can persist after iridotomy Associated with angle closure glaucoma without pupillary block
Other factors: Family history Diabetes mellitus Certain medications (e.g., topiramate)	3. Phacomorphic mechanism: (Option B ruled out) Swollen lens contributes to angle closure Can cause pupillary block or push peripheral iris forward Can occur in acute or chronic forms Associated with intumescent or other lens morphological abnormalities

Solution for Question 15:

Correct Answer: B) Zonular instability and phacodonesis.

Explanation:

- Zonular instability is a hallmark of pseudoexfoliation syndrome, where the accumulation of pseudoexfoliative material weakens the zonular fibers.
- This can lead to phacodonesis (lens wobbling).
- Ocular manifestation of Pseudoexfoliation Syndrome (XFS) is characterized by the accumulation of fibrillar, proteinaceous material on various ocular structures, including the lens, iris, and trabecular meshwork.
- This material can be seen as flaky white deposits on the anterior lens capsule and pupillary margin, often with a distinct clear zone at the center of the lens.
- It causes secondary open-angle glaucoma, known as pseudoexfoliative glaucoma.
- The excessive material and pigment clog the trabecular meshwork, increasing intraocular pressure (IOP) and causing damage to the optic nerve, which leads to glaucoma.
- It also predisposes patients to complications during cataract surgery due to weakened zonules, resulting in lens instability or phacodonesis.

Features of Pseudoexfoliation Syndrome:

- White flaky deposits on the anterior lens capsule and iris
- Increased trabecular meshwork pigmentation
- Elevated IOP, leading to secondary glaucoma
- Zonular instability and risk of lens dislocation during cataract surgery
- Mutations in the LOXL1 gene have been strongly associated with PXF, although not all individuals with these mutations develop the disease.
- More prevalent in elderly patients.

Posterior synechiae (Option A) is incorrect because while inflammation can lead to synechiae, it is not a primary feature of pseudoexfoliation syndrome.

Corneal guttata (Option C) is associated with Fuchs' endothelial dystrophy, not pseudoexfoliation syndrome.

Iris heterochromia (Option D) is typically seen in Fuchs' heterochromic iridocyclitis, not pseudoexfoliation syndrome.

Solution for Question 16:

Correct Answer: C) Topical prostaglandin analogue

Explanation:

This patient's presentation is consistent with pigmentary glaucoma.

The key features are:

1. Young, myopic male
2. Elevated IOP
3. Krukenberg spindle (vertical line of pigment on corneal endothelium)
4. Heavily pigmented trabecular meshwork
5. Iris transillumination defects
6. Symptoms exacerbated by exercise (due to pupillary movement causing increased pigment release)

Pigmentary glaucoma:

- Pigmentary glaucoma is a form of secondary open-angle glaucoma characterized by clogging up of the trabecular meshwork caused by the pigment particles.
- Pigment release may be caused by mechanical rubbing of the posterior pigment layer of the iris with the zonular fibrils.
- It is more common in young, myopic males.
- Pigment granules are deposited in anterior segment structures, including the iris, the posterior surface of the cornea (Krukenberg's spindle), the trabecular meshwork, ciliary zonules, and the crystalline lens.
- On gonioscopy, pigment is seen accumulating along Schwalbe's line, particularly in the inferior region (Sampaolesi's line).
- Iris transillumination reveals radial, slit-like defects in the mid-periphery, which is a characteristic feature.

- Treatment is along the lines of POAG management, which initially includes medical treatment, with prostaglandin analogues often being the first choice due to their IOP-lowering efficacy and convenient dosing schedule. (Option C)

Nd:YAG laser peripheral iridotomy (Option A) is not the first-line treatment.

- It is generally reserved for cases with pronounced iris concavity or those not responding to medical therapy.

Selective laser trabeculoplasty (SLT) (Option B) is typically considered after initial medical therapy or in combination with it.

Argon laser peripheral iridoplasty (Option D) is primarily used in angle-closure glaucoma to open the angle.

- It is not typically used in pigmentary glaucoma, where the angle is already open.

Solution for Question 17:

Correct Answer: B) Pigment dispersion syndrome

Explanation:

- Pigment dispersion syndrome occurs when pigment granules from the iris epithelium are released into the anterior chamber and deposit on various structures, including the corneal endothelium and trabecular meshwork.
- This can lead to trabecular outflow obstruction and subsequent glaucoma.
- Symmetric Iris transillumination defects in the mid-peripheral iris, is characteristic of Pigment Dispersion Syndrome (PDS).

Secondary Glaucoma:

- Secondary glaucoma refers to elevated intraocular pressure resulting from an identifiable ocular or systemic disorder.
- Unlike primary glaucomas, secondary glaucomas have a clear underlying cause that alters the aqueous humor dynamics, leading to increased IOP and subsequent optic nerve damage.

Causes of Secondary Glaucoma:

- Lens-induced glaucomas (phacomorphic, phacolytic, lens-particle)
- Pigmentary glaucoma

- Pseudoexfoliative glaucoma
- Neovascular glaucoma
- Inflammatory glaucoma (uveitic)
- Traumatic glaucoma
- Steroid-induced glaucoma
- Glaucoma-in-aphakia
- Iridocorneal endothelial syndrome (ICE syndrome)
- Aqueous misdirection syndrome (malignant glaucoma)
- Glaucomas associated with intraocular hemorrhage.
- Glaucoma associated with intraocular tumors.

Pseudoexfoliation syndrome (Option A) presents with whitish, dandruff-like material on the lens capsule and pupillary margin. Transillumination defects can also be seen in Pseudoexfoliation syndrome but they are usually patchy and irregular. Gonioscopy may show increased pigment deposition on the trabecular meshwork.

Neovascular glaucoma (Option C) is characterized by new blood vessel growth over the iris and anterior chamber angle, often secondary to ischemic retinal diseases. It typically presents with a red eye, severe pain, and very high IOP.

Lens-particle glaucoma (Option D) occurs when lens material enters the anterior chamber, usually after cataract surgery or lens injury.

Solution for Question 18:

Correct Answer: C) Cyclodestructive procedure

Explanation:

- Absolute glaucoma is the final phase of Primary Angle Closure Glaucoma.

Clinical features of Absolute Glaucoma are:

Management of absolute glaucoma

- In absolute glaucoma, the goal of treatment is to relieve pain and reduce IOP, as vision cannot be restored.

- Retrobulbar alcohol injection - given to relieve pain. First, 1 mL of 2% xylocaine is injected followed after about 5-10 minutes by 1 mL of 80% alcohol. It suppresses the ciliary ganglion.
- A cyclodestructive procedure, such as cyclocryotherapy or cyclophotocoagulation, is the most appropriate management option in this scenario as it reduces aqueous production by damaging the secretory ciliary processes aiming to reduce IOP. (Option C)
- If pain persists despite cyclodestruction, enucleation/evisceration may be considered. (Option D)

Topical timolol and brimonidine (Option A) are incorrect because medical management with topical medications is insufficient for absolute glaucoma.

Trabeculectomy (Option B) is incorrect because filtration surgery is not indicated in absolute glaucoma.

- The eye is already blind, and there is a high risk of complications such as hypotony, choroidal effusion, or suprachoroidal hemorrhage in eyes with such advanced disease.

Solution for Question 19:

Correct Answer: D) Progressive retinal damage

Explanation:

- Vogt's triad is a clinical set of findings typically associated with acute primary angle-closure glaucoma (APAC) after an attack has been resolved.
- The triad helps in identifying damage caused during the acute episode.

Features of APAC:

- Descemet membrane folds
- Iris atrophy
- Irregular pupil
- Glaucomflecken
- Cataract
- Cupping +/-

Vogt's triad: (Option A, B, C)

It may be seen in both treated and non-treated cases after an attack of acute PAC. It is characterized by:

Solution for Question 20:

Correct Answer: B) Prophylactic peripheral iridotomy

Explanation:

- PACS is characterized by an occludable anterior chamber angle as seen during gonioscopy but without evidence of increased IOP or glaucomatous damage.
- This indicates a risk for potential future angle closure but does not yet present as acute angle-closure glaucoma.

PACS: Primary Angle Closure Suspect

- Iridotrabecular contact or occludable angle of more than 3 quadrants on gonioscopy
- No peripheral anterior synechiae
- Intraocular pressure is normal
- Eclipse sign of PACS - convex iris lens diaphragm
- Treatment: Prophylactic Peripheral iridotomy (Option B) is done in both eyes with lens extraction using NdYAG laser

PAC: Primary Angle Closure

- Iridotrabecular contact or occludable angle of more than 3 quadrants
- PAS+/- (by indentation gonioscopy)
- Raised IOP
- Treatment: Prophylactic Peripheral iridotomy and control of IOP

I.V Mannitol/Acetazolamide (Option A) is treatment of choice to decrease the pressure in cases of acute primary angle closure (APAC) / Acute congestive glaucoma

Filtration surgery (Option C) i.e. trabeculectomy should be performed in cases where IOP is not controlled with the maximum medical therapy following an attack of acute PAC or when gonioscopy reveals PAS $>270^\circ$ angle and also when peripheral iridotomy is not effective.

Topical beta-blocker therapy (Option D) is used to lower IOP in patients with glaucoma but is not appropriate for a patient with narrow angles without elevated IOP.

Solution for Question 21:

Correct Answer: B) The width of the anterior chamber angle

Explanation:

- The VAN Herick technique is an eye examination method used to determine the width of the anterior chamber angle of the eye using the ratio of the anterior chamber depth and corneal section which gives the status of the angle as per the chart given below.

Width of the empty space (LACD) as compared to the corneal thickness	VAN Herick Grade	Angle status
No black space observed	0	Closed
<1/4 corneal thickness	1	Extremely narrow
1/4 of corneal thickness	2	Narrow
>1/4 to 1/2 of corneal thickness	3	Open
≥ of corneal thickness	4	Wide open

LACD= Limbal anterior Chamber depth

The curvature of the cornea (Option A): The curvature of the cornea is assessed using a keratometer or topographer.

The condition of the retina (Option C): VAN Herick technique is used to estimate the anterior chamber angle (ACA) and not the condition of the retina.

The intraocular pressure (Option D): Measuring intraocular pressure is done with a tonometer.

Solution for Question 22:

Correct Answer: D) Grade 3

Explanation:

- The Shaffer system is one of the various methods used to grade the angle during the gonioscopy.
- The angle in degree between two imaginary lines is tangential to the inner surface of trabecular meshwork and the interior surface of the iris (1/3rd distance from the periphery)
- When the iris is in a normal position it will open up to 45 degrees, and it decreases with various problems. When it keeps on decreasing the angle structures will be different on gonioscopy.
- They are graded according to the following chart. Grade 0 (Option A): The angle is closed with no visible structures; only the iris and cornea are seen. Grade 1 (Option B): The angle is very narrow, and only a small portion of the angle structure, such as the iris insertion, is visible. Grade 2 (Option C): The angle is moderately open; structures like the trabecular meshwork are partially visible.
- Grade 0 (Option A): The angle is closed with no visible structures; only the iris and cornea are seen.
- Grade 1 (Option B): The angle is very narrow, and only a small portion of the angle structure, such as the iris insertion, is visible.
- Grade 2 (Option C): The angle is moderately open; structures like the trabecular meshwork are partially visible.
- Grade 0 (Option A): The angle is closed with no visible structures; only the iris and cornea are seen.
- Grade 1 (Option B): The angle is very narrow, and only a small portion of the angle structure, such as the iris insertion, is visible.
- Grade 2 (Option C): The angle is moderately open; structures like the trabecular meshwork are partially visible.

ACA in degrees	ACA grade	Angle status	Visible structures
0	0	Closed	No structures visible
≤ 10	1	Extremely narrow	Schwalbe's line
11-19	2	Narrow	Trabecular meshwork
20-34	3	Open	Scleral spur
35-45	4	Wide open	Ciliary body

ACA= Anterior Chamber Angle

Solution for Question 23:

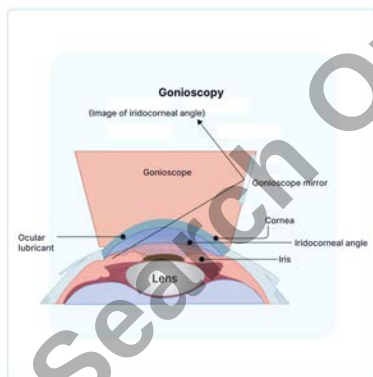
Correct Answer: B) It is a technique for visualizing the anterior chamber angle of the eye using a special lens.

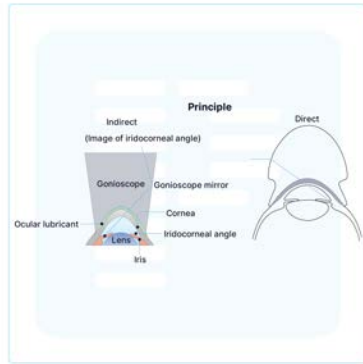
Explanation:

- Gonioscopy is the main method used to measure the angle of the anterior chamber
- The angled structure is not visible to the examiner because of the total internal reflection inside the anterior chamber.
- Principle: It overcomes total internal reflection
- Critical angle = 46 degree
- A Gonioscope with the same refractive index as the cornea and mirror is placed over the cornea.
- Light from the anterior chamber strikes the cornea and travels through and gets reflected by the mirror in the gonioscope, thus reaching our eye.

Structures in the angle of the anterior chamber that can be visualized by gonioscope

- Ciliary body band
- Scleral spur
- Trabecular meshwork
- Schwalbe's line (peripheral termination of descemet's membrane)





- Measuring intraocular pressure is done with a tonometer, not a gonioscopy. (Option A)
- It is a technique for visualizing the anterior chamber angle of the eye using a special lens and not the posterior chamber. (Option C)
- The retina is evaluated with a fundus camera or other retinal imaging techniques, not gonioscopy. (Option D)

Solution for Question 24:

Correct Answer: B) It is more commonly seen in older patients with primary angle closure glaucoma

Explanation:

- Plateau iris configuration is actually more common in younger patients with primary angle closure glaucoma, not older patients.

Plateau Iris Configuration

Anatomy	Plateau iris is characterized by an anteriorly positioned ciliary body, which pushes the peripheral iris forward. This creates a "table-top" or plateau-like appearance of the iris.
Age of presentation	It is more commonly seen in younger patients with primary angle closure glaucoma. This is in contrast to the pupillary block, which is more common in older individuals.
Diagnostic tools	Ultrasound biomicroscopy (UBM) is particularly useful in diagnosing plateau iris configuration. (Option D)

Management	Due to its persistence after iridotomy, patients with plateau iris may require additional treatments such as: (Option C) Argon laser peripheral iridoplasty Long-term use of miotic drops
------------	--

Solution for Question 25:

An angle is considered occludable if the posterior trabecular meshwork is not visible in more than 180 degrees, not less than 180 degrees.

Occludable angle:

- Defined as when the posterior trabecular meshwork is not visible in more than 180 degrees without indentation or manipulation
- Associated with a shallow anterior chamber
- The angle recess in such eyes is generally less than 20°
- Can be further evaluated with a dark room prone provocative test

Tools for anterior chamber angle assessment include:

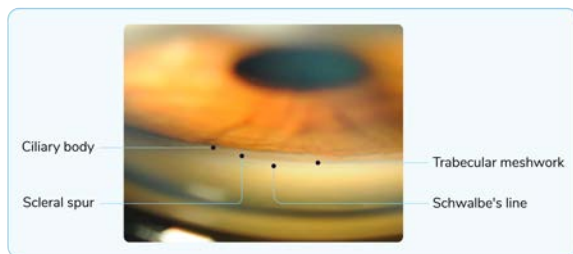
- Gonioscopy- Definitive Diagnosis (Option A ruled out)
- Van-Herick Method
- Grading the angle - Shaffer system
- Ultrasound biomicroscopy (UBM)-50 MHz
- AS-OCT: Non-Contact Method

There are 2 types of gonioscopy lenses:

Indirect (Option D ruled out)	Direct
Provides a mirror image of the opposite angle	Provides a direct view of the angle
Example : 1. Goldmann lenses Single mirror Surger Two mirror Three mirror 2. Zeiss 3. Susman 4. Posner – Ritch	Example: 1. Koeppe 2. Swan – Jacob

Normal angle structures from anterior to posterior: (Option B ruled out)

- Schwalbe's line
- Non-Pigmented trabecular meshwork
- Pigmented Trabecular meshwork
- Scleral spur
- Ciliary body band
- Iris



Glaucoma - POAG, Tonometry, Perimetry

1. Match the following: 1) Perkins Tonometer a) {{caption_text}} 2) Tonopen b) {{caption_text}} 3) Pulsair tonometer c) {{caption_text}} 4) Rebound Tonometer d) {{caption_text}}

1) Perkins Tonometer	 a) {{caption_text}}
2) Tonopen	 ... (content truncated)
3) Pulsair tonometer	 ... (content truncated)

4) Rebound Tonometer



d)
{{caption_text}}

- A. 1-a, 2-c, 3-d, 4-b
- B. 1-c, 2-b, 3-d, 4-a
- C. 1-d, 2-a, 3-b, 4-c
- D. 1-c, 2-d, 3-a, 4-b

2. A 62-year-old man with a history of Glaucoma undergoes visual field testing to assess his peripheral vision. The test involves moving a stimulus of constant size and intensity from a non-seeing area into a seeing area to determine the boundaries of his visual field. Which type of perimetry is being described?

- A. Goldmann perimetry
- B. Automated perimetry
- C. Frequency doubling perimetry
- D. Short-wavelength automated perimetry

3. A 55-year-old patient with primary open-angle glaucoma undergoes visual field testing. Which of the following visual field defects is considered the earliest clinically significant field defect occurring in glaucoma?

- A.
- B.
- C.
- D.

4. A 55-year-old patient undergoes a comprehensive eye examination including visual field testing. Which of the following statements is incorrect?

- A. A scotoma is an area of decreased vision surrounded by normal vision
 - B. An isopter is a line connecting points of equal light sensitivity and size detection in the visual field
 - C. The blind spot is a type of absolute scotoma
 - D. The visual field extends maximally in the superior direction
-

5. A 58-year-old patient is being evaluated for suspected primary open-angle glaucoma. The ophthalmologist wants to perform a thorough examination of the optic nerve head to detect glaucomatous changes. Which of the following techniques is considered the examination of choice for evaluating fundus changes in this patient?

- A. Direct ophthalmoscopy
 - B. Slit-lamp biomicroscopic examination with non-contact lens
 - C. Optical coherence tomography (OCT)
 - D. Indirect ophthalmoscopy
-

6. A 62-year-old woman with a family history of glaucoma presents with mild visual disturbances. Her intraocular pressure is measured to be 24 mmHg in both eyes. On dilated fundus examination, the following findings are noted: The cup-to-disc ratio of 0.5 Splinter hemorrhages at the optic disc margin Wedge-shaped retinal nerve fiber layer defects on red-free light examination What stage of glaucomatous optic disc changes is most likely present in this patient?

- A. Glaucomatous optic atrophy
- B. Advanced glaucomatous changes

- C. Early glaucomatous changes
 - D. Physiological cupping
-

7. A 52-year-old male presents with progressive vision loss. Fundoscopic examination reveals a vertical elongation of the optic disc and diffuse thinning of the retinal nerve fiber layer (RNFL). Imaging confirms significant damage to the peripapillary region. Which of the following best describes the characteristic peripapillary change seen in glaucomatous optic neuropathy?

- A. Increased retinal pigment epithelium (RPE) density in the peripapillary area
 - B. Presence of alpha zone peripapillary atrophy (PPA)
 - C. Thickening of peripapillary RNFL on optical coherence tomography (OCT)
 - D. Uniform retinal nerve fiber layer thickness without focal loss
-

8. A 58-year-old woman with a long history of untreated open-angle glaucoma presents with gradual peripheral vision loss. Fundoscopic examination reveals advanced optic disc cupping. Which of the following vascular changes is most characteristic of progressive glaucomatous optic neuropathy?

- A. Inferior displacement of retinal vessels
 - B. Nasal displacement of retinal vessels
 - C. Arterial attenuation with increased vein diameter
 - D. Radial peripapillary capillary obliteration
-

9. A 58-year-old male with a family history of glaucoma presents for a comprehensive eye exam. His visual acuity is 20/20 in both eyes, and his intraocular pressure (IOP) is measured at 23 mmHg in the right eye and 21 mmHg in the left eye. Fundoscopic examination reveals the following optic disc findings

in the right eye: Cup-to-disc ratio: 0.7 vertically Neuroretinal rim thinning: More pronounced in the superior and inferior quadrants Peripapillary atrophy present No optic disc hemorrhages noted All of the following statements are correct regarding the optic disc assessment in this patient except?

- A. The optic disc appearance follows the ISNT rule, indicating a healthy optic nerve.
- B. The cup-to-disc ratio indicates early glaucomatous damage.
- C. The absence of optic disc hemorrhages does not rule out glaucoma.
- D. Peripapillary atrophy is commonly associated with glaucoma and warrants further monitoring.

10. Which device is a non contact tonometer that is based on the principle of corneal hysteresis?

- A. Schiottz tonometer
- B. Goldmann applanation tonometer
- C. Ocular Response Analyzer
- D. Pneumatonometer

**11. Which of the following statements about aqueous humor are incorrect?
Aqueous humor is formed from the non-pigmented epithelium of ciliary processes
Aqueous constitutes 99% water The non-pigmented epithelium has a Na/Cl symporter responsible for aqueous humor secretion. The rate of formation of Aqueous humor is 2.3 microlitres/min**

- A. 1, 2, 4 are incorrect
- B. Only 3 is incorrect
- C. 1,3 is incorrect
- D. Only 1 is incorrect

12. Which type of tonometer measures intraocular pressure (IOP) by assessing how much a plunger indents the eye?

- A. Schiøtz Tonometer
 - B. Goldmann Tonometer
 - C. Perkins Tonometer
 - D. Draeger Tonometer
-

13. A 58-year-old man presents for a routine eye examination. His intraocular pressure (IOP) is consistently measured at 23 mm Hg on multiple occasions. He reports no visual symptoms, and his fundus examination shows no abnormalities. Visual field testing reveals no defects. Based on these findings, which of the following diagnoses is most appropriate?

- A. Normal tension glaucoma
 - B. Ocular hypertension
 - C. Acute angle-closure glaucoma
 - D. Secondary glaucoma
-

14. A 65-year-old woman with a history of glaucoma reports worsening vision, headaches, and difficulty with reading. Her IOP shows significant fluctuations throughout the day, and fundus examination reveals optic disc cupping with visual field defects. Which of the following about IOP changes in POAG is false?

- A. Diurnal variation includes a morning rise in IOP in 20% of cases and a biphasic rise in 55% of cases.
- B. In early POAG, IOP is consistently above 21 mm Hg.
- C. IOP is higher in winter due to seasonal variations.

D. Strenuous exercise temporarily lowers IOP.

15. Which gene mutation is linked to the development of primary open angle glaucoma?

- A. MYOC gene
 - B. LOXL1 gene
 - C. PAX6 gene
 - D. RB1 gene
-

16. Which of the following is not a recognized risk factor for Primary Open-Angle Glaucoma?

- A. Elevated intraocular pressure
 - B. Diastolic perfusion pressure of more than 55 mm Hg
 - C. Family history of glaucoma
 - D. High myopia
-

17. All the following statements about trabecular meshwork are true except;

- A. It regulates the production of aqueous humor
 - B. Trabecular meshwork secrete Glycosaminoglycans
 - C. It facilitates the drainage of aqueous humor from the anterior chamber of the eye
 - D. Trabecular meshwork has 3 parts and among the three, juxtacanalicular has the maximum resistance
-

18. Which of the following best describes the role of uveoscleral outflow in intraocular pressure regulation?

- A. It involves the drainage of aqueous humor through the trabecular meshwork and Schlemm's canal.
 - B. It accounts for the drainage of aqueous humor through the ciliary body and sclera, providing an alternative pathway to the conventional outflow route.
 - C. It is the process by which aqueous humor exits the eye through the vitreous body.
 - D. It describes the absorption of aqueous humor by the corneal endothelium.
-

19. All the following constituents are more in aqueous humor compared to that of plasma except

- A. Lactate
 - B. Ascorbate
 - C. Pyruvate
 - D. Glucose
-

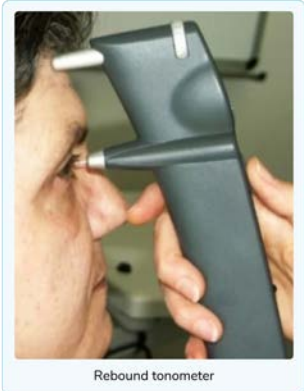
Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	3
Question 4	4
Question 5	2
Question 6	3
Question 7	2
Question 8	2
Question 9	1
Question 10	3
Question 11	2
Question 12	1
Question 13	2
Question 14	2
Question 15	1
Question 16	2
Question 17	1
Question 18	2
Question 19	4

Solution for Question 1:

Correct Answer: C) 1-d, 2-a, 3-b, 4-c

Explanation:

Perkin's applanation tonometer It is portable and does not require a slit lamp, but accurate readings need significant practice. Hand held version of Goldmann.	applanation tonometer (labeled)" data-author="Paolo Brusini ,Maria Letizia Salvetat and Marco Zeppieri " data-hash="7268" data-license="CC BY" data-source="https://www.mdpi.com/2077-0383/10/17/3860" data-tags="April2025" src="https://image.prepladder.com/notes/uAZzf1Uju74fIT4VwbUm1743750272.png" /> {{caption_text}}
Pulsair (non-contact tonometer) A noncontact device that measures intraocular pressure.	 Pulsair tonometer ... (content truncated)
Tonopen (based on both indentation and applanation) A portable, computerized tonometer that uses a microscopic transducer to applanate the cornea and convert intraocular pressure into electronic signals.	 Tonopen {{caption_text}}
Rebound tonometer Used for self-measurement	 Rebound tonometer

Solution for Question 2:

Correct Answer: A) Goldmann perimetry

Explanation:

- Goldmann perimetry is a type of manual kinetic perimetry.
- In this technique, a stimulus of constant size and intensity is moved from a non-seeing area (usually the far periphery) towards the center of vision until the patient detects it.
- This method allows for a detailed mapping of the visual field boundary.



In kinetic perimetry, the stimulus of known luminance is moved from a peripheral non seeing point towards the center till it is perceived to establish the isopters.

Various methods of kinetic perimetry:

Confrontation method	Confrontation visual field testing involves having the patient look directly at the examiner's eye or nose. It compares the visual field of the patient and the physician.
Lister's perimetry	It has a metallic semicircular arc, graded in degrees, with a white dot for fixation in the center. The arc can be rotated in different meridians. It is useful for peripheral field charting. Not used nowadays
Tangent screen scotometry (Bjerrum Screen)	It is useful to evaluate the central and paracentral area (30°) of the visual field
Goldmann's perimetry	It consists of a hemispherical dome. Its main advantage over the tangent screen is that the test conditions and the intensity of the target are always

the same.
It permits greater reproducibility.

Automated perimetry (Option B):

- Static perimetry where stationary stimuli of varying intensities are presented at fixed locations in the visual field.
- It doesn't involve moving stimuli.

Frequency doubling perimetry (Option C):

- Static perimetry that uses a specific stimulus (a sinusoidal grating) to test contrast sensitivity and is particularly useful for detecting early glaucomatous damage.

SWAP (Option D):

- Static perimetry technique that uses a blue light stimulus on a yellow background to selectively test the function of blue-yellow ganglion cells, which may be affected early in glaucoma.

Solution for Question 3:

Correct Answer: C) Small ring-shaped paracentral scotoma-

paracentral scotoma (unlabeled)" data-author="" data-hash="7284" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/cRZQpzjRWifwS OsOYeuT1743753652.png" />

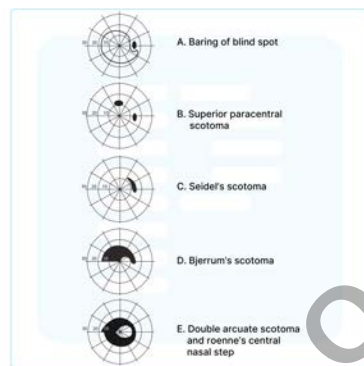
Explanation:

- The earliest clinically significant field defect to appear in patients of glaucoma is small of fixation point ring-shaped paracentral scotoma.
- It may appear either below or above the blind spot in Bjerrum's area (an arcuate area extending above and below the blind spot between 10° and 20°)

Visual Field Defects in Primary Open Angle Glaucoma

1. Isopter contraction	Mild generalized constriction of central as well as peripheral field
2. Baring of blind spot	Early glaucomatous change, but nonspecific and of limited diagnostic value

3. Small ring-shaped paracentral scotoma	Earliest clinically significant field defect; appears in Bjerrum's area
4. Seidel's scotoma	Paracentral scotoma joins the blind spot to form a sickle-shaped defect
5. Arcuate or Bjerrum's scotoma	Formed at a later stage by extension of Seidel's scotoma in an area above or below the fixation point
6. Ring or double arcuate scotoma	Develops when the two arcuate scotomas join together
7. Roenne's central nasal step	An area of depression above or below the horizontal meridian on the nasal side. This is due to the two arcuate scotomas running in different arcs and meet to form a sharp right-angled gap.
8. Peripheral nasal step	Appear sometimes at an early stage; peripheral nasal step results from unequal contraction of the peripheral isopter
9. Advanced glaucomatous field defects	Visual field loss gradually spreads peripherally, eventually only the central and temporal island of vision is left behind.



Solution for Question 4:

Correct Answer: D) The visual field extends maximally in the superior direction

Explanation:

- Visual field is the peripheral vision.
- It extends maximally in the temporal direction and minimally in the superior direction.

The extent of the visual field: (mnemonic: TINS)

- Temporally: 90° to 100°
- Inferiorly: 70°
- Nasally: 60°
- Superiorly: 50°

Scotoma (Option A) is an area of decreased vision surrounded by normal vision.

- Relative scotoma is related to a decrease in sensitivity of the retina (one is no longer able to perceive some or all colors except for white).
- Absolute scotoma is the total absence of visual perception, e.g., physiological blind spot.

Isopter (Option B):

- Line connecting points of equal light sensitivity and size detection in the visual field.

Physiological blind spot (Option C):

- Corresponds to the area where the optic nerve exits the eye, which lacks photoreceptors.
- It is a type of absolute scotoma.
- It is located approximately 10 to 20 degrees from fixation and 1.5 degrees below the horizontal meridian.

Solution for Question 5:

Correct Answer: B) Slit-lamp biomicroscopic examination with non-contact lens

Explanation:

- The examination of choice for detecting fundal changes in a glaucoma suspect is Slit-lamp biomicroscopic examination with non-contact lens.
- The non-contact lenses of +78 or +90 dioptres are commonly used.

Investigations to examine fundus changes in Glaucoma

Slit Lamp Biomicroscopy	Investigation of choice
	Lenses used: Contact Lens: Goldman three mirror Non Contact Lens: Convex (+78D, +90D) or Concave (Hruby Lens)

Direct Ophthalmoscopy (Option A)	No depth perception as the examiner uses only one eye.
Indirect Ophthalmoscopy (Option D)	Magnification is less, hence it is not used to examine the disc and macula.
OCT (Optical coherence tomography) (Option C)	Useful method to measure optic disc diameter, CDR, NRR thickness and optic disc area
Scanning Laser Polarimetry	Nerve fiber analyzer- To measure thickness of nerve fiber layer

Solution for Question 6:

Correct Answer: C) Early glaucomatous changes

Explanation:

- Early glaucomatous changes are indicated by signs such as: Cup-to-disc ratio of >0.3 . Asymmetry of the cups: A cup-to-disc ratio difference greater than 0.2 between two eyes. Splinter hemorrhages near the optic disc margin. Retinal nerve fiber layer (RNFL) defects - wedge-shaped dark areas seen on the red-free light examination - an early indicator of glaucomatous damage.
- Cup-to-disc ratio of >0.3 .
- Asymmetry of the cups: A cup-to-disc ratio difference greater than 0.2 between two eyes.
- Splinter hemorrhages near the optic disc margin.
- Retinal nerve fiber layer (RNFL) defects - wedge-shaped dark areas seen on the red-free light examination - an early indicator of glaucomatous damage.
- Advanced glaucomatous changes Cup-to-disc ratio of 0.7–0.9 Significant neuroretinal rim thinning Vessel bayonetting
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- Significant neuroretinal rim thinning
- Vessel bayonetting
- Glaucomatous optic atrophy Complete destruction of the neural tissue Deeply excavated and white optic disc. Cavernous optic atrophy - large caverns or lacunae are formed due to: Raised IOP leading to modulation of lamina cribrosa. Ischaemic atrophy of the nerve fibers without a corresponding increase of supporting glial tissue.

- Complete destruction of the neural tissue
- Deeply excavated and white optic disc.
- Cavernous optic atrophy - large caverns or lacunae are formed due to: Raised IOP leading to modulation of lamina cribrosa. Ischaemic atrophy of the nerve fibers without a corresponding increase of supporting glial tissue.
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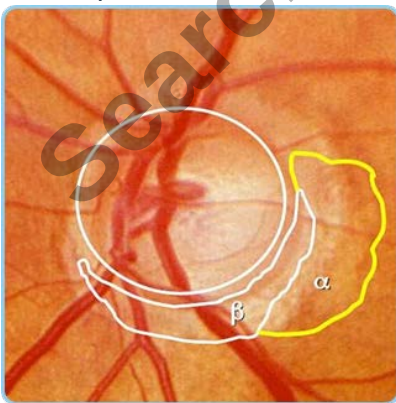
Solution for Question 7:

Correct Answer: B) Presence of alpha zone peripapillary atrophy (PPA)

Explanation:

Peripapillary atrophy (PPA) in the region around the optic nerve head - is indicative of early damage in glaucoma. It has 2 zones:

- Alpha zone (outer): Characterized by irregularities in the retinal pigment epithelium (RPE). Larger in size and more commonly observed in glaucomatous eyes.
- Characterized by irregularities in the retinal pigment epithelium (RPE).
- Larger in size and more commonly observed in glaucomatous eyes.
- Beta zone (inner): Chorioretinal atrophy. A visible white band of exposed sclera adjacent and distinct to the scleral rim. Larger and more frequently seen in glaucoma. Considered a risk factor for glaucoma progression. The location of the beta zone often correlates with patterns of visual field loss in glaucoma patients.
- Chorioretinal atrophy.
- A visible white band of exposed sclera adjacent and distinct to the scleral rim.
- Larger and more frequently seen in glaucoma.
- Considered a risk factor for glaucoma progression.
- The location of the beta zone often correlates with patterns of visual field loss in glaucoma patients.
- Characterized by irregularities in the retinal pigment epithelium (RPE).
- Larger in size and more commonly observed in glaucomatous eyes.
- Chorioretinal atrophy.
- A visible white band of exposed sclera adjacent and distinct to the scleral rim.
- Larger and more frequently seen in glaucoma.
- Considered a risk factor for glaucoma progression.
- The location of the beta zone often correlates with patterns of visual field loss in glaucoma patients.



Increased retinal pigment epithelium (RPE) density in the peripapillary area (Option A):

- The peripapillary area often shows signs of degeneration, not hyperpigmentation.

Thickening of peripapillary RNFL on optical coherence tomography (OCT) (Option C):

- In glaucoma, the RNFL undergoes thinning, especially in the superior and inferior regions, not thickening, due to the loss of retinal ganglion cells.

Uniform retinal nerve fiber layer thickness without focal loss (Option D):

- Glaucoma is characterized by focal thinning of the RNFL, particularly around the optic nerve head.
- Uniform thickness is more typical in healthy eyes, whereas focal thinning is a sign of glaucomatous damage.

Solution for Question 8:

Correct Answer: B) Nasal displacement of retinal vessels

Explanation:

- In advanced glaucoma, there is progressive optic disc cupping.
- A characteristic feature of this process is the nasal displacement of retinal vessels, as the optic nerve head undergoes excavation.
- This displacement is caused by the pulling of the retinal vessels toward the nasal side as the cupping advances, changing the shape of lamina cribrosa.

Vascular Blood Vessel Alignment Changes in Glaucoma:

- Early glaucomatous changes: Splinter hemorrhages on or near the optic disc margin with pallor areas on the disc.
- Advanced glaucomatous changes: Bayonetting sign - Double angulation of a blood vessel. Nasal shifting of retinal vessels. Pulsations of the retinal arterioles are seen at the disc margin - a pathognomonic sign.
- Bayonetting sign - Double angulation of a blood vessel.
- Nasal shifting of retinal vessels.
- Pulsations of the retinal arterioles are seen at the disc margin - a pathognomonic sign.
- Glaucomatous optic atrophy: the nerve fibers undergo ischemic atrophy, but there is no proportional increase in the supporting glial tissue.

- Bayonetting sign - Double angulation of a blood vessel.
- Nasal shifting of retinal vessels.
- Pulsations of the retinal arterioles are seen at the disc margin - a pathognomonic sign.

Inferior displacement of retinal vessels (Option A):

- This is not characteristic of glaucomatous optic neuropathy; vessels typically shift nasally.

Arterial attenuation with increased vein diameter (Option C):

- While arterial narrowing can be seen in certain conditions, this pattern is more typical of hypertensive retinopathy.

Radial peripapillary capillary obliteration (Option D):

- This finding is more associated with ischemic optic neuropathy, not typical in glaucoma.

Solution for Question 9:

Correct Answer: A) The optic disc appearance follows the ISNT rule, indicating a healthy optic nerve.

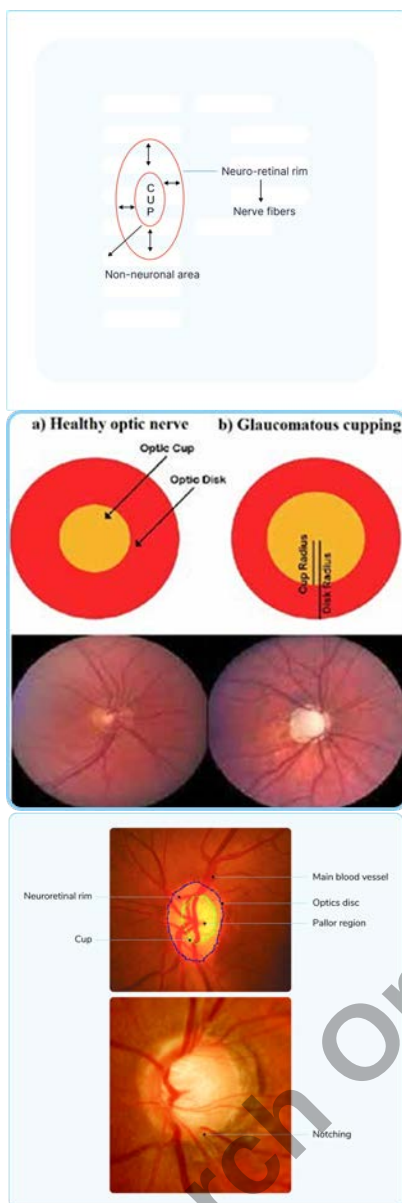
Explanation:

- All of the findings are consistent with early glaucomatous changes except that the disc shows superior and inferior neuroretinal rim thinning violating the ISNT rule.

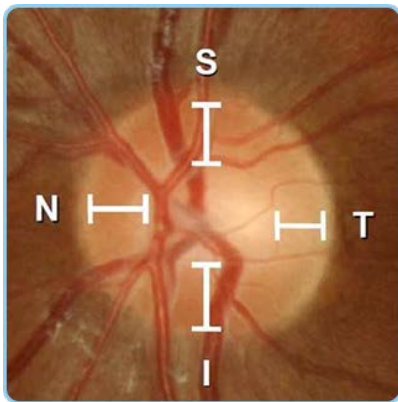
The patient's optic disc shows several characteristic features of glaucomatous optic neuropathy:

1. Increased cup-to-disc ratio (0.7): A ratio greater than 0.6 is suspicious for glaucoma.
2. Superior and inferior neuroretinal rim thinning: This violates the ISNT rule.
3. Peripapillary atrophy present

In glaucoma, there is a progressive loss of retinal nerve fibers, leading to enlargement of the cup and thinning of the neuroretinal rim.



- The ISNT rule (Option A) states that, in a healthy optic nerve, the neuroretinal rim thickness typically follows a pattern, being the broadest in: Inferior > Superior > Nasal > Temporal.
- Changes in this rule, particularly inferior or superior rim thinning, is often an early sign of glaucomatous damage, and does not indicate a healthy optic nerve, making this an incorrect statement.



The cup-to-disc ratio indicates early glaucomatous damage (Option B) is correct.

- A cup-to-disc ratio of 0.7 is considered enlarged and suspicious for glaucoma.
- Normal cup-to-disc ratio is ≤ 0.3 .

The absence of optic disc hemorrhages does not rule out glaucoma (Option C) is correct.

- Their presence is a risk factor for the development and progression of glaucoma.
- They are more common in NTG, but can also occur in healthy individuals as well as patients with systemic vascular disease.

Peripapillary atrophy is commonly associated with glaucoma and warrants further monitoring (Option D) is correct.

- Peripapillary atrophy surrounding the optic nerve head may be of significance in glaucoma and may be a sign of early damage in patients with ocular hypertension.

Solution for Question 10:

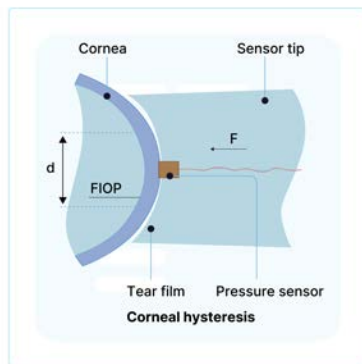
Correct Answer: C) Ocular Response Analyzer

Explanation:

The Ocular Response Analyzer (ORA):

- The Ocular Response Analyzer (ORA) is a non-contact tonometer designed to measure intraocular pressure (IOP) and corneal biomechanics.
- It uses air impulses to deform the cornea, recording the inward and outward movement to assess corneal elasticity or hysteresis.

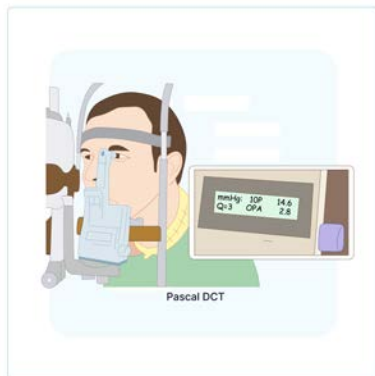
- The device calculates two corneal parameters: corneal hysteresis and corneal resistance factor, providing insights into the cornea's viscoelastic properties.
- ORA's corneal-compensated IOP measurements are less influenced by corneal properties compared to traditional methods like Goldmann applanation tonometry (GAT), but it may overestimate IOP, especially at elevated levels.



Other Latest Tonometer:

Pascal Dynamic Contour Tonometry (DCT) is a contact tonometer designed to measure intraocular pressure (IOP) by directly conforming to the corneal surface.

- It uses a contoured, pressure-sensitive tip that matches the shape of the cornea, allowing continuous and dynamic monitoring of IOP.
- It additionally measures a parameter called ocular pulse amplitude (OPA), which reflects the difference between systolic and diastolic IOP during the cardiac cycle
- DCT readings are generally considered more independent of corneal properties compared to Goldmann applanation tonometry (GAT).
- The reading does not depend on CCT (central corneal thickness).



Fixed force tonometers:

- Fixed force tonometers use a constant force applied to the cornea and measure the area of appplanation.
- Two notable examples of fixed-force tonometers are:

1. MaklaKov tonometer:



2. Barraquer tonometer:

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Schiotz tonometer (Option A) is an indentation tonometer. It doesn't measure corneal hysteresis.

Goldmann applanation tonometer (Option B) is the gold standard for IOP measurement and is based on Imbert-Fick law.

Pneumatonometer (Option D) is a non-contact applanation tonometer, not based on corneal hysteresis.

Solution for Question 11:

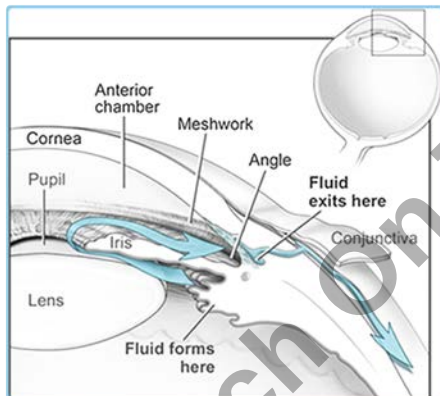
Correct Answer: B) Only 3 is incorrect

Explanation:

The non-pigmented epithelium has a Na/K/ATPase pump responsible for aqueous humor secretion.

Aqueous humor :

- Aqueous humor is formed from the non-pigmented epithelium of ciliary processes
- Ciliary processes have blood vessels, stroma, and two layers of epithelium.
- The inner epithelium is pigmented and the outer epithelium is the non-pigmented epithelium
- The non-pigmented epithelium has a Na/K/ATPase pump and is responsible for aqueous humor secretion.
- Aqueous constitutes 99.9% water and 0.1% solutes.



- The diagram above shows the outflow.
- The aqueous that is formed in the posterior chamber goes through the pupil into the anterior chamber, and through the trabecular pathway.
- When it passes through the uveo-sclera, it is called the uveoscleral pathway.

Process of aqueous humor production:

- Diffusion
- Ultrafiltration
- Secretion (maximum formation)

The rate of formation of aqueous humor is 2.3 microlitres/min

The total volume of aqueous humor is 0.31mL

Solution for Question 12:

Correct Answer: A) Schiottz Tonometer

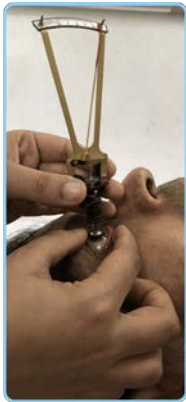
Explanation:

- A tonometer is an instrument used to measure intraocular pressure (IOP) within the eye.
- This measurement is crucial in diagnosing and managing glaucoma and other eye conditions.
- Schiottz Tonometer measures IOP by the amount of corneal indentation caused by a plunger.
- It is a type of indentation tonometer.

Types of Tonometry:

1. Indentation Tonometry:

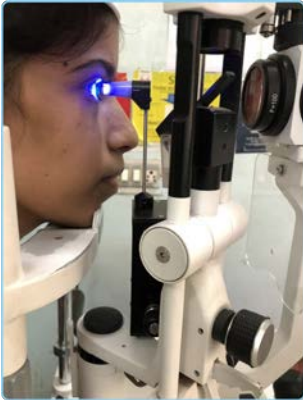
- Concept: Measures IOP by how much a plunger indents the eye; softer eyes indent more than harder eyes.
- Schiottz Tonometer (Option A): Most widely used tonometer in the world. Advantages: Inexpensive, simple, and reliable. Disadvantages: Gives false readings in eyes with abnormal scleral rigidity (e.g., high myopes, post-surgery).
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- Advantages: Inexpensive, simple, and reliable.
- Disadvantages: Gives false readings in eyes with abnormal scleral rigidity (e.g., high myopes, post-surgery).
- Most widely used tonometer in the world.
- Advantages: Inexpensive, simple, and reliable.
- Disadvantages: Gives false readings in eyes with abnormal scleral rigidity (e.g., high myopes, post-surgery).



2. Applanation Tonometry:

- Concept: Measures IOP based on the force needed to flatten the cornea's surface (Imbert-Fick Law: $P = W/A$).
- Goldmann Tonometer (Option B): Most popular and accurate; uses a double prism on a slit lamp. Technique: Cornea is anesthetized and stained with fluorescein. Cornea and biperisms are illuminated with cobalt blue light. Prism touches the cornea, and the inner edges of two semicircles are adjusted to just touch. IOP is read by multiplying the dial reading by ten.
- Most popular and accurate; uses a double prism on a slit lamp.
- Technique: Cornea is anesthetized and stained with fluorescein. Cornea and biperisms are illuminated with cobalt blue light. Prism touches the cornea, and the inner edges of two semicircles are adjusted to just touch. IOP is read by multiplying the dial reading by ten.
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- IOP is read by multiplying the dial reading by ten.
- Cornea is anesthetized and stained with fluorescein.
- Cornea and biperisms are illuminated with cobalt blue light.
- Prism touches the cornea, and the inner edges of two semicircles are adjusted to just touch.

- IOP is read by multiplying the dial reading by ten.



Variable force tonometers:

- Draeger tonometer is similar to Perkins
- It has a different set of prisms
- It operates with a motor

Fixed force tonometer:

Solution for Question 13:

Correct Answer: B) Ocular Hypertension

Explanation:

- The patient has elevated IOP consistently above 21 mm Hg without any visual field defects or fundus changes.
- These characteristics are consistent with ocular hypertension.

ASPECT	OCULAR HYPERTENSION (Option B)	NORMAL TENSION GLAUCOMA (Option A)
Characteristic Features	IOP consistently above 21 mm Hg (No Damage) No Fundus findings No visual field defects	IOP is normal. Fundus changes Present Visual field defect present The anterior chamber angle is open on gonioscopy, and there is no secondary cause for the changes.

Risk Factors	Consistent IOP >30 mm Hg. Central corneal thickness <550 µm. Vertical cup-to-disc ratio >0.7. Increased age. Increased pattern standard deviation (PSD) on Humphrey visual field test. Disc hemorrhages (splinter hemorrhages near or over the disc).	Raynaud phenomenon Migraine: Increased risk. Nocturnal systemic hypotension: Low blood pressure during the night, or overtreated systemic hypertension. Reduced blood flow velocity in the ophthalmic artery
Management	When these risk factors are present, treatment may be initiated similarly to open-angle glaucoma, but with a less aggressive approach.	Medical Treatment: Target the IOP levels to be around 12-14 mm Hg. Prostaglandin analogues (e.g., latanoprost) are preferred for their effective ocular hypotensive effect. Beta blockers and adrenergic drugs should be avoided due to potential adverse effects on optic nerve perfusion.

Acute Angle-Closure Glaucoma (Option C):

- This condition typically presents with symptoms such as sudden eye pain, blurred vision, and possibly nausea, and usually shows a significant increase in IOP along with a closed anterior chamber angle.
- The patient's IOP is elevated but there are no acute symptoms or angle closure present.

Secondary Glaucoma (Option D):

- This type of glaucoma results from an underlying condition that causes increased IOP.
- There is no indication in this case of a secondary cause for the elevated IOP, such as trauma, inflammation, or a systemic condition, so secondary glaucoma is not the most appropriate diagnosis.

Solution for Question 14:

Correct Answer: B) In early POAG, IOP is consistently above 21 mm Hg.

Explanation:

- The diagnosis of Primary Open-Angle Glaucoma (POAG) in this case is supported by significant diurnal fluctuations in intraocular pressure (IOP), characteristic optic disc cupping, and specific visual field defects.

- In the initial stages of POAG, IOP may not be consistently elevated and often shows exaggerated diurnal variation.

Non-Specific Symptoms of POAG

- Mild headache and eye ache.
- Difficulty with reading and close work due to accommodative failure and frequent changes in presbyopic glasses.
- Delayed dark adaptation, becoming problematic in later stages.
- Significant vision loss and potential blindness in untreated cases.

On Examination- 3 Salient features:

- IOP Changes
- Fundus Changes
- Visual field defect

Intraocular Pressure (IOP) Changes in POAG

- Initial Stage: IOP may not be consistently high; diurnal variation (fluctuation throughout the day) is exaggerated. Repeated measurements (every 3-4 hours) are needed to detect this. (Option B)
- Diurnal Variation Patterns: Morning rise in IOP: 20% of cases Afternoon rise in IOP: 25% of cases Biphasic rise in IOP (morning and afternoon): 55% of cases (Option A)
- Morning rise in IOP: 20% of cases
- Afternoon rise in IOP: 25% of cases
- Biphasic rise in IOP (morning and afternoon): 55% of cases (Option A)
- Suspicious Variation: A variation in IOP of over 5 mm Hg is suspicious, and over 8 mm Hg is diagnostic of glaucoma.
- Later Stages: IOP becomes consistently elevated above 21 mm Hg, typically ranging from 30 to 45 mm Hg.
- Morning rise in IOP: 20% of cases
- Afternoon rise in IOP: 25% of cases
- Biphasic rise in IOP (morning and afternoon): 55% of cases (Option A)

General Factors Affecting Intraocular Pressure (IOP)

- Heredity: Influences IOP through multifactorial mechanisms.
- Age: IOP increases after age 40 due to reduced aqueous outflow.
- Sex: IOP is equal between sexes from ages 20-40; however, older females tend to have a higher increase in IOP compared to males.

- Diurnal Variation: Higher IOP in the morning and lower in the evening; glaucomatous eyes show greater fluctuation than normal eyes.
- Postural Variations: IOP increases when moving from a sitting to a supine position.
- Seasonal Variations: Higher IOP is observed in winter months. (Option C)
- Blood Pressure: No long-term effect on IOP, but glaucoma prevalence is slightly higher in hypertensives.
- Osmotic Pressure: Increased plasma osmolarity lowers IOP and vice versa.
- Drugs: General anesthetics, alcohol, tobacco, caffeine, and steroids increase IOP; some antiglaucoma drugs lower IOP.
- Exercise: Strenuous exercise lowers IOP temporarily. (Option D)

Solution for Question 15:

Correct Answer: A) MYOC gene

Explanation:

- The MYOC gene (myocilin) is most commonly associated with primary open angle glaucoma (POAG).

POAG has been associated with at least 20 loci in the human genome, the most important ones being:

- Myocillin - MYOC gene
- Optineurin gene - OPTN gene
- WTR - 36

LOXL1 gene (Option B):

- LOXL1 mutations are associated with pseudoexfoliation syndrome, which is the commonest cause of secondary open angle glaucoma.

PAX6 gene (Option C):

- PAX6 gene is crucial for eye development and is associated with congenital eye disorders, including aniridia.

RB1 gene (Option D):

- RB1 gene is primarily associated with retinoblastoma.

- While it plays a role in tumor suppression, it is not related to the pathogenesis of primary open-angle glaucoma.

Solution for Question 16:

Correct Answer: B) Diastolic perfusion pressure of more than 55 mm Hg

Explanation:

- A diastolic perfusion pressure of more than 55 mm Hg is generally not considered to be a risk factor for POAG. Lower diastolic perfusion pressure (<55 mm Hg) is a significant risk factor for glaucoma.

Primary Open-Angle Glaucoma (POAG)

Characteristics:

- Intraocular Pressure: Elevated (>21 mm Hg) on multiple occasions.
- Anterior Chamber Angle: Open and normal in appearance.
- Optic Disc: Characteristic cupping.
- Visual Field: Specific defects present.

Risk Factors:

- Elevated IOP: Asymmetry of IOP of ≥ 4 mmHg is considered significant. (Option A)
- Family History: Higher risk in siblings (10%) and offspring (4%). Involves polygenic inheritance with genes like MYOC, OPTN, and WDR36. (Option C)
- Age: Increased risk with age, especially between the 5th and 7th decades of life.
- Race: Higher prevalence and severity in Black individuals compared to Whites.
- Myopia: Increased risk in myopic individuals. (Option D)
- Thinner Central Corneal Thickness (CCT) (<555 μm) is an independent risk factor.
- Cigarette Smoking: Increased risk
- Blood Pressure: Higher prevalence in hypertensives; diastolic perfusion pressure <55 mm Hg is a significant risk factor. (Option B)
- Thyrotoxicosis: Increased in Graves' ophthalmic disease.
- Corticosteroid Responsiveness: Greater risk of IOP rise with topical steroid therapy.

Pathogenesis:

- The rise in IOP in POAG occurs due to decreased aqueous outflow facility.
- This is caused by trabecular meshwork stiffening and apposition of Schlemm's canal wall.
- Contributing factors include thickening and sclerosis of the trabecular meshwork, narrowing of intertrabecular spaces, deposition of amorphous material in the juxtacanalicular space, and collapse of Schlemm's canal.

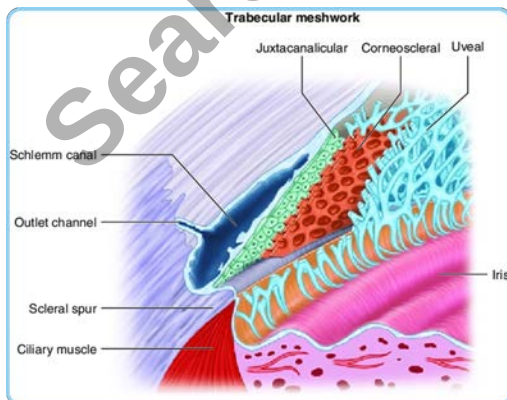
Solution for Question 17:

Correct Answer: A) It regulates the production of aqueous humor

Explanation:

Trabecular meshwork has no role in the production of aqueous humor.

- Aqueous humor is produced from the non-pigmented epithelium of the ciliary process.
- Trabecular meshwork facilitates the drainage of aqueous humor from the anterior chamber of eye. (Option C)
- It has 3 parts: (Option D) Uveal trabecular meshwork Corneoscleral trabecular meshwork Juxtacanalicular trabecular meshwork
- Uveal trabecular meshwork
- Corneoscleral trabecular meshwork
- Juxtacanalicular trabecular meshwork
- Uveal trabecular meshwork
- Corneoscleral trabecular meshwork
- Juxtacanalicular trabecular meshwork



- Trabecular meshwork secrete Glycosaminoglycans (Option B)
- Maximum resistance in trabecular meshwork is seen in Juxtacanalicular trabecular meshwork (Option D)
- Contractile properties are present due to the presence of actin and myosin

Solution for Question 18:

Correct Answer: B) It accounts for the drainage of aqueous humor through the ciliary body and sclera, providing an alternative pathway to the conventional outflow route.

Explanation:

- Uveoscleral outflow is an alternative route for aqueous humor drainage, distinct from the conventional trabecular meshwork and Schlemm's canal system.
- In this process, aqueous humor drains through the ciliary body and into the sclera.
- This route helps regulate intraocular pressure by providing an additional means for aqueous humor to exit the eye.

(Options A, C, and D) describe different aspects of ocular physiology but do not accurately reflect the uveoscleral outflow pathway.

aqueous humor" data-author="" data-hash="7259" data-license="" data-source="" data-tags="April2025" src="https://image.prepladder.com/notes/2AGm4wQ4OMoCnMKn1w3M1743748042.png" />

The flow of aqueous humor:

- Non-pigmented epithelium of ciliary processes secreted in the posterior chamber via the pupil enters the anterior chamber.
- Through the angle of the anterior chamber, 90% (IOP dependent) Trabecular outflow: starts from trabecular meshwork via Schlemm's canal reaches the aqueous vein, and finally to the episcleral vein.
- 10% (IOP independent) Uveo-scleral outflow drains into supraciliary space.
- The aqueous enters the ciliary body through the iris root, ciliary body face, and uveal trabecular meshwork.
- Aqueous passes across the ciliary body between the bundles of ciliary muscles into the suprachoroidal space and is drained by the venous circulation of the ciliary body, choroid,

and sclera.

Solution for Question 19:

Correct Answer: D) Glucose

Explanation:

- Due to the difficulty of substances entering the eye through the blood-aqueous barrier and the ease of their exit via drainage channels, many constituents of the aqueous humor are found in lower concentrations compared to blood, except for those that are actively secreted.
- Glucose levels in the aqueous humor are lower than those in plasma.

Constituents that are more in aqueous than in plasma are:

- Lactate (due to anaerobic glycolysis) (Option A)
- Ascorbate (antioxidant) (Option B)
- Chloride
- Pyruvate (Option C)

Blood Aqueous Barrier is formed by:

- Non-pigmented epithelium (Tight junction)
 - Vascular endothelium of the iris
-

Previous Year Questions

1. A patient complains of eye pain and difficulty seeing after watching a movie. What is the first step in management?

- A. Mannitol with moxifloxacin
- B. Mannitol with pilocarpine
- C. Mannitol with atropine
- D. Mannitol with carboxymethylcellulose

2. A patient present with sudden painful decrease in vision and fixed vertically oval pupil. What is the likely diagnosis?

- A. Macular degeneration
- B. Optic neuritis
- C. Acute congestive glaucoma
- D. Acute anterior uveitis

3. A patient presents with a vision problem in the right eye. On examination, the intraocular pressure (IOP) of the Right eye is 50 mmHg, while the IOP of the Left eye is 16 mmHg. Slit lamp examination of the right eye reveals flare cells in the anterior chamber and a cataractous lens. What is the diagnosis?

- A. Phacolytic glaucoma
- B. Malignant glaucoma
- C. Fuch heterochromic iridocyclitis
- D. Phacomorphic glaucoma

4. Which of the following instruments are considered as gold standard for measuring intraocular pressure?

- A. Gonioscope
 - B. Goldmann applanation tonometer
 - C. Hertel's exophthalmometer
 - D. Tonopen
-

5. What is the gold standard method for intraocular pressure measurement?

- A. Hertel's exophthalmometer
 - B. Gonioscope
 - C. Goldmann applanation tonometer
 - D. Tonopen
-

6. The first line of treatment for glaucoma in adults is?

- A. PG analogues
 - B. Atropine
 - C. Acetazolamide
 - D. Pilocarpine
-

7. Please arrange the following ocular medications according to their respective side effects. Column A Column B 1. Amiodarone 2. Digoxin 3. Systemic steroids 4.

Hydroxychloroquine A. Retinopathy B. Cataract C. Yellow vision D. Optic neuropathy

Column A	Column B
1. Amiodarone 2. Digoxin 3. Systemic steroids 4. Hydroxychloroquine	A. Retinopathy B. Cataract C. Yellow vision D. Optic neuropathy

- A. 1-C, 2-D, 3-B, 4-A
- B. 1-A, 2-C, 3-B, 4-D
- C. 1-D, 2-C, 3-B, 4-A
- D. 1-B, 2-A, 3-D, 4-C

8. In a patient with primary open-angle glaucoma, which of the following pupillary abnormalities is most commonly seen?

- A. Marcus Gunn pupil
- B. Adie's tonic pupil
- C. Argyll-Robertson pupil
- D. Hutchinson's pupil

9. Which of the following statements regarding lens induced glaucoma is not incorrect?

- A. Phacomorphic glaucoma is an open angle glaucoma
- B. In phacolytic glaucoma lens capsule is intact.
- C. PPhacomorphic glaucoma has a shallow anterior chamber
- D. Mature cataract is an example of phacolytic glaucoma

10. A 6-year- old child is admitted for evaluation for recurrent seizures. He has a large port-wine stain on the left side of her face as shown below. The MRI of the head shows leptomeningeal angioma. What is the most ocular manifestation of the underlying condition ?

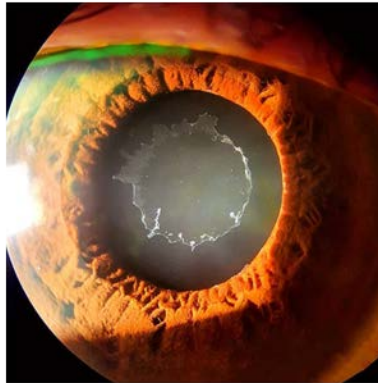


- A. Uveitis
- B. Retinitis pigmentosa
- C. Keratitis
- D. Glaucoma

11. In which of the following conditions is 100-day glaucoma observed?

- A. Central retinal vein occlusion
- B. Eale's disease
- C. Central retinal artery occlusion
- D. Steroid – induced glaucoma

12. What is the diagnosis?



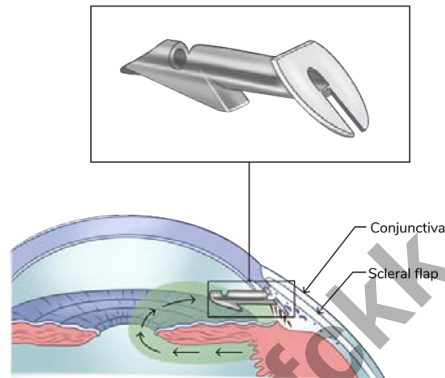
- A. Intraocular foreign body
- B. Pseudoexfoliation syndrome
- C. Ocular trauma
- D. Vossius ring

13. What is the probable diagnosis of a 35-year-old male with a history of prolonged corneal ulcer who now presents with a deformed eye as shown in the image?



- A. Megalocornea
- B. Conjunctivitis
- C. Buphthalmos
- D. Anterior staphyloma

14. What is the name of the device as shown in the image, used for the management of the glaucoma?



- A. Ex-Press TM implant
- B. Ahmed glaucoma valve
- C. Molteno implant
- D. Baerveldt implant

15. Which of the following medications should not be administered to a patient with a known case of angle closure glaucoma who has also been diagnosed with major depressive disorder?

- A. Amitriptyline
- B. Sertraline
- C. Mirtazapine
- D. Fluvoxamine

16. Which medication should be avoided in a 30-year-old female presenting with redness, pain, increased intraocular pressure, keratic precipitates, and aqueous flare?

- A. Beta-blockers
 - B. Mannitol
 - C. Carbonic anhydrase inhibitors
 - D. Prostaglandin analogs
-

17. The drug increasing uveoscleral flow is?

- A. Timolol
 - B. Latanoprost
 - C. Pilocarpine
 - D. Carbonic anhydrase inhibitor
-

18. What is the likely diagnosis for a 2-month-old infant who was brought in by their mother due to complaints of light sensitivity, difficulty opening the eyes, and excessive tearing?



- A. Ophthalmia neonatorum

- B. Congenital glaucoma
 - C. Congenital Dacryocoele
 - D. Congenital Cataract
-

19. What is the earliest indication of glaucoma on perimetry?

- A. Isopter contraction
 - B. Complete field loss
 - C. Tunnel vision defect
 - D. Complete vision loss
-

20. In which of the following conditions is atropine not recommended for use?

- A. Acute congestive glaucoma
 - B. Acute anterior uveitis
 - C. Malignant glaucoma
 - D. Corneal ulcer
-

21. After an acute episode of angle-closure glaucoma, which triad of symptoms is typically observed?

- A. Vogt's triad
 - B. Virchow triad
 - C. Hutchinson's triad
 - D. Gaucher triad
-

22. What is the probable diagnosis for an infant brought in with symptoms of excessive tear production, sensitivity to light, and blepharospasm?



- A. Congenital glaucoma
- B. Congenital cataract
- C. Congenital Dacryocystitis
- D. Ophthalmia neonatorum

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	1
Question 4	2
Question 5	3
Question 6	1
Question 7	3
Question 8	1
Question 9	3
Question 10	4
Question 11	1
Question 12	2
Question 13	4
Question 14	1
Question 15	1
Question 16	4
Question 17	2
Question 18	2
Question 19	1
Question 20	1
Question 21	1
Question 22	1

Solution for Question 1:

Correct Option B - Mannitol with pilocarpine:

- The scenario describes a patient presenting with symptoms after watching a movie, which is suggestive of acute primary angle-closure glaucoma (APACG). Mannitol helps to reduce intraocular pressure rapidly, and pilocarpine is a miotic agent that constricts the pupil and pulls the iris away from the trabecular meshwork, thereby opening the drainage angle.

Incorrect Options:

Options A and D: Moxifloxacin and carboxymethylcellulose have no effect on intraocular pressure.

Option C: Atropine is a mydriatic agent that dilates the pupil, which can worsen angle closure by increasing the pupillary block.

Solution for Question 2:

Correct Option C - Acute congestive glaucoma:

- Acute congestive glaucoma is characterized by sudden painful decrease in vision and fixed vertically oval pupil

Incorrect Options:

Option A - Macular degeneration:

- It is characterized by irreversible loss of vision
- Fixed oval pupils is not a feature of this condition
- So this cannot be the answer

Option B - Optic neuritis:

- Pupillary reactions are normal in Optic neuritis
- So this cannot be the answer

Option D - Acute anterior uveitis:

- Anterior uveitis is characterized by a dilated, festooned-shaped pupil

Solution for Question 3:

Correct Option A - Phacolytic glaucoma:

- Flare cells in the anterior chamber of the eye, cataractous lens, and unequal IOP point towards the diagnosis of phacolytic glaucoma.
- It is a type of secondary open-angle glaucoma in which trabecular meshwork is clogged by the lens proteins, macrophages which have phagocytosed the lens proteins, and inflammatory debris.
- Leakage of the lens proteins occurs through an intact capsule in the hypermature (Morgagnian) cataractous lens.
- The anterior chamber may become deep, and the aqueous may contain fine white protein particles, which settle down as pseudohypopyon.
- Anterior chamber angle is open on gonioscopy, and aqueous flare or cells may be seen in the anterior chamber.

Incorrect Options:

Option B - Malignant glaucoma:

- In Malignant glaucoma, shallowing of the anterior chamber is noticed and Seidel's test is negative.

Option C - Fuch heterochromic iridocyclitis:

- In Fuch heterochromic iridocyclitis, low-grade flare and cells, with stellate keratic precipitates and fine filaments may be seen but IOP is not very high as in the above case.

Option D - Phacomorphic glaucoma:

- In Phacomorphic glaucoma, ciliary congestion, dilated pupil, and intumescent senile cataractous lens is noted.

Solution for Question 4:

Correct Option B - Goldmann applanation tonometer:

- The gold standard instrument for measuring intraocular pressure is Goldmann applanation tonometer. It is based on Imbert Fick law according to which the pressure inside a sphere (P) is equal to the force (F) required to flatten its surface divided by the area of flattening (A).
- Pressure (P) = Force (F)/Area (A)

Incorrect Options:

Option A - Gonioscope: Gonioscope is used to visualize the angle of the anterior chamber.

Option C - Hertel's exophthalmometer: Hertel's exophthalmometer is used to measure the protrusion of the apex of the cornea from the outer orbital margin.

Option D - Tonopen: Tonopen is used for measuring the intraocular pressure in a scarred cornea.

Solution for Question 5:

Correct Option C- Goldmann applanation tonometer:

- Goldmann applanation tonometer is considered the gold standard for measuring intraocular pressure.
- It works by flattening a specific area of the cornea, and the force required to flatten this area is directly proportional to the IOP.
- This method provides accurate and reproducible measurements, making it the preferred choice in clinical settings.
- It is also less affected by corneal thickness variations compared to other methods.

Incorrect Options:

Option A- Hertel's exophthalmometer: The Hertel exophthalmometer is used to measure the degree of exophthalmos (protrusion of the eyeball) in conditions such as thyroid eye disease. It is not designed for measuring IOP.

Option B- Gonioscope: A gonioscope is a specialized lens used to visualize the iridocorneal angle, which is important in assessing angle-closure glaucoma. It is not used for direct

measurement of IOP.

Option D- Tonopen: The Tonopen is a handheld device that measures IOP by applanation, similar to the Goldmann tonometer. It is used as a convenient screening tool but may not provide as reliable results as the Goldmann tonometer.

Solution for Question 6:

Correct Option A - PG analogues:

Glaucoma

- Glaucoma is a group of condition that have in common - a chronic progressive optic neuropathy that results in characteristic morphological changes at the optic nerve head and in retinal nerve fiber layer
- IOP is a key modifiable factor
- PG analogues are the first line of treatment for glaucoma in adults
- Medical treatment: the drugs either decrease the formation of aqueous or increase the drainage to ultimately decrease the IOP
- This is of two types topical and systemic
- Topical Beta blockers - decrease the formation of aqueous Alpha agonists- decrease the formation of aqueous Miotics- increase the trabecular outflow (pilocarpine- option D) PGF2a agonists: increases the uveoscleral outflow Topical carbonic anhydrase inhibitors: decrease the formation of aqueous Newer antiglaucoma drugs: increase the trabecular outflow
- Beta blockers - decrease the formation of aqueous
- Alpha agonists- decrease the formation of aqueous
- Miotics- increase the trabecular outflow (pilocarpine- option D)
- PGF2a agonists: increases the uveoscleral outflow
- Topical carbonic anhydrase inhibitors: decrease the formation of aqueous
- Newer antiglaucoma drugs: increase the trabecular outflow
- Systemic Carbonic anhydrase inhibitor: acetazolamide (option C) Hyperosmotic agents
- Carbonic anhydrase inhibitor: acetazolamide (option C)
- Hyperosmotic agents
- Beta blockers - decrease the formation of aqueous

- Alpha agonists- decrease the formation of aqueous
- Miotics- increase the trabecular outflow (pilocarpine- option D)
- PGF2a agonists: increases the uveoscleral outflow
- Topical carbonic anhydrase inhibitors: decrease the formation of aqueous
- Newer antiglaucoma drugs: increase the trabecular outflow
- Carbonic anhydrase inhibitor: acetazolamide (option C)
- Hyperosmotic agents

Incorrect Options:

Option B - atropine:

- It is the first line of treatment in malignant glaucoma

Option C & D is described in the Correct answer.

Solution for Question 7:

Correct Option C - 1-D, 2-C, 3-B, 4-A:

- It accurately matches the ocular drugs with their corresponding side effects. Amiodarone (1) is associated with optic neuropathy (D). Digoxin (2) is associated with yellow vision (C). Systemic steroids (3) are associated with cataract formation (B). Hydroxychloroquine (4) is associated with retinopathy (A).
- Amiodarone (1) is associated with optic neuropathy (D).
- Digoxin (2) is associated with yellow vision (C).
- Systemic steroids (3) are associated with cataract formation (B).
- Hydroxychloroquine (4) is associated with retinopathy (A).
- Amiodarone (1) is associated with optic neuropathy (D).
- Digoxin (2) is associated with yellow vision (C).
- Systemic steroids (3) are associated with cataract formation (B).
- Hydroxychloroquine (4) is associated with retinopathy (A).

Incorrect Options:

- Option A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 8:

Correct Option A - Marcus Gunn pupil:

- In a patient with primary open-angle glaucoma, the most likely pupillary abnormality to be observed is a Marcus Gunn pupil, also known as an afferent pupillary defect or relative afferent pupillary defect (RAPD). This abnormality is characterized by a diminished or absent pupillary response to light stimulation.
- It is caused by incomplete optic nerve lesions (glaucoma and optic neuritis,) and severe retinal diseases like rhegmatogenous retinal detachment.

Incorrect Options:

Option B - Adie's tonic pupil: This is characterized by a dilated pupil that reacts slowly and poorly to light but shows a stronger response to near vision or accommodation. It is typically caused by damage to the postganglionic parasympathetic pupillomotor damage.

Option C - Argyll-Robertson pupil: It is associated with neurosyphilis characterized by absent light reflex and positive accommodation reflex..

Option D - Hutchinson's pupil: It is characterized by initial ipsilateral miosis followed by dilatation with no light reflex due to raised intracranial pressure due to surgery or trauma affecting pial blood vessels.

Solution for Question 9:

Correct Option C - Phacomorphic glaucoma has a shallow anterior chamber:

- Phacomorphic glaucoma is a type of secondary angle-closure glaucoma caused by a intumescent cataract, anterior subluxation of lens, and spherophakia.
- In phacomorphic glaucoma, the lens swells, causing a forward displacement of the iris and narrowing or closure of the anterior chamber angle. This leads to increased intraocular pressure.

- One of the characteristic features of phacomorphic glaucoma is a shallow anterior chamber, which is a consequence of the forward movement of the lens-iris diaphragm.

Incorrect Options:

Option A - Phacomorphic glaucoma is an open angle glaucoma: Phacomorphic glaucoma is a type of angle-closure glaucoma.

Option B - In phacolytic glaucoma, the lens capsule is intact: Phacolytic glaucoma occurs due to leakage of lens protein through an intact lens capsule, leading to an inflammatory response and increased intraocular pressure.

Option D - Mature cataract is an example of phacolytic glaucoma: Phacolytic glaucoma is more common in hypermature cataracts where the lens protein leaks through the intact lens capsule, triggering inflammation and increased intraocular pressure.

Solution for Question 10:

Correct Option D - Glaucoma:

- The clinical presentation of the seizures, port wine stain on face, and leptomeningeal angioma is suggestive of Sturge-Weber syndrome.
- The ocular manifestation of Sturge-Weber syndrome is congenital glaucoma..

Incorrect Options:

- Options A, B, and C are incorrect. Sturge-Weber syndrome is not associated with these diseases.

Solution for Question 11:

Correct Option A - Central retinal vein occlusion:

- "100-day glaucoma," also known as neovascular glaucoma, can occur as a complication of ischemic central retinal vein occlusion (CRVO).

- Ischaemic CRVO (hemorrhagic retinopathy) refers to acute (sudden) complete occlusion of the central retinal vein.

Incorrect Options:

- Options B, C, and D are incorrect. 100 days glaucoma is not associated with these diseases.

Solution for Question 12:

Correct Option B - Pseudoexfoliation syndrome:

- The image reveals gray-white fibrillary material on the anterior surface of lens derived from extracellular matrix metabolism is suggestive of Pseudoexfoliation syndrome.
- Pseudoexfoliation syndrome is characterized by the deposition of fibrillar material on various ocular structures, including the lens capsule, zonular fibers, iris, trabecular, and conjunctiva.

Incorrect Options:

Option A - Intraocular foreign body: It is characterized by burning, foreign body sensation, conjunctival congestion and presence of a foreign body in the eye.

Option C - Fleischer's ring: Fleischer's ring: It is characterized by iron deposition at the base of an epithelial layer. It is associated with keratoconus.

Option D - Vossius ring: It is characterized by pigment deposition of the iris on the anterior lens capsule surface as a result of blunt trauma.

Solution for Question 13:

Correct Option D - Anterior staphyloma:

- The history of a prolonged corneal ulcer is suggestive of low body immune response or infection by a virulent strain of bacteria leading to sloughing of the whole cornea, leaving a

narrow rim behind along with iris prolapse resulting in the false cornea that cannot withstand raised intraocular pressure and results in forward bulging of the cornea with iris tissue. As a result an ectatic cicatrix forms called anterior staphyloma as seen in the image.

Incorrect Options:

Option A - Megalocornea: Also known as anterior megalophthalmos, is a rare bilateral nonprogressive congenital defect that is characterized by an increased corneal diameter greater than 12.5 to 13 mm at birth and a deep anterior chamber with normal intraocular pressures and clear cornea.

Option B - Conjunctivitis: Is the inflammation or infection of the transparent membrane that lines your eyelid and eyeball. Typical symptoms include redness and a gritty sensation in your eye, along with itching.

Option C - Buphthalmos: Is visible enlargement of the eyeball detected at birth or soon after, due to any uncontrolled glaucoma in early childhood.

Solution for Question 14:

Correct Option A - Ex-Press™ implant:

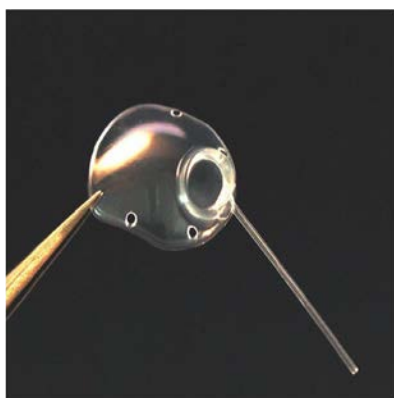
- The Ex-Press™ (mini shunt) implant is a small stainless steel device that is used in glaucoma surgery to facilitate drainage and lower intraocular pressure (IOP).
- It is implanted under the partial-thickness scleral flap as a modification of trabeculectomy. It provides a more favorable safety and postoperative recovery profile than standard trabeculectomy.

Incorrect Options:

Option B - Ahmed glaucoma valve: It consists of a small tube connected to a plate that is implanted in the eye to facilitate drainage.



Option C - Molteno implant: It involves the placement of a silicone tube with a plate to facilitate drainage.



Option D - Baerveldt implant: It consists of a tube connected to a silicone plate that is implanted in the eye to facilitate drainage.

Solution for Question 15:

Correct Option A - Amitriptyline:

- The drug that should be avoided in a patient with angle closure glaucoma who has been diagnosed with major depressive disorder is Amitriptyline. This medication which is classified as a tricyclic antidepressant (TCA) has anticholinergic effects that can provoke the onset of acute angle closure glaucoma by causing pupillary dilation and obstructing the outflow of the aqueous humor.

The drug that should be avoided in a patient with angle closure glaucoma who has been diagnosed with major depressive disorder is Amitriptyline. This medication which is classified as a tricyclic antidepressant (TCA) has anticholinergic effects that can provoke the onset of acute angle closure glaucoma by causing pupillary dilation and obstructing the outflow of the aqueous humor.

Incorrect Options:

Option B - Sertraline: Is a Selective Serotonin Reuptake Inhibitor (SSRI) commonly used in treating depression. It induces and exacerbates depression-related dry eye by interference with aqueous and mucus secretion.

Option C - Mirtazapine: Which is an atypical antidepressant, it is rarely associated with angle closure glaucoma.

Option D - Fluvoxamine: Is also a SSRI associated with development of cataract formation.

Solution for Question 16:

Correct Option D - Prostaglandin analogs:

- The clinical presentation of redness, pain, keratic precipitates, aqueous flare, and increased intraocular pressure is suggestive of anterior uveitis induced glaucoma.
- Prostaglandin analogs are medications often used to treat glaucoma by reducing intraocular pressure by increasing uveoscleral outflow. However, prostaglandins should be used cautiously in patients with anterior uveitis as they can exacerbate the infection..

Incorrect Options:

Option A - Beta-blockers: These are typically safe for use in uveitis patients. They lower IOP by decreasing the production of aqueous humor however should be used cautiously particularly metipranolol in patients with granulomatous uveitis.

Option B - Mannitol: This is an osmotic diuretic that can be used in acute cases of elevated IOP, such as in acute glaucoma. It's not associated with causing or worsening uveitis.

Option C - Carbonic anhydrase inhibitors: Similar to beta-blockers, these agents decrease IOP by reducing aqueous humor production and do not worsen uveal inflammation

Solution for Question 17:

Correct Option B - Latanoprost:

- Latanoprost is a prostaglandin analog commonly used in the treatment of glaucoma. It works by increasing the outflow of aqueous humor through the uveoscleral pathway, which helps reduce intraocular pressure (IOP).
- The uveoscleral pathway refers to the flow of aqueous humor through the ciliary body and into the supraciliary and suprachoroidal spaces before ultimately exiting the eye. By increasing uveoscleral outflow, Latanoprost effectively lowers IOP by enhancing the drainage of aqueous humor.

Incorrect Options:

Option A - Timolol: It is a beta-blocker that reduces IOP by decreasing aqueous humor production.

Option C - Pilocarpine: It is a cholinergic agonist that primarily acts on the trabecular meshwork to enhance aqueous humor outflow through the conventional pathway (trabecular meshwork and Schlemm's canal).

Option D - Carbonic anhydrase inhibitors: Dorzolamide and brinzolamide are carbonic anhydrase inhibitors which work by reducing the production of aqueous humor by inhibiting the enzyme carbonic anhydrase.

Solution for Question 18:

Correct Option B - Congenital glaucoma:

- The clinical presentation of excessive tear production, photophobia, and blepharospasm are the classic triad of symptoms of congenital glaucoma.
- Large cornea in an infant along with buphthalmos (enlargement of eye) as shown in the image is suggestive of congenital glaucoma.

Incorrect Options:

Option A - Ophthalmia neonatorum: Refers to conjunctivitis occurring within the first month of life. It presents with redness, pain, photophobia, foreign body sensation, and subconjunctival hemorrhage.

Option C - Congenital dacryoceles: It is caused due to entrapment of amniotic fluid and mucous in the lacrimal sac characterized by bluish cystic swelling and epiphora.

Option D - Congenital Cataract: Congenital cataract is a clouding of the eye's natural lens that is present at birth or develops in the first year of life and is usually asymptomatic.

Solution for Question 19:

Correct Option A - Isopter contraction:

- The earliest visual field change in glaucoma on perimetry is typically an isopter contraction.
- A line connecting a series of points in a visual field at which a person can detect a certain size and intensity of light is known as isopter.
- An isopter contraction refers to the narrowing or constriction of the visual field isopter.
- The isopter is contracted in glaucoma, but does not involve the blind spot. Hence, it is also known as the Baring of blind spot.

Incorrect Options:

- Options B, C, and D are incorrect. These are late features of glaucoma.

Solution for Question 20:

Correct Option A - Acute congestive glaucoma:

- Acute congestive glaucoma, also known as acute angle-closure glaucoma, is a contraindication for atropine use. Atropine is a mydriatic and cycloplegic drug that dilates the pupil and paralyzes the ciliary muscle.

- In acute congestive glaucoma, there's a rapid rise in intraocular pressure due to obstruction of aqueous humor outflow, often associated with a shallow anterior chamber and narrow angles. Pupil dilation from atropine can exacerbate this condition by further narrowing the angle, thus increasing intraocular pressure.

Incorrect Options:

Option B - Acute anterior uveitis: Atropine can be beneficial in this condition as it helps to relieve ciliary muscle spasms, reduces pain, prevents the formation of posterior synechiae, and keeps the pupil mobile.

Option C - Malignant glaucoma: Atropine is the first line of treatment for management of malignant glaucoma. It dilates the ciliary ring and relaxes the ciliary muscles which causes stretching of zonules that will pull back the lens, thus opening the block.

Option D - Corneal ulcer: Atropine can be used in this case to help relieve pain associated with ciliary muscle spasms, reduces pain, prevents the formation of posterior synechiae, and improve vascularity.

Solution for Question 21:

Correct Options A - Vogt's triad:

- Vogt's Triad is the combination of clinical findings typically seen following an acute episode of angle-closure glaucoma. This triad consists of: Dilated non-reactive pupil. Glaukomflecken (anterior subcapsular opacity of lens secondary to ischemia). Patches of iris atrophy
- Dilated non-reactive pupil.
- Glaukomflecken (anterior subcapsular opacity of lens secondary to ischemia).
- Patches of iris atrophy
- Dilated non-reactive pupil.
- Glaukomflecken (anterior subcapsular opacity of lens secondary to ischemia).
- Patches of iris atrophy

Incorrect Options:

Option B - Virchow's triad: Refers to the three broad categories contributing to thrombosis: endothelial injury, abnormal blood flow, and hypercoagulability.

Option C - Hutchinson's triad: It is a set of symptoms of congenital syphilis. It comprises interstitial keratitis, Hutchinson's teeth (notched incisors), and deafness.

Option D - Gaucher triad: It is associated with Gaucher's disease, a genetic disorder. It consists of hepatosplenomegaly, erosion of femoral heads, and pancytopenia. This is not related to glaucoma.

Solution for Question 22:

Correct Option A - Congenital glaucoma:

- The clinical presentation of excessive tear production, photophobia, blepharospasm are the classic triad of symptoms of congenital glaucoma and large cornea in an infant as shown in the image is suggestive of congenital glaucoma.

Incorrect Options:

Option B - Congenital cataract: Congenital cataract is a clouding of the eye's natural lens that is present at birth or develops in the first year of life and is usually asymptomatic.

Option C - Congenital dacryocystitis: It usually presents with excessive tearing and purulent discharge from the puncta

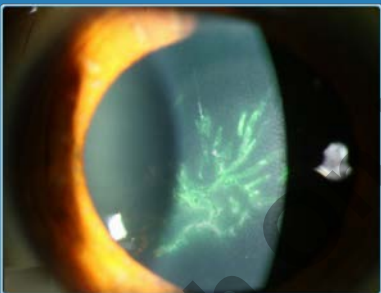
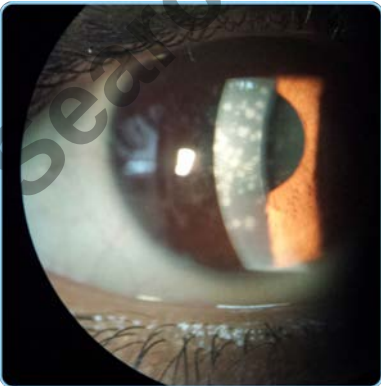
Option D - Ophthalmia neonatorum: Ophthalmia neonatorum aka neonatal conjunctivitis is characterized by redness, pain, photophobia, foreign body sensation, and subconjunctival hemorrhage.

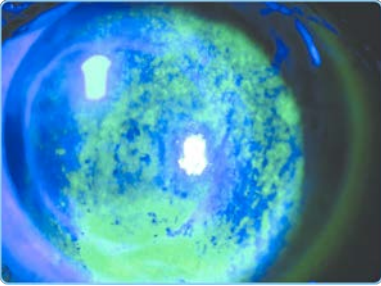
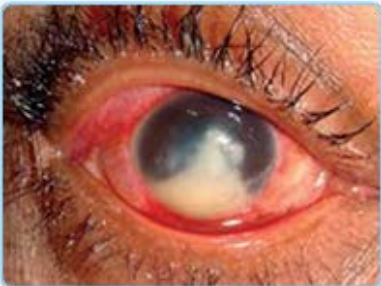
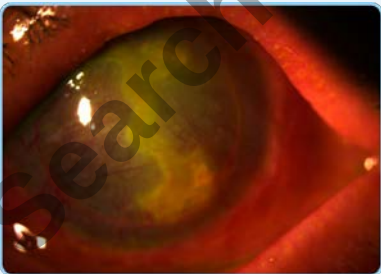
Cornea

1. Which of the following is not an indication of Penetrating Keratoplasty (PK)?

- A. Corneal opacity
- B. Cosmetic improvement of the eye
- C. Advanced keratoconus
- D. Secondary glaucoma

2. Match the Following: 1) a) Bacterial Keratitis 2) {{caption_text}} b) Superficial Punctate Keratitis 3) {{caption_text}} c) Pseudodendritis 4) {{caption_text}} d) Dendritic Ulcer 5) e) Nummular Keratitis 6) {{caption_text}} f) Fungal Keratitis

 1)	a) Bacterial Keratitis
 ... (content truncated)	b) Superficial Punctate Keratitis

 3) {{caption_text}}	c) Pseudodendritis
 4) {{caption_text}}	d) Dendritic Ulcer
 5) {{caption_text}}	e) Nummular Keratitis
 6) {{caption_text}}	f) Fungal Keratitis

A. 1-b; 2-e; 3-c; 4-f; 5-d; 6-a

B. 1-c; 2-e; 3-d; 4-f; 5-a; 6-b

C. 1-e; 2-c; 3-b; 4-f; 5-d; 6-a

D. 1-c; 2-e; 3-b; 4-f; 5-d; 6-a

3. A 28-year-old female contact lens wearer with severe eye pain and blurred vision shows a ring-shaped corneal infiltrate, and radial keratoneuritis on examination. Corneal scraping showed the presence of cysts. What is the most appropriate initial treatment?

- A. Topical moxifloxacin
- B. Oral acyclovir
- C. Topical polyhexamethylene biguanide (PHMB)
- D. Topical corticosteroids

4. A 23 y/o male presents with B/L eye redness, pain, and severe blurring of vision. He has a history of congenital syphilis. O/E, there is a characteristic pinkish-red discolouration of the cornea with deep stromal vascularization. Which of the following findings is most likely associated with this patient's condition during the healing stage?

- A. Pseudodendrites
- B. Ghost vessels
- C. Ring-shaped abscess
- D. Radial keratoneuritis

5. A 19 y/o female presents with a history of progressive worsening of vision that is not fully corrected by glasses. After evaluation, she is diagnosed with keratoconus. Which of the following examination findings are seen in keratoconus? Munson's sign Distorted window reflex Vogt striae Fleischer ring

- A. 1,2,3,4
 - B. 1,3,4
 - C. 2,3,4
 - D. 1,4
-

6. Question: A 22 y/o male with a history of worsening vision despite wearing rigid gas-permeable lenses now complains of increased visual distortion. O/E, his corneal topography shows progressive ectasia, and he is advised to consider treatment options to halt disease progression. Which of the following treatments is most appropriate to stabilize the condition and improve contact lens tolerance in this patient?

- A. Spectacle correction
 - B. Intracorneal ring segments (INTACS)
 - C. Corneal collagen cross-linking (CXL)
 - D. Penetrating keratoplasty
-

7. Which of the following conditions is not commonly associated with keratoconus?

- A. Marfan's syndrome
 - B. Turner's syndrome
 - C. Down syndrome
 - D. Atopic keratoconjunctivitis
-

8. A 50 y/o patient presents for a corneal transplant evaluation. The patient has a history of systemic infections, including tuberculosis, and is seeking a corneal graft for a therapeutic indication. Which of the following is an absolute

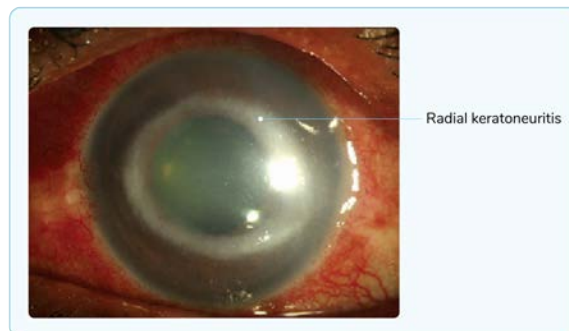
contraindication for cornea donation in this patient?

- A. Death due to an unknown cause
- B. Previous history of ocular surgery
- C. Ocular inflammation
- D. Retinoblastoma

9. A donor cornea is harvested and needs to be preserved for a planned corneal transplant for 10 days. Which of the following storage methods is most appropriate for preserving the donor cornea?

- A. Moist Chamber Method
- B. McCarey Kaufman (M-K) Medium
- C. Optisol GS
- D. Cryopreservation

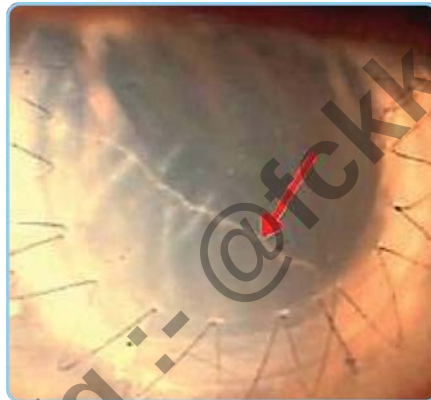
10. A 30 y/o contact lens wearer presents with very severe eye pain, blurred vision, and discomfort in his right eye. O/E, the following corneal finding is noted. Based on the image and the clinical presentation, which of the following is the most likely diagnosis? {{caption_text}}



- A. Herpes Simplex Keratitis

- B. Fungal Keratitis
- C. Acanthamoeba Keratitis
- D. Bacterial Keratitis

11. A 45 y/o patient presents with worsening vision, mild pain, and redness 6 months after Penetrating Keratoplasty. Examination reveals the following. Which feature is most indicative of endothelial rejection? {{caption text}}



- A. Increased Flare and Cells
- B. Kayes dots
- C. Khodadoust Line
- D. Epithelial Rejection Line

12. All of the following statements about lamellar keratoplasty are true except:

- A. It is indicated in pseudophakic bullous keratopathy
- B. It has lower risk of graft rejection compared to lamellar keratoplasty
- C. Deep Anterior Lamellar Keratoplasty (DALK) replaces 80-90% of the anterior stroma and is suitable when the endothelium is normal.

D. Descemet's Stripping Automated Endothelial Keratoplasty (DSAEK) is performed when the endothelium is defective.

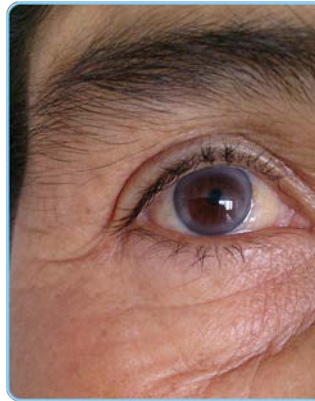
13. Match the following: 1) Arcus senilis a) {{caption_text}} 2) Band shaped Keratopathy b) {{caption_text}} 3) Oil droplet keratopathy c) {{caption_text}} 4) Salzmann nodular degeneration d) {{caption_text}}

1) Arcus senilis



a) {{caption_text}}

2) Band shaped Keratopathy



... (content truncated)

3) Oil droplet keratopathy



... (content truncated)

4) Salzmann nodular degeneration



d)
{{caption_text}}

A. 1-b, 2-d, 3-a, 4-c

B. 1-d, 2-a, 3-c, 4-b

C. 1-c, 2-d, 3-b, 4-a

D. 1-a, 2-b, 3-c, 4-d

14. Which of the following statements about corneal degeneration is incorrect?

- A. Salzmann nodular degeneration affects the superficial stroma and Bowman's membrane.
- B. Vortex keratopathy is generally asymptomatic and associated with indomethacin intake.
- C. Lipid keratopathy involves both intracellular and extracellular fat deposits.

D. Vortex keratopathy remains permanent even after discontinuing the offending medication.

15. A 45 y/o male presents with progressively worsening vision in his left eye over the past year. O/E, the following findings were observed which caused significant blurring of his vision. Which of the following corneal opacities is most likely causing the greatest reduction in his vision?



- A. Nebular corneal opacity
 - B. Macular corneal opacity
 - C. Leucomatous corneal opacity
 - D. Adherent leucoma
-

16. A 25-year-old patient presents with a history of gradual vision loss. O/E, multiple milky, granular deposits are seen in the anterior corneal stroma, with clear cornea between the deposits. Which type of stromal corneal dystrophy is most likely?

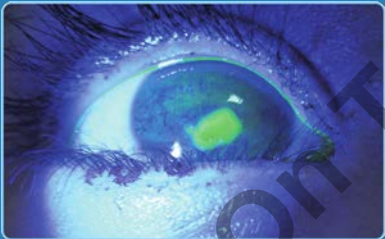
- A. Granular Corneal Dystrophy, Type 1 (GCD1)
- B. Lattice Corneal Dystrophy, Type 1 (LCD1)
- C. Macular Corneal Dystrophy (MCD)

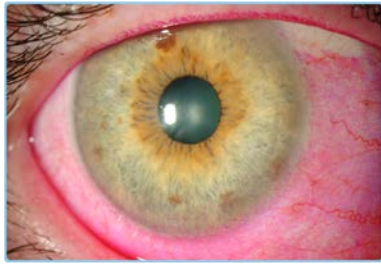
D. Schnyder Corneal Dystrophy (SCD)

17. Which of the following conditions is most commonly associated with the development of a "beaten-metal" appearance on the cornea?

- A. Posterior Polymorphous Corneal Dystrophy (PPCD)
- B. Congenital Hereditary Endothelial Dystrophy (CHED)
- C. Keratoconus
- D. Fuchs Endothelial Corneal Dystrophy (FECD)

18. Match the following. A. {{caption_text}} 1. Rose bengal B. {{caption_text}} 2. Lissamine green C. {{caption_text}} 3. Fluorescein sodium

 A. {{caption_text}}	1. Rose bengal
 B. {{caption_text}}	2. Lissamine green



C.
{{caption_text}}

3. Fluorescein sodium

- A. A-2, B-3, C-1
- B. A-1, B-2, C-3
- C. A-3, B-2, C-1
- D. A-1, B-3, C-2

19. A 55 y/o male with poorly controlled diabetes presents with a non-healing corneal ulcer in his right eye, despite two weeks of appropriate antibiotic therapy. The ulcer shows signs of impending perforation with severe thinning of the cornea. Which of the following is the most appropriate next step in the management of this patient?

- A. Increase the frequency of antibiotic eye drops.
- B. Apply a pressure bandage and recommend strict bed rest.
- C. Perform mechanical debridement and apply tissue adhesive glue.
- D. Start topical steroids to reduce inflammation.

20. Which layer of the cornea is primarily responsible for maintaining its transparency by regulating fluid content?

- A. Stroma
- B. Endothelium

- C. Epithelium
 - D. Bowman's Membrane
-

21. What is the primary metabolism of cornea?

- A. Anaerobic
 - B. Aerobic.
 - C. Both anaerobic and aerobic
 - D. None of the above
-

22. Which of the following factors is not responsible for maintaining corneal transparency?

- A. Corneal Epithelium and Tear Film
 - B. Avascularity of the Cornea
 - C. Corneal nerve supply
 - D. Endothelial Pumps
-

23. Match the following corneal assessment techniques with their correct descriptions: 1) Keratometry a) Measures both anterior and posterior corneal surfaces. 2) Placido's Disc b) Assesses the smoothness and curvature of the corneal surface. 3) Corneal Topography c) Evaluate the anterior curvature of the cornea and generate colour-coded maps. 4) Corneal Tomography (Pentacam) d) Measures the curvature of the central cornea.

1) Keratometry

a) Measures both anterior and posterior corneal surfaces.

2) Placido's Disc	b) Assesses the smoothness and curvature of the corneal surface.
3) Corneal Topography	c) Evaluate the anterior curvature of the cornea and generate colour-coded maps.
4) Corneal Tomography (Pentacam)	d) Measures the curvature of the central cornea.

- A. 1-a, 2-c, 3-d, 4-b
- B. 1-d, 2-b, 3-c, 4-a
- C. 1-d, 2-a, 3-b, 4-c
- D. 1-b, 2-a, 3-c, 4-d

24. Which of the following statements is false?

- A. Pachymetry measures corneal thickness and is used for diagnosing glaucoma
- B. Specular Microscopy is used for photographing the corneal endothelium
- C. Aesthesiometer measures corneal sensation and is used to assess neurotrophic keratitis
- D. Aesthesiometer measures corneal sensation by evaluating the second cranial nerve

25. A 32 y/o contact lens wearer presents with a red, painful eye, blurred vision, and increased sensitivity to light. Examination reveals lid swelling, conjunctival chemosis, and a large corneal ulcer with greenish exudate, rapidly expanding with hypopyon. What is the most likely cause?

- A. Staphylococcus aureus
- B. Streptococcus pneumoniae
- C. Pseudomonas aeruginosa
- D. Enterobacteriaceae

26. A 60 y/o woman presents with a painful red eye, photophobia, and a history of chronic use of corticosteroid eye drops. Examination reveals a corneal ulcer with necrosis extending deep into the corneal stroma, approaching Descemet's membrane. Which type of bacterial keratitis is this patient most at risk for?

- A. Localized Ulcer
 - B. Perforating Ulcer
 - C. Adherent Leucoma
 - D. Sloughing Ulcer
-

27. Clinical question: Which of the following statements about the treatment of bacterial keratitis is incorrect?

- A. Cycloplegics like atropine can be used to reduce ciliary spasms and pain.
 - B. Ciprofloxacin (750mg twice daily) is given when the sclera gets involved.
 - C. Topical steroids should be started immediately to reduce inflammation.
 - D. Duo therapy with fortified cefazolin and tobramycin provides broad-spectrum coverage.
-

28. What is the average thickness of the central cornea?

- A. 1 to 1.2 mm
 - B. 0.8 to 0.9 mm
 - C. 0.5 to 0.6 mm
 - D. 7.8 mm
-

29. A 60 y/o man from a tropical region presents with a painful, red eye and a corneal ulcer following recent exposure to decaying vegetable matter. Which is the most common fungal infection affecting the cornea?

- A. *Candida albicans*
 - B. *Fusarium*
 - C. Mucormycosis
 - D. *Aspergillus fumigatus*
-

30. A 45 y/o man from a tropical region presents with a painful red eye, significant watering, and sensitivity to light after working in a garden with decaying plant material. The patient is diagnosed with keratomycosis. Which of the following is not an examination finding in keratomycosis?

- A. Feathery extensions at the margins of the corneal ulcer
 - B. Extensive corneal vascularization
 - C. Sterile immune ring
 - D. Endothelial plaque
-

31. A 45 y/o male presents with redness, pain, and decreased vision in his right eye. He reports a history of trauma to the eye with a plant material three days ago. O/E, there is a corneal ulcer with feathery edges and satellite lesions. Hypopyon is also noted. Which of the following is the most appropriate initial treatment for this patient?

- A. Topical Moxifloxacin
- B. Topical Natamycin
- C. Oral Acyclovir

D. Intravitreal Amphotericin B

32. A 45 y/o patient presents with eye redness, excessive tearing, and sensitivity to light. O/E you observe dendritic ulcers, and decreased corneal sensation. Based on these findings, which virus is most likely responsible for the patient's condition?

- A. Adenovirus
 - B. Varicella Zoster Virus (VZV)
 - C. Herpes Simplex Virus (HSV)
 - D. Chlamydia trachomatis
-

33. A 35 y/o patient presents with pain, redness, photophobia, and decreased vision in the right eye. Slit-lamp examination reveals a dendritic ulcer on the cornea. Which of the following is the most appropriate initial treatment for this patient?

- A. Topical steroids and antibiotics
 - B. Oral acyclovir and lubricants
 - C. Topical acyclovir and cycloplegics
 - D. Oral steroids and topical antibiotics
-

34. A 50 y/o male presents with decreased vision, redness, and a dull sensation in his R/E. He has a history of trigeminal neuralgia for which he underwent a Gasserian ganglion block. O/E, there is decreased corneal sensation and epithelial desquamation. Which of the following is the most likely cause of his symptoms?

- A. Keratomalacia
- B. Herpes Simplex Virus (HSV) Infection

- C. Neurotrophic Keratopathy
 - D. Neuroparalytic Keratopathy
-

35. A 65 y/o woman presents with a painful rash on her forehead and eyelid. She also reports blurred vision and photophobia. O/E, she has conjunctivitis, punctate epithelial keratitis, and iridocyclitis. Which of the following is a chronic ocular complication she is at risk of developing?

- A. Episcleritis
 - B. Corneal Scarring
 - C. Conjunctivitis
 - D. Secondary Glaucoma
-

36. A 70 y/o woman with a history of hypertension presents with severe pain in her right eye and forehead, along with a rash that started 5 days ago. She was diagnosed with Herpes Zoster Ophthalmicus (HZO). Which of the following statements is true regarding the management of her condition?

- A. Systemic steroids alone are sufficient to manage HZO
 - B. Antiviral therapy is started within 72 hours of rash onset.
 - C. Postherpetic neuralgia is less common in patients over 70 years of age.
 - D. Topical antiviral therapy is not recommended to manage eye involvement in HZO.
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	3
Question 4	2
Question 5	1
Question 6	3
Question 7	2
Question 8	1
Question 9	3
Question 10	3
Question 11	3
Question 12	1
Question 13	1
Question 14	4
Question 15	1
Question 16	1
Question 17	4
Question 18	3
Question 19	3
Question 20	2
Question 21	2
Question 22	3

Question 23	2
Question 24	4
Question 25	3
Question 26	2
Question 27	3
Question 28	3
Question 29	4
Question 30	2
Question 31	2
Question 32	3
Question 33	3
Question 34	3
Question 35	2
Question 36	2

Solution for Question 1:

Correct Answer: D) Secondary glaucoma

Explanation:

- Secondary glaucoma is not an indication of PK but rather an early complication of the procedure.
- It is a condition where increased intraocular pressure results from another underlying issue, and it typically requires treatment that addresses the underlying cause of the increased pressure, not corneal replacement.
- PK addresses corneal diseases and issues directly, not intraocular pressure problems or secondary glaucoma.

Penetrating Keratoplasty (PK):

- It is a type of Allografting (Allo-Keratoplasty) which is the replacement of a patient's diseased cornea with a healthy donor cornea.
- Full-thickness grafting.



Indications for Penetrating Keratoplasty:

Optical	Corneal opacity (Option A ruled out) Bullous keratopathy Corneal dystrophies Advanced keratoconus (Option C ruled out)
Therapeutic	To replace an inflamed cornea that does not respond to conventional therapy
Tectonic Graft	To restore the integrity of the eyeball, e.g., after corneal perforation or in cases of significant corneal thinning.
Cosmetic	To improve the appearance of the eye (Option B ruled out)

Complications: (ak142)

- Early Complications: Flat anterior chamber Iris prolapse Infection Secondary glaucoma (Option D) Epithelial defects Primary graft failure
- Flat anterior chamber
- Iris prolapse
- Infection
- Secondary glaucoma (Option D)
- Epithelial defects
- Primary graft failure
- Late Complications: Graft rejection Recurrence of disease Astigmatism
- Graft rejection
- Recurrence of disease

- Astigmatism
- Flat anterior chamber
- Iris prolapse
- Infection
- Secondary glaucoma (Option D)
- Epithelial defects
- Primary graft failure
- Graft rejection
- Recurrence of disease
- Astigmatism

Solution for Question 2:

Correct Answer: D) 1-c; 2-e; 3-b; 4-f; 5-d; 6-a.

Explanation:

Image	Description
[Image not available] Bacterial Keratitis: {{caption_text}}	A corneal epithelial defect with an infiltrate (a larger area of corneal involvement) and significant circumcorneal injection, often with a hypopyon.
[Image not available] Fungal Keratitis: {{caption_text}}	Grey or yellow-white stromal infiltrate with indistinct fluffy margins, often with satellite lesions, or a yellow-white densely suppurative infiltrate in Candida keratitis.
Superficial Punctate Keratitis: Punctate Keratitis" data-author="Okoro, Chiemela & Amiebenomo, Onyeka & Aruotu, Nwakuso" data-hash="2973" data-license="CC BY 4.0" data-source="Extensive superficial punctate staining; this could be caused by... Download Scientific Diagram (researchgate.net)" data-tags="March2025" src="https://image.prepladder.com/notes/WMt60FWyxjtEfSC5QjyZ1743055288.png" /> {{caption_text}}	This term describes any corneal epithelial disturbance with a dot-like morphology, including conditions like Thygeson superficial punctate keratitis.
[Image not available] Nummular Keratitis: {{caption_text}}	Characterized by fine granular subepithelial deposits surrounded by a halo of stromal haze.

[Image not available] Dendritic Ulcer:	Linear branching ulcer with terminal bulbs, the base of which stains well with fluorescein, and where the virus-laden cells at the margin of the ulcer stain with rose Bengal.
[Image not available] Pseudodendritis:	Corneal epithelial lesions that are smaller and finer than herpes simplex dendrites, have tapered ends without terminal bulbs, and stain better with rose Bengal than with fluorescein.

Solution for Question 3:

Correct Answer: C) Topical polyhexamethylene biguanide (PHMB)

Explanation:

- The clinical features and findings point to Acanthamoeba keratitis.
- PHMB is an amoebicidal agent effective against both the trophozoites and cysts of Acanthamoeba.
- It is the treatment of choice for Acanthamoeba keratitis, especially when cysts are identified.

Investigations and Treatment of Acanthamoeba Keratitis:

Investigations:

- Corneal Scraping: The primary method for obtaining tissue for examination. Scrape the margins and base of the lesion after applying a non-preserved topical anaesthetic.
- The primary method for obtaining tissue for examination.
- Scrape the margins and base of the lesion after applying a non-preserved topical anaesthetic.
- Staining: PAS (Periodic acid–Schiff) and Calcofluor white: Highly effective for visualizing Acanthamoeba cysts. Gram and Giemsa stains: Less sensitive but can also demonstrate cysts.
- PAS (Periodic acid–Schiff) and Calcofluor white: Highly effective for visualizing Acanthamoeba cysts.
- Gram and Giemsa stains: Less sensitive but can also demonstrate cysts.
- Culture: Definitive diagnosis by culturing on non-nutrient agar with dead E. coli.
- Additional Investigations: Immunohistochemistry: Uses antibodies to detect specific Acanthamoeba antigens. PCR: Useful for detecting Acanthamoeba DNA. In Vivo Confocal

Microscopy: Provides detailed images of Acanthamoeba cysts within the cornea.

- Immunohistochemistry: Uses antibodies to detect specific Acanthamoeba antigens.
- PCR: Useful for detecting Acanthamoeba DNA.
- In Vivo Confocal Microscopy: Provides detailed images of Acanthamoeba cysts within the cornea.
- Corneal Biopsy: Considered if other diagnostic methods are inconclusive.
- Considered if other diagnostic methods are inconclusive.
- The primary method for obtaining tissue for examination.
- Scrape the margins and base of the lesion after applying a non-preserved topical anaesthetic.
- PAS (Periodic acid–Schiff) and Calcofluor white: Highly effective for visualizing Acanthamoeba cysts.
- Gram and Giemsa stains: Less sensitive but can also demonstrate cysts.
- Immunohistochemistry: Uses antibodies to detect specific Acanthamoeba antigens.
- PCR: Useful for detecting Acanthamoeba DNA.
- In Vivo Confocal Microscopy: Provides detailed images of Acanthamoeba cysts within the cornea.
- Considered if other diagnostic methods are inconclusive.

Treatment Modalities:

- Debridement: Removal of infected epithelium to enhance topical medication penetration.
- Topical Amoebicides: PHMB (polyhexamethylene biguanide) 0.02% and Chlorhexidine 0.02%: Effective against both trophozoites and cysts. Hexamidine: More effective than Propamidine (Brolene).
- PHMB (polyhexamethylene biguanide) 0.02% and Chlorhexidine 0.02%: Effective against both trophozoites and cysts.
- Hexamidine: More effective than Propamidine (Brolene).
- Topical Steroids: Considered for persistent inflammation after two weeks of anti-amoebic treatment. Caution against use if corneal thinning is present.
- Considered for persistent inflammation after two weeks of anti-amoebic treatment.
- Caution against use if corneal thinning is present.
- Pain Management: Oral NSAIDs are recommended for pain relief.
- Therapeutic Keratoplasty: Required for cases resistant to medical therapy or when corneal perforation occurs.
- PHMB (polyhexamethylene biguanide) 0.02% and Chlorhexidine 0.02%: Effective against both trophozoites and cysts.

- Hexamidine: More effective than Propamidine (Brolene).
- Considered for persistent inflammation after two weeks of anti-amoebic treatment.
- Caution against use if corneal thinning is present.

Topical moxifloxacin (Option A):

- Moxifloxacin is an antibiotic used to treat bacterial keratitis.
- It is not effective against Acanthamoeba, which is the suspected pathogen in this case based on clinical findings and the presence of cysts.

Oral acyclovir (Option B):

- Acyclovir is an antiviral agent used primarily for herpes simplex keratitis.
- It is ineffective against Acanthamoeba keratitis.

Topical corticosteroids (Option D):

- While corticosteroids may be used cautiously for persistent inflammation after initiating anti-amoebic treatment, they are not first-line therapy and should be avoided initially due to the risk of worsening the infection if used prematurely.

Solution for Question 4:

Correct Answer: B) Ghost vessels

Explanations:

- The given features describe Syphilitic Interstitial Keratitis.
- Ghost vessels are remnants of previously filled blood vessels that become empty after the active inflammation of syphilitic interstitial keratitis resolves.
- They are a characteristic finding during the healing stage of the condition.

Interstitial Keratitis (IK): Inflammation of the corneal stroma. Typically does not affect the corneal epithelium or endothelium initially.

Causes of IK:

- Syphilis
- Other infectious causes include tuberculosis, Lyme disease, parasitic infections (common in endemic areas)

[Image not available]

Clinical Presentation of Syphilitic Interstitial Keratitis:

Aspect	Details
Symptoms	Acute anterior uveitis: Pain, redness, photophobia, blurred vision. Severe blurring of vision during the active phase.
Clinical Signs: Active Stage	Salmon Patch Appearance: Pinkish-red discoloration due to deep stromal vascularization and clouding. Limbitis: Inflammation at the corneal-scleral junction. Granulomatous Anterior Uveitis: Large inflammatory deposits (keratic precipitates) on the corneal endothelium.
Clinical Signs: Healing Stage	Ghost Vessels: Empty blood vessels leaving faint tracks. Stromal Scarring: Feathery appearance from scarring. Corneal Thinning: This may lead to irregular astigmatism and vision impairment. Band Keratopathy: Calcium deposition as a late complication.

[Image not available]

Pseudodendrites (Option A) are typically associated with herpetic keratitis, not with interstitial keratitis caused by syphilis.

Ring-shaped abscess (Option C): Ring-shaped abscesses are seen in advanced *Acanthamoeba* keratitis, not in syphilitic interstitial keratitis.

Radial keratoneuritis (Option D) is a hallmark feature of *Acanthamoeba* keratitis and is not seen in syphilitic interstitial keratitis.

Solution for Question 5:

Correct Answer: A) 1,2,3,4

Explanation: Keratoconus

[Image not available]

- Definition: Progressive, non-inflammatory bilateral ectatic corneal condition affecting the axial part.

- Pathology: Involves progressive thinning and ectasia of the cornea due to defective synthesis of mucopolysaccharides and collagen.

Clinical Features

- Symptoms: Defective vision due to progressive myopia and irregular astigmatism, uncorrectable by glasses. Blurred and distorted vision, glare, halos around lights. Light sensitivity and ocular irritation. Visual loss mainly from irregular astigmatism and myopia; secondary loss due to corneal scarring.
- Defective vision due to progressive myopia and irregular astigmatism, uncorrectable by glasses.
- Blurred and distorted vision, glare, halos around lights.
- Light sensitivity and ocular irritation.
- Visual loss mainly from irregular astigmatism and myopia; secondary loss due to corneal scarring.
- Defective vision due to progressive myopia and irregular astigmatism, uncorrectable by glasses.
- Blurred and distorted vision, glare, halos around lights.
- Light sensitivity and ocular irritation.
- Visual loss mainly from irregular astigmatism and myopia; secondary loss due to corneal scarring.

Examination Findings

- Corneal Thinning & Bulging: The Cornea thins near the centre and bulges forward, with the cone apex slightly below the corneal centre.
- Specific Signs: Distorted Window Reflex: Visual irregularity on corneal reflection. Placido Disc Irregularity: Circles appear irregular. Slit-Lamp Examination: Thinning and ectasia of the central cornea. Opacity at the apex, Fleischer Ring: Brownish ring around the cone, possibly due to haemosiderin in the epithelium.
- Distorted Window Reflex: Visual irregularity on corneal reflection.
- Placido Disc Irregularity: Circles appear irregular.
- Slit-Lamp Examination: Thinning and ectasia of the central cornea. Opacity at the apex, Fleischer Ring: Brownish ring around the cone, possibly due to haemosiderin in the epithelium.
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- Opacity at the apex, Fleischer Ring: Brownish ring around the cone, possibly due to haemosiderin in the epithelium.
- Distorted Window Reflex: Visual irregularity on corneal reflection.

- Placido Disc Irregularity: Circles appear irregular.
- Slit-Lamp Examination: Thinning and ectasia of the central cornea. Opacity at the apex, Fleischer Ring: Brownish ring around the cone, possibly due to haemosiderin in the epithelium.
- Thinning and ectasia of the central cornea.
- Opacity at the apex, Fleischer Ring: Brownish ring around the cone, possibly due to haemosiderin in the epithelium.
- Thinning and ectasia of the central cornea.
- Opacity at the apex, Fleischer Ring: Brownish ring around the cone, possibly due to haemosiderin in the epithelium.

[Image not available]

- Folds in Descemet's and Bowman's membranes. Vogt Striae: Fine, parallel striae at the apex, best seen with a slit lamp, disappearing with external pressure (vs Haab striae: They are curvilinear breaks in Descemet's membrane, resulting acutely from stretching of the cornea in primary congenital glaucoma).
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- Vogt Striae: Fine, parallel striae at the apex, best seen with a slit lamp, disappearing with external pressure (vs Haab striae: They are curvilinear breaks in Descemet's membrane, resulting acutely from stretching of the cornea in primary congenital glaucoma).
- Retinoscopy: Yawning reflex (scissor reflex) with high oblique/irregular astigmatism. Distant Direct Ophthalmoscopy: Annular dark shadow (oil droplet reflex) separating central and

peripheral cornea. A ring of shadow concentric with the margin in the red reflex, resembling an oil droplet. Ophthalmoscope Findings: Irregular Astigmatism: Progressive myopia and irregular astigmatism in later stages, poorly corrected by glasses. Munson's Sign: Localized bulging of the lower lid when looking down. Keratometer Findings: Malformed, malaligned, and malfocused mires. Corneal Topography: The most sensitive method for early detection (Forme Fruste Keratoconus) shows the shape of the corneal surface.

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- Corneal Topography: The most sensitive method for early detection (Forme Fruste Keratoconus) shows the shape of the corneal surface.

Solution for Question 6:

Correct Answer: C) Corneal collagen cross-linking (CXL)

Explanation:

- Corneal collagen cross-linking (CXL) is the most appropriate treatment for stabilizing keratoconus progression and improving contact lens tolerance in this patient.
- It works by strengthening the corneal stroma through the cross-linking of collagen fibres, which helps to halt the progression of ectasia, thereby preserving vision and enhancing the effectiveness of contact lenses.

Treatment of Keratoconus

- Spectacle Correction: Effective in the early stages, but contact lenses are more beneficial. Less effective in later stages due to irregular astigmatism.
- Effective in the early stages, but contact lenses are more beneficial.
- Less effective in later stages due to irregular astigmatism.
- Contact Lenses: Rigid gas permeable lenses improve vision in early cases. Specially designed scleral lenses (e.g., Rose-K) are useful in early to moderate cases.
- Rigid gas permeable lenses improve vision in early cases.
- Specially designed scleral lenses (e.g., Rose-K) are useful in early to moderate cases.
- Corneal Collagen Cross-Linking (CXL): New treatment modality to slow disease progression. Involves riboflavin (0.1%) eye drops and UVA radiation. Riboflavin induces cross-linking of corneal collagen to stabilize the stroma and improve contact lens tolerance.
- New treatment modality to slow disease progression.
- Involves riboflavin (0.1%) eye drops and UVA radiation.
- Riboflavin induces cross-linking of corneal collagen to stabilize the stroma and improve contact lens tolerance.
- Keratoplasty: Recommended if disease progresses or in cases of acute hydrops (Descemet's membrane tear). Reserved for more advanced diseases or when other treatments fail. Penetrating keratoplasty is common, but Deep Anterior Lamellar Keratoplasty (DALK) is preferred if Descemet's membrane and endothelium are intact, reducing rejection risk.
- Recommended if disease progresses or in cases of acute hydrops (Descemet's membrane tear).
- Reserved for more advanced diseases or when other treatments fail.
- Penetrating keratoplasty is common, but Deep Anterior Lamellar Keratoplasty (DALK) is preferred if Descemet's membrane and endothelium are intact, reducing rejection risk.

- Intracorneal Ring Segments (INTACS) : Effective in early to moderate cases by flattening the cornea. Less effective in halting progression.
- Effective in early to moderate cases by flattening the cornea.
- Less effective in halting progression.
- Effective in the early stages, but contact lenses are more beneficial.
- Less effective in later stages due to irregular astigmatism.
- Rigid gas permeable lenses improve vision in early cases.
- Specially designed scleral lenses (e.g., Rose-K) are useful in early to moderate cases.
- New treatment modality to slow disease progression.
- Involves riboflavin (0.1%) eye drops and UVA radiation.
- Riboflavin induces cross-linking of corneal collagen to stabilize the stroma and improve contact lens tolerance.
- Recommended if disease progresses or in cases of acute hydrops (Descemet's membrane tear).
- Reserved for more advanced diseases or when other treatments fail.
- Penetrating keratoplasty is common, but Deep Anterior Lamellar Keratoplasty (DALK) is preferred if Descemet's membrane and endothelium are intact, reducing rejection risk.
- Effective in early to moderate cases by flattening the cornea.
- Less effective in halting progression.

Spectacle correction (Option A):

- Glasses are ineffective in stabilizing or treating corneal ectasia; they only correct refractive errors.

Intracorneal ring segments (INTACS) (Option B):

- INTACS can improve vision by reshaping the cornea, but they do not stop disease progression.
- They may be used in conjunction with CXL to achieve better outcomes.

Penetrating keratoplasty (Option D):

- This is a full-thickness corneal transplant typically reserved for advanced cases where other treatments have failed.
- It is not a first-line treatment for early or moderate ectasia.

Solution for Question 7:

Correct Answer: B) Turner's syndrome

Explanation:

- Turner's syndrome is not commonly associated with keratoconus.

Conditions Associated with Keratoconus

Ocular Conditions	Systemic Conditions
Ectopia lentis Congenital cataract Aniridia Retinitis pigmentosa Vernal keratoconjunctivitis (VKC) Atopic keratoconjunctivitis (Option D) Leber's congenital amaurosis Floppy eyelid syndrome Corneal endothelial dystrophy	Marfan's syndrome (Option A) Atopy (asthma, eczema, hay fever) Down's syndrome (Option C) Ehlers-Danlos syndrome Osteogenesis imperfecta Mitral valve prolapse

Solution for Question 8:

Correct Answer: A) Death due to an unknown cause

Explanation:

- Death due to an unknown cause: This is an absolute contraindication as it could signal serious undiagnosed conditions.

Keratoplasty

- Definition: Surgical procedure where the patient's diseased cornea is replaced with a healthy clear cornea.

Types of Keratoplasty:

- Penetrating Keratoplasty (PK): Full-thickness grafting.
- Lamellar Keratoplasty (LK): Partial-thickness grafting.

Donor Tissue

- Timing: Donor eye removal should be within 6 hours of death (up to 12 hours in cold climates).
- Storage: Stored under sterile conditions.
- Evaluation: Biomicroscopic Examination: Assess the condition of corneal epithelium, stroma, Descemet's membrane, and endothelium. Grades: excellent to poor. Specular Microscopy: Check endothelial cell count and morphology. Microbial Evaluation: Culture from conjunctival swab. Serological Screening: Test for HIV, hepatitis B, hepatitis C, and syphilis.
- Biomicroscopic Examination: Assess the condition of corneal epithelium, stroma, Descemet's membrane, and endothelium. Grades: excellent to poor.
- Specular Microscopy: Check endothelial cell count and morphology.
- Microbial Evaluation: Culture from conjunctival swab.
- Serological Screening: Test for HIV, hepatitis B, hepatitis C, and syphilis.
- Biomicroscopic Examination: Assess the condition of corneal epithelium, stroma, Descemet's membrane, and endothelium. Grades: excellent to poor.
- Specular Microscopy: Check endothelial cell count and morphology.
- Microbial Evaluation: Culture from conjunctival swab.
- Serological Screening: Test for HIV, hepatitis B, hepatitis C, and syphilis.

Contraindications for cornea donation

Absolute contraindications	Relative contraindications
1) Death due to an unknown cause 2) Systemic infections HIV Hepatitis Congenital rubella TB Syphilis 3) CNS infection or diseases Multiple sclerosis Rabies Creutzfeldt-Jakob disease 4) Most haematological malignancies Leukaemia ... (content truncated)	Ocular inflammation (Option C ruled out) Any congenital or acquired disorders of the Eye or previous ocular surgery (including corneal laser surgery) (Option B ruled out) Retinoblastoma (Option D ruled out) Malignant tumours or ocular metastases in the anterior segment of the eye.

Solution for Question 9:

Correct Answer: C) Optisol GS

Explanation:

- Optisol GS is designed for intermediate-term storage, effective for up to 2 weeks, making it suitable for preserving the cornea for up to 10 days.

Storage of Cornea

Short-term Storage (24 to 96 hours)	Intermediate-term Storage (up to 2 weeks)	Long-term Preservation (months to years):
Moist Chamber Method: Preserve the whole globe at 4°C in a refrigerator for up to 24 hours. McCarey Kaufman (M-K) Medium: Pink-colored medium containing TC-199, Dextran-40, HEPES buffer, and gentamicin.	Optisol GS: Contains chondroitin sulfate, dextran, ascorbic acid, vitamin B12, ATP precursors, gentamicin, and streptomycin. Other Media: Cornisol, Eusol, and Life-GE medium also available for up to 2 weeks.	Glycerine Preservation: Effective for 3-4 weeks. Organ Culture Method: Preserves cornea at 30°C to 37°C for up to 35 days. Cryopreservation: Preserves corneal button at -70°C for up to 1 year.

Moist Chamber Method (Option A): Suitable only for short-term storage up to 24 hours, so it is not appropriate for a 10-day storage period.

McCarey Kaufman (M-K) Medium (Option B): Effective for short-term storage up to 96 hours, which is insufficient for 10 days.

Cryopreservation (Option D): Although it preserves corneal tissue for up to 1 year, it is not typically used for the 10-day storage period due to its specific requirements and more complex handling.

Solution for Question 10:

Correct Answer: C) Acanthamoeba Keratitis

Explanations:

- Acanthamoeba keratitis is characterized by severe pain that is often out of proportion to the clinical findings, along with features such as perineural infiltrates and, in advanced cases, ring-shaped abscesses.

- The image provided shows these hallmark signs.

Examination Findings in Acanthamoeba Keratitis:

Symptoms:

- Blurred Vision: Common in corneal conditions due to the disruption of the cornea's transparent surface.
- Discomfort: Significant discomfort in the affected eye.
- Severe Pain: Pain is often out of proportion to clinical signs, a key clinical clue.

Signs:

Early Disease:

- Irregular and Greyish Epithelium: Early sign indicating amoebic invasion.
- Pseudodendrites: Lesions resembling herpetic keratitis.
- Limbitis and Stromal Infiltrates: Inflammation at the corneal-scleral junction with diffuse or focal infiltrates in the anterior stroma.
- Perineural Infiltrates (Radial Keratoneuritis): Hallmark sign, radial streaks extending towards the cornea's centre, pathognomonic for Acanthamoeba keratitis.

Advanced Disease:

- Ring Abscess: Characteristic of advanced infection, where infiltrates form a ring-shaped abscess.
- Stromal Opacification and Vascularisation: Corneal opacity and new blood vessel growth, leading to significant vision impairment.
- Corneal Melting: Serious complication where corneal tissue breaks down, often at the infiltrate edges.
- Scleritis: Reactive inflammation of the sclera, not a direct infection extension.

Herpes Simplex Keratitis (Option A) typically presents with dendritic ulcers and less severe pain relative to Acanthamoeba keratitis.

- The image and symptoms described, especially the severe pain out of proportion to clinical signs, are more indicative of Acanthamoeba keratitis.

Fungal Keratitis (Option B) usually presents with feathery-edged infiltrates and satellite lesions.

- While it can cause significant discomfort, the pain level and clinical features (e.g., perineural infiltrates) seen in Acanthamoeba keratitis are distinctive.

Bacterial Keratitis (Option D) typically presents with a more prominent purulent discharge and a dense, localized infiltrate.

- The disproportionate pain and the clinical signs in the image are more consistent with Acanthamoeba keratitis.

Solution for Question 11:

Correct Answer: C) Khodadoust Line

Explanation:

- Khodadoust Line: This is a characteristic sign of endothelial rejection, marked by a demarcation line between healthy and damaged endothelium.
- It is a key feature in diagnosing endothelial rejection.

[Image not available]

CORNEAL GRAFT REJECTION

Incidence:

Less common than other organ transplants but can occur as early as 2 weeks and up to several years after keratoplasty.

Types of Rejection:

- Epithelial Rejection
- Stromal Rejection
- Endothelial Rejection (most common and most likely to lead to graft failure)

Clinical Presentations of Corneal Graft Rejection:

- Epithelial Rejection: Elevated epithelial rejection line staining with fluorescein. Epithelial Rejection Line: Irregular elevated line in the epithelium. (Option D ruled out)
- Epithelial Rejection Line: Irregular elevated line in the epithelium. (Option D ruled out)
- Subepithelial Infiltrates: Known as Kayes dots. (Option B ruled out)
- Known as Kayes dots. (Option B ruled out)
- Stromal Rejection: Sudden onset of full-thickness stromal haze in a previously clear graft. Cellular Infiltration or Edema: In the stroma.
- Sudden onset of full-thickness stromal haze in a previously clear graft.
- Cellular Infiltration or Edema: In the stroma.
- Epithelial Rejection Line: Irregular elevated line in the epithelium. (Option D ruled out)

- Known as Kayes dots. (Option B ruled out)
- Sudden onset of full-thickness stromal haze in a previously clear graft.
- Cellular Infiltration or Edema: In the stroma.
- Endothelial Rejection: Khodadoust Line: Demarcation line between healthy and damaged endothelium. Diffuse Endothelial Rejection: Characterized by numerous keratic precipitates. Endothelial Rejection Line: Linear deposition of white blood cells on the corneal endothelium.
- Khodadoust Line: Demarcation line between healthy and damaged endothelium.
- Diffuse Endothelial Rejection: Characterized by numerous keratic precipitates.
- Endothelial Rejection Line: Linear deposition of white blood cells on the corneal endothelium.
- Additional Signs: Circumcorneal Congestion Fresh Keratic Precipitates Epithelial Edema Increased Flare and Cells: In the anterior chamber. (Option A ruled out)
- Circumcorneal Congestion
- Fresh Keratic Precipitates
- Epithelial Edema
- Increased Flare and Cells: In the anterior chamber. (Option A ruled out)
- Khodadoust Line: Demarcation line between healthy and damaged endothelium.
- Diffuse Endothelial Rejection: Characterized by numerous keratic precipitates.
- Endothelial Rejection Line: Linear deposition of white blood cells on the corneal endothelium.
- Circumcorneal Congestion
- Fresh Keratic Precipitates
- Epithelial Edema
- Increased Flare and Cells: In the anterior chamber. (Option A ruled out)

Solution for Question 12:

Correct Answer: A) It is indicated in pseudophakic bullous keratopathy

Explanation:

- Keratoplasty: Commonly known as corneal transplantation, where a patient's diseased cornea is replaced by healthy clear cornea.

[Image not available]

Indications of Penetrating Keratoplasty:

- Pseudophakic bullous keratopathy
- Corneal opacity
- Corneal dystrophies
- Advanced keratoconus
- Marked corneal thinning

Indications of Endothelial Keratoplasty:

- Fuchs endothelial dystrophy
- Pseudophakic corneal edema
- Posterior polymorphous dystrophy
- Aphakic corneal edema
- Iridocorneal endothelial syndrome
- Failed corneal graft

KERATOPROSTHESIS

- Keratoprosthesis is an artificial corneal device used for patients unsuitable for conventional keratoplasty.
- Indications: Bilateral blindness due to corneal diseases not amenable to keratoplasty
Stevens-Johnson syndrome Chemical burns
- Bilateral blindness due to corneal diseases not amenable to keratoplasty
- Stevens-Johnson syndrome
- Chemical burns
- Bilateral blindness due to corneal diseases not amenable to keratoplasty
- Stevens-Johnson syndrome
- Chemical burns
- Types of Keratoprosthesis: Boston Keratoprosthesis (Boston K-Pro)
- Boston Keratoprosthesis (Boston K-Pro)
- Boston Keratoprosthesis (Boston K-Pro)

[Image not available]

- Alfa Cor Keratoprosthesis Modified Osteo-Odonto-Keratoprosthesis (MOOKP)
- Alfa Cor Keratoprosthesis
- Modified Osteo-Odonto-Keratoprosthesis (MOOKP)
- Alfa Cor Keratoprosthesis

- Modified Osteo-Odonto-Keratoprosthesis (MOOKP)

[Image not available]

Solution for Question 13:

Correct Answer: A) 1-b, 2-d, 3-a, 4-c

Explanation:

CORNEAL DEGENERATION	IMAGE
Arcus Senilis Lipoid infiltration of the cornea. It is common in the elderly. Begins as a crescentic grey or whitish arc around the corneal margins. Extends to form a complete opaque ring around the cornea. Thicker at the top and bottom; sharply defined on the peripheral side, fading towards the centre. Typically no more than 1 mm wide. Does not affect vision or corneal vitality. Unrelated to secondary hypercholesterolemia. Round opacity at a young age is called arcus juvenilis, caused by dyslipidemia.	[Image not available] {{caption_text}}
Band Shaped Keratopathy Common in chronic uveitis, aphakic eyes with prolonged silicone oil, and occasionally linked to hyperparathyroidism or vitamin D poisoning. Characterized by a whitish band across the cornea with round 'holes' or cleaves. Caused by hyaline infiltration and deposition of calcareous salts in the superficial stroma. Treatment options include scraping off the opacity, dissolving it with sodium edetate, or using an excimer laser.	keratopathy obscuring the visual axis with no view of pupil... Download Scientific Diagram (researchgate.net)" data-tags="March2025" src="https://image.prepladder.com/notes/NjlpPTDP20x9dFTvnoHW1743062255.png" /> {{caption_text}}

Oil droplet keratopathy Common in individuals exposed to hot, dry, dusty environments, such as farmers. An exposed interpalpebral portion of the cornea (sparing the limbal area) is affected. Superficial, nonvascularized corneal opacity. Focal lesions in the epithelium resembling droplets of oil.	Droplet Keratopathy" data-author="Yasmine Maher Shaaban, Ahmed Mohamed Reda, Somaia Ahmed Saad El Din" data-hash="3142" data-license="CC BY 4.0" data-source="Using Immunofluorescence to Investigate an Associated Specific Humoral Immune Response Role in Climatic Droplet Keratopathy (symbiosisonlinepublishing.com)" data-tags="March2025" src="https://image.prepladder.com/notes/Z6I35KkF3nuGZZcTc6AR1743062274.png" /> {{caption_text}}
Salzmann nodular degeneration A degenerative condition of the cornea. Bluish-white, avascular nodules. Located in the superficial stroma and Bowman's membrane. Typically occurs in individuals with a history of previous corneal disease.	nodular degeneration " data-author="S. V. Trufanov, I. A. Riks, M. Ezugbaya" data-hash="3143" data-license="CC BY 4.0" data-source="Salzmann's nodular degeneration. Nodules with vascularization and... Download Scientific Diagram (researchgate.net)" data-tags="March2025" src="https://image.prepladder.com/notes/NNrfkYc3LpB8Ccr4UlxK1743062291.png" /> {{caption_text}}

Solution for Question 14:

Correct Answer: D) Vortex keratopathy remains permanent even after discontinuing the offending medication.

Explanation:

- The condition of vortex keratopathy is reversible upon stopping the offending medication.

CORNEAL DEGENERATION	IMAGE
Vortex Keratopathy / Cornea verticillata Whorl-like opacity in the corneal epithelium. Causes: Long-term use of medications like amiodarone, chloroquine, phenothiazines, netrasudil and indomethacin, or associated with Fabry disease. (Option B ruled out) Symptoms: Generally asymptomatic and harmless; can cause glare and surface discomfort. Reversibility: The condition is reversible upon stopping the offending medication. (Option D) Treatment: Surface discomfort may be managed with topical lubricants.	Cornea verticillata 01 - Cornea verticillata - Wikipedia" data-tags="March2025" src="https://image.prepladder.com/notes/uIMHf5GMDvyY7qeisOUb1743062896.png" /> {{caption_text}}

Lipid Keratopathy Whitish or yellowish deposits in the cornea, composed mainly of cholesterol and fatty acids. Primary Lipid Keratopathy: Rare, serum lipid levels are normal. Secondary Lipid Keratopathy: Occurs in vascularized corneas due to diseases like corneal infections, interstitial keratitis, ocular trauma, glaucoma, or chronic iridocyclitis. Fat Deposits: Initially intracellular but may become extracellular with necrosis of stromal cells. (Option C ruled out)	Lipid Keratopathy Due to Presumed Herpes Simplex Keratitis: Case Report Semantic Scholar" data-tags="March2025" src="https://image.prepladder.com/notes/9MhboyeO9bO77RPZHmHg1743062991.png" /> {{caption_text}}
Salzmann nodular degeneration A degenerative condition of the cornea. Bluish-white, avascular nodules. Located in the superficial stroma and Bowman's membrane. (Option A ruled out) Typically occurs in individuals with a history of previous corneal disease. Tends to be slowly progressive. May be treated with lamellar keratoplasty.	nodular degeneration " data-author="S. V. Trufanov, I. A. Riks, M. Ezugbaya" data-hash="3143" data-license="CC BY 4.0" data-source="Salzmann's nodular degeneration. Nodules with vascularization and... Download Scientific Diagram (researchgate.net)" data-tags="March2025" src="https://image.prepladder.com/notes/NNrfkYc3LpB8Ccr4UlxK1743062291.png" /> {{caption_text}}

Solution for Question 15:

Correct answer: A) Nebular corneal opacity

Explanation:

- The image shows a faint opacity covering the central cornea which suggests a Nebular opacity.
- Although faint, a nebular opacity can severely impact vision, particularly if it covers the pupillary area.
- The irregular refraction caused by the nebula distorts the image more than a denser opacity like leucoma, leading to greater visual discomfort and blurring.

Macular corneal opacity (Option B):

- This semi-dense opacity affects the cornea less than a nebula regarding irregular refraction, but it still causes significant visual impairment.

Leucomatous corneal opacity (Option C):

- While this dense white opacity blocks light more completely, it causes less irregular refraction than a nebula.
- If it doesn't cover the entire pupillary area, it may not affect vision as much as a nebula.

[Image not available]

Adherent leucoma (Option D):

- This is a dense opacity associated with iris incarceration and is typically more significant in light blockage, but its effect on vision varies depending on the extent of iris involvement.

[Image not available]

Management of corneal opacity:

- Optical Iridectomy: This procedure can be used for central macular or leucomatous opacities when vision improves with pupil dilation.
- Phototherapeutic Keratectomy (PTK): PTK uses an excimer laser, and is effective for treating superficial corneal opacities, such as nebulae.
- Keratoplasty
- Cosmetic-Coloured Contact Lenses: These lenses offer a cosmetic solution for unsightly scars in the eyes without visual potential.
- Corneal Tattooing: It is mainly reserved for firm scars in eyes with no useful vision.

Solution for Question 16:

Correct answer: A) Granular Corneal Dystrophy, Type 1 (GCD1)

Explanation:

- Multiple milky, granular deposits in the anterior corneal stroma, with a clear cornea between the deposits, indicate Granular corneal dystrophy Type 1.

Stromal Dystrophies	
TGFB1 Corneal Dystrophies These dystrophies are associated with mutations in the TGFB1 gene	
Lattice Corneal Dystrophy (LCD) (Option B ruled out)	Marked by amyloid deposits that form a lattice-like pattern within the corneal stroma. There are two subtypes: LCD1: Typically presents in early childhood with lattice lines that spare the peripheral cornea, leading to

	recurrent erosions and visual impairment. LCD2 (Gelsolin Type): Appears later in life and begins with lattice lines in the peripheral cornea, gradually affecting the centre. Lattice corneal dystrophy (uiowa.edu)" data-tags="March2025" src="https://image.prepladder.com/notes/lWfAdcFCrK87hOqGr6lO1743067499.png" /> {{caption_text}}
Granular Corneal Dystrophy (GCD) (Option A)	It is characterized by the deposition of hyaline material within the corneal stroma, with the following subtypes: GCD1 (Classic): Presents in childhood with milky, granular deposits in the anterior stroma. GCD2 (Granular-Lattice): Features both granular and lattice-like deposits, usually appearing in the first decade of life. Granular corneal dystrophy (uiowa.edu)" data-tags="March2025" src="https://image.prepladder.com/notes/6Kh7ugTr5oeWG1QnYwAk1743067515.png" /> {{caption_text}}
Other Stromal Dystrophies	
Macular Corneal Dystrophy (MCD) (Option C ruled out)	Inherited in an autosomal recessive pattern, this dystrophy is characterized by diffuse stromal haze and macular-shaped opacities, often causing significant visual impairment in childhood or early adulthood.
Schnyder Corneal Dystrophy (SCD) (Option D ruled out)	An autosomal dominant condition marked by the deposition of cholesterol crystals in the corneal stroma, leading to ring-shaped opacities and progressive vision loss.

Solution for Question 17:

Correct answer: D) Fuchs Endothelial Corneal Dystrophy (FECD)

Explanation:

- Fuchs Endothelial Corneal Dystrophy is strongly associated with the "beaten-metal" appearance on Descemet's membrane.
- It is a common type of endothelial dystrophy, particularly affecting older adults.

Posterior Polymorphous Corneal Dystrophy (PPCD) (Option A):

- PPCD is a corneal dystrophy that often presents with unique lesions on the endothelium, such as nodules, vesicles, or band-like patterns that resemble railroad tracks.

Congenital Hereditary Endothelial Dystrophy (CHED) (Option B):

- CHED is a genetic disorder where the endothelium is abnormal from birth or early childhood with diffuse corneal clouding, thickening, and decreased vision.

Keratoconus (Option C):

- Keratoconus is a condition where the cornea progressively thins and bulges into a cone-like shape, causing irregular astigmatism and vision distortion.
- It does not involve Descemet's membrane or produce the "beaten-metal" appearance.

Solution for Question 18:

Correct Answer: C) A-3, B-2, C-1

Explanation:

Fluorescein sodium

[Image not available]

- Fluorescein penetrates poorly into the lipid layer of the corneal epithelium, and therefore, it does not stain normal cornea.
- Although fluorescein is a very effective stain for the diseased cornea, it is more difficult to detect fluorescein staining of the conjunctiva because of the poor scleral contrast.
- Fluorescein absorbs light in the blue spectrum (490 nm) and emits yellow-green light of a higher wavelength (530 nm), therefore staining can be more readily viewed if a cobalt blue or yellow (blue-free) filter is used.

Rose bengal

- Rose bengal is a derivative of fluorescein and is used to detect ocular surface damage.
- It was initially theorized that rose bengal stains dead and devitalized tissue, but it is now thought to stain any part of the ocular surface which is not protected by an overlying tear film, specifically in areas lacking membrane-associated mucins.

Lissamine green

- has a high affinity for staining dead and degenerated cells, including mucous strands.

- The dye appears to stain areas where there are disrupted intercellular junctions.
- Use of a green (red-free) filter shows the stained areas to appear black in color.
- As it is better tolerated than rose bengal, it is the preferred dye for staining the bulbar conjunctiva.

Solution for Question 19:

Correct answer: C) Perform mechanical debridement and apply tissue adhesive glue

Explanation:

- Mechanical debridement is necessary to remove necrotic material and promote healing in a non-healing ulcer.
- Tissue adhesive glue is indicated to prevent perforation, especially when the perforation is less than 2mm.

Management of Non-Healing Ulcers	Management of Impending Perforation
Identify and Remove Causes: Local factors (e.g., ↑ IOP, foreign bodies, dacryocystitis) and systemic factors (e.g., diabetes, malnutrition) causing non-healing. Mechanical Debridement: Scrape the ulcer floor to remove necrotic material under local anaesthesia to promote healing. Cauterization: Using pure carbolic acid or 10-20% trichloroacetic acid in non-responding cases. Bandage Soft Contact Lens: To aid healing by protecting the corneal surface. Peritomy: Sever perilimbal conjunctival vessels to reduce excessive vascularization hindering healing.	Avoid Strain: Avoid sneezing, coughing, and straining, and maintain strict bed rest. Pressure Bandage: To provide external support to the cornea. Lower Intraocular Pressure: Use acetazolamide, mannitol, oral glycerol, timolol eye drops, and paracentesis if needed. Tissue Adhesive Glue: Apply cyanoacrylate glue to prevent perforation, if perforation <2mm. Bandage Soft Contact Lens: Used to support and protect the cornea. Conjunctival Flap: Cover the cornea partially or completely with a conjunctival flap for additional support. Amniotic Membrane Transplantation Penetrating Therapeutic Keratoplasty: Perform a tectonic graft if suitable and available to prevent perforation or repair a perforated cornea if perforation >2-4mm.

Increasing the frequency of antibiotic eye drops (Option A) alone may not be sufficient in a case of impending perforation.

Applying a pressure bandage and bed rest (Option B) helps, but alone it is not sufficient without addressing the structural integrity of the cornea.

Topical steroids (Option D) are contraindicated as they can worsen the infection.

Solution for Question 20:

Correct answer: B) Endothelium

Explanation:

- The innermost layer of the cornea is the endothelium which consists of a single layer of hexagonal cells.
- These cells are crucial for maintaining corneal transparency by regulating fluid balance within the cornea through an active pump mechanism.

Layers of Cornea	
Epithelium (Option C ruled out)	The outermost layer comprises 5-6 layers of non-keratinized, stratified squamous epithelial cells. Acts as a barrier, protecting the cornea and preventing tear fluid from entering the underlying stroma. The basal cells at the limbus are stem cells responsible for regenerating the corneal epithelium.
Bowman's Membrane (Option D ruled out)	Located just beneath the epithelium, this acellular layer is 12 µm thick and composed of densely packed collagen fibrils. It is a protective barrier to the stroma but cannot regenerate if damaged and heals by scarring.
Stroma (Option A ruled out)	Making up about 90% of the cornea's thickness (approximately 0.5 mm thick), the stroma consists of collagen fibrils arranged in parallel layers. These fibrils are embedded in a hydrated matrix of proteoglycans - Chondroitin and Keratan Sulphate (keratan sulphate maintains corneal transparency).
Pre-Descemet's Membrane (Dua's Layer)	This layer, discovered in 2013, is about 15 µm thick, acellular, and lies just above Descemet's membrane. It is strong and resistant to air penetration.
Descemet's Membrane	A robust basement membrane of the corneal endothelium is made of collagen and glycoproteins. It is tense, but when torn, curls inwards on itself. It can regenerate if damaged and is resistant to trauma and pathological processes.
Endothelium (Option B)	The innermost layer consists of a single layer of flat, hexagonal cells. These cells are crucial for maintaining corneal transparency by regulating fluid balance within the

cornea through an active pump mechanism.

[Image not available]

Solution for Question 21:

Correct Answer: B) Aerobic

Explanation:

- The primary metabolism of the cornea is aerobic, where oxygen is used to generate energy, mainly through oxidative phosphorylation.
- This is possible due to the oxygen supply from the tear film and the atmosphere, despite the cornea's lack of blood vessels.
- While the cornea does have some anaerobic metabolic processes, its primary metabolism is aerobic.
- Activation of anaerobic glycolysis leads to lactic acid formation, resulting in metabolic acidosis causing inhibition of sodium- potassium pump, in turn leading to corneal edema. (Option A)
- The primary metabolism of the lens is anaerobic, as it lacks blood vessels, and it relies on anaerobic glycolysis to generate ATP.
- While both anaerobic and aerobic pathways are involved in corneal metabolism, aerobic metabolism is dominant. (Option C)

Solution for Question 22:

Correct Answer: C) Corneal Nerve Supply

Explanation:

- Corneal Nerve Supply is not directly involved in maintaining transparency.
- It provides sensation to the cornea but does not influence the clarity of the corneal tissue.

Factors Responsible for Corneal Transparency:

- Corneal Epithelium and Tear Film: Ensures a uniform refractive index. (Option A ruled out)
- Arrangement of Corneal Lamellae: Lattice structure (Maurice's theory) and specific refractive index variations (Goldmann and Benedek theory).
- Avascularity: Absence of blood vessels in the cornea. (Option B ruled out)
- Barrier Function: Limiting layers (epithelium and endothelium) maintain relative dehydration.
- Endothelial Pumps: Control fluid transport through Na⁺/K⁺-ATPase, bicarbonate-dependent ATPase, and Na⁺/H⁺ pumps, crucial for maintaining corneal dehydration. (Option D ruled out)
- Evaporation: From the corneal surface.
- Normal Intraocular Pressure (IOP): Helps maintain corneal shape.
- Swelling Pressure (SP): Counteracts the imbibition effect of IOP.
- Corneal Crystals/Proteins: Water-soluble proteins in keratocytes (e.g., transketolase, aldehyde dehydrogenase class 1A) contribute to transparency at the cellular level.

Solution for Question 23:

Correct Answer: B) 1-d, 2-b, 3-c, 4-a

Explanation:

INSTRUMENT	IMAGE
Keratometry An objective method for measuring the curvature of the central cornea to estimate corneal astigmatism. Principle of Keratometer: Measures corneal curvature by analyzing the size of the image formed by the anterior corneal surface, utilizing visible doubling. Normal Keratometric Values: Average: 45 D Keratoconus >45 D Types of Keratometers: Javal-Schiotz Model Bausch & Lomb Model	Refractive Keratometer" data-author="Mass Communication Specialist 2nd Class Jonathen E. Davis" data-hash="1996" data-license="Open access" data-source="https://commons.wikimedia.org/wiki/File:U S_Navy_101019-N-1531D-054_Air_Force_Master_Sgt._Breck_Downey_uses_a_auto-refractive_keratometer_to_determine_an_eye_prescription_during_a_Continui.jpg" data-tags="March2025" src="https://image.prepladder.com/notes/GbtpVnhiiknN7ulfqJZF1742990992.png" /> {{caption_text}} Auto refractive Keratometer

Keratotomy or placido's disc A disc with alternating black and white circles. Assesses the smoothness and curvature of the corneal surface. Normal Observation: Uniform, sharp image of circles on the cornea. Irregularities on cornea: Distort the circles	[Image not available]
Corneal topography Evaluates the anterior curvature of the cornea. Generates colour-coded maps of the corneal surface's shape, curvature, and elevation. Diagnoses conditions like keratoconus. Assists in planning refractive surgeries. Does not provide information about the posterior corneal surface.	Topography - Corneal topography - Wikipedia" data-tags="March2025" src="https://image.prepladder.com/notes/JWice7GDHn1Vummu47qT1742991030.png" /> {{caption_text}}
Corneal tomography (Pentacam) Measures both anterior and posterior corneal surfaces. Provides a three-dimensional view of corneal structure. Identifies subclinical keratoconus. Assesses candidates for keratorefractive surgery. Uses Scheimpflug imaging for high accuracy in corneal measurements. Valuable for preoperative evaluations and monitoring disease progression.	Tomography (Pentacam)" data-author="DEXi" data-hash="2001" data-license="CC0 1.0" data-source="Oculus Pentacam Corneal Eye Scan - Corneal topography - Wikipedia" data-tags="March2025" src="https://image.prepladder.com/notes/Uy91b12Jyol3po1eYEiD1742991053.png" /> {{caption_text}}

Solution for Question 24:

Correct Answer: D) Aesthesiometer measures corneal sensation by evaluating the second cranial nerve

Explanation:

- Aesthesiometer (Esthesiometer) measures corneal sensation by evaluating the ophthalmic branch of the fifth cranial nerve (trigeminal nerve), not the second cranial nerve (optic nerve).
- The fifth cranial nerve is responsible for sensory input from the cornea, which includes detecting tactile sensations.
- The second cranial nerve (optic nerve) is primarily involved in vision, not sensation.

INSTRUMENT

IMAGE

Pachymetry Measures corneal thickness. Used for diagnosing and treating glaucoma. (Option A ruled out) Assesses corneal thickness for refractive errors and LASIK surgery. Evaluate corneal diseases such as corneal hydrops, keratoconus, and Fuchs' dystrophy. Monitors eye injuries.	[Image not available] {{caption_text}}
Specular Microscopy: Noninvasive technique for photographing and assessing the corneal endothelium. (Option B ruled out) Evaluates preoperative endothelial health before high-risk surgeries. Assesses the impact of lasers during refractive surgery. Examines donor cornea before transplantation. Pathologies Assessed Include Fuchs endothelial dystrophy, corneal dystrophies, posterior polymorphous dystrophy, congenital hereditary endothelial dystrophy, uveitis, viral endothelitis, pseudophakic bullous keratopathy, ICE syndrome, trauma, pharmacological disruptions.	[Image not available] [Image not available] {{caption_text}} {{caption_text}}
Esthesiometer (Aesthesiometer): Measures corneal sensation (tactile sensation). Evaluates the ophthalmic branch of the fifth cranial nerve (trigeminal nerve). (Option D ruled out) Uses a cotton-tipped applicator for qualitative testing. The handheld esthesiometer (Cochet-Bonnet) features a retractable nylon monofilament, adjustable up to 6 cm, with pressure increasing from 11 mm/gm to 200 mm/gm as the length decreases. Diagnoses neurotrophic keratitis and assesses corneal health in long-term contact lens wearers. (Option C ruled out)	[Image not available] {{caption_text}}

Solution for Question 25:

Correct Answer: C) *Pseudomonas aeruginosa*

Explanation:

- The patient's clinical presentation, including the irregular corneal ulcer with greenish exudate and hypopyon, along with a history of contact lens use, suggests a bacterial corneal ulcer.

- *Pseudomonas aeruginosa* is known for causing irregular corneal ulcers with greenish mucopurulent exudate, rapid progression, and hypopyon.
- It is also common in contact lens-related infections.

Aetiology of Purulent Corneal Ulcer:

- **Corneal Epithelial Damage:** Due to Corneal abrasion, epithelial drying (xerosis, exposure keratitis), necrosis (keratomalacia), desquamation (bullous keratopathy), and trophic changes (neuroparalytic keratitis).
- **Due to Corneal abrasion, epithelial drying (xerosis, exposure keratitis), necrosis (keratomalacia), desquamation (bullous keratopathy), and trophic changes (neuroparalytic keratitis).**
- **Pathogens Invading Intact Epithelium:** *Neisseria gonorrhoeae*, *Corynebacterium diphtheriae*, *Neisseria meningitidis*, *Haemophilus aegyptius*, *Listeria* species.
- *Neisseria gonorrhoeae*, *Corynebacterium diphtheriae*, *Neisseria meningitidis*, *Haemophilus aegyptius*, *Listeria* species.
- **Sources of Infection: Exogenous Sources:** Conjunctival sac, lacrimal sac (dacryocystitis), infected foreign bodies, vegetative material, waterborne or airborne infections.
- **Exogenous Sources:** Conjunctival sac, lacrimal sac (dacryocystitis), infected foreign bodies, vegetative material, waterborne or airborne infections.
- **Common Bacteria Associated with Corneal Ulceration:** General: Staphylococci, *Pseudomonas*, *Streptococcus pneumoniae*, Enterobacteriaceae, *Neisseria*. In Contact Lens Wearers: *Pseudomonas* is the most common cause. (Option C). *Acanthamoeba* is another differential in contact lens misuse.
- General: Staphylococci, *Pseudomonas*, *Streptococcus pneumoniae*, Enterobacteriaceae, *Neisseria*.
- In Contact Lens Wearers: *Pseudomonas* is the most common cause. (Option C). *Acanthamoeba* is another differential in contact lens misuse.
- **All causative organisms:** Gram-positive Cocci: *Staphylococcus aureus*, *Staphylococcus epidermidis*, *Streptococcus pneumoniae*. Gram-negative Cocci: *Neisseria gonorrhoeae*, *Neisseria meningitidis*. Gram-positive Bacilli: *Corynebacterium diphtheriae*, *C. xerosis*, *Bacillus cereus*, *Propionibacterium acne*, *Listeria*, *Clostridium*. Gram-negative Bacilli: *Pseudomonas aeruginosa*, Enterobacteriaceae (*Klebsiella*, *Proteus*, *E. coli*, *Serratia*), *Moraxella lacunata*, *Haemophilus influenzae*. Gram-positive Filamentous Bacteria: *Actinomyces*, *Nocardia*. Mycobacteria: Non-tuberculous mycobacteria, *M. tuberculosis*.
- Gram-positive Cocci: *Staphylococcus aureus*, *Staphylococcus epidermidis*, *Streptococcus pneumoniae*.
- Gram-negative Cocci: *Neisseria gonorrhoeae*, *Neisseria meningitidis*.

- Gram-positive Bacilli: *Corynebacterium diphtheriae*, *C. xerosis*, *Bacillus cereus*, *Propionibacterium acne*, *Listeria*, *Clostridium*.
- Gram-negative Bacilli: *Pseudomonas aeruginosa*, Enterobacteriaceae (*Klebsiella*, *Proteus*, *E. coli*, *Serratia*), *Moraxella lacunata*, *Haemophilus influenzae*.
- Gram-positive Filamentous Bacteria: *Actinomyces*, *Nocardia*.
- Mycobacteria: Non-tuberculous mycobacteria, *M. tuberculosis*.
- Due to Corneal abrasion, epithelial drying (xerosis, exposure keratitis), necrosis (keratomalacia), desquamation (bullous keratopathy), and trophic changes (neuroparalytic keratitis).
- *Neisseria gonorrhoeae*, *Corynebacterium diphtheriae*, *Neisseria meningitidis*, *Haemophilus aegyptius*, *Listeria* species.
- Exogenous Sources: Conjunctival sac, lacrimal sac (dacryocystitis), infected foreign bodies, vegetative material, waterborne or airborne infections.
- General: Staphylococci, *Pseudomonas*, *Streptococcus pneumoniae*, Enterobacteriaceae, *Neisseria*.
- In Contact Lens Wearers: *Pseudomonas* is the most common cause. (Option C). *Acanthamoeba* is another differential in contact lens misuse.
- Gram-positive Cocci: *Staphylococcus aureus*, *Staphylococcus epidermidis*, *Streptococcus pneumoniae*.
- Gram-negative Cocci: *Neisseria gonorrhoeae*, *Neisseria meningitidis*.
- Gram-positive Bacilli: *Corynebacterium diphtheriae*, *C. xerosis*, *Bacillus cereus*, *Propionibacterium acne*, *Listeria*, *Clostridium*.
- Gram-negative Bacilli: *Pseudomonas aeruginosa*, Enterobacteriaceae (*Klebsiella*, *Proteus*, *E. coli*, *Serratia*), *Moraxella lacunata*, *Haemophilus influenzae*.
- Gram-positive Filamentous Bacteria: *Actinomyces*, *Nocardia*.
- Mycobacteria: Non-tuberculous mycobacteria, *M. tuberculosis*.

Clinical Features of Bacterial Corneal Ulcers:

Symptoms:

- Pain and foreign body sensation.
- Watery eye (reflex hyper lacrimation).
- Photophobia (light intolerance).
- Blurred vision.
- Redness of the eye.

Signs:

- Lid swelling.
- Blepharospasm (lid twitching).
- Conjunctival chemosis and hyperemia.
- Corneal ulcer characteristics: Early stage: Epithelial defect with greyish-white infiltrate. Established: Yellowish-white ulcer, swollen margins, necrotic floor, stromal oedema.
- Early stage: Epithelial defect with greyish-white infiltrate.
- Established: Yellowish-white ulcer, swollen margins, necrotic floor, stromal oedema.
- Specific bacterial features: Staphylococcus aureus and Streptococcus pneumoniae: Oval, yellowish-white ulcer surrounded by clear cornea. (Options A & B ruled out) Pseudomonas species: Irregular ulcer with greenish exudate, rapid spread, often with hypopyon. (Option C) Enterobacteriaceae: Shallow ulcer with greyish-white suppuration, ring-shaped infiltrate. (Option D ruled out)
- Staphylococcus aureus and Streptococcus pneumoniae: Oval, yellowish-white ulcer surrounded by clear cornea. (Options A & B ruled out)
- Pseudomonas species: Irregular ulcer with greenish exudate, rapid spread, often with hypopyon. (Option C)
- Enterobacteriaceae: Shallow ulcer with greyish-white suppuration, ring-shaped infiltrate. (Option D ruled out)
- The anterior chamber may contain pus (hypopyon).
- Muddy iris colour.
- Small pupil due to iritis.
- Possible raised intraocular pressure (inflammatory glaucoma).
- Early stage: Epithelial defect with greyish-white infiltrate.
- Established: Yellowish-white ulcer, swollen margins, necrotic floor, stromal oedema.
- Staphylococcus aureus and Streptococcus pneumoniae: Oval, yellowish-white ulcer surrounded by clear cornea. (Options A & B ruled out)
- Pseudomonas species: Irregular ulcer with greenish exudate, rapid spread, often with hypopyon. (Option C)
- Enterobacteriaceae: Shallow ulcer with greyish-white suppuration, ring-shaped infiltrate. (Option D ruled out)

Solution for Question 26:

Correct answer: B) Perforating Ulcer

Explanation:

- Chronic use of corticosteroid eye drops can lead to thinning and weakening of the corneal tissue, making it more susceptible to deep infections.
- When a corneal ulcer extends deep into the stroma, nearing Descemet's membrane, there is a high risk of perforation.
- A perforating ulcer is a severe form of bacterial keratitis where the ulceration penetrates deeply, potentially leading to full-thickness corneal perforation.
- The description of the ulcer penetrating deeply into the corneal stroma, approaching Descemet's membrane, suggests a Perforating Ulcer.

Types of Bacterial Keratitis

Localized Ulcer	Perforating Ulcer	Adherent Leucoma	Sloughing Ulcer
Begins with infiltration of inflammatory cells. Leads to necrosis and sloughing of corneal tissue. Healing occurs with epithelialization and scarring.	Deep penetration of ulcer reaching Descemet's membrane. Risk of perforation with any strain, leading to intraocular complications. Small perforations opposite the iris may heal rapidly, forming an adherent leucoma	Occurs when a small perforation is plugged by the iris. Results in scar formation with permanent opacity.	Rapidly spreads across the cornea due to virulent organisms or weak immunity. This leads to complete corneal destruction, iris prolapse, and formation of an anterior staphyloma with a pseudocornea.

Investigations for Bacterial Keratitis

- Corneal Scraping: Obtained from ulcer edges and base after local anaesthesia and using a Kimura spatula.
- Culture Media: Blood, chocolate, and thioglycollate broth for bacteria; non-nutrient agar for Acanthamoeba.
- Staining: Gram and Giemsa stains to identify pathogens and analyze cell morphology.

Localized Ulcers (Option A) do not typically penetrate as deeply and remain more superficial.

Adherent Leucoma (Option C) refers to a corneal opacity that adheres to the underlying iris, often resulting from the healing of a corneal ulcer with iris involvement.

- It is more of a sequela of a previous ulcer rather than an active form of bacterial keratitis.

Sloughing Ulcer (Option D) refers to the shedding of necrotic corneal tissue, often leading to a large area of ulceration.

- While severe, it generally describes the process of tissue loss rather than the specific risk of perforation that occurs with deeper stromal involvement, as seen in perforating ulcers.

Solution for Question 27:

Correct answer: C) Topical steroids should be started immediately to reduce inflammation.

Explanation:

Topical steroids should not be used in bacterial keratitis as they can exacerbate the infection and delay healing.

Treatment of Bacterial Keratitis

- Empirical Monotherapy: A single fluoroquinolone (e.g., ofloxacin) is recommended but not standard practice due to high resistance rates in some regions. Perforation is more likely in ulcers treated with ofloxacin.
- A single fluoroquinolone (e.g., ofloxacin) is recommended but not standard practice due to high resistance rates in some regions.
- Perforation is more likely in ulcers treated with ofloxacin.
- Duo Therapy (Option D ruled out): Use a combination of fortified cefazoline 5% and fortified tobramycin 1.3% to cover both gram-positive and gram-negative bacteria.
- Use a combination of fortified cefazoline 5% and fortified tobramycin 1.3% to cover both gram-positive and gram-negative bacteria.
- Ointments: Ocular ointments can be beneficial at bedtime for milder cases and may serve as useful adjunctive therapy.
- Ocular ointments can be beneficial at bedtime for milder cases and may serve as useful adjunctive therapy.
- Cycloplegics (Option A ruled out): Administer 1% atropine eye drops to reduce ciliary spasm pain, prevent posterior synechiae, increase blood supply, and decrease exudation. 2% homatropine is an alternative.
- Administer 1% atropine eye drops to reduce ciliary spasm pain, prevent posterior synechiae, increase blood supply, and decrease exudation.
- 2% homatropine is an alternative.
- Vitamin Supplements: Vitamins A, B-complex, and C are recommended to promote early healing of the corneal ulcer.

- Vitamins A, B-complex, and C are recommended to promote early healing of the corneal ulcer.
- Scleral involvement: (Option B ruled out): In Scleral or systemic involvement cases, systemic antibiotics like oral ciprofloxacin (750 mg twice daily) or a combination of cephalosporin and aminoglycoside may be necessary.
- In Scleral or systemic involvement cases, systemic antibiotics like oral ciprofloxacin (750 mg twice daily) or a combination of cephalosporin and aminoglycoside may be necessary.
- Avoid Sponge/Pad and steroids. Avoid using an eye sponge or pad to prevent infection from worsening by trapping bacteria. Steroids should be avoided as they may exacerbate the infection and delay the healing process.
- Avoid using an eye sponge or pad to prevent infection from worsening by trapping bacteria.
- Steroids should be avoided as they may exacerbate the infection and delay the healing process.
- A single fluoroquinolone (e.g., ofloxacin) is recommended but not standard practice due to high resistance rates in some regions.
- Perforation is more likely in ulcers treated with ofloxacin.
- Use a combination of fortified cefazoline 5% and fortified tobramycin 1.3% to cover both gram-positive and gram-negative bacteria.
- Ocular ointments can be beneficial at bedtime for milder cases and may serve as useful adjunctive therapy.
- Administer 1% atropine eye drops to reduce ciliary spasm pain, prevent posterior synechiae, increase blood supply, and decrease exudation.
- 2% homatropine is an alternative.
- Vitamins A, B-complex, and C are recommended to promote early healing of the corneal ulcer.
- In Scleral or systemic involvement cases, systemic antibiotics like oral ciprofloxacin (750 mg twice daily) or a combination of cephalosporin and aminoglycoside may be necessary.
- Avoid using an eye sponge or pad to prevent infection from worsening by trapping bacteria.
- Steroids should be avoided as they may exacerbate the infection and delay the healing process.

Solution for Question 28:

Correct answer: C) 0.5 to 0.6 mm

Explanation:

- 0.5 to 0.6 mm is the correct average thickness of the cornea at its central region, where it is thinnest.

CORNEA	
Diameter	11-11.7 mm
Refractive Index	1.376
Refractive Power	45 Dioptre
Radius of curvature	Anterior - 7.8mm (Option D ruled out) Posterior - 6.5mm
Thickness at the centre	0.5-0.6 mm (Option C)
Thickness at the periphery	1-1.2 mm (Option A ruled out)

0.8 to 0.9 mm (Option B): This thickness is greater than the actual average central thickness, suggesting a thicker cornea.

Solution for Question 29:

Correct answer: D) *Aspergillus fumigatus*

Explanation:

- *Aspergillus fumigatus* is a common causative organism of keratomycosis in tropical and subtropical regions and is often associated with exposure to decaying organic matter.
- Most Common Causative Organisms: *Aspergillus fumigatus*: Most common in India and tropical regions. *Fusarium* (Option B ruled out): Common in warm climates like Brazil. *Candida albicans* (Option A ruled out): More frequent in temperate climates.
- *Aspergillus fumigatus*: Most common in India and tropical regions.
- *Fusarium* (Option B ruled out): Common in warm climates like Brazil.
- *Candida albicans* (Option A ruled out): More frequent in temperate climates.
- Mode of Infection: Injury with Vegetative Material: Typically occurs after trauma with objects like thorns, wooden sticks, crop leaves, or decaying vegetable matter. Common in Field Workers: Especially during harvesting season in rural and tropical areas.

Immunocompromised Individuals: Higher risk for developing fungal keratitis.

- Injury with Vegetative Material: Typically occurs after trauma with objects like thorns, wooden sticks, crop leaves, or decaying vegetable matter.
- Common in Field Workers: Especially during harvesting season in rural and tropical areas.
- Immunocompromised Individuals: Higher risk for developing fungal keratitis.
- Special Considerations: Candida albicans: Leading cause of endogenous endophthalmitis. Mucormycosis (Option C ruled out) & Invasive Aspergillosis: Considered as the cause of orbital cellulitis in immunocompromised patients.
- Candida albicans: Leading cause of endogenous endophthalmitis.
- Mucormycosis (Option C ruled out) & Invasive Aspergillosis: Considered as the cause of orbital cellulitis in immunocompromised patients.
- Aspergillus fumigatus: Most common in India and tropical regions.
- Fusarium (Option B ruled out): Common in warm climates like Brazil.
- Candida albicans (Option A ruled out): More frequent in temperate climates.
- Injury with Vegetative Material: Typically occurs after trauma with objects like thorns, wooden sticks, crop leaves, or decaying vegetable matter.
- Common in Field Workers: Especially during harvesting season in rural and tropical areas.
- Immunocompromised Individuals: Higher risk for developing fungal keratitis.
- Candida albicans: Leading cause of endogenous endophthalmitis.
- Mucormycosis (Option C ruled out) & Invasive Aspergillosis: Considered as the cause of orbital cellulitis in immunocompromised patients.

Solution for Question 30:

Correct answer: B) Extensive corneal vascularization

Explanation:

- Extensive corneal vascularization is generally absent in pure fungal keratitis, which distinguishes it from bacterial infections where vascularization is more common.

Keratomycosis :

[Image not available]

- Pain & Foreign Body Sensation

- Watering of the eye
- Photophobia
- Blurred Vision
- Redness.
- Corneal Ulcer: Greyish-white, dry-looking with elevated, rolled-out margins.
- Pigmented Ulcer: Brownish appearance, possible with dematiaceous fungi.
- Feathery Extensions (Option A ruled out): Delicate, finger-like projections into the surrounding stroma.
- Sterile Immune Ring (Option C ruled out): Yellow line of demarcation where fungal antigen and host antibodies meet.
- Satellite Lesions: Multiple small lesions around the main ulcer.
- Hypopyon: Large, non-sterile, even with a small ulcer.
- Endothelial Plaque (Option D ruled out): Located under the stromal lesion, made of fibrin and leucocytes.
- Corneal Vascularization: Typically absent in pure fungal ulcers.
- Perforation: Rare but possible.
- Corneal Scraping: Performed under topical anaesthesia for microbiological examination.
- Wet KOH Mount: Reveals filamentous fungi (branched, septate hyphae) or non-filamentous yeast-like fungi (Candida).
- Gram's & Giemsa Stains: To detect fungal hyphae and inflammatory cells.
- Culture: Sabouraud's agar medium is used for fungal culture.
- PCR: Recommended for rapid identification of fungal pathogens.
- Anterior Chamber Paracentesis: Considered if there is an intraocular extension.
- Corneal Biopsy: Indicated for deep stromal abscess or when cultures are negative.
- Confocal Microscopy: Used for identifying actual fungi within the cornea.

Solution for Question 31:

Correct Answer: B) Topical Natamycin

Explanation:

- The history of trauma with plant material, along with the presence of a corneal ulcer with feathery edges and satellite lesions, is highly suggestive of fungal keratitis.
- Natamycin is the first-line treatment for fungal keratitis, especially caused by filamentous fungi like *Fusarium* and *Aspergillus*.
- It is effective in treating the type of fungal infection that is suggested by the clinical presentation in this patient.

Treatment of Keratomycosis (Fungal Corneal Ulcer):

Primary Treatment: Antifungal Medications:

- Topical Antifungal Agents: Natamycin Voriconazole Amphotericin B (effective against *Aspergillus* and *Fusarium*) Nystatin (effective against *Candida*)
- Natamycin
- Voriconazole
- Amphotericin B (effective against *Aspergillus* and *Fusarium*)
- Nystatin (effective against *Candida*)
- Oral Antifungal Agents (for severe ulcers with specific features): Ketoconazole Voriconazole Indications for Oral Therapy: Hypopyon Perforation Suspicion of endophthalmitis
- Ketoconazole
- Voriconazole
- Indications for Oral Therapy: Hypopyon Perforation Suspicion of endophthalmitis
- Hypopyon
- Perforation
- Suspicion of endophthalmitis
- Natamycin
- Voriconazole
- Amphotericin B (effective against *Aspergillus* and *Fusarium*)
- Nystatin (effective against *Candida*)
- Ketoconazole
- Voriconazole
- Indications for Oral Therapy: Hypopyon Perforation Suspicion of endophthalmitis
- Hypopyon
- Perforation
- Suspicion of endophthalmitis
- Hypopyon

- Perforation
- Suspicion of endophthalmitis

Surgical Management (if unresponsive to medical therapy):

- Repeated Scraping: Removal of necrotic material
- Cauterization Options: Pure carboic acid (100%) Trichloroacetic acid (10%–20%) Povidone-iodine 5%
- Pure carboic acid (100%)
- Trichloroacetic acid (10%–20%)
- Povidone-iodine 5%
- Therapeutic Keratoplasty: Corneal transplantation
- Pure carboic acid (100%)
- Trichloroacetic acid (10%–20%)
- Povidone-iodine 5%

Management of Suspected Endophthalmitis:

- Ultrasound: To check for exudates in the vitreous
- Vitreous Tap & intravitreal injections of antibiotics and antifungals (amphotericin B) if confirmed
- Surgical Options (if medical therapy fails): Corneal graft Conjunctival flap
- Corneal graft
- Conjunctival flap
- Corneal graft
- Conjunctival flap

Topical Moxifloxacin (Option A):

- Moxifloxacin is an antibiotic effective against bacterial infections, not fungal infections.
- Given the history of trauma with plant material and the appearance of a corneal ulcer with feathery edges and satellite lesions, the most likely diagnosis is fungal keratitis.
- Therefore, moxifloxacin would not be appropriate.

Oral Acyclovir (Option C):

- Acyclovir is an antiviral medication used primarily for herpes simplex virus infections.
- It would not be effective for a fungal corneal ulcer, making it an inappropriate choice in this scenario.

Intravitreal Amphotericin B (Option D):

- Intravitreal Amphotericin B is used in cases of fungal endophthalmitis, not as an initial treatment for keratitis.
- Although hypopyon suggests severe infection, the primary treatment should still be topical antifungal agents unless there is confirmed or high suspicion of endophthalmitis requiring intravitreal therapy.

Solution for Question 32:

Correct Answer: C) Herpes Simplex Virus (HSV)

Explanations:

- HSV keratitis is characterized by dendritic ulcers on the cornea, which are branching, tree-like lesions.
- HSV also commonly causes decreased corneal sensation.

Examination Findings in Viral Keratitis:

Signs:

- Swollen Epithelial Cells: Early signs presenting as swollen, opaque cells in a coarse punctate or stellate pattern.
- Dendritic Ulcer: Characteristic linear, branching ulcer with terminal bulbs; stains with fluorescein.
- Geographic Ulcer: Enlarged amoeboid-shaped ulcer, often due to inappropriate steroid use.
- Rose Bengal Staining: Stains virus-laden cells at the ulcer edges, aiding in differentiation from other conditions.
- Reduced Corneal Sensation: Result of corneal nerve damage.
- Subepithelial Haze: Mild haziness was observed in the subepithelial layer.
- Anterior Chamber Activity: Mild inflammatory reaction in the anterior chamber.
- Elevated IOP: Increased intraocular pressure, requiring monitoring.
- Persistent Punctate Erosions (PEE): Residual erosions and epithelial irregularities after healing, resolving spontaneously without indicating recurrence.

Specific Findings Based on Virus Type:

Herpes Simplex Virus (HSV) Keratitis:

- Dendritic Ulcers: Branching, tree-like ulcers.
- Geographic Ulcers: Larger, amoeboid-shaped ulcers.
- Stromal Keratitis: Inflammation of the corneal stroma, leading to oedema, infiltrates, and scarring.
- Disciform Keratitis: Central, disc-shaped inflammation in the stroma.

Herpes Zoster Ophthalmicus (HZO): (Option B)

- Vesicles and Scars: Painful rash and blisters following the involvement of ophthalmic division of the trigeminal nerve.
- Nummular Keratitis: Large, coin-shaped, subepithelial corneal infiltrates.
- Scleritis: Inflammation of the sclera, causing pain and redness.
- Iritis: Inflammation of the iris leading to pain, photophobia, and blurred vision.

Adenovirus (Option A):

- Adenoviruses are known for causing epidemic keratoconjunctivitis (EKC) and may present with conjunctivitis and keratitis.
- They are less likely to cause dendritic ulcers, which are specific to HSV.

Chlamydia trachomatis (Option D):

- Chlamydia typically causes trachoma or inclusion conjunctivitis, which may involve conjunctival scarring and corneal pannus, but not dendritic ulcers.

Solution for Question 33:

Correct Answer: C) Topical acyclovir and cycloplegics

Explanation:

- The patient presents with pain, redness, photophobia, and decreased vision in the right eye, along with a dendritic ulcer on the cornea.
- This clinical picture is classic for Herpes Simplex Keratitis (HSK), an infection of the cornea caused by the herpes simplex virus (HSV).

Treatment:

- Topical Acyclovir: This antiviral medication is the first-line treatment for HSV keratitis. It directly targets the herpes simplex virus, reducing viral replication and promoting healing of

the corneal ulcer.

- Cycloplegics: These medications help to relieve pain caused by ciliary muscle spasms and prevent the formation of synechiae (adhesions) in the iris.

Treatment of Herpes Simplex Virus (HSV) Keratitis:

Topical Therapy:

- Antivirals: Common Agents: Aciclovir 3% ointment: 5 times daily. Ganciclovir 0.15% gel: 5 times daily. Trifluridine: Up to 9 times daily. Older Agents: Idoxuridine and vidarabine (less common due to lower efficacy/higher toxicity).
- Common Agents: Aciclovir 3% ointment: 5 times daily. Ganciclovir 0.15% gel: 5 times daily. Trifluridine: Up to 9 times daily.
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- Ganciclovir 0.15% gel: 5 times daily.
- Trifluridine: Up to 9 times daily.
- Older Agents: Idoxuridine and vidarabine (less common due to lower efficacy/higher toxicity).
- Other Topical Medications: Antibiotics: Prevent secondary bacterial infections, especially with epithelial defects. Cycloplegics: (e.g., Homatropine 1%) Relieve pain and photophobia by paralyzing ciliary muscles and dilating the pupil. Steroids: Epithelial Keratitis: Contraindicated due to the risk of worsening infection and corneal perforation. Stromal Keratitis: Used cautiously with antivirals to control inflammation and reduce scarring.
- Antibiotics: Prevent secondary bacterial infections, especially with epithelial defects.
- Cycloplegics: (e.g., Homatropine 1%) Relieve pain and photophobia by paralyzing ciliary muscles and dilating the pupil.
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- Older Agents: Idoxuridine and vidarabine (less common due to lower efficacy/higher toxicity).
- Aciclovir 3% ointment: 5 times daily.
- Ganciclovir 0.15% gel: 5 times daily.
- Trifluridine: Up to 9 times daily.
- Antibiotics: Prevent secondary bacterial infections, especially with epithelial defects.
- Cycloplegics: (e.g., Homatropine 1%) Relieve pain and photophobia by paralyzing ciliary muscles and dilating the pupil.
- Steroids: Epithelial Keratitis: Contraindicated due to the risk of worsening infection and corneal perforation. Stromal Keratitis: Used cautiously with antivirals to control inflammation and reduce scarring.
- Epithelial Keratitis: Contraindicated due to the risk of worsening infection and corneal perforation.
- Stromal Keratitis: Used cautiously with antivirals to control inflammation and reduce scarring.
- Epithelial Keratitis: Contraindicated due to the risk of worsening infection and corneal perforation.
- Stromal Keratitis: Used cautiously with antivirals to control inflammation and reduce scarring.

Oral Therapy:

- Antivirals: Indications: immunocompromised patients, children, severe ocular surface disease, intolerance/resistance to topical treatment. Prophylaxis: Low-dose oral acyclovir (400 mg twice daily) reduces recurrence. Alternatives include valaciclovir (500 mg once daily) and famciclovir.
- Indications: immunocompromised patients, children, severe ocular surface disease, intolerance/resistance to topical treatment.
- Prophylaxis: Low-dose oral acyclovir (400 mg twice daily) reduces recurrence. Alternatives include valaciclovir (500 mg once daily) and famciclovir.
- Low-dose oral acyclovir (400 mg twice daily) reduces recurrence.
- Alternatives include valaciclovir (500 mg once daily) and famciclovir.
- Steroids: Generally avoided but may be used with caution in severe stromal inflammation as an adjunct to topical therapy.
- Indications: immunocompromised patients, children, severe ocular surface disease, intolerance/resistance to topical treatment.
- Prophylaxis: Low-dose oral acyclovir (400 mg twice daily) reduces recurrence. Alternatives include valaciclovir (500 mg once daily) and famciclovir.

- Low-dose oral acyclovir (400 mg twice daily) reduces recurrence.
- Alternatives include valaciclovir (500 mg once daily) and famciclovir.
- Low-dose oral acyclovir (400 mg twice daily) reduces recurrence.
- Alternatives include valaciclovir (500 mg once daily) and famciclovir.

Surgical Management:

- Debridement: Removal of infected epithelial cells, helpful in resistant epithelial keratitis.
- Keratoplasty: Indication: Severe corneal scarring after at least a year of infection-free status. Risk of Recurrence: HSV keratitis may recur post-keratoplasty. Eyes with extensive corneal vascularization are at higher risk of graft rejection and recurrence. Prophylactic oral acyclovir is recommended.
- Indication: Severe corneal scarring after at least a year of infection-free status.
- Risk of Recurrence: HSV keratitis may recur post-keratoplasty. Eyes with extensive corneal vascularization are at higher risk of graft rejection and recurrence. Prophylactic oral acyclovir is recommended.
- HSV keratitis may recur post-keratoplasty. Eyes with extensive corneal vascularization are at higher risk of graft rejection and recurrence.
- Prophylactic oral acyclovir is recommended.
- Indication: Severe corneal scarring after at least a year of infection-free status.
- Risk of Recurrence: HSV keratitis may recur post-keratoplasty. Eyes with extensive corneal vascularization are at higher risk of graft rejection and recurrence. Prophylactic oral acyclovir is recommended.
- HSV keratitis may recur post-keratoplasty. Eyes with extensive corneal vascularization are at higher risk of graft rejection and recurrence.
- Prophylactic oral acyclovir is recommended.
- HSV keratitis may recur post-keratoplasty. Eyes with extensive corneal vascularization are at higher risk of graft rejection and recurrence.
- Prophylactic oral acyclovir is recommended.

Topical steroids and antibiotics (Option A):

- Topical steroids are contraindicated in the treatment of active epithelial HSV keratitis, as they can suppress the immune response and worsen the viral infection, potentially leading to complications like corneal perforation.
- Antibiotics are not needed unless there is a secondary bacterial infection, which is not indicated here.

Oral acyclovir and lubricants (Option B):

- While oral acyclovir is used in more severe or recurrent cases of HSV keratitis, topical acyclovir is preferred as the initial treatment.
- Lubricants can be supportive but are not the primary treatment.

Oral steroids and topical antibiotics (Option D):

- Oral steroids are contraindicated in HSV keratitis, similar to topical steroids, as they can exacerbate the infection.
- Topical antibiotics are unnecessary unless there's evidence of bacterial superinfection.

Solution for Question 34:

Correct Answer: C) Neurotrophic Keratopathy

Explanation:

- Neurotrophic keratopathy is characterized by decreased corneal sensation and epithelial desquamation due to trigeminal nerve damage or dysfunction.
- The patient's history of Gasserian ganglion block, a procedure known to impair the trigeminal nerve, strongly supports this diagnosis.

Causes of Decreased Corneal Sensation:

- HSV keratitis typically presents with dendritic ulcers on the cornea and is associated with pain, photophobia, and possibly vesicular skin lesions.
- However, the history of a Gasserian ganglion block and the description of decreased corneal sensation point more toward neurotrophic keratopathy.
- Caused by varicella zoster virus.
- Leads to decreased corneal sensation.
- Ocular complications include: Punctate epithelial keratitis. Nummular keratitis. Stromal keratitis. Iridocyclitis. Scleritis.
- Punctate epithelial keratitis.
- Nummular keratitis.
- Stromal keratitis.
- Iridocyclitis.
- Scleritis.

- Punctate epithelial keratitis.
- Nummular keratitis.
- Stromal keratitis.
- Iridocyclitis.
- Scleritis.
- Result of vitamin A deficiency.
- The cornea becomes dull and insensitive.
- Signs include haziness and yellow infiltrates (not seen in this patient)
- Can progress to corneal necrosis.
- Caused by trigeminal nerve paralysis.
- Decreased corneal sensation and epithelial desquamation.
- May follow procedures like Gasserian ganglion removal or blockage, often used in treating trigeminal neuralgia.
- This condition occurs due to facial nerve paralysis, leading to lagophthalmos (inability to close the eye) and subsequent corneal exposure.
- The patient's history does not suggest facial nerve paralysis, making this less likely.

Solution for Question 35:

Correct Answer: B) Corneal Scarring

Explanations:

- This is a chronic complication that can develop after the acute phase of herpes zoster ophthalmicus.
- It results from damage to the cornea due to keratitis and can lead to vision impairment.

Ocular Manifestations of Herpes Zoster Ophthalmicus (HZO):

- Hutchinson's rule: The eye will be involved if the nose tip is involved. As it indicates nasociliary nerve involvement.
- Vesicles are present on the nose. Whenever there is involvement of the nose and skin lesions, this is called the Hutchinson's sign.

Acute Ocular Manifestations:

- Conjunctivitis: Redness and inflammation of the conjunctiva. (Option C ruled out)
- Keratitis: Punctate Epithelial Keratitis: Coarse and irregular lesions. Nummular Keratitis: Larger, disc-shaped lesions. Disciform keratitis.
- Punctate Epithelial Keratitis: Coarse and irregular lesions.
- Nummular Keratitis: Larger, disc-shaped lesions.
- Disciform keratitis.
- Iridocyclitis: Inflammation of the iris and ciliary body, causing pain, photophobia, and blurred vision.
- Keratouveitis and endothelitis: Leads to acute endothelial cell loss
- Episcleritis and Scleritis: Inflammation of the episclera or sclera, resulting in pain and redness. (Option A ruled out)
- Punctate Epithelial Keratitis: Coarse and irregular lesions.
- Nummular Keratitis: Larger, disc-shaped lesions.
- Disciform keratitis.

Chronic Ocular Manifestations:

- Post-herpetic neuralgia: When pain persists despite the resolution of the eruptive phase. In the long term, this leads to some anaesthesia in the affected skin, known as anaesthesia dolorosa.
- When pain persists despite the resolution of the eruptive phase.
- In the long term, this leads to some anaesthesia in the affected skin, known as anaesthesia dolorosa.
- Corneal lesions: Neuroparalytic ulceration. Neurotrophic Keratopathy Corneal scarring Exposure keratitis Mucous plaque keratitis (Sudden development of elevated mucous plaque which stains on rose bengal)
- Neuroparalytic ulceration.
- Neurotrophic Keratopathy
- Corneal scarring
- Exposure keratitis
- Mucous plaque keratitis (Sudden development of elevated mucous plaque which stains on rose bengal)
- When pain persists despite the resolution of the eruptive phase.
- In the long term, this leads to some anaesthesia in the affected skin, known as anaesthesia dolorosa.
- Neuroparalytic ulceration.
- Neurotrophic Keratopathy

- Corneal scarring
- Exposure keratitis
- Mucous plaque keratitis (Sudden development of elevated mucous plaque which stains on rose bengal)
- Lid lesions: D/t scarring. Include ptosis, trichiasis & entropion.
- Secondary Glaucoma: Increased intraocular pressure (IOP) leading to optic nerve damage and vision loss.

Recurrent Infections: Less common and usually milder than the initial episode.

Secondary Glaucoma (Option D):

- While secondary glaucoma is a chronic complication of herpes zoster ophthalmicus, it is more related to increased intraocular pressure due to inflammation or damage.
- Corneal scarring is a direct result of keratitis and is specifically related to the chronic effects on the cornea, making it a more direct chronic complication from the given options.

Solution for Question 36:

Correct Answer: B) Antiviral therapy is started within 72 hours of rash onset

Explanation:

- Antiviral therapy is most effective when started within 72 hours of rash onset, its effectiveness diminishes significantly after this period.
- However, treatment may still be considered beyond 72 hours, especially in severe cases, but the benefits are reduced.

Treatment of Herpes Zoster Ophthalmicus (HZO):

- Goals: Reduce viral replication, manage inflammation, and alleviate symptoms.

1. Antiviral Therapy:

- Early Intervention: Start within 72 hours of rash onset to reduce severity, duration, postherpetic neuralgia, and late ophthalmic complications.
- Drugs of Choice: Aciclovir: 800 mg five times daily for 7-10 days. Valaciclovir: 1 g three times daily for 7-10 days. Famciclovir: 500 mg three times daily for 7 days.
- Aciclovir: 800 mg five times daily for 7-10 days.

- Valaciclovir: 1 g three times daily for 7-10 days.
- Famciclovir: 500 mg three times daily for 7 days.
- Aciclovir: 800 mg five times daily for 7-10 days.
- Valaciclovir: 1 g three times daily for 7-10 days.
- Famciclovir: 500 mg three times daily for 7 days.

2. Systemic Steroids:

- Usage: Often used for moderate to severe HZO, especially with neurological complications; always paired with antivirals.
- Benefits: Reduces acute pain and expedites skin healing.
- Drawbacks: No effect on postherpetic neuralgia; avoid immunodeficiency.

3. Topical Treatments:

- Skin Lesions: Use drying agents, antiseptic solutions, and cold compresses.
- Eye Involvement: Specific treatments depend on the chronic phase and manifestation.

4. Immunocompromised Patients:

- Require specialized care.
- Extended antiviral treatment and possibly intravenous administration.
- Systemic steroids are generally not recommended.

Post-Herpetic Neuralgia:

- Definition: Pain persisting over a month after rash healing.
- Prevalence: Up to 75% in patients over 70. (Option C ruled out)
- Treatment: Stepwise approach with analgesics, tricyclic antidepressants, anticonvulsants like gabapentin, and potentially opioids.

Systemic steroids alone are sufficient to manage HZO (Option A):

- Systemic steroids should not be used as a monotherapy for HZO.
- They must be combined with antiviral therapy to manage inflammation and pain effectively, as steroids alone do not reduce viral replication or prevent complications.

Topical antiviral therapy is not recommended to manage eye involvement in HZO (Option D):

- Topical antiviral therapy in combination with systemic antiviral therapy is the standard of care for managing the viral infection.

Previous Year Questions

1. A patient presents with eyelid crusting and a thready sensation between the cornea and lids. What is the most likely diagnosis?

- A. Chronic Blepharitis
- B. Dry Eye Syndrome
- C. Meibomian Gland Dysfunction
- D. Conjunctivitis

2. The following shows?

[Image not available]

- A. Cobalt blue- Tryptan blue - Corneal ulcer
- B. Cobalt green- Rose bengal stain- Corneal ulcer
- C. Cobalt blue- Fluorescein stain- Corneal ulcer
- D. Cobalt blue- Lissamine green- Corneal ulcer

3. A 40-year-old male patient, who works in an iron factory, presents to the clinic with complaints of visual disturbances. He reports that his vision has been gradually worsening and he has noticed a change in the color perception. On examination, the ophthalmologist observes the presence of iron deposits in the eye. What is the most likely ocular condition responsible for the iron deposits in the eye and the patient's visual disturbances?

- A. Siderosis bulbi
- B. Hemochromatosis
- C. Retinitis pigmentosa

D. Age-related macular degeneration

4. A 45-year-old male patient presents to the ophthalmology clinic with a history of eye trauma with vegetative material. He complains of severe eye pain, redness, tearing, and photophobia. Microscopic examination of the corneal scraping reveals septate hyphae at an acute angle. What is the most likely causative agent responsible for this condition?

- A. Fungal corneal ulcer
 - B. Bacterial corneal ulcer
 - C. Viral corneal ulcer
 - D. Acanthamoeba corneal ulcer
-

5. Arlt's line is seen in which condition?

- A. Vernal keratoconjunctivitis
 - B. Phlyctenular conjunctivitis
 - C. Ophthalmia neonatorum
 - D. Trachoma
-

6. A patient presents with redness, extreme eye pain, and diminished vision. The patient frequently wears contact lenses. Which of the following organisms is most likely responsible for her condition?

- A. Staphylococcus aureus
- B. Streptococcus pyogenes
- C. Acanthamoeba
- D. Listeria

7. A corneal wisp test was performed, and the corneal reflex was elicited. Which of the following nerves is responsible for the afferent limb of this reflex?

- A. Facial Nerve
 - B. Trigeminal Nerve
 - C. Oculomotor Nerve
 - D. Abducens Nerve
-

8. A patient sustained blunt trauma to the eye 6 months ago and now presents with blurring of vision. What is the most likely condition?

[Image not available]

- A. Iridodialysis
 - B. Ankyloblepharon
 - C. Vossius Ring
 - D. Cycloiriditis
-

9. Which of the following layers is associated with the beaten bronze appearance?

- A. Endothelial Layer
 - B. Stroma
 - C. Sub-Epithelial Layer
 - D. Epithelial Layer
-

10. Shield ulcer is seen in which of the following ?

- A. Spring catarrh
 - B. Phlyctenular conjunctivitis
 - C. Atopic keratoconjunctivitis
 - D. Autoimmune Conditions
-

11. The following image shows:

[Image not available]

- A. Bacterial keratitis
 - B. Fungal keratitis
 - C. Viral keratitis
 - D. Syphilitic keratitis
-

12. Which drug is contraindicated in infectious keratitis?

- A. Tear drops
 - B. Steroids
 - C. Cycloplegics
 - D. Timolol
-

13. What is the origin of corneal endothelium?

- A. Neuroectoderm
- B. Neural crest cells
- C. Mesoderm

D. Surface ectoderm

14. Which of the following statements is incorrect regarding the corneal epithelium?

- A. Regeneration is limited to the limbus
 - B. Bowman's membrane regenerates
 - C. Lined by stratified squamous epithelium
 - D. Apical cells have microvilli
-

15. What is the proper sequence for administering eye drops and eye ointment?

- A. Eye drops first, ointment 15 min later
 - B. Ointment first, eye drops 15 min later
 - C. Eye drops first, then ointment immediately
 - D. Ointment first, then eye drops immediately.
-

16. A 64-year-old woman with a history of trauma to the left eye 5 days back now presented with complaints of complete loss of vision. Upon examination the following findings were seen. What should be the subsequent course of action to manage this condition?

[Image not available]

- A. Keratoplasty
 - B. Keratotomy
 - C. LASIK
 - D. DCR
-

17. Which glycosaminoglycan is responsible for the transparency of the cornea?

- A. Keratan sulfate
 - B. Hyaluronic acid
 - C. Heparan sulfate
 - D. Dermatan sulfate
-

18. Assertion: Topical steroids should not be used in the treatment of viral corneal ulcers. Reason: Topical steroids can cause increased intraocular pressure What is the correct option?

- A. Both assertion and reasoning are correct and reason is the correct explanation of the assertion
 - B. Both assertion and reasoning are independently correct and reason is not correct explanation of the assertion
 - C. Assertion is independently a correct statement, but the reason is false
 - D. Both assertion and reason are false
-

19. Which of the following statements about the instrument shown in the image given below is false?

[Image not available]

- A. Used to measure the curvature of the cornea.
 - B. Assess corneal astigmatism
 - C. Depends on purkinje image I
 - D. Measure intraocular pressure
-

20. What's the main risk for a 37-year-old man with blurred vision and severe pain in his right eye. He revealed that recently, he went swimming in a pond. During examination, a ring-like lesion was found, as seen in the image.

[Image not available]

- A. Diabetes
 - B. Hypertension
 - C. Steroids
 - D. Soft Contact lenses
-

21. What is the diagnosis for a female patient, aged 65, who has a history of chickenpox and presents at the clinic with the following findings?

[Image not available]

- A. Chicken pox exacerbation
 - B. Drug allergy to acyclovir
 - C. Herpes zoster ophthalmicus
 - D. Herpes zoster oticus
-

22. What is the likely reason for a 21-year-old student to have ocular redness, feeling of a foreign object in the eye, and excessive tearing after wearing a soft contact lens, as indicated by the corneal scraping interface contrast microscopy image?

[Image not available]

- A. Acanthamoeba
- B. Naegleria fowleri
- C. Entamoeba histolytica

D. Balamuthia mandrillaris

23. Keratometer is used to assess

- A. Curvature of cornea
 - B. Diameter of cornea
 - C. Thickness of cornea
 - D. Curvature of lens
-

24. Please analyze the image provided and determine the specific abnormality or injury present.

[Image not available]

- A. Viral keratitis
 - B. Fungal keratitis
 - C. Bacterial keratitis
 - D. Interstitial keratitis
-

25. What is the probable diagnosis for a 27-year-old male patient who experiences eye trauma from vegetative matter, and currently exhibits symptoms including a corneal ulcer, hypopyon, satellite lesions, mild pain, photophobia, and redness?

- A. Fungal keratitis
 - B. Acanthamoeba keratitis
 - C. Viral keratitis
 - D. Bacterial keratitis
-

26. A patient presents to you with pain and redness in the right eye following an injury during welding. On examination, the image as seen below is observed. Which of the following is the appropriate next step in managing this patient?

[Image not available]

- A. Steroids and antibiotics
 - B. Surgical exploration
 - C. Keratoplasty
 - D. Removal by 26G needle
-

27. The eye examination of a 20-year-old male shows a golden ring on the cornea. What of the following is the investigation of choice to diagnose the condition?

- A. Alpha-1-antitrypsin
 - B. Alpha-fetoprotein
 - C. Serum iron
 - D. Serum ceruloplasmin
-

28. What dye and filter are used in the image, as shown below?

[Image not available]

- A. Lissamine dye green filter
 - B. Lissamine dye cobalt blue filter
 - C. Fluorescein dye green filter
 - D. Fluorescein dye cobalt blue filter
-

29. Which of the following is not characteristic of pterygium?

- A. Rate of recurrence is 95%
 - B. It is a fibrovascular growth
 - C. The parts of a pterygium are the head, neck and body
 - D. Has progressive and regressive types
-

30. Which of the subsequent options does not exhibit a characteristic of keratoconus?

- A. Haab's striae
 - B. Munson's sign
 - C. Irregular astigmatism
 - D. Scissoring reflex
-

31. A farmer presents with pain and discharge in his left eye. He says that he sustained an injury caused by a leaf a few days back. What is the finding seen in the image?

[Image not available]

- A. Hyphema
 - B. Aqueous flare
 - C. Arlt line
 - D. Hypopyon
-

32. The person in the image is at risk of developing?

[Image not available]

- A. Exposure keratopathy
 - B. Restricted eye movements
 - C. Diplopia
 - D. Amblyopia
-

33. What is the diagnosis?

[Image not available]

- A. Anterior uveitis
 - B. Glaucoma
 - C. Optic neuritis
 - D. Keratoconus
-

34. What does this image indicate?

[Image not available]

- A. Keratoconus
 - B. Vogt's limbal girdle
 - C. Keratoglobus
 - D. Corneal dystrophy
-

35. Fleischer's ring is seen in

- A. Keratoconus
- B. Keratomalacia
- C. Keratoglobus

D. Trachoma

36. A 20-year-old male presents with recurrent fluctuations in his vision prescription. What is the probable diagnosis?

- A. Keratoconus
 - B. Pathological myopia
 - C. Glaucoma
 - D. Cataract
-

37. Which layer is accountable for preserving the moisture and clarity of the cornea?

- A. Descemet's membrane
 - B. Endothelium
 - C. Stroma
 - D. Corneal epithelium
-

38. What is the distinguishing characteristic of a fungal ulcer?

- A. Hypopyon
 - B. Dendritic ulcer on a fluorescein dye
 - C. Ring abscess
 - D. Satellite lesion
-

39. For which of the subsequent diseases is collagen cross-linking therapy utilized?

- A. Keratoconus
 - B. Corneal ulcer
 - C. Band keratopathy
 - D. Uveitis
-

40. Arlt's line is seen in?

- A. Vernal keratoconjunctivitis
 - B. Phlyctenular keratoconjunctivitis
 - C. Trachoma
 - D. Pterygium
-

41. In the image provided, which refractive error is observed in the depicted condition?

[Image not available]

- A. Myopia and astigmatism
 - B. Hypermetropia and astigmatism
 - C. Myopia and hypermetropia
 - D. Hypermetropia
-

42. What is the adjuvant therapy in fungal corneal ulcer?

- A. Pilocarpine eye drops
- B. Lignocaine eye drops

- C. Dexamethasone eye drops
 - D. Atropine Sulfate eye ointment
-

43. A 30 year old man presents with a history of injury to the eye caused by a leaf 5 days ago, resulting in pain, photophobia, and redness of the eye for 2 days. Upon examination, corneal ulcer with feathery ends and satellite lesions around it are seen. What would be the most likely diagnosis?

- A. Anterior uveitis
 - B. Conjunctivitis
 - C. Fungal corneal ulcer
 - D. HSV corneal ulcer
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	1
Question 4	1
Question 5	4
Question 6	3
Question 7	2
Question 8	1
Question 9	1
Question 10	1
Question 11	3
Question 12	2
Question 13	2
Question 14	2
Question 15	1
Question 16	1
Question 17	1
Question 18	2
Question 19	4
Question 20	4
Question 21	3
Question 22	1

Question 23	1
Question 24	1
Question 25	1
Question 26	4
Question 27	4
Question 28	4
Question 29	1
Question 30	1
Question 31	4
Question 32	1
Question 33	4
Question 34	1
Question 35	1
Question 36	1
Question 37	2
Question 38	4
Question 39	1
Question 40	3
Question 41	1
Question 42	4
Question 43	3

Solution for Question 1:

Chronic Blepharitis is an inflammation of the eyelid margins that can cause symptoms such as eyelid crusting and a thready sensation between the cornea and eyelids, often due to the

accumulation of secretions along the lash line. The eyelids may feel gritty or have a foreign body sensation.

Bacterial blepharitis:

Table rendering failed

Dry Eye Syndrome (Option B): While dry eyes can cause a foreign body sensation, it typically does not cause eyelid crusting. Dry eye symptoms are more related to discomfort from insufficient tear production, but the crusting is not a hallmark feature.

Meibomian Gland Dysfunction (Option C): This condition can lead to dry eyes and eyelid discomfort due to insufficient oil production, but it doesn't usually cause eyelid crusting in the way that blepharitis does.

Conjunctivitis (Option D): Conjunctivitis (inflammation of the conjunctiva) causes symptoms such as redness, discharge, and itching

Solution for Question 2:

Correct Option C - Cobalt blue - Fluorescein stain- corneal ulcer:

- Epithelial defects are stained by Fluorescein stain, Fluorescein stain can be best seen under blue light under a slit lamp.

Incorrect Options:

Option A - Cobalt blue - Trypan blue- corneal ulcer: This combination is incorrect.

Option B - Cobalt green- rose bengal stain- corneal ulcer: This option is incorrect as these dyes do not stain dry tissue.

Option D - Cobalt blue- Lissamine green- corneal ulcer: This option is incorrect as the stain used is Fluorescein stain.

Solution for Question 3:

Correct Option A - Siderosis bulbi

- This is the correct answer. Siderosis bulbi refers to the condition in which iron particles, usually from a previous trauma, are deposited in the eye's intraocular tissues leading to visual disturbances and discoloration. In this case, the patient's line of work in an iron factory could have exposed him to iron particles.

Incorrect Options:

Option B - Hemochromatosis: This is a genetic disorder that leads to excessive iron absorption from the diet. It affects various organs in the body, including the liver, heart, and pancreas. While it can cause a number of health problems, it is not typically associated with iron deposits in the eye or the visual disturbances described by the patient.

Option C - Retinitis pigmentosa: This is a group of rare, genetic disorders that involve a breakdown and loss of cells in the retina—which is the light sensitive tissue that lines the back of the eye. It is not typically associated with iron deposits in the eye.

Option D - Age-related macular degeneration: This is a common eye condition among people over 50. It causes blurred or reduced central vision, due to thinning of the macula, the part of the retina responsible for clear vision in your direct line of sight. It is not associated with iron deposits in the eye.

Solution for Question 4:

Correct Option A - Fungal Corneal Ulcer

- Fungal corneal ulcers are typically caused by filamentous fungi, and they present with symptoms such as severe eye pain, redness, tearing, and photophobia. Microscopic examination of corneal scrapings often reveals septate hyphae at an acute angle. This presentation is consistent with the information provided in the question.

Incorrect Options:

Option B - Bacterial Corneal Ulcer: Bacterial corneal ulcers can also cause similar symptoms, including eye pain, redness, and tearing. However, bacterial ulcers are usually associated with a purulent discharge and the appearance of neutrophils on microscopic examination of corneal scrapings.

Option C - Viral Corneal Ulcer: Viral corneal ulcers, such as those caused by herpes simplex virus (HSV), typically present with a dendritic ulcer pattern on the cornea. They may cause symptoms like eye pain and photophobia, but they usually lack the septate hyphae seen in fungal infections.

Option D - Acanthamoeba Corneal Ulcer: Acanthamoeba corneal ulcers are caused by a protozoan parasite. They often occur in contact lens wearers or individuals exposed to contaminated water. These ulcers can cause eye pain, redness, and photophobia, but they are not characterized by septate hyphae

Solution for Question 5:

Correct Option D - Trachoma:

- Arlt's line is a finding seen in trachoma, a chronic infectious disease caused by Chlamydia trachomatis. It is a horizontal white line of inflammatory exudate at the upper tarsal conjunctiva, associated with scarring in the advanced stages of trachoma.

Incorrect Options:

Option A - Vernal keratoconjunctivitis: Arlt's line is not typically associated with vernal keratoconjunctivitis, which is characterized by papillary hypertrophy and a gelatinous limbal infiltrate.

Option B - Phlyctenular conjunctivitis: Arlt's line is not commonly seen in phlyctenular conjunctivitis. This condition is characterized by small nodules on the conjunctiva.

Option C - Ophthalmia neonatorum: Arlt's line is not typically associated with ophthalmia neonatorum, which refers to conjunctivitis occurring in the first month of life.

Solution for Question 6:

Correct Option C – Acanthamoeba:

- Acanthamoeba is most likely responsible for this patient's condition. Acanthamoeba keratitis is associated with improper use of contact lenses, and the patients exhibit extreme pain due to perineural invasion.

Incorrect Options:

Options A, B, and D are not responsible for causing this patient's condition, as explained above.

Solution for Question 7:

The corneal reflex (blink reflex) is a protective response that involves both sensory (afferent) and motor (efferent) pathways:

- Afferent limb (sensory input): Trigeminal nerve (CN V1 - Ophthalmic division) The cornea is innervated by the nasociliary branch of the ophthalmic division (V1) of the trigeminal nerve. A light touch to the cornea (e.g., with a cotton wisp) stimulates sensory fibers, sending signals to the trigeminal sensory nucleus in the brainstem.
- The cornea is innervated by the nasociliary branch of the ophthalmic division (V1) of the trigeminal nerve.
- A light touch to the cornea (e.g., with a cotton wisp) stimulates sensory fibers, sending signals to the trigeminal sensory nucleus in the brainstem.
- Efferent limb (motor response): Facial nerve (CN VII) The facial nerve (CN VII) innervates the orbicularis oculi muscle, which causes eyelid closure in response to the stimulus.
- The facial nerve (CN VII) innervates the orbicularis oculi muscle, which causes eyelid closure in response to the stimulus.
- The cornea is innervated by the nasociliary branch of the ophthalmic division (V1) of the trigeminal nerve.
- A light touch to the cornea (e.g., with a cotton wisp) stimulates sensory fibers, sending signals to the trigeminal sensory nucleus in the brainstem.
- The facial nerve (CN VII) innervates the orbicularis oculi muscle, which causes eyelid closure in response to the stimulus.

Reflex	Afferent Nerve	Efferent Nerve
Accommodation	II (Optic nerve)	III (Oculomotor nerve)

Corneal	V■ (Ophthalmic, nasociliary branch)	Bilateral VII (Facial nerve, temporal branch - orbicularis oculi)
Cough	X (Vagus nerve)	X (Vagus nerve, also phrenic and spinal nerves)
Gag	IX (Glossopharyngeal nerve)	X (Vagus nerve)
Jaw Jerk	V■ (Mandibular branch, sensory from masseter)	V■ (Mandibular branch, motor to masseter)
Lacrimation	V■ (Ophthalmic branch) (loss of reflex does not preclude emotional tears)	VII (Facial nerve)
Pupillary	II (Optic nerve)	III (Oculomotor nerve)

Facial Nerve (CN VII) (Option A): The facial nerve is responsible for the efferent limb (motor response), not the afferent limb. It causes eyelid closure by contracting the orbicularis oculi muscle.

Oculomotor Nerve (CN III) (Option C): CN III controls extraocular muscles and the levator palpebrae superioris (eyelid elevation), but it does not mediate the corneal reflex.

Abducens Nerve (CN VI) (Option D): CN VI controls the lateral rectus muscle for eye abduction but is not involved in the corneal reflex.

Solution for Question 8:

Iridodialysis is detachment of iris from its root at the ciliary body. It results in D shaped pupil and a lack biconvex area seen at the periphery.

[Image not available]

Blunt trauma (also known as non-penetrating or blunt force trauma)

- Cause: A strong impact, a fall, or a physical assault with a blunt object, is most commonly the result of sports injuries and assaults.
- Features: Conjunctiva: Subconjunctival hemorrhage
- Conjunctiva: Subconjunctival hemorrhage
- Conjunctiva: Subconjunctival hemorrhage

- Cornea: Abrasions, corneal edema, Descemet's membrane tears. Hyphaema: Blood in the anterior chamber leading to corneal staining. Iris: Iridodialysis (dehiscence of the iris from the ciliary body at its root), iris sphincter tear, Ciliary body: Ciliary body tear leading to angle recession. Lens: Vossious Ring (in the shape of the pupil) (Option C)
- Cornea: Abrasions, corneal edema, Descemet's membrane tears.
- Hyphaema: Blood in the anterior chamber leading to corneal staining.
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- Vossious Ring (in the shape of the pupil) (Option C)

[Image not available]

- Vossious Ring (in the shape of the pupil) (Option C)

[Image not available]

- Rosette-Shaped Cataract (flower-shaped opacity), subluxation of lens.
- Rosette-Shaped Cataract (flower-shaped opacity), subluxation of lens.
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- Rosette-Shaped Cataract (flower-shaped opacity), subluxation of lens.
- Rosette-Shaped Cataract (flower-shaped opacity), subluxation of lens.

[Image not available]

- Globe rupture (caused by severe blunt trauma, indicated by iris prolapse) Vitreous haemorrhage: Pigment cells ('tobacco dust') can be seen floating in the anterior vitreous. Choroidal rupture Retina: Commotio Retinae (also known as Berlin's edema), retinal dialysis (disinsertion from ora serrata), giant retinal Tear (horseshoe shaped), macular hole.
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- Choroidal rupture
- Retina: Commotio Retinae (also known as Berlin's edema), retinal dialysis (disinsertion from ora serrata), giant retinal Tear (horseshoe shaped), macular hole.

[Image not available]

[Image not available]

[Image not available]

- Injury to optic nerve Traumatic neuropathy with chorioretinal scarring (decreased visual acuity, visual field defect, afferent pupillary defect, color, and brightness).
- Injury to optic nerve Traumatic neuropathy with chorioretinal scarring (decreased visual acuity, visual field defect, afferent pupillary defect, color, and brightness).
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- Injury to optic nerve
- Traumatic neuropathy with chorioretinal scarring (decreased visual acuity, visual field defect, afferent pupillary defect, color, and brightness).

Ankyloblepharon (Option B): This refers to the abnormal fusion of the eyelids, typically a congenital issue or sometimes a result of trauma leading to scarring of the eyelids.

[Image not available]

Cycloiriditis (Option D): This is inflammation of the ciliary body and iris. While trauma can trigger this, it usually presents with pain, redness, and photophobia (sensitivity to light) in addition to

blurred vision.

Solution for Question 9:

Correct Option A - Endothelial Layer:

- The beaten bronze appearance of the cornea is a characteristic finding associated with the endothelial layer. This appearance is seen in conditions like Fuchs' Endothelial Dystrophy

Incorrect Options:

- Options B, C, and D do not contribute to the beaten bronze appearance.
-

Solution for Question 10:

Correct Option A - Spring catarrh:

Spring catarrh/ Vernal keratoconjunctivitis

- It is usually seen in summer season
- It is type 1 hypersensitivity reaction caused by exogenous allergens, such as pollen, dust etc
- Common in male children
- Clinical features: itching, watering, and ropy discharge containing mucin, keratoconus
- Forms: palpebral and limbal
- In later stages, it can progress to the development of a shield ulcer, characterized by the deposition of mucus and calcium phosphates in micro-erosions.

[Image not available]

- Treatment: antihistamines, mast cell stabilizer, topical steroids

Incorrect Options:

Option B - Phlyctenular conjunctivitis:

- Allergy caused by endogenous antigens → Staph. aureus and TB infection
- It presents with watering and itching in the eyes

Option C - Atopic keratoconjunctivitis:

- Common in temperate regions and winters
- Dennie morgan folds present
- Presence of Madarosis and Hertoghe's sign

Option D. Autoimmune Conditions:

- Not typically associated with Shield Ulcer: Various autoimmune diseases can affect the eyes, such as rheumatoid arthritis, systemic lupus erythematosus, and Sjögren's syndrome, leading to conditions like dry eye, scleritis, and uveitis. However, shield ulcers are not commonly seen as a direct manifestation of these autoimmune conditions.

Solution for Question 11:

Correct Option C - Viral keratitis:

- Viral keratitis is a condition characterized by inflammation of the cornea caused by a viral infection. One of the classic manifestations of viral keratitis is the presence of a dendritic ulcer, which is a branching pattern of corneal epithelial lesions resembling a tree branch or dendrite. These ulcers are typically caused by the herpes simplex virus (HSV) and are commonly seen in cases of herpetic keratitis.

Incorrect Options:

Option A - Bacterial keratitis: It typically presents with a more localized and suppurative (pus-forming) corneal ulcer along with a sterile hypopyon in the anterior chamber.

Option B - Fungal keratitis: It is characterized by dry/rough white ulcer with feathery ends, satellite nodules around it and a non sterile hypopyon.

Option D - Syphilitic keratitis: It typically presents as an interstitial keratitis and absent light reflex with positive accommodation reflex in neurosyphilis.

Solution for Question 12:

Correct Option B - Steroids:

- Steroids are contraindicated in the treatment of infectious keratitis. Infectious keratitis is usually caused by bacteria, viruses, fungi, or parasites, and it is essential to avoid using steroids as they can exacerbate the infection and potentially lead to severe complications.
- Steroids have immunosuppressive and anti-inflammatory properties, which can suppress the body's immune response to the infection. This suppression can hinder the ability to fight off the invading microorganisms, leading to prolonged infection, increased tissue damage, and potentially worsening vision loss.

Incorrect Options:

Option A - Teardrops: Tear drops, or artificial tears, are often used to provide lubrication and relieve dryness or discomfort in the eyes due to non-healing ulcer

Option C - Cycloplegics: Cycloplegic eye drops are used to temporarily paralyze the muscles of the eye, primarily to dilate the pupil and relieve pain associated with spasm, prevents posterior synechiae formation, and increase blood supply.

Option D - Timolol: Timolol is a medication commonly used to lower intraocular pressure in conditions such as glaucoma. It is used in keratitis with impending perforation.

Solution for Question 13:

Correct Option B - Neural crest cells:

- The corneal endothelium is derived from neural crest cells during embryonic development. It plays a crucial role in maintaining corneal transparency by regulating the fluid balance in the cornea.

Incorrect Options:

Option A - Neuroectoderm: Secondary vitreous, tertiary vitreous, optic nerve, retina, smooth muscles of the iris (iris sphincter and dilator pupillae), and epithelium lining of iris and ciliary body are derived from neuroectoderm.

Option C - Mesoderm: Primary vitreous, temporal part of sclera, extra ocular parts, and endothelial lining of the blood vessels are derived from mesoderm.

Option D - Surface ectoderm: The skin of eyelids and appendages, lens, lacrimal gland, epithelium of the cornea, and epithelium of conjunctiva are derived from surface ectoderm.

Solution for Question 14:

Correct Option B - Bowman's membrane regenerates:

- The Bowman layer is the acellular superficial layer of the stroma and is formed from collagen fibers. Bowman's membrane, also known as the anterior limiting lamina, is a non-regenerating layer. It is located between the corneal epithelium and the stroma.

Incorrect Options:

Option A - Regeneration is limited to the limbus: The corneal epithelium has a remarkable regenerative capacity, and its regeneration primarily occurs at the limbus due to presence of corneal stem cells.

Option C - Lined by stratified squamous epithelium: The corneal epithelium is lined by stratified squamous non-keratinized epithelium.

Option D - Apical cells have microvilli: The apical cells of the corneal epithelium, which are the outermost layer of cells, often possess microvilli and microplacae that facilitate the attachment of the tear film and mucin.

Solution for Question 15:

Correct Option A - Eye drops first, ointment 15 minutes later:

- When administering both eye drops and eye ointment, it is generally recommended to follow a specific order to ensure optimal effectiveness and minimize potential interactions.

- Eye drops are usually prescribed to be administered before applying eye ointment. By instilling the eye drops first, they can penetrate the eye and exert their therapeutic effect. Waiting for about 15 minutes before applying the eye ointment allows for adequate absorption and action of the eye drops.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 16:

Correct Option A - Keratoplasty:

- The history of trauma to the eye and complete loss of vision and examination finding revealing congestion and corneal perforation is suggestive of uncontrolled keratitis commonly seen with bacterial keratitis.
- The choice of treatment for corneal perforation with uncontrolled infection penetrating keratoplasty is preferred. If small corneal perforation with controlled infection bandage contact lenses are used.

Incorrect Options:

Option B - Keratotomy: Keratotomy is used for correcting refractive error causing flattening of the cornea and reducing its focusing power.

Option C - LASIK: LASIK surgery is commonly used for keratomileusis, which is altering of the curvature of the cornea, done for refractive correction. .

Option D - DCR: Dacryocystorhinostomy (DCR) is a surgical procedure done for dacryocystitis.

Solution for Question 17:

Correct Option A - Keratan sulfate:

- Keratan sulfate and chondroitin sulfate are the major glycosaminoglycan present in the corneal stroma. They exist in the proteoglycan form and are responsible for maintaining the space between the regularly oriented layers of collagen fibrils in the stroma. Maintenance of the regular arrangement and spacing of the collagen is critical to optical clarity.

Incorrect Options:

Options B, C, and D are incorrect. These glycosaminoglycans are not present in the eye.

Solution for Question 18:

Correct Option B - Both assertion and reasoning are independently correct and reason is not correct explanation of the assertion:

- Both assertion and reasoning are independently correct, but the reason is not the correct explanation of the assertion: The assertion that topical steroids are contraindicated in viral corneal ulcers is true. This is because topical steroids in such cases can hinder the body's immune response to the infection and results in perforated corneal ulcer..
- The reasoning stating that topical steroids can cause increased intraocular pressure is also true but does not directly explain the contraindication in viral corneal ulcers.

Incorrect Option:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 19:

Correct Option D - Measure intraocular pressure:

- The instrument shown in the image is a Keratometer used to assess the corneal curvature responsible for causing astigmatism. It is based on the size of the image formed by the anterior surface of the cornea i.e., purkinje image I.
- Intraocular pressure is measured by using a tonometer.

Incorrect Options:

Options A, B, and C are incorrect. refer to Option D for an explanation.

Solution for Question 20:

Correct Option: D

- The clinical presentation of blurred vision, severe pain after swimming in a pond with ring like lesion on examination is suggestive of Acanthamoeba keratitis.
- The most important risk factor for the developing Acanthamoeba keratitis is usage of Soft contact lenses. The users who wash soft contact lenses with tap water or swimming with contact lens carries the risk of contamination of the lens with the acanthamoeba cyst which later turn into trophozoites for tissue penetration and destruction characterized by pain (severe pain can be associated due to perineural invasion).

Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 21:

Correct Option C - Herpes zoster ophthalmicus:

- The clinical presentation, age, and a history of chickenpox is suggestive of Herpes zoster ophthalmicus.
- It is a localized reactivation of the varicella-zoster virus seen in elderly and immunocompromised patient involving the ophthalmic division of the trigeminal nerve. It typically presents with vesicular rash, erythema, and swelling in the distribution of the affected nerve, as seen in the image.

Incorrect Options:

Option A - Chicken pox exacerbation: Chickenpox, caused by the varicella-zoster virus (VZV), typically presents with a generalized vesicular rash.

Option B - Drug allergy to acyclovir: Acyclovir is an antiviral medication commonly used to treat herpes zoster (shingles) and other viral infections caused by herpesviruses.

Option D - Herpes zoster oticus: Herpes zoster oticus, also known as Ramsay Hunt syndrome, is a specific form of herpes zoster that involves the facial nerve. It presents with vesicles and rash in the ear canal, earlobe, and sometimes on the face.

Solution for Question 22:

Correct Option A - Acanthamoeba:

- The clinical presentation of ocular redness, foreign body sensation, epiphora, history of using a soft contact lens, and image showing a cyst of acanthamoeba characterized by outer wrinkled wall is suggestive of acanthamoeba keratitis caused by acanthamoeba a free living protozoan.
- It is prevalent in individuals using soft contact lens and using tap water for cleaning or swimming with the contact lens in a swimming pool.

Incorrect Options:

Option B - Naegleria fowleri: Naegleria fowleri is a free-living amoeba that can cause a rare and often fatal brain infection known as primary amoebic meningoencephalitis (PAM).

Option C - Entamoeba histolytica: Entamoeba histolytica is a parasitic amoeba that can cause amoebiasis, a gastrointestinal infection.

Option D - Balamuthia mandrillaris: Balamuthia mandrillaris is a free-living amoeba that can cause a rare and often fatal brain infection known as granulomatous amoebic encephalitis (GAE).

Solution for Question 23:

Correct Option A - Curvature of cornea:

- A keratometer is an ophthalmic instrument used to measure the curvature of the cornea.. It is primarily used to assess the central corneal curvature and determine the refractive power of the cornea.
- The keratometer is based on the fact that the anterior surface of the cornea acts as a convex mirror, so the size of the image produced varies with its curvature.
- Therefore, from the size of the image formed by the anterior surface of the cornea (first Purkinje image), the radius of curvature of the cornea can be calculated.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 24:

Correct Option A - Viral keratitis:

- The lesion in viral keratitis is typically characterized by dendritic or branching patterns on the cornea, as seen in the image.
- Viral keratitis is a type of corneal infection caused by a virus, such as herpes simplex virus (HSV) or varicella-zoster virus (VZV).

Incorrect Options:

Option B - Fungal keratitis: The lesion in fungal keratitis often presents as an infiltrate or ulcer with feathery edges (hyphae) and satellite lesions.

Option C - Bacterial keratitis: The lesion in bacterial keratitis often presents with ulcer, stromal edema, inflammation, and keratic precipitates due to anterior uveitis.

Option D - Interstitial keratitis: This is a condition in which there is inflammation of the cornea that affects the middle layer (stroma) of the cornea. The lesion in interstitial keratitis is typically characterized by a hazy appearance of the cornea, with no obvious ulcer or infiltrate.

Solution for Question 25:

Correct Option A - Fungal keratitis:

- The history of trauma with vegetative matter and the clinical features of a corneal ulcer, hypopyon, and satellite lesions are highly suggestive of fungal keratitis.
- Fungal keratitis is a serious ocular infection that can result from trauma with vegetative matter, such as plant material, which can introduce fungal spores into the eye.
- The symptoms of fungal keratitis include pain, photophobia, redness, and a corneal ulcer with a feathery or irregular border.
- Hypopyon (pus in the anterior chamber of the eye) and satellite lesions (smaller ulcers around the main ulcer) are also common in fungal keratitis.

Incorrect Options:

Option B - Acanthamoeba keratitis: Acanthamoeba keratitis is a rare but serious eye infection caused by a free-living amoeba. It is typically associated with contact lens wear using tap water to clean and can present as painful, red eye with photophobia, blurred vision and ring like lesion in cornea. However, the presence of hypopyon and satellite lesions is not typical of Acanthamoeba keratitis.

Option C - Viral keratitis: Viral keratitis is an infection of the cornea caused by a virus, such as herpes simplex virus. It can cause a painful, red eye with photophobia, stellate lesion, dendritic ulcer or geographical ulcer.

Option D - Bacterial keratitis: Bacterial keratitis is characterized by a dry/rough ulcer, with hypopyon (pus cells), stromal edema, descemet's membrane fold, and keratic precipitates due to anterior uveitis.

Solution for Question 26:

Correct Option D - Removal by 26G needle:

- Based on the image and history of welding, is suggestive of superficial foreign body in the eye presenting with pain and redness.
- To remove the foreign body and alleviate the symptoms following steps are taken. First, add some anesthetic drops (proparacaine) to the injured eyes. Then, take a 26-gauge needle

and scrape it out if it's not too deep. After that, apply the antibiotic ointment and bandage

- First, add some anesthetic drops (proparacaine) to the injured eyes.
- Then, take a 26-gauge needle and scrape it out if it's not too deep.
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Incorrect Options: Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 27:

Correct Option D - Serum ceruloplasmin:

- A golden ring on the cornea of a 20-year-old male is suggestive of a Kayser Fleischer ring seen in Wilson's disease, an inherited disorder caused by a mutation in the ATP7B gene, in which there is difficulty metabolizing copper, leading to a buildup of copper in the liver, brain, and other organs. The investigation of choice to diagnose this condition is serum ceruloplasmin.
- Serum ceruloplasmin is a protein that carries copper in the bloodstream; a low ceruloplasmin level suggests the patient has Wilson's disease (normal values in adults is 20 to 35 mg/dl).

[Image not available]

Incorrect Options:

Option A - Alpha-1-antitrypsin: Alpha-1-antitrypsin is a protein produced by the liver to protect the lungs from damage. This test is typically used to diagnose and monitor conditions such as emphysema and chronic obstructive pulmonary disease (COPD), which can cause a deficiency in alpha-1-antitrypsin.

Option B - Alpha-fetoprotein: Alpha-fetoprotein is a protein that is normally produced during fetal development. Elevated levels of alpha-fetoprotein in a non-pregnant female and adult male can be a sign of liver cancer, liver cirrhosis, hepatitis, and germ cell tumors.

Option C - Serum iron: Serum iron is typically used to diagnose and monitor conditions such as iron-deficiency anemia and hemochromatosis, a condition in which the body absorbs too much

iron from food.

Solution for Question 28:

Correct Option D - Fluorescein dye - visualized under cobalt blue filter:

- The dye used in the image is fluorescein dye, which is viewed under a cobalt blue filter. Fluorescein takes the small wavelength eye and transmits a high wavelength light color. It is used for diagnosis of corneal ulcers and in certain tests like the Seidel's test, rigid contact lens fitting, and tear film break up time.
- Fluorescein is a water-soluble dye that is injected into the eye and absorbed by the corneal tissue. When viewed under a cobalt blue filter, the dye appears bright (high wavelength) green and highlights any areas of corneal defects, indicating the presence of an ulcer.

Incorrect Options:

Option A - Lissamine dye green filter: Lissamine green is a dye used to stain dead or devitalized cells on the ocular surface and is commonly used in the diagnosis of dry eye disease, i.e., keratoconjunctivitis sicca. It is visualized under a green filter (red-free).

[Image not available]

Options B and C are incorrect. Refer to Option A and D for an explanation.

Solution for Question 29:

Correct Option A - Rate of recurrence is 95%:

- Pterygium, an eye condition, is characterized by a benign growth of the conjunctiva over the cornea.
- The recurrence rate after surgical excision is around 30 - 50%. Recurrence rates vary, depending on the surgical technique used for pterygium removal. With modern surgical techniques and the use of adjunctive therapies such as Mitomycin C or autografts, the recurrence rate can be as low as 5%.

Incorrect Options:

Option B - It is a fibrovascular growth: A pterygium is a triangular fibrovascular sub-epithelial ingrowth of degenerative bulbar conjunctival tissue over the limbus on to the cornea.

Option C - The parts of a pterygium are head, neck and body: The parts of the pterygium are commonly classified into the head (part over the cornea), neck (area of transition), and body (part on the conjunctiva).

Option D - Has progressive and regressive types: Pterygium can present in both progressive and regressive forms. Progressive pterygium actively grows, extending over the cornea, while the regressive type has halted growth and may even shrink over time.

Solution for Question 30:

Correct Option A - Haab's striae:

- Keratoconus is a non-inflammatory eye condition where the normally round dome-shaped cornea progressively thins and weakens, leading to the development of a cone-like bulge. Haab's striae, horizontal lines on the Descemet's membrane of the cornea, are not found in keratoconus. Instead, they are typically associated with congenital glaucoma.

Incorrect Option:

Option B - Munson's sign: Munson's sign is the v-shaped indentation on the lower eyelid when the patient looks downwards. It's a classic clinical sign of advanced keratoconus.

Option C - Irregular astigmatism: In keratoconus, the cornea thins and bulges out into a cone shape, leading to irregular astigmatism. This results in distorted and blurred vision.

Option D - Scissoring reflex: A scissoring reflex observed during retinoscopy is seen in keratoconus.

Solution for Question 31:

Correct Option D - Hypopyon:

- The farmer's presentation of pain and discharge following a recent eye injury with organic material is suggestive of fungal keratitis.
- Fungal keratitis presents with a rough dry ulcer with feathery margins (hyphae), pain, redness, and an unsterile hypopyon (containing fungal hyphae).

Incorrect Options:

Option A - Hyphema: Hyphema refers to the presence of blood in the eye's anterior chamber. This can occur due to trauma, but it would appear as a red, rather than white, layer in the anterior chamber.

Option B - Aqueous flare: Aqueous flare is the appearance of 'flakes' or 'sparkles' seen in the beam of a slit lamp due to the presence of protein and inflammatory cells in the aqueous humor seen in anterior uveitis.. It's an indicator of inflammation in the eye but it doesn't produce a visible layer of pus.

Option C - Arlt line: Arlt line is a line of scar tissue seen on the upper palpebral conjunctiva seen in trachoma, a chronic inflammatory condition caused by Chlamydia trachomatis.

Solution for Question 32:

Correct Option A - Exposure keratopathy:

- The person is at risk of developing Exposure keratopathy is a condition where the cornea becomes dry and damaged due to inadequate eyelid closure or incomplete blinking, leading to exposure of the cornea to the environment.
- It manifests as irritation , pain, foreign body sensation, and blurred vision.
- In severe cases, it can cause corneal ulceration and scarring.

Incorrect Options:

Options B, C, and D are incorrect. The person is not at risk of developing these conditions.

Solution for Question 33:

Correct Option D - Keratoconus:

- The given image is of V-shaped deformity on looking down is characteristic of keratoconus.
- Keratoconus is a progressive eye disorder characterized by thinning and bulging of the cornea, which results in a cone-shaped deformity.
- This condition can cause visual distortion, blurry vision, and astigmatism.

Incorrect Options:

Option A - Anterior uveitis: Anterior uveitis is inflammation of the front portion of the uvea, which includes the iris and ciliary body. It is characterized by aqueous cells, flare, keratic precipitates, and photophobia.

Option B - Glaucoma: Glaucoma is a group of eye conditions characterized by damage to the optic nerve, often caused by increased intraocular pressure.

Option C - Optic neuritis: Optic neuritis refers to inflammation of the optic nerve, which can cause vision loss, reduced visual activity, RAPD (relative afferent pupillary defect), dyschromatopsia, decreased contrast sensitivity, nerve fiber bundle defect, and visual field

Solution for Question 34:

Correct Option A - Keratoconus:

- The image reveals the presence of the intracorneal ring segment aka INTACS used to treat early or moderate visual impairment caused by keratoconus, characterized by corneal thinning and bulging in the center, by flattening effect.

Incorrect Options:

Option B - Vogt's limbal girdle: Vogt's limbal girdle is a condition of the cornea that most frequently affects the elderly. It consists of whitish crescentic limbal bands composed of chalk-like flecks centred at 9 and/or 3 o'clock, more often nasally.

Option C - Keratoglobus: It is a familial and hereditary bilateral congenital disorder characterized by thinning and hemispherical protrusion of the whole cornea.

Option D - Corneal dystrophy: Corneal dystrophy is a group of disorders characterized by the non-inflammatory, inherited, bilateral opacity of the cornea with no vascularisation, treated by lubricants, bandage, and contact lens.

Solution for Question 35:

Correct Option A - Keratoconus:

- Fleischer's ring is a characteristic finding observed in keratoconus.
- Fleischer's ring appears as a dark, iron-colored ring around the base of the cone-shaped cornea when seen under cobalt blue filter. It is caused by the deposition of iron within the cornea due to mechanical stress and changes in the corneal structure.

[Image not available]

Incorrect Options:

Options B, C, and D are incorrect. Fleischer's ring is not associated with these conditions.

Solution for Question 36:

Correct Option A - Keratoconus:

- Keratoconus is a progressive eye condition characterized by thinning and bulging of the cornea, resulting in a cone-shaped corneal surface. This abnormal corneal shape can cause frequent changes in the refractive error of the eye in young adults leading to the repeated changes in glasses prescription.

Incorrect Options:

Option B - Pathological myopia: Pathological myopia refers to severe nearsightedness caused due to overstretching of retina, excessive axial length, overstretching of the eyeball, it will not lead to frequent changes of glass.

Option C - Glaucoma: Glaucoma is a condition that damages the optic nerve, often due to increased pressure within the eye presenting with headache, and visual field defect.

Option D - Cataract: Cataract refers to the clouding of the eye's natural lens, leading to blurry vision, it will not lead to frequent changes of glass..

Solution for Question 37:

Correct Option B - Endothelium:

- The corneal endothelium is primarily responsible for maintaining the hydration and transparency of the cornea.

Incorrect Options:

Option A - Descemet's membrane: This layer is located between the corneal stroma and the corneal endothelium. It provides structural support to the cornea.

Option C - Stroma: The stroma is the thickest layer of the cornea and is composed of collagen fibers and water. It contributes to the cornea's strength and shape.

Option D - Corneal epithelium: This is the outermost layer of the cornea and serves as a protective barrier against foreign substances and pathogens.

Solution for Question 38:

Correct Option D - Satellite lesion:

- Fungal infection typically presents with dry/rough central corneal ulcer with feathery edges (hyphae) and it may present as a satellite (single) lesion which are infective and immune reactive..
- These satellite lesions are often seen as smaller infiltrates or white fluffy patches adjacent to the primary ulcer.

Incorrect Options:

Option A - Hypopyon: A hypopyon is the accumulation of pus or inflammatory cells in the anterior chamber of the eye. It is commonly seen in bacterial and fungal keratitis.

Option B - Dendritic ulcer on a fluorescein dye: A dendritic ulcer on fluorescein dye is a characteristic finding of herpetic keratitis caused by the herpes simplex virus.

Option C - Ring abscess: A ring abscess is seen in Acanthamoeba keratitis commonly seen in patients using contact lens and using tap water to clean it.

Solution for Question 39:

Correct Option A - Keratoconus:

- Collagen cross-linking treatment is primarily used for the management of Keratoconus, a progressive eye disorder that causes thinning and bulging of the cornea. Collagen cross-linking involves the application of riboflavin (vitamin B2) eye drops to the cornea, followed by exposure to ultraviolet (UV) light. This procedure stabilizes or reverses the ectasia resulting in slow progression of keratoconus..

Incorrect Options:

Options B, C, and D are incorrect. Collagen cross-linking therapy is not used for the treatment of the conditions.

Solution for Question 40:

Correct Option C - Trachoma:

- Arlt's line is seen in chronic cicatricial Trachoma. Trachoma is a chronic infectious disease caused by the bacterium Chlamydia trachomatis.
- It primarily affects conjunctiva and is characterized by follicles in the upper palpebral region leading to scarring in a horizontal line, recognized as Arlt's line.

[Image not available]

Incorrect Options:

Option A - Vernal keratoconjunctivitis: It is characterized by intense itching, redness, watery discharge, and cobblestone appearance of upper palpebral conjunctiva.

Option B - Phlyctenular keratoconjunctivitis: It is characterized by the formation of small nodules called as phlycten near limbus or ulcers on the cornea.

Option D - Pterygium: It is conjunctival degeneration characterized by triangular fibrovascular subepithelial ingrowth of bulbar conjunctiva over the limbus.

Solution for Question 41:

Correct Option A - Myopia and astigmatism:

- The image given above is that of Munson's sign which is the V-shaped conformation of the lower lid produced by the ectatic cornea in downgaze seen in keratoconus.
- The refractive error seen in keratoconus is myopia and irregular astigmatism leading to defective vision which does not improve fully despite full correction with glasses..

Incorrect Options:

Options B, C, and D are incorrect. Myopia with irregular astigmatism is seen in Keratoconus.

Solution for Question 42:

Correct Option D - Atropine Sulfate eye ointment:

- Atropine Sulfate eye ointment is the adjuvant therapy commonly used in fungal corneal ulcers. To reduce pain from ciliary spasms. To prevent the formation of posterior synechiae. It increases the blood supply to the anterior uvea by relieving pressure on the anterior ciliary arteries and brings more antibodies into the aqueous humor. It also reduces exudation by decreasing hyperemia and vascular permeability.
- To reduce pain from ciliary spasms.

- To prevent the formation of posterior synechiae.
- It increases the blood supply to the anterior uvea by relieving pressure on the anterior ciliary arteries and brings more antibodies into the aqueous humor.
- It also reduces exudation by decreasing hyperemia and vascular permeability.
- To reduce pain from ciliary spasms.
- To prevent the formation of posterior synechiae.
- It increases the blood supply to the anterior uvea by relieving pressure on the anterior ciliary arteries and brings more antibodies into the aqueous humor.
- It also reduces exudation by decreasing hyperemia and vascular permeability.

Incorrect Option:

Options A, B, and C are incorrect. These are not used as the adjunctive therapy for fungal keratitis.

Solution for Question 43:

Correct Option C - Fungal corneal ulcer:

- The most likely pathology in this case is a fungal corneal ulcer. The history of an eye injury caused by a leaf, along with symptoms of pain, photophobia, redness, and examination revealing corneal ulcer with feathery edges (hyphae) surrounded by satellite lesions are suggestive of fungal keratitis..

Incorrect Options:

Option A - Anterior uveitis: It typically presents with pain, redness, photophobia, and blurred vision, keratic precipitates, and aqueous flare.

Option B - Conjunctivitis: It is characterized by redness, pain, photophobia, discharge depending on the infecting organism, follicles or papillae in the palpebral conjunctiva.

Option D - HSV corneal ulcer: Ulcer caused by herpes simplex virus usually presents with superficial punctate keratitis and dendritic ulcer which may progress to geographical ulcer.

Sclera

1. Which of the following correctly lists the layers of the sclera from outermost to innermost?

- A. Lamina fusca > Sclera proper > Episclera
- B. Episclera > Sclera proper > Lamina fusca
- C. Sclera proper > Episclera > Lamina fusca
- D. Episclera > Lamina fusca > Sclera proper

2. Match the scleral apertures with the structures they transmit

Scleral Apertures	Structures Transmitted
A. Anterior scleral apertures	1. Short and long ciliary arteries
B. Middle scleral apertures	2. Central retinal artery and vein
C. Posterior scleral apertures	3. Anterior ciliary arteries and veins
D. Lamina cribrosa	4. Vortex veins

	Scleral Apertures		Structures Transmitted
A.	Anterior scleral apertures	1.	Short and long ciliary arteries
B.	Middle scleral apertures	2.	Central retinal artery and vein
C.	Posterior scleral apertures	3.	Anterior ciliary arteries and veins
D.	Lamina cribrosa	4.	Vortex veins

- A. A-3, B-4, C-1, D-2
- B. A-2, B-1, C-4, D-3
- C. A-3, B-2, C-1, D-4
- D. A-3, B-1, C-4, D-2

3. A 28-year-old patient presents with painless redness of the eye. The ophthalmologist performed a diagnostic test using 2.5% phenylephrine, and no change was noticed. When 10% phenylephrine is used, the following change occurs. What is the most likely diagnosis? Before application of 10% phenylephrine: {{caption_text}} After application of 10% phenylephrine: {{caption_text}}

[Image not available]

[Image not available]

- A. Conjunctivitis
- B. Scleritis
- C. Episcleritis
- D. Uveitis

4. A 40-year-old patient presents with recurrent episodes of episcleritis. After taking a detailed history and performing a thorough systemic examination, which of the following investigations would you consider ordering?

- A. Full blood count and rheumatoid factor (RF)
- B. Mantoux test and antinuclear antibody (ANA)
- C. VDRL and X-rays of the chest and sacroiliac joints
- D. All of the above

5. A 45-year-old woman with a 15-year history of rheumatoid arthritis (RA) presents with severe ocular pain, redness, and progressively worsening vision in her left eye for the past 2 weeks. Examination reveals a localized area of scleral thinning with adjacent congestion, and fluorescein staining highlights areas of avascular necrosis. There is no posterior involvement on B-scan ultrasound. Which of the

following is the most likely diagnosis?

- A. Anterior staphyloma
 - B. Necrotizing scleritis with inflammation
 - C. Nodular scleritis
 - D. Scleromalacia perforans
-

6. A 38-year-old male presents with persistent eye discomfort, redness, and watering. On examination, the left eye is diffusely tender, sclera appears uniformly reddened and swollen without localized thinning areas. There are no systemic signs of vasculitis or autoimmune disease on further investigation. Which type of anterior scleritis is most consistent with this presentation?

- A. Necrotizing anterior scleritis with inflammation
 - B. Non-necrotizing anterior diffuse scleritis
 - C. Non-necrotizing anterior nodular scleritis
 - D. Scleromalacia perforans
-

7. A 45-year-old woman presents with severe pain in her right eye, worsened by eye movements, and reduced vision. On examination, proptosis and restricted ocular movements were observed. Ophthalmoscopy shows disc, macular edema, and choroidal folds. Which of the following B-scan ultrasonography findings is most indicative of this disease?

- A. T sign
 - B. Moth-Eaten Appearance
 - C. Sunray Sign
 - D. Kiss Sign
-

8. A 60-year-old woman with a 20-year history of rheumatoid arthritis has a painless, progressively thinning area in the sclera of her left eye. The area appears as shown in the image below. Which of the following best describes the pathophysiological mechanism responsible for this condition? {{caption_text}}

[Image not available]

- A. Type III hypersensitivity
 - B. Type II hypersensitivity
 - C. Type IV hypersensitivity
 - D. Infective etiology
-

9. Depending on the location of the staphyloma in the eye which among the following options is matched correctly with its cause?

- A. Anterior staphyloma - peripheral corneal ulcer
 - B. Intercalary staphyloma - degenerative myopia
 - C. Equatorial staphyloma - scleritis
 - D. Ciliary - pseudocornea
-

10. Blue sclera is commonly associated with all of the following conditions except:

- A. Osteogenesis Imperfecta
 - B. Ehlers-Danlos Syndrome
 - C. Alkaptonuria
 - D. Marfan Syndrome
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	3
Question 4	4
Question 5	2
Question 6	2
Question 7	1
Question 8	1
Question 9	3
Question 10	3

Solution for Question 1:

Correct Answer: B) Episclera > Sclera proper > Lamina fusca

Explanation:

- The sclera consists of three ill-defined layers, namely the sclera proper with the episclera on the outside and the lamina fusca interiorly.

Layers of the Sclera:

Layer	Description	Composition	Remarks
Episclera	Thin vascularized. connective tissue.	Collagen fibers Fibroblasts Blood vessels Nerves	Located just outside the sclera proper

Sclera proper	Relatively avascular. Dense tissue.	Dense irregular bundles of collagen (compared to the orderly arrangement in the cornea) Elastic fibers Fibroblasts, Proteoglycans	The thickest layer of the sclera Provides structural support and integrity
Lamina fusca	Innermost layer. Loose connective tissue layer.	Melanocytes (migrated from choroid)	It contains pigmented cells derived from the choroid

[Image not available]

Solution for Question 2:

Correct Answer: A) A-3, B-4, C-1, D-2

Explanation:

- The anterior scleral apertures transmit the anterior ciliary arteries and veins, while the middle scleral apertures allow passage for the vortex veins.
- The posterior scleral apertures transmit the short and long ciliary arteries.
- The Lamina cribrosa is where the central retinal artery and vein enter the sclera.

Apertures of the sclera:

Apertures	Location	Structures transmitted
Anterior	3-4 mm posterior to the limbus	Anterior ciliary vessels, nerves, and lymphatics
Middle	4 to 7 mm behind the equator	4 vortex veins (Vena verticose)
Posterior	Around the optic nerve	8 to 20 short ciliary vessels (circle of Zinn) and nerves 2 long posterior ciliary arteries anterior to short ciliary
Lamina cribrosa	Posterior most	Optic nerve fibers Central retinal artery and veins.

[Image not available]

Solution for Question 3:

Correct Answer: C) Episcleritis

Explanation:

- The patient has painless redness of the eye which blanches (as shown in the image) following the application of 10% phenylephrine, suggesting episcleritis.

Episcleritis	
Definition	Episcleritis refers to inflammation of the loose connective tissue between the sclera and the conjunctiva
Etiology	Idiopathic Systemic diseases - gout, rosacea, psoriasis, and connective tissue diseases. Hypersensitivity reaction to endogenous tubercular or streptococcal toxins. Infectious episcleritis - herpes zoster virus, syphilis, Lyme disease, and tuberculosis.
Clinical features	Common in young females Acute onset redness Discomfort - gritty/foreign body sensation Mild to no pain Resolves spontaneously Recurrence is common
Clinical types	Simple episcleritis: Sectorial (occasionally diffuse) inflammation of episclera. The engorged episcleral vessels are large and run in a radial direction beneath the conjunctiva Nodular episcleritis: Pink or purple flat nodule surrounded by injection, usually situated 2-3 mm away from the limbus. The nodule is firm, tender can be moved separately from the sclera and the overlying conjunctiva also moves freely.
Diagnosis	10% Topical phenylephrine blanches overlying conjunctival vessels and inflamed episcleral vessels.
Differential diagnosis	Conjunctivitis, which is more superficial; blanching occurs with 2.5% phenylephrine (Option A ruled out) Phlyctenulosis Inflamed pinguecula Scleritis, which is deeper and more painful; does not blanch with 10% phenylephrine (Option B ruled out)

Treatment	Topical NSAIDs Topical mild corticosteroid eye drops Topical artificial tears, Cold compresses applied to the closed lids may offer symptomatic relief from ocular discomfort. Systemic NSAIDs
-----------	--

Uveitis (Option D) is associated with pain, redness, photophobia, and excess lacrimation, with associated findings in slit lamp examination, which are not seen here.

[Image not available]

Solution for Question 4:

Correct Answer: D) All of the above

Explanation:

- Episcleritis can be associated with systemic conditions such as rheumatoid arthritis (RF), systemic lupus erythematosus (ANA), and other systemic infections or inflammatory diseases that may be assessed with the Mantoux test (Tuberculosis) and VDRL (Syphilis).
- X-rays of the chest and sacroiliac joints can help rule out conditions like ankylosing spondylitis, which can also present with episcleritis.
- Therefore, a comprehensive approach including all listed investigations is appropriate.

Conditions associated with Episcleritis:

- Idiopathic: The exact etiology is unknown in many cases.
- Systemic Diseases: Gout Rosacea Psoriasis Connective tissue diseases (e.g., rheumatoid arthritis, lupus)
 - Gout
 - Rosacea
 - Psoriasis
- Connective tissue diseases (e.g., rheumatoid arthritis, lupus)
- Hypersensitivity Reactions: Endogenous tubercular toxins Streptococcal toxins
- Endogenous tubercular toxins
- Streptococcal toxins
- Infectious Episcleritis: Herpes zoster virus Syphilis Lyme disease Tuberculosis

- Herpes zoster virus
- Syphilis
- Lyme disease
- Tuberculosis
- Gout
- Rosacea
- Psoriasis
- Connective tissue diseases (e.g., rheumatoid arthritis, lupus)
- Endogenous tubercular toxins
- Streptococcal toxins
- Herpes zoster virus
- Syphilis
- Lyme disease
- Tuberculosis

Solution for Question 5:

Correct Answer: B) Necrotizing scleritis with inflammation

Explanation:

- Severe pain, scleral thinning, and necrosis are hallmark signs of necrotizing scleritis, the most severe form of RA-associated scleritis.
- RA leads to immune-mediated vasculitis, causing ischemic necrosis of the sclera.

Classification of Scleritis:

[Image not available]

Anterior Staphyloma (Option A):

- Anterior staphyloma is a late-stage complication, where uveal tissue bulges through a thinned, scarred sclera.
- This patient has active inflammation, whereas staphyloma occurs after chronic necrosis and healing.

Nodular Scleritis (Option C):

- Nodular scleritis presents with discrete, firm nodules on the sclera, without necrosis.
- This patient has scleral thinning and avascular necrosis, which rules out nodular scleritis.

Scleromalacia Perforans (Option D):

- Scleromalacia perforans is a form of necrotizing scleritis WITHOUT inflammation (painless, non-red eye).
- This patient has severe pain and congestion, so active necrotizing scleritis is the correct choice.

Solution for Question 6:

Correct Answer: B) Non-necrotizing anterior diffuse scleritis

Explanation:

- Clinical presentation of persistent eye discomfort, watering, diffuse redness, and tenderness without localized necrosis or thinning is characteristic of non-necrotizing anterior diffuse scleritis.

Anterior Scleritis:

- Most common in elderly female patients (40-70 years)
- Etiology: Associated with connective tissue disease (50% of cases)

Clinical classification:

- Non-necrotizing anterior scleritis
 - Diffuse
 - Nodular
- Necrotizing anterior scleritis
 - With inflammation
 - Without inflammation
- Diffuse
 - Nodular
 - With inflammation
 - Without inflammation

Non-necrotizing anterior scleritis:

- Pain, redness, watering, discharge, and bluish or brownish discoloration.
- Nodular: One or more dark red or bluish nodules, which later become purple and semitransparent like porcelain. (Option C ruled out)
- Diffuse (most common): Diffuse or localized congestion (Option B)

Treatment:

- First-line Topical steroids eye drops Oral NSAIDs
- Topical steroids eye drops
- Oral NSAIDs
- Second-line Systemic steroids
- Systemic steroids
- Topical steroids eye drops
- Oral NSAIDs
- Systemic steroids

[Image not available]

[Image not available]

Necrotizing anterior scleritis:

- Necrosis of sclera
- Severe thinning, and melting in severe cases

With inflammation: (Option A ruled out)

- Severe form of scleritis
- Red, painful eye, and progressive worsening of symptoms

Without inflammation (Scleromalacia perforans): (Option D ruled out)

Painless scleral thinning

- Topical steroids eye drops
- High dose of oral steroids
- Non-responsive cases
- Immunosuppressive agents
- Surgical treatment: scleral patch graft in severe cases to preserve the integrity of the globe due to severe thinning and melting.

[Image not available]

Solution for Question 7:

Correct Answer: A) T sign

Explanation:

- Given the clinical presentation of severe pain in the eye, proptosis, restricted ocular movements, reduced vision, disc, and macular edema, and the potential findings of choroidal folds, the diagnosis most consistent with these symptoms is posterior scleritis.
- Thickening of the posterior sclera, retrobulbar thickening, and fluid surrounding the optic nerve creates a characteristic "T" sign in B-scan ultrasound.

Posterior scleritis:

- Inflammation with thickening of the posterior sclera

Clinical manifestations:

- Decreased vision, with or without pain
- Proptosis
- Restricted ocular movements
- Posterior vitreous
- Disc and macular oedema
- Choroidal folds
- Choroidal detachment
- Uveal effusion syndrome
- Exudative retinal detachment

Diagnosis:

- USG-B scan shows T sign (Pathognomonic finding): (Option A) It represents the fluid accumulation between the optic nerve and the sclera. It is the name given to the shape of the scleral thickening: Each arm of the T represents the posterior sclera with the stem as the optic nerve.
- It represents the fluid accumulation between the optic nerve and the sclera
- It is the name given to the shape of the scleral thickening: Each arm of the T represents the posterior sclera with the stem as the optic nerve.
- It represents the fluid accumulation between the optic nerve and the sclera

- It is the name given to the shape of the scleral thickening: Each arm of the T represents the posterior sclera with the stem as the optic nerve.
- Others-CT, MRI

[Image not available]

Treatment:

- Systemic NSAIDs
- Systemic steroids
- Immunosuppressive therapy
- Biologics: Infliximab and rituximab

Moth-Eaten Appearance (Option B): Seen in Retinoblastoma, characterized by intraocular calcifications, typically in children, not adults.

Sunray Sign (Option C): Seen in Choroidal Melanoma, due to radial vascularity, but this patient lacks an intraocular mass.

Kiss Sign (Option D): Seen in Total Retinal Detachment, where the detached retina folds inward, but this case has no retinal involvement.

Solution for Question 8:

Correct Answer: A) Type III hypersensitivity

Explanation:

- The diagnosis in this scenario is scleromalacia perforans.
- It is usually associated with long-standing rheumatoid arthritis and presents with scleral thinning appearing bluish and without signs of inflammation.
- The underlying cause of scleral thinning is probably ischemia because of vessel damage (type III hypersensitivity).

Scleromalacia Perforans:

Most common in elderly females

Pathophysiology:

- Associated with long-standing rheumatoid arthritis

- Melting of the sclera is due to obliteration of arterial supply causing ischemia
- Vessel damage is mediated by type III hypersensitivity (Option A)

Clinical manifestations:

- Painless scleral thinning
- Underlying dark uveal tissue is visible
- Rarely perforation of the globe

[Image not available]

Treatment:

- No specific therapy
- Protection of the eye from trauma is very crucial
- Plenty of topical lubrication
- Oral NSAIDs
- Oral steroids
- Immunomodulatory agents
- Surgery-scleral patch grafting

Type II hypersensitivity (Option B), type IV hypersensitivity (Option C), and infective etiology (Option D): There is no known association between these and scleromalacia perforans in its pathophysiology.

Solution for Question 9:

Correct Answer: C) Equatorial staphyloma - scleritis

Explanation:

- The causes of equatorial staphyloma are scleritis, degenerative myopia, and chronic uncontrolled glaucoma.

Staphyloma:

- It is ectasia of the outer coats of the eye (cornea, sclera, or both) with an incarceration of the uveal tissue.

Based on the site affected, staphyloma can be classified as

[Image not available]

Types of staphyloma	Causes
Anterior (Option A ruled out)	Pseudocornea
Intercalary (Option B ruled out)	Perforating injuries of the peripheral cornea, marginal corneal ulcer, anterior scleritis, and scleromalacia perforans
Ciliary (Option D ruled out)	Glaucoma, scleritis, and trauma to the ciliary region
Equatorial (approx 14 mm behind the limbus) (Option C)	Scleritis, degenerative myopia, and chronic uncontrolled glaucoma
Posterior	Degenerative high axial myopia

Treatment:

- Treat underlying condition
- Local excision and repair with a corneal and scleral patch graft
- Large staphyloma: Staphylectomy and keratoplasty, or enucleated and replaced with an implant

Solution for Question 10:

Correct Answer: C) Alkaptonuria

Explanation:

- Yellow-brown or blue-black pigment deposition occurs in different parts of the eye in alkaptonuria.
- Classic scleral pigment deposits are called “Osler’s sign.”

[Image not available]

Blue sclera:

Uveal pigment shines through the thin sclera and produces the blue color

Causes:

- Osteogenesis imperfecta (Option A ruled out)
- Ehlers–Danlos syndrome (Option B ruled out)
- Marfan syndrome (Option D ruled out)

- Pseudoxanthoma elasticum
 - Scleromalacia perforans
-

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Uvea

1. A 42-year-old African American woman presents with visual disturbances. Eye examination reveals vitreous cells, snowball opacities, and candle wax drippings. Which type of uveitis is seen in the underlying disease?

- A. Anterior uveitis
- B. Panuveitis
- C. Posterior uveitis
- D. Peripheral multifocal chorioretinitis

2. Which of the following conditions can cause panuveitis?

- A. VKH syndrome
- B. Sympathetic Ophthalmia
- C. Sarcoidosis
- D. All of the above

3. A 45-year-old male presents with vision loss in his right eye following a traumatic injury that resulted in a penetrating ocular wound two months ago. Recently, he has developed redness, photophobia, and decreased vision in his left eye as well. On examination, both eyes show signs of bilateral granulomatous uveitis. Which of the following best explains the pathogenesis of the patient's condition?

- A. Type I hypersensitivity reaction to lens proteins
- B. Delayed hypersensitivity reaction to melanocytes
- C. Direct bacterial infection spreading from the injured eye

D. Immune complex deposition in the retinal vasculature

4. A 28-year-old woman diagnosed with Vogt–Koyanagi–Harada (VKH) disease, Fundus examination shows the following finding. What does it indicate?

[Image not available]

- A. Prodromal phase
 - B. Uveitic phase
 - C. Convalescent phase
 - D. Recurrent phase
-

5. A 35-year-old male presents with progressively worsening blurry vision, eye pain, and redness in the right eye. Examination reveals restricted eye movement, and fundoscopic evaluation identifies a well-defined cystic lesion in the vitreous humor, accompanied by surrounding inflammation. What is the most probable diagnosis in this patient?

- A. Ocular toxoplasmosis
 - B. Endophthalmitis
 - C. Ocular cysticercosis
 - D. Posterior uveitis due to tuberculosis
-

6. Which of the following ocular conditions is most commonly associated with a positive Amsler sign?

- A. Cataract
- B. Ocular histoplasmosis syndrome
- C. Fuchs heterochromatic uveitis

D. Acute retinal necrosis

7. A 35-year-old male with a recent travel history to Africa presents with intense itching, skin nodules, and progressive vision loss. On examination, multiple palpable nodules are found on his skin, and there is early corneal clouding. Microfilariae are observed in a skin snip biopsy. What is the most appropriate treatment for this condition?

- A. Clindamycin + sulfadiazine
 - B. Diethylcarbamazine
 - C. Ivermectin
 - D. Self-limiting condition, no treatment needed
-

8. A 32-year-old immunocompetent woman living with cats presents with decreased vision in her left eye. On examination, she has mild anterior uveitis with vitritis and the funduscopy image is shown below. Which of the following is the most likely causative organism?

[Image not available]

- A. Cytomegalovirus (CMV)
 - B. Candida albicans
 - C. Toxoplasma gondii
 - D. Histoplasma capsulatum
-

9. Match bacterial uveitis in column A with corresponding clinical manifestation in column B

Bacterial uveitis	Clinical features
1. Tubercular uveitis	a. Iris pearls
2. Leprotic uveitis	b. Iris rosette
3. Spirochetal uveitis	c. Headlight in fog appearance
d. Koeppe's nodules	

Bacterial uveitis	Clinical features
1. Tubercular uveitis	a. Iris pearls
2. Leprotic uveitis	b. Iris roceola
3. Spirochetal uveitis	c. Headlight in fog appearance
d. Koeppe's nodules	

- A. 1-d, 2-a, 3-b
- B. 1-b, 2-c, 3-d
- C. 1-c, 2-d, 3-a
- D. 1-b, 2-a, 3-d

10. A 34-year-old woman with a history of Crohn's disease presents with sudden onset of eye pain, redness, and blurred vision. Slit-lamp examination reveals fine, grey keratic precipitates, anterior chamber cells, and flare. Which type of uveitis is most consistent with her presentation?

- A. Granulomatous uveitis
- B. Non-granulomatous anterior uveitis
- C. Posterior uveitis
- D. Both A and C are correct

11. A 30-year-old patient with HIV presents with vision loss and floaters. On examination, the patient has a CD4+ count of 40 cells/mm³. Fundoscopy reveals areas of necrosis with the following appearance. What is the most likely diagnosis?

[Image not available]

- A. Acute Retinal Necrosis
- B. Herpes Simplex Virus (HSV) Retinitis
- C. Progressive Outer Retinal Necrosis

D. Herpes Zoster Ophthalmicus

12. Which of the following statements about the uveal tract is false?

- A. The choroid connects anteriorly to the iris via the ciliary body.
 - B. The ciliary body contains both radial and circular smooth muscle fibers.
 - C. The posterior layer of the iris is continuous with the ciliary body
 - D. The constrictor pupillae muscle is controlled by sympathetic nerves.
-

13. A 30-year-old man presents with symptoms of recurrent bilateral eye inflammation, including redness, pain, and vision loss. He also has a history of frequent oral and genital ulcers and skin rashes. He is HLA-B51 positive. What would be the most appropriate approach for managing his uveitis?

- A. Start with systemic corticosteroids and consider long-term immunosuppressive therapy
 - B. Immediate initiation of antiviral therapy
 - C. Use of topical antibiotics for ocular symptoms
 - D. Prescription of oral antihistamines for symptom relief
-

14. A 30-year-old man with a recent history of urethritis and arthritis presents with eye symptoms. He is positive for HLA-B27. What is the most appropriate initial treatment?

- A. Oral antibiotics
- B. Topical corticosteroids
- C. Systemic tetracycline
- D. Immunosuppressive agents

15. A 45-year-old male presents with unilateral eye pain, photophobia, and redness for the past five days. Slit-lamp examination reveals circumcorneal injection, anterior chamber cells and flare, and posterior synechiae. Despite initial treatment with topical corticosteroids and cycloplegics, his symptoms persist, and intraocular pressure (IOP) is found to be elevated. Which of the following is the most likely complication of the condition he is suffering from?

- A. Retinal detachment
 - B. Secondary glaucoma
 - C. Vitreous hemorrhage
 - D. Optic neuritis
-

16. Which of the following is not a component of Posterior uveitis?

- A. Retinitis
 - B. Choroiditis
 - C. Iridocyclitis
 - D. Retinal vasculitis
-

17. A 35-year-old male presents with bilateral, painless vision blurring. He has no significant medical history. On examination, slit-lamp findings are unremarkable, but indirect ophthalmoscopy reveals snowball opacities in the vitreous near the ora serrata. Further evaluation for systemic diseases is negative. Which of the following is the most appropriate initial treatment?

- A. Pars plana vitrectomy
- B. Systemic steroids

- C. Periocular injection of triamcinolone
 - D. Cryotherapy
-

18. A 25-year-old female presents with floaters and gradual deterioration of vision. On examination, there is anterior vitritis with white snowball-like exudates and mild peripheral periphlebitis. No other relevant history or clinical findings. Which of the following is the most likely diagnosis?

- A. CMV Retinitis
 - B. Tuberculosis
 - C. Intermediate uveitis
 - D. Lyme Uveitis
-

19. A 40-year-old patient presents with eye pain, redness, and a constricted pupil. During the examination, fine, grey deposits on the corneal endothelium are noted. Which of the following is the most appropriate next step in investigating and treating this condition?

- A. Perform a chest X-ray and initiate treatment with topical cycloplegics.
 - B. Order a serum ACE level and start treatment with non-steroidal voclosporin.
 - C. Conduct an HLA typing and begin treatment with topical prednisolone.
 - D. Arrange for an OCT scan and administer systemic interferon-alpha.
-

20. A 35-year-old patient presents with a 1-week history of worsening eye pain and redness. On examination, a constricted pupil, a hazy anterior chamber, and deposits on the back of the cornea as shown below. Based on these findings, which of the following is the most likely diagnosis?

[Image not available]

- A. Herpes Simplex Keratitis
 - B. Conjunctivitis
 - C. Anterior Uveitis
 - D. Acute Angle Closure Glaucoma
-

21. A 50-year-old patient presents with anterior uveitis. After a detailed evaluation, the ophthalmologist considers various systemic associations and possible masquerades for the condition. Which of the following conditions is a known systemic association with anterior uveitis, but not a masquerade?

- A. Intraocular Foreign Body
 - B. Behcet's Disease
 - C. Retinoblastoma
 - D. Pigment Dispersion Syndrome
-

22. Which of the following statements about the classification of uveitis based on the time of inflammatory activity is false?

- A. Acute uveitis is defined as having a duration of less than 3 months and typically has a sudden onset.
 - B. Chronic uveitis is classified as lasting more than 6 months without interruption.
 - C. Recurrent uveitis involves repeated episodes with periods of inactivity between them.
 - D. Remission in uveitis is defined as no inflammatory activity for 3 months or more.
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	2
Question 4	3
Question 5	3
Question 6	3
Question 7	3
Question 8	3
Question 9	1
Question 10	2
Question 11	3
Question 12	4
Question 13	1
Question 14	2
Question 15	2
Question 16	3
Question 17	3
Question 18	3
Question 19	3
Question 20	3
Question 21	2
Question 22	2

Solution for Question 1:

Correct Answer: B) Panuveitis

Explanation:

The given clinical scenario suggests ocular sarcoidosis (granulomatous panuveitis).

Ocular Sarcoidosis Overview:

- Disease Characteristics: Granulomatous, multi-system disease with unknown cause
Non-caseating epithelioid cell granulomas Common in young adults (20-50 years), more frequent in women and individuals of African descent
- Granulomatous, multi-system disease with unknown cause
- Non-caseating epithelioid cell granulomas
- Common in young adults (20-50 years), more frequent in women and individuals of African descent
- Common Presentations: Bilateral hilar lymphadenopathy Pulmonary infiltration Skin and eye lesions
- Bilateral hilar lymphadenopathy
- Pulmonary infiltration
- Skin and eye lesions
- Ocular Lesions: Occur in 20-50% of cases Types: Sarcoid uveitis, conjunctival lesions, lacrimal gland involvement
- Occur in 20-50% of cases
- Types: Sarcoid uveitis, conjunctival lesions, lacrimal gland involvement
- Granulomatous, multi-system disease with unknown cause
- Non-caseating epithelioid cell granulomas
- Common in young adults (20-50 years), more frequent in women and individuals of African descent
- Bilateral hilar lymphadenopathy
- Pulmonary infiltration
- Skin and eye lesions
- Occur in 20-50% of cases
- Types: Sarcoid uveitis, conjunctival lesions, lacrimal gland involvement

Sarcoid Uveitis

- Prevalence: Accounts for 2% of uveitis cases
- Types and Signs: Sarcoidosis is associated with Panuveitis

Type of Uveitis	Signs
Anterior (Option A)	Granulomatous iridocyclitis, iris nodules, large mutton fat KPs, anterior chamber cells and flare, posterior synechiae
Intermediate	Vitreous cells, snowball opacities, snowbanking
Posterior (Option C)	Choroidal and retinal granulomas, cystoid macular edema, periphlebitis retinae with candle wax droppings appearance
Peripheral Multifocal Chorioretinitis (Option D)	Small, punched-out atrophic spots, highly suggestive of sarcoidosis
Uveoparotid Fever (Heerfordt's Syndrome)	Bilateral granulomatous panuveitis, painful parotid gland enlargement, cranial nerve palsies, skin rashes, fever, malaise

- Complications: Complicated cataract Inflammatory glaucoma Cystoid macular edema
- Complicated cataract
- Inflammatory glaucoma
- Cystoid macular edema
- Complicated cataract
- Inflammatory glaucoma
- Cystoid macular edema

Other Ocular Lesions

- Conjunctival Lesions: Sarcoid nodules Keratoconjunctivitis sicca
- Sarcoid nodules
- Keratoconjunctivitis sicca
- Lacrimal Glands: Enlargement can lead to Mickulicz's syndrome (with diffuse salivary gland swelling)
- Enlargement can lead to Mickulicz's syndrome (with diffuse salivary gland swelling)
- Sarcoid nodules
- Keratoconjunctivitis sicca
- Enlargement can lead to Mickulicz's syndrome (with diffuse salivary gland swelling)

Solution for Question 2:

Correct Answer: D) All of the above

Explanation:

Panuveitis can be caused by various systemic and ocular conditions, including VKH syndrome, Sympathetic Ophthalmia, and Sarcoidosis.

Causes of Panuveitis:

Autoimmune	Infectious
Sarcoidosis. Birdshot retinochoroidopathy. Sympathetic ophthalmia. Polyarteritis nodosa. Lens-induced.	Infections (syphilis, TB, HSV/VZV, toxoplasmosis, cryptococcosis).

Solution for Question 3:

Correct Answer: B) Delayed hypersensitivity reaction to melanocytes

Explanation:

The patient is likely experiencing sympathetic ophthalmia, a condition characterized by bilateral granulomatous uveitis following trauma to one eye. The pathogenesis involves a delayed hypersensitivity reaction (T-cell mediated) to tyrosinase peptide antigens or melanocytes, leading to non-necrotizing granulomatous inflammation in both eyes.

Sympathetic ophthalmia (SO)	
Definition	Bilateral, granulomatous, panuveitis occurring after penetrating trauma where uveal prolapse may have been a feature of the trauma

Etiology and pathogenesis	Trauma to one eye (exciting eye) — 2 weeks to 3 months — Uveitis in the uninjured eye (sympathizing eye). T-cell mediated autoimmune hypersensitivity reaction to Retinal S antigen. (Option B) Maximum risk upon injury to the ciliary body.
Histopathology	Diffuse lymphocytic infiltration of the choroid. Scattered aggregates of epithelioid cells are seen, many of which contain fine granules of melanin. Dalen–Fuchs nodules - granulomas located between Bruch membrane and the RPE (also seen in VKH syndrome)
Clinical features	The exciting eye is frequently red and irritable. The sympathizing eye develops irritation, blurred vision, photophobia and loss of accommodation causing difficulty in near vision (first symptom) Anterior uveitis develops in both eyes, which may be mild or severe and is usually granulomatous
Investigations	Fundus exam: Dalen–Fuchs nodules, Exudative retinal detachment, vasculitis and optic disc swelling. As inflammation settles, residual chorioretinal scarring may confer a 'sunset glow' appearance similar to VKH. B-scan ultrasonography: Choroidal thickening and retinal detachment
Treatment	Enucleation of a severely injured eye within 10 to 14 days of the injury is effective in preventing or reducing the severity of SO Steroids are the basis of initial treatment. Intravenous methylprednisolone followed by high-dose oral prednisolone is given for several months and gradually tapered according to response. Supplementary topical steroids and cycloplegics may be given

Solution for Question 4:

Correct Answer: C) Convalescent phase

Explanation:

The image shown is of Sunset Glow Fundus in the convalescent Phase of Vogt–Koyanagi–Harada Disease. The fundus has a slightly reddish appearance, mainly due to the disappearance of melanocytes in the choroid.

Vogt–Koyanagi–Harada (VKH) Syndrome	
Definition	Idiopathic multisystem autoimmune disease featuring inflammation of melanocyte-containing tissues such as the uvea, ear, and meninges.
Etiology	Idiopathic. HLA associations: HLA-DR1 and HLA-DR4.
Classification	Vogt–Koyanagi disease: Skin changes and anterior uveitis. Harada disease: Neurological features and exudative retinal detachments.
Clinical features	Clinically, the disease is categorized into four phases: Prodromal phase (lasts for a few days): Neurological (meningitis, cranial nerve paresis) and auditory manifestations (tinnitus, vertigo and deafness). Acute uveitic phase: Bilateral granulomatous anterior and multifocal posterior uveitis with diffuse choroidal infiltration, vitritis, papillitis and exudative retinal detachments. Dalen–Fuchs nodules can be seen in this phase. Convalescent phase (follows several weeks later): Localized alopecia, poliosis and vitiligo, depigmented fundus appearance ('sunset glow' fundus) and depigmented limbal lesions (Sugiura sign). (Option C) Chronic recurrent phase: smouldering anterior uveitis with exacerbations
Investigations	Lumbar puncture: CSF shows a transient lymphocytic pleocytosis and melanin-containing macrophages. Fundus Autofluorescence (FAF): Areas of serous detachment. Ultrasonography: Diffuse choroidal thickening. FA(Fluorescein Angiography) of the acute phase shows multifocal hyperfluorescent dots at the level of the retinal pigment epithelium. OCT: Quantification of subretinal fluid. ICGA: For monitoring
Complications and Prognosis	Ocular complications include choroidal neovascularization, subretinal fibrosis, preretinal and disc new vessels and vitreous hemorrhage, cataract and glaucoma. Neurological and auditory manifestations tend to resolve but skin, lash and hair changes usually persist.

Treatment	Intravenous methylprednisolone pulse therapy (500–1000 mg/day). High-dose (1–2 mg/kg/day) oral prednisolone, tapered over 3–6 months. Topical steroids and cycloplegics are used for anterior uveitis. Steroid-resistant patients may require immunosuppressives or biological blockers such as infliximab
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Solution for Question 5:

Correct Answer: C) Ocular cysticercosis

Explanation:

The presence of a well-defined cystic lesion in the vitreous humor with surrounding inflammation, along with restricted eye movement, is characteristic of ocular cysticercosis, caused by the larval stage of *Taenia solium*. Inflammation occurs mainly due to the release of toxins from the dead cysts.

Ocular cysticercosis	
Definition	Infestation caused by <i>Cysticercus cellulosae</i> , the larval form of the pork tapeworm <i>Taenia solium</i> .
Pathophysiology	Humans ingest eggs of <i>T. solium</i>, leading to larvae formation (cysticercus) in various tissues. Inflammation is triggered by antigens released from dead larvae, leading to tissue damage. It is not contracted by eating undercooked pork.
Clinical Features	Systemic Involvement: Larvae can infest the CNS (neurocysticercosis), muscles, lungs, and subcutaneous tissues. Neurological manifestations include seizures and headaches.

Eye involvement: Orbit and adnexal cysticercosis: The extraocular muscles (superior rectus- most common) form is the most common type of orbital and adnexal cysticercosis. Presents as periocular swelling, proptosis, ptosis, pain, diplopia, restriction of ocular motility, strabismus, decreased vision, lid edema and orbital cellulitis. Ocular cysticercosis: Intraocular cysticercosis can involve either the anterior or the posterior segment (more common). Presents as blurring of vision, uveitis, vitritis, scotoma, and photopsia.	
Diagnosis	B-scan Ultrasound: Diagnostic for intraocular cysticercosis; scolex within the cyst can be identified. Serology: Detection of anti-cysticercus antibodies via ELISA. Ophthalmic Examination: Identification of cysts in the AC or vitreous, signs of inflammation or retinal detachment.
Treatment	Ocular cysticercosis: Surgical Treatment: Early removal of cysts via pars plana vitrectomy (PPV) or transscleral approach depending on cyst location. Steroids: Systemic corticosteroids are used to control inflammation post-surgery. Medical Management: Albendazole and corticosteroids are used for systemic infestation but with caution in intraocular cases to avoid severe inflammatory reactions caused by dying cysts.

Ocular Toxoplasmosis (Option A): Fundoscopy shows focal necrotizing retinitis surrounded by a distinctive “headlight in the fog” appearance.

Endophthalmitis (Option B): It is usually associated with post-surgical infection, trauma, or endogenous spread. It presents with diffuse inflammation, hypopyon, and vitritis, not a well-defined cystic lesion.

Posterior uveitis due to TB (Option D): Tubercular uveitis often presents with granulomas, choroiditis, or vasculitis rather than a single cystic lesion with inflammation.

Solution for Question 6:

Correct Answer: C)Fuchs heterochromic uveitis

Explanation:

Amsler sign (aka Amsler-Verrey sign) is commonly seen in Fuchs heterochromic uveitis (FHU). It is characterized by hemorrhage in the anterior chamber after anterior chamber paracentesis or entry with a blade during anterior segment surgeries

Fuchs heterochromic uveitis:

Non-granulomatous anterior uveitis

Clinical manifestation:

- Typically unilateral
- Atypical anterior uveitis
- Posterior uveitis- less common
- Lightening of the colour of the affected iris- Heterochromia

[Image not available]

- Stellate keratic precipitates

keratic precipitates" data-author="Zemba, Mihail & Rogoz, Roxana-Elena & Zaharia, Alexandra & Dimirache, Andreea & Dumitrescu, Otilia-Maria & D■r■bu■, Diana-Maria." data-hash="4186" data-license="CC BY 2.0" data-source="https://www.researchgate.net/figure/Slit-lamp-image-of-stellate-keratic-precipitates-aright-eye-b-left-eye_fig2_355219657" data-tags="March2025" height="461" src="https://image.prepladder.com/notes/qds01CqZ7km8jvJupmLi1743127162.png" width="500" />

- Trabecular meshwork and iris display abnormal vessels with variable vitreous inflammation
- Amsler sign (aka Amsler-Verrey sign)- hemorrhage in the anterior chamber after anterior chamber paracentesis or entry with the blade during anterior segment surgeries. Small twig-like vessels in the angle likely give rise to the Amsler sign.

[Image not available]

- Posterior subcapsular cataract

Treatment:

- Anterior uveitis- topical steroid eye drops
- Posterior uveitis- systemic steroids

Solution for Question 7:

Correct Answer: C) Ivermectin

Explanation:

The patient's travel history, symptoms, and findings are consistent with onchocerciasis, caused by *Onchocerca volvulus*. The primary treatment for onchocerciasis is ivermectin, which kills the microfilariae and reduces the intensity of symptoms.

Onchocerciasis uveitis:

- It is parasitic infestation- filarial nematode worm
- Organism: *Onchocerca volvulus*
- Vector- black fly
- Also called river blindness

Pathophysiology:

- Microfilariae are mobile and reach the eyes
- Alive organisms cause little or no reaction
- When dead, they produce focal inflammation with reactive destruction of the tissues.

Whereas in case of loa loa alive, adult worms produce a non inflammatory reaction.

Clinical manifestation: (Option D)

- Sclerosing keratitis with anterior uveitis is common and leads to blindness
- Oncosercoma.
- Microfilariae is visualized in the anterior chamber
- Acute papular onchodermatitis

Acute papular onchodermatitis

acute papular onchodermatitis" data-author="Bari, Arfan & Rahman, S.B.. " data-hash="4178" data-license="NA" data-source="https://www.researchgate.net/figure/Acute-papular-onchodermatitis-in-a-young-child_fig6_287579898" data-tags="March2025" height="610" src="https://image.prepladder.com/notes/q3TujPtju59KTWDDReMr1743126813.png" width="500" />

keratitis with anterior uveitis" data-author="Brad H. Feldman, M.D., Paul O. Phelps, MD, FACS, Vinay A. Shah M.D., Vatinee Y. Bunya, MD, MSCE, Maria A. Woodward, MD, Colleen Halfpenny, M.D., Alan Palestine, MD, Jessica Shantha, MD, Marissa Larochelle, MD, Hubert H. Pham, MD MS" data-hash="4181" data-license="NA"

data-source="https://eyewiki.org/Onchocerciasis_%28African_River_Blindness%29"
data-tags="March2025" height="369" src="https://image.prepladder.com/notes/Hg8H8mWG7kQ0
1Dgm5vIN1743126881.png" width="500" />

Microfilariae

[Image not available]

Treatment:

- Ivermectin- reduces the occurrence of blindness and skin disease (Option C)
- Older treatments consisting of suramin and diethylcarbamazine are avoided (Option B)

Clindamycin + sulfadiazine (Option A): Used in Toxoplasmosis uveitis in combination or as part of quadruple therapy.

Solution for Question 8:

Correct Answer: C) Toxoplasma gondii

Explanation:

- The most probable diagnosis in this scenario is toxoplasmosis uveitis, which is caused by Toxoplasma gondii
- Fundoscopy shows focal necrotizing retinitis surrounded by a distinctive “headlight in the fog” appearance
- An immunocompromised person living with cats is a risk factor for toxoplasmosis

Toxoplasmosis uveitis:

- It is a protozoan infestation.
- Organism: Toxoplasma gondii
- Definitive host - cats
- Intermediate hosts - humans
- Systemic lesions are more common in immunocompromised patients

Clinical manifestation:

- Most common cause of posterior uveitis and focal necrotizing retinitis (90% of cases)

- Congenital Inactive disease - Healed punched out heavily pigmented chorioretinal scars in the macular area
- Inactive disease - Healed punched out heavily pigmented chorioretinal scars in the macular area
- Inactive disease - Healed punched out heavily pigmented chorioretinal scars in the macular area

[Image not available]

- In active inflammation - "headlight in the fog" appearance due to intense vitritis

[Image not available]

- Acquired Asymptomatic Same as congenital Punctate outer retinal toxoplasmosis
- Asymptomatic
- Same as congenital
- Punctate outer retinal toxoplasmosis
- Asymptomatic
- Same as congenital
- Punctate outer retinal toxoplasmosis

Treatment:

- Quadruple therapy: Pyrimethamine + Sulfadiazine + Clindamycin + Oral steroid
- Steroid eye drops for anterior uveitis

Cytomegalovirus (CMV) (Option A): It causes viral retinitis. The characteristic picture is areas of necrosis with areas of vasculitis, retinal hemorrhages, and loss of fundal details giving a pizza pie appearance.

Candida albicans (Option B): It causes multifocal chorioretinitis. The characteristic picture is multiple small, round, whitish areas, which may be associated with areas of hemorrhages with pale centers (Roth's spots).

albicans chorioretinitis" data-author="Jennifer I Lim MD, Monique Munro MD FRCSC, Pooja Bhat MD, Ghazala D. O'Keefe, MD, Katherine Chen, MD MS, Neelakshi Bhagat, MD, FACS" data-hash="4175" data-license="NA" data-source="https://eyewiki.org/Ocular_Candidiasis" data-tags="March2025" height="388" src="https://image.prepladder.com/notes/9U61ccYzc08qWJs26U981743126615.png" width="500" />

Histoplasma capsulatum (Option D): It causes fungal uveitis. The characteristic picture is choroidal neovascular membranes or disciform lesions at the macula with well-circumscribed peripheral chorioretinal scars in the absence of inflammatory signs in the aqueous and vitreous, and associated with peripapillary pigment epithelial atrophy.

Solution for Question 9:

Correct Answer: A) 1-d, 2-a, 3-b

Explanation:

Table rendering failed

Bacterial uveitis:

- Tuberculosis
- Syphilis
- Leprosy

Tubercular uveitis:

- Causes both anterior, intermediate, posterior, and panuveitis
- Occurs as acute non-granulomatous iridocyclitis or granulomatous anterior uveitis
- Granulomatous inflammation causes iris nodules in the pupillary margin (Koeppe nodule) or anterior mid-iris (Busacca nodule)
- Exudative non-granulomatous type of iritis is probably allergic or immunoinflammatory in nature

Koeppe nodule

[Image not available]

[Image not available]

Treatment: Rifampicin, isoniazid, ethambutol and pyrazinamide

Spirochetal uveitis:

- Organism: *Treponema pallidum*
- Anterior uveitis: It can present as granulomatous or non-granulomatous iritis typically in the secondary syphilis stage Iris roseola - dilated iris vessels that develop into nodules 'Plastic' iritis occurs in congenital syphilis and Jarisch–Herxheimer reaction
- It can present as granulomatous or non-granulomatous iritis typically in the secondary syphilis stage
- Iris roseola - dilated iris vessels that develop into nodules
- 'Plastic' iritis occurs in congenital syphilis and Jarisch–Herxheimer reaction

- It can present as granulomatous or non-granulomatous iritis typically in the secondary syphilis stage
- Iris roseola - dilated iris vessels that develop into nodules
- 'Plastic' iritis occurs in congenital syphilis and Jarisch–Herxheimer reaction

[Image not available]

- Posterior uveitis: It occurs as disseminated, peripheral, or diffuse choroiditis.
- It occurs as disseminated, peripheral, or diffuse choroiditis.
- Treatment: Systemic penicillin
- It occurs as disseminated, peripheral, or diffuse choroiditis.

Leprotic uveitis:

- Organism-*Mycobacterium leprae*
- It occurs as acute non granulomatous iritis or chronic granulomatous iritis
- Chronic granulomatous iritis occurs due to direct invasion of the organismal and is characterized by 'iris pearls' near the pupillary margin in a necklace form.

[Image not available]

- Small pearls enlarge and coalesce to form large pearls
- Other ocular manifestations: Conjunctivitis, episcleritis, exposure keratopathy because of involvement of the facial nerve, and neurotrophic keratopathy due to involvement of the trigeminal nerve
- Treatment: Dapsone Rifampicin, ofloxacin, clofazimine and minocycline in combination with dapsone
- Dapsone
- Rifampicin, ofloxacin, clofazimine and minocycline in combination with dapsone
- Dapsone
- Rifampicin, ofloxacin, clofazimine and minocycline in combination with dapsone

Solution for Question 10:

Correct Answer: B)Non-granulomatous anterior uveitis

Explanation:

The most likely diagnosis is anterior uveitis, and the extraintestinal manifestations of IBD (Crohn's disease here) include ocular complications. It is non-granulomatous type.

Uveitis in bowel disease:

- Inflammatory bowel disease Most common in Crohn's disease Ocular complications-conjunctivitis, episcleritis, scleritis, marginal keratitis, anterior uveitis, retinitis, retinal vascular occlusive disease, optic neuritis, and orbital inflammatory disease. Type: Non-granulomatous
- Most common in Crohn's disease
- Ocular complications-conjunctivitis, episcleritis, scleritis, marginal keratitis, anterior uveitis, retinitis, retinal vascular occlusive disease, optic neuritis, and orbital inflammatory disease.
- Type: Non-granulomatous
- Whipple's disease
- Most common in Crohn's disease
- Ocular complications-conjunctivitis, episcleritis, scleritis, marginal keratitis, anterior uveitis, retinitis, retinal vascular occlusive disease, optic neuritis, and orbital inflammatory disease.
- Type: Non-granulomatous

Posterior uveitis (Option C): Though IBD can also present with panuveitis or posterior uveitis, Presence of fine, grey keratic precipitates, anterior chamber cells, and flare on slit lamp examination is consistent with anterior uveitis.

Solution for Question 11:

Correct Answer: C)Progressive Outer Retinal Necrosis

Explanation:

Progressive Outer Retinal Necrosis is common in immunocompromised patients (e.g., those with AIDS and low CD4+ counts). Characterized by painless, rapidly spreading outer retinal necrosis with minimal vasculitis or vitritis, and the "pizza pie" appearance.

Retinal Necrosis" data-author="S Sudharshan, Sudha K Ganesh, Jyotirmay Biswas" data-hash="4167" data-license="CC BY 2.0" data-source="https://upload.wikimedia.org/wikipedia/commons/8/8d/PMID20029144_06_cytomegalovirus_retinitis_-_pizza_pie.png" data-tags="March2025" height="399" src="https://image.prepladder.com/notes/OT8RgBb2syVNX

gyQzMeh1743125826.png" width="500" />

Progressive Outer Retinal Necrosis (Option C)

- Severe necrotizing retinitis due to VZV.
- Occurs in immunocompromised patients, often with AIDS (CD4+ T-cells $<50/\text{mm}^3$).
- Symptoms: painless, rapid vision loss, rapidly spreading outer retinal necrosis with minimal vasculitis or vitritis.
- Complication: retinal detachment.

Viral Uveitis Overview

Herpes Zoster Ophthalmicus (HZO) (Option D)

- Affects the ophthalmic branch of the trigeminal nerve.
- Anterior uveitis occurs in 40-50% of cases, usually within 2 weeks of skin rash onset.
- Symptoms: mild iritis, vesicular rash on the nose, small KPs, mild aqueous flare, possible hemorrhagic hypopyon.
- Complications: iris atrophy, secondary glaucoma, complicated cataracts.

Herpes Simplex Virus (HSV) (Option B)

- Often linked with keratitis.
- May present with dendritic or geographical corneal ulcers, or disciform keratitis.
- Anterior uveitis can occur independently, presenting as mild non-granulomatous iridocyclitis due to hypersensitivity.

Cytomegalovirus (CMV)

- Common in immunocompromised individuals (e.g., AIDS, chemotherapy).
- Prevalence: 40% in AIDS patients.
- Occurs with CD4+ counts <50 cells/ mm^3 .
- HAART has reduced CMV retinitis incidence.
- Symptoms: may be asymptomatic, or cause decreased vision and floaters.
- Forms: Hemorrhagic Retinitis: Necrotic areas, vasculitis, retinal hemorrhages, "pizza pie" appearance. Granular Retinitis: Peripheral granular opacities, possible hemorrhage.
- Hemorrhagic Retinitis: Necrotic areas, vasculitis, retinal hemorrhages, "pizza pie" appearance.
- Granular Retinitis: Peripheral granular opacities, possible hemorrhage.
- Hemorrhagic Retinitis: Necrotic areas, vasculitis, retinal hemorrhages, "pizza pie" appearance.
- Granular Retinitis: Peripheral granular opacities, possible hemorrhage.

Acute Retinal Necrosis (ARN) (Option A)

- Rare; caused by VZV, HSV 1 & 2, or CMV (rarely).
- VZV is more common in older patients; HSV 1 & 2 in younger patients.
- Symptoms: decreased vision, floaters, ocular discomfort, pain, photophobia, anterior segment inflammation.
- Features: marked vitritis, focal peripheral necrotizing retinitis, occlusive arteritis, posterior pole typically spared.

Additional Notes

- Viral uveitis may be linked to bacterial uveitis and other infections such as chikungunya, dengue, West Nile virus, and rubella.

Solution for Question 12:

Correct Answer: D) The constrictor pupillae muscle is controlled by sympathetic nerves.

Explanation:

The constrictor pupillae muscle is controlled by parasympathetic nerves, not sympathetic.

Middle Coat of the Eye (Uveal Tract)

- General Characteristics: Contains most of the blood vessels in the eye. Contains melanin-producing cells. Consists of three parts: choroid, ciliary body, and iris.
- Contains most of the blood vessels in the eye.
- Contains melanin-producing cells.
- Consists of three parts: choroid, ciliary body, and iris.
- Contains most of the blood vessels in the eye.
- Contains melanin-producing cells.
- Consists of three parts: choroid, ciliary body, and iris.

Choroid:

- Origin: Neuroectoderm
- Location: Posterior part of the vascular coat.
- Characteristics: Lines the inner surface of the sclera. Connected anteriorly to the iris via the ciliary body. (Option A) Pierced posteriorly by the optic nerve.

- Lines the inner surface of the sclera.
- Connected anteriorly to the iris via the ciliary body. (Option A)
- Pierced posteriorly by the optic nerve.
- Arteries: Short ciliary arteries from the optic nerve area.
- Veins: Whorled arrangement forming 4-5 venae vorticosae, draining into ophthalmic veins.
- Attachment: Firmly attached to the retina; nourishes photoreceptors by diffusion.
- Layers: Suprachoroid Lamina (Lamina Fusca): Contains long posterior ciliary vessels and nerves. Vascular Lamina: Loose tissue with pigment cells, contains branches of ciliary arteries and veins forming venae vorticosae. Capillary Lamina: Fine capillary network nourishing retinal photoreceptors. Basal Lamina (Membrane of Bruch): Thin, transparent membrane
- Suprachoroid Lamina (Lamina Fusca): Contains long posterior ciliary vessels and nerves.
- Vascular Lamina: Loose tissue with pigment cells, contains branches of ciliary arteries and veins forming venae vorticosae.
- Capillary Lamina: Fine capillary network nourishing retinal photoreceptors.
- Basal Lamina (Membrane of Bruch): Thin, transparent membrane

Ciliary Body:

- Origin: Neuroectoderm
- Location: Posterior to the corneoscleral junction, anterior to the ora serrata.
- Ciliary Muscles: Smooth muscle with radial and circular fibers. (Option B)
- Function: Focuses the lens for near vision by altering tension on the suspensory ligaments, allowing the lens to become more spherical.

Iris:

- Origin: Neural crest cells and Neuroectoderm
- Location: Between the cornea and lens, with a central opening called the pupil.
- Attachment: Attached peripherally to the ciliary body; forms the iridocorneal angle (Option C).
- Actions: Constrictor Pupillae: Constricts the pupil. Dilator Pupillae: Dilates the pupil.
- Constrictor Pupillae: Constricts the pupil.
- Dilator Pupillae: Dilates the pupil.
- Constrictor Pupillae: Constricts the pupil.
- Dilator Pupillae: Dilates the pupil.

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Solution for Question 13:

Correct Answer: A) Start with systemic corticosteroids and consider long-term immunosuppressive therapy

Explanation:

Systemic corticosteroids can provide initial relief for uveitis symptoms in Behcet's disease, but for long-term management, immunosuppressive therapy is necessary due to the disease's chronic nature and poor long-term prognosis with corticosteroids alone.

Behcet's Disease

- Nature: Idiopathic multisystem disease
- Characteristics: Recurrent non-granulomatous uveitis Aphthous and genital ulcers Erythema multiforme
- Recurrent non-granulomatous uveitis
- Aphthous and genital ulcers
- Erythema multiforme
- Pathology: Obliterative vasculitis, likely due to circulating immune complexes
- Demographics: Common in young men with HLA-B51
- Recurrent non-granulomatous uveitis
- Aphthous and genital ulcers
- Erythema multiforme

Uveitis in Behcet's Disease

- Presentation: Bilateral and acute Recurrent iridocyclitis with hypopyon May include posterior uveitis, vitritis, periphlebitis retinae, and retinitis (white necrotic infiltrates)
- Bilateral and acute
- Recurrent iridocyclitis with hypopyon
- May include posterior uveitis, vitritis, periphlebitis retinae, and retinitis (white necrotic infiltrates)
- Bilateral and acute
- Recurrent iridocyclitis with hypopyon

- May include posterior uveitis, vitritis, periphlebitis retinae, and retinitis (white necrotic infiltrates)

Treatment

- Corticosteroids: Effective initially but not long-term
- Immunosuppressive Drugs: Methotrexate, azathioprine, cyclosporine for resistant or severe cases

Immediate initiation of antiviral therapy (Option B): Antiviral therapy is not appropriate as Behcet's disease is not caused by viral infection.

Use of topical antibiotics for ocular symptoms (Option C): Topical antibiotics are not effective since the uveitis is due to an autoimmune process.

Prescription of oral antihistamines for symptom relief (Option D): Oral antihistamines are not suitable for treating the autoimmune-related uveitis in Behcet's disease.

Solution for Question 14:

Correct Answer: B) Topical corticosteroids

Explanation:

Topical corticosteroids are the initial treatment for acute iridocyclitis.

Reiter's Syndrome and Uveitis

- Characteristics: Reiter's syndrome includes urethritis, arthritis, and conjunctivitis, often with iridocyclitis. Common in young men, especially those positive for HLA-B27.
- Ocular Features: Acute mucopurulent conjunctivitis, possibly with superficial punctate keratitis. Acute, non-granulomatous iridocyclitis occurs in 20-30% of cases.
- Acute mucopurulent conjunctivitis, possibly with superficial punctate keratitis.
- Acute, non-granulomatous iridocyclitis occurs in 20-30% of cases.
- Treatment: Iridocyclitis: Managed with topical, periocular, and systemic corticosteroids. Post Venereal form (likely due to Chlamydia): Treated with systemic tetracycline (250 mg, four times daily for 10 days).
- Iridocyclitis: Managed with topical, periocular, and systemic corticosteroids.

- Post Venereal form (likely due to Chlamydia): Treated with systemic tetracycline (250 mg, four times daily for 10 days).
- Acute mucopurulent conjunctivitis, possibly with superficial punctate keratitis.
- Acute, non-granulomatous iridocyclitis occurs in 20-30% of cases.
- Iridocyclitis: Managed with topical, periocular, and systemic corticosteroids.
- Post Venereal form (likely due to Chlamydia): Treated with systemic tetracycline (250 mg, four times daily for 10 days).

Solution for Question 15:

Correct Answer: B)Secondary glaucoma

Explanation:

- Secondary glaucoma is a known complication of anterior uveitis, often resulting from inflammatory trabeculitis, corticosteroid response, or synechial angle closure. In this case, the patient has posterior synechiae, which can lead to pupillary block and increased intraocular pressure (IOP).
- Chronic inflammation can also cause damage to the trabecular meshwork, leading to impaired aqueous outflow and secondary glaucoma.

Complication	Description	Additional Details
Complicated Cataract	Common in iridocyclitis, characterized by posterior subcapsular opacities with "polychromatic luster" and "bread-crumbs" appearance.	Matures faster with posterior synechiae presence.
Secondary Glaucoma	Can occur early or late due to iridocyclitis.	Early: Usually Open-angle. Blockage of trabecular meshwork by exudates/inflammatory cells. Late: Usually Closed. Pupillary block causes impaired aqueous humor flow.
Cyclitic Membrane	Late complication where exudates behind the lens become organized in acute plastic iridocyclitis.	
Choroiditis	Long-standing iridocyclitis can lead to choroiditis due to the continuity of	

	the iris, ciliary body, and choroid.	
Retinal Complications	Includes cystoid macular edema, macular scar/hole, epiretinal membrane, exudative retinal detachment, secondary periphlebitis retinae, retinal scars, sub-retinal fibrosis.	
Papillitis	Inflammation of the optic disc, seen in severe cases of iridocyclitis.	
Band-shaped Keratopathy	Common in children with Still's disease and chronic uveitis.	
Phthisis Bulbi	Final stage of chronic uveitis where the eye becomes soft, shrunken, and atrophic.	Without Shrinkage: Globe shape intact, vision loss, cataractous lens, retinal detachment, raised IOP early. With Shrinkage: Lowered IOP, collapsed anterior chamber, shrunken eyeball. With Disorganization: Significant eyeball reduction, scleral thickening, possible intraocular ossification.

- Retinal detachment is more commonly associated with posterior uveitis or severe panuveitis, where inflammatory damage to the retina and choroid leads to traction or exudative detachment.
- Anterior uveitis typically does not cause retinal detachment unless it is part of a more widespread inflammatory process.
- Vitreous hemorrhage occurs due to retinal neovascularization, trauma, or severe posterior uveitis.
- Anterior uveitis mainly affects the anterior segment of the eye, and there is no mention of retinal or choroidal involvement in this patient.
- Optic neuritis is typically associated with multiple sclerosis (MS), neuromyelitis optica (NMO), or infections, and it presents with retrobulbar pain and vision loss rather than the pain and photophobia seen in anterior uveitis.
- Anterior uveitis alone does not usually cause optic nerve inflammation.

Solution for Question 16:

Correct Answer: C) Iridocyclitis

Explanation:

Posterior uveitis is defined as the inflammation of the choroid and the retina. The components of posterior uveitis include the following,

- Choroiditis
- Retinitis
- Retinal vasculitis Periphlebitis - Venous sheathing Periarteritis - Hemorrhages
- Periphlebitis - Venous sheathing
- Periarteritis - Hemorrhages
- Periphlebitis - Venous sheathing
- Periarteritis - Hemorrhages

Causes of Posterior Uveitis:

Category	Examples
Infectious Causes	Viruses: CMV, HSV, herpes zoster, rubella, measles
Bacteria: Mycobacterium tuberculosis, Brucella spp., Borrelia burgdorferi, other gram-positive and gram-negative bacteria	
Fungi: Candida spp., Histoplasma capsulatum, Blastomyces spp., Cryptococcus neoformans, Aspergillus spp.	
Parasites: Toxoplasma gondii, Toxocara spp., Onchocerca volvulus, Acanthamoeba spp.	
Non-Infectious Causes	Autoimmune: Behçet's disease, VKH syndrome, SLE, Wegener's granulomatosis, sympathetic ophthalmia
Malignancy: Intraocular lymphoma, malignant melanoma, leukemia	
Unknown Etiology: Sarcoidosis, serpiginous choroiditis, Birdshot chorioretinopathy, retinal pigmented epitheliopathy	

Clinical Features of Posterior Uveitis

Type	Features
Symptoms	Defective Vision: Ranges from mild blurriness to severe vision loss
Photopsia: Flashes of light from retinal irritation	
Floaters: Black spots or cobwebs due to vitreous debris	

Metamorphopsia: Distorted vision, with straight lines appearing wavy	
Scotoma: Blind spot or reduced vision area in the visual field	
Signs	Vitreous Opacities: Fine or coarse opacities, "snowballs", or strands in the vitreous humor
Active Chorioretinal Lesions: Pale yellow or white, raised areas with blurred margins; possible retinal haziness and vasculitis	
Healed Chorioretinal Lesions: White, atrophic areas with choroidal thinning; black pigment clumping at edges	

Note: Posterior uveitis is typically painless, unlike anterior uveitis, which often presents with pain, redness, and photophobia.

Solution for Question 17:

Correct Answer: C) Periocular injection of triamcinolone

Explanation:

The most appropriate initial treatment for intermediate uveitis, especially when it is idiopathic and presents with significant vitreous inflammation, is the periocular injection of triamcinolone. This approach targets inflammation locally with fewer systemic side effects compared to systemic steroids.

Intermediate Uveitis: Overview, Investigations, and Treatment

- Intermediate uveitis is an inflammation affecting the pars plana, peripheral retina, choroid, and vitreous base, comprising 10% of all uveitis cases, and 20% in children.
- It's bilateral in 80% of cases and equally affects both genders.
- The cause is idiopathic in 85% of cases, typically occurring in the 2nd and 4th decades of life.
- Known causes include tuberculosis, syphilis, sarcoidosis, Lyme disease, and multiple sclerosis.

Investigations

- Slit-lamp examination: Detects subtle anterior segment signs like low-grade flare, cells in the aqueous humor, and keratic precipitates.
- Indirect ophthalmoscopy: Visualizes the retina and vitreous, identifying snowball opacities, vitreous condensation, and snowbanking near the ora serrata.
- Goldmann three-mirror examination: Detailed peripheral retina and pars plana examination for snowbanking.
- Fluorescein angiography: Assesses retinal vessel leakage, especially in snowbanking or macular edema.
- Systemic investigations: Blood tests, chest X-ray, and lumbar puncture to identify underlying conditions like sarcoidosis, tuberculosis, and multiple sclerosis.

Treatment (Kaplan's Modified 4-Step Protocol)

Step	Treatment	Details
1	Periocular & Systemic Steroids	Triamcinolone injection: 40 mg, 3 injections spaced 3 weeks apart. Oral corticosteroids: Typically considered if periocular steroids are insufficient. Gradual tapering is done. (Option B)
2	Immunosuppressive Drugs	Agents: Cyclosporine, azathioprine, methotrexate, cyclophosphamide. Close monitoring due to potential side effects.
3	Cryotherapy/Indirect Laser Photocoagulation (Option D)	Cryotherapy: Controlled cold-induced injury. Used for snowbanking when less invasive treatments fail. Laser photocoagulation: Focused light beam treatment.
4	Pars Plana Vitrectomy (Option A)	Surgery: For severe cases or complications (e.g., tractional retinal detachment). Removes vitreous gel, and clears debris.

Complications to Monitor

- Cataract formation
- Cystoid macular edema
- Band keratopathy
- Secondary glaucoma
- Vitreous hemorrhage

- Tractional retinal detachment

Regular follow-up is essential to monitor and adjust treatment, minimizing complications.

Solution for Question 18:

Correct Answer: C) Intermediate uveitis

Explanation:

The presence of floaters, anterior vitritis, and white snowball-like exudates near the ora serrata with mild peripheral periphlebitis are characteristic of intermediate uveitis (pars planitis).

Intermediate Uveitis: Causes and Features

Definition: Intermediate uveitis, also known as pars planitis, primarily affects the pars plana of the ciliary body and the peripheral choroid. It mostly occurs in children and young adults, with a higher incidence in females.

Causes

- Etiology: Nonspecific inflammation with an unknown cause.
- Onset: Insidious

Features

Common Features	Less Frequent Features
Floaters	Macular edema
Vision deterioration due to anterior vitreous opacities	Papillitis or disc oedema
Minimal aqueous flare with occasional keratic precipitates	Retrolenticular cyclitic membranes
Anterior Vitritis	Vitreous hemorrhage
White snowball-like exudates near the ora serrata	Tractional retinal detachment (rare)
Mild peripheral periphlebitis	

uveitis snowballs and snowbanking" data-author="NA" data-hash="4157" data-license="NA" data-source="NA" data-tags="March2025" height="293" src="https://image.prepladder.com/notes/RB2UOY6dT4uLxhsiudaz1743125257.png" width="500" />

- Snowball-like exudates may coalesce into a "snowbank" appearance.
- 80% of cases do not require treatment.
- Chronic cases may require corticosteroids; however, there is poor response to therapy.
- Immunosuppressants are reserved for severe cases unresponsive to steroids.

Differential Diagnosis

- CMV Retinitis: Typically presents with focal retinochoroiditis rather than the diffuse features seen in intermediate uveitis. (Option A)
- Peripheral toxocariasis
- Tuberculosis: Typically causes anterior uveitis. (Option B)
- Lyme Uveitis: May cause uveitis but typically presents with more acute dermatological symptoms, and the ocular findings are not as specific to intermediate uveitis as in this case. More commonly seen in the relapsing-remitting type. (Option D)
- Sarcoidosis

Solution for Question 19:

Correct Answer: C) Conduct an HLA typing and begin treatment with topical prednisolone.

Explanation:

The findings are consistent with a case of anterior uveitis.

- Investigation: HLA typing is important in identifying associated systemic conditions that may cause anterior uveitis, such as Behçet's disease or VKH syndrome.
- Treatment: Topical prednisolone is a standard treatment for reducing inflammation in anterior uveitis.

Perform a chest X-ray and initiate treatment with topical cycloplegics (Option A): While a chest X-ray can be useful for diagnosing sarcoidosis, it usually presents with mutton fat KP's not fine ones. Cycloplegics are used to relieve pain and prevent synechiae but do not address inflammation directly.

Order a serum ACE level and start treatment with non-steroidal voclosporin (Option B): Serum ACE levels are specific for diagnosing sarcoidosis, and voclosporin is used for non-steroidal treatment but is not the first-line treatment.

Arrange for an OCT scan and administer systemic interferon alpha (Option D): OCT scans are useful for checking Cystoid Macular Edema. Systemic interferon alpha is used in severe, chronic cases.

Investigations for Anterior Uveitis:

Ocular Investigations:

- USG-B-Scan: To assess the posterior segment of the eye for any abnormalities.
- Fundus Fluorescein Angiography (FFA): To evaluate the retinal and choroidal circulation and detect any leakage or abnormalities.
- Optical Coherence Tomography (OCT)

Systemic Investigations:

- Syphilis Tests: VDRL TPHA: Treponema pallidum Hemagglutination Assay.
- VDRL
- TPHA: Treponema pallidum Hemagglutination Assay.
- Sarcoidosis: Serum ACE Levels: Elevated in sarcoidosis. Chest X-Ray: To identify any pulmonary involvement or sarcoidosis.
- Serum ACE Levels: Elevated in sarcoidosis.
- Chest X-Ray: To identify any pulmonary involvement or sarcoidosis.
- Mantoux Test / QuantiFERON Test: To detect tuberculosis infection.
- X-Ray Spine: To check for ankylosing spondylitis or other spinal abnormalities.
- HLA Typing: HLA-B27: Associated with various types of arthritis, including ankylosing spondylitis. HLA-B5 and HLA-B51: Associated with Behçet's disease. HLA-DR4: Associated with VKH syndrome and sympathetic ophthalmia.
- HLA-B27: Associated with various types of arthritis, including ankylosing spondylitis.
- HLA-B5 and HLA-B51: Associated with Behçet's disease.
- HLA-DR4: Associated with VKH syndrome and sympathetic ophthalmia.
- VDRL
- TPHA: Treponema pallidum Hemagglutination Assay.
- Serum ACE Levels: Elevated in sarcoidosis.
- Chest X-Ray: To identify any pulmonary involvement or sarcoidosis.
- HLA-B27: Associated with various types of arthritis, including ankylosing spondylitis.
- HLA-B5 and HLA-B51: Associated with Behçet's disease.
- HLA-DR4: Associated with VKH syndrome and sympathetic ophthalmia.

Treatment for Anterior Uveitis:

- Topical Steroids (Treatment of Choice): Prednisolone: Commonly used for reducing inflammation. Loteprednol: Less potential for increasing intraocular pressure.
- Prednisolone: Commonly used for reducing inflammation.
- Loteprednol: Less potential for increasing intraocular pressure.
- Cycloplegics (Topical): Rest the ciliary muscles to relieve pain. Prevent and break posterior synechiae (adhesions of the iris to the lens). Improve blood supply, leading to better healing.
- Rest the ciliary muscles to relieve pain.
- Prevent and break posterior synechiae (adhesions of the iris to the lens).
- Improve blood supply, leading to better healing.
- Non-Steroidal Treatment: Voclosporin: Part of the Luminata program, used as a non-steroidal option.
- Voclosporin: Part of the Luminata program, used as a non-steroidal option.
- For Chronic or Vision-Threatening Uveitis: Monoclonal Antibodies: Adalimumab: TNF-alpha inhibitor, used for severe or chronic cases. Infliximab: TNF-alpha inhibitor. Interferon Alpha: An option for severe cases, helps modulate the immune response.
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- Adalimumab: TNF-alpha inhibitor, used for severe or chronic cases.
- Infliximab: TNF-alpha inhibitor.
- Interferon Alpha: An option for severe cases, helps modulate the immune response.
- Prednisolone: Commonly used for reducing inflammation.
- Loteprednol: Less potential for increasing intraocular pressure.
- Rest the ciliary muscles to relieve pain.
- Prevent and break posterior synechiae (adhesions of the iris to the lens).
- Improve blood supply, leading to better healing.
- Voclosporin: Part of the Luminata program, used as a non-steroidal option.
- Monoclonal Antibodies: Adalimumab: TNF-alpha inhibitor, used for severe or chronic cases. Infliximab: TNF-alpha inhibitor.
- Adalimumab: TNF-alpha inhibitor, used for severe or chronic cases.
- Infliximab: TNF-alpha inhibitor.
- Interferon Alpha: An option for severe cases, helps modulate the immune response.
- Adalimumab: TNF-alpha inhibitor, used for severe or chronic cases.
- Infliximab: TNF-alpha inhibitor.

Solution for Question 20:

Correct Answer: C) Anterior Uveitis

Explanation:

- Anterior uveitis is characterized by the presence of fine, grey keratic precipitates (KPs) on the corneal endothelium, a hazy anterior chamber (aqueous flare), and a constricted pupil.

Anterior Uveitis

Definition: Most common type of uveitis affecting the anterior segment of the eye.

Clinical Features:

- Pain
- Redness
- Photophobia (sensitivity to light)
- Blepharospasm (involuntary blinking)
- Serous discharge
- Blurring and diminished vision

Examination Findings:

- Keratic Precipitates (KPs): Non-Granulomatous: Fine, grey deposits; composed of lymphocytes Granulomatous: Large, greasy-looking deposits known as "mutton fat KPs"; composed of macrophages Location: Typically found in the lower cornea due to convection currents in aqueous humor; this area is known as Arlt's triangle
- Non-Granulomatous: Fine, grey deposits; composed of lymphocytes
- Granulomatous: Large, greasy-looking deposits known as "mutton fat KPs"; composed of macrophages
- Location: Typically found in the lower cornea due to convection currents in aqueous humor; this area is known as Arlt's triangle
- Non-Granulomatous: Fine, grey deposits; composed of lymphocytes
- Granulomatous: Large, greasy-looking deposits known as "mutton fat KPs"; composed of macrophages
- Location: Typically found in the lower cornea due to convection currents in aqueous humor; this area is known as Arlt's triangle

[Image not available]

- Aqueous Cells: Indicator of active inflammation
- Indicator of active inflammation
- Aqueous Flare: Haziness in the anterior chamber due to leakage of proteins from inflamed tissues
- Haziness in the anterior chamber due to leakage of proteins from inflamed tissues
- Muddy Iris: Edema in the iris caused by protein leakage
- Edema in the iris caused by protein leakage
- Miosis: Constricted pupil due to toxins released from inflamed tissue
- Constricted pupil due to toxins released from inflamed tissue
- Nodules on the Iris: Koeppe's Nodules: At the pupillary border; seen in both granulomatous and non-granulomatous uveitis Busacca Nodules: At the base of the iris; seen in granulomatous uveitis
- Koeppe's Nodules: At the pupillary border; seen in both granulomatous and non-granulomatous uveitis
- Busacca Nodules: At the base of the iris; seen in granulomatous uveitis
- Indicator of active inflammation
- Haziness in the anterior chamber due to leakage of proteins from inflamed tissues
- Edema in the iris caused by protein leakage
- Constricted pupil due to toxins released from inflamed tissue
- Koeppe's Nodules: At the pupillary border; seen in both granulomatous and non-granulomatous uveitis
- Busacca Nodules: At the base of the iris; seen in granulomatous uveitis

Koeppe's Nodules

[Image not available]

Complications: (Findings)

- Posterior Synechiae: Iris adheres to the lens; may lead to festooned-shaped pupil

[Image not available]

- Seclusio Pupil/Ring Synechiae: Adhesion along the entire pupillary border; can lead to Iris Bombe and peripheral anterior synechiae, causing angle closure glaucoma
- Occlusio Pupil: Covers the whole surface of the lens
- Cyclitic Membrane
- Rubeosis Iridis: New vessel formation on the iris

Herpes Simplex Keratitis (Option A): Presents with dendritic corneal ulcers, not fine, grey keratic precipitates or a hazy anterior chamber. It also usually does not involve a constricted pupil.

Conjunctivitis (Option B): Does not cause keratic precipitates , hazy anterior chamber and a constricted pupil.

Acute Angle Closure Glaucoma (Option D): Presents with a mid-dilated or fixed, mid-dilated pupil, severe pain, redness, and vision changes.

Solution for Question 21:

Correct Answer: B) Behcet's Disease

Explanation:

Behcet's Disease is a known systemic association with anterior uveitis, characterized by recurrent oral and genital ulcers, and uveitis.

Anterior uveitis, also known as iritis, is an inflammatory condition of the eye, including the iris and ciliary body.

Systemic Associations and Masquerades of Anterior Uveitis

- Infectious Causes: Herpes Infections: Varicella Zoster, Herpes Simplex Virus (HSV)
Tuberculosis (TB) Syphilis Leprosy
- Herpes Infections: Varicella Zoster, Herpes Simplex Virus (HSV)
- Tuberculosis (TB)
- Syphilis
- Leprosy
- Herpes Infections: Varicella Zoster, Herpes Simplex Virus (HSV)
- Tuberculosis (TB)
- Syphilis
- Leprosy
- Non-Infectious Causes: With Systemic Association: Sarcoidosis Behçet's Disease HLA-B27 Associated Arthritis Inflammatory Bowel Disease: Ulcerative Colitis (UC) > Crohn's Disease
Without Systemic Association: Drug-Induced: Pilocarpine, prostaglandin analogues
- With Systemic Association: Sarcoidosis Behçet's Disease HLA-B27 Associated Arthritis Inflammatory Bowel Disease: Ulcerative Colitis (UC) > Crohn's Disease
- Sarcoidosis

- Behçet's Disease
- HLA-B27 Associated Arthritis
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- Without Systemic Association: Drug-Induced: Pilocarpine, prostaglandin analogues
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- Inflammatory Bowel Disease: Ulcerative Colitis (UC) > Crohn's Disease
- Sarcoidosis
- Behçet's Disease
- HLA-B27 Associated Arthritis
- Inflammatory Bowel Disease: Ulcerative Colitis (UC) > Crohn's Disease
- Without Systemic Association: Drug-Induced: Pilocarpine, prostaglandin analogues
- Drug-Induced: Pilocarpine, prostaglandin analogues
- Sarcoidosis
- Behçet's Disease
- HLA-B27 Associated Arthritis
- Inflammatory Bowel Disease: Ulcerative Colitis (UC) > Crohn's Disease
- Drug-Induced: Pilocarpine, prostaglandin analogues
- Masquerade Syndromes: Definition: Conditions that mimic the symptoms of anterior uveitis but are not primarily inflammatory in nature. Neoplastic Masquerades: Lymphomas Anterior Segment Melanoma Retinoblastoma (Option C) Metastasis Non-Neoplastic Masquerades: Intraocular Foreign Body (Option A) Old Retinal Detachment Pigment Dispersion Syndrome (Option D) Coats' Disease
- Definition: Conditions that mimic the symptoms of anterior uveitis but are not primarily inflammatory in nature.
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- Metastasis
- Non-Neoplastic Masquerades: Intraocular Foreign Body (Option A) Old Retinal Detachment Pigment Dispersion Syndrome (Option D) Coats' Disease
- Intraocular Foreign Body (Option A)
- Old Retinal Detachment

- Pigment Dispersion Syndrome (Option D)
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- Intraocular Foreign Body (Option A)
- Old Retinal Detachment
- Pigment Dispersion Syndrome (Option D)
- Coats' Disease
- Lymphomas
- Anterior Segment Melanoma
- Retinoblastoma (Option C)
- Metastasis
- Intraocular Foreign Body (Option A)
- Old Retinal Detachment
- Pigment Dispersion Syndrome (Option D)
- Coats' Disease

Solution for Question 22:

Correct Answer: B) Chronic uveitis is classified as lasting more than 6 months without interruption.

Explanation:

Chronic uveitis is defined as uveitis that lasts more than 3 months, not 6 months.

Classification Based on Time of Inflammatory Activity

Acute Uveitis: (Option A)

- Duration: < 3 months
- Onset: Sudden

Chronic Uveitis:

- Duration: > 3 months

Recurrent Uveitis:(Option C)

- Characterized by repeated episodes
- Includes periods of inactivity between episodes

Remission:

- No inflammatory activity for ≥ 3 months (Option D)
-

Previous Year Questions

1. Which structures are affected in a pregnant woman who exhibits fever, retroorbital pain, headache, pulsatile proptosis of the right eye, and tinnitus, while having normal blood pressure and fundus? Abducens nerve Oculomotor nerve Internal carotid artery Cavernous sinus

- A. A,B,C
- B. B,C,D
- C. A,C,D
- D. A,B,C,D

2. Iritis is least associated with all of the following diseases:

- A. Systemic Lupus Erythematosus
- B. Behcet's Disease
- C. Psoriatic Arthropathy
- D. Ulcerative colitis

3. A patient presents with sudden diminution of vision, difficulty looking towards the light, and circumcorneal congestion with a hypopyon. What is the most likely diagnosis?

- A. Acute conjunctivitis
- B. Acute angle closure glaucoma
- C. Episcleritis
- D. Acute anterior uveitis

4. What will be observed in a patient diagnosed with sympathetic ophthalmitis, and a history of penetrating eye injury?

- A. Acute anterior uveitis
 - B. Pars planitis
 - C. Panuveitis
 - D. Chronic anterior uveitis
-

5. On ophthalmologic examination, the presence of Koeppe's nodules is observed in a patient diagnosed with sarcoidosis. In which area of the eye are these nodules typically observed?

- A. Cornea
 - B. Conjunctiva
 - C. Iris
 - D. Retina
-

6. A 25-year-old male patient with a penetrating injury to the left eye was treated with corneoscleral suturing. Two months later, he returned with complaints of reduced near vision, pain, redness, and sensitivity to light in his right eye. Direct ophthalmoscopy revealed nodules in the fundus. Which part of the eye is affected?

- A. Uvea
- B. Cornea
- C. Conjunctiva
- D. Sclera

7. Which medication can be administered to a patient diagnosed with acute anterior uveitis, presenting with symptoms of photophobia and blurred vision?

- A. Moxifloxacin
 - B. Atropine
 - C. Acetazolamide
 - D. Methotrexate
-

8. What is the specific sign that is pathognomonic for acute anterior uveitis among the options provided?

- A. Aqueous flare
 - B. Keratic precipitates
 - C. Festooned pupil
 - D. Bussaca's nodules
-

9. An 11-month-old baby was brought by his mother with complaints of the left upper eyelid moving up and down while breastfeeding that disappears when the baby is not breastfeeding. She gave a history of the child being diagnosed with left eye ptosis. What is this condition called?

- A. Symblepharon
 - B. Blepharophimosis syndrome
 - C. Lagophthalmos
 - D. Marcus -Gunn jaw-winking
-

10. Which of the following is not an ocular manifestation of dengue?

- A. Cataract
 - B. Subconjunctival hemorrhage
 - C. Uveitis
 - D. Retinal hemorrhage
-

11. Which eye condition is commonly associated with sarcoidosis?

- A. Cataract
 - B. Ectopia lentis
 - C. Anterior uveitis
 - D. Glaucoma
-

12. A 26-year-old patient came with complaints of blurring of vision, photophobia, redness, pain in the left eye, and joint pain. Upon examination keratic precipitates and aqueous cells were observed. Which of the following major histocompatibility complex is be associated with this condition?

- A. HLA B27
 - B. HLA B51
 - C. HLA B47
 - D. HLA B57
-

13. What is the likely diagnosis in a patient complaining of blurred vision along with aqueous flares and keratic precipitates found on examination?

- A. Intermediate uveitis
 - B. Posterior uveitis
 - C. Iridocyclitis
 - D. Toxoplasma uveitis
-

14. Sympathetic ophthalmia is seen after _____?

- A. Penetrating trauma
 - B. Blunt trauma
 - C. Chemical injury
 - D. Ocular infection
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	4
Question 4	3
Question 5	3
Question 6	1
Question 7	2
Question 8	2
Question 9	4
Question 10	1
Question 11	3
Question 12	1
Question 13	3
Question 14	1

Solution for Question 1:

Correct Option D - A,B,C,D:

- The scenario suggests a condition known as Cavernous Sinus Thrombosis (CST). This is a rare but serious condition often caused by the spread of a bacterial infection. The cavernous sinuses are hollow spaces located at the base of the brain, behind each eye socket. Blood clot formation in these spaces leads to CST.

- Abducens Nerve (A): The Abducens nerve runs through the cavernous sinus. This nerve controls the lateral rectus muscle, which turns the eye outward. Damage to this nerve can cause double vision, a symptom consistent with the presentation.
- Oculomotor Nerve (B): The Oculomotor nerve also traverses the cavernous sinus and controls most of the eye muscles. Impairment can lead to drooping eyelids and dilated pupils.
- Internal Carotid Artery (C): This major blood vessel supplies blood to the brain and passes through the cavernous sinus. Inflammation or a blood clot can cause pulsatile proptosis (bulging of the eye) and tinnitus (ringing in the ears).
- Cavernous Sinus (D): As the site of the suspected thrombosis, the cavernous sinus is the structure directly involved. This condition can lead to fever, headache, and retroorbital pain.
- The correct answer, therefore, is option D (A,B,C,D) as the symptoms presented suggest a condition involving the Abducens nerve, Oculomotor nerve, Internal Carotid artery, and Cavernous sinus.

Incorrect Choices:

Option A - A,B,C: This option neglects to mention the cavernous sinus (D), which is the primary site of involvement.

Option B - B,C,D: This selection overlooks the involvement of the Abducens nerve (A), which is a key element in this case.

Option C - A,C,D: This option disregards the Oculomotor nerve (B), a critical structure in this scenario.

Solution for Question 2:

Correct Option A- Systemic Lupus Erythematosus: SLE is the disease least commonly associated with iritis. Rheumatoid arthritis has the 2nd least chance of showing iritis in systemic diseases after SLE

Incorrect Options:

Option B- Behcet's Disease: It is not associated with arthritis but has HLA B5 positive. Patients present with a triad of oral ulcers, genital ulcers, and uveitis(which will further involve iris)

Option C- Psoriatic Arthropathy: This is incorrect as patients with psoriatic arthropathy will have iridocyclitis.

Option D- Ulcerative colitis: This is incorrect as patients with ulcerative colitis will have iridocyclitis.

Solution for Question 3:

Correct Option D - Acute anterior uveitis:

- The clinical presentation of the patient is suggestive of acute anterior uveitis. The presence of a hypopyon is often seen in conditions like Behcet disease, which can typically cause non-granulomatous anterior uveitis.

Incorrect Options:

Options A, B, and C do not present with a hypopyon.

Solution for Question 4:

Correct Option C - Panuveitis:

- Sympathetic ophthalmitis is a bilateral granulomatous panuveitis occurring after penetrating trauma to one eye.
- Panuveitis involves all three uveal tissues: the iris, ciliary body, and choroid.
- The primary site of inflammation are Anterior chamber Vitreous Retina Choroid
- Anterior chamber
- Vitreous
- Retina
- Choroid
- Anterior chamber
- Vitreous
- Retina
- Choroid

Incorrect Options:

- Options A, B, and D are incorrect. Sympathetic Ophthalmitis is a granulomatous panuveitis.

Solution for Question 5:

Correct Option C: Iris

- Koeppe's nodules are small, yellow-white nodules that appear on the surface of the iris observed in granulomatous uveitis like sarcoidosis, tuberculosis, Vogt-Kanayagi-Harada's disease, and sympathetic ophthalmitis .
- They are typically located at the pupillary margin, which is the border between the iris and the pupil.
- These nodules are composed of collections of inflammatory cells.

[Image not available]

Incorrect Options:

- Options A, B, and D are incorrect. Koeppe's nodules are present at the pupillary margin of the iris.

Solution for Question 6:

Correct Option A - Uvea:

- The clinical presentation of penetrating injury to the left eye and subsequent reduced near vision, pain, redness, and photophobia along with nodules on fundus i.e., Dalen-Fuchs nodules are suggestive of Sympathetic ophthalmitis.
- Dalen -Fuchs nodules are proliferation of the pigment epithelium of the iris and ciliary body to form nodular aggregations in sympathetic ophthalmitis.
- This is due to an autoimmune reaction towards uveal tissue and the antigen responsible is the retinal S antigen.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 7:

Correct Option B - Atropine:

Treatment of acute anterior uveitis includes the following.

- Treatment of choice: Topical steroids like prednisolone, difluprednate, and loteprednol to reduce inflammation.
- Cycloplegics: Topical cycloplegics like atropine, homatropine, or cyclopentolate. Cycloplegics give rest to ciliary muscles and, therefore, relieve pain. Prevents posterior synechiae. Improves blood supply- Leads to better healing
- Cycloplegics give rest to ciliary muscles and, therefore, relieve pain.
- Prevents posterior synechiae.
- Improves blood supply- Leads to better healing
- Cycloplegics give rest to ciliary muscles and, therefore, relieve pain.
- Prevents posterior synechiae.
- Improves blood supply- Leads to better healing

Incorrect Options:

Option A - Moxifloxacin: Moxifloxacin is an antibiotic commonly used to treat bacterial infections.

Option C - Acetazolamide: Acetazolamide is a medication primarily used to treat glaucoma by reducing the production of aqueous humor and lowering intraocular pressure.

Option D - Methotrexate: Methotrexate is an immunosuppressive medication commonly used in the treatment of various autoimmune and inflammatory conditions. It may be prescribed for severe or chronic cases of uveitis that do not respond to other treatments.

Solution for Question 8:

Correct Option B - Keratic precipitates:

- Keratic precipitates (KPs) are the pathognomonic sign of anterior uveitis.
- Keratic precipitates are proteinaceous cellular deposits occurring at the back of the cornea.
- Mostly, these are arranged in a triangular fashion (Arlt's triangle), occupying the centre and inferior part of the cornea due to convection currents in the aqueous humor

Incorrect Options:

Option A - Aqueous flare: Aqueous flare refers to the presence of protein and inflammatory cells in the anterior chamber of the eye. It is the earliest sign of acute anterior uveitis.

Option C - Festooned pupil: Festooned pupil, also known as an irregular or distorted pupil is observed in acute anterior uveitis results from segmental posterior synechiae formation, It is not a pathognomonic sign of acute anterior uveitis.

Option D - Bussaca's nodules: Bussaca's nodules are nodular formations on the anterior surface of the iris. They are typically associated with chronic uveitis, particularly in conditions such as granulomatous uveitis.

Solution for Question 9:

Correct Option D - Marcus -Gunn jaw-winking:

- The clinical presentation is suggestive of Marcus-Gunn jaw-winking characterized by retraction of the ptotic lid in conjunction with stimulation of the ipsilateral pterygoid muscles by chewing, sucking, opening the mouth, or contralateral jaw movement. Marcus Gunn Jaw winking syndrome (MGJWS) is a congenital ptosis with an aberrant connection between the LPS and Lateral pterygoid muscle. Treatment is disinsertion of LPS with sling operation.

Incorrect Options:

Option A - Symblepharon: Symblepharon is the adhesion between the palpebral and bulbar conjunctiva due to burns, chemical injuries, surgery, and ulcer.

Option B - Blepharophimosis syndrome: Blepharophimosis syndrome is a congenital anomaly characterized by blepharophimosis (decreased horizontal length), ptosis, congenital ectropion, and telecanthus.

Option C - Lagophthalmos: Lagophthalmos is the inability to close the eyelids voluntarily.

Solution for Question 10:

Correct Option A - Cataract:

- "Cataract" is not an ocular manifestation typically associated with dengue fever.

Incorrect Options:

Option B - Subconjunctival hemorrhage: Subconjunctival hemorrhage is a common ocular manifestation seen in dengue fever. It is characterized by the presence of bright red patches or spots on the white part of the eye (conjunctiva).

Option C - Uveitis: Uveitis, which refers to inflammation of the uvea, the middle layer of the eye, can occur in dengue fever. Uveitis can present with symptoms such as eye pain, redness, sensitivity to light, and blurred vision.

Option D - Retinal hemorrhage: Retinal hemorrhage can occur in dengue fever, particularly in severe cases or in the presence of associated complications like dengue hemorrhagic fever (DHF) or dengue shock syndrome (DSS).

Solution for Question 11:

Correct option: C

- Sarcoidosis is a systemic inflammatory disease that can affect multiple organs, including the eyes. Anterior uveitis, characterized by inflammation of the anterior chamber of the eye, is a common ocular manifestation of sarcoidosis.
- Other Ocular Features of Sarcoidosis. Sarcoid nodules on the sclera or episclera Interstitial keratitis Band-shaped keratopathy because of hypercalcemia. Candle wax dripping: Segmental or nodular periphlebitis. Lander's sign: Pre-retinal nodules. Heerfordt syndrome: Sarcoidosis with uveo-parotitis. Lofgren syndrome: Sarcoidosis with anterior uveitis, bilateral hilar lymphadenopathy, erythema nodosum.
- Sarcoid nodules on the sclera or episclera

- Interstitial keratitis
- Band-shaped keratopathy because of hypercalcemia.
- Candle wax dripping: Segmental or nodular periphlebitis.
- Lander's sign: Pre-retinal nodules.
- Heerfordt syndrome: Sarcoidosis with uveo-parotitis.
- Lofgren syndrome: Sarcoidosis with anterior uveitis, bilateral hilar lymphadenopathy, erythema nodosum.
- Sarcoid nodules on the sclera or episclera
- Interstitial keratitis
- Band-shaped keratopathy because of hypercalcemia.
- Candle wax dripping: Segmental or nodular periphlebitis.
- Lander's sign: Pre-retinal nodules.
- Heerfordt syndrome: Sarcoidosis with uveo-parotitis.
- Lofgren syndrome: Sarcoidosis with anterior uveitis, bilateral hilar lymphadenopathy, erythema nodosum.

Incorrect Options:

- Options A, B, and D are incorrect. These ocular features are not associated with sarcoidosis.

Solution for Question 12:

Correct Option A: HLA B27

- The clinical presentation of blurring vision, photophobia, redness, pain, and keratic precipitates are suggestive of anterior uveitis.
- Patients with acute anterior uveitis with joint pain is suggestive of arthritis associated with uveitis. The disease affects young males who are positive for HLA B27. Among patients with ankylosing spondylitis, 90% of them belong to the HLA-B27 antigenic group.

Incorrect Options:

Option B: HLA B51: Is associated with Behcet's syndrome with anterior uveitis. Behcet's syndrome is characterized by bilateral recurrent oral ulcers, genital ulcers, uveitis, and skin lesions.

Option C: HLA B47: Is associated with congenital adrenal hyperplasia due to 21-hydroxylase deficiency. It is characterized by symptoms like diarrhea, vomiting, arrhythmias, and low blood pressure.

Option D: HLA B57: Is associated with the detection of abacavir hypersensitivity and slow progression of HIV.

Solution for Question 13:

Correct Option C - Iridocyclitis:

- The clinical features in the question stem is consistent with the diagnosis of Iridocyclitis
- Anterior Uveitis, involves Iris and pars plicata, this condition is also referred to as Iridocyclitis
- Clinical features: Pain, redness, photophobia, blepharospasm, blurring and diminished vision
- Keratic precipitates: Proteinaceous or cellular deposits at back of the cornea
- Aqueous flare: Haziness in the anterior chamber, due to the leakage of proteins from the inflamed tissue
- Investigation: Ocular: USG-B-Scan, fundus fluorescein angiography, optical coherence tomography
- Treatment: topical steroids, topical cycloplegics, nonsteroidal treatment, monoclonal antibodies, interferon alpha
- Keratic precipitates: Proteinaceous or cellular deposits at back of the cornea
- Aqueous flare: Haziness in the anterior chamber, due to the leakage of proteins from the inflamed tissue

Incorrect Options:

Option A - Intermediate uveitis:

- Inflammation of vitreous, peripheral retina and pars plana
- It is a chronic disease with insidious onset

Option B - Posterior uveitis:

- Components: Choroiditis, retinitis, retinal vasculitis
- Tx: local steroids, systemic steroids, immunosuppressives

Option D - Toxoplasma uveitis:

- Definitive host: Cat; Intermediate host: Human
- Ocular features: granulomatous or non-granulomatous uveitis
- Primarily it is posterior uveitis and spreads to anterior

Solution for Question 14:

Correct Option A - Penetrating trauma:

- Sympathetic ophthalmia is most commonly observed as a complication following penetrating ocular trauma, in which intraocular antigens are exposed to the immune system, triggering an autoimmune response that affects both eyes.

Incorrect Options:

Options A, C, and D are incorrect. Sympathetic Ophthalmitis is caused due to penetrating injury to the eye.

Neuro Ophthalmology

1. A 50-year-old patient presents with worsening headaches and visual disturbances. Fundoscopic examination reveals optic disc swelling. Which of the following is least likely to be associated with papilledema?

- A. Increased cup-to-disc ratio
- B. Enlargement of the blind spot
- C. Blurring of the optic disc margins
- D. Engorged and tortuous retinal veins

2. A 30-year-old patient comes to your clinic complaining of dizziness. During the examination, you observe that the patient's eyes exhibit involuntary, brief, and rhythmic eye movements when they gaze to the extreme left or right. These eye movements stop once the patient returns to a neutral position. This type of nystagmus is most likely:

- A. Endpoint nystagmus
- B. Central nystagmus
- C. Peripheral vestibular nystagmus
- D. Gaze-evoked nystagmus

3. A 50-year-old woman presents with impaired adduction of the left eye on rightward gaze, along with overshooting of the right eye during leftward gaze. There is no noticeable ptosis, and the convergence remains intact. MRI shows a lesion in the brainstem affecting the medial longitudinal fasciculus. Which of the following is the most likely diagnosis?

- A. Internuclear ophthalmoplegia

- B. One-and-a-half syndrome
 - C. Oculomotor nerve palsy
 - D. Third nerve palsy
-

4. A 60-year-old woman presents with difficulty moving her eyes to the right. On examination, she is unable to gaze to the right, but has normal eye movements when gazing to the left, up, and down. There is no weakness in the limbs or facial muscles. Which of the following is the most likely location of the lesion?

- A. Left abducens nucleus
 - B. Paramedian pontine reticular formation
 - C. Midbrain reticular formation
 - D. Right oculomotor nucleus
-

5. A 32-year-old woman presents with blurred vision in her right eye and notices that her right pupil appears larger than the left. On examination, her right pupil is dilated and reacts sluggishly to light but constricts normally with accommodation. The left pupil appears normal. Which of the following is the most likely diagnosis?

- A. Horner's syndrome
 - B. Adie's tonic pupil
 - C. Third nerve palsy
 - D. Acute angle-closure glaucoma
-

6. A 30-year-old male has difficulty distinguishing between red and green colours. His occupation requires accurate colour perception. Which of the following is the most appropriate test to quantify and classify the severity of his red-green colour

vision deficiency?

- A. Ishihara colour plates
 - B. Farnsworth-Munsell 100-Hue Test
 - C. Nagel Anomaloscope
 - D. Lantern Test
-

7. A 30-year-old male presents with difficulty in colour vision. On further examination, he has a complete inability to perceive the red end of the spectrum. Which of the following is the most likely diagnosis?

- A. Protanopia
 - B. Deutanopia
 - C. Tritanopia
 - D. Total Color Blindness
-

8. On examination, a 45-year-old male was found to have repetitive, involuntary horizontal eye movements after looking at a moving striped pattern. He reports no dizziness, and his neurological examination is otherwise unremarkable. The phenomenon he is experiencing is most likely due to which of the following?

- A. Gaze-evoked nystagmus
 - B. Vestibular Nystagmus
 - C. Optokinetic Nystagmus
 - D. Pendular Nystagmus
-

9. A 45-year-old male presents with difficulty moving his eyes upward and convergence-retraction nystagmus triggered by an upward gaze. His pupils react

poorly to light but constrict normally when focusing on near objects. Which of the following syndromes is most likely?

- A. Parinaud syndrome
 - B. Weber syndrome
 - C. Benedict syndrome
 - D. Millard-Gubler syndrome
-

10. A 50-year-old female patient with anisocoria had cocaine drops applied to both eyes. After 60 minutes, the anisocoria increased. To determine whether the lesion is preganglionic or postganglionic, what is the next step in the diagnostic process?

- A. Reapply cocaine drops
 - B. 0.5% Apraclonidine
 - C. Adrenaline
 - D. 1% hydroxyamphetamine
-

11. A 55-year-old male presents with a severe headache and neck stiffness that began a few days ago. His right pupil is smaller than the left, and he has drooping of the right eyelid and reduced sweating on the right side of his face. What is the most likely cause of these symptoms?

- A. Cluster headaches
 - B. Pancoast tumour
 - C. Carotid artery dissection
 - D. Cavernous sinus tumor
-

12. A 45-year-old male reports a gradual decrease in vision in his left eye over the past two months, describing it as a "dimming" rather than a complete loss. He also has difficulty reading and seeing clearly. As per the image below, which test is most likely being conducted?

[Image not available]

- A. Swinging flashlight test
- B. Perimetry
- C. Tonometry
- D. Visual acuity test

13. A 48-year-old male with a history of untreated syphilis from several years ago now presents with memory loss, irritability, difficulty walking, blurred vision, and occasional double vision. What type of pupil abnormality could be observed in this patient?

- A. Marcus Gunn pupil
- B. Amaurotic Pupil
- C. Adie's Pupil
- D. Argyll Robertson Pupil

14. A 45-year-old man presents with difficulty focusing on near objects. Which of the following is responsible for controlling ciliary muscle contraction and pupillary constriction during accommodation?

- A. Lateral Geniculate Nucleus
- B. Edinger-Westphal Nucleus
- C. Superior Colliculus
- D. Visual Cortex

15. A 60-year-old patient exhibits the following pupillary responses: Light is shown in the left eye; the right pupil constricts, but the left pupil remains dilated. Light is shown in the right eye; only the right pupil constricts, while the left pupil remains dilated. Which of the following lesions is most likely responsible for this pattern of pupillary response?

- A. Left optic nerve
 - B. Right Edinger-Westphal nucleus
 - C. Left oculomotor nerve
 - D. Medial chiasma lesion
-

16. A 62-year-old patient presents with unilateral optic atrophy and contralateral papilledema. On fundoscopic examination of the eye with papilledema, obscuration of blood vessels is found on the disc & vessels leaving the disc, with a significant halo around the disc. According to the Frisen grading system, what is the grade of papilledema?

- A. Grade 5
 - B. Grade 2
 - C. Grade 3
 - D. Grade 4
-

17. A 62-year-old patient presents with difficulty seeing objects in the right half of the visual field in both eyes. Based on this clinical presentation, which of the following is the most likely location of the lesion affecting the visual pathway?

- A. Left optic nerve
- B. Optic chiasma

- C. Left optic tract
 - D. Right Lateral Geniculate Body
-

18. A 45-year-old patient presents with headaches and transient blurred vision. Fundoscopy reveals optic disc swelling with indistinct margins and engorged veins. What is the most accurate explanation for the pathogenesis of this condition?

- A. Compression of the central retinal vein.
 - B. Increased CSF pressure causes axoplasmic stasis.
 - C. Inflammation of the optic nerve head
 - D. Increased intraocular pressure.
-

19. A 35-year-old patient with multiple sclerosis (MS) notices that her vision becomes temporarily blurred whenever she exercises. Which of the following phenomena is most likely explained by this temporary worsening of neurological symptoms due to increased body temperature?

- A. Riddoch's Phenomenon
 - B. Uhthoff's Phenomenon
 - C. Pulfrich's Phenomenon
 - D. Cogan's Reflex
-

20. A 25-year-old man presents with a sudden onset of painless, bilateral vision loss that started in the right eye and later affected the left eye. Genetic testing reveals a mutation in the mitochondrial DNA. Which of the following is the most likely diagnosis?

- A. Kjer dominant Optic Neuropathy
 - B. Leber Hereditary Optic Neuropathy (LHON)
 - C. Wolfram Syndrome
 - D. Behr syndrome
-

21. A 45-year-old man presents with progressive bilateral vision loss over the past few months. He reports a history of heavy smoking and alcohol consumption. Examination reveals a centrocecal scotoma in both eyes, and fundoscopy shows mild temporal pallor of the optic discs, as shown in the image. Which of the following substances is most likely responsible for his condition?

[Image not available]

- A. Methanol
 - B. Cyanide
 - C. Arsenic
 - D. Ethambutol
-

22. A 72-year-old patient presents with sudden vision loss in one eye, severe headache, scalp tenderness, and jaw claudication. Fundoscopic examination reveals a pale, swollen optic disc with haemorrhages. Laboratory tests show elevated ESR and CRP levels. Anterior Ischemic Optic Neuropathy (AION) is suspected. What is the type of AION, and what is the appropriate management?

- A. Arteritic AION; Lifestyle modifications
 - B. Non-arteritic AION; Lifestyle modifications
 - C. Arteritic AION; High dose corticosteroids
 - D. Non-arteritic AION; High dose corticosteroids
-

23. A 28-year-old woman presents with a history of decreased vision in her right eye. She reports pain around the eye, especially with movement. On examination, her left visual acuity is reduced, and there is an afferent pupillary defect (APD) on the right side. What is the most appropriate management?

- A. Conservative management
- B. High-dose Corticosteroids
- C. Oral antibiotics
- D. Surgical intervention

24. A 40-year-old woman presents with a 3-week history of difficulty in vision in her left eye, along with persistent headaches and occasional eye pain. She also reports trouble with colour perception. Fundoscopic examination reveals the findings shown in the image. Which type of optic neuritis is most likely to be seen?

[Image not available]

- A. Papillitis
- B. Retrobulbar neuritis
- C. Neuroretinitis
- D. Papilledema

25. A 34-year-old woman has a 6-month history of worsening vision and difficulty with detailed tasks, along with occasional eye pain and blurred vision in her left eye. Fundoscopy shows a chalky white optic disc with normal vessels in her left eye. What is the most common cause of these findings?

- A. Papilledema
- B. Multiple sclerosis
- C. Retinitis pigmentosa
- D. Glaucoma

26. A 45-year-old female reports a gradual decline in vision in her right eye over the past three months, accompanied by blurry vision and impaired colour perception in that eye. Upon examination, the ophthalmologist suspected optic nerve damage and advised further tests. Which of the following manifestations does not indicate optic nerve dysfunction?

- A. Increase in dark adaptation
 - B. Dyschromatopsia
 - C. Central scotoma
 - D. Decrease in contrast sensitivity
-

27. Ophthalmic artery is located Inferiolateral to which part of the optic nerve??

- A. Intraocular part
 - B. Intraorbital part
 - C. Intracanalicular part
 - D. Intracranial part
-

28. Which of the following arteries contributes to the formation of the Circle of Zinn-Haller?

- A. Central retinal artery
 - B. Ophthalmic artery
 - C. Short posterior ciliary arteries
 - D. All of the above
-

29. A 62-year-old patient presents with a visual field defect following a stroke. Neuroimaging reveals a lesion in the occipital lobe related to the territory of the posterior cerebral artery affecting the calcarine cortex. Which of the following visual field characteristics is most likely associated with this type of lesion?

- A. Congruous Homonymous Hemianopia with Macular Sparing
 - B. Congruous Homonymous Hemianopia without Macular Sparing
 - C. Incongruous homonymous hemianopia with Macular sparing
 - D. Incongruous homonymous hemianopia without Macular sparing
-

30. A 58-year-old patient presents with visual disturbances. Neuroimaging reveals a lesion in the temporal lobe affecting the inferior optic radiations. Which visual field defects are most likely associated with this finding?

- A. Inferior homonymous quadrantanopia
 - B. Superior homonymous quadrantanopia
 - C. Central scotoma
 - D. Homonymous hemianopia
-

31. A 58-year-old patient presents with right-sided visual field loss in both eyes, resulting in a homonymous hemianopic defect. Neuroimaging and clinical examination suggest a lesion affecting the visual pathway. Based on the above findings, which anatomical location is most likely implicated in this patient's condition?

- A. Right optic tract
- B. Left optic tract
- C. Right optic nerve

D. Optic chiasma

32. A 64-year-old patient presents with difficulty seeing objects. MRI findings reveal a pituitary adenoma causing compression at the optic chiasma. Which visual field defect is most likely associated with this condition?

- A.
 - B.
 - C.
 - D.
-

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Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	1
Question 4	2
Question 5	2
Question 6	3
Question 7	1
Question 8	3
Question 9	1
Question 10	4
Question 11	3
Question 12	1
Question 13	4
Question 14	2
Question 15	3
Question 16	4
Question 17	3
Question 18	2
Question 19	2
Question 20	2
Question 21	2
Question 22	3

Question 23	2
Question 24	3
Question 25	2
Question 26	1
Question 27	3
Question 28	3
Question 29	1
Question 30	2
Question 31	2
Question 32	1

Solution for Question 1:

Correct Answer: A) Increased cup-to-disc ratio

Explanation:

- Papilledema typically presents with blurring of the optic disc margins, enlargement of the blind spot, and engorged and tortuous retinal veins. As the disc swells, the physiological cup size usually decreases.
- An increased cup-to-disc ratio is more indicative of glaucomatous changes rather than papilledema.

Papilloedema:

- Swelling of the optic disc due to increased intracranial pressure
- Characteristic Features: Passive oedema without inflammation. Swelling primarily in the optic nerve head in front of the lamina cribrosa. Engorged, tortuous small blood vessels (Option D) and swollen nerve fibres. Axoplasmic stasis with swollen, disrupted mitochondria. Retinal changes include macular star and cotton-wool spots.
- Passive oedema without inflammation.
- Swelling primarily in the optic nerve head in front of the lamina cribrosa.
- Engorged, tortuous small blood vessels (Option D) and swollen nerve fibres.

- Axoplasmic stasis with swollen, disrupted mitochondria.
- Retinal changes include macular star and cotton-wool spots.
- Signs: Initial blurring of optic disc margins. (Option C) Enlargement of the blind spot due to the separation of the retina. (Option B) Hyperaemia, swelling, and engorgement of disc vessels. Potential for haemorrhages, exudates, and macular star formation. Chronic cases may lead to optic atrophy and visual loss.
- Initial blurring of optic disc margins. (Option C)
- Enlargement of the blind spot due to the separation of the retina. (Option B)
- Hyperaemia, swelling, and engorgement of disc vessels.
- Potential for haemorrhages, exudates, and macular star formation.
- Chronic cases may lead to optic atrophy and visual loss.
- Passive oedema without inflammation.
- Swelling primarily in the optic nerve head in front of the lamina cribrosa.
- Engorged, tortuous small blood vessels (Option D) and swollen nerve fibres.
- Axoplasmic stasis with swollen, disrupted mitochondria.
- Retinal changes include macular star and cotton-wool spots.
- Initial blurring of optic disc margins. (Option C)
- Enlargement of the blind spot due to the separation of the retina. (Option B)
- Hyperaemia, swelling, and engorgement of disc vessels.
- Potential for haemorrhages, exudates, and macular star formation.
- Chronic cases may lead to optic atrophy and visual loss.
- Stages of Papilledema:

Stages	Description
Early Papilledema	Blurring of optic disc margins, disc hyperemia, and mild disc swelling. Normal visual field & acuity.
Fully Developed Papilledema	Significant optic disc elevation, engorged veins, and peripapillary haemorrhages. Transient disturbance of vision.
Chronic Papilledema	Disc cup obliteration, reduced hyperemia, and hard exudates Visual acuity decreases, & Constriction of the visual field.
Late Papilledema	Optic atrophy, reduced disc swelling, and a grey, blurred disc appearance.

Solution for Question 2:

Correct Answer: A) Endpoint nystagmus

Explanation:

Endpoint nystagmus is a benign, physiologic form of nystagmus that occurs when the eyes are held at the extreme gaze position. It is typically mild and resolves once the eyes return to a neutral position.

Classification of Nystagmus:

Physiological Nystagmus

Optokinetic nystagmus:

- A physiological jerk nystagmus induced by moving visual stimuli. The eyes track a moving object, then jerk back to fixate on a new one. This occurs when looking at objects from a moving train or a rotating drum with stripes.

Endpoint nystagmus:

- It is a fine, horizontal jerk nystagmus that occurs in normal individuals when gazing far to the extreme right or left.

Physiological vestibular nystagmus (vestibulo-ocular reflex, i.e. VOR):

- It is a jerk nystagmus elicited by stimulating the tympanic membrane with hot or cold water. Cold water in the right ear causes left jerk nystagmus (opposite side), while warm water causes right jerk nystagmus (same side). The mnemonic is COWS: Cold - Opposite, Warm - Same.

Pathological Nystagmus

Early onset (childhood) Nystagmus

Infantile nystagmus syndrome:

- Refers to congenital nystagmus, which can be idiopathic or associated with sensory deprivation due to conditions like ocular albinism, aniridia, or Leber's congenital amaurosis.
- It is characterized by erratic waveform with or without roving eye movements associated with reduced visual acuity.

Fusion maldevelopment nystagmus syndrome (FMNS):

- Infantile onset and is associated with strabismus (e.g., congenital esotropia and dissociated vertical deviation), decreases in intensity with age, and appears when one eye is covered.

Spasmus Nutans syndrome (SNS):

- Infantile onset, associated with abnormal head posture and oscillation that improve or disappear during childhood, and often spontaneously remits clinically between 2 to 8 years of age.

Acquired Nystagmus

Nystagmus due to disorders of visual fixation:

- May occur due to diseases affecting any part of the visual system, from the retina to the visual cortex, or conditions interrupting visual projections to the pons and cerebellum.

Vestibular nystagmus:

- Peripheral vestibular nystagmus: Characterized by conjugate horizontal jerk nystagmus with the fast phase away from the side of the lesion, which improves with fixation and worsens when gazing towards the fast phase.
- Central vestibular nystagmus: Upbeat nystagmus is characterized by an upward fast component in primary gaze and is typically seen in lesions of the central tegmentum of the brainstem. Downbeat nystagmus has a downward fast component in primary gaze and is commonly associated with posterior fossa diseases, including cerebellar lesions and Arnold-Chiari syndrome.

Nystagmus due to disorders of gaze holding:

- Gaze-evoked nystagmus: A slow, conjugate horizontal jerk nystagmus that occurs in the direction of gaze at smaller angles (around 45°), and is absent in primary gaze.
- Dissociated or disconjugate nystagmus (Ataxic nystagmus): A type of pathologic gaze-evoked nystagmus. It presents as unilateral or asymmetric nystagmus, typically affecting the abducting eye, and occasionally the adducting eye.

Solution for Question 3:

Correct answer: A) Internuclear ophthalmoplegia

Explanation:

This patient's presentation, with impaired adduction of the left eye and dysmetria (overshooting) of the right eye during lateral gaze, along with the intact convergence and MRI findings, is consistent with internuclear ophthalmoplegia (INO).

- Internuclear Ophthalmoplegia (INO) Injury to the MLF between the abducens nucleus and the contralateral oculomotor nerve's medial rectus subnucleus. Impaired adduction of the ipsilateral eye, with slow or absent adduction during lateral gaze. The abducting eye overshoots the target (dysmetria), resulting in disconjugate nystagmus. Convergence is typically preserved.
- Injury to the MLF between the abducens nucleus and the contralateral oculomotor nerve's medial rectus subnucleus.
- Impaired adduction of the ipsilateral eye, with slow or absent adduction during lateral gaze.
- The abducting eye overshoots the target (dysmetria), resulting in disconjugate nystagmus.
- Convergence is typically preserved.
- Bilateral INO Features Abduction saccades may be slow due to impaired inhibition of medial rectus resting tone. Associated with upward beating and torsional nystagmus.
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- Associated with upward beating and torsional nystagmus.
- Associated Clinical Features Skew deviation. Defective vertical smooth pursuit. Impairment of the vertical VOR and difficulty suppressing it.
- Skew deviation.
- Defective vertical smooth pursuit.
- Impairment of the vertical VOR and difficulty suppressing it.
- Causes of INO Vascular, demyelinating, or metastatic brainstem lesions. Must be differentiated from pseudo-INO caused by myasthenia gravis, paraneoplastic strabismus, or long-standing exotropia.
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- Must be differentiated from pseudo-INO caused by myasthenia gravis, paraneoplastic strabismus, or long-standing exotropia.
- Injury to the MLF between the abducens nucleus and the contralateral oculomotor nerve's medial rectus subnucleus.
- Impaired adduction of the ipsilateral eye, with slow or absent adduction during lateral gaze.
- The abducting eye overshoots the target (dysmetria), resulting in disconjugate nystagmus.
- Convergence is typically preserved.
- Abduction saccades may be slow due to impaired inhibition of medial rectus resting tone.
- Associated with upward beating and torsional nystagmus.

- Skew deviation.
- Defective vertical smooth pursuit.
- Impairment of the vertical VOR and difficulty suppressing it.
- Vascular, demyelinating, or metastatic brainstem lesions.
- Must be differentiated from pseudo-INO caused by myasthenia gravis, paraneoplastic strabismus, or long-standing exotropia.

One-and-a-Half Syndrome (Option B): This syndrome results from damage to the caudal pons, which involves both the MLF and either the PPRF or abducens nucleus. It causes gaze palsy (inability to look to one side) in addition to INO. This patient doesn't have a gaze palsy, making one-and-a-half syndrome less likely. Common causes of One-and-a-Half Syndrome include multiple sclerosis, brainstem stroke, primary or metastatic brainstem tumors.

Oculomotor nerve palsy (Option C): Would typically present with ptosis, impaired eye movements (including both vertical and horizontal), and possibly dilated pupil.

Third nerve palsy (Option D): Third nerve palsy usually presents with a combination of ptosis, diplopia, and impaired eye movement. Additionally, it can cause a dilated pupil due to parasympathetic involvement.

Solution for Question 4:

Correct answer: B) Paramedian pontine reticular formation

Explanation:

The paramedian pontine reticular formation (PPRF) is responsible for coordinating ipsilateral horizontal gaze. A lesion in the right PPRF would impair rightward horizontal gaze, while preserving other directions of eye movement, including vertical gaze.

- The brainstem contains gaze centers, which are the final common pathways for eye movements.
- The paramedian pontine reticular formation (PPRF) controls ipsilateral horizontal gaze. PPRF is managed by the frontal eye field (FEF).
- The midbrain reticular formation (MRF) (Option C) controls vertical gaze, vergence eye movements, and ocular counter-rolling.

- These gaze centers receive input from higher brain regions such as the cerebral hemispheres, superior colliculus, vestibular nuclei, and cerebellum.
- The PPRF and MRF send signals to the three ocular motor nuclei that control eye movements.
- Supranuclear gaze palsies result from disruption of pathways that carry voluntary saccade and pursuit commands before they reach the brainstem eye movement centers.
- PPRF lesion: Will lead to ipsilateral horizontal gaze palsy.
- FEF lesion: Will lead to contralateral horizontal gaze palsy.
- MLF lesion: It is also called Internuclear ophthalmoplegia (INO), it is between the third, fourth and sixth nuclei

Left abducens nucleus (Option A): The abducens nucleus controls the lateral rectus muscle of the eye, which is responsible for moving the eye laterally. A lesion in the left abducens nucleus would impair the movement of the left eye to the left, but it wouldn't cause an issue with moving both eyes to the right. This would be inconsistent with the patient's inability to move both eyes to the right.

Right oculomotor nucleus (Option D): The oculomotor nucleus controls the medial rectus (responsible for convergence) and the other muscles that move the eye. A lesion in the right oculomotor nucleus would result in pupil dilation (due to the parasympathetic fibers being affected) and ptosis, as well as problems with eye movement, but not isolated difficulty with horizontal gaze.

Solution for Question 5:

Correct Answer: B) Adie's tonic pupil

Explanation:

- In Adie's tonic pupil, the affected pupil is abnormally dilated (resulting in isocoria).
- In this condition, the pupil's response to light is absent, and its reaction to near stimuli is slow and tonic.
- It is caused by damage to the post-ganglionic parasympathetic pupillomotor fibers and is usually unilateral.
- The affected pupil constricts when treated with weak pilocarpine (0.125%) drops, while the normal pupil does not, due to the denervated iris sphincter becoming supersensitive to

parasympathomimetic agents.

Pharmacological tests in suspected parasympathetic palsy: The pilocarpine test is used to distinguish between a dilated pupil caused by third nerve palsy, pharmacological mydriasis, or tonic pupil.

- If the pupil constricts with a low concentration of pilocarpine (0.125%), it indicates Adie's tonic pupil.
- If the pupil constricts with the usual concentration of pilocarpine (1-2%), it suggests third nerve palsy.
- If the pupil does not constrict with pilocarpine, it points to mydriasis due to parasympathetic drug use. Other conditions where the dilated pupil does not react to pilocarpine include traumatic mydriasis and fixed pupils in iritis.

For suspected sympathetic palsy, pharmacological tests include:

- Cocaine test (2-4%): A small pupil that fails to dilate is indicative of Horner's syndrome.
- Hydroxyamphetamine test (1%): In central or pre-ganglionic Horner's syndrome, the small pupil will dilate, while in post-ganglionic Horner's syndrome, it will not dilate.

Solution for Question 6:

Correct Answer: C) Nagel Anomaloscope

Explanation:

- Nagel Anomaloscope is the gold standard for diagnosing and quantifying red-green colour vision deficiencies.
- The test involves adjusting red and green lights to match a yellow field, providing a precise classification of the type and severity of red-green colour blindness.
- It quantifies both the severity and type of the colour vision defect.

Test	Procedure	Evaluation	Applications
[Image not available] Lantern Tests (Option D)	Individuals identify various coloured lights emitted from a lantern.	The accuracy of colour identification is evaluated.	Used in occupations requiring accurate colour vision (e.g., aviation, maritime). Example: Edridge-Green lantern.

[Image not available] Farnsworth-Munsell 100-Hue Test (Option B)	Measures ability to differentiate between subtle colour differences using hue arrangements.	A higher error score indicates poorer colour vision.	Detects fine colour discrimination errors. Often used to assess toxic optic neuropathy.
[Image not available] Ishihara Color Plates (Option A)	Coloured dots form numbers or patterns designed to challenge individuals with specific deficiencies.	Based on the ability to correctly identify numbers or patterns formed by dots of different colours.	Primarily detects red-green color blindness. Hardy-Rand-Ritter (H-R-R) plates can detect blue-yellow deficiencies.
[Image not available] Anomaloscope (Nagel, Pickford-Nicolson)	Split visual field: adjust red and green lights to match a yellow field.	Determines severity and type of red-green colour vision deficiency through light adjustments.	Nagel anomaloscope for red-green anomalies. Pickford-Nicolson for blue-yellow anomalies.
Holmgren's Wools Test	Subject is asked to make a series of colour-matches from a selection of skeins of coloured wools.	Based on the ability to correctly match the samples with the reference colours provided.	Originally designed to screen red-green colour blindness. Although not a suitable screening test.

Solution for Question 7:

Correct Answer: A) Protanopia

Explanation:

The patient has difficulty in colour vision. Specifically, the absence of perception of the red end of the spectrum is consistent with protanopia.

Red-green colour blindness is the most common type of partial colour blindness and is usually congenital. It can be divided into two main types:

- Protanopia: Characterized by complete inability to perceive red light due to a deficiency or absence of the red cone photopigment.
- Deuteranopia (Option B): Involves inability to perceive green light due to a deficiency of the green cone photopigment.

- Protanomalous and Deuteranomalous (milder forms)

[Image not available]

- Blue blindness: Occurs with ageing due to increased lens pigment.
- Black Cataract: Blue vision decline in individuals with sclerosing lenses.

Solution for Question 8:

Correct Answer: C) Optokinetic Nystagmus

Explanation:

Optokinetic nystagmus is a physiological response that occurs when the eyes follow a moving object, like a striped pattern, and then rapidly return to their original position. It is a normal and non-pathological condition.

- Assessing Visual Acuity: Optokinetic nystagmus (OKN) is an objective method for assessing visual acuity, especially in uncooperative patients. A rotating drum with black stripes is shown to the patient, and their eyes follow the stripes with slow, smooth movements (slow phase), snapping back when the stripe disappears (fast phase). This involuntary reflex indicates proper visual function, and adjustments in stripe width or distance can assess varying acuity levels.
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- This involuntary reflex indicates proper visual function, and adjustments in stripe width or distance can assess varying acuity levels.

[Image not available]

- Diagnosing Neurological Conditions: Individuals with homonymous hemianopia (loss of vision in half of the visual field) due to parietal lobe lesions do not exhibit a normal optokinetic nystagmus response. This finding indicates that an intact parietal lobe is crucial for the functioning of the optokinetic reflex pathway. Therefore, the absence of OKN in these patients provides insight into potential damage to the neurological pathways involved in visual processing.
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Gaze-evoked nystagmus (Option A) occurs when the eye is held in an eccentric position, and it is typically seen in cerebellar dysfunction.

Vestibular Nystagmus (Option B) is an involuntary movement of the eyes due to vestibular system disturbances, often associated with dizziness or vertigo.

Pendular Nystagmus (Option D) is characterised by smooth, sinusoidal oscillations without a slow or fast phase and is usually associated with congenital conditions or visual loss. It does not result from external visual stimuli like a moving striped pattern.

Solution for Question 9:

Correct Answer: A) Parinaud syndrome

Explanation:

- Parinaud syndrome, also known as dorsal midbrain syndrome, is a neurological condition caused by damage to the dorsal midbrain, specifically affecting the pretectal area and superior colliculus.
- It results in a distinct set of eye movement abnormalities and is commonly linked to lesions, tumours, or strokes affecting this area.

- Clinical features and signs: Upgaze Palsy (Vertical Gaze Palsy): Difficulty or inability to move their eyes upward (vertical gaze palsy). Convergence-Retraction Nystagmus: When the patient attempts to look upward, the eyes may instead converge (move inward toward each other) and retract into the orbits Pupillary Light-Near Dissociation: The pupils do not constrict to light but react normally when focusing on a near object (accommodation). Eyelid Retraction (Collier's sign): Abnormally high upper eyelids due to damage in the midbrain.
- Upgaze Palsy (Vertical Gaze Palsy): Difficulty or inability to move their eyes upward (vertical gaze palsy).
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- Pupillary Light-Near Dissociation: The pupils do not constrict to light but react normally when focusing on a near object (accommodation).
- Eyelid Retraction (Collier's sign): Abnormally high upper eyelids due to damage in the midbrain.

[Image not available]

- Skew Deviation: Misalignment of the eyes, often vertically.
- Downward ocular deviation (setting sun sign)

ocular deviation" data-author="Nikki Lindbauer, MD, Volker Strenger, MD, and Christian Urban, MD" data-hash="4223" data-license="Open access"

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- Pseudoabducens palsy

Weber syndrome (Option B) is caused by a midbrain lesion involving the oculomotor nerve and corticospinal tract, leading to ipsilateral oculomotor palsy and contralateral hemiparesis.

Benedict syndrome (Option C) involves a lesion in the midbrain affecting the oculomotor nerve and red nucleus, resulting in ipsilateral oculomotor nerve palsy, contralateral tremor, and ataxia.

Millard-Gubler syndrome (Option D) occurs due to a lesion in the ventral pons, causing ipsilateral facial palsy and contralateral hemiparesis.

Solution for Question 10:

Correct Answer: D) 1% hydroxyamphetamine

Explanation:

The given scenario is most likely Horner's syndrome, and to determine the lesion's location, the next best step is to apply 1% hydroxyamphetamine drops.

Algorithm for determining the site of lesion in Horner's syndrome:

- Cocaine (4%): Block the reuptake of norepinephrine (noradrenaline) normally released from the nerve endings. Hence it dilates the normal pupil (Option A).
- Apraclonidine (Option B): α 2-adrenergic agonist: Miosis in the normal eye. α 1-adrenergic agonist: Mydriasis in Horner's syndrome (presence of adrenergic denervation supersensitivity)
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- α 1-adrenergic agonist: Mydriasis in Horner's syndrome (presence of adrenergic denervation supersensitivity)
- Hydroxyamphetamine (1%): releases norepinephrine from storage vesicles in the postganglionic sympathetic nerve endings at the iris dilator muscle (Option D).
Postganglionic: No dilation (3rd order nerve is dead). Preganglionic lesion: Dilation
- Postganglionic: No dilation (3rd order nerve is dead).
- Preganglionic lesion: Dilation
- α 2-adrenergic agonist: Miosis in the normal eye.
- α 1-adrenergic agonist: Mydriasis in Horner's syndrome (presence of adrenergic denervation supersensitivity)
- Postganglionic: No dilation (3rd order nerve is dead).
- Preganglionic lesion: Dilation

Adrenaline (Option C) or Adrenaline 1 in 1000 if used in postganglionic cases then pupils dilate due to denervation supersensitivity.

Solution for Question 11:

Correct Answer: C) Carotid artery dissection

Explanation:

The above case suggests features of Horner's syndrome, which is most likely due to carotid artery dissection since there are complaints of neck stiffness and severe headache and no other significant history.

Horner's Syndrome

- Horner's syndrome is caused by a lesion of the sympathetic chain, which clinically manifests as ipsilateral miosis, facial anhidrosis, and ptosis (drooping of the eyelid).
- Causes:

Table rendering failed

- Clinical manifestations: Miosis, ptosis, enophthalmos. Anhidrosis (loss of sweating): It is seen in preganglionic lesions. Absence of cilio-spinal reflex (involuntary response where the pupil of the eye dilates in response to painful stimuli applied to the skin of the neck or upper back)
- Miosis, ptosis, enophthalmos.
- Anhidrosis (loss of sweating): It is seen in preganglionic lesions.
- Absence of cilio-spinal reflex (involuntary response where the pupil of the eye dilates in response to painful stimuli applied to the skin of the neck or upper back)
- Acute Horner's syndrome with associated ipsilateral neck or facial pain requires urgent imaging of the neck for evaluation of a carotid dissection (Option C).
- Miosis, ptosis, enophthalmos.
- Anhidrosis (loss of sweating): It is seen in preganglionic lesions.
- Absence of cilio-spinal reflex (involuntary response where the pupil of the eye dilates in response to painful stimuli applied to the skin of the neck or upper back)

Cluster headache (Option A)) causes severe, unilateral periorbital pain with autonomic features, such as ipsilateral tearing, nasal congestion and conjunctival injection along with ptosis, miosis and other features of Horner's syndrome.

Pancoast tumour (Option B) may compress the brachial plexus, leading to symptoms like localised pain around the shoulder and upper back, which might be associated with weakness or atrophy of the muscles in the arm on the affected side.

Cavernous sinus tumour (Option D) causes ocular pain and diplopia along with other symptoms of Horner's syndrome, and it is gradual in onset, unlike the above case.

Solution for Question 12:

Correct Answer: A) Swinging flashlight test

Explanation:

The above given scenario is suspicious of optic nerve dysfunction and the image depicted above is describing RAPD (Relative Afferent Pupillary Defect) which is tested by swinging flashlight test.

Swinging flashlight test:

The swinging flashlight test is used to detect a relative afferent pupillary defect (RAPD), often seen with unilateral optic nerve lesions, even when visual acuity is normal.

Procedure:

[Image not available]

Normal response: Both pupils constrict equally, and the pupil receiving the light remains tightly constricted.

Abnormal response (RAPD): If there is a relative afferent pathway defect, the affected pupil will dilate when the light is moved from the normal eye to the abnormal eye.

Significance: The presence of Marcus Gunn pupil (RAPD) is an early indication of optic nerve disease, even when visual acuity appears normal.

Causes:

Table rendering failed

Perimetry (Option B) measures and maps the visual field, assessing peripheral vision to detect defects or abnormalities. It aids in diagnosing eye conditions like glaucoma, retinal diseases, and neurological disorders affecting visual pathways.

Tonometry (Option C) is a diagnostic test used in Glaucoma to measure intraocular pressure (IOP).

The visual acuity test (Option D) measures the sharpness or clarity of a person's vision. It evaluates how well a person can see details at a specific distance, typically assessing the ability to read letters or symbols on Snellen's chart.

Solution for Question 13:

Correct Answer: D) Argyll Robertson Pupil

Explanation:

The given scenario of cognitive and psychotic symptoms with a history of untreated syphilis suggests Neurosyphilis, where Argyll Robertson's pupil is a clinical finding.

Abnormal Pupillary Reactions

Reaction	Description	Causes
Amaurotic light reflex (Option B)	Absence of direct light reflex on the affected side and an absence of consensual light reflex on the normal side.	Optic and retinal lesions on the affected side
Efferent pathway defect	Absence of both direct and consensual light reflex on the affected side and presence of both direct and consensual light reflex on the normal side. Near reflex is also absent on the affected side.	Parasympatholytic drugs (e.g. atropine, homatropine) Internal ophthalmoplegia Third-nerve paralysis.
Marcus Gunn pupil or RAPD (Option A)	The pupils paradoxically dilate when the flashlight is swung from the unaffected to the affected eye due to the decreased light perception. It is the first sign of optic nerve disorder. Tested by a swinging flashlight test.	Incomplete optic nerve lesions (Optic neuritis) and severe retinal diseases (Retinopathy, CRVO)

Adie's Pupil (Option C)	Reaction to light is absent, and the near reflex is very slow and tonic. It is usually unilateral, associated with absent knee jerk and occurs more often in young women. It shows pupillary constriction to 0.125% Pilocarpine due to denervation supersensitivity.	Postganglionic parasympathetic pupillomotor damage.
Argyll Robertson Pupil (ARP) (Option D)	Reaction to near reflex is present, but light reflex (both direct and consensual) is absent, i.e. there is light near dissociation.	Neurosyphilis
Wernicke's hemianopic pupil	Light reflex is absent when light is thrown on the temporal half of the retina of the affected side and the nasal half of the opposite side, while it is present when the light is thrown on the nasal half of the affected side and the temporal half of the opposite side.	Optic tract lesion

Solution for Question 14:

Correct Answer: B) Edinger-Westphal Nucleus

Explanation:

The Edinger-Westphal Nucleus is responsible for the accommodation reflex, controlling the ciliary muscles to adjust the lens and sphincter pupillae constriction for near vision.

Accommodation Reflex (Adjustments made by the eye to focus on near objects):

- It integrates three key responses: Accommodation: Adjusting the lens curvature for clear near vision. Convergence: Both eyes turn inward to align with the near object. Miosis: Pupils constrict to improve depth of field.
- Accommodation: Adjusting the lens curvature for clear near vision.
- Convergence: Both eyes turn inward to align with the near object.
- Miosis: Pupils constrict to improve depth of field.
- Accommodation: Adjusting the lens curvature for clear near vision.
- Convergence: Both eyes turn inward to align with the near object.

- Miosis: Pupils constrict to improve depth of field.

[Image not available]

- The Edinger-Westphal Nucleus is the central structure crucial for the near reflex.
Convergence: Signals from the medial rectus muscles travel to the Edinger-Westphal nucleus, which then influences the eye muscles to converge. Accommodation: Visual input from the cortex is relayed to the Edinger-Westphal nucleus, which adjusts the ciliary muscles for proper lens focus. Miosis: The Edinger-Westphal nucleus sends parasympathetic signals via the oculomotor nerve to the iris sphincter muscle, causing pupil constriction.
- Convergence: Signals from the medial rectus muscles travel to the Edinger-Westphal nucleus, which then influences the eye muscles to converge.
- Accommodation: Visual input from the cortex is relayed to the Edinger-Westphal nucleus, which adjusts the ciliary muscles for proper lens focus.
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- Miosis: The Edinger-Westphal nucleus sends parasympathetic signals via the oculomotor nerve to the iris sphincter muscle, causing pupil constriction.

Lateral Geniculate Nucleus (LGN) (Option A) is involved in visual processing but not directly in the accommodation reflex.

Superior Colliculus (Option C) involves visual reflexes and eye movements but does not control lens adjustment.

Visual Cortex (Option D) processes visual information but does not directly manage the accommodation reflex.

Solution for Question 15:

Correct Answer: C) Left oculomotor nerve

Explanation:

A lesion in the left oculomotor nerve would disrupt parasympathetic signals to the left iris muscles, preventing the left pupil from constricting. This results in dilating the left pupil regardless of light stimuli, while the right eye's responses remain unaffected.

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Light Reflex:

- Direct Light Reflex: Constriction of the pupil in the same eye that is directly exposed to light.
- Indirect (Consensual) Light Reflex: Constriction of the pupil in the opposite eye when light is shone into one eye.

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Lesions and their effects of Light Reflex:

Optic Nerve (Option A)	Direct and consensual reactions are lost on the affected side Both are retained on the opposite side. Near reflex and lid reflex remain intact.
Medial Chiasma (Option D)	Bitemporal hemianopia
Lateral Chiasma	Binasal hemianopia.
Optic Tract	Contralateral hemianopia; Wernicke reaction.
Proximal Optic Tract	Normal pupillary reactions
Central Decussation	Bilateral reflex paralysis with no reaction to light but retention of near reflex, lid reflexes, and psycho-sensory reactions Argyll Robertson pupil.
Nuclear / Supranuclear Lesion (Option B)	Ipsilateral absolute pupillary paralysis.
Third Nerve Lesion	Absolute pupillary paralysis.
Ciliary Ganglion Lesion	The light reflex is abolished, but the retention of the near reflex is sometimes followed by tonic contraction (Adie pupil).

Solution for Question 16:

Correct Answer: D) Grade 4

Explanation:

- In this clinical scenario, the patient exhibits unilateral optic atrophy and contralateral papilledema (due to a space occupying lesion), characteristic of Foster-Kennedy syndrome.
- In the Frisen grading system, Grade 4 is characterised by the obscuration of major vessels on the disc. Although vessels that leave the disc are also obscured, which corresponds to Grade 3, the overall presentation falls under Grade 4.

Grading of Papilledema (Modified Frisen Scale)	
Grade 1: C-shaped halo nasally with a temporal gap.	[Image not available]
Grade 2: Circumferential halo around the optic disc.	[Image not available]
Grade 3: Obscuration of one or more major vessels leaving the disc.	[Image not available]
Grade 4: Obscuration of major vessels on the disc.	[Image not available]
Grade 5: Complete obscuration of all vessels on and leaving the disc	

Solution for Question 17:

Correct Answer: C) Left Optic tract

Explanation:

The clinical presentation of difficulty seeing objects in the right half of the visual field in both eyes indicates a visual field defect known as Right Homonymous Hemianopia, which occurs due to a defect in the Left Optic Tract.

Visual Pathway:

- Temporal fibers carry impulses from the Nasal field, and Nasal fibres carry the Temporal field.

[Image not available]

Components of Visual Pathway	
Retina	First-order Neurons: Bipolar cells in the inner nuclear layer of the retina. Second-order Neurons: Ganglion cells in the retina; their axons form the optic nerve.
Optic Nerve	Carries visual information from the retina to the optic chiasm. Temporal and Nasal fibres of the same eye.
Optic Chiasma	Partial decussation occurs Nasal fibres cross to the opposite side, while temporal fibres remain ipsilateral.
Optic Tract	Carries visual information from the optic chiasms to the Lateral Geniculate body (LGB). Temporal fibres of the same eye and Nasal fibres of the opposite eye. The lesion causes Homonymous Hemianopia in the opposite eye.
Lateral Geniculate Body (LGB)	Third Order Neurons: Relay visual information to the visual cortex.
Optic Radiations	Meyer's Loop: Carries information from lower retinal quadrants to the inferior part of the visual cortex. Baum's Loop (Dorsal): Carries information from upper retinal quadrants to the superior part of the visual cortex.
Occipital Cortex	The primary visual cortex (V1), corresponding to Brodmann area 17, is situated on the medial side of the occipital lobe, particularly along the borders of the calcarine fissure.

Solution for Question 18:

Correct Answer: B) Increased CSF pressure causes axoplasmic stasis.

Explanation:

- The clinical presentation suggests Papilledema, a hydrostatic, non-inflammatory condition characterised by swelling of the optic disc due to increased intracranial pressure.
- Typically bilateral, it occurs because the optic nerve, encased in the meningeal sheaths, is affected by changes in cerebrospinal fluid (CSF) pressure.

- Pathogenesis: Initially thought to be caused by compression of the central retinal vein, leading to its collapse while the artery remains intact. (Option A ruled out) Current Understanding: Blockage of axoplasmic transport in the optic nerve head due to elevated CSF pressure results in optic disc swelling and vascular changes on the disc surface.
- Initially thought to be caused by compression of the central retinal vein, leading to its collapse while the artery remains intact. (Option A ruled out)
- Current Understanding: Blockage of axoplasmic transport in the optic nerve head due to elevated CSF pressure results in optic disc swelling and vascular changes on the disc surface.
- Initially thought to be caused by compression of the central retinal vein, leading to its collapse while the artery remains intact. (Option A ruled out)
- Current Understanding: Blockage of axoplasmic transport in the optic nerve head due to elevated CSF pressure results in optic disc swelling and vascular changes on the disc surface.
- Aetiology (Common causes)
 - Intracranial tumors (e.g., mid-brain, parieto-occipital region, cerebellum).
 - Brain abscesses, cavernous sinus thrombosis, aneurysms, subarachnoid haemorrhage.
 - Pseudotumour cerebri (idiopathic intracranial hypertension).
 - Malignant hypertension is typically accompanied by increased intracranial pressure.
 - Decrease in intraocular pressure (Option D ruled out)
- Symptoms: Early symptoms: Transient blurred vision or blackouts. Progression: Enlarged blind spots, visual field contraction, and severe vision loss. Headache often worsens in the morning and improves during the day. Nausea, vomiting, and diplopia may also occur.
- Early symptoms: Transient blurred vision or blackouts.
- Progression: Enlarged blind spots, visual field contraction, and severe vision loss.
- Headache often worsens in the morning and improves during the day.
- Nausea, vomiting, and diplopia may also occur.
- Early symptoms: Transient blurred vision or blackouts.
- Progression: Enlarged blind spots, visual field contraction, and severe vision loss.
- Headache often worsens in the morning and improves during the day.
- Nausea, vomiting, and diplopia may also occur.

Inflammation of the optic nerve head (Option C) describes optic neuritis, which involves inflammation leading to swelling and visual symptoms. In contrast, papilledema is caused by increased intracranial pressure affecting axoplasmic flow rather than direct inflammation of the optic nerve head.

Solution for Question 19:

Correct Answer: B) Uhthoff's Phenomenon

Explanation:

- Uhthoff's Phenomenon refers to the temporary worsening of neurological symptoms in individuals with multiple sclerosis (MS) due to increased body temperature.
- Mechanism: In MS, the immune system attacks the myelin sheath, leading to demyelination and impaired nerve conduction. Elevated body temperature exacerbates this impairment by further affecting the conduction of nerve signals through heat-sensitive demyelinated axons, resulting in a transient conduction block.
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- Elevated body temperature exacerbates this impairment by further affecting the conduction of nerve signals through heat-sensitive demyelinated axons, resulting in a transient conduction block.
- Symptoms: Blurring of vision during activities like hot showers or exercise. Increased muscle weakness during a fever. Worsening of balance or coordination after exposure to heat.
- Blurring of vision during activities like hot showers or exercise.
- Increased muscle weakness during a fever.
- Worsening of balance or coordination after exposure to heat.
- Resolution: Symptoms typically resolve once body temperature returns to normal.
- Blurring of vision during activities like hot showers or exercise.
- Increased muscle weakness during a fever.
- Worsening of balance or coordination after exposure to heat.

Riddoch's Phenomenon (Option A) refers to a visual disturbance where a patient can perceive movement but not static objects, typically resulting from damage to the occipital lobe.

Pulfrich's Phenomenon (Option C) is a visual anomaly where objects moving in a straight line appear to follow a curved trajectory due to delayed visual processing. It is often associated with optic neuritis but is not related to heat or temperature changes.

Cogan's Reflex (Option D) refers to an involuntary upward eye movement when looking down, seen in progressive supranuclear palsy or other neurological disorders such as Parkinson's disease and certain forms of ataxia.

Solution for Question 20:

Correct Answer: B) Leber Hereditary Optic Neuropathy (LHON)

Explanation:

Leber Hereditary Optic Neuropathy (LHON) is a mitochondrial genetic disorder that typically presents with subacute, bilateral vision loss, often starting in one eye and affecting the other eye within a few weeks, and it is associated with mutations in mitochondrial DNA.

Leber Hereditary Optic Neuropathy	
Inheritance	Mitochondrial DNA disorder; Maternally inherited.
Variable Penetrance	50%–70% of males and 10%–15% of females
Age of Onset	Typically between 15-30 years, later onset is possible, especially in females (40s-50s).
Vision Loss	Sudden, painless, bilateral, central vision loss, central scotoma common.

Kjer dominant Optic Neuropathy (Option A) is an autosomal dominant condition with gradual vision loss beginning in childhood, not sudden onset in young adulthood. It typically does not present with sudden, painless vision loss or significant optic disc pallor, and visual impairment usually does not progress to severe loss.

Wolfram syndrome (Option C): A rare genetic disorder that includes optic atrophy along with diabetes insipidus, diabetes mellitus, and hearing loss. It is inherited in an autosomal recessive pattern.

Behr Syndrome (Option D) is a rare autosomal recessive type of spinocerebellar ataxia that presents with Optic Atrophy, Cerebellar Ataxia, and Spastic Paraplegia. Symptoms usually begin in childhood or early adulthood.

Solution for Question 21:

Correct Answer: B) Cyanide

Explanation:

Cyanide, particularly from tobacco smoke, is a well-known cause of toxic amblyopia, especially in cases of "tobacco-alcohol amblyopia." Given the patient's history of heavy smoking and alcohol consumption, cyanide exposure is highly likely.

Table rendering failed

Methanol (Option A) poisoning can lead to toxic optic neuropathy, similar to the presentation described. Methanol toxicity typically causes acute onset of vision loss, often accompanied by systemic symptoms like metabolic acidosis.

Arsenic poisoning (Option C): Chronic arsenic exposure can lead to optic nerve damage and toxic amblyopia.

Ethambutol (Option D) is known to cause optic neuropathy, often presenting with vision loss and colour vision abnormalities. However, it is typically associated with patients receiving treatment for tuberculosis.

Solution for Question 22:

Correct Answer: C) Arteritic AION; High dose corticosteroids

Explanation:

- Arteritic AION is often related to Giant Cell Arteritis (GCA) and presents with systemic symptoms such as severe headache, scalp tenderness, and jaw claudication.

- It requires prompt treatment with high-dose corticosteroids to manage inflammation and prevent further vision loss.

[Image not available]

Feature	Arteritic AION (AAION)	Non-Arteritic AION (NAION)
Cause	Inflammation and potential blood clot formation due to Giant Cell Arteritis (GCA)	No inflammatory process; unclear reason for blood vessel blockage
Prevalence	5-10% of AION cases	90-95% of AION cases
Typical Age	Around 70 years	Middle-aged to older adults (usually over 50)
Risk Factors	Age > 50 Northern European descent Other autoimmune diseases	Diabetes Hypertension Atherosclerosis Crowded optic disc
Symptoms	Severe headache Scalp tenderness Jaw claudication Systemic symptoms like fever and fatigue	Sudden vision loss (painless) Altitudinal field defect Relative Afferent Pupillary Defect (RAPD)
Ophthalmoscopic Findings	Pale and swollen optic disc Cotton wool spots Splinter haemorrhages	Swollen and hyperemic optic disc
Visual Field Defect	Inferior altitudinal defects are common	
Diagnostic Tests	Elevated ESR and CRP Temporal artery biopsy	Fluorescein angiography Visual field testing
Management	High-dose corticosteroids Long-term low-dose corticosteroids	No proven treatment Manage risk factors through lifestyle modifications Low-dose aspirin (can be used sometimes)

Solution for Question 23:

Correct Answer: B) High-dose Corticosteroids

Explanation:

- The patient's symptoms and clinical findings are consistent with acute retrobulbar optic neuritis, commonly associated with demyelination in multiple sclerosis.
- While spontaneous recovery can occur, IV methylprednisolone is recommended to reduce inflammation, shorten symptom duration, and decrease the risk of future attacks, especially in high-risk patients.

Table rendering failed

Oral antibiotics (Option C) are typically used to treat bacterial infections. While optic neuritis can be associated with infections, such as neuroretinitis, antibiotics would only be appropriate if there is clear evidence of a bacterial infection.

Surgical intervention (Option D) is not indicated in the management of optic neuritis. While surgery, such as optic nerve sheath decompression, may be considered in other optic nerve conditions like papilledema, it is not appropriate for optic neuritis.

Solution for Question 24:

Correct Answer: C) Neuroretinitis

Explanation:

The clinical scenario with a macular star in fundoscopy suggests Neuroretinitis, a subtype of optic neuritis.

Optic neuritis is an inflammatory condition of the optic nerve that can cause vision loss, with symptoms including decreased vision, pain in eye movement, and visual field defects.

Classification of Optic neuritis

	Papillitis (Option A)	Retrobulbar Neuritis (Option B)	Neuroretinitis (Option C)
Site	Optic disc	Optic nerve behind the eyeball.	Optic disc and surrounding retina in the macular area.
Characteristic features	[Image not available] Fundoscopy reveals a blurred disc margin due to leakage and increased capillary permeability, edematous and	Pain on eye elevation RAPD (relative afferent pupillary defect) is observed. Fundoscopic examination typically reveals normal	[Image not available] A macular star is seen in fundoscopy (accumulation of lipids).

	hyperaemic disc, and physiological cup is obliterated.	optic disc, at least initially, which is why the classic phrase, "the patient sees nothing, and the doctor sees nothing" is often used to describe retrobulbar neuritis.	
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Clinical features,

It may be asymptomatic or associated with,

- Visual loss: Monocular sudden, progressive and profound visual loss is the hallmark of acute optic neuritis.
- Dark adaptation may be lowered.
- Visual obscuration in bright light is a typical symptom of acute optic neuritis.
- Impairment of color vision is always present in optic neuritis.
- Visual acuity is usually reduced markedly.
- Visual field changes: The most common field defect in optic neuritis is a relative central or centrocecal scotoma.
- Contrast sensitivity is impaired.

Papilledema (Option D) is bilateral optic disc swelling due to increased intracranial pressure. Findings include elevated optic disc, blurred disc margins, retinal hemorrhages, and venous congestion. It can be associated with optic neuritis.

[Image not available]

Solution for Question 25:

Correct Answer: B) Multiple sclerosis

Explanation:

The given scenario suggests Primary optic atrophy, with multiple sclerosis being the most common cause.

Optic Atrophy: It is the degeneration of the optic nerve, which occurs as an end result of any pathologic process that damages axons in the anterior visual system.

Classification of Optic atrophy:

	Primary	Secondary	Consecutive	Glaucomatous
Pathology	Refers to the degeneration of the nerve fibres Seen in the retrolaminar area.	Occurs following any pathologic condition or disease.	Occurs following destruction of ganglion cells secondary to lesions of the choroid and/ or retina	Occurs due to long-standing glaucoma.
Etiology	Multiple sclerosis (Option B) Retrobulbar neuritis Leber's hereditary optic atrophies Intracranial tumours (e.g. pituitary tumours) Traumatic severance or avulsion of the optic nerve Toxic amblyopia (chronic retrobulbar neuritis) Tabes dorsalis.	Long-standing optic nerve disease. Papilledema (Option A)	Diffuse chorioretinitis Retinal pigmentary dystrophies (e.g., retinitis pigmentosa (Option C), pathological myopia and occlusion of the central retinal artery)	Glaucoma (Option D)
Fundoscopy	Chalky white Clear disc margin Lamina cribrosa well seen Normal vessels.	Dirty grey Clear disc margin Lamina cribrosa obscured Peripapillary sheathing over vessels Hyaline bodies on the retina.	Waxy pallor Blurred disc margin Lamina cribrosa well seen Attenuated vessels	Characterized by deep and wide cupping of the optic disc and nasal shift of the blood vessels.
optic atrophy" data-author="Sean Collol, James Gilman" data-hash="4206" data-license="NA" data-source="https://morancore.utah.edu/basic-ophthalmology-review/optic-atrophy/" data-tags="March2025" height="415" src="https://image.prepladder.com/notes/z6KJcvvW0ejYph1ItqDE1743131175.png" width="500" />	optic atrophy" data-author="Sean Collol, James Gilman" data-hash="4207" data-license="NA" data-source="https://morancore.utah.edu/basic-ophthalmology-review/optic-atrophy/" data-tags="March2025" height="499" src="https://image.prepladder.com/notes/EGfpiddMbWuiw0m22Kf1743131253.png" width="500" />	optic atrophy" data-author="V. Osaguona" data-hash="4208" data-license="NA" data-source="https://www.semanticscholar.org/paper/Differential-diagnoses-of-the-pale-white-atrophic-Osaguona/9ef64b2cbe3b7e8541a3deede266744dd069bf04" data-tags="March2025" height="341" src="https://image.prepladder.com/notes/IGLRyjFHxZYd6NM9uCJ01743131	optic atrophy" data-author="Sean Collol, James Gilman" data-hash="4209" data-license="NA" data-source="https://morancore.utah.edu/basic-ophthalmology-review/optic-atrophy/" data-tags="March2025" height="500" src="https://image.prepladder.com/notes/JQeO7hZDIbwid3RnoZ71743131440.png" width="500" />	

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Solution for Question 26:

Correct Answer: A) Increase in dark adaptation

Explanation:

With optic nerve damage, dark adaptation decreases, along with dyschromatopsia, central scotoma, and contrast sensitivity decreases.

Manifestations of Optic nerve dysfunction

- Visual loss can be gradual and sudden, depending on the cause.
- Relative Afferent Pupillary Defect (RAPD) or Marcus Gunn Pupil The pupils paradoxically dilate when the flashlight is swung from the unaffected to the affected eye due to the decreased light perception. It is caused by incomplete optic nerve lesions and severe retinal diseases. It is the first sign of optic nerve disorder and can be tested by swinging flashlight test.
- The pupils paradoxically dilate when the flashlight is swung from the unaffected to the affected eye due to the decreased light perception.
- It is caused by incomplete optic nerve lesions and severe retinal diseases.
- It is the first sign of optic nerve disorder and can be tested by swinging flashlight test.
- The pupils paradoxically dilate when the flashlight is swung from the unaffected to the affected eye due to the decreased light perception.
- It is caused by incomplete optic nerve lesions and severe retinal diseases.
- It is the first sign of optic nerve disorder and can be tested by swinging flashlight test.

[Image not available]

- Dyschromatopsia: Impairment of colour vision (Option B).
- There is a decrease in, Visual activity Dark adaptation (Option A) Brightness Contrast sensitivity (Option D)
- Visual activity
- Dark adaptation (Option A)
- Brightness

- Contrast sensitivity (Option D)
- Visual field defects include,
- Visual activity
- Dark adaptation (Option A)
- Brightness
- Contrast sensitivity (Option D)

Central scotoma (most common) (Option C)	[Image not available]
Centrocecal scotoma	[Image not available]
Altitudinal field defect	[Image not available]

Solution for Question 27:

Correct Answer: C) Intracanalicular part

Explanation:

The intracanalicular part of the optic nerve is the segment located within the optic canal. The ophthalmic artery enters the optic canal alongside the optic nerve and lies inferolateral to the optic nerve.

Parts of the optic nerve:

Part of the optic nerve	Description
Intraocular (Option A)	It starts from the optic disc, pierces the choroid and sclera and becomes continuous with the intraorbital part at the back of the eyeball.
Intra-orbital (Option B)	Extends from the back of the eyeball to the optic foramina and runs parallel to the ophthalmic artery.
Intra-canalicular (Option C)	It is closely related to the ophthalmic artery, which lies inferolateral to optic nerve. It crosses obliquely over ophthalmic artery as it enters the orbit to lie on its medial side.
Intracranial (Option D)	It lies above the cavernous sinus and converges with its fellow to form the optic chiasma.

optic nerve" data-author="NA" data-hash="4201" data-license="NA" data-source="NA" data-tags="March2025" height="500" src="https://image.prepladder.com/notes/mX1MlwHoKh4oVDKxscMk1743130886.png" width="500" />

Solution for Question 28:

Correct Answer: C) Short posterior ciliary artery

Explanation:

The Circle of Zinn or Circle of Zinn-Haller is primarily formed by the short posterior ciliary arteries, which supply the optic nerve head.

Optic nerve:

- The optic nerve (2nd Cranial nerve) starts from the optic disc and extends up to the optic chiasma, where the two nerves meet.
- Blood supply of optic nerve: Circle of Zinn (also called Circle of Zinn-Haller): Formed by anastomosis of the short posterior ciliary artery. Central retinal artery (branch of ophthalmic artery)
- Circle of Zinn (also called Circle of Zinn-Haller): Formed by anastomosis of the short posterior ciliary artery.
- Central retinal artery (branch of ophthalmic artery)
- Circle of Zinn (also called Circle of Zinn-Haller): Formed by anastomosis of the short posterior ciliary artery.
- Central retinal artery (branch of ophthalmic artery)

retinal artery" data-author="NA" data-hash="4200" data-license="NA" data-source="NA" data-tags="March2025" height="500" src="https://image.prepladder.com/notes/UqSzREbutE6UWLjDb0JU1743130817.png" width="500" />

The central retinal artery (Option A) supplies the inner 6 layers of the optic disc. It does not contribute to the formation of the Circle of Zinn.

The ophthalmic artery (Option B), along with the choroidal arteries, circle of Zinn, and central retinal artery, contributes to the formation of the pial plexus.

Solution for Question 29:

Correct Answer: A) Congruous Homonymous Hemianopia with Macular Sparing

Explanation:

- A lesion in the occipital lobe, specifically in the Posterior Cerebral Artery (PCA) territory affecting the calcarine cortex, typically causes congruous homonymous hemianopia with macular sparing.
- This characteristic defect arises because the posterior cerebral artery (PCA) is the major artery supplying the primary visual cortex, while the macula is protected by its blood supply from middle cerebral artery (MCA).

Lesions in Visual Cortex:

- Typically present as congruous homonymous hemianopia (visual field loss is consistent in both eyes).

[Image not available]

Blood supply of Visual Cortex:

[Image not available]

- The primary visual cortex, mainly supplied by the posterior cerebral artery (PCA), processes visual information, and a PCA stroke can lead to homonymous hemianopia.
- Macular Area: Located at the posterior tip of the visual cortex and is crucial for detailed central vision. Its dual blood supply (from both PCA and MCA) helps to protect it from complete loss even when there is an isolated PCA lesion. (Macular Sparing).

Injury to the tip of the visual cortex leads to Macular Homonymous Hemianopia.

Macular Homonymous Hemianopia" data-author="RobertB3009" data-hash="7122" data-license="CC BY SA 4.0" data-source="https://commons.wikimedia.org/wiki/File:Visual_field_homonymous_hemianopia.png" data-tags="April2025" src="https://image.prepladder.com/notes/co4YxQ2ymMx4MAfUyE181743687227.png" />

Solution for Question 30:

Correct Answer: B) Superior homonymous quadrantanopia

Explanation:

A lesion affecting the inferior optic radiations, particularly in Meyer's loop within the temporal lobe, typically results in superior homonymous quadrantanopia.

[Image not available]

Optic Radiation:

- A neural pathway that transmits visual information from the lateral geniculate body to the primary visual cortex in the occipital lobe.

Aspect	Inferior Optic Radiations (Meyer's Loop)	Superior Optic Radiations (Baum's Loop)
Location of Lesion	Temporal lobe	Parietal lobe
Affected Visual Field	Superior quadrants of the contralateral visual field of both eyes	Inferior quadrants of the contralateral visual field of both eyes.
Visual Defect	Superior homonymous quadrantanopia	Inferior homonymous quadrantanopia (Option A)
Associated Symptoms	Memory loss Hearing loss Auditory hallucinations	Contralateral hemiplegia Hemianesthesia Optokinetic nystagmus
	[Image not available]	[Image not available]

Central scotoma (Option C) is a defect in central vision that is typically associated with lesions in the retina, optic nerve, and macula rather than the specific pathways in the temporal lobe.

[Image not available]

Homonymous hemianopia (Option D) occurs in lesions involving the optic tract, lateral geniculate body (LGB), or both superior and inferior optic radiations.

[Image not available]

Solution for Question 31:

Correct Answer: B) Left optic tract

Explanation:

A lesion in the left optic tract would cause right (contralateral) homonymous hemianopia, where the right visual field is lost in both eyes.

Location of the lesion	Characteristic features
Optic nerve	Marked loss of vision or complete blindness on the affected side associated. Near (accommodation) reflex is present.
Optic chiasma	Bitemporal hemianopia
Optic tract	Contralateral homonymous hemianopia Contralateral hemianopic pupillary responses, i.e. Wernicke's reaction.
Optic radiations	Superior quadrantic hemianopia (pie in the sky) Produced when the inferior fibres (Meyer's loop) are involved in temporal lobe lesions) Inferior quadrantic hemianopia (pie on the floor) Produced when superior fibres (Baum's loop) are involved in parietal lobe lesions.
Visual cortex	Homonymous hemianopia, often with macular sparing

[Image not available]

Solution for Question 32:

Correct Answer: A) Bitemporal Hemianopia

Explanation:

A pituitary adenoma compressing the optic chiasm typically results in Bitemporal hemianopia due to the involvement of crossing nasal fibres of the optic nerves, causing loss of vision in the outer portions of both visual fields.

Visual Pathway Lesions and Defects		
Site of Lesion	Fibers Affected	Field Defect
Optic Nerve	Ipsilateral Temporal + Nasal Fibers	Anopia

Optic Chiasma	Bilateral Nasal Fibers	[Image not available] Bitemporal Hemianopia (Heteronymous Hemianopia)
Optic Tract Lateral Geniculate Body (LGB Lesion) Optic Radiation	Ipsilateral Temporal Fibers + Contralateral Nasal Fibers	[Image not available] Homonymous Hemianopia (Option C)
Visual Cortex without the involvement of tip (Post. Cerebral Artery)	Ipsilateral Temporal Fibers + Contralateral Nasal Fibers except macular representation	Homonymous Hemianopia with macular sparing" data-author="NA" data-hash="4192" data-license="NA" data-source="NA" data-tags="March 2025" height="500" src="https://imag e.prepladder.com/notes/475PyWODI Hb1bh0szWR41743129664.png" width="500" /> Homonymous Hemianopia with Macular Sparing (Option D)
Tip of Visual Cortex (Middle Cerebral Artery)	Macular Representation of Ipsilateral Temporal and Contralateral Nasal Fibers.	Homonymous Hemianopia" data-author="NA" data-hash="4193" data-license="NA" data-source="NA" data-tags="March2025" height="500" src="https://image.prep ladder.com/notes/BbGiSI040JK03Ge tGyL21743129916.png" width="500" /> Macular Homonymous Hemianopia

Pie in the Sky (Option B) (Superior Homonymous Quadrantanopia) occurs in Lesion of Inferior Optic Radiation.

[Image not available]

Previous Year Questions

1. A 5-month-old baby is brought by the mother with complaints of the left upper eyelid moving up and down during breastfeeding or thumb sucking, which disappears when the baby is not being fed. What is the most likely diagnosis?

- A. Marcus Gunn Jaw-Winking Syndrome
- B. Lagophthalmos
- C. Myasthenia Gravis
- D. 3rd CN palsy

2. A patient presents with breathlessness and is diagnosed with a Pancoast tumor. Which of the following findings is not typically associated with this condition?

- A. Anhidrosis
- B. Miosis
- C. Ptosis
- D. Exophthalmos

3. Identify the defect shown in the image given below:

[Image not available]

- A. Binasal hemianopia
- B. Bitemporal hemianopia
- C. Homonymous hemianopia
- D. Altitudinal defect

4. Identify the visual field defect:

[Image not available]

- A. Altitudinal defect
 - B. Roennes nasal step
 - C. Blind spot enlargement
 - D. Arcuate scotoma
-

5. A patient on antitubercular therapy (ATT) started experiencing visual problems. Which ATT drug is most likely responsible for this adverse effect?

- A. Rifampicin
 - B. Streptomycin
 - C. Ethambutol
 - D. Pyrazinamide
-

6. Which nerve defect causes lagophthalmos?

- A. 7th nerve
 - B. 6th nerve
 - C. 4th nerve
 - D. 5th nerve
-

7. A 39-year-old woman arrives at the emergency department with a gradual loss of vision in her left eye over the past three months, as well as blurry vision in her

right eye for the past 10 days. She has been experiencing frequent headaches for the past six months, which are worse in the morning and accompanied by nausea and vomiting. She has been taking ondansetron and paracetamol, which have provided some relief. Her visual acuity in her right eye is 20/30, while she can only count fingers with her left eye. An MRI scan reveals a frontal brain lesion on the left side. These findings suggest a diagnosis of Foster-Kennedy syndrome. Which of the following features is not typically seen in Foster-Kennedy syndrome?

[Image not available]

- A. Ipsilateral optic atrophy, ipsilateral central scotoma, bilateral progressive optic neuropathy
- B. Contralateral papilledema, ipsilateral central scotoma, bilateral progressive optic neuropathy
- C. Ipsilateral optic atrophy, contralateral papilledema, ipsilateral central scotoma
- D. Contralateral anosmia, Contralateral central scotoma, bilateral progressive optic neuropathy

8. Sudden painful loss of vision in one eye is due to the lesion at?

- A. Optic nerve
- B. Optic chiasma
- C. Optic tract
- D. Visual cortex

9. What is the probable diagnosis for a patient who exhibits miosis, anhidrosis, mild ptosis, and a persistent small pupil even in low light conditions?

- A. Adie's tonic people
- B. Horner syndrome
- C. Marcus gunn pupil
- D. Argyll Robertson pupil

10. Optic nerve glioma is seen in?

- A. NF-1
 - B. Tuberous sclerosis
 - C. NF-2
 - D. Schwannoma
-

11. Which tuberculosis drug causes impairment of color vision and optic neuritis?

- A. Rifampicin
 - B. Ethambutol
 - C. Pyrizinamide
 - D. Streptomycin
-

12. A 40-year-old male patient presents with altered sensation in the extremities, difficulty with coordination, and speech problems. On examination, it is noted that the pupillary light reaction is absent, but the accommodation reaction is present. Which condition is most likely responsible for this presentation?

- A. Neurosyphilis
 - B. Horner's syndrome
 - C. Papilledema
 - D. Optic Neuritis
-

13. A patient presented with tabes dorsalis. What will be observed when his pupils are exposed to light?

- A. Pupils react but do not accommodate
 - B. Pupils accommodate but do not react
 - C. Pupils neither react nor accommodate
 - D. Pupils react and accommodate
-

14. In a patient undergoing visual field testing, the observed loss suggests a potential site of lesion within the optic pathway.

[Image not available]

- A. Frontal visual field
 - B. Occipital lobe
 - C. Optic radiation
 - D. Optic chiasma
-

15. A patient presents to the OPD with the given visual field effect. What is the probable site of the lesion based on the image displayed?

[Image not available]

[Image not available]

- A. 3
 - B. 1
 - C. 4
 - D. 6
-

16. What is the probable diagnosis for a man of 45 years who arrives at the emergency department with an abrupt loss of vision and experiencing painful movements of the eye? Upon examination, a relative afferent pupillary defect is observed in the affected eye. Further visual field testing demonstrates a central scotoma in the same eye.

- A. Optic nerve glioma
 - B. Optic neuritis
 - C. Eale's disease
 - D. Neuroretinitis
-

17. Where is the location of the lesion in Wernicke's hemianopic pupil?

- A. Optic nerve
 - B. Optic tract
 - C. Optic radiation
 - D. Optic chiasma
-

18. What is the clinical presentation of a patient with a tumor located in the anterior pituitary gland causing compression on the optic chiasma?

- A. Homonymous hemianopia
 - B. Inferior quadrantanopia hemianopia
 - C. Bitemporal hemianopia
 - D. Superior quadrantanopia hemianopia
-

19. A 45-year-old male arrived at the emergency ward with sudden vision loss and experiencing painful eye movements. Upon examination, the affected eye shows a

relative afferent pupillary defect, a normal optic disc, and a central scotoma. What is the likely diagnosis?

- A. Optic nerve glioma
 - B. Optic neuritis
 - C. Eale's disease
 - D. Optic atrophy
-

20. What is the correct information about the image provided below?

[Image not available]

- A. It is called Argyl Robertson pupil
 - B. Both eyes are abnormal
 - C. Right eye is normal
 - D. Seen in optic neuritis
-

21. Which of the following conditions is characterized by the presence of ipsilateral third nerve palsy and contralateral hemiparesis?

- A. Weber syndrome
 - B. Benedikt syndrome
 - C. Millard-Gubler syndrome
 - D. Foville's syndrome
-

22. What type of field defects arise from a lesion located at the optic chiasma?

- A. Bitemporal hemianopia
- B. Unilateral blindness

- C. Pie on the floor
 - D. Homonymous hemianopia Bilateral homonymous macular defect
-

23. Where is the location of the lesion resulting in the following visual field defect?

[Image not available]

- A. Anterior parietal radiations
 - B. Optic tract
 - C. Optic chiasma
 - D. Temporal radiations
-

24. Which of the following is the site of lesion in internuclear ophthalmoplegia?

- A. Medial Longitudinal Fasciculus
 - B. 6th nerve nucleus
 - C. Pontine paramedian reticular formation
 - D. A&C
-

25. Which disease is associated with bilateral optic neuritis and transverse myelitis?

- A. Schilder disease
 - B. Parainfectious optic neuritis
 - C. Neuromyelitis optica
 - D. Multiple sclerosis
-

26. In a patient diagnosed with a stroke, their visual field testing indicated macula sparing homonymous hemianopia. Where is the probable location of the lesion within the optic pathway?

- A. Optic tract
 - B. Occipital lobe
 - C. Optic radiation
 - D. Optic chiasma
-

27. Identify the site of the lesion.

[Image not available]

- A. Left occipital lobe
 - B. Right occipital lobe
 - C. Right optic tract
 - D. Optic chiasma
-

28. Based on the image provided, what is the likely location of the lesion in a patient who is experiencing difficulty perceiving the lateral fields?

[Image not available]

- A. Optic chiasma
 - B. Right optic tract
 - C. Left optic tract
 - D. Right optic radiation
-

29. The purpose of the photostress test is to distinguish between:

- A. Macula and optic nerve disease
 - B. Cataract and glaucoma
 - C. Lens and cornea
 - D. Retina and vitreous pathologies
-

30. In a patient, a 33-year-old female presents with symptoms of gradual vision loss specifically affecting the right halves of both eyes. The question pertains to the location of the lesion in the optic pathway.

- A. Left optic tract
 - B. Right visual cortex
 - C. Optic chiasma
 - D. Right optic nerve
-

31. Which of the following statements accurately describes optic neuritis?

- A. Relative afferent pupillary defect is seen
 - B. There is no pain on ocular movement
 - C. Multiple myeloma is a common cause
 - D. Oral corticosteroids are the mainstay of treatment
-

32. What would be the next investigative step for a 27-year-old female patient who presents with sudden vision loss accompanied by a relative afferent pupillary defect in her right eye, while her left eye appears normal on examination? MRI brain & orbit Myelin oligodendrocyte glycoprotein (MOG) and neuromyelitis optica (NMO) antibodies Blood sugar Thyroid profile

- A. 1, 2 and 3
 - B. 1 and 2
 - C. 2 and 3
 - D. 3 and 4
-

33. In a female patient, there is a complaint of vision loss, specifically affecting the right halves of both eyes. Where is the lesion located?

- A. Optic nerve
 - B. Left optic tract
 - C. Optic chiasma
 - D. Right optic tract
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	3
Question 4	3
Question 5	3
Question 6	1
Question 7	4
Question 8	1
Question 9	2
Question 10	1
Question 11	2
Question 12	1
Question 13	2
Question 14	2
Question 15	1
Question 16	2
Question 17	2
Question 18	3
Question 19	2
Question 20	4
Question 21	1
Question 22	1

Question 23	4
Question 24	1
Question 25	3
Question 26	2
Question 27	1
Question 28	1
Question 29	1
Question 30	1
Question 31	1
Question 32	1
Question 33	2

Solution for Question 1:

- Marcus Gunn Jaw-Winking Syndrome (also known as the jaw-winking reflex) is a congenital condition where there is an abnormal connection between the trigeminal nerve (supplying Lateral pterygoid) and the oculomotor nerve (supplying LPS).
- This results in the upper eyelid (ptosis) lifting when the jaw moves, such as during chewing, sucking, or other movements involving the jaw.
- The eye movement (or "winking") is typically unilateral and can occur during activities like breastfeeding or thumb-sucking, as in this case.
- This reflex is not present when the baby is not feeding, which is a key feature of this condition.

Congenital Ptosis: Associated with congenital weakness or maldevelopment of the levator palpebrae superioris (LPS) muscle.

Congenital ptosis probably results from a failure of neuronal migration or development with muscular sequelae secondary to this. A minority of patients have a family history.

Characteristic features:

- Drooping of one or both upper lids.

- Lid crease is either diminished or absent.
- Lid lag on downgaze due to tethering effect of abnormal LPS muscle.

Types of congenital ptosis:

1. Simple congenital ptosis:

- Unilateral or bilateral
- Non-progressive
- No fluctuation: Unlike myogenic ptosis, congenital ptosis does not worsen throughout the day or improve with rest.

2. Congenital ptosis with associated weakness of superior rectus muscle.

3. Blepharophimosis syndrome:

- Congenital ptosis
- Blepharophimosis
- Telecanthus
- Epicanthus inversus.

4. Congenital synkinetic ptosis (Marcus Gunn jaw-winking ptosis)

- Retraction of the ptotic lid with jaw movements due to stimulation of ipsilateral pterygoid muscle.
- It occurs due to trigemino-oculomotor synkinesis. The pathophysiology of Marcus Gunn Jaw winking phenomenon is a congenital misdirection of mandibular branch of trigeminal nerve, i.e., this synkinetic movement results from a congenital aberrant connection between motor branches of the trigeminal nerve controlling muscles of mastication and superior division of the oculomotor nerve controlling the levator palpebrae superioris.

[Image not available]

Lagophthalmos (Option B): This condition is characterized by an inability to completely close the eyelids, usually due to facial nerve (VII) dysfunction or other causes of eyelid weakness.

Myasthenia Gravis (Option C): Myasthenia gravis typically presents with fatigable weakness of the muscles, including the eyelids (ptosis), which worsens with activity.

3rd CN Palsy (Option D): A third cranial nerve palsy typically causes ptosis (drooping of the eyelid) and ocular motility issues (like strabismus or limited eye movement).

Solution for Question 2:

A Pancoast tumor is a type of lung cancer that occurs at the apex of the lung and can invade nearby structures, including the brachial plexus, sympathetic chain, and adjacent ribs. The tumor is commonly associated with a condition known as Horner's syndrome due to its involvement with the sympathetic nervous system.

Horner's Syndrome

- Horner's syndrome is caused by a lesion of the sympathetic chain, which clinically manifests as ipsilateral miosis, facial anhidrosis, and ptosis (drooping of the eyelid).
- Causes:

Table rendering failed

- Clinical manifestations: Miosis, ptosis, enophthalmos. Anhidrosis (loss of sweating): It is seen in preganglionic lesions. Absence of cilio-spinal reflex (involuntary response where the pupil of the eye dilates in response to painful stimuli applied to the skin of the neck or upper back)
- Miosis, ptosis, enophthalmos.
- Anhidrosis (loss of sweating): It is seen in preganglionic lesions.
- Absence of cilio-spinal reflex (involuntary response where the pupil of the eye dilates in response to painful stimuli applied to the skin of the neck or upper back)
- Acute Horner's syndrome with associated ipsilateral neck or facial pain requires urgent imaging of the neck for evaluation of a carotid dissection
- Miosis, ptosis, enophthalmos.
- Anhidrosis (loss of sweating): It is seen in preganglionic lesions.
- Absence of cilio-spinal reflex (involuntary response where the pupil of the eye dilates in response to painful stimuli applied to the skin of the neck or upper back)

Solution for Question 3:

Correct Option C - Homonymous hemianopia:

- The given image shows homonymous hemianopia. Homonymous hemianopia refers to the loss of the same half of the visual field in both eyes, meaning that the patient loses vision on

either the left or right side in both eyes.

Incorrect Options:

Option A - Binasal hemianopia: Involves the loss of the nasal (inner) half of the visual field in both eyes.

Option B - Bitemporal hemianopia: Involves the loss of the temporal (outer) half of the visual field in both eyes.

Option D - Altitudinal defect: Refers to the loss of the upper or lower half of the visual field in one or both eyes.

Solution for Question 4:

Correct Option C - Blind spot enlargement:

The given image shows blind spot enlargement.

Incorrect Options:

Option A - Altitudinal defect:

[Image not available]

Option B - Roennes nasal step:

[Image not available]

Option D - Arcuate scotoma:

[Image not available]

Solution for Question 5:

Correct Option C - Ethambutol:

- Ethambutol can cause Toxic optic neuropathy or Toxic amblyopia
- Toxic Amblyopia can lead to partial loss of vision.

Incorrect Options:

- Options A, B & D: Rifampicin, Streptomycin and Pyrazinamide do not cause visual problems

Solution for Question 6:

Correct Option A - 7th nerve

- Lagophthalmos is the inability to close the eye.
- It is a feature of 7th nerve palsy.
- Neuroparalytic keratitis can occur due to seventh nerve palsy, resulting in lagophthalmos from affected orbicularis oculi and leading to exposure keratitis

Incorrect Options:

- Options B, C & D: 6th, 4th and 5th nerves are not involved in Lagophthalmos

Solution for Question 7:

Correct option D.

- Contralateral anosmia, contralateral central scotoma, bilateral progressive optic neuropathy. This option is incorrect because anosmia is not typically associated with Foster-Kennedy syndrome. Additionally, bilateral progressive optic neuropathy is not a characteristic feature of this syndrome.
- Foster-Kennedy syndrome is a clinical syndrome characterized by a combination of optic atrophy in one eye and papilledema in the other eye. It is typically associated with a frontal brain lesion, such as a tumor or meningioma, which can cause compression or infiltration of the optic nerves and optic chiasm.

Incorrect options:

Option A: Ipsilateral optic atrophy, ipsilateral central scotoma, bilateral progressive optic neuropathy. This option is incorrect.

Option B: Contralateral papilledema, ipsilateral central scotoma, bilateral progressive optic neuropathy. This option is incorrect .

Option C: Ipsilateral optic atrophy, contralateral papilledema, ipsilateral central scotoma. This option is incorrect.

Solution for Question 8:

Correct Option A - Optic Nerve:

- Prechiasmatic lesions typically have a traumatic or inflammatory origin with concentric narrowing of the visual field after the compression of the outer portions of the optic nerve fibers.
- It presents with unilateral sudden painful loss of vision.

Incorrect Options:

Option B - Optic chiasma: Lesion at optic chiasma will result in bitemporal hemianopia.

Option C - Optic tract: Lesion at optic tract will result in homonymous hemianopia.

Option D - Visual cortex: Lesion at visual cortex will result in homonymous hemianopia with macular sparing.

Solution for Question 9:

Correct Option B - Horner syndrome:

- The clinical presentation of the patient is suggestive of horner syndrome. Signs seen are Mild ptosis as a result of the weakness of Müller muscle and miosis due to the unopposed action of the sphincter pupillae with resultant anisocoria. A key examination finding - anisocoria is accentuated in dim light since, in contrast to a normal fellow pupil, the Horner pupil will dilate only very slowly. Reduced ipsilateral sweating is seen in the lesion below the superior cervical ganglion.

- Mild ptosis as a result of the weakness of Müller muscle and miosis due to the unopposed action of the sphincter pupillae with resultant anisocoria.
- A key examination finding - anisocoria is accentuated in dim light since, in contrast to a normal fellow pupil, the Horner pupil will dilate only very slowly.
- Reduced ipsilateral sweating is seen in the lesion below the superior cervical ganglion.
- Mild ptosis as a result of the weakness of Müller muscle and miosis due to the unopposed action of the sphincter pupillae with resultant anisocoria.
- A key examination finding - anisocoria is accentuated in dim light since, in contrast to a normal fellow pupil, the Horner pupil will dilate only very slowly.
- Reduced ipsilateral sweating is seen in the lesion below the superior cervical ganglion.

Incorrect Options:

Option A - Adie's tonic people: It is a unilateral condition and is characterized by anisocoria (different size of pupil) and direct light reflex is sluggish or absent.

Option C - Marcus Gunn pupil: Aka relative afferent pupillary defect is characterized by paradoxical dilatation of both the pupil when light is flashed on the diseased eye.

Option D - Argyll Robertson pupil: It is a specific sign of late-stage syphilis. The bilaterally small pupils do not constrict when exposed to bright light.

Solution for Question 10:

Correct Option A - NF-1:

- Optic nerve glioma is a slow growing tumor that typically affects children associated with neurofibromatosis 1.

Incorrect Options:

- Options B, C, and D are incorrect. Optic nerve glioma is not associated with these constions.
-

Solution for Question 11:

Correct Option B - Ethambutol:

- Ethambutol can cause acute or chronic optic neuritis characterized by painless bilateral blurring and impairment of color vision.

Incorrect options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.
-

Solution for Question 12:

Correct Option A - Neurosyphilis:

- Neurosyphilis is a bacterial infection of the brain or spinal cord. It occurs in people who have had untreated syphilis for many years. One of the signs of neurosyphilis can be Argyll Robertson pupil, where the pupils do not react to light (absent pupillary light reaction) but do react to accommodation (preserved accommodation reaction).

Incorrect Options:

Option B - Horner's syndrome: It is characterized by miosis, ptosis, and decreased sweating anhidrosis on the affected side of your face.

Option C - Papilledema: This is a condition in which increased pressure in or around the brain causes the part of the optic nerve inside the eye to swell. It can cause visual disturbances but doesn't typically result in an absent pupillary light reaction with preserved accommodation reaction.

Option D - Optic Neuritis: It presents with Marcus Gunn pupil, i.e., paradoxical constriction of both the pupil when light is flashed to the diseased eye.

Solution for Question 13:

Correct Option B - Pupils accommodate but do not react:

- Tabes dorsalis is a slowly progressive parenchymatous degenerative disease of the dorsal column and dorsal root of the spinal cord is seen in neurosyphilis caused by *Treponema pallidum*.
- The ophthalmic feature seen in neurosyphilis is the Argyll Robertson pupil, characterized by a positive accommodative reflex and absent light reflex.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 14:

Correct Option B - Occipital lobe:

- The occipital lobe is primarily responsible for processing visual information received from the eyes.
- The occipital lobe lesion is 90% of the time due to stroke causing congruous homonymous hemianopia.
- When PCA is blocked the macula is spared. This is what is called macula sparing hemianopia or also called key hole vision.

Incorrect options

Option A - Frontal visual field:

- Lesion in the frontal visual field refers to damage or impairment in the area of the visual field that is processed by the frontal part of the brain.

Option C - Optic radiation:

- A lesion in the optic radiation can lead to visual field defects that are specific to the affected area. For example, a lesion in the temporal optic radiation may result in a contralateral superior quadrantanopia.

Option D - Optic chiasm:

- Lesion at the optic chiasm causes visual field defects involving the nasal visual field of each eye, such as bitemporal hemianopia.

Solution for Question 15:

Correct Option A - 3:

- The lateral visual field defect, as shown in the image, is of bitemporal hemianopia, i.e., the nasal fibers to the retina of both eyes are damaged, leading to loss of vision on the temporal side of the visual field.
- The lesion at optic chiasma, i.e., 3 is responsible for bitemporal hemianopia as the nasal fibers from the retina decussate at optic chiasma.

Incorrect Options: Option B - 1: Lesion at 1 i.e., optic nerve will result in left eye loss of vision.

Option C - 4: Lesion at 4, i.e., optic tract will result in right sided homonymous hemianopia.

Option D - 6: Lesion at 6 i.e., anterior parietal radiation will result in right side inferior quadrantanopia aka pie on the floor.

Solution for Question 16:

Correct Option B - Optic neuritis:

- The sudden loss of vision, painful eye movements, relative afferent pupillary defects, and central scotoma are characteristic of optic neuritis.
- It typically presents with marked sudden vision loss, dark adaptation may be lowered, and decreased contrast sensitivity.
- On examination, RAPD (relative afferent pupillary defect), central scotoma, and centrocecal scotoma can be seen

Incorrect Options:

Option A - Optic nerve glioma is a tumor that can affect the optic nerve, but it is more commonly seen in children and young adults.

Option C - Eale's disease, also known as retinal vasculitis, typically presents with recurrent episodes of retinal hemorrhages, floaters, and vision loss.

Option D - Neuroretinitis is characterized by monocular visual decrease, vitritis, papillitis, retinal vasculitis, optic atrophy, retinal vascular attenuation and diffuse RPE degeneration.

Solution for Question 17:

Correct Option B - Optic tract:

- In Wernicke's hemianopic pupil, the site of the lesion is typically in the optic tract.
- The optic tracts contain both visual and pupillomotor fibers. The visual fibers terminate in the lateral geniculate body, and pupillary fibers leave the optic tract and terminate in the pretectal nuclei.
- Wernicke hemianopic pupil is characterized by normal pupillary light reflex when the unaffected hemiretina is stimulated and absent when the involved hemiretina is stimulated.

Incorrect Options:

- Options A, C, and D are incorrect. Lesions at these sites will only result in visual abnormalities.
-

Solution for Question 18:

Correct Option C - Bitemporal hemianopia:

- Bitemporal hemianopia is usually caused by tumors in the region of the sella turcica, pressure by a suprasellar aneurysm, or chronic arachnoiditis; these press upon the chiasma, so that the fibers going to the nasal halves of each retina are destroyed.
- So a tumor in the anterior pituitary pressing on the optic chiasma will present as a case of Bitemporal hemianopia.

Incorrect Options:

Option A: Homonymous hemianopia: Homonymous hemianopia is caused by post chiasmal lesions like optic tract, optic radiations, and visual cortex lesions.

Option B: Inferior quadrantanopia hemianopia: It is caused by a lesion of the fibers of the optic radiations in the anterior parietal lobe.

Option D: Superior quadrantanopia hemianopia: It is caused by a lesion of the fibers of the optic radiations in the temporal lobe.

Solution for Question 19:

Correct Option B - Optic neuritis:

- The sudden loss of vision, painful eye movements, relative afferent pupillary defects, and central scotoma are characteristic of optic neuritis.
- It typically presents with marked sudden vision loss, dark adaptation may be lowered, and decreased contrast sensitivity.
- On examination RAPD (relative afferent pupillary defect), central scotoma, and centrocecal scotoma can be seen.

Incorrect Options:

Option A - Optic nerve glioma: Optic nerve glioma is a type of brain tumor that can affect the optic nerve, but it is relatively rare in adults and typically presents with symptoms of optic nerve with proptosis.

Option C - Eale's disease: Eale's disease is a condition that causes inflammation and occlusion of the retinal blood vessels, leading to vision loss and floaters.

Option D - Optic atrophy: It is characterized by atrophy of the all damaged nerve fibers with total afferent pupillary defect and loss of perception of light.

Solution for Question 20:

Correct Option D - Seen in optic neuritis:

- The image represents the relative afferent pupillary defect (RAPD) characterized by weak pupillary response of the diseased and brisk response by the normal eye seen in incomplete optic nerve lesion (optic neuritis) or severe retinal disease (rhegmatogenous retinal detachment). The difference between the two pupillary reactions is highlighted by the swinging flash test.
- The image reveals the following findings: Both pupils dilated without the flash light. Both pupils contract when light is flashed to the left eye i.e., normal eye. Both pupils show paradoxical dilatation when light is flashed to the right eye, i.e., showing relative afferent pupillary defect. Again both the pupils contract when light is flashed to the left eye i.e., normal eye. Again both pupils show paradoxical dilatation when light is flashed to the right eye.
- Both pupils dilated without the flash light.
- Both pupils contract when light is flashed to the left eye i.e., normal eye.
- Both pupils show paradoxical dilatation when light is flashed to the right eye, i.e., showing relative afferent pupillary defect.
- Again both the pupils contract when light is flashed to the left eye i.e., normal eye.
- Again both pupils show paradoxical dilatation when light is flashed to the right eye.
- Both pupils dilated without the flash light.
- Both pupils contract when light is flashed to the left eye i.e., normal eye.
- Both pupils show paradoxical dilatation when light is flashed to the right eye, i.e., showing relative afferent pupillary defect.
- Again both the pupils contract when light is flashed to the left eye i.e., normal eye.
- Again both pupils show paradoxical dilatation when light is flashed to the right eye.

Incorrect Option:

Option A - It is called Argyl Robertson pupil: Argyl Robertson pupil is seen in neurosyphilis, where the pupils shows positive accommodation reflex and absent light reflex,

Option B - Both eyes are abnormal: Left eye is normal and right eye is affected with optic nerve lesion.

Option C - Right eye is normal: An optic nerve lesion or severe retinal disease is associated with right eye resulting in RAPD.

Solution for Question 21:

Correct Option A - Weber syndrome:

- Weber syndrome, also known as midbrain stroke syndrome, is a neurological condition characterized by the presence of ipsilateral third nerve palsy (cranial nerve III) and contralateral hemiparesis.
- In Weber syndrome, there is damage or compression of the midbrain, specifically in the region where the third nerve passes through the cerebral peduncle (crus cerebri).

Incorrect Options:

Option B - Benedikt syndrome: It is characterized by ipsilateral third nerve palsy and contralateral extrapyramidal signs such as hemitremors.

Option C - Millard-Gubler syndrome: Millard-Gubler syndrome is a rare condition characterized by ipsilateral sixth nerve palsy, contralateral hemiplegia, and ipsilateral LMN facial nerve palsy due to a lesion involving the ventral pons.

Option D - Foville's syndrome: Foville's syndrome is a rare brainstem syndrome that involves ipsilateral involvement of the fifth to eighth cranial nerves, central sympathetic fibres (Horner syndrome) and horizontal gaze palsy.

Solution for Question 22:

Correct Option A - Bitemporal hemianopia:

- A lesion at the optic chiasm, which is located at the base of the brain, can lead to a specific visual field defect called bitemporal hemianopia. In this condition, there is a loss of vision in the outer (temporal) half of the visual field of both eyes.
- This occurs because the visual field coordinates are opposite to the retinal field coordinates i.e., temporal visual field is supplied by the nasal retina which cross over at the optic chiasm resulting in damage to the nasal retinal nerve fibers only.

Incorrect Options:

Option B - Unilateral blindness: Unilateral blindness refers to the loss of vision in one eye and occurs due to ipsilateral optic nerve lesion.

Option C - Pie on the floor: Pie on the floor aka inferior quadrantanopia is suggestive of lesion at anterior parietal radiations.

Option D - Homonymous hemianopia: Homonymous hemianopia is suggestive of lesion at the optic tract.

Solution for Question 23:

Correct Option D - Temporal radiations:

- The visual field defect shown in the image is of the right side superior homonymous quadrantanopia, i.e., pie in the sky.
- The lesion will be in the temporal radiations because the inferior fibers of the optic radiations, which serve the upper visual fields, first sweep anteroinferior into the temporal lobe around the anterior tip of the temporal horn of the lateral ventricle called Meyer's loop.

Incorrect Options:

Option A - Anterior parietal radiations: Lesion at anterior parietal lesion will result in inferior quadrantanopia, i.e., pie on the floor.

Option B - Optic tract: Lesion at optic tract will result in homonymous hemianopia.

Option C - Optic chiasma: Lesion at optic chiasma will result in bitemporal hemianopia.

Solution for Question 24:

Correct Option A - Medial Longitudinal Fasciculus:

- "Medial Longitudinal Fasciculus (MLF)" is the site of the lesion in internuclear ophthalmoplegia (INO), it is between the third, fourth and sixth nuclei.
- It is characterized by defective ipsilateral adduction, and the contralateral abducting eye showing ataxic nystagmus.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for reference.

Solution for Question 25:

Correct Option C - Neuromyelitis optica:

- Neuromyelitis optica (NMO), also known as Devic's disease, is an autoimmune disorder that primarily affects the optic nerves and the spinal cord. It is characterized by recurrent episodes of bilateral optic neuritis, and can present with transverse myelitis, inflammation of the spinal cord, is also commonly associated with NMO.

Incorrect Options:

Option A - Schilder disease: It is characterized by breakdown of the myelin over nerves in the brain and spinal cord and present with bilateral optic neuritis.

Options B - Parainfectious optic neuritis: Optic neuritis with infectious cause is known as parainfectious disease. It is not associated with transverse myelitis.

Option D - Multiple sclerosis: Multiple sclerosis is the most common cause of optic neuritis and presents unilaterally.

Solution for Question 26:

Correct Option B - Occipital lobe:

- Homonymous hemianopia with macular sparing after a stroke which is indicative of the lesion being at the occipital lobe (visual cortex).
- It is because they are responsible for macular vision has dual blood supply via posterior cerebral arteries and middle cerebral artery.

Incorrect Options:

Option A - Optic tract: Lesion at optic tract will result in homonymous hemianopia involving the macular area.

Option C - Optic radiation: Lesions to optic radiations result in homonymous contralateral quadrantanopia. Those affecting Meyer's loop in the temporal lobe result in a homonymous upper quadrantanopia ("pie in the sky"), while those affecting the parietal lobe cause a lower quadrantanopia ("pie on the floor").

Option D - Optic chiasma: The visual field defect seen in the chiasmal syndrome is typically a bitemporal hemianopia.

Solution for Question 27:

Correct Option A - Left occipital lobe:

- The patient is presenting with loss of vision on the right side of both eyes with macular sparing, i.e., right-side homonymous hemianopia with macular hemianopia. So, the lesion will be on the left side because the visual field coordinates are opposite to the retinal field coordinates, i.e., the temporal part of the right eye will give vision toward the left and vice versa.
- So, the right side homonymous hemianopia means the temporal fibers from the right eye and nasal fibers from the left eye are affected. This is because the fibers from the nasal side of the right eye retina cross at the optic chiasm and continue as the optic tract with temporal fibers from the left eye temporal retina.
- Lesion at left visual cortex in temporal lobe is responsible for causing homonymous hemianopia with macular sparing effect due to dual supply to the area (watershed zone) responsible for the macular vision.

Incorrect Options:

Option B - Right occipital lobe: Lesion at right occipital lobe will result in left sided homonymous hemianopia with macular sparing effect.

Option C - Right optic tract: Lesion at right optic tract will result in left sided homonymous hemianopia.

Option D - Optic chiasma: Lesion at optic chiasma will result in bitemporal hemianopia.

Solution for Question 28:

Correct Option A - Optic chiasma:

- The lateral visual field defect, as shown in the image, is of bitemporal hemianopia, i.e., the nasal fibers to the retina of both the eyes are damaged, leading to loss of vision on the temporal side of the visual field.
- The lesion at optic chiasma is responsible for bitemporal hemianopia as the nasal fibers from the retina decussate at optic chiasma.

Incorrect Options:

Option B - Right optic tract: Lesion at right optic tract will result left sided homonymous hemianopia.eyes.

Option C - Left optic tract: Lesion at left optic tract will result in right sided homonymous hemianopia.

Option D - Right optic radiation: Lesion at right optic radiation will result in left side homonymous hemianopia.

Solution for Question 29:

Correct Option A - Macula and optic nerve disease:

- Photostress Test helps in this differentiation: Macular Diseases: If the macula is affected by conditions such as age-related macular degeneration (AMD), macular edema, or macular dystrophies, the recovery time after the photostress exposure may be significantly prolonged around 60 sec. Optic Nerve Diseases: Optic nerve diseases, such as optic neuritis, optic nerve ischemia, or optic nerve degeneration. In these conditions, the macula is relatively spared, and the recovery time after the photostress exposure remains within the normal range i.e., around 30 to 50 seconds.
- Macular Diseases: If the macula is affected by conditions such as age-related macular degeneration (AMD), macular edema, or macular dystrophies, the recovery time after the photostress exposure may be significantly prolonged around 60 sec.
- Optic Nerve Diseases: Optic nerve diseases, such as optic neuritis, optic nerve ischemia, or optic nerve degeneration. In these conditions, the macula is relatively spared, and the

recovery time after the photostress exposure remains within the normal range i.e., around 30 to 50 seconds.

- **Macular Diseases:** If the macula is affected by conditions such as age-related macular degeneration (AMD), macular edema, or macular dystrophies, the recovery time after the photostress exposure may be significantly prolonged around 60 sec.
- **Optic Nerve Diseases:** Optic nerve diseases, such as optic neuritis, optic nerve ischemia, or optic nerve degeneration. In these conditions, the macula is relatively spared, and the recovery time after the photostress exposure remains within the normal range i.e., around 30 to 50 seconds.

Incorrect Options:

Option B, C and D: Photostress test do not help in detecting these conditions

Solution for Question 30:

Correct Option A - Left optic tract:

- The patient is presenting with loss of vision on the right side of both eyes, i.e., right-side homonymous hemianopia. So, the lesion will be on the left side because the visual field coordinates are opposite to the retinal field coordinates, i.e., the temporal part of the right eye will give vision toward the left and vice versa.
- So, the right side homonymous hemianopia means the temporal fibers from the right eye and nasal fibers from the left eye are affected. This is because the fibers from the nasal side of the right eye retina cross at the optic chiasm and continue as the optic tract with temporal fibers from the left eye temporal retina.
- Lesion at 3 sites of the visual pathway is responsible for right homonymous hemianopia:
Left optic tract Left optic radiation Left visual cortex
- Left optic tract
- Left optic radiation
- Left visual cortex
- Among the given options, the correct answer is a lesion in the left optic tract, which would result in loss of vision in the right visual field of both eyes.
- Left optic tract
- Left optic radiation

- Left visual cortex

Incorrect Options:

Option B - Right visual cortex: Lesions at the optic chiasm often result in bitemporal hemianopia, where there is loss of vision in the outer (temporal) visual fields of both eyes.

Option C - Optic chiasma: The right visual cortex is responsible for processing visual information from the left visual field of both eyes. Lesions in the right visual cortex can lead to loss of vision in the left visual field of both eyes..

Option D - Right optic nerve: The right optic nerve carries visual information from the right eye to the brain. Lesions in the right optic nerve would result in loss of vision in the right eye only

Solution for Question 31:

Correct Option A - Relative afferent pupillary defect is seen:

- Optic neuritis is an optic nerve inflammation, usually causing sudden, reduced vision in the affected eye. A relative afferent pupillary defect (RAPD), Marcus Gunn pupil, is a clinical sign observed in optic neuritis. This occurs due to an impaired neurological response in one eye, where the affected eye shows a reduced or absent pupillary response to light stimulus compared to the normal eye.

Incorrect Options:

Option B - There is no pain on ocular movement: In optic neuritis, pain on ocular movement is a common symptom.

Option C - Multiple myeloma is a common cause: Multiple myeloma, a type of blood cancer, is not considered a common cause of optic neuritis. Most cases of optic neuritis are associated with autoimmune conditions, especially multiple sclerosis.

Option D - Oral corticosteroids are the mainstay of treatment: Intravenous (not oral) corticosteroids are typically the first line of treatment for acute optic neuritis to speed up recovery, though they do not affect the ultimate degree of recovery.

Solution for Question 32:

Correct Option A - 1, 2 and 3:

- The sudden decrease in vision and relative afferent pupillary defect (RAPD) suggests a problem with the optic nerve in the right eye, most likely due to optic neuritis, which can be associated with demyelinating diseases like multiple sclerosis (MS) or neuromyelitis optica (Devic's disease). MRI scan of the brain and orbit should preferably be performed in each case presenting with acute optic neuritis to check for nerve swelling, optic nerve glioma, cerebral, and brainstem lesion.. MOG and NMO antibodies: These antibodies are seen in autoimmune conditions affecting the central nervous system, like Multiple sclerosis and neuromyelitis optica (NMO). Detection of these antibodies can help confirm the diagnosis. Blood sugar levels have to be checked. Because Optic neuritis is Primarily treated with Steroids as Steroids may precipitate hyperglycemia in individuals with DM. A thyroid profile could be relevant in cases of thyroid eye disease, but this typically presents different symptoms, such as eye protrusion, redness, and double vision. Given the scenario, thyroid dysfunction is less likely.
- MRI scan of the brain and orbit should preferably be performed in each case presenting with acute optic neuritis to check for nerve swelling, optic nerve glioma, cerebral, and brainstem lesion..
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- MRI scan of the brain and orbit should preferably be performed in each case presenting with acute optic neuritis to check for nerve swelling, optic nerve glioma, cerebral, and brainstem lesion..
- MOG and NMO antibodies: These antibodies are seen in autoimmune conditions affecting the central nervous system, like Multiple sclerosis and neuromyelitis optica (NMO). Detection of these antibodies can help confirm the diagnosis.
- Blood sugar levels have to be checked. Because Optic neuritis is Primarily treated with Steroids as Steroids may precipitate hyperglycemia in individuals with DM.
- A thyroid profile could be relevant in cases of thyroid eye disease, but this typically presents different symptoms, such as eye protrusion, redness, and double vision. Given the scenario,

thyroid dysfunction is less likely.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 33:

Correct Option B - Left optic tract:

- The patient is presenting with loss of vision on the right side of both eyes, i.e., right-side homonymous hemianopia. So, the lesion will be on the left side because the visual field coordinates are opposite to the retinal field coordinates, i.e., the temporal part of the right eye will give vision toward the left and vice versa.
- So, the right side homonymous hemianopia means the temporal fibers from the right eye and nasal fibers from the left eye are affected. This is because the fibers from the nasal side of the right eye retina cross at the optic chiasm and continue as the optic tract with temporal fibers from the left eye temporal retina.
- Lesion at 3 sites of the visual pathway is responsible for right homonymous hemianopia:
Left optic tract Left optic radiation Left visual cortex
- Left optic tract
- Left optic radiation
- Left visual cortex
- Among the given options, the correct answer is a lesion in the left optic tract, which would result in loss of vision in the right visual field of both eyes.
- Left optic tract
- Left optic radiation
- Left visual cortex

Incorrect Options:

Option A - Optic nerve: Lesions at the optic nerve often result in Monocular vision loss.

Option C - Optic chiasma: Lesions at the optic chiasm often result in bitemporal hemianopia, where there is loss of vision in the outer (temporal) visual fields of both eyes.

Option D - Right optic tract: If the lesion was in the right optic tract, it would cause loss of vision in the left visual field of both eyes, i.e., left sided homonymous hemianopia.

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Conjunctiva

1. A 2-day-old infant is evaluated for discharge from the eyes. The discharge was initially serous, but it quickly turned into mucopurulent and then purulent. On examination, the eyes are bright red and swollen. What is the most likely causative agent here?

- A. Chlamydia trachomatis
- B. Silver nitrate
- C. Treponema pallidum
- D. Neisseria gonorrhoeae

2. A 65-year-old man presents with severe dry eye symptoms unresponsive to artificial tears and warm compresses. What is the most appropriate next step in managing his severe dry eye?

- A. Timolol eye drops
- B. Moisture chamber spectacles
- C. Punctal occlusion with silicone plugs
- D. Punctal occlusion with collagen plugs

3. A 64-year-old patient with dry eyes undergoes several tests. Which finding best supports the diagnosis of dry eye syndrome?

- A. Tear film break-up time of 15 seconds
- B. Schirmer test showing 12 mm wetting
- C. Extensive Rose Bengal staining
- D. Normal tear osmolarity

4. A 60-year-old patient diagnosed with dry eye syndrome asks about the typical features of the condition. Which of the following is not commonly associated with dry eye?

- A. Frothy tears along the lid margin
 - B. Punctate epithelial erosions on the cornea
 - C. Conjunctival xerosis and keratinization
 - D. Follicles on the conjunctiva
-

5. A 65-year-old woman with rheumatoid arthritis presents with dry eyes and mouth. Rose bengal staining shows corneal damage, and tear analysis reveals a tear-to-lysozyme ratio of 0.1. Which tear film layer is primarily deficient in this condition?

- A. Mucinous layer
 - B. Aqueous layer
 - C. Lipid layer
 - D. All layers are equally affected
-

6. A 70-year-old man with extensive sun exposure has a yellowish, triangular patch on the nasal conjunctiva, as shown below. The lesion does not extend onto the cornea. What is the most likely diagnosis?

[Image not available]

- A. Pterygium
- B. Pinguecula
- C. Dermoid

D. Malignant melanoma

7. A 60-year-old man comes to the hospital complaining of a small fleshy triangular mass on the nasal side of his left eye, which causes excessive blurry or distorted vision. He is advised for surgical excision. Which of the following statements is true about preventing the recurrence?

- A. Mitomycin C drops are the best choice for postoperative recurrence prevention.
 - B. An amniotic membrane graft is used during the surgery to prevent recurrence.
 - C. Using an auto-conjunctival graft can lead to scleral necrosis, cataracts, and iritis.
 - D. Limbal conjunctival tissue must be taken from the same eye for grafting.
-

8. A 50-year-old male comes to the clinic with complaints of a small fleshy triangular mass in the eye. There is no history of pain or discharge. On examination, it is as in the image. What is true about this lesion?

[Image not available]

- A. This is a non-progressive lesion
 - B. It can be present in any quadrant of the eye
 - C. It is caused by thermochemical burn, trauma, and chronic inflammation
 - D. Histology shows collagenous degenerative changes in the vascularized subepithelial stroma
-

9. A 25-year-old male presents with difficulty seeing at night. Which of the following is not a cause of nyctalopia (night blindness)?

- A. Gyrate atrophy
- B. Pathological myopia

- C. Cone dystrophy
 - D. Choroideremia
-

10. A 3-year-old child presents with symptoms of night blindness and dry eyes. On examination, there are Bitot's spots on the temporal side of both eyes. The child's guardian reports that the child has had a history of frequent respiratory infections. What is the most appropriate treatment regimen to prevent further ocular complications?

- A. 100,000 IU of vitamin A orally on diagnosis, next day, and after 2 weeks.
 - B. Vitamin A eye drops and 10,000 IU orally for 1 month.
 - C. 200,000 IU of vitamin A orally on diagnosis, next day, and after 2 weeks.
 - D. Oral multivitamins with increased vitamin A-rich foods for 3 months.
-

11. A 5-year-old boy from a rural area has difficulty seeing at night and irritated eyes. Upon examination, you notice the presence of triangular, dry-white spots on the bulbar conjunctiva in both eyes. Which of the following represents the correct diagnosis and grade of xerophthalmia based on the clinical findings?

[Image not available]

- A. Night Blindness (XN)
 - B. Conjunctival Xerosis (X1A)
 - C. Bitot's Spots (X1B)
 - D. Corneal Xerosis (X2)
-

12. A 23-year-old male presents with a history of atopic dermatitis and asthma. He complains of tearing, burning, and blurry vision for the past week. On examination, his eyelids are thickened and swollen, with signs of meibomitis. Slit-lamp

examination reveals anterior subcapsular cataracts. What is the most appropriate first-line treatment for this patient's ocular symptoms?

- A. Topical corticosteroids
- B. Topical dual-acting antihistamine/mast cell stabilizer
- C. Topical cyclosporine
- D. Oral antihistamines

13. A 10-year-old child presents with itching, watering, and redness in both eyes. On examination, small round yellow nodules are observed near the limbus, and conjunctival congestion is noted. The child has a history of recurrent styes and has a past h/o TB. Which of the following best explains the pathophysiology of this condition?

- A. Type I hypersensitivity reaction caused by environmental allergens
- B. Type IV hypersensitivity reaction to endogenous bacterial antigens
- C. Type II hypersensitivity reaction triggered by viral infections
- D. Autoimmune reaction against corneal epithelial cells

14. A 6-year-old boy presents with a 2-week history of worsening bilateral eye redness, itching, and frequent lacrimation. On examination, hypertrophied, cobblestone-like papillae on the palpebral conjunctiva. What is the most likely diagnosis?

[Image not available]

- A. Atopic Keratoconjunctivitis
- B. Vernal Keratoconjunctivitis
- C. Bacterial Conjunctivitis
- D. Chlamydial Conjunctivitis

15. Which of the following conjunctival structures connects Palpebral and Bulbar conjunctiva?

- A. Marginal Conjunctiva
 - B. Tarsal conjunctiva
 - C. Conjunctival fornix
 - D. Orbital conjunctiva
-

16. A patient with characteristic features of trachoma is being evaluated. Which of the following would be the most appropriate systemic antibiotic to be used for the treatment of trachoma?

- A. Erythromycin
 - B. Doxycycline
 - C. Amoxicillin
 - D. Azithromycin
-

17. A patient presents with irritation of the eyes and foreign body sensation in the eyes. He is examined by the ophthalmologist, who observes the following in his upper eyelid. According to the WHO classification, which grade is the disease here?

[Image not available]

- A. TF
 - B. TI
 - C. CO
 - D. TS
-

18. A patient is being evaluated for conjunctivitis. She gives a history of mild foreign body sensation in the eyes, stickiness of the lids, and mucoid discharge from the eyes. On examination, there is conjunctival congestion. Which of the following signs would be seen in this stage of the disease?

- A. Pannus siccus
 - B. Arlt's line
 - C. Herbert's follicle
 - D. Tylosis
-

19. Which of the following statements about chlamydial conjunctivitis is false?

- A. It is caused by two species: Chlamydia and Chlamydophilia
 - B. Transmission in neonates is from the mother when follicles are less evident
 - C. The reticulate body is the infectious form that constitutes the inclusion bodies
 - D. Chlamydia is a gram-negative bacteria containing both RNA and DNA
-

20. A 28-year-old man develops sudden redness, pain, and subconjunctival hemorrhages in both eyes after returning from a tropical region. His visual acuity is normal. What is the most likely causative agent?

- A. Epstein-Barr virus
 - B. Coxsackievirus
 - C. Herpes simplex virus
 - D. Molluscum Contagiosum
-

21. A 30-year-old male presents with irritation and redness at both eye corners, foamy discharge, and skin maceration as shown below. He also has chronic nasal discharge. Which organism is most likely causing his symptoms?

[Image not available]

- A. Staphylococcus aureus
- B. Moraxella axenfeld
- C. Neisseria gonorrhoeae
- D. Haemophilus influenzae

22. A 7-year-old child presents with the eye condition shown in the image. The child reports severe eye pain, redness, and eyelid swelling. Examination reveals an inflamed conjunctiva with thick, adherent membranes that bleed when removed. Which of the following organisms is most likely responsible for this presentation?

[Image not available]

- A. Adenovirus
- B. Neisseria gonorrhoeae
- C. Beta-hemolytic streptococci
- D. Corynebacterium diphtheria

23. A 25-year-old male presents with severe eye pain, intense redness, and copious purulent discharge from his right eye. He also has significant lid swelling and tender preauricular lymphadenopathy. Gram stain of the purulent discharge reveals Gram-negative intracellular diplococci. What is the next best step in management?

- A. Intravenous ampicillin
- B. Eye irrigation with saline
- C. Intramuscular Ceftriaxone

D. Topical antibiotic drops

24. A 25-year-old woman with a history of soft contact lens use for the past 8 months presents with bilateral itching, redness, and watery discharge. Examination shows large, polygonal papillae on the upper tarsal conjunctiva. What is the most likely diagnosis?

- A. Vernal Keratoconjunctivitis
 - B. Giant Papillary Conjunctivitis
 - C. Phlyctenular Conjunctivitis
 - D. Papillary Conjunctivitis
-

25. A 28-year-old woman reports bilateral itching, redness, and watery discharge that worsens in spring. Examination revealed mild conjunctival congestion with small papillae on the upper tarsal conjunctiva. What is the most likely diagnosis?

- A. Viral Conjunctivitis
 - B. Allergic Conjunctivitis
 - C. Bacterial Conjunctivitis
 - D. Chlamydial Conjunctivitis
-

26. A 27-year-old male presents with redness, mild swelling, and foreign body sensation in the right eye for 2 days. On examination, no discharge or visual impairment is noted, and a diagnosis of acute conjunctivitis is made. Which of the following is not a common sign of conjunctivitis?

- A.
- B.

C.

D.

27. Which of the following glands are primarily responsible for secreting mucus to maintain ocular surface moisture?

A. Glands of Krause

B. Goblet Cells

C. Glands of Wolfring

D. Crypts of Moll

28. Which of the following statements is true regarding the epithelium of the conjunctiva?

A. The epithelium of the limbal conjunctiva is thinner than that of the fornices.

B. The conjunctival epithelium is lined by stratified non-keratinized epithelium.

C. The adenoid layer of the conjunctiva is present at birth and plays a role in neonatal immunity.

D. The dense fibrous layer is superficial to the adenoid layer and contains goblet cells.

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	3
Question 4	4
Question 5	2
Question 6	2
Question 7	2
Question 8	4
Question 9	3
Question 10	3
Question 11	3
Question 12	2
Question 13	2
Question 14	2
Question 15	3
Question 16	4
Question 17	4
Question 18	3
Question 19	3
Question 20	2
Question 21	2
Question 22	3

Question 23	3
Question 24	2
Question 25	2
Question 26	4
Question 27	2
Question 28	2

Solution for Question 1:

Correct Answer: D) Neisseria gonorrhoeae

Explanation:

The clinical scenario points to the diagnosis of ophthalmia neonatorum caused by Neisseria gonorrhea and it presents as an inflamed, red, and swollen conjunctiva with discharge that changes fast.

Definition	Ophthalmia neonatorum or neonatal conjunctivitis is a mucoid, mucopurulent, or purulent discharge from one or both eyes in the first month of life. Any discharge, even watery, from a baby's eyes in the first week of life must be viewed under suspicion as tears are not secreted so early in life.
Aetiology of ophthalmia neonatorum	Mode of infection: Before birth: Very rare through infected liquor amnii During birth: From an infected birth canal After birth: During the first bath of a newborn Soiled clothes Fingers with infected lochia Source of infection: Neisseria gonorrhoeae: Transmitted at the time of delivery Presents between 2-5 days of birth ... (content truncated)
Prophylaxis	Antenatal measures: Treatment of genital infections and care of mother Surveillance of pregnant women for STDs Natal measures:

	Aseptic, sterile precautions to be followed during deliveries Clean the infant's closed lids with sterile cotton soaked in normal saline and dried Post-natal measures: Povidone-iodine 2.5% solution If the mother is under suspicion, 1% tetracycline or 0.5% erythromycin eye ointment is used in the baby immediately after birth and watched for a week. Ceftriaxone 50mg/kg IM or IV single dose for infants born to mothers with untreated gonococcal infection. Crede's method: Old method using 1% silver nitrate applied to infants' eyes right after birth It is not used now as it can cause chemical neonatal conjunctivitis.
Diagnosis	Treatment
Neisseria gonorrhoea	Ceftriaxone IM, gentamicin drops, bacitracin eye ointment
Chemical	Wash eyes, erythromycin ointment, observe
Other bacteria	Neomycin-bacitracin ointment, gentamicin, or tobramycin drops
Herpes simplex virus	Acyclovir 3% eye ointment, systemic acyclovir (systemic involvement)
Chlamydia trachomatis	Erythromycin or chlortetracycline eye ointment, oral erythromycin (systemic infection)

Treponema pallidum (Option C):

- Causes syphilis and rarely manifests in the conjunctiva as a primary chancre.
- It usually presents with bilateral interstitial keratitis by the age of 5-15 years.
- In the conjunctiva, it usually occurs as a less indurated chancre and is transmitted by contact with an infected oral lesion.
- Regional lymph node enlargement, along with a chronic ulcer or gummatous ulcer of the conjunctiva, suggests the disease.

Solution for Question 2:

Correct Answer: C) Punctal occlusion with silicone plugs

Explanation:

Permanent occlusion by inserting silicone plugs is effective for severe dry eye as it increases tear retention by blocking puncta, thereby providing sustained relief.

Severity Level	Treatment Recommendations
Mild	Education and environmental modifications Elimination of offending topical or systemic medications Aqueous enhancement using artificial tear substitutes, gels/ointments (e.g., 0.25 to 0.7% methylcellulose, 0.3% hypromellose or polyvinyl alcohol 1.4%) Eyelid therapy (warm compresses and eyelid hygiene) Treatment of contributing ocular factors such as blepharitis or meibomianitis
Moderate	In addition to the above-mentioned treatments: Anti-inflammatory agents (topical cyclosporine and corticosteroids), systemic omega-3 fatty acid supplements Punctal plugs (collagen plugs) (Option D) Spectacle side shields and moisture chambers (Option B)
Severe	In addition to the above-mentioned treatments: Systemic cholinergic agonists Systemic anti-inflammatory agents Mucolytic agents (5% acetylcysteine eye drops) Autologous serum tears Contact lenses Correction of eyelid abnormalities Permanent punctal occlusion (silicone plugs or thermal cautery) (Option C) Tarsorrhaphy Oral mucous membrane grafting (MMG) Minor salivary gland grafting (MSGG) Amniotic membrane grafting

Surgery:

punctal occlusion" data-author="NA" data-hash="4588" data-license="NA" data-source="NA" data-tags="March2025" height="218" src="https://image.prepladder.com/notes/ks9IUy57mRxuuR4xMKeH1743178303.png" width="500" />

Timolol (Option A) is a beta-blocker used for glaucoma to reduce intraocular pressure. However, it worsens dry eye by decreasing aqueous production, making it contraindicated in dry eye management.

Solution for Question 3:

Correct Answer: C) Extensive Rose Bengal staining

Explanation:

Extensive Rose Bengal staining indicates significant damage to the corneal and conjunctival epithelium, which is a feature of dry eye.

Slit-lamp examination	Superficial corneal erosions Inadequate tear lake volume Early tear film break-up time Conjunctival hyperemia Conjunctival surface irregularities Meibomian gland dysfunction
Schirmer's Test:	[Image not available] Schirmer's test is used to diagnose aqueous deficiency dry eye disease. Schirmer's Test 1 Measures both basal and reflex tear secretions Uses Whatman-4 filter paper, kept in the lower fornix at the junction of lateral one-third and medial two-thirds. After 5 minutes, wetting of the filter strip from the bent end is measured. Normal value: >15 mm 5-10 mm: Mild to moderate keratoconjunctivitis sicca (KCS). < 5 mm: Severe KCS (Option B) Schirmer's Test 2 Measures only basal secretion under anesthesia
Phenol Red Test	[Image not available] A modified Schirmer's test using a cotton thread impregnated with phenol red dye. The thread is placed in the lower fornix for 15 seconds. Phenol dye changes from yellow to red depending on the pH of the tears. Wet length: >20 mm: Normal. <10 mm: Severe dry eye. ≤19 mm as borderline dry eye.

Tear Film Break-Up Time (TBUT)	[Image not available] Tear film stained with sodium fluorescein 1% is observed under a slit lamp, and time is noted after the patient blinks. The interval between the complete blink and the first appearance of a 'dark' dry spot on the cornea is recorded as TBUT. It is an indicator of the adequacy of the mucin component of tears. Normal: 15-35 seconds < 10 seconds indicates unstable tear film. (Option A)
Rose Bengal or Lissamine Green	[Image not available] [Image not available] Stain devitalized, desiccated corneal and conjunctival epithelium red or green. (Option C) Rose Bengal stain Lissamine Green
Alcian Blue	Stains particulate matter in the tear film due to mucus
Tear Osmolarity	Increased in dry eye (Option D)
Tear to Lysozyme ratio	Between 0.9 and 0.6 indicates dry eye.

Solution for Question 4:

Correct Answer: D) Follicles on the conjunctiva

Explanation:

Follicles on the conjunctiva are typically seen in viral or chlamydial infections, not in dry eye.

Etiology of Dry eye:

[Image not available]

Symptoms:

- Irritation
- Foreign body (sandy) sensation
- Eye dryness
- Reduced vision
- Itching

- Nonspecific ocular discomfort

Signs:

- Tear Film Signs: Presence of stingy mucus and particulate matter Reduced or absent marginal tear strip Frothy tears along the lid margin indicate meibomian gland dysfunction (Option A) Posterior Blepharitis (Meibomitis): Inflammation of meibomian glands.
- Presence of stingy mucus and particulate matter
- Reduced or absent marginal tear strip
- Frothy tears along the lid margin indicate meibomian gland dysfunction (Option A)
- Posterior Blepharitis (Meibomitis): Inflammation of meibomian glands.
- Presence of stingy mucus and particulate matter
- Reduced or absent marginal tear strip
- Frothy tears along the lid margin indicate meibomian gland dysfunction (Option A)
- Posterior Blepharitis (Meibomitis): Inflammation of meibomian glands.

[Image not available]

- Conjunctival Signs: Lustreless and mildly congested conjunctiva Conjunctival xerosis and keratinization may occur (Option C) Positive Rose Bengal or Lissamine green staining
- Lustreless and mildly congested conjunctiva
- Conjunctival xerosis and keratinization may occur (Option C)
- Positive Rose Bengal or Lissamine green staining
- Corneal Signs: Punctate epithelial erosions (Option B) Filaments and mucus plaques Loss of corneal luster Lesions delineated by fluorescein, Rose Bengal, or Lissamine green staining
- Punctate epithelial erosions (Option B)
- Filaments and mucus plaques
- Loss of corneal luster
- Lesions delineated by fluorescein, Rose Bengal, or Lissamine green staining
- Lustreless and mildly congested conjunctiva
- Conjunctival xerosis and keratinization may occur (Option C)
- Positive Rose Bengal or Lissamine green staining
- Punctate epithelial erosions (Option B)
- Filaments and mucus plaques
- Loss of corneal luster
- Lesions delineated by fluorescein, Rose Bengal, or Lissamine green staining

Solution for Question 5:

Correct Answer: B) Aqueous layer

Explanation:

Keratoconjunctivitis sicca (Sjögren syndrome) is the most likely diagnosis due patient's history of RA and the tear-to-lysozyme ratio of 0.1, dry eyes and mouth, and corneal damage shown by Rose Bengal staining, which indicates a deficiency in the aqueous layer of the tear film.

Tear Film Layer	Secreted By	Function	Defective in
Innermost Mucinous Layer (Option A)	Goblet cells, crypts of Henle, Manz's glands	Convert hydrophobic corneal surface into hydrophilic one	Vitamin A deficiency
Middle Aqueous Layer (thickest layer) (Option B)	Lacrimal and accessory conjunctival glands (Krause, Wolfring/Ciaccio)	Contain tears Also contains antibacterial substances like lysozyme, beta-lysin, and lactoferrin.	Sjogren's syndrome
Outermost Lipid Layer (thinnest layer) (Option C)	Meibomian glands	Prevent overflow of tears. Retard evaporation. Lubricates the eyelids.	Contact lens wear Meibomian gland defect

[Image not available]

Keratoconjunctivitis sicca (sjögren syndrome)	
Definition	Condition caused by deficiency of the aqueous component of tears due to impaired lacrimal secretion.
Pathology	Lacrimal glands show fibrosis and lymphocytic infiltration.
Occurs in	Postmenopausal women are often associated with rheumatoid arthritis.
Symptoms	Eye dryness Irritative symptoms Epithelial erosions or filaments on the corneal surface
Clinical Diagnosis	Damage to the corneal and conjunctival epithelium can be seen with Rose bengal staining.

Tear Composition	Typical tear to lysozyme ratio of 0.1.
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Primary Sjögren syndrome = Keratoconjunctivitis sicca + Xerostomia (dryness of mouth)

Secondary Sjögren syndrome = Primary Sjögren syndrome + Autoimmune disease (like Rheumatoid arthritis)

All layers equally affected (Option D): In Sjögren syndrome, the primary issue is a deficiency in the aqueous layer, not an equal impact on all layers.

Solution for Question 6:

Correct Answer: B) Pinguecula

Explanation:

A yellowish, triangular patch on the conjunctiva near the limbus, typically in the elderly, associated with sun exposure and not extending onto the cornea, points towards the diagnosis of Pinguecula.

Pinguecula	
Definition	A triangular patch on the conjunctiva appears yellow and resembles fat (pinguis).
Pathology	Caused by hyaline infiltration and elastotic degeneration of submucous tissue.
Etiology	Common in elderly people, particularly those exposed to strong sunlight, dust, and wind.
Clinical Feature	Bilateral Usually stationary Yellowish triangular patch near the limbus Apex of the triangle is away from the cornea Affects the nasal side first, followed by the temporal side
Image	[Image not available]
Complication	Inflammation of pinguecula Intraepithelial abscess formation Rarely calcification

Treatment	Generally, it requires no treatment. When inflamed, treated with topical steroids. May be excised when cosmetically unacceptable.
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Pterygium (Option A) is a triangular, vascularized growth extending onto the cornea, whereas the lesion described is a yellowish, triangular patch on the conjunctiva that does not encroach onto the cornea, fitting the description of pinguecula.

[Image not available]

Dermoid (Option C) appears as a lenticular yellow tumor near the corneal margin and consists of epidermoid epithelium with sebaceous glands and hair, whereas the lesion described is a yellowish, triangular patch on the conjunctiva that does not involve the corneal margin or have these characteristics.

[Image not available]

Malignant melanoma (Option D) typically occurs at the limbus, is usually pigmented, and can spread or metastasize, whereas the described lesion is a yellowish, triangular patch on the conjunctiva without these features or aggressive behavior.

[Image not available]

Solution for Question 7:

Correct Answer: B) An amniotic membrane graft is used during the surgery to prevent recurrence

Explanation:

An amniotic membrane graft or a piece of the limbal conjunctival tissue from the same or other eye can be used to prevent recurrence.

Management of pterygium:

Management	Best left alone Medical treatment: Tear substitutes for small regressive pterygium with dry eye Topical steroids for associated inflammation Protection from UV rays with sunglasses Surgical excision for: Progression of lesion towards the pupillary area causing excessive astigmatism and visual impairment Restricted ocular movements Disfiguring Diplopia
Procedures	Bare sclera technique: The pterygium is excised from the cornea and conjunctiva, leaving a bare sclera Associated with a high recurrence rate Autoconjunctival graft A piece of limbal conjunctival tissue from the same or the other eye (Option D) or An amniotic membrane graft is used to cover the bare sclera Sutured at the limbus to cover the defect Surgical excision with lamellar keratectomy and lamellar keratoplasty for deeply infiltrating recurrent recalcitrant pterygia Old methods (not used now): McRaynold's operation Postoperative use of beta irradiations
Post-operative therapy and care	Prevent recurrence: Mitomycin C 0.02% drops Complications like scleral necrosis, cataracts, and iritis (Option A&C)

Solution for Question 8:

Correct Answer: D) Histology shows collagenous degenerative changes in the vascularized subepithelial stroma

Explanation:

The above clinical scenario points to the diagnosis of pterygium.

Pterygium/ Surfer's eye	Pseudopterygium	
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Definition	[Image not available] Pterygium is a triangular encroachment of vascularized granulation tissue covered by conjunctiva in the interpalpebral area. Parts: Cap, head, neck, body Degenerative process	[Image not available] Pterygium like lesion Due to cicatrizing conjunctival inflammatory overgrowth.
Aetiology	Common in dry, sunny climates with sandy soils Ultraviolet light is a probable factor Usually elderly persons	Seen at any age Caused by (Option C) Trauma Chronic conjunctivitis Thermochemical burn
Pathology	[Image not available] Histology shows collagenous degenerative changes in the vascularized subepithelial stroma.	
Pathogenesis is	Degenerative subconjunctival proliferation Granulation tissue proliferates as avascularized tissue encroaches on the cornea. Destroys the superficial layers of the stroma and Bowman's membrane	
Symptoms	[Image not available] Frequently follows a pinguecula Often bilateral Usually on the nasal side (Option B) Impairment of vision due to: Progression into the pupillary area of the cornea Induced astigmatism by corneal flattening in the axis of pterygium Progressive pterygium: thick and vascular Regressive pterygium: thin, pale, and atrophic Ceases to grow but never disappears A probe can be passed under the lesion Can present as inflamed pterygium	Non-progressive (Option A) Can be at any quadrant of the corneal limbus Located away from the horizontal Attached to the cornea due to scar tissue Adhered to the adjacent lid Always stationary

Signs	[Image not available] A probe cannot be passed under the lesion Fusch's spots/Islets of Vogt: small discrete whitish flecks made of pterygial epithelial cells seen in progressive pterygium and are also called the cap of the pterygium. Stocker line: linear epithelial iron deposition, seen in regressive pterygium.	A probe can be passed under the neck.
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Solution for Question 9:

Correct Answer: C)Cone dystrophy

Explanation:

Cone dystrophy is primarily associated with day blindness (hemeralopia), which is the inability to see clearly in bright light due to poor light adaptation.

Nyctalopia/ Night Blindness is the inability to see in low-light conditions. It is attributed to interference with the functions of the retinal rods.

Causes of Nyctalopia:

- Retinitis pigmentosa
- Xerophthalmia
- Pathological myopia (Option B)
- Advanced stage of primary open-angle glaucoma
- Peripheral cortical cataract
- Diffuse choroidal dystrophy
- Choroideremia (Option D)
- Gyrate atrophy (Option A): Due to a deficiency of the enzyme ornithine aminotransferase, plasma ornithine levels are elevated. Characterized by progressive patches of atrophy of choroid and retinal pigment epithelium (RPE). Management: Restriction of arginine (precursor amino acid for ornithine) in diet. Vitamin B6 supplementation lowers plasma ornithine levels.

- Due to a deficiency of the enzyme ornithine aminotransferase, plasma ornithine levels are elevated.
- Characterized by progressive patches of atrophy of choroid and retinal pigment epithelium (RPE).
- Management: Restriction of arginine (precursor amino acid for ornithine) in diet. Vitamin B6 supplementation lowers plasma ornithine levels.
- Due to a deficiency of the enzyme ornithine aminotransferase, plasma ornithine levels are elevated.
- Characterized by progressive patches of atrophy of choroid and retinal pigment epithelium (RPE).
- Management: Restriction of arginine (precursor amino acid for ornithine) in diet. Vitamin B6 supplementation lowers plasma ornithine levels.

[Image not available]

- Congenital Stationary Night Blindness (CSNB): Inherited retinal disease that causes night blindness in childhood. 4 types of CSNB Fundus Albipunctatus: Small white dots scattered across the posterior pole, sparing the fovea. Oguchi Disease: Gray-white metallic sheen that disappears after dark adaptation (Mizuo-Nakamura phenomenon). Riggs: Normal fundus Schubert-Bornschein: Normal fundus
- Inherited retinal disease that causes night blindness in childhood.
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- Oguchi Disease: Gray-white metallic sheen that disappears after dark adaptation (Mizuo-Nakamura phenomenon).
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- Riggs: Normal fundus
- Schubert-Bornschein: Normal fundus

adaptation test" data-author="Nadia K Waheed, Ahmed Hassaan Qavi, Sarah N Malik, Maleeha Maria, Moeen Riaz, Frans P M Cremers, Maleeha Azam, Raheel Qamar" data-hash="4430" data-license="CC BY 3.0" data-source="https://www.researchgate.net/figure/Dark-adaptation-test-of-the-patient-Fundus-photographs-of-the-patient-before-A-and_fig2_225186862" data-tags="March2025" height="194" src="https://image.prepladder.com/notes/jJMmUieVrVhUdHLkdd6S1743162036.png" width="500" />

Solution for Question 10:

Correct Answer: C) 200,000 IU of vitamin A orally on diagnosis, next day, and after 2 weeks.

Explanation:

- This is the recommended treatment for children aged 1 year and older with xerophthalmia.
- The WHO advises administering 200,000 IU of vitamin A immediately, then repeating the dose on the following day and again 2 weeks later.
- This regimen rapidly replenishes vitamin A stores, which is crucial for reversing ocular damage and preventing severe complications such as corneal ulceration and blindness.

Pathological Changes in Epithelium and Tear Film:

1. Alterations in Conjunctival Epithelium:

- Loss of Mucin Secretion: Leads to tear film instability.
- Epidermoid Metaplasia: Epithelium transforms into stratified squamous epithelium with granular, keratinized and horny layers.

2. Impact on Tear Film and Ocular Surface:

- Increased Tear Film Breakup: Exposes ocular surface to dryness.
- Appearance: Conjunctiva becomes dry and loses its lustre.

Management:

Vitamin A Supplementation: The cornerstone of treatment is high-dose vitamin A supplementation. Dosages vary by age and severity of the condition:

Population	Vitamin A Dose	Treatment regimen
Children (≥ 1 year)	200,000 IU orally on diagnosis	Day1, Day 2, and 2 weeks later (Option C)
Children (Infants under 12 months)	100,000 IU orally	Day1, Day 2, and 2 weeks later (Options A and B)
Adults (non-pregnant)	200,000 IU orally	200,000 IU orally on Day 1, Day 2, and after 2 weeks.
Pregnant Women	10,000 IU orally daily	Daily oral administration of 10,000 IU for 1 month (due to the risk of teratogenicity, higher doses are avoided during pregnancy).
Lactating Women	200,000 IU orally	One dose of 200,000 IU orally

Symptomatic Relief:

Artificial Tears: Lubricate the ocular surface, Frequency adjustment (1 to 6 hourly)

Mucomimetic Agents: Mimic natural mucin to stabilize the tear film; Used alongside artificial tear

Solution for Question 11:

Correct Answer: C) Bitot's Spots (X1B)

Explanation:

Bitot's spots are characteristic of xerophthalmia and appear as triangular, dry-white lesions on the bulbar conjunctiva, a hallmark of grade X1B xerophthalmia.

Grade	Symptoms	Signs	Image
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XN (Night Blindness)	Difficulty seeing in low light (night blindness).	Normal conjunctiva, early visual impairment in low light conditions.	Nyctalopia: The earliest sign of xerophthalmia
X1A (Conjunctival Xerosis)	Dry eyes, lusterless appearance.	Non-wettable, dry conjunctiva affecting the interpalpebral area, resembling "sandbanks".	[Image not available]
X1B (Bitot's Spots)	Mild irritation, possibly no symptoms.	Triangular, dry-white masses on the bulbar conjunctiva, usually bilateral and temporal. Bitot's spot consists of keratinized epithelium. Associated with <i>Corynebacterium</i> infection. More common on the temporal side	[Image not available]
X2 (Corneal Xerosis)	Blurred vision, and eye dryness.	Punctate keratopathy, hazy cornea with granular and xerotic changes, beginning from the nasal quadrant.	[Image not available]
X3 (Corneal Ulceration & Scarring)	Severe eye pain, and visual impairment.	Collagen necrosis leads to sharply demarcated corneal ulcers, corneal scarring, and opacity.	[Image not available]
XF (Xerophthalmic Fundus)	Visual impairment, difficulty with central vision.	Whitish, seed-like, raised lesions scattered in the posterior pole of the retina near the optic disc.	[Image not available]

(Night Blindness - XN): Night blindness does not present with the visible, triangular, dry spots seen on the conjunctiva. Night blindness occurs in dim light, with no obvious external eye changes. (Option A)

(Conjunctival Xerosis - X1A): Conjunctival xerosis presents as dry, lusterless conjunctiva (Option B)

(Corneal Xerosis - X2): Corneal xerosis affects the cornea, typically presenting with haziness and granular or xerotic changes in the cornea (Option D)

Solution for Question 12:

Correct Answer: B) Topical dual-acting antihistamine/mast cell stabilizer

Explanation:

Topical dual-acting antihistamine/mast cell stabilizer is the first-line treatment for atopic keratoconjunctivitis. It addresses both histamine release and mast cell stabilization, helping control inflammation and allergic responses.

Atopic Keratoconjunctivitis	
Patient demographics	Common in young adults, often associated with a family history of atopic conditions such as asthma, allergic rhinitis, or atopic dermatitis. Prevalent during winter months.
Disease characteristics	Severe, chronic ocular surface disorder with a range of severity from mild to persistent inflammation and extends into adulthood.
Characteristic findings	[Image not available] [Image not available] [Image not available] Symptoms: Tearing, burning, clear mucus discharge, Blurry vision, photophobia, foreign body sensation Shield cataract/Atopic cataract (anterior or posterior subcapsular) Eyelid Changes: Thickened, swollen eyelids (intermittent due to chronic inflammation). Lichenified and woody eyelids leading to Cicatricial ectropion and lagophthalmos Madarosis. Meibomitis and blepharitis Dennie Morgan folds: Skin folds under the eyes due to excessive rubbing. Hertoghe's sign: Loss of the lateral third of the eyebrows
Treatment	First-line therapy: Topical dual-acting antihistamine/mast cell stabilizer If there is no improvement in 2-3 weeks: A brief course of topical corticosteroids. (Option A) For moderate-to-severe disease requiring frequent corticosteroids - Use topical cyclosporine (2%, compounded). (Option C) Eyelid Dermatitis: Topical tacrolimus is preferred over corticosteroids.

Oral antihistamines (Option D) may help with systemic symptoms, but they are not as effective for treating ocular manifestations directly.

Solution for Question 13:

Correct Answer: B) Type IV hypersensitivity reaction to endogenous bacterial antigens

Explanation:

- The above presentation is a case of phlyctenular keratoconjunctivitis primarily mediated by a delayed-type (Type IV) hypersensitivity reaction triggered by bacterial proteins of *Staphylococcus aureus* or tuberculosis. The child's history of TB is significant, as TB antigens can trigger this hypersensitivity response, leading to the characteristic ocular findings.
- The immune system reacts to these antigens, causing the formation of nodules (phlyctens) and subsequent inflammation around the limbus.
- This explains the chronic and localized inflammatory response seen in this condition.

Phlyctenular keratoconjunctivitis	
Cause	[Image not available] Allergic reactions to endogenous antigens are often linked to bacterial infections such as <i>Staphylococcus aureus</i> (M/C/C in Western countries) or tuberculosis (M/C/C in India).
Symptoms & signs	Watering, itching Reflex blepharospasm (involuntary blinking) Conjunctival congestion Phlycten - Small, round, grayish, or yellowish nodules on the bulbar conjunctiva near the limbus. Corneal involvement can lead to fascicular ulcers or ring ulcers.
Pathophysiology	Type IV hypersensitivity reaction.
Common Complications	Corneal involvement: Ulcers form on the cornea, leading to potential scarring.
Associated Conditions	Long-standing mild infections like tonsillitis or adenoiditis, and in populations with a high prevalence of tuberculosis.

Treatment	Local therapy: Corticosteroids: Drops/ointment to reduce inflammation. Address underlying cause (e.g., tuberculosis). Antibiotics Cycloplegics: Paralyze ciliary muscle, relieve pain, and prevent complications. Dark Glasses/Eye Shade: For photophobia relief. General therapy: Oral tetracycline or doxycycline is used in recurrent cases linked to Staphylococcal blepharitis. Supportive: Dark glasses or eye shade.
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Type I hypersensitivity reaction caused by environmental allergens(Option A):

Type I hypersensitivity from environmental allergens, like pollen, causes rapid IgE-mediated allergic reactions.

Type II hypersensitivity reaction triggered by viral infections(Option C):

Type II hypersensitivity, seen in viral infections, involves antibody-mediated cell destruction, not related to phlyctenular keratoconjunctivitis.

Autoimmune reaction against corneal epithelial cells(Option D):

Phlyctenular keratoconjunctivitis is an immune response to bacterial proteins, not an autoimmune attack on corneal epithelial cells

Solution for Question 14:

Correct Answer: B) Vernal Keratoconjunctivitis

Explanation:

- Vernal Keratoconjunctivitis is the most likely diagnosis given the patient's age, the characteristic findings of cobblestone-like papillae on the palpebral conjunctiva
- It commonly affects young children and adolescents, presenting with symptoms such as itching, burning, and lacrimation, and often worsens during warm weather.

Vernal keratoconjunctivitis or Spring catarrh	
It is a recurrent bilateral conjunctivitis that tends to occur with the onset of warm weather. It primarily affects young children and adolescents, particularly boys.	

Course	The condition typically starts in childhood and may resolve on its own over 5–10 years.
Symptoms	Burning, Itching, Photophobia (sensitivity to light), and lacrimation (excessive tearing).
Signs	[Image not available] [Image not available] [Image not available] [Image not available] Palpebral Form: Hypertrophied palpebral conjunctiva with a cobblestone appearance; dense fibrous tissue with thickened epithelium. Limbal or Bulbar Form: Opacification at the limbus, gelatinous thickening, and Horner–Trantas dots Pseudogerontoxon/cupid's bow (which resembles arcus senilis) is a grayish-white ring at the limbus seen in severe limbal vernal keratoconjunctivitis (VKC). Shield ulcers are a significant complication of VKC, particularly in severe cases. These microerosions typically form on the upper third of the cornea. Mixed form: Exhibits characteristics of both the tarsal (palpebral) and limbal subtypes, typically presenting in only one eye. Maxwell Lyon's sign is a pseudomembrane formed due to excessive mucus deposition on the papilla resulting in cobblestone papillae.
Histology	In the palpebral form, hypertrophied papillae are observed, rich in eosinophils and other inflammatory cells.
Treatment	Topical therapy: Antihistamines - Levocabastine & emedastine Mast cell stabilizers - Sodium cromoglicate Mild steroids - Fluorometholone and loteprednol etabonate, Acetylcysteine can control excess mucus. Local therapy: Subtarsal steroid injections and cryotherapy for severe cases. Surgical treatment: Excision of giant papillae and treatment of shield ulcers with debridement and amniotic membrane or bandage contact lens. Cold compresses and tinted glasses for comfort. Avoid eye rubbing

Atopic Keratoconjunctivitis (Option A) presents as a severe ocular surface disorder associated with a history of atopic diseases. It is more chronic and typically affects older children and adults.

Bacterial Conjunctivitis (Option C) presents with purulent discharge and often includes a fever or systemic signs. The given presentation does not favor this diagnosis.

Chlamydial Conjunctivitis (Option D) often involves a history of sexual contact or transmission and typically affects older children and adults. It usually presents with severe symptoms and sometimes includes a discharge with lymphadenopathy.

Solution for Question 15:

Correct Answer: C) Conjunctival fornix

Explanation:

- The Conjunctival fornix is a continuous, circular cul-de-sac broken only on the medial side by the caruncle and the plica semilunaris.
- It joins the Bulbar and Palpebral conjunctiva.
- It is divided into superior, inferior, medial, and lateral fornices.

[Image not available]

Parts of Conjunctiva:

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plica semilunaris" data-author="Biswadip Barman" data-hash="4231" data-license="CC BY SA 4.0" data-source="https://commons.wikimedia.org/wiki/File:Visible-human-eye.jpg" data-tags="March2025" height="282" src="https://image.prepladder.com/notes/h9k6UILJlQsIF8fmyDPD1743135616.png" width="500" />

[Image not available]

Solution for Question 16:

Correct Answer: D) Azithromycin

Explanation:

Azithromycin stands out as the preferred systemic antibiotic or treatment of choice, given its effectiveness and convenient dosing regimen, particularly when taken orally either as a single

dose or as part of a mass drug administration approach.

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Erythromycin (Option A) or doxycycline (Option B) may be considered, although caution is warranted with tetracyclines in pregnancy, breastfeeding, and among children, and hence, it is not preferred over azithromycin.

Amoxicillin (Option C) is not recommended as it would need a longer duration of administration and follow-ups.

Solution for Question 17:

Correct Answer: D) TS

Explanation:

The above scenario is of trachoma wherein the picture represents a linear scar in the sulcus sub tarsalis called Arlt's line, seen in trachomatous scarring stage.

WHO grading system of trachoma:

WHO Classification of Trachoma (FISTO)			
Stage	Description	Image	Management

TF: Trachomatous Inflammation-Follicular (Option A)	The presence of five or more follicles in the upper tarsal conjunctiva Active stage of trachoma	[Image not available]	Antibiotics to eliminate the organism Topical: Tetracycline 1% or Erythromycin 1% Sulfacetamide Systemic: Tetracycline or erythromycin orally Doxycycline orally Azithromycin single oral dose Combined systemic and topical therapy: For TI or when an associated genital infection is present Tetracycline 1% or erythromycin 1% eye ointment Azithromycin orally
TI: Trachomatous Inflammation-Intense (Option B)	Pronounced inflammatory thickening of the tarsal conjunctiva that obscures more than half of the normal deep tarsal vessels.	[Image not available]	
TS: Trachomatous Scarring	Presence of scarring in tarsal conjunctiva White, bands or sheets of fibrosis in tarsal conjunctiva	[Image not available]	Concretions removed with a hypodermic needle Conjunctival xerosis with artificial tears (lubricating drops)
TT: Trachomatous Trichiasis	At least one eyelash rub on eyeball or evidence of recent removal of intumed eyelashes.	[Image not available]	Permanent lash removal: Electrolysis Cryolysis Radiofrequency epilation Bilamellar tarsal resection Cicatricial entropion surgery
CO: Corneal Opacity (Option C)	Easily visible corneal opacity over the pupil Part of the pupil margin is blurred when seen through the opacity as it is very dense corneal scarring	[Image not available]	After treatment of other trachoma sequelae Penetrating keratoplasty for significant corneal scarring Keratoprosthesis for bilateral blind cases with extensive scarring and surface problems Punctal occlusion and lateral tarsorrhaphy for coexistent eye surface

problems

Solution for Question 18:

Correct Answer: C) Herbert's follicles

Explanation:

The above clinical scenario points to an active trachomatous infection, and Herbert's follicles are present in the limbal area of the eye in the active stage of the disease.

Signs of active trachoma	Signs of cicatricial trachoma
Conjunctival signs: Conjunctival congestion Follicles in the upper tarsal conjunctiva: Look like sago-grains Papillary enlargement: red, velvety	Conjunctival signs: Scarring It may be irregular, star-shaped, or linear Linear scar in the sulcus subtarsalis is called Arlt's line (Option B) accumulation of dead epithelial cells and inspissated mucus in the glands of Henle Pseudocyst Xerosis Symblepharon
Corneal signs: Superficial keratitis on slit lamp and fluorescein Herbert's follicles Progressive pannus/Trachomatous pannus: The upper part of the cornea is cloudy Corneal ulcers are common at the advancing edge of the pannus Shallow, little infiltrated Cause lacrimation and photophobia	Corneal signs: Regressive pannus or pannus siccus (Option A): Receding infiltration Vessels extend beyond the area of infiltration and haze Herbert pits: oval or circular pitted scars formed after the healing of Herbert's follicles in the limbal area Vascularized and opaque cornea. Corneal xerosis Corneal ectasia Total corneal pannus (blinding sequelae)
	Lid signs: Distortion of the lids Tylosis (Option D): Thickening of the tarsal plate/lid margin Trichiasis (Option C) due to rubbing of lashes on the cornea Trachomatous ptosis (sleepy appearance) Entropion Madarosis Ankyloblepharon

Lacrimal apparatus sequelae: Chronic dacryocystitis Chronic dacryoadenitis	
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Images:

Active Trachoma	[Image not available] Follicles in upper palpebral conjunctiva	[Image not available] Trachomatous pannus	Progressive pannus pannus N" data-author="N A" data-hash="4287" data-license="NA" data-source="NA" data-tags="March2025" height="500" src="https://image.prepladder.com/notes/grHGvINFMKmtSUTBTJcX1743139498.png" width="500" />
Cicatricial Trachoma	[Image not available] Arlt's line	[Image not available] Herbert's pits	[Image not available] Regressive pannus
[Image not available] Conjunctival concretions	[Image not available] Trichiasis	[Image not available] Tylosis (thickening of tarsal border of eyelid)	

Solution for Question 19:

Correct Answer: C) The reticulate body is the infectious form that constitutes the inclusion bodies

Explanation:

The reticulate bodies are the intracellular metabolically active form that is non-infectious and differentiates to form elementary bodies (infectious), which constitute the inclusion bodies.

Conjunctivitis refers to the inflammation of the conjunctiva.

Chlamydial conjunctivitis:

Causative organism	Family: Chlamydiaceae Chlamydia is a Small, gram-negative, obligate intracellular bacteria with both RNA and DNA. (Option D) Two genera: Chlamydia and Chlamydophila (Option A) Present in two morphologically distinct forms: Elementary body: Infectious, extracellular. Reticulate body: Non-infectious, intracellular.
Pathology	Characteristic intracytoplasmic inclusion bodies from scrapings of the palpebral conjunctiva formed by Chlamydia oculogenitalis. These perinuclear inclusion bodies contain 100-500 elementary bodies Elementary bodies attach to the host and form reticulate bodies, which grow and produce new elementary bodies.
Mode of transmission (Chlamydia inclusion conjunctivitis)	Sexual transmission from a genital reservoir Adults: via fingers but commonly through the water in swimming pools leading to local epidemics (swimming pool conjunctivitis/adult inclusion conjunctivitis) Mother to newborn when follicle formation is less evident (Option B)

Trachoma:

Other name	Egyptian ophthalmia
Aetiology	Chlamydia trachomatis. It is a bacteria that is a prokaryotic, obligatory intracellular parasite. Serovars D-K: Inclusion conjunctivitis or oculogenital chlamydial disease (adult), ophthalmia neonatorum Serotypes A, B, Ba, and C: Trachoma Two species: C. trachomatis C. psittaci
Predisposing factors	Age: children in the first few years of life Gender: female > male Environmental factors include lack of hygiene, exposure to smoke, dust, and crowded areas.
Route of transmission	Direct spread: airborne or waterborne Vector transmission: by flies Material transfer: Conjunctival secretions through fingers, towels, and flies (profuse ocular discharge attracts flies)

[Image not available]

Repeated infections lead to a chronic immune reaction caused by periodic exposure to chlamydial antigens. This reaction manifests as a delayed-type hypersensitivity (type 4) response. The ongoing exposure to these antigens results in scarring and other types of lasting damage.

Solution for Question 20:

Correct Answer: B) Coxsackievirus

Explanation:

Sudden redness, pain, and subconjunctival hemorrhages, especially after travel to a tropical region, are characteristic of acute hemorrhagic conjunctivitis, caused by Coxsackievirus.

Viral Conjunctivitis:

- Causes: Adenovirus Herpes simplex Herpes zoster Molluscum contagiosum Poxvirus Myxovirus Paramyxovirus
- Adenovirus
- Herpes simplex
- Herpes zoster
- Molluscum contagiosum
- Poxvirus
- Myxovirus
- Paramyxovirus
- Adenovirus
- Herpes simplex
- Herpes zoster
- Molluscum contagiosum
- Poxvirus
- Myxovirus
- Paramyxovirus

Adenovirus Conjunctivitis

[Image not available]

[Image not available]

[Image not available]

Epstein-Barr virus (Option A) – Causes mild conjunctivitis but not acute hemorrhagic conjunctivitis.

Epstein-Barr virus (Option A) – Causes mild conjunctivitis but not acute hemorrhagic conjunctivitis.

Herpetic Conjunctivitis (Option C)	Hemorrhagic Conjunctivitis/ Apollo Conjunctivitis (Option B)	Molluscum Contagiosum (Option D)
Commonly caused by herpes simplex type 1. Seen in young children infected by contagion carriers of the virus. Corneal vesicles merge to form dendritic figures. Treatment: Topical lubricants If corneal is involved: Acyclovir 3% or vidarabine 3% eye ointment or trifluorothymidine 1% drops 5 times a day.	Cause: Coxsackievirus Enterovirus 70 Occurs in the pandemic form of violent inflammatory conjunctivitis. Clinical Features: Lacrimation Photophobia Subconjunctival hemorrhage Enlarged preauricular lymph nodes Cornea is usually affected	Caused by poxvirus. Small, white, umbilicated swelling, generally multiple. Produces severe conjunctivitis and occasionally keratitis. Treatment: Surgical excision.
[Image not available]	[Image not available]	[Image not available]

Solution for Question 21:

Correct Answer: B) Moraxella axenfeld

Explanation:

The presence of localized inflammation, redness, and foamy discharge confined to the angles of the eyes, along with skin maceration around these areas, points towards the diagnosis of angular conjunctivitis. The most common cause is Moraxella axenfeld.

Angular Conjunctivitis	Type of chronic conjunctivitis characterized by mild inflammation confined to the conjunctiva and lid margins near the angles of the eyes, associated with maceration of the surrounding skin.
Causative Agent	Moraxella axenfeld is the most common cause, leading to the term "diplobacillary conjunctivitis" due to the bacilli being arranged end-to-end. Rarely Staphylococci.
Source of Infection	Nasal Cavity

Mode of Transmission	Infection is transmitted from the nasal cavity to the eyes via contaminated fingers or handkerchiefs.
Pathology	The bacillus produces proteolytic enzymes that macerate the epithelium, resulting in mild chronic inflammation and eczematous changes in the skin around the eye angles.
Symptoms	Irritation Burning sensation Discomfort in the eyes
Signs	Hyperemia of the bulbar conjunctiva and lid margins near the angles Excoriation of the skin around the angles Foamy mucopurulent discharge at the angles
Image	[Image not available]
Complications	Blepharitis Shallow marginal catarrhal corneal ulceration
Treatment	Oxytetracycline (1%) eye ointment applied 2-3 times a day for 9-14 days. Zinc lotion instilled during the day, and zinc oxide ointment at bedtime to inhibit the proteolytic enzyme and reduce maceration.

Solution for Question 22:

Correct Answer: C) Beta-hemolytic streptococci

Explanation:

Child symptoms like thick, adherent membranes that bleed upon removal, along with the image showing inflamed conjunctiva with adherent membrane, suggest the diagnosis of membranous conjunctivitis. Beta-hemolytic streptococci are the most common cause of membranous conjunctivitis.

Aspect	Membranous Conjunctivitis	Pseudomembranous Conjunctivitis
Characteristics	True membranes grow into the conjunctival epithelium and are more adherent. Removal of these membranes often results in significant bleeding.	More superficial, with no growth into conjunctival epithelium, and are less adherent than true membranes. These can be removed with minimal bleeding.

Caus es	Bacterial: Beta-hemolytic streptococci (M/c), Staphylococcus aureus, Neisseria gonorrhoeae, Corynebacterium diphtheria. Viral: Adenovirus, Herpes simplex virus, Epstein-Barr virus. Drug Reactions: Hypersensitivity reactions, Stevens-Johnson syndrome (SJS). Chemical or thermal injury.	Infectious: Corynebacterium diphtheria, Neisseria gonorrhoeae, Streptococcus pyogenes, adenovirus, COVID-19. Inflammatory: Hypersensitivity reactions, foreign body reactions. Ligneous Conjunctivitis
Imag e	[Image not available]	[Image not available]
Sym ptom s	Eye redness Eyelid swelling Watery discharge Visual disturbances Foreign body sensation	
Sign s	Conjunctival hyperemia Chemosis Eyelid edema Corneal edema Mucopurulent discharge	
Man age ment	Topical steroids and antibiotics. Lubricants improve comfort and rehydrate the eye. Membranes can dissolve on their own.	Peeling of pseudomembrane. Topical steroids and antibiotics. Removal of foreign bodies if present. Washing the ocular surface to remove chemical irritants.

Solution for Question 23:

Correct Answer: C) Intramuscular Ceftriaxone

Explanation:

The presence of copious purulent discharge, tender preauricular lymphadenopathy, and Gram-negative intracellular diplococci strongly suggests gonococcal conjunctivitis, caused by *Neisseria gonorrhoeae* also termed as acute blennorrhoea/ acute purulent conjunctivitis/ hyperacute conjunctivitis, treated with single dose Ceftriaxone IM and oral Azithromycin or doxycycline (twice a day for 7 days).

Acute Mucopurulent Conjunctivitis:

- Symptoms: Redness Mucus discharge Eyelids stick together, especially in the morning
- Redness
- Mucus discharge
- Eyelids stick together, especially in the morning
- Redness
- Mucus discharge
- Eyelids stick together, especially in the morning

[Image not available]

- Treatment: Topical Antibiotics: Broad-spectrum antibiotics (e.g., chloramphenicol, lomefloxacin, ofloxacin, ciprofloxacin) applied 4-6 times daily. Antibiotic ointment for lid application to prevent sticking and manage discharge.
- Broad-spectrum antibiotics (e.g., chloramphenicol, lomefloxacin, ofloxacin, ciprofloxacin) applied 4-6 times daily.
- Antibiotic ointment for lid application to prevent sticking and manage discharge.
- Broad-spectrum antibiotics (e.g., chloramphenicol, lomefloxacin, ofloxacin, ciprofloxacin) applied 4-6 times daily.
- Antibiotic ointment for lid application to prevent sticking and manage discharge.

Acute Purulent Conjunctivitis/ Hyperacute conjunctivitis/ Acute blennorrhoea:

- Symptoms: Fiery red conjunctiva ('pink eye') Severe pain Lid swelling Copious purulent discharge Pus seen in the fornices and margin of the lid Matting of lashes with dirty yellow crust Coloured halos Tender preauricular lymphadenopathy
- Fiery red conjunctiva ('pink eye')
- Severe pain
- Lid swelling
- Copious purulent discharge
- Pus seen in the fornices and margin of the lid
- Matting of lashes with dirty yellow crust
- Coloured halos
- Tender preauricular lymphadenopathy
- Fiery red conjunctiva ('pink eye')
- Severe pain
- Lid swelling
- Copious purulent discharge
- Pus seen in the fornices and margin of the lid

- Matting of lashes with dirty yellow crust
- Coloured halos
- Tender preauricular lymphadenopathy

[Image not available]

- Treatment: Systemic Antibiotics: Single dose Ceftriaxone IM and oral Azithromycin or doxycycline (twice a day for 7 days)
- Single dose Ceftriaxone IM and oral Azithromycin or doxycycline (twice a day for 7 days)
- Single dose Ceftriaxone IM and oral Azithromycin or doxycycline (twice a day for 7 days)

Intravenous ampicillin (Option A) is not the first-line treatment for hyperacute conjunctivitis and is not commonly used.

Eye irrigation with saline (Option B) might provide temporary relief but does not address the underlying infection. Systemic antibiotics are needed to prevent complications.

Topical antibiotic drops (Option D) are generally insufficient for treating hyperacute conjunctivitis caused by *Neisseria gonorrhoeae*; systemic antibiotics are required.

Solution for Question 24:

Correct Answer: B) Giant Papillary Conjunctivitis

Explanation:

A history of prolonged soft contact lens use, along with symptoms like bilateral itching, redness, watery discharge, and the presence of large, polygonal papillae on the upper tarsal conjunctiva, points towards a diagnosis of giant papillary conjunctivitis.

Giant Papillary Conjunctivitis	
Definition	Specific allergic inflammation affecting the superior tarsal conjunctiva.
Etiology	Soft hydrophilic contact lenses Protruding suture ends Ocular prostheses
Underlying Mechanism	Hypersensitivity reactions (types I and IV)
Symptoms	Itching

	Watering Foreign body sensation Occasionally blurred vision
Signs	Conjunctival congestion predominantly in the upper palpebral region Large polygonal papillae on superior tarsal conjunctiva Macropapillae (0.3-1.0 mm) and giant papillae (1-2 mm)
Management	Discontinue soft contact lenses, remove offending sutures, clean and polish ocular prosthesis; replace with biocompatible material (e.g., Biocoat). Ancillary therapy: Mast cell stabilizers (e.g., lodoxamide, cromolyn sodium, olopatadine) along with artificial tears, antihistaminic drops, and decongestants.
Image	papillary conjunctivitis" data-author="Mpopi Lenake & Nagib du Toit" data-hash="4244" data-license="CC BY-NC-ND 4.0" data-source="https://www.tandfonline.com/doi/figure/10.1080/20786190.2014.977017?scroll=top&needAccess=true" data-tags="March2025" height="353" src="https://image.prepladder.com/notes/xqINwelHQcEvsqTa1n2k1743136949.png" width="500" />

Vernal Keratoconjunctivitis (Option A) is characterized by seasonal symptoms, cobblestone-like papillae, limbal thickening with Horner-Trantas dots, and potential corneal involvement, unlike giant papillary conjunctivitis, which typically presents with large papillae associated with contact lens use.

Phlyctenular Conjunctivitis (Option C) shows small nodules near the limbus with localized congestion, not large papillae on the tarsal conjunctiva, as seen in giant papillary conjunctivitis.

Papillary Conjunctivitis (Option D) is a broad term and lacks the specificity of the large, polygonal papillae and pronounced symptoms seen in Giant Papillary Conjunctivitis, which is linked to contact lens use.

Solution for Question 25:

Correct Answer: B) Allergic Conjunctivitis

Explanation:

Given the seasonal pattern of symptoms, bilateral itching, watery discharge, mild conjunctival congestion, and the presence of small papillae point toward the diagnosis of allergic

conjunctivitis.

Conjunctival inflammatory reactions can be in the form of:

Follicles	Papillae	Granuloma
[Image not available] Yellowish-white, round elevations, 1-2 mm in diameter, due to lymphocyte aggregations in the subepithelial adenoid layer. Conjunctiva of newborns cannot produce follicles until 2-3 months of age.	[Image not available] Hyperplasia of the normal vascular system with glomerulus-like bunches of capillaries growing into the epithelium.	[Image not available] Fleshy, sessile, or pedunculated lesions with inflammatory tissue. May or may not be truly granulomatous. Resulting from chronic local irritation, such as foreign bodies.

Viral Conjunctivitis (Option A) typically presents with watery discharge and moderate conjunctival congestion. It is often associated with follicles rather than papillae, and a seasonal pattern is less likely than that of allergic conjunctivitis.

Bacterial Conjunctivitis (Option C) is characterized by purulent discharge and marked conjunctival congestion. While bacterial conjunctivitis can sometimes present with papillae, these are not usually as prominent or marked as those in allergic conjunctivitis.

Chlamydial Conjunctivitis (Option D) typically presents with mucopurulent discharge and moderate conjunctival congestion. It is often associated with follicles, and papillae may or may not be present. The lack of prominent papillae and specific seasonal patterns make it less likely than allergic conjunctivitis.

Solution for Question 26:

Correct Answer: D)

[Image not available]

Explanation:

Ciliary congestion, also known as ciliary injection, is not a sign of conjunctivitis. It refers to deep, radial vessel congestion around the limbus, usually associated with intraocular conditions like iritis, keratitis, or acute glaucoma. The vessels here are part of the deeper structure of the eye, unlike the superficial vessels involved in conjunctivitis.

Symptoms and Signs of Conjunctivitis:

- Hyperemia or redness of the conjunctiva (Option B)
- Pain
- Photophobia (rare, associated corneal involvement or iritis can be present)
- Blepharospasm
- Discharge (prominent feature)
- Foreign body sensation
- Subconjunctival hemorrhage (Option C)
- Chemosis or conjunctival edema (Option A)
- Inflammatory reactions: Follicles (localized aggregation of lymphocytes), papillae (hyperplasia of the normal vascular system), granulomas

Feature	Conjunctival Congestion	Ciliary Congestion
Location of Congestion	Confined to the conjunctiva, often more marked in the fornices.	Involves the area around the corneal margin, known as the limbal region.
On moving conjunctiva	Congested vessels also move	Congested vessels do not move
Common Causes	Bacterial conjunctivitis, allergic conjunctivitis, viral conjunctivitis	Acute iridocyclitis, acute glaucoma, keratitis, corneal ulcer.
Image	[Image not available]	[Image not available]

Solution for Question 27:

Correct Answer: B) Goblet Cells

Explanation:

Goblet Cells are unicellular mucous-secreting glands located within the epithelium of the conjunctiva. They produce mucus and, hence, maintain the ocular surface's moisture.

Glands of the Eyelid, Mucus glands, and Accessory glands of Conjunctiva:

Gland Type	Name	Location
Mucous Secretory Glands	Goblet Cells	Within the epithelial layer of the conjunctiva

Crypts of Henle	Tarsal conjunctiva	
Glands of Manz	Limbal conjunctiva	
Accessory Lacrimal Glands	Glands of Krause	In the subconjunctival connective tissue of fornices; 42 in the upper fornix, 8 in the lower fornix
Glands of Wolfring	Upper border of the superior tarsus and lower borders of the inferior tarsus	
Glands of the Eyelid	Meibomian Glands	Stroma of the tarsal plate, lid margin
Glands of Zeis	Along hair follicles of eyelashes	
Glands of Moll	Near hair follicles of eyelashes	

[Image not available]

Solution for Question 28:

Correct Answer: B) The conjunctival epithelium is lined by stratified non-keratinized epithelium.

Explanation:

The conjunctival epithelium is composed of stratified non-keratinized epithelium.

Layers of Conjunctiva	
Outer Epithelial Layer	Lined by stratified squamous epithelium. The epithelium gradually thickens from the fornices to the limbus (Option A), becoming stratified and nonkeratinized near the corneal margin, while the palpebral conjunctiva lining the eyelids remains thinner and nonkeratinized. It contains: Goblet cells Melanocytes Langerhans cells
Adenoid Layer (or lymphoid layer)	It is located below the epithelium. This layer develops lymphoid tissue only after 3-4 months of age, which explains why newborns cannot produce a follicular conjunctival reaction. (Option C)

Dense Fibrous Layer	It is present deep in the adenoid layer, merging into the underlying tissues (lid or sclera). This layer contains the vessels and nerves of the conjunctiva.(Option D)
[Image not available]	

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Previous Year Questions

1. Which of the following is seen in the given image?

[Image not available]

- A. Horner trantas spots
- B. Herbert pits
- C. Pannus
- D. Corneal dystrophy

2. A patient presents to the outpatient department with the following findings. What is the most likely diagnosis?

[Image not available]

- A. Pterygium
- B. Limbal dermoid
- C. Pyogenic granuloma
- D. Pinguecula

3. A young boy from poor socioeconomic background has hyperemia and yellowish nodules at limbus. What could be the likely diagnosis?

[Image not available]

- A. Bitot's spots
- B. Pterygium
- C. Pinguecula
- D. Phlyctenular conjunctivitis

4. Which of the following is the normal conjunctival flora?

- A. Staph aureus
 - B. Staph Epidermidis
 - C. Strep. Pneumonia
 - D. H. influenzae
-

5. Which of the following is associated with Vitamin A deficiency?

- A. Pinguecula
 - B. Bitot's spots
 - C. Pterygium
 - D. Stockers line
-

6. Which microorganism is responsible for angular conjunctivitis?

- A. Moraxella lacunata
 - B. Haemophilus influenzae
 - C. Adenovirus type 32
 - D. Coxsackie B virus
-

7. What is the most frequently observed eye symptom associated with COVID-19?

- A. Uveitis

- B. Mucormycosis
 - C. Follicular conjunctivitis
 - D. Papillary conjunctivitis
-

8. What could be the most likely diagnosis for a 5-year-old boy presenting with severe itching, sensitivity to light, and sticky discharge from his eyes. Upon examination the following findings were seen?

[Image not available]

- A. Trachoma
 - B. Vernal keratoconjunctivitis
 - C. Phlyctenular conjunctivitis
 - D. Xerophthalmia
-

9. Identify the clinical condition

[Image not available]

- A. Herpes simplex keratitis
 - B. Bacterial keratitis
 - C. Mycotic keratitis
 - D. Vernal keratoconjunctivitis
-

10. In the management of Trachoma, the SAFE strategy, which is advocated by the World Health Organization (WHO), serves as the foundation. In the acronym SAFE, what does the letter S stand for?

- A. Surgery

- B. Symptom
 - C. Spectacles
 - D. Symbol
-

11. What is the causative agent of the ocular condition as shown in the image?

[Image not available]

- A. Chlamydia
 - B. Mycoplasma
 - C. Pseudomonas
 - D. Staphylococcus
-

12. True about pterygium:

- A. Inflammation of cornea
 - B. Fatty deposition of sclera
 - C. Conjunctival invasion making a flap over cornea
 - D. Dead epithelial debris accumulation
-

13. Identify the test shown below:

[Image not available]

- A. Schirmer test
- B. Tear film break-up test (BUT)
- C. Rose Bengal staining
- D. Tear osmolarity

14. What is the underlying factor responsible for the development of acute hemorrhagic conjunctivitis?

- A. Enterovirus 68
 - B. Enterovirus 72
 - C. Enterovirus 70
 - D. Enterovirus 71
-

15. In developed countries, what is the primary etiology responsible for ophthalmia neonatorum?

- A. Chlamydia
 - B. Gonorrhoea
 - C. Chemical conjunctivitis
 - D. Staphylococcus aureus
-

16. What is the recommended approach for managing a middle-aged woman with a progressive pterygium that is extending towards the pupillary area?

- A. Keratoplasty
 - B. Excision with conjunctival autograft
 - C. Observation
 - D. Beta-radiation
-

17. What could be the probable diagnosis for the eye lesion observed in a 21-year-old male patient?

[Image not available]

- A. Bitot's spots
 - B. Pterygium
 - C. Phlyctenular conjunctivitis
 - D. Pinguecula
-

18. What is the causative agent of the sago-grain appearance of the conjunctiva?

[Image not available]

- A. Chlamydia trachomatis
 - B. Neisseria gonorrhoeae
 - C. Haemophilus influenzae
 - D. Vitamin A deficiency
-

19. Which of the following treatment option is most frequently used for recurrent pterygium

- A. Conjunctival autograft
 - B. Beta-radiation
 - C. Observation
 - D. Bare sclera technique
-

20. In a patient who has had contact with COVID-19 patient present with symptoms such as redness, sensitivity to light, excessive tearing, and overall discomfort. A

thorough examination was conducted with no eye abnormality. Based on this information, which specific part of the eye is most likely to be affected in this patient?

- A. Conjunctiva
 - B. Uvea
 - C. Choroid
 - D. Retina
-

21. A 20-year-old male presents with complaints fever and swelling of the left eyelid for the past two days. He did not have any history of trauma, but experienced a mild flu-like illness one week ago. Upon examination, there was no chemosis or protrusion of the eyeball, and his eyes appeared as follows. Based on these observations, what is the most likely diagnosis?

[Image not available]

- A. Panophthalmitis
 - B. Orbital cellulitis
 - C. Preseptal cellulitis
 - D. Lid tumor
-

22. What is the likely diagnosis from the image given below?

[Image not available]

- A. Pinguecula
 - B. Concretions
 - C. Pterygium
 - D. Bitot's spots
-

23. Which organism is most likely causing the infection in a 14-day-old neonate with bilateral conjunctivitis and lung infiltrates, as found in a chest X-ray performed at the outpatient department?

- A. Chlamydia trachomatis
 - B. Hemophilus influenzae
 - C. Streptococcus
 - D. Mycoplasma
-

24. Which of the following is not included in the SAFE strategy for trachoma?

- A. Surgery for trichiasis
 - B. Antibiotics
 - C. Facial cleanliness
 - D. Evaluation of control program
-

25. What is the term for the additional row of eyelashes located behind the grey line?

- A. Trichiasis
 - B. Distichiasis
 - C. Tylosis
 - D. Madarosis
-

26. What is the cause of phlyctenular conjunctivitis?

- A. Atropine
 - B. Asthma
 - C. Contact lens
 - D. Tuberculous proteins
-

27. A contact lens user presents with the following clinical picture. He has had watering, redness, and foreign body sensation in the eye for the past 2 months. What is the most probable diagnosis?

[Image not available]

- A. Trachoma
 - B. Follicular conjunctivitis
 - C. Giant papillary conjunctivitis
 - D. Spring catarrh
-

28. What is the likely diagnosis for a 5-year-old boy who is experiencing severe itching, and thick discharge from his eyes. Upon examination the following finding as shown in the image was observed.

[Image not available]

- A. Trachoma
 - B. Vernal keratoconjunctivitis
 - C. Phlyctenular conjunctivitis
 - D. Xerophthalmia
-

29. What is the most effective treatment for a patient who experiences considerable discomfort when reading text from paper or a screen, despite having normal visual

acuity and other eye examination findings, except for a Schirmer's test result of 5mm?

- A. Steroids
 - B. Anti-histamine
 - C. Lubricating eye drops
 - D. Prescribed spectacles
-

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	4
Question 4	2
Question 5	2
Question 6	1
Question 7	3
Question 8	2
Question 9	4
Question 10	1
Question 11	1
Question 12	3
Question 13	1
Question 14	3
Question 15	1
Question 16	2
Question 17	1
Question 18	1
Question 19	1
Question 20	1
Question 21	3
Question 22	3

Question 23	1
Question 24	4
Question 25	2
Question 26	4
Question 27	3
Question 28	2
Question 29	3

Solution for Question 1:

Correct Option B - Herbert pits:

- The image shows scarring in the upper limbus, which is referred to as Herbert's pits. It is seen in the chronic cicatricial stage of trachoma.

Incorrect Options:

Option A - Horner trantas spots:

[Image not available]

Option C – Pannus:

[Image not available]

Option D - Corneal dystrophy:

[Image not available]

Microcystic/Map dot/ Fingerprint Dystrophy

[Image not available]

Meesmann dystrophy

[Image not available]

Reis-Bucklers corneal dystrophy

Solution for Question 2:

Correct Option B - Limbal dermoid:

- Limbal dermoids are congenital benign tumors that cause astigmatism, encroachment on the visual axis, and fatty component infiltration into the cornea. They consist of overgrowths of epidermal appendages, connective tissues, skin, fat, sweat glands, lacrimal glands, and neurologic tissues located partly over the cornea and partly over the sclera.

Incorrect Options:

Option A - Pterygium: A pterygium is a fibrovascular growth of the conjunctiva that extends onto the cornea.

Option C - Pyogenic granuloma: A pyogenic granuloma is a red, fleshy, vascular lesion that usually arises after trauma or surgery.

Option D - Pinguecula: A pinguecula is a yellowish, slightly raised lesion on the conjunctiva, usually near the limbus.

Solution for Question 3:

Correct Option D - Phlyctenular conjunctivitis:

- Poor socioeconomic status suggests that the person is prone to develop infections
- The above image is of Phlyctenular conjunctivitis
- In Phlyctenular conjunctivitis, nodule of phlycten (yellowish nodules) near the limbus and conjunctival congestion is seen
- Allergy caused by endogenous antigens → Staphylococcus aureus and tuberculosis infections.

Incorrect Options:

Option A - Bitot's spots: Bitot's spots are seen in Vitamin A deficiency

Option B - Pterygium: It is conjunctival degeneration characterized by triangular fibrovascular subepithelial ingrowth of bulbar conjunctiva over the limbus

Option C - Pinguecula: It is a yellowish-white mound near the limbus formed by elastotic degeneration of conjunctival stroma.

Solution for Question 4:

Correct Option B - Staph Epidermidis:

- The normal conjunctival flora is Staph Epidermidis
- Staph Epidermidis is the most common organism which is responsible for Acute Onset Endophthalmitis.

Incorrect Options:

- Options A, C & D: Staph aureus, Strep. Pneumonia and H. influenzae do not form the normal conjunctival flora
-

Solution for Question 5:

Correct Option B - Bitot's spots:

- Vitamin A deficiency is associated with Bitot's spots

Incorrect Options:

- Options A, C & D: These are not associated with Vitamin A deficiency
-

Solution for Question 6:

Correct Option A - Moraxella lacunata:

- Angular conjunctivitis, also known as angular blepharitis or angular keratoconjunctivitis, is commonly caused by the bacterium *Moraxella lacunata*.
- It is characterized by inflammation and infection of the canthi(angles) of the eyes, resulting in redness, discharge, and crusting.
- Other pathogens causing angular conjunctivitis are: *Moraxella axenfeld* (most common). *Moraxella catarrhalis*. *Staphylococcus aureus*.
- *Moraxella axenfeld* (most common).
- *Moraxella catarrhalis*.
- *Staphylococcus aureus*.
- *Moraxella axenfeld* (most common).
- *Moraxella catarrhalis*.
- *Staphylococcus aureus*.

Incorrect Options:

- Options B, C, and D are incorrect. These organisms do not cause angular conjunctivitis.

Solution for Question 7:

Correct Option C - Follicular conjunctivitis:

- The most common ocular manifestation of COVID-19 is follicular conjunctivitis. Conjunctivitis, also known as "pink eye," is an inflammation or infection of the conjunctiva. It is distinguished by the presence of follicles (small, raised, white or yellowish areas) on the conjunctiva.

[Image not available]

Incorrect Options:

Option A - Uveitis: Uveitis is not commonly associated with COVID-19. Uveitis refers to inflammation of the uvea, the middle layer of the eye.

Option B - Mucormycosis: Mucormycosis is a serious fungal infection caused by mucorales. It is associated with covid 19 due to overuse of steroids.

Option D - Papillary conjunctivitis: Papillary conjunctivitis, marked by the presence of small, round bumps (papillae) on the inner side of the eyelids. It is caused by mechanical trauma and subsequent response to the antigens deposited on the surface.

Solution for Question 8:

Correct Option B - Vernal keratoconjunctivitis:

- The patient's symptoms of intense itching and sticky (ropy) discharge, along with the presence of hard, flat-topped papillae on the upper tarsal conjunctiva with a cobblestone appearance as shown in the image are characteristic of vernal keratoconjunctivitis (VKC).

Incorrect Options:

Option A - Trachoma: It is characterized by sago-grain follicles on the upper tarsal conjunctiva, Herbert's pit, conjunctival scarring, art's line, trichiasis, and corneal opacity, which can lead to blindness if left untreated.

Option C - Phlyctenular conjunctivitis: Phlyctenular conjunctivitis is a type of hypersensitivity reaction to endogenous antigens like bacterial antigens, staphylococcal & tuberculosis antigens. It presents as a small, white, or yellowish nodule (phlyctenule) near the limbus and conjunctival congestion.

Option D - Xerophthalmia: Xerophthalmia is a spectrum of ocular disease caused by lack of vitamin A. It is characterized by night blindness (earliest feature), conjunctival xerosis, Bitot's spot, corneal xerosis, ulcer, keratomelacia.

Solution for Question 9:

Correct Option D - Vernal keratoconjunctivitis:

- The given image is suggestive of conjunctival congestion and is suggestive of vernal keratoconjunctivitis, a chronic allergic condition that primarily affects the conjunctiva.
- It is characterized by recurrent episodes of inflammation, itching, redness, and a cobblestone-like appearance on the upper palpebral conjunctiva..
- This condition is more common in young individuals, particularly boys, and is often associated with a history of allergies.

Incorrect Options:

Option A - Herpes simplex keratitis: Herpes simplex keratitis is a viral infection caused by the herpes simplex virus characterized by redness, and dendritic or branching pattern of corneal ulcers.

Option B - Bacterial keratitis: It usually presents with symptoms such as eye redness, pain, discharge, hypopyon, and a corneal infiltrate or ulcer.

Option C - Mycotic keratitis: It usually presents with redness, pain, discharge, unsterile hypopyon and dry corneal ulcer with feathery ends.

Solution for Question 10:

Correct Option A - Surgery:

- Trachoma is the fourth most common cause of blindness caused by Chlamydia trachomatis. According to the SAFE strategy, S is for Surgery.
- SAFE strategy: S - Surgery (for trichiasis-posterior misdirection of eyelids) A - Antibiotics: DOC-Azithromycin 1g tablet - To treat and prevent active infection. F - Facial cleanliness: To prevent disease transmission. E - Environmental hygiene: To increase access to water and sanitation.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 11:

Correct Option A - Chlamydia:

- The image shows follicles resembling sago grain on the upper palpebral conjunctiva is suggestive of trachoma and the organism causing this condition is Chlamydia trachomatis.

- Trachoma, a chronic childhood conjunctivitis caused by Chlamydia trachomatis, is a leading preventable cause of irreversible blindness worldwide. It manifests in different strains: A, B, Ba, and C. It commonly affects children aged 1-9.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 12:

Correct option C - Conjunctival invasion making a flap over cornea:

- A pterygium is a fibrovascular sub-epithelial ingrowth of degenerative bulbar conjunctival tissue over the limbus onto the cornea.
- It appears as a triangular-shaped tissue with blood vessels and can cause discomfort, redness, and visual disturbances. It is characterized by the invasion of conjunctival tissue over the cornea, forming a flap-like structure.

Incorrect Options:

- Options A, B, and D are incorrect. Pterygium is a sub epithelial of conjunctiva on the cornea like a flap.

Solution for Question 13:

Correct Option A - Schirmer test:

- The Schirmer test is a diagnostic test used to measure the quantity of tear production and assess for dry eye syndrome.
- The test involves measuring the amount of wetting of a special (no. 41 Whatman) filter paper, 5 mm wide and 35 mm long, bent at one end and kept in the lower fornix and the patient is asked to look up and not to blink eye.
- After 5 minutes wetting of the filter paper from the bent end is measured and results are interpreted as Normal values - more than 15 mm of wet paper. 5 to 10 mm is suggestive of

mild dry eye/keratoconjunctivitis sicca. Less than 5 mm is suggestive of severe dry eye.

- Normal values - more than 15 mm of wet paper.
- 5 to 10 mm is suggestive of mild dry eye/keratoconjunctivitis sicca.
- Less than 5 mm is suggestive of severe dry eye.
- Test performed with an anesthetic is known as Schirmer 2 test - measures only basal secretion.
- Test performed without anesthesia is known as Schirmer 1 test - measures both basal and reflex secretion.
- Normal values - more than 15 mm of wet paper.
- 5 to 10 mm is suggestive of mild dry eye/keratoconjunctivitis sicca.
- Less than 5 mm is suggestive of severe dry eye.

Incorrect Options:

Option B - Tear film break-up test: It involves using fluorescein to stain cornea and check for dry spot under the blue cobalt light.

Option C - Rose Bengal staining: It is also used to diagnose dry eye syndrome, by staining the dead cells and mucus. Three patterns has been described i.e., A (severe disease), B (moderated cases), and C (mild cases.)

Option D - Tear osmolarity: It is used to diagnose dry eye with a device, a high value indicates dre eye.

Solution for Question 14:

Correct Option C - Enterovirus 70: Viral causes of hemorrhagic conjunctivitis are:

- Enterovirus - 70
- Adenovirus
- Coxsackie - 24
- Echovirus - 34

Incorrect Options: Options A, B, and D do not cause hemorrhagic conjunctivitis

Solution for Question 15:

Correct Option A - Chlamydia:

- Chlamydia trachomatis is the most common cause of ophthalmia neonatorum worldwide..
- Ophthalmia neonatorum refers to conjunctivitis that occurs in newborn babies within the first month of life.
- Chlamydial ophthalmia neonatorum is typically acquired during delivery when the baby comes into contact with the infected birth canal.

Incorrect Options:

Option B - Gonorrhoea: Neisseria gonorrhoeae can manifest with severe symptoms of ophthalmia neonatorum, but is less common than chlamydia.

Option C - Chemical conjunctivitis: Chemical conjunctivitis refers to conjunctival inflammation caused by exposure to irritants or chemicals which are given as a prophylaxis for conjunctivitis.

Option D - Staphylococcus aureus: Staphylococcus aureus can cause various types of infections, including conjunctivitis, but it is not the most common cause of ophthalmia neonatorum.

Solution for Question 16:

Correct Option B - Excision with conjunctival autograft:

- The most appropriate management for pterygium encroaching the pupillary area is surgical excision with conjunctival autograft.
- This involves removing the pterygium tissue and replacing it with a graft of healthy conjunctival tissue taken from another area of the patient's own eye.
- This procedure has been shown to have a low recurrence rate and a high success rate in preventing visual disturbance caused by the pterygium.

Incorrect Options:

Option A - Keratoplasty: Keratoplasty involves the surgical replacement of the cornea with a healthy donor cornea. Indicated for corneal opacities, bullous keratoplasty, corneal dystrophies, and keratoconus.

Option C - Observation: If the pterygium is not causing any visual disturbance, observation may be appropriate. However, if it is steadily progressing towards the pupillary area, it is likely to cause visual disturbances and should be managed surgically.

Option D - Beta-radiation: Beta-radiation therapy involves using low doses of radiation to prevent recurrence of the pterygium, not recommended.

Solution for Question 17:

Correct Option A - Bitot's spots:

- It is a triangular patch of foamy keratinized epithelium in the interpalpebral zone.
- It is a characteristic feature of xerophthalmia and is most commonly found temporally.

Incorrect Options:

Option B - Pterygium: Pterygium is a degenerative, subconjunctival fibrovascular tissue encroaching the cornea.

Option C - Phlyctenular conjunctivitis: Phlyctenular conjunctivitis has a white pinkish nodule (phlycten) in bulbar conjunctiva often caused due to endogenous microbial proteins..

Option D - Pinguecula: Pinguecula is a yellowish-white mound near the limbus formed by elastotic degeneration of conjunctival stroma.

Solution for Question 18:

Correct Option A - Chlamydia trachomatis:

- The sago grain appearance of the follicles in conjunctiva is seen in active stage of trachoma and the organism responsible is chlamydia trachomatis. It is a follicular reaction.

Incorrect Options:

- Options B, C, and D are incorrect. These organisms are not associated with sago grain appearance of the follicles.

Solution for Question 19:

Correct Option A - Conjunctival autograft:

- The most common and effective treatment procedure for recurrent pterygium is the conjunctival autograft. Here, limbal tissue from the same eye or the other eye is grafted and secured with fibrin glue or sutures.

Incorrect Options:

Option B - Beta-radiation: Beta radiation is no longer commonly used as it causes late scleral necrosis.

Option C - Observation: Observation is preferred only for cases which do not encroach the cornea, for recurrent pterygium and pterygium encroaching cornea surgical management is required.

Option D - Bare sclera technique: The bare sclera technique involves excising the head and body of the pterygium while allowing the bare scleral bed to re-epithelialize, recurrence rate is high with this technique.

Solution for Question 20:

Correct Option A - Conjunctiva:

- The most common ocular manifestation of COVID-19 is follicular conjunctivitis. It is distinguished by the presence of follicles (small, raised, white or yellowish areas) on the conjunctiva.
- Therefore, most commonly involved part of the eye in Covid 19 is conjunctiva.

Incorrect Options:

- Options B, C, and D are incorrect. These are not the most commonly involved part of the eye in a Covid 19 infection.

Solution for Question 21:

Correct Option C - Preseptal cellulitis:

- Preseptal cellulitis is an infection of the eyelid and surrounding skin anterior to the orbital septum. It is characterized by eyelid swelling, erythema, fever, and tenderness, often following a minor trauma or spreading from a nearby infection, or spreading from a remote infection of middle ear and upper respiratory tract infection.. In this case, the absence of significant ocular findings, such as chemosis or proptosis, along with the mild flu-like illness history, is more consistent with preseptal cellulitis

Incorrect Options:

Option A - Panophthalmitis: It is an intense purulent inflammation of the whole eyeball, including the tenon's capsule, characterized by severe pain, fever, malaise complete loss of vision, and purulent discharge.

Option B - Orbital cellulitis: Orbital cellulitis is an infection involving the soft tissues within the orbit (behind the septum), typically associated with severe eyelid swelling, unilateral and painful proptosis, restricted eye movements, and decreased visual acuity.

Option D - Lid tumor: Lid tumors refer to abnormal growths or masses in the eyelid.

Solution for Question 22:

Correct Option C - Pterygium:

- The given image reveals fleshy growth extending onto the cornea.
- Pterygium is a subconjunctival fibrovascular tissue encroaching the cornea. It appears as a triangular or wedge-shaped fleshy growth on the inner or outer side of the eye, usually

starting from the conjunctiva near the nasal or temporal side (nasal more common).

Incorrect Options:

Option A - Pinguecula: A pinguecula is a yellowish, elevated growth on the conjunctiva, but it does not extend onto the cornea..

Option B - Concretions: Concretions refer to tiny calcium or protein deposits that can form on the conjunctiva.

Option D - Bitot's Spots: Bitot's spots are a specific manifestation of vitamin A deficiency. They appear as small, foamy, irregularly shaped white or grayish patches on the conjunctiva, typically located on the temporal side of the eye.

Solution for Question 23:

Correct Option A - Chlamydia trachomatis:

- The bilateral neonate conjunctivitis aka ophthalmia neonatorum and lung infiltrates is suggestive of pneumonia.
- Chlamydia trachomatis is the most common bacterial sexually transmitted infection and infants born vaginally to infected mothers with genital disease are at risk for acquiring C. trachomatis, which usually presents as conjunctivitis and/or pneumonia.

Incorrect Options:

- Options B, C, and D are incorrect. These are not the most common infections responsible for bilateral conjunctivitis in neonate.

Solution for Question 24:

Correct Option D - Evaluation of control program:

- SAFE Strategy: WHO programme to control trachoma. It includes the following: Surgery (Trichiasis) Antibiotics (oral azithromycin > topical tetracycline 1% ointment) Azithromycin:

1gm in adults and 20 mg/kg in children. Facial hygiene. Environmental changes

- Surgery (Trichiasis)
- Antibiotics (oral azithromycin > topical tetracycline 1% ointment) Azithromycin: 1gm in adults and 20 mg/kg in children.
- Azithromycin: 1gm in adults and 20 mg/kg in children.
- Facial hygiene.
- Environmental changes
- Surgery (Trichiasis)
- Antibiotics (oral azithromycin > topical tetracycline 1% ointment) Azithromycin: 1gm in adults and 20 mg/kg in children.
- Azithromycin: 1gm in adults and 20 mg/kg in children.
- Facial hygiene.
- Environmental changes
- Azithromycin: 1gm in adults and 20 mg/kg in children.

Incorrect Options:

- Options A, B, and C are incorrect. These are included in the safe strategy to prevent Trachoma.

Solution for Question 25:

Correct Option B - Distichiasis:

- Distichiasis is a condition in which the eyelashes grow abnormally and emerge from the Meibomian gland openings behind the normal lash line.
- These extra lashes can cause irritation and damage to the cornea and conjunctiva, leading to symptoms such as eye discomfort, redness, tearing, and sensitivity to light.
- Treatment options for distichiasis include manual plucking of the extra lashes, cryotherapy (freezing the hair follicles), or various surgical procedures to remove or redirect the aberrant eyelashes.

[Image not available]

Incorrect Options:

Option A - Trichiasis: Trichiasis is a condition in which the eyelashes grow inwards towards the eye, causing them to rub against the cornea or conjunctiva.

Option C - Tylosis: Tylosis is a genetic condition characterized by thickening of the lid margin.

Option D - Madarosis: Madarosis refers to the loss or absence of eyelashes and/or eyebrows.

Solution for Question 26:

Correct Option D - Tuberculous proteins:

- Phlyctenular conjunctivitis is most commonly associated with a delayed hypersensitivity reaction to endogenous microbial proteins.
- Endogenous microbial proteins are: Tuberculous proteins. Staphylococcal proteins. Moraxella Axenfeld bacillus and certain parasites.
- Tuberculous proteins.
- Staphylococcal proteins.
- Moraxella Axenfeld bacillus and certain parasites.
- Tuberculous proteins.
- Staphylococcal proteins.
- Moraxella Axenfeld bacillus and certain parasites.

Incorrect Options:

Option A - Atropine: It is known to cause contact dermatitis.

Option B - Asthma: Asthma (atopy) is known to cause atopic keratoconjunctivitis and some cases of vernal keratoconjunctivitis.

Option C - Contact lens: It is known to cause giant papillary conjunctivitis.

Solution for Question 27:

Correct Option C - Giant papillary conjunctivitis:

- The clinical presentation of watering, redness, and foreign body sensation along with image showing large papillae in the upper tarsal conjunctiva in a contact lens wearer is suggestive of Giant papillary conjunctivitis.
- Giant Papillary Conjunctivitis (GPC) aka mechanically-induced papillary conjunctivitis is a localised allergic response to a physically rough or deposited surface (contact lens, prosthesis, nylon suture, and scleral buckle).

Incorrect Options:

Option A - Trachoma: Trachoma is a chronic, infectious form of conjunctivitis caused by the bacterium *Chlamydia trachomatis*. It is characterized by follicles in the upper palpebral conjunctiva, conjunctival scarring, trichiasis, and corneal opacity, which can lead to blindness if left untreated.

Option B - Follicular conjunctivitis: Follicular conjunctivitis typically presents with superficial punctate keratitis and multiple small, elevated follicles on the fornix and palpebral conjunctiva. It is often caused by viral (adenovirus) or bacterial infections and is not specifically associated with contact lens wear..

Option D - Spring catarrh: Spring catarrh, also known as vernal keratoconjunctivitis, is a seasonal allergic condition that primarily affects young males. It is characterized by intense itching, mucus discharge, and cobblestone-like papillae on the upper tarsal conjunctiva.

Solution for Question 28:

Correct Option B - Vernal keratoconjunctivitis:

- The patient's symptoms of intense itching and thick ropy discharge, along with the presence of hard, flat-topped papillae on the upper tarsal conjunctiva with a cobblestone appearance, are characteristic of vernal keratoconjunctivitis (VKC).

Incorrect Options:

Option A - Trachoma: Trachoma is a chronic, infectious form of conjunctivitis caused by the bacterium *Chlamydia trachomatis*. It is characterized by conjunctival scarring, trichiasis, and corneal opacity, which can lead to blindness if left untreated.

Option C - Phlyctenular conjunctivitis: Phlyctenular conjunctivitis is a type of hypersensitivity reaction to endogenous antigens like staphylococcal & tuberculosis antigen, typically seen in children. It presents as a small, white, or yellowish nodule (phlyctenule) near limbus and conjunctival congestion..

[Image not available]

Option D - Xerophthalmia: Xerophthalmia is a condition caused by severe vitamin A deficiency. It is characterized by dryness and thickening of the conjunctiva and cornea, which can lead to corneal ulceration and blindness.

Solution for Question 29:

Correct Option C - Lubricating eye drops:

- A low Schirmer's test value of 5 mm (15 mm normal value) indicates reduced tear production and dry eye syndrome.
- Lubricants such as artificial tears like cellulose derivatives alleviate dry eye symptoms and improve visual comfort. They work by supplementing the natural tears, improving tear film stability, and reducing ocular surface inflammation.

Incorrect Options:

- Options A, B, and D are incorrect. They are not used for the management of Scirmer's test.
-

Lacrimal Drainage System

1. Which structure within the lacrimal passages prevents the backflow of tears from the nasal cavity into the nasolacrimal duct?

- A. Valve of Hasner
- B. The valve of Rosenmüller
- C. Orbital lobe
- D. Glands of Krause

2. Which of the following is a cause of lacrimal pump failure, leading to excessive tearing (epiphora)?

- A. Trauma to the canaliculus
- B. Bell's palsy
- C. Obstruction of the nasolacrimal duct
- D. Narrowing of the punctum

3. A 65 y/o male presents with persistent tearing in his right eye that has worsened over the past year. The physician orders an imaging study to pinpoint the exact location of the blockage and assess the condition of the lacrimal sac. Which investigation is most appropriate for this purpose?

- A. Dacryocystography
- B. Fluorescein Dye Disappearance Test (FDDT)
- C. Dacryoscintigraphy
- D. Lacrimal Syringing

4. A 50 y/o man undergoes Jones Test 1 for chronic tearing, which shows no dye on the cotton bud placed in the inferior meatus. Following this, Jones Test 2 is performed, and fluorescein dye is detected after lacrimal syringing. What does this result suggest?

- A. Partial obstruction within the lacrimal passages
- B. Complete obstruction of the nasolacrimal duct
- C. Lacrimal pump failure
- D. Primary hypersecretion of tears

5. A 9-month-old infant is brought to the clinic with ongoing tearing and discharge from the right eye. Despite regular lacrimal sac massage and topical antibiotics, the symptoms persist. What is the next best step in management?

- A. Probing of the nasolacrimal duct
- B. Probing of the nasolacrimal duct
- C. Dacryocystorhinostomy (DCR)
- D. Balloon catheter dilation

6. Which of the following types of Dacryocystorhinostomy (DCR) is considered for treating nasolacrimal duct obstruction, offering the highest success rate?

- A. External Dacryocystorhinostomy (External DCR)
- B. Endonasal Dacryocystorhinostomy (Endonasal DCR)
- C. Endocanalicular Laser Dacryocystorhinostomy (Laser DCR)
- D. Conjunctivodacryocystorhinostomy (CDCR)

7. A 2-week-old infant presents with a bluish swelling in the inner corner of the right eye. The parents report excessive tearing and some slimy discharge from the eye. There is no history of trauma or illness. What is the most likely diagnosis?

- A. Congenital Dacryoceles
 - B. Dacryoadenitis
 - C. Dacryops
 - D. Chronic Dacryocystitis
-

8. A 55 y/o woman presents with a persistent painful swelling below the inner corner of her left eye. She also reports a history of watery eyes and recurrent conjunctivitis. On examination, there is fluctuant swelling at the lacrimal sac, and mucoid fluid can be expressed. What is the most likely diagnosis and appropriate management?

- A. Chronic Dacryocystitis; Probing and syringing.
 - B. Chronic Dacryocystitis; Dacryocystorhinostomy (DCR).
 - C. Acute Dacryocystitis; Systemic antibiotics and abscess drainage.
 - D. Acute Dacryocystitis; Dacryocystectomy (DCT).
-

9. A 60 y/o man presents with watery eyes, redness, swelling at the inner corner of his right eye, and persistent thick yellow discharge. O/E there is pouting of lacrimal puncta and yellow granules from the punctum upon massage. Which organism is most likely responsible, and what is the definitive treatment?

- A. Eikenella corrodens; Oral antibiotics and steroid drops.
- B. Staphylococcus aureus; Topical antibiotics and warm compresses

- C. Actinomyces israelii; Canaliculotomy
 - D. Chlamydia trachomatis; Irrigation and syringing of the tear duct.
-

10. A 27 year old man presents with a painless, slow-growing, firm mass on the upper eyelid that has been gradually increasing in size over the past year. On examination, the lesion is well-circumscribed, mobile, and non-tender, with no signs of ulceration or inflammation. Imaging reveals a well-defined, lobulated mass without bony invasion. What is the most likely diagnosis?

- A. Pleomorphic Adenoma
 - B. Lymphoma
 - C. Lacrimal Gland Carcinoma
 - D. Dacryocystitis
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	1
Question 4	1
Question 5	2
Question 6	1
Question 7	1
Question 8	2
Question 9	3
Question 10	1

Solution for Question 1:

- The valve of Hasner is a crucial structure located at the lower end of the nasolacrimal duct, where it opens into the inferior meatus of the nose.
- Its primary function is to prevent the retrograde flow of fluid from the nasal cavity into the nasolacrimal duct, which is particularly important during actions like nose blowing.
- This valve ensures that tears drain properly from the eye into the nasal cavity without backflow.
- The valve of Rosenmüller (Option B) is a mucosal fold located at the junction where the common canaliculus (formed by the superior and inferior canaliculi) opens into the lacrimal sac. This valve is responsible for preventing the backward flow of tears from the lacrimal sac into the canaliculi, thereby maintaining the forward movement of tears toward the nasolacrimal duct. Orbital Lobe (Option C) is one of the two lobes of the main lacrimal gland, which is involved in the production of reflex tears. It is located in the lacrimal fossa of the

frontal bone and plays no role in the prevention of backflow within the lacrimal passages.

- Glands of Krause (Option D) are accessory lacrimal glands found in the conjunctival fornices. They are responsible for producing basal tears, which help maintain eye moisture. These glands do not participate in preventing backflow within the lacrimal drainage system.

[Image not available]

The valve of Rosenmüller (Option B) is a mucosal fold located at the junction where the common canaliculus (formed by the superior and inferior canaliculi) opens into the lacrimal sac. This valve is responsible for preventing the backward flow of tears from the lacrimal sac into the canaliculi, thereby maintaining the forward movement of tears toward the nasolacrimal duct.

Orbital Lobe (Option C) is one of the two lobes of the main lacrimal gland, which is involved in the production of reflex tears. It is located in the lacrimal fossa of the frontal bone and plays no role in the prevention of backflow within the lacrimal passages.

[Image not available]

Solution for Question 2:

Correct answer: B) Bell's palsy

Explanation:

- The lacrimal pump mechanism relies on the proper function of the orbicularis oculi muscle and the correct positioning of the punctum.
- In Bell's palsy, Weakness of the orbicularis oculi muscle prevents proper eyelid closure and tear drainage, as the orbicularis oculi muscle is involved in both the tear suction and drainage phase of the lacrimal pump mechanism.
- This leads to epiphora (excessive tearing).

Clinical evaluation for Watering Eye (Epiphora):

- Eyelid and Punctal Examination: Eversion of the punctum can be observed in ectropion or lax eyelids.

Fluorescein Dye Test: This test can help assess the tear drainage system by observing the time it takes for dye to disappear from the eye.

[Image not available]

- **Lacrimal Syringing test:** Testing with probing or syringing may help identify blockages in the tear drainage system, such as in the canaliculi or nasolacrimal duct.

Trauma to the canaliculus (Option A): While an injury to the canaliculus can block the tear drainage system, it does not directly disrupt the lacrimal pump mechanism. Trauma may cause scar formation or narrowing, but it primarily impacts tear flow through mechanical obstruction rather than pump function.

Obstruction of the nasolacrimal duct (Option C): This condition involves a blockage in the nasolacrimal duct, preventing proper tear drainage. However, it is considered a mechanical obstruction rather than a lacrimal pump failure.

Narrowing of the punctum (Option D): Punctal stenosis refers to a narrowing or closure of the punctum (the opening at the corner of the eyelid that drains tears). While this causes tear accumulation, it results from a physical obstruction at the punctum, not from a dysfunction of the lacrimal pump mechanism.

Solution for Question 3:

Dacryocystography(DCG) provides highly detailed anatomical information about the entire lacrimal drainage system. By injecting contrast dye into the lacrimal sac and taking X-rays, DCG can precisely identify the site, extent, and nature of the blockage, making it the investigation of choice when anatomical clarity is required.

[Image not available]

[Image not available]

Fluorescein Dye Disappearance Test (FDDT) (Option B):

- 2 drops of the Fluorescein dye are injected into both the conjunctival sacs.
- It usually disappears in 2 minutes in the normal eye.
- Longer retention of the dye indicates any obstruction or dysfunction in tear flow.
- It is mainly used to assess the overall efficiency of tear drainage and does not provide information on the exact location or nature of any obstruction within the lacrimal system.

Dacryoscintigraphy (Option C) or lacrimal scintillography is a functional test that evaluates the rate at which tears flow through the lacrimal drainage system by tracking a radioactive tracer. While it is useful for identifying functional delays or blockages, it lacks the anatomical precision needed to determine the exact location or cause of the obstruction.

[Image not available]

Lacrimal Syringing (Option D):

- Normal saline is injected into the lacrimal sac using a syringe.
- No obstruction: Free passage through the lacrimal passages & the nose
- Partial obstruction: Saline passes with resistance and pressure.
- Complete obstruction: No fluid passes. If fluid refluxes through the same punctum: Obstruction in the same or common canaliculus. If fluid refluxes through the opposite punctum: Obstruction in the lower sac or NLD.
- If fluid refluxes through the same punctum: Obstruction in the same or common canaliculus.
- If fluid refluxes through the opposite punctum: Obstruction in the lower sac or NLD.
- If fluid refluxes through the same punctum: Obstruction in the same or common canaliculus.
- If fluid refluxes through the opposite punctum: Obstruction in the lower sac or NLD.

Although it can identify the degree of blockage & approximate blockage site, it does not offer detailed anatomical information.

Solution for Question 4:

A positive Jones Test 2 indicates that the fluorescein dye was able to enter the lacrimal sac but could not reach the nasal cavity without syringing. This finding is typical of a partial blockage somewhere along the lacrimal drainage system, allowing some tear flow but not enough for normal drainage.

[Image not available]

Jones Test 1:

- Purpose: To distinguish between partial obstruction and excessive tear production.
- Procedure: Fluorescein dye is placed in the conjunctival sac, and after 5 minutes, a cotton bud soaked in anaesthetic is placed in the inferior meatus inside the nose. The cotton bud is checked for dye.
- Interpretation: Positive: Dye on the cotton bud indicates patent tear drainage and suggests hypersecretion. Negative: No dye indicates potential drainage issues, requiring further testing with Jones Test.
- Positive: Dye on the cotton bud indicates patent tear drainage and suggests hypersecretion.

- Negative: No dye indicates potential drainage issues, requiring further testing with Jones Test.
- Positive: Dye on the cotton bud indicates patent tear drainage and suggests hypersecretion.
- Negative: No dye indicates potential drainage issues, requiring further testing with Jones Test.

Jones Test 2:

- Purpose: Performed if Jones Test 1 is negative to differentiate between partial obstruction and pump failure.
- Procedure: Lacrimal syringing is done while the cotton bud remains in the nose.
- Interpretation: Positive: Dye flushed onto the cotton bud indicates partial obstruction. Negative: No dye suggests lacrimal pump failure.
- Positive: Dye flushed onto the cotton bud indicates partial obstruction.
- Negative: No dye suggests lacrimal pump failure.
- Positive: Dye flushed onto the cotton bud indicates partial obstruction.
- Negative: No dye suggests lacrimal pump failure.

[Image not available]

Complete obstruction of the nasolacrimal duct (Option B): In the case of a complete obstruction, the dye would not move at all, even with syringing. Therefore, no dye would be detected in the nose after Jones Test 2, which is different from the positive result in this patient.

Lacrimal pump failure (Option C) would result in a negative Jones Test 2, meaning that the dye wouldn't move into the nose even with syringing, because the pumping action of the lacrimal system is impaired.

Primary hypersecretion of tears (Option D) would present with a positive Jones Test 1, where the dye would naturally pass through the lacrimal system and into the nose without any obstruction. A negative Jones Test 1 followed by a positive Jones Test 2 makes this diagnosis unlikely.

Solution for Question 5:

Probing is the recommended next step for a child in this age group (> 6-9 months) who has not responded to conservative treatment. This procedure uses a small probe to mechanically clear the blockage in the nasolacrimal duct and is typically done under anaesthesia.

In case of failure, it is repeated at an interval of 3-4 weeks.

Continue lacrimal sac massage and antibiotics (Option A): Continuing with conservative management at this point is unlikely to be effective, as the infant has already been treated with massage and antibiotics for several months without symptom improvement. The blockage persists, so further intervention is needed.

Dacryocystorhinostomy (DCR) (Option C): DCR is a more invasive surgical option used in older children (>4 y/o) when less invasive treatments like probing and balloon dilation have failed. It is not usually the next step for infants at this age who have not yet undergone probing.

Balloon catheter dilation (Option D): Balloon dilation may be considered if repeated probing fails. It involves inserting a small balloon into the nasolacrimal duct to widen the passage. However, it is not the first choice of treatment after conservative measures fail.

Solution for Question 6:

External Dacryocystorhinostomy (External DCR) procedure is performed through an incision near the inner corner of the eye, providing direct access to the lacrimal sac and nasal mucosa. It is widely done due to its high success rate, which exceeds 90%. However, it leaves a visible scar and may result in greater blood loss compared to other methods.

[Image not available]

Endonasal Dacryocystorhinostomy (Endonasal DCR) (Option B) involves using an endoscope inserted through the nasal cavity to create a drainage opening, thus avoiding any external incision. While it has the advantage of being scarless and typically involves less bleeding, its success rate is slightly lower, ranging between 70-90%.

Endocanalicular Laser Dacryocystorhinostomy (Laser DCR) (Option C): This method utilizes a laser passed through the canaliculi to create a drainage opening. It is minimally invasive and can be performed quickly under local anaesthesia. However, its success rate is lower than external or endonasal DCR, around 70%, and patients often require additional surgery if the initial procedure fails.

[Image not available]

Conjunctivodacryocystorhinostomy (CDCR) (Option D): CDCR is specifically used when both the canaliculi and the nasolacrimal duct are blocked. It is a more specialized procedure compared to standard DCR and is not typically the first choice for isolated nasolacrimal duct obstruction.

- CDCR involves creating a passage directly from the conjunctival sac to the nasal cavity, often using a Jones tube and is reserved for more complex cases where the canaliculi are also involved.

Solution for Question 7:

The bluish swelling in the inner corner of the eye, along with excessive tearing and mucous discharge, is characteristic of congenital dacryoceles.

Congenital Dacryocel

A condition where obstruction of the nasolacrimal duct causes amniotic fluid accumulation in the lacrimal sac, leading to a cystic swelling below the inner corner of the eye.

[Image not available]

Aetiology:

- Obstruction: Blockage at the Hasner valve (distal end) and Rosenmuller valve (proximal end) of the nasolacrimal duct.
- Demographics: More common in females; can be unilateral or bilateral.

Clinical Features:

- Bluish or greyish cystic swelling in the medial canthal area, noticeable at birth or shortly thereafter.
- Epiphora: Excessive tearing due to blocked tear drainage.
- Dacryocystitis: Inflammation or infection of the lacrimal sac.
- Conjunctival Discharge: Mucous or purulent discharge from the eye, with possible redness and irritation.

Treatment:

- Conservative Management: Digital Massage: Gentle massage to relieve obstruction. Topical Antibiotics: To prevent or treat infections. Warm Compresses: Applied to promote drainage and comfort.
- Digital Massage: Gentle massage to relieve obstruction.
- Topical Antibiotics: To prevent or treat infections.
- Warm Compresses: Applied to promote drainage and comfort.

- Surgical Intervention: Probing of the Nasolacrimal Duct: Clears obstruction, typically performed under general anaesthesia. Nasal Endoscopy: Checks for and addresses any nasal cysts or obstructions if needed.
- Probing of the Nasolacrimal Duct: Clears obstruction, typically performed under general anaesthesia.
- Nasal Endoscopy: Checks for and addresses any nasal cysts or obstructions if needed.
- Digital Massage: Gentle massage to relieve obstruction.
- Topical Antibiotics: To prevent or treat infections.
- Warm Compresses: Applied to promote drainage and comfort.
- Probing of the Nasolacrimal Duct: Clears obstruction, typically performed under general anaesthesia.
- Nasal Endoscopy: Checks for and addresses any nasal cysts or obstructions if needed.

Dacryoadenitis (Option B): Dacryoadenitis typically presents with a painful, red swelling of the outer third of the upper eyelid, often associated with systemic infections. The described swelling is not painful and is located in the medial canthal area.

Dacryops (Option C): Dacryops presents as a cystic swelling in the upper fornix due to blocked lacrimal ducts and would typically not be described as bluish or associated with mucous discharge in the medial canthal area.

Chronic Dacryocystitis (Option D): Chronic dacryocystitis usually occurs when a blockage leads to recurrent infections of the lacrimal sac over time. It is less likely in a 2-week-old infant presenting for the first time with these symptoms. Additionally, chronic dacryocystitis would typically involve more prolonged symptoms and recurrent infections, which are not indicated in this scenario.

Solution for Question 8:

The presence of a mucocele suggests chronic dacryocystitis where the lacrimal sac has been obstructed for a prolonged period, leading to the accumulation of mucoid fluid.

Milky or gelatinous mucoid fluid expressed from the lower punctum on pressing the swelling (positive Regurgitation test) indicates lacrimal mucocele.

Dacryocystorhinostomy (DCR) is the definitive surgical procedure designed to create a new drainage pathway between the lacrimal sac and the nasal cavity, thus resolving the obstruction

and restoring proper tear drainage. This approach addresses the underlying blockage and prevents recurrence.

Table rendering failed

Chronic Dacryocystitis; Probing and syringing (Option A): Probing and syringing are not typically definitive for a mucocele, which suggests more advanced or longstanding dacryocystitis.

Acute Dacryocystitis; Systemic antibiotics and abscess drainage (Option C): The scenario describes a mucocele, indicative of a chronic issue, not acute dacryocystitis. Management of acute dacryocystitis would involve different treatments.

Acute Dacryocystitis; Dacryocystectomy (DCT) (Option D): DCT is generally reserved for chronic cases or significant fibrosis, not for the management of a mucocele in an acute setting.

Solution for Question 9:

- Canaliculitis is diagnosed based on symptoms and clinical findings.
- In this case, the patient's symptoms include watery eyes, redness, swelling at the corner of the eye, and a thick, yellow discharge.
- On examination, the presence of pouting of the lacrimal puncta and yellow granules expressed from the punctum when massaged are characteristic signs.
- *Actinomyces israelii* is the most common cause of canaliculitis with yellow granules.
- Canaliculotomy is the definitive treatment to remove concretions and resolve the infection.
- It involves accessing and cleaning out the canaliculus to remove concretions and ensure thorough treatment of the infection.

Canaliculitis

Canaliculitis is an inflammation or infection of the lacrimal canaliculus.

Causes:

- Bacterial infection- most commonly *Actinomyces israelii*
- Other bacteria: *Staphylococcus*, *Streptococcus*, *Eikenella*, *Lactococcus*, *Nocardia*
- Fungal infections.
- Viral infections.

Symptoms:

- Eye-watering & discharge.
- Swelling and redness near the corner of the eye
- Pouting or swollen lacrimal punctum
- Mattering on the eyelids
- Expression of yellow granules, concretions, or stones from the punctum when massaged
- Treatment:
 - Initial: Topical and systemic antibiotics, warm compresses, local massage, irrigation and syringing of the tear duct
 - Definitive: Surgery often required for effective treatment Canaliculotomy for direct access and curettage. Safe and effective procedure. Punctal dilation or punctoplasty with curettage Canaliculoplasty with intubation to avoid scarring
 - Canaliculotomy for direct access and curettage. Safe and effective procedure.
 - Punctal dilation or punctoplasty with curettage
 - Canaliculoplasty with intubation to avoid scarring
 - Follow-up with antibiotic treatment and addressing underlying causes, such as a loose punctal plug
 - *Eikenella corrodens*; Oral antibiotics and steroid drops. (Option A): *Eikenella corrodens* is less common than *Actinomyces israelii* for canaliculitis. The treatment listed (oral antibiotics and steroid drops) is not definitive for this condition but rather a part of initial management. *Staphylococcus aureus*; Topical antibiotics and warm compresses. (Option B): *Staphylococcus aureus* can cause canaliculitis, but *Actinomyces israelii* is more common. The treatment mentioned (topical antibiotics and warm compresses) is not sufficient for definitive management but is a part of initial management. *Chlamydia trachomatis*; Irrigation and syringing of the tear duct. (Option D): *Chlamydia trachomatis* is not typically responsible for canaliculitis with yellow granules. Irrigation and syringing alone are part of initial management.

[Image not available]

Treatment:

- Canaliculotomy for direct access and curettage. Safe and effective procedure.
- Punctal dilation or punctoplasty with curettage
- Canaliculoplasty with intubation to avoid scarring

Eikenella corrodens; Oral antibiotics and steroid drops. (Option A): *Eikenella corrodens* is less common than *Actinomyces israelii* for canaliculitis. The treatment listed (oral antibiotics and steroid drops) is not definitive for this condition but rather a part of initial management.

Staphylococcus aureus; Topical antibiotics and warm compresses. (Option B): Staphylococcus aureus can cause canaliculitis, but Actinomyces israelii is more common. The treatment mentioned (topical antibiotics and warm compresses) is not sufficient for definitive management but is a part of initial management.

Chlamydia trachomatis; Irrigation and syringing of the tear duct. (Option D): Chlamydia trachomatis is not typically responsible for canaliculitis with yellow granules. Irrigation and syringing alone are part of initial management.

Solution for Question 10:

Correct answer: A) Pleomorphic Adenoma

Explanation:

Pleomorphic Adenoma:

- Pleomorphic adenoma is the most common tumor of the lacrimal gland.
- It typically presents as a benign, painless, firm, nontender and slow-growing mass.
- Location: Upper eyelid or superotemporal orbit.
- Occurs predominantly in young adult males.

Option C) Lacrimal gland carcinoma:

- Lacrimal gland carcinoma is a rare but aggressive malignancy with high morbidity and mortality.
- The most common type is adenoid cystic carcinoma (50%), followed by pleomorphic adenocarcinoma, mucoepidermoid carcinoma, and squamous cell carcinoma.
- Rapidly growing, painful mass in the superotemporal orbit (months).
- Intraocular globe displacement, proptosis, ophthalmoplegia (if posterior extension).
- Hypoesthesia in the lacrimal nerve distribution (perineural spread).

Diagnosis:

- CT Scan: Heterogeneous lesion with irregular serrated edges, bone invasion, and calcifications.
- Biopsy: Confirms histology.
- Neurological assessment: Necessary due to perineural spread.

Management:

- Wide surgical excision/orbital exenteration if extensive.
- Radiotherapy $\pm$ intra-arterial chemotherapy/brachytherapy for palliation.
- Prognosis: Poor, especially with perineural invasion.

Option B) Lymphoma: Lymphoma can present as a mass in the lacrimal gland but often has associated systemic symptoms and can cause more diffuse swelling rather than a distinct mass.

Option D) Dacryocystitis: It refers to inflammation of the nasolacrimal sac and would typically present with pain, redness, and possibly discharge, along with swelling in the lower eyelid.

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Previous Year Questions

1. A 3-month-old child was brought to the outpatient department (OPD) with symptoms of photophobia and watering of the eyes. Which of the following is not a cause of this condition?

- A. Congenital nasolacrimal duct obstruction
- B. Congenital entropion
- C. Congenital glaucoma
- D. Congenital distichiasis

2. A 3-year-old child presents with epiphora, and probing reveals a soft block at 7 mm. What is the best treatment option for this condition?

- A. Endonasal Dacryocystorhinostomy (DCR)
- B. Conjunctival Dacryocystorhinostomy (DCR)
- C. External Dacryocystorhinostomy (DCR)
- D. Dacryocystectomy

3. Lacrimal gland is supplied through a parasympathetic system. It is supplied by which ganglion?

- A. Ciliary ganglion
- B. Otic ganglion
- C. Pterygopalatine ganglion
- D. Submandibular ganglion

4. In Dacryocystorhinostomy (DCR), the point of opening of the lacrimal sac is into:

- A. Inferior meatus
 - B. Middle meatus
 - C. Supreme meatus
 - D. Superior meatus
-

5. A 55-year-old woman is being counseled for surgery for her long-standing dacryocystitis. Which of the following is the surgery of choice?

- A. Dacryocystorhinostomy
 - B. Dacryocystectomy
 - C. Probing
 - D. Balloon catheter dilation
-

6. A 60-year-old patient presents with chronic dacryocystitis. She complains of recurrent tearing, discharge, and mild swelling in the inner corner of the eye. The clinical findings and history are consistent with chronic dacryocystitis. What should be the treatment of choice for this patient?

- A. Lacrimal sac massage
 - B. Dacryocystorhinostomy
 - C. Syringing
 - D. Probing
-

7. Which bones are cut during/removed during external Dacryocystorhinostomy?

- A. Zygomatic
 - B. Maxillary
 - C. Lacrimal
 - D. Frontal
-

8. Which of the following treatments is appropriate for a 3-month-old child with a mucopurulent discharge from the right puncta and a history of increased watering in the right eye, as noticed by the mother?

- A. Syringing
 - B. Probing with syringing
 - C. Sac massage with topical antibiotics
 - D. Dacryocystorhinostomy
-

9. What is the most appropriate treatment for a 55-year-old woman who is experiencing long-term mucopurulent discharge from her left eye and swelling near the inner corner of the eye?

- A. Dacryocystorhinostomy
 - B. Dacryocystectomy
 - C. Incision and drainage
 - D. Balloon catheter dilation
-

10. Which of the following surgeries use this instrument?

[Image not available]

- A. Evisceration
 - B. Enucleation
 - C. Cataract surgery
 - D. Dacryocystorhinostomy
-

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Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	3
Question 4	2
Question 5	1
Question 6	2
Question 7	3
Question 8	3
Question 9	1
Question 10	4

Solution for Question 1:

Correct Option D - Congenital distichiasis:

- The hallmark of congenital distichiasis is an extra row of eyelashes that typically emerge from the openings of the meibomian glands along the posterior margin of the eyelid.
- It does not necessarily touch the cornea like entropion therefore it's less likely to cause lacrimation and photophobia among the given options

Incorrect Options:

- Options A, B, and C: All of these conditions present with photophobia and watering of the eye.

Solution for Question 2:

- Epiphora (excessive tearing) and a soft block at 7 mm on probing indicate that the level of obstruction is canalicular (in the canaliculus, not the nasolacrimal duct).
- For canalicular obstruction, the preferred treatment is Conjunctival DCR (also known as Lester Jones tube placement). This procedure involves creating a bypass from the conjunctival sac to the nasal cavity using a glass tube (Lester Jones tube) to restore tear drainage.
- Conjunctival DCR is specifically designed for proximal obstructions (e.g., canalicular blocks), whereas Endonasal DCR or External DCR is used for distal obstructions (e.g., nasolacrimal duct blocks).

[Image not available]

Endonasal Dacryocystorhinostomy (Option A): This is used for nasolacrimal duct obstructions, not for canalicular blockages.

External Dacryocystorhinostomy (Option C): This is more suitable for distal nasolacrimal duct obstructions and is not the first choice for canalicular obstructions.

Dacryocystectomy (Option D): This is a last-resort procedure used for chronic infections or tumors and is not indicated for a functional lacrimal system with a canalicular block.

Solution for Question 3:

Correct Option C - Pterygopalatine ganglion:

- The lacrimal gland, responsible for producing tears, is innervated by the parasympathetic nervous system. The postsynaptic parasympathetic fibers that reach the lacrimal gland originate from the Pterygopalatine ganglion.
- The preganglionic parasympathetic fibers synapse in the Pterygopalatine ganglion, and the postganglionic fibers then travel to form zygomatic and zygomaticotemporal nerve and reach the lacrimal gland to stimulate tear production.

Incorrect Options:

Option A - Ciliary ganglion: The ciliary ganglion is associated with the innervation of pupil constrictor and ciliary muscle responsible for controlling the pupillary constriction and lens

accommodation.

Option B - Otic ganglion: The otic ganglion is involved in the innervation of the parotid salivary gland.

Option C - Submandibular ganglion: The submandibular ganglion plays a role in the innervation of the submandibular and sublingual salivary glands.

Solution for Question 4:

Correct Option B - Middle meatus:

- Dacryocystorhinostomy is a surgical procedure used when there's an obstruction beyond the medial opening of the common canaliculus. This procedure involves connecting the lacrimal sac to the mucosa of the middle nasal meatus.
- To perform this procedure, a vertical incision is made in the skin, 10 mm medial to the inner canthus. Then, the medial canthal tendon and lacrimal sac are exposed and reflected. The intervening bone is removed, and the sac is incised. Finally, the sac is attached to an opening created in the nasal mucosa.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 5:

Correct Option A - Dacryocystorhinostomy:

- The treatment of choice for chronic dacryocystitis is dacryocystorhinostomy (DCR) which helps in lacrimal drainage.
- The procedure involves the removal of the bone near the nasolacrimal sac and the creation of an opening of the lacrimal sac into the mucosa of the middle meatus in the nose. This makes way for drainage from the sac to the middle meatus.

Incorrect Options:

Option B - Dacryocystectomy: This procedure involves the removal of the lacrimal sac and is usually reserved for lacrimal sac fibrosis..

Option C - Probing: It is a common treatment approach for congenital nasolacrimal duct obstruction in infants.

Option D - Balloon catheter dilation: This procedure is used in some cases to dilate and open a partially blocked nasolacrimal duct but may not be the primary choice for chronic dacryocystitis.

Solution for Question 6:

Correct Option B - Dacryocystorhinostomy:

- The treatment of choice for chronic dacryocystitis is dacryocystorhinostomy (DCR) which helps in lacrimal drainage.
- The procedure involves the removal of the bone near the nasolacrimal sac and the creation of an opening of the lacrimal sac into the mucosa of the middle meatus in the nose.

Incorrect Options:

Option A - Lacrimal sac massage: Lacrimal sac massage involves gently massaging the area over the lacrimal sac to try and open the blocked nasolacrimal duct in an infant of less than 9 months with congenital dacryocystitis.

Option C - Syringing: Syringing, or irrigation, involves flushing the nasolacrimal system with fluid to clear any obstructions. While it can be useful for diagnosing the location of the obstruction, it is not a definitive treatment for chronic dacryocystitis.

Option D - Probing: Probing is a procedure in which a thin probe is inserted into the nasolacrimal duct to try and open the blockage. It is a common treatment approach for congenital nasolacrimal duct obstruction in infants of age more than 9 months, success rate is till 2 years.

Solution for Question 7:

Correct Option C - Lacrimal:

Dacryocystorhinostomy (DCR)

- It is one of the most common oculoplastics surgery done to create an anastomosis between the lacrimal sac and the nasal mucosa of middle meatus via a bony ostium.
- To create a bony ostium bone is cut at the junction of the lamina papyracea of the ethmoid and lacrimal bone

Incorrect Options:

- Options A, B, and D are incorrect. These bones are not involved in the creation of bony ostium in dacryocystorhinostomy.

Solution for Question 8:

Correct Option C - Sac massage with topical antibiotics:

- In the above case, there is increased watering in his right eye or epiphora and mucopurulent discharge from the punctum. It is suggestive of congenital dacryocystitis which occurs due to congenital blockage of the nasolacrimal duct (membranous occlusion) which leads to stasis of secretions resulting in chronic inflammation. The child is 3 months old hence the treatment of choice here is sac massage with topical antibiotics.

Incorrect Options:

Option A - Syringing: Syringing is performed to assess the passage of nasolacrimal duct and can be performed for adults with early chronic dacryocystitis.

Option B - Probing with syringing: Probing is performed after the age of 6 months till 4 years.

Option D - Dacryocystorhinostomy: Dacryocystorhinostomy is done after the age of 4 years.

Solution for Question 9:

Correct Option A - Dacryocystorhinostomy:

- The clinical presentation of mucopurulent discharge from her left eye along with presence of a swelling, i.e., mucocele at inner canthus is suggestive of chronic dacryocystitis.
- The treatment of choice for chronic dacryocystitis is dacryocystorhinostomy (DCR) which helps in lacrimal drainage. The procedure involves the removal of the bone near the nasolacrimal sac and creating an opening of the lacrimal sac into the mucosa of the middle meatus in the nose. This makes way for drainage from the sac to the middle meatus.

[Image not available]

Incorrect Options:

Option B - Dacryocystectomy is the treatment for lacrimal sac fibrosis.

Option C - Incision & drainage will result in formation of lacrimal fistula.

Option D - Balloon catheter dilation is another form of treatment is intubation and balloon dilatation of NLD but is not the treatment of choice.

Solution for Question 10:

Correct Option D - Dacryocystorhinostomy:

- The instrument shown is Bone Punch, composed of a sturdy spring handle and two blades that are joined at a right angle. The upper blade features a sharp cutting edge and a small hole, while the lower blade has a cup-shaped indentation.
- This tool is utilized in DCR (dacryocystorhinostomy) surgery to expand the bony opening by punching the bone from the edges of the opening.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Orbit

1. A 9-year-old boy presents with a 2-day history of swelling, redness, and pain around his left eye, worsening with eye movements. He had a recent upper respiratory infection. On examination, there is proptosis, restricted eye movement, and chemosis. Vision is intact. CT scan confirms orbital cellulitis. Which staging system is used to classify this condition?

- A. Chandler's classification
- B. Chandler's classification
- C. Mackiewicz classification
- D. Shaffer grading

2. A 45-year-old patient presents with a large, painless orbital mass that has been causing progressive proptosis and visual disturbance. After imaging and biopsy, the mass is identified as malignant. The surgical team is discussing potential treatment options. Which of the following statements regarding surgical procedures for orbital masses is true?

- A. Evisceration is a radical procedure removing all orbital contents including the eyeball and periorbita.
- B. Enucleation is less radical than evisceration
- C. Enucleation involves removal of the entire eyeball including the sclera.
- D. Exenteration is indicated for benign tumors

3. A 12-year-old patient presents to the OPD with a typical S-shaped deformity of the eyelid. On examination, the patient is found to have multiple café-au-lait spots and axillary freckling. After further evaluation, the child is diagnosed with

Neurofibromatosis Type 1 (NF1). Which of the following ocular findings is most common to be observed in this condition?

- A. Early-onset cataracts
 - B. Lisch nodules
 - C. Combined pigment epithelial and retinal hamartomas
 - D. Optic nerve hamartomas
-

4. A 6-year-old child presents with gradual, painless unilateral proptosis and vision loss. Fundoscopic examination reveals optic atrophy. Imaging studies show fusiform growth along the optic nerve. What is the most appropriate management for this child, considering the tumor is stationary and the vision is still present?

- A. Immediate surgical excision of the tumor
 - B. Radiotherapy to manage the tumor
 - C. Observation and regular monitoring
 - D. Chemotherapy as the first-line treatment
-

5. A 52-year-old woman presents with progressive, painless proptosis of the left eye and swelling in the temporal region of the face. On examination, she has boggy eyelid swelling. A CT scan reveals hyperostosis of the sphenoid bone. Based on these findings she is diagnosed with secondary orbital meningioma. Identify the false statement regarding the same.

- A. They arise from or involve the sphenoid bone.
 - B. They are more common in middle-aged women.
 - C. Meningioma en plaque is common
 - D. They are slow-growing tumors from astrocytes in the optic nerve.
-

6. Which of the following is not part of the classical triad of Sturge-Weber syndrome?

- A. Port-wine stain (nevus flammeus) in the distribution of the trigeminal nerve
 - B. Secondary congenital glaucoma
 - C. Bilateral optic nerve glioma
 - D. Ipsilateral meningeal hemangiomatosis
-

7. A 7-year-old male child presents with rapidly progressing proptosis in the right eye, which starts suddenly and mimics an acute inflammatory process. Imaging shows a well-defined orbital mass with adjacent bone destruction. A biopsy confirms the diagnosis of rhabdomyosarcoma. What is the most common histopathological subtype of this tumor?

- A. Alveolar sarcoma
 - B. Embryonal sarcoma
 - C. Botryoid sarcoma
 - D. Spindle cell sarcoma
-

8. A 50-year-old woman presents with painless proptosis of the right eye for the past year. She denies any visual changes, pain, or other ocular symptoms. On examination, there is mild proptosis. MRI reveals a well-defined, intraconal mass with homogeneous enhancement, suggestive of a cavernous hemangioma. What is the most appropriate management for this patient?

- A. Surgical excision
- B. Radiotherapy
- C. Steroid injections

D. Observation and regular follow-up

9. Which among the following statements is/are true regarding orbital varices? Proptosis worsens with the Valsalva maneuver. It is thin-walled distensible high-flow veins. Venous stasis can lead to the formation of phlebolith. Endovascular embolization is contraindicated.

- A. 1 and 3
 - B. 1, 3, and 4
 - C. Only 2
 - D. All of the above
-

10. A 35-year-old man presents with a progressive painless swelling in the superolateral part of his left orbit over the past year. Examination reveals mild proptosis and downward and inward displacement of the eye. CT of the orbit shows a well-circumscribed, smooth, encapsulated mass in the lacrimal gland region. Excisional biopsy is shown below. Which of the following is the most likely diagnosis?

[Image not available]

- A. Adenoid cystic carcinoma
 - B. Squamous cell carcinoma
 - C. Pleomorphic adenoma
 - D. Lymphoma
-

11. Which is the most common non-epithelial malignant lacrimal gland tumor?

- A. Pleomorphic adenoma-carcinoma

- B. Adenoid cystic carcinoma
 - C. Lymphoma
 - D. Mucoepidermoid carcinoma
-

12. A 45-year-old male presents with double vision, drooping of the right eyelid, and difficulty moving his right eye in multiple directions. On examination, the patient has normal visual acuity, ptosis, limited eye movement in all directions, decreased sensation over the forehead, and absent corneal reflex on the same side. What is the most likely diagnosis and which cranial nerve is/are most likely affected in this condition?

- A. Superior orbital fissure syndrome; III, IV, V, and VI nerves
 - B. Orbital apex syndrome; optic nerve
 - C. Tolosa Hunt syndrome; II, III and IV nerves
 - D. Rothen-Duvigneaud syndrome; III and VI nerves
-

13. A 45-year-old diabetic male presents with complaints of headache, nasal congestion, and facial pain for the past week. On examination, there is periorbital swelling, proptosis, and black eschar on the nasal mucosa. His blood sugar levels are poorly controlled. CT scan of the paranasal sinuses reveals extensive sinus involvement, and biopsy confirms the diagnosis of rhino-orbital-cerebral mucormycosis (ROCM). Which of the following is NOT a known complication of rhino-orbital-cerebral mucormycosis?

- A. Meningitis
 - B. Brain abscess
 - C. Retinal vascular occlusion
 - D. Subarachnoid hemorrhage
-

14. A 50-year-old male, recently recovered from COVID-19, presents with complaints of severe facial pain, nasal congestion, and black eschar on the right eye with periorbital pain. He has a history of uncontrolled diabetes mellitus. On examination the following is seen. Which of the following is the most likely causative organism?

[Image not available]

- A. Aspergillus species
- B. Rhizopus species
- C. Candida species
- D. Cryptococcus neoformans

15. A 32-year-old male had blunt trauma to the face during a sports event. He reports difficulty moving his right eye upward and experiences double vision when looking up. On examination, there is tenderness at the right infraorbital rim, and the patient exhibits noticeable swelling and bruising around the right eye. Which wall of the orbit is most likely affected in this case?

- A. Medial wall
- B. Lateral wall
- C. Floor
- D. Roof

16. A 10-year-old girl presents with a swollen, painful right eye that started after a recent cold. She is febrile and on examination, her right eyelid is erythematous and swollen, with limited ocular movement but normal visual acuity. What is the most common primary source of this eye infection in the child?

- A. Ethmoid sinusitis

- B. Maxillary sinusitis
 - C. Frontal sinusitis
 - D. Otitis media
-

17. A 25-year-old male presents with intermittent bulging of his right eye, worsening with cough and bending forward. His vision is intact, and there is no history of trauma or infection. What is the most likely cause of his condition?

- A. Cavernous sinus thrombosis
 - B. Orbital varix
 - C. Thyroid orbitopathy
 - D. Orbital cellulitis
-

18. A 48-year-old man presents with a complaint of eye protrusion. He describes experiencing a rhythmic, pulsatile sensation in his right eye for the past few weeks. Which of the following is NOT a likely cause ?

- A. Carotico-cavernous fistula
 - B. Saccular aneurysm of the ophthalmic artery
 - C. Congenital meningocele
 - D. Orbital cellulitis
-

19. If a patient with thyroid eye disease exhibits stable symptoms with no acute inflammation and has experienced a gradual improvement in symptoms over the past 1-2 years, which stage of Rundle's Curve is most likely being described?

- A. Initial Phase
- B. Peak Phase

C. Improvement Phase

D. Chronic Phase

20. A 48-year-old woman with Graves' disease presents with worsening proptosis, diplopia, diminution of vision and difficulty closing her eyes despite treatment with antithyroid medications and corticosteroids. On examination afferent pupillary defects are observed. CT scan shows orbital congestion and muscle enlargement. What is the most appropriate next step in management?

A. Adalimumab.

B. Systemic steroids

C. Radiotherapy.

D. Decompression Surgery.

21. A 52-year-old woman with a history of thyroid disease presents with difficulty closing her eyes, noticeable eye protrusion and prominent bilateral lid retraction giving a staring and frightened appearance of the eyes as shown in the image below. Which of the following best describes the sign observed in this patient?

[Image not available]

A. Von Graefe's Sign

B. Kocher's sign

C. Stellwag's Sign

D. Joffroy's Sign

22. A 45-year-old woman with a history of thyroid disease presents with bilateral eye bulging and difficulty closing her eyes. On examination, there is noticeable eye protrusion, eyelid swelling, and restricted extraocular movements. Which of the

following best describes the pathogenesis of her eye condition?

- A. Extraocular muscles are the primary target of inflammation.
 - B. The earliest defective ocular movement seen in this patient is depression.
 - C. T-cells attack the orbital fibroblasts and adipocytes in the orbit.
 - D. Inflammation is primarily due to neutrophils.
-

23. Which of the following is not a cause of pseudoproptosis?

- A. High Myopia
 - B. Eyelid Retraction
 - C. Contralateral Enophthalmos
 - D. Orbital Cellulitis
-

24. A 34-year-old woman with a history of hyperthyroidism presents with bilateral proptosis and difficulty closing her eyes. To assess the degree of eye protrusion, a Hertel's exophthalmometer is used. Which of the following statements regarding the use of the exophthalmometer is correct?

- A. Normal values for exophthalmometer readings are typically between 20 and 30 mm.
 - B. Difference of 2–3 mm or more between the two eyes is suspicious regardless of the absolute values
 - C. A reading greater than 10 mm in both eyes is indicative of proptosis.
 - D. Hertel's exophthalmometer can only be used to measure the degree of proptosis in one eye at a time.
-

25. Which condition is most commonly associated with bilateral proptosis in adults?

- A. Orbital cellulitis
 - B. Thyroid ophthalmopathy
 - C. Rhabdomyosarcoma
 - D. Neuroblastoma
-

26. Which surgical space is typically used for administering a retrobulbar injection?

- A. Sub-Tenon Surgical Space
 - B. Extraconal Surgical Space
 - C. Intraconal Surgical Space
 - D. Subarachnoid Surgical Space
-

27. Which of the following statements is TRUE regarding the orbital foramina and fissures?

- A. Superior orbital fissure is between the medial and lateral walls.
 - B. Inferior orbital fissure is between the lesser wing of the sphenoid and the maxillary bone.
 - C. Optic foramen is between the lateral wall and the orbital floor.
 - D. Optic nerve and ophthalmic artery pass through the optic foramen.
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	2
Question 4	3
Question 5	4
Question 6	3
Question 7	2
Question 8	4
Question 9	1
Question 10	3
Question 11	3
Question 12	1
Question 13	4
Question 14	2
Question 15	3
Question 16	1
Question 17	2
Question 18	4
Question 19	4
Question 20	4
Question 21	2
Question 22	3

Question 23	4
Question 24	2
Question 25	2
Question 26	3
Question 27	4

Solution for Question 1:

Orbital cellulitis	
Definition	Orbital cellulitis is an acute infection of the soft tissues of the orbit located behind the orbital septum.
Complications	Disc edema RAPD (Relative Afferent Pupillary Defect) due to optic nerve dysfunction Cavernous sinus thrombosis CRAO (Central Retinal Artery Occlusion) or CRVO (Central Retinal Venous Occlusion) Generalized septicemia or pyemia
Investigations	White cell count. Cultures: Blood, nasal discharge. X-ray PNS (sinusitis) High-resolution CT of the orbit, sinuses and brain (exclude a subperiosteal or intracranial abscess). MRI Lumbar puncture (if meningeal or cerebral signs develop).
Staging	[Image not available] Modified Chandler staging (Option A) system: Stage I: Preseptal cellulitis/ periorbital cellulitis (limited anterior to orbital septum) Stage II: Inflammatory orbital oedema/ Orbital cellulitis (Posterior to orbital septum) Stage III: Subperiosteal abscess Stage IV: Orbital abscess Stage V: Cavernous sinus thrombosis

Treatment	Antibiotics (IV) Empirical therapy treats orbital cellulitis by giving Cefazidime, Vancomycin, and Metronidazole. Monitoring of optic nerve function Surgery Drainage of an orbital abscess or infected sinuses should be considered if there is a lack of response to antibiotics. Biopsy Severe optic nerve compression may warrant an emergency canthotomy/ cantholysis
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Beck's classification (Option B) is used for optic nerve gliomas, assessing the severity of visual loss and optic nerve involvement.

Mackiewicz classification (Option C) is used in thyroid eye disease (Graves' orbitopathy) to assess severity and disease progression, not orbital cellulitis.

Shaffer grading (Option D) is used to describe anterior chamber angle status in glaucoma, categorizing the risk of angle closure.

Solution for Question 2:

Debulking is a less radical procedure than exenteration and is often preferred when treating orbital masses, as it aims to reduce the size of the mass and alleviate symptoms while preserving as much of the orbital structure as possible.

Surgical Procedures for Orbital Masses: Definitions and Differences

Exenteration

- It involves removal of the globe together with the soft tissues of the orbit (eyeball and periorbita).
- Incision made along orbital margins. Indicated for malignant tumors or extensive disease. (Option D ruled out)
- Debulking of the orbit is now often preferred over exenteration.

Evisceration

- Removal of eyeball contents while preserving the scleral shell and extraocular muscles. (Option A ruled out)
- Less radical than enucleation; not typically used for malignancy. (Option B ruled out)

Enucleation

- Removal of the entire eyeball (globe), indicated in: Primary intraocular malignancies After severe trauma where the risk of sympathetic ophthalmitis may outweigh any prospect of visual recovery Blind painful or unsightly eyes
- Primary intraocular malignancies
- After severe trauma where the risk of sympathetic ophthalmitis may outweigh any prospect of visual recovery
- Blind painful or unsightly eyes
- Contraindication: Panophthalmitis (due to the risk of infection spreading to the brain).
- Primary intraocular malignancies
- After severe trauma where the risk of sympathetic ophthalmitis may outweigh any prospect of visual recovery
- Blind painful or unsightly eyes

Key Differences

Procedure	Extent of Removal	Indications	Cosmetic and Functional Outcomes
Exenteration	All orbital contents (Eyeball + EOM + eyelids)	Malignant tumors, extensive disease	Significant cosmetic and functional deficits
Evisceration	Contents of the eyeball, sclera preserved	Not typically used for malignancy	Preserves some eye movement
Enucleation	Entire eyeball including sclera	Malignant tumours, less radical than exenteration	Allows for ocular prosthesis placement

Solution for Question 3:

Lisch nodules are iris hamartomas and a well-established diagnostic criteria for NF1.

Ocular Manifestations in Neurofibromatosis

- Lisch Nodules: Iris hamartomas, key diagnostic feature. (Option D)
- Glaucoma: Common, especially with orbital-facial involvement; may lead to asymmetric globe enlargement.
- Optic Nerve Glioma: Can cause strabismus and proptosis.

- Palpebral Neurofibroma: S-shaped deformity of the lid
- Choroidal Alterations: Detected via near-infrared reflectance imaging; potential diagnostic criteria.
- Early-Onset Cataracts: Present in up to 80% of cases. (Option A ruled out)
- Optic Nerve Hamartomas: Found in some cases. (Option D ruled out)
- Combined Pigment Epithelial and Retinal Hamartomas: Also observed. (Option C ruled out)

Solution for Question 4:

Observation is appropriate for stationary tumors with good vision and non-disfiguring proptosis, allowing for regular monitoring of the tumor's progress.

Optic Nerve Gliomas

- A slow-growing, tumor of astrocytes (Option D)
- It is generally a unilateral tumor that typically affects children (more common in females).
- More common in patients with neurofibromatosis type 1 (NF1).

Clinical features:

- Presents with gradual diminution of vision and increasing axial proptosis.
- RAPD (relative afferent pupillary defect) and optic atrophy, disc oedema.
- Gradual, painless, unilateral proptosis.
- Fundus findings: optic atrophy, papilledema, or venous engorgement.

Investigations:

- X-rays: Rounded enlargement of the optic foramen
- CT scan & MRI: Fusiform growth along the optic nerve.

Treatment:

- Observation to preserve the nerve as long as possible.
- Radiotherapy: For inoperable cases. (Option B ruled out)
- Chemotherapy: Used in specific cases when other treatments are not suitable. (Option D ruled out)
- Surgery (cosmetically unacceptable proptosis in a blind eye). (Option A ruled out)

Solution for Question 5:

Orbital meningiomas are not tumors derived from astrocytes in the optic nerve. Instead, they originate from the meninges, the protective coverings of the brain and spinal cord. Astrocytes are the cells from which optic nerve gliomas arise, not meningiomas.

Optic Nerve Tumor Overview

- A tumor from the meningo-epithelial cells from the arachnoid layer.
- More common in middle-aged women. (Option B ruled out)

Clinical features:

- The Hyot-Spencer triad includes progressive vision loss, optic atrophy, and opto-ciliary shunt vessels.
- Proptosis late features
- Restricted-up gaze (involvement of superior rectus).
- Gaze evoked amaurosis fugax (transient loss of vision).
- Temporal fossa fullness due to the hyperostosis of the sphenoid bone.
- Types: Optic Nerve Gliomas (ectodermal tumors of the nerve) Meningiomas (mesodermal tumors of the sheath of the nerve).
- Optic Nerve Gliomas (ectodermal tumors of the nerve)
- Meningiomas (mesodermal tumors of the sheath of the nerve).
- Optic Nerve Gliomas (ectodermal tumors of the nerve)
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Meningioma

- Arise from or involve the sphenoid bone) (Option A ruled out).
- Mostly spread extensively along the dura in forms of sheets rather than a globular mass (meningioma en plaque) (Option C ruled out)
- Predominant feature is early visual loss.
- Patients present with swollen or atrophic optic discs, restriction of eye movements.
- Opticociliary shunt vessels are present
- Biopsy: or any surgery which transgresses the dura is to be avoided unless the rate of growth suggests a malignant type of meningioma, when biopsy is indicated.

- High-resolution CT scans show a small anterior tumour and useful vision is retained, the orbit may be explored and the lesion removed without endangering vision.

Solution for Question 6:

Sturge–Weber syndrome (SWS) is a dermato-oculo-neural syndrome characterized by:

- cutaneous facial nevus flammeus in the distribution of the branches of the trigeminal nerve
- Secondary congenital glaucoma
- ipsilateral diffuse cavernous hemangioma of the choroid
- ipsilateral meningeal hemangiomas

The lesions in the eye, skin, and brain are always present at birth.

- Cutaneous feature: Facial nevus flammeus (port-wine stain): Unilateral lesion, typically on the face, often involving the first (V1) branch of the trigeminal nerve, occasionally extending to the second (V2) and rarely the third (V3) branches. May also affect ipsilateral nasal and buccal mucosa. Localized hypertrophy of affected tissues can be present.
- Unilateral lesion, typically on the face, often involving the first (V1) branch of the trigeminal nerve, occasionally extending to the second (V2) and rarely the third (V3) branches.
- May also affect ipsilateral nasal and buccal mucosa.
- Localized hypertrophy of affected tissues can be present.
- Unilateral lesion, typically on the face, often involving the first (V1) branch of the trigeminal nerve, occasionally extending to the second (V2) and rarely the third (V3) branches.
- May also affect ipsilateral nasal and buccal mucosa.
- Localized hypertrophy of affected tissues can be present.
- CNS feature: Ipsilateral leptomeningeal hemangiomas: Causes atrophy of cortical brain tissue, seizures, and mental retardation. Lesions are progressive and may become calcified over time, visible on skull radiographs.
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- Lesions are progressive and may become calcified over time, visible on skull radiographs.
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- Lesions are progressive and may become calcified over time, visible on skull radiographs.

- Choroidal hemangioma: Diffuse thickening of the choroid with mature blood vessels. Appears as saturated red color on ophthalmoscopy (better detected by comparing both eyes).
- Diffuse thickening of the choroid with mature blood vessels.
- Appears as saturated red color on ophthalmoscopy (better detected by comparing both eyes).
- Secondary glaucoma.
- Additional ocular findings: Telangiectasia of the conjunctiva and episclera. Congenital/infantile/juvenile glaucoma in some cases. Secondary serous retinal detachment may develop, leading to profound visual loss if untreated.
- Telangiectasia of the conjunctiva and episclera.
- Congenital/infantile/juvenile glaucoma in some cases.
- Secondary serous retinal detachment may develop, leading to profound visual loss if untreated.
- Treatment:
- Seizures are treated medically. Intractable seizures and progressive mental deterioration are sometimes treated surgically by techniques such as subtotal hemispherectomy.
- The facial nevus flammeus can be treated by dermatological laser therapy.
- Treatment of the ophthalmic lesions and complications of SWS include medical and surgical treatments for glaucoma and photodynamic therapy or low-dose external beam or plaque radiotherapy for non-rhegmatogenous retinal detachment secondary to the hemangioma.
- Unfortunately, the visual prognosis for affected eyes is usually quite poor.
- Bilateral optic nerve glioma (Option C) is a finding specific to Neurofibromatosis Type 1, not Sturge-Weber syndrome.
- Diffuse thickening of the choroid with mature blood vessels.
- Appears as saturated red color on ophthalmoscopy (better detected by comparing both eyes).
- Telangiectasia of the conjunctiva and episclera.
- Congenital/infantile/juvenile glaucoma in some cases.
- Secondary serous retinal detachment may develop, leading to profound visual loss if untreated.

Treatment:

Bilateral optic nerve glioma (Option C) is a finding specific to Neurofibromatosis Type 1, not Sturge-Weber syndrome.

Solution for Question 7:

Embryonal sarcoma is the most common subtype of rhabdomyosarcoma, accounting for 60% of cases, particularly in children.

Rhabdomyosarcoma Overview

- Most common soft tissue sarcoma of childhood (90% of cases are seen in children under 16 year of age).
- It is the most common primary orbital malignancy in children
- Derived from undifferentiated mesenchymal cells, it is a connective tissue tumor with the potential to differentiate into striated muscle cells.

Diagnosis Methods

- X-rays show bone destruction
- Definitive diagnosis: Biopsy.

Histopathological Types

Type	Prevalence (%)
Embryonal (Option B)	60%
Alveolar (Option A ruled out)	31%
Botryoid (Option C ruled out)	6%
Spindle cells (Option D ruled out)	3%
Anaplastic	Rare

Treatment Options

- Chemotherapy (Vincristine and actinomycin D)
- A combination of radiotherapy, chemotherapy and sometimes surgical debulking.

[Image not available]

[Image not available]

Solution for Question 8:

When orbital cavernous hemangioma is small, asymptomatic, and does not affect vision or cause significant cosmetic concerns, observation with regular follow-up is the preferred management.

Cavernous hemangioma of orbit:

- It is a low-flow, non-distensible orbital venous malformation
- The primary location is intraconal space
- Common in middle-aged females

Clinical manifestation:

- Variable size, smooth contours, and noninvasive
- Gradual-onset axial proptosis, blurred vision, pain, or diplopia
- Compressive optic neuropathy

[Image not available]

Diagnosis:

- USG- Well-circumscribed, round-to-oval lesion with high internal reflectivity, a regular acoustic structure with poor vascular flow
- Contrast-enhanced CT
- Valsalva-augmented MRI

Treatment: (Option D)

- Asymptomatic cases: Observation and periodic follow-up
- Indications for surgical excision are: (Option A ruled out) Symptomatic CVMs Visual loss due to optic nerve compression Disfiguring proptosis Dystopia
- Symptomatic CVMs
- Visual loss due to optic nerve compression
- Disfiguring proptosis
- Dystopia
- Other treatment options are stereotactic radiotherapy, pre-operative embolization percutaneous sclerotherapy, and intralesional injection of pingyangmycin (bleomycin A5) (Options B & C ruled out)
- Symptomatic CVMs
- Visual loss due to optic nerve compression

- Disfiguring proptosis
- Dystopia

Solution for Question 9:

Correct statements are:

3. Venous stasis can lead to the formation of phlebolith.

Orbital varices:

It is a plexus of thin-walled dilated low-flow vein (Statement 2 ruled out)

Clinical manifestation:

- Soft-tissue compressible swelling with a 'bag of worms'
- Proptosis, diplopia, visual disturbances, or ecchymosis
- Distension on dependent position, coughing or Valsalva maneuver (Statement 1)

Diagnosis:

- Dynamic investigations, such as Valsalva-augmented MRI/CT
- Venous stasis can lead to the formation of a phlebolith, which is visualized on CT or plain X-ray (Statement 3)

Treatment:

- Endovascular embolization (Statement 4 ruled out)
- CO2 laser ablation
- Surgical excision is very difficult due to the friable nature with the tendency to bleed

Solution for Question 10:

Pleomorphic Lacrimal Gland Adenoma: (Option C)

- It is a benign mixed tumor

- Most common benign lacrimal gland tumor
- Incidence- young adult males

Clinical manifestation:

- Slow-growing painless swelling in the upper-outer quadrant of the orbit
- Displaces the eyeball downwards and inwards

Diagnosis:

- Imaging: CT- well-defined mass in superotemporal quadrant
- CT- well-defined mass in superotemporal quadrant
- Histology: The inner layer of cells forms glandular tissue associated with squamous differentiation and keratin production. The outer cells undergo metaplastic change leading to the formation of myxoid tissue Treatment: Adenoid cystic carcinoma (Option A), Squamous cell carcinoma (Option B), and Lymphoma (Option D): All of them are lacrimal gland malignancies typically present as a painful, rapidly growing mass in the superotemporal quadrant of the orbit, causing displacement of the eye in an inferonasal direction. CT scans often reveal an infiltrative tumor. Complete excision with the capsule
- The inner layer of cells forms glandular tissue associated with squamous differentiation and keratin production.
- The outer cells undergo metaplastic change leading to the formation of myxoid tissue Treatment:
- Adenoid cystic carcinoma (Option A), Squamous cell carcinoma (Option B), and Lymphoma (Option D): All of them are lacrimal gland malignancies typically present as a painful, rapidly growing mass in the superotemporal quadrant of the orbit, causing displacement of the eye in an inferonasal direction. CT scans often reveal an infiltrative tumor. Complete excision with the capsule
- CT- well-defined mass in superotemporal quadrant
- The inner layer of cells forms glandular tissue associated with squamous differentiation and keratin production.
- The outer cells undergo metaplastic change leading to the formation of myxoid tissue Treatment:
- Adenoid cystic carcinoma (Option A), Squamous cell carcinoma (Option B), and Lymphoma (Option D): All of them are lacrimal gland malignancies typically present as a painful, rapidly growing mass in the superotemporal quadrant of the orbit, causing displacement of the eye in an inferonasal direction. CT scans often reveal an infiltrative tumor. Complete excision with the capsule

[Image not available]

Treatment:

[Image not available]

Adenoid cystic carcinoma (Option A), Squamous cell carcinoma (Option B), and Lymphoma (Option D): All of them are lacrimal gland malignancies typically present as a painful, rapidly growing mass in the superotemporal quadrant of the orbit, causing displacement of the eye in an inferonasal direction. CT scans often reveal an infiltrative tumor.

Solution for Question 11:

Lymphoma is the most common non-epithelial malignant lacrimal gland tumor.

[Image not available]

Lacrimal gland tumors:

- Swelling of the gland may cause limitations in eye movements.
- Inferonasal dystopia (globe is pushed downwards and inwards)
- Proptosis

[Image not available]

Most common lacrimal gland tumor	Pleomorphic adenoma
Most common malignant lacrimal gland tumor	Adenoid cystic carcinoma (Option B ruled out)
Most common non-epithelial malignant lacrimal gland tumor	Lymphoma (Option C)

Pleomorphic adenoma-carcinoma (Option A): It is an epithelial malignant tumor.

Mucoepidermoid carcinoma (Option D): It is an epithelial malignant tumor and accounts for only 5% of the malignant epithelial lacrimal gland tumors.

Solution for Question 12:

Superior orbital fissure syndrome typically involves cranial nerves III (oculomotor), IV (trochlear), V1 (ophthalmic branch of the trigeminal nerve), and VI (abducens), which pass through the

superior orbital fissure. Clinical manifestation of ptosis (III), ophthalmoplegia (III, IV, VI), sensory loss over the forehead, and absent corneal reflex (V1) points towards the diagnosis of this syndrome.

Superior orbital fissure syndrome:

- Also known as the Rochon-Duvigneaud syndrome. (Option D ruled out)
- It is caused by compression of structures anterior to the orbital apex.

Etiology:

- Most common cause is trauma
- Idiopathic
- Neoplasias
- Meningitis, syphilis, sinusitis, herpes zoster
- Inflammation - Lupus, sarcoidosis, Tolosa-Hunt syndrome (Episodes of unilateral orbital pain, cranial nerve involved and granulomatous inflammation)
- Vascular phenomena - carotid-cavernous fistulas, retro-orbital hematoma, and carotid aneurysms

Pathophysiology: Involvement of the structures passing through the superior orbital fissure (III, IV, VI, and the ophthalmic division of the trigeminal nerve). (Option A)

Clinical features:

- Pain (retro-orbital ache)
- Sensory loss in the distribution of ophthalmic division of the 5th cranial nerve
Absent corneal reflex
Decreased sensation over the forehead
- Absent corneal reflex
- Decreased sensation over the forehead
- Ipsilateral ophthalmoplegia due to 3rd, 4th, and 6th cranial nerve involvement
- Ptosis due to 3rd nerve paralysis
- Absent corneal reflex
- Decreased sensation over the forehead

Investigations:

- CT, MRI
- Angiography- vascular causes of the syndrome

Orbital apex syndrome; optic nerve (Option B):

- It is a symptom complex due to the involvement of structures present at the apex of the orbit (structures passing through the superior orbital fissure and the optic canal)

- So the cranial nerves involved are II, III, IV, V, and VI
- Optic nerve (II) involvement causes visual loss and afferent pupillary defect

Tolosa Hunt syndrome; II, III, and IV nerves (Option C):

- It is idiopathic nonspecific granulomatous inflammation involving the cavernous sinus and orbital apex causing compression of surrounding nerves.
- Commonly manifests as painful ophthalmoplegia and presents as superior orbital fissure syndrome or orbital apex syndrome depending upon the extent of involvement
- Clinical manifestations in this scenario suggest there is no involvement of the optic nerve (II)

Solution for Question 13:

Mucormycosis commonly causes meningitis, brain abscess, retinal vascular occlusion, and multiple cranial nerve palsies due to its aggressive spread. However, subarachnoid hemorrhage is not a typical complication of ROCM.

Rhino Orbital Cerebral Mucormycosis	
Etiology	Fungal infection (Mucoraceae, primarily Mucor and Rhizopus species) in immunocompromised individuals.
Clinical Features	Facial and periorbital swelling, pain, proptosis, diplopia, fever. (similar to orbital cellulitis) Tissue necrosis (tissue ischemia).
Diagnosis	Diagnosis is clinical, Biopsy: Non-septate broad branching hyphae (Option C) CT or MRI may show mild findings despite severe infection.
Treatment	Correction of the underlying metabolic defect. Antifungal (intravenous) Daily packing and irrigation of the involved areas with antifungal agent. Wide excision of devitalized and necrotic tissues. Exenteration may be required in unresponsive cases in order to reduce the risk of death. Adjunctive hyperbaric oxygen may be helpful.
Prognosis	The prognosis is poor due to rapid progression to cerebral involvement. Recurrences are common even with treatment.

Complications	If untreated, mucormycosis can rapidly progress to Meningitis (Option A ruled out) Brain abscess (Option B ruled out) Retinal vascular occlusion (Option C ruled out) Multiple cranial nerve palsies Death
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Solution for Question 14:

This scenario describes Rhino-Orbital-Cerebral Mucormycosis, commonly caused by Rhizopus, particularly in immunocompromised patients with uncontrolled diabetes.

Rhino Orbital Cerebral Mucormycosis	
Etiology	Fungal infection (Mucoraceae, primarily Mucor and Rhizopus species). (Option B)
Risk factors	Immunocompromised individuals
Clinical Features	Facial and periorbital swelling, pain, proptosis, diplopia, fever. (similar to orbital cellulitis) Tissue necrosis (tissue ischemia)

Aspergillus species, Candida species, Cryptococcus neoformans (Options A, C and D ruled out) do not cause Mucormycosis.

Solution for Question 15:

- In the presented case, the patient exhibits difficulty moving the right eye upward (limitation of upward gaze) and diplopia (double vision) when looking up. These symptoms are characteristic of a blowout fracture.

Medial Wall (Thinnest):

- Frontal process of maxilla (anterior-most)
- Lacrimal bone

- Orbital plate of ethmoid (posterior to lacrimal process)
- Body of sphenoid (posterior-most)

Lateral Wall (Strongest): (Option B ruled out)

- Orbital surface of zygomatic bone (anterior part)
- Orbital surface of greater wing of sphenoid (posterior part)

Floor:

- Orbital surface of body of maxilla (medial part)
- Orbital surface of zygomatic bone (anterolateral part)
- Orbital process of the palatine bone (posteromedial part)

Roof:

- Orbital plate of the frontal bone (anterior part)
- Lesser wing of the sphenoid (posterior part)

[Image not available]

Key features:

- Weakest Wall of the Orbit: Medial wall is the thinnest due to the cribriform plate of ethmoid.(Option A)
- Medial wall is the thinnest due to the cribriform plate of ethmoid.(Option A)
- Blowout Fracture: Fracture of floor of the orbit due to blunt trauma. (Option C)
- Weakest Part of the Floor: Posteromedial Part of the orbit.
- Posteromedial Part of the orbit.
- Medial wall is the thinnest due to the cribriform plate of ethmoid.(Option A)
- Posteromedial Part of the orbit.

Roof (Option D): Symptoms and examination findings are more suggestive of orbital floor fracture due to blunt trauma in this scenario.

Solution for Question 16:

In children, ethmoid sinusitis is the most common cause of orbital cellulitis. Cribriform plate of ethmoid sinus is part of the medial wall.

Orbital cellulitis	
Definition	[Image not available] Orbital cellulitis is an acute infection of the soft tissues of the orbit located behind the orbital septum.
Etiology	Sinusitis: Ethmoid sinusitis (most common) carries a high risk of causing orbital cellulitis. (Option A) Underlying dental problem. Dacryocystitis (inflammation of the lacrimal sac). Endogenous cause: Septicemia Exogenous cause: Fracture or surgery of the orbit.
Causative Organisms	Staphylococcus aureus (most common) Haemophilus influenzae (most common in children) Streptococcus pneumoniae Streptococcus pyogenes
Clinical Features	Painful, unilateral Proptosis (swelling and severe pain worsened by eye movement or pressure) Restriction of eye movements (due to proptosis) Decreased visual acuity Lid edema conjunctival congestion / chemosis Systemic symptoms: like fever, headache, nausea
Complications	Disc edema RAPD (Relative Afferent Pupillary Defect) due to optic nerve dysfunction Cavernous sinus thrombosis CRAO (Central Retinal Artery Occlusion) or CRVO (Central Retinal Venous Occlusion) Generalized septicemia or pyemia.

[Image not available]

Preseptal Cellulitis

Solution for Question 17:

The intermittent proptosis triggered by cough and bending forward, along with the absence of pain or inflammation, suggests a vascular cause like orbital varix.

Causes of intermittent proptosis:

- Orbital varix (Option B)
- Periodic orbital oedema

- Recurrent orbital hemorrhage
- Highly vascular tumors

Orbital Varices	
Dilated venous channels in the orbit (part of the spectrum of developmental venous anomalies)	
Clinical features	Most cases are unilateral and the most frequent site is upper nasal. Slowly progressive intermittent proptosis which worsens with head position changes, bending forward, or Valsalva maneuver.
Pathology	May involve one large vessel or multiple ectatic veins with fibrotic walls. Thrombosed lumens can contain phleboliths from calcified thrombi.
Imaging	CT - serpiginous, dense structures that enlarge with Valsalva, enhances with contrast administration. MRI - T1 images show low isointense signal, T2 shows dark signal void from flowing blood. Enhancement with Gadolinium is marked, except in areas of thrombosis.
Treatment	No treatment needed for mild symptoms; Surgical or embolization options for severe cases.
Prognosis	Good unless large lesions compress the optic nerve, risking vision loss.

Cavernous sinus thrombosis (Option A) typically presents with severe pain, ophthalmoplegia, and rapid onset symptoms, unlike this slow intermittent presentation.

Thyroid orbitopathy (Option C) usually causes bilateral, progressive proptosis, not intermittent bulging.

Orbital cellulitis (Option D) presents with acute onset, pain, fever, and signs of infection, which are absent here.

Solution for Question 18:

- This presentation suggests pulsatile proptosis. Pulsatile proptosis, also known as pulsatile exophthalmos, is a condition that causes the eyeball to protrude and pulsate.

- Orbital cellulitis typically causes proptosis due to inflammation and swelling within the orbit. Orbital cellulitis often presents with pain, redness, swelling, and fever, without the rhythmic pulsation characteristic of pulsatile proptosis.

Causes of Pulsatile Proptosis

- Carotico-cavernous Fistula: Abnormal connection between the carotid artery and the cavernous sinus (Option A ruled out).
- Saccular Aneurysm of the Ophthalmic Artery: Aneurysm located on the ophthalmic artery causing pulsatile proptosis (Option B ruled out).
- Congenital Meningocele or Meningoencephalocele: Protrusion of the meninges or brain through a defect in the orbital roof (Option C ruled out).
- Neurofibromatosis: Genetic disorder causing tumors on nerve tissue, which can affect orbital structures.
- Traumatic or Operative Hiatus: Defects in the orbital roof resulting from trauma or surgery, allowing transmitted cerebral pulsations

Solution for Question 19:

- Since the patient's disease has stabilized and no longer exhibits significant progression or acute inflammation, they are in the Chronic or "Burnt-Out" Stage, where further changes are unlikely.
- In the chronic or "burnt-out" stage of thyroid eye disease, the condition has stabilized after reaching its peak severity. This stage is characterized by stable symptoms that have ceased worsening, the absence of acute inflammation indicating the end of the most active phase, and a gradual improvement in symptoms over the preceding months to years, reflecting the natural progression towards stabilization.

Rundle's Curve and Thyroid Eye Disease:

Phase	Details
Initial Phase (Option A ruled out)	
Duration	A few months.
Characteristics	Progressive worsening of symptoms.

Peak Phase (Option B ruled out)	
Characteristics	The disease reaches its maximum severity.
Improvement Phase (Option C ruled out)	
Duration	1-2 years.
Characteristics	Symptoms start to improve spontaneously, transitioning from acute to chronic.
Chronic or "Burnt-Out" Stage (Option D)	
Characteristics	Disease stabilizes; further changes are unlikely.
Treatment Focus	Typically requires surgical intervention.
Active Disease	
Features	Acute inflammation.
Timing	Early phase of Rundle's curve.
Treatment	Potentially responsive to medical treatments.
Inactive Disease	
Features	Stabilized disease with no acute inflammation.
Treatment	Requires surgical options for altering outcomes.
Significance of Rundle's Curve	Defines the natural progression of thyroid eye disease and guides treatment choices based on disease phase and severity.

[Image not available]

Solution for Question 20:

- This case is of severe thyroid eye disease. Despite initial treatments, her condition has not improved, indicating a need for more advanced management.
- Orbital decompression surgery is specifically designed to address proptosis by increasing the volume of the orbit.
- Decompression surgery is the most appropriate treatment for severe thyroid eye disease.

Treatment of Thyroid Eye Disease:

Management Option	Details
Systemic Steroids	Help reduce inflammation but do not cure the disease.
Immunosuppressors	Tocilizumab- Targets IL-6. Infliximab- Targets TNF- α Adalimumab- Targets TNF- α (not specifically for TGF) Teprotumumab- An anti-IGF-1 receptor antibody, effective in treating thyroid eye disease
Radiotherapy	Anti-inflammatory radiotherapy helps manage symptoms but is not a cure.
Decompression Surgery	Performed in severe cases. Orbital Decompression Surgery- Addresses proptosis by increasing the orbital volume. Indications for Surgery- Sight-threatening compressive optic neuropathy, subsided inflammation, failure of initial treatments (Steroids, Immunosuppressors, and Radiotherapy). (Options A B & C ruled out) Decompression Surgery Sequence- Deep lateral wall → Medial wall → Inferior wall → Lateral wall.

Solution for Question 21:

The Kocher sign describes a staring and frightened appearance of the eyes with severe bilateral lid retraction which is particularly marked on attentive fixation. This sign directly correlates with the observed prominent lid retraction in the patient.

[Image not available]

Clinical Features:

Clinical Features	Details
Lid Signs	
Dalrymple's Sign	Upper lid retraction causing a staring appearance (90% of cases).
Von Graefe's Sign	Lid lag when the globe moves downward (50% of cases). (Option A ruled out)
Enroth's Sign	Puffy, oedematous swelling of the eyelids.

Gifford's Sign	Difficulty in eversion of the upper lid.
Stellwag's Sign	Infrequent blinking. (Option C ruled out)
Joffroy's Sign	Poor forehead wrinkling on looking up. (Option D ruled out)
Kocher's sign	Increased lid retraction with visual fixation (staring look)
Conjunctival Signs	
Deep Injection	Pronounced redness.
Chemosis	Swelling of the conjunctiva.
Pupillary Signs	Less significant; may show unequal dilatation of pupils.
Ocular Motility Defects	
Mobius's Sign	Convergence weakness.
Restrictive Myopathy	Limited movement in the direction opposite to the affected muscle.
Exophthalmos	Both eyes are often affected with possible unilateral proptosis.

Solution for Question 22:

This scenario is related to thyroid eye disease (Graves' orbitopathy), where T-cells attack orbital fibroblasts and adipocytes. This autoimmune response causes inflammation, increased fat formation, and swelling, leading to eye protrusion and eyelid swelling. The extraocular muscles are affected secondarily.

Pathogenesis of Thyroid Eye Disease (Graves' Orbitopathy)

- It involves an organ-specific autoimmune reaction in which anti-thyrotropin-receptor antibody reacts against thyroid gland cells and orbital fibroblasts.
- Insulin-like growth factor 1 receptor (IGF-1R) and the thyroid-stimulating hormone receptor (TSHR) are upregulated in orbital and brow fat, leading to inflammation of extraocular muscles, interstitial tissues, orbital fat and lacrimal glands.
- Osmotic imbibition of water (pleomorphic cellular infiltration (Option D ruled out) and increased secretion of glycosaminoglycans) and lipogenesis induced by fibroblasts and preadipocytes, which causes orbital fat enlargement.

- Optic nerve may be compression (increased IOP).
- Subsequent degeneration of muscle fibres results in restrictive myopathy and diplopia.
- Primary Target: Orbital fibroblasts
- Secondary Involvement: Extraocular muscles (Option A ruled out).

Optic nerve compression in thyroid ophthalmopathy is typically due to increased orbital volume (from inflammation and muscle swelling), not from vascular occlusion. The increase in orbital contents can compress the optic nerve, leading to visual disturbances.

The earliest defective ocular movement seen in this patient is depression. (Option B) Elevation is the earliest defective movement seen in this condition

Solution for Question 23:

Orbital cellulitis is an infection and inflammation of the orbital tissues that causes actual protrusion of the eye (exophthalmos) due to swelling. This is not an example of pseudoproptosis, which involves the appearance of protrusion without actual displacement of the eye.

Pseudoproptosis:

- Pseudoproptosis is a condition in which the eyeball appears to be proptosed but actually there is no forward displacement.

Causes of Pseudoproptosis

- It is seen in high myopia (due to increase in size of eyeball) (Option A ruled out).
- Lid retraction (Option B ruled out).
- Shallow orbits (exorbitism)
- Contralateral ptosis
- Unilateral enophthalmos. (Option C ruled out).

Solution for Question 24:

This is the most commonly used instrument for measuring proptosis. It is preferred because it measures the forward displacement of both eyes simultaneously (Option D), providing a more accurate and consistent measurement of proptosis and identifying asymmetry between the eyes.

[Image not available]

Clinical evaluation of proptosis

- Differentiate proptosis from pseudoproptosis Pseudoproptosis (the false impression of proptosis) may be due to facial asymmetry, enlargement of the globe (high myopia, buphthalmos), lid retraction or contralateral enophthalmos.
- Pseudoproptosis (the false impression of proptosis) may be due to facial asymmetry, enlargement of the globe (high myopia, buphthalmos), lid retraction or contralateral enophthalmos.
- Asymmetrical proptosis is readily detected by looking down at the patient from above and behind
- Globe location: If it is misplaced superiorly, inferiorly, medially, laterally, or straight out (axially) by an exophthalmometer. An exophthalmometer is used to quantify the distance from the orbital rim to the corneal surface of each eye in order to assess the symmetry of globe protrusion. Readings >20 mm are indicative of proptosis (Option C) and a difference of 2–3 mm or more between the two eyes is suspicious regardless of the absolute values. (Option B) Hertel's exophthalmometer: In adults. Luedde exophthalmometer: In children. Naugle exophthalmometer: Non-axial proptosis
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- An exophthalmometer is used to quantify the distance from the orbital rim to the corneal surface of each eye in order to assess the symmetry of globe protrusion. Readings >20 mm are indicative of proptosis (Option C) and a difference of 2–3 mm or more between the two eyes is suspicious regardless of the absolute values. (Option B)
- Hertel's exophthalmometer: In adults.
- Luedde exophthalmometer: In children.
- Naugle exophthalmometer: Non-axial proptosis

Solution for Question 25:

Thyroid ophthalmopathy is also known as Graves' orbitopathy or thyroid eye disease. It is the most common cause of bilateral proptosis in adults. NO SPECS classification is used in thyroid ophthalmopathy.

Classification of Proptosis

Unilateral proptosis	Bilateral proptosis
Orbital cellulitis Developmental anomalies Inflammatory conditions Vascular anomalies Neoplasms Metabolic diseases	Thyroid eye disease (thyroid ophthalmopathy) Orbital myositis Cavernous sinus thrombosis Metastatic neuroblastoma
Axial Proptosis	Nonaxial or Abaxial Proptosis
Lesion will originate in the intraconal space, e.g., optic nerve gliomas, meningiomas and cavernous haemangiomas.	The eye is displaced from its central location, opposite to the site of the spaceoccupying lesion, e.g., superotemporally located dermoids and lacrimal gland tumours.
In Adults	In Children
Cavernous haemangiomas Mucocoeles Lacrimal gland tumours Lymphoid tumours Meningiomas Metastases from the breast, lung and gastrointestinal tract cancers.	Capillary haemangiomas Venolymphatic malformations Optic nerve glioma Rhabdomyosarcoma (Option C ruled out) Leukaemias Metastatic neuroblastoma Plexiform neurofibromatosis Congenital causes, e.g. teratoma and vascular lesions.

- Most common cause of Unilateral proptosis in children is orbital cellulitis; in adults, thyroid ophthalmopathy. (Option A ruled out)
- Most common cause of bilateral proptosis in children is neuroblastoma and leukemia (chloroma) (Option D ruled out); in adults, thyroid ophthalmopathy. (Option B)

Solution for Question 26:

This space is located within the muscle cone and is the appropriate site for retrobulbar injections to provide anesthesia or other medications directly behind the globe, within the muscle cone.

Subperiosteal Surgical Space

- Location: Between the bone and the periorbita (periosteum)

Extraconal Surgical Space (Option B ruled out)

- Location: Between the periorbita and the muscle cone
- Site for: Peribulbar injection

Sub-Tenon Surgical Space (Option A ruled out)

- Location: Between the Tenon capsule and the globe

Intraconal Surgical Space (Option C)

- Location: Within the muscle cone
- Site for: Retrobulbar injection

Subarachnoid Surgical Space (Option D ruled out)

- Location: Between the optic nerve and the nerve sheath

Solution for Question 27:

The optic foramen, which is part of the roof of the orbit, allows passage of the optic nerve and the ophthalmic artery

Superior Orbital Fissure:

- A bony cleft found at the orbital apex between the roof and lateral wall.
- Located between the lesser and greater wings of the sphenoid bone. (Option A ruled out)

Inferior Orbital Fissure:

- Defined as a space between the lateral wall and the orbital floor.
- Present between the greater wing of the sphenoid and the maxillary bone. (Option B ruled out)

Optic Foramen:

- Part of the roof of the orbit.
- It is between lesser wing and body of sphenoid. (Option C ruled out)
- Essential for the morphogenesis of the eye during embryologic development.

Structures Passing Through the Optic Foramen: (Option D)

- Optic Nerve
- Ophthalmic Artery

[Image not available]

[Image not available]

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Previous Year Questions

1. A patient presents with proptosis that increases when bending down. What is the most likely diagnosis?

- A. Orbital Varices
- B. Thyroid Eye Disease
- C. Cavernous Sinus Thrombosis
- D. Orbital Cellulitis

2. Which of the following statements is false regarding the condition shown in the image?

[Image not available]

- A. NOSPEC classification is used
- B. Management corresponds to an improvement in the thyrotoxic state
- C. Vision loss may occur
- D. It is not seen in 10% of patients with hyperthyroidism

3. What is the most common cause of proptosis in a middle-aged person?

- A. Orbital varix
- B. Thyroid eye disease
- C. Retinoblastoma
- D. Cavernous hemangioma

4. Blow out fracture can be due to?

- A. Tennis ball injury
 - B. Chisel and hammer injury
 - C. Sudden fall
 - D. Penetrating injury by knife
-

5. What is the first sign to be seen in a patient with sympathetic ophthalmitis?

- A. Circumcorneal congestion
 - B. Hypopyon
 - C. Retrolental flare
 - D. Iris nodule
-

6. A man suffered trauma to the eye from a punch while fighting with another person. What is the most common site of fracture in orbit?

- A. Medial wall
 - B. Lateral wall
 - C. Roof
 - D. Floor
-

7. Diagnose the clinical condition as shown in photo:

[Image not available]

- A. Keratoglobus

- B. Keratoconus
 - C. Limbal dermoid
 - D. Pinguecula
-

8. What is the most common primary intraocular malignancy in adults?

- A. Retinoblastoma
 - B. Malignant Choroidal melanoma
 - C. Reticulum cell sarcoma
 - D. Neurofibroma
-

9. A 35-year-old female reports recurring episodes of discomfort, pain, and a sensation of pressure behind her left eye. She exhibits intermittent reducible proptosis, and her symptoms intensify when she bends forward. What is the most likely diagnosis?

- A. Orbital hemangioma
 - B. Orbital varices
 - C. Orbital metastasis
 - D. Orbital dermoid cyst
-

10. A middle-aged woman presented with bilateral proptosis, restriction of eye movements, and chemosis. Her thyroid profile showed she was hyperthyroid. What is the most probable cause?

- A. Orbital pseudotumor
- B. Grave's ophthalmopathy

- C. Orbital lymphoma
 - D. Orbital cellulitis
-

11. Which of the following statements is true regarding a patient who has a history of falling on a symmetrical surface and presents with enophthalmos, diplopia on upward gaze, and loss of sensitivity over the cheek?

- A. Maxillary fracture
 - B. It is a blow out fracture
 - C. Zygomatic bone is most likely injured
 - D. Frontal bone fracture
-

12. During evisceration of the right eyeball, which layer of the eyeball is eliminated?

- A. Middle and inner
 - B. Outer and middle
 - C. Outer and inner
 - D. All the layers of eyeball
-

13. A 5-year-old was brought with the complaints of protrusion of the right eye with no h/o fever. CT scan showed a well – defined mass in the orbit with an irregular border and adjacent bony destruction. Biopsy showed small, round cells which were positive for Desmin in immunohistochemistry. What is the probable diagnosis ?

- A. Retinoblastoma
- B. Cavernous hemangioma
- C. Orbital Cellulitis

D. Rhabdomyosarcoma

14. A 56-year-old male presented with concerns about his eyes bulging and becoming more prominent over the past month. He has previously been diagnosed with Grave's disease. The image below shows the patient's eyes. Based on these findings, what is the most probable diagnosis?

[Image not available]

- A. Orbital cellulitis
 - B. Preseptal cellulitis
 - C. Thyroid eye disease
 - D. Capillary hemangioma
-

15. What is the most common site of an orbital blowout fracture?

- A. Medial wall
 - B. Lateral wall
 - C. Roof
 - D. Floor
-

16. A child was brought to the outpatient department with a history of trauma near the eye and now presents with symptoms of fever, severe pain, and swelling in the left eye. On examination, no abnormal findings were observed. What is the likely diagnosis?

[Image not available]

- A. Orbital cellulitis
- B. Endophthalmitis

- C. Panophthalmitis
 - D. Preseptal cellulitis
-

17. What would be your diagnosis for a patient who has a previous trauma history and is currently experiencing diplopia and following findings on CT?

[Image not available]

- A. Le-fort fracture
 - B. Fracture maxilla
 - C. Fracture zygomatic
 - D. Blow out fracture
-

18. What is the most frequently occurring tumor found in the lacrimal gland?

- A. Adenoid cystic carcinoma
 - B. Lymphoma
 - C. Pleomorphic adenoma
 - D. Mucoepidermoid carcinoma
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	2
Question 4	1
Question 5	3
Question 6	4
Question 7	3
Question 8	2
Question 9	2
Question 10	2
Question 11	2
Question 12	1
Question 13	4
Question 14	3
Question 15	4
Question 16	4
Question 17	4
Question 18	3

Solution for Question 1:

Orbital Varices are abnormal dilations of veins in the orbit. They often present with proptosis (bulging of the eye) that can worsen with increased venous pressure, such as when the patient bends down or performs actions that increase intra-abdominal or intra-thoracic pressure.

Causes of intermittent proptosis:

- Orbital varix
- Periodic orbital oedema
- Recurrent orbital hemorrhage
- Highly vascular tumors

Orbital Varices	
Dilated venous channels in the orbit (part of the spectrum of developmental venous anomalies)	
Clinical features	Most cases are unilateral and the most frequent site is upper nasal. Slowly progressive intermittent proptosis which worsens with head position changes, bending forward, or Valsalva maneuver.
Pathology	May involve one large vessel or multiple ectatic veins with fibrotic walls. Thrombosed lumens can contain phleboliths from calcified thrombi.
Imaging	CT - serpiginous, dense structures that enlarge with Valsalva, enhances with contrast administration. MRI - T1 images show low isointense signal, T2 shows dark signal void from flowing blood. Enhancement with Gadolinium is marked, except in areas of thrombosis.
Treatment	No treatment needed for mild symptoms; Surgical or embolization options for severe cases.
Prognosis	Good unless large lesions compress the optic nerve, risking vision loss.

Thyroid Eye Disease (Option B): This condition typically causes bilateral proptosis due to inflammation and fibrosis of the extraocular muscles. While proptosis occurs, it does not usually worsen with bending down and is often associated with thyroid dysfunction. It has a more gradual onset compared to conditions like orbital varices.

Cavernous Sinus Thrombosis (Option C): This serious condition presents with painful proptosis, usually accompanied by headache, cranial nerve palsies, and vision changes. Proptosis does not typically worsen with bending down, and the condition is associated with acute symptoms and systemic signs of infection.

Orbital Cellulitis (Option D): This is an infection of the orbit, causing painful proptosis, redness, and swelling, often accompanied by fever. Pr

Solution for Question 2:

Correct Option B - Management corresponds to an improvement in the thyrotoxic state:

- The given image shows Kocher's sign, which is seen in thyroid eye disease. The resection of overacting muscle LPS and resecting of underacting muscle will help to resolve the patient's symptoms, not just the management of the thyrotoxic state.

Incorrect Options:

Options A, C, and D are true about thyroid eye diseases.

Solution for Question 3:

Correct Option B - Thyroid eye disease:

- The most common cause of both unilateral and bilateral proptosis in adults is thyroid ophthalmopathy.
- In children, the most common cause of unilateral proptosis is orbital cellulitis and bilateral proptosis is neuroblastoma.

Incorrect Options:

Option A - Orbital varix:

- Orbital varix causes intermittent proptosis.

Option C - Retinoblastoma:

- Retinoblastoma does not cause proptosis.

Option D - Cavernous hemangioma:

- Cavernous hemangioma causes proptosis but it is not the most common cause.
-

Solution for Question 4:

Correct Option A - Tennis ball injury:

- Blow-out orbital fractures occur due to trauma to the orbit, which is often caused by a large and rounded object like a tennis ball, cricket ball, human fist, or part of an automobile.
- The force of the blow causes the eyeball to be displaced backwards and increases intraorbital pressure. This results in a fracture at the weakest point of the orbital wall, which is typically the orbital floor but can also be the medial wall.

Incorrect Options:

Option B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 5:

Correct Option C - Retrolental flare:

- Retrolental flare refers to the presence of protein and inflammatory cells in the vitreous cavity, which can be seen on fundus examination. It is the earliest finding of sympathetic ophthalmia and indicates the presence of intraocular inflammation.

Incorrect Options:

Options A, B, and D are incorrect.

Solution for Question 6:

Correct Option D - Floor:

- The punch to the eye can result in a blowout fracture.

- The most common site of a blowout fracture is the floor of the orbit, leading to herniation of orbital content into the maxillary antrum resulting in teardrop sign in CT.. The floor is made of the zygomatic, maxillary and the palatine bones.

Incorrect Options:

Options A, B, and C are incorrect.

Solution for Question 7:

Correct Option C - Limbal dermoid:

- The image revealed a raised, yellowish-white lesion with hair-like structures in the limbal area (the junction between the cornea and sclera) is suggestive of limbal dermoid, which is a congenital benign tumor is a choristoma consisting of a mass of collagenous tissue containing dermal elements, which is covered by stratified squamous epithelium.

[Image not available]

Incorrect Options:

Option A - Keratoconus: Keratoconus is a progressive eye condition characterized by thinning and bulging of the central cornea, resulting in a cone-shaped corneal surface. It typically presents as a V-shaped deformity of the cornea (munson's sign).

Option B - Keratoglobus: It is a familial and hereditary bilateral congenital disorder characterized by thinning and hemispherical protrusion of the whole cornea.

Option D - Pinguecula: Pinguecula is a common degenerative condition of the conjunctiva characterized by a yellowish-white mound near the limbus formed by elastotic degeneration of conjunctival stroma seen in a person exposed to strong sunlight, dust, and wind.

Solution for Question 8:

Correct Option B - Malignant Choroidal melanoma:

- Malignant choroidal melanoma is the most common primary intraocular malignancy in adults. It originates from the pigment-producing cells (melanocytes) in the choroid, which is a vascular layer beneath the retina in the eye. Choroidal melanoma is more prevalent in adults, particularly in individuals with fair skin and light-colored eyes.

Incorrect Options:

Option A – Retinoblastoma: It is the most common primary intraocular malignancy in Children.

Option C - Reticulum cell sarcoma: Reticulum cell sarcoma, or primary intraocular lymphoma, is a type of non-Hodgkin lymphoma that can occur in the eye. While it is a primary intraocular malignancy, it is less common than malignant choroidal melanoma.

Option D – Neurofibroma: Neurofibromas are tumors that arise from nerves. They are often associated with neurofibromatosis type 1 (NF1), a genetic condition.

Solution for Question 9:

Correct Option B - Orbital varices:

- Orbital varices are vascular anomalies characterized by dilated and tortuous veins in the orbit, which are associated with intermittent proptosis.
- Proptosis may worsen when the patient is in an upright position or during activities that increase intra-abdominal pressure, such as coughing or bending forward.

Incorrect Options:

Option A - Orbital hemangioma: Orbital hemangiomas are benign vascular tumors of the orbit. They typically present as painless, slow-growing masses without the characteristic features of intermittent proptosis.

Option C - Orbital metastasis: Orbital metastasis is less probable because of younger age and the recurrent nature of the symptom.

Option D - Orbital dermoid cyst: Orbital dermoid cyst is characterized by a painless nodule or proptosis.

Solution for Question 10:

Correct Option B - Grave's ophthalmopathy:

Grave's ophthalmopathy also known as thyroid-associated ophthalmopathy, is an autoimmune condition commonly associated with Graves' disease, which is a form of hyperthyroidism.

- It is characterized by inflammation and swelling of the tissues surrounding the eyes, including the muscles and fatty tissues within the orbit.
- The condition often presents with the following features:
- It is the most common cause of Proptosis - Unilateral/bilateral, painful/non-painful, axial/non-axial.
- Grittiness and red eyes.
- Lid retraction.

Incorrect Options:

Option A: Orbital pseudotumor: It is characterized by congestive proptosis, ophthalmoplegia, and optic nerve dysfunction and generally present as a unilateral condition in adults.

Option C: Orbital lymphoma: It is characterized by discomfort, bulging eye, diplopia, and visible mass and can be unilateral or bilateral..

Option D: Orbital cellulitis: It is the most common cause of unilateral proptosis in children and is characterized by unilateral and painful proptosis.

Solution for Question 11:

Correct Option B - It is a blow out fracture:

- The patient's symptoms, including enophthalmos (sunken eye), diplopia on upward gaze, and loss of sensitivity over the cheek, are consistent with a blowout fracture.
- A blowout fracture refers to a fracture of the orbital floor or medial wall caused by trauma. It can result in displacement of orbital contents and lead to various symptoms like periorbital ecchymosis, enophthalmos, diplopia, and loss of sensation over the cheek because of the involvement of the infraorbital nerve.

Incorrect Options:

Options A, C, and D are incorrect. described symptoms are not associated with these fractures.

Solution for Question 12:

Correct Option A - Middle and inner:

- In evisceration, the middle layer (uvea) and inner layer (retina) of the eyeball are typically removed while preserving the outer layer (sclera). The procedure involves the removal of the intraocular contents, including the iris, lens, and vitreous humor, while leaving the outer structure intact.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 13:

Correct Option D - Rhabdomyosarcoma:

- "Rhabdomyosarcoma" is the probable diagnosis based on the given information.
- Rhabdomyosarcoma is a malignant tumor that arises from embryonal cells of skeletal muscle origin. It is the most common soft tissue sarcoma in children and can occur in various locations, including the orbit. The clinical presentation of an orbital rhabdomyosarcoma often includes proptosis (protrusion of the eye), without a history of fever.
- The CT scan findings of a well-defined mass in the orbit with an irregular border and adjacent bony destruction are consistent with the characteristics of rhabdomyosarcoma..
- The biopsy findings of small, round cells that are positive for Desmin, and Desmin is a marker of rhabdomyosarcomas.

Incorrect Options:

Option A - Retinoblastoma: Retinoblastoma is a malignant tumor of the retina, which is the inner layer of the eye. It typically presents as leukocoria , strabismus , or visual impairment.

Option B - Cavernous hemangioma: Cavernous hemangioma is a benign vascular tumor that can occur in the orbit. It is characterized by a collection of dilated blood vessels.

Option C - Orbital Cellulitis: Orbital cellulitis is an infection involving the soft tissues around the eye. It typically presents with eyelid swelling, erythema, pain, fever, proptosis, restricted eye movements.

Solution for Question 14:

Correct Option C - Thyroid eye disease:

- The clinical presentation, history of grave's disease and the image is suggestive of thyroid eye disease aka Grave's ophthalmopathy..
- The image reveals bilateral proptosis, periorbital swelling caused by edema and infiltration behind the orbital septum. This may be associated with chemosis and prolapse of retroseptal fat into the eyelids, dalrymple sign is seen, i.e., lid retraction in primary gaze, and Kocher's sign, characterized by staring and frightened appearance of the eyes are characteristic of thyroid eye disease.

Incorrect Options:

Option A - Orbital cellulitis: Orbital cellulitis is the inflammation behind the orbital septum caused by bacteria or fungi and characterized by unilateral proptosis, chemosis, restricted eye movements.

Option B - Preseptal cellulitis: It is characterized by a swollen, often firm, tender red eyelid caused by staphylococcus aureus and streptococcus pyogenes after a trauma near the eye.

Option D - Capillary hemangioma: Capillary hemangioma is the benign tumor of the orbit commonly affecting infants and is caused due to incomplete blood vessel formation.

Solution for Question 15:

Correct Option D - Floor:

- The blowout fracture most frequently involves the floor of the orbit along the thin bone covering the infraorbital canal.

Incorrect Options:

Options A, B, and C are incorrect. These walls of the orbit are not commonly involved in the blowout fracture.

Solution for Question 16:

Correct Option D - Preseptal cellulitis:

- Preseptal cellulitis is an infection of the subcutaneous tissues anterior to the orbital septum. Typically, *Staphylococcus aureus* and *Streptococcus pyogenes* are responsible.
- Causes include skin trauma such as laceration or insect bites, spread from focal ocular or periocular infection.
- It is characterized by a swollen, often firm, tender red eyelid. Proptosis and chemosis are absent, and visual acuity, pupillary reactions, and ocular motility are unimpaired.

Incorrect Options:

Option A - Orbital cellulitis: Orbital cellulitis is characterized by proptosis, restricted eye movements, and chemosis.

Option B - Endophthalmitis is the inflammation of the inner structures of the eyeball and is characterized by pain, redness, worsening vision, discharge, and progressive vitritis.

Option C - Panophthalmitis is an intense purulent inflammation of the entire eyeball characterized by severe ocular pain, complete vision loss, marked redness, swelling, and purulent discharge.

Solution for Question 17:

Correct Option D - Blowout Fracture:

- The clinical presentation and CT showing Teardrop sign is suggestive of blowout fracture. A blowout fracture typically occurs when there is blunt trauma to the orbit resulting in fracture of the thin bones surrounding the orbit leading to teardrop sign due to prolapse of the soft tissue content into maxillary antrum.
- It is characterized by ecchymosis, enophthalmos, and diplopia due to entrapment of inferior rectus and inferior oblique muscle in the fracture.

Incorrect Options:

Options A, B, and C are incorrect. Teardrop sign is not seen in these conditions.

Solution for Question 18:

Correct Option C - Pleomorphic adenoma:

- The most common lacrimal gland tumor is pleomorphic adenoma.

Incorrect Options:

Option A - Adenoid cystic carcinoma is the most common malignant tumor of the lacrimal gland.

Option B - Lymphoma are the most common non-epithelial malignant lacrimal gland tumor..

Option D - Mucoepidermoid carcinoma can occur in the lacrimal gland it is quite uncommon and certainly not the most common tumor of the lacrimal gland.

Ocular Injuries

1. Which of the following types of ocular injuries can result from blunt trauma to the eye? 1) 2) 3) 4)

1)	[Image not available]
2)	[Image not available]
3)	[Image not available]
4)	[Image not available]

- A. 2 and 4
- B. 1 and 2
- C. 3 and 1
- D. All of the Above

2. A patient presents with a full-thickness corneal wound caused by a nail. Which of the following is the most likely complication if the injury is left untreated?

- A. Ring abscess of the cornea
- B. Retinal detachment
- C. Glaucoma
- D. Strabismus

3. A 40-year-old construction worker presents with eye pain and blurred vision after metal struck his eye. CT scan reveals an intraocular foreign body lodged in the vitreous cavity. What is the most appropriate initial treatment step?

- A. Observation and follow-up in one week
 - B. Corticosteroid injection without IOFB removal
 - C. Immediate surgery with pars plana vitrectomy and IOFB removal
 - D. Application of topical antibiotics and patching
-

4. A 28-year-old metalworker presents with gradual vision loss, a greenish discoloration of his iris, and a history of metallic foreign body injury 4 months ago. Examination reveals rusty deposits on the lens and pigmentary changes in the retina. What is the most likely cause of his condition?

- A. Chalcosis
 - B. Retinitis pigmentosa
 - C. Siderosis bulbi
 - D. Glaucoma
-

5. A 35-year-old male presents to the emergency department after an accidental chemical exposure to his eyes while working in a lab. He reports immediate burning and decreased vision. Upon examination, there is corneal epithelial loss and conjunctival swelling. What is the most appropriate immediate management step?

- A. Necrotic tissue removal and administer topical antibiotics
 - B. Apply corticosteroids to reduce inflammation
 - C. Immediate irrigation with water or saline for 20-30 minutes
 - D. Perform corneal transplant
-

6. A 28-year-old male presents to the emergency department 10 days after sustaining blunt trauma to the face during a basketball game. He complains of diplopia and difficulty moving his left eye upwards. On examination, there is significant periorbital swelling. The X-ray is shown below. What does this X-ray indicate in this patient, and what is the most appropriate next step in management?

[Image not available]

- A. Orbital fat herniation; apply ice and prescribe analgesics
 - B. Muscle entrapment due to an orbital floor fracture; surgical evaluation
 - C. Orbital haemorrhage; immediate CT scan
 - D. Maxillary sinus fracture; follow-up with ENT specialist
-

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Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	3
Question 4	3
Question 5	3
Question 6	2

Solution for Question 1:

Correct Answer: D) All of the Above

Explanation:

All of the given presentations can result from Blunt trauma to the eye.

Hyphema	[Image not available]
2. Descemet's membrane tear	[Image not available]
3. Vossius ring	[Image not available]
4. Lens subluxation	[Image not available]

Blunt trauma (also known as non-penetrating or blunt force trauma)

Cause: A strong impact, a fall, or a physical assault with a blunt object resulting from sports injuries and assaults.

Features:

- Conjunctiva: Subconjunctival hemorrhage
- Cornea: Abrasions, corneal oedema, Descemet's membrane tears. (Option D)
- Hyphaema: Blood in the anterior chamber leading to corneal staining. (Option D)

- Iris: Iridodialysis (dehiscence of the iris from the ciliary body at its root), iris sphincter tear,
- Ciliary body: Ciliary body tear leading to angle recession.
- Lens: Vossious Ring (in the shape of the pupil), Rosette-shaped cataract (flower-shaped opacity), subluxation of lens. (Option D)
- Globe rupture (caused by severe blunt trauma, indicated by iris prolapse)
- Vitreous haemorrhage: Pigment cells ('tobacco dust') can be seen floating in the anterior vitreous.
- Choroidal rupture
- Retina: Commotio Retinae (also known as Berlin's oedema), retinal dialysis (disinsertion from ora serrata), giant retinal Tear (horseshoe shaped), macular hole.
- Injury to optic nerve
- Traumatic neuropathy with chorioretinal scarring (decreased visual acuity, visual field defect, afferent pupillary defect, colour, and brightness).

Solution for Question 2:

Correct Answer: A) Ring abscess of the cornea

Explanation:

Untreated penetrating injuries can introduce microorganisms, leading to severe infections like a ring abscess of the cornea, a sight-threatening condition.

Penetrating Eye Injury	
Definition	A penetrating injury enters the eye without exiting, resulting in no exit wound.
Causes	Sharp objects such as sharpened pencils (most common cause in children), needles, knives, nails, screws, glass, etc.
Damage Mechanisms	[Image not available] Tearing and structural disruption Pathogens enter through open wounds, leading to ring abscesses, corneal sloughing, iridocyclitis, endophthalmitis, panophthalmitis, etc. (Option A)

Prognosis	It depends on the severity of the injury and the complications that arise secondary to the injury. Influenced by: - Initial visual acuity - Presence of a relative pupillary defect (RAPD) - Presence of an intraocular foreign body - Presence of infection - Presence of a retinal injury or detachment.
-----------	--

- Retinal detachment (Option B) is more commonly associated with blunt trauma rather than penetrating injuries.
- Glaucoma (Option C) is a potential late complication but not the immediate concern in this scenario.
- Strabismus (Option D) refers to eye misalignment, which arises from muscle or nerve dysfunction, not corneal trauma.

Solution for Question 3:

Correct Answer: C) Immediate surgery with pars plana vitrectomy and IOFB removal

Explanation:

Pars plana vitrectomy with IOFB removal is essential to prevent complications and preserve vision, and it is ideally done during the initial closure.

Intraocular Foreign Bodies (IOFBs)	
Description	An IOFB can lodge in any part of the anterior or posterior segments, causing mechanical damage, infection, or toxic effects on intraocular structures.
Types	Inert: Glass, wood, plastic, rubber, gold, silver, platinum Reactive: Organic and Inorganic (iron causing siderosis, copper causing chalcosis).
Diagnosis	History. Examination (wound examination, foreign body location, associated signs such as lid laceration and anterior segment damage should be noted). CT (investigation of choice) with axial and coronal cuts detects and localizes a metallic IOFB. MRI is contraindicated (metallic IOFB).

Treatment	Magnetic removal (metallic IOFB) Forceps removal through pars plana vitrectomy (Option C) Prophylaxis against infection (topical/systemic antibiotics) (Option D ruled out) Conservative and supportive management (topical corticosteroids, analgesics, etc). (Option B ruled out)
-----------	--

- Observation and follow-up in a week (Option A) in the presence of a foreign body in the vitreous cavity could lead to infection and further complications.

Solution for Question 4:

Correct Answer: C) Siderosis bulbi

Explanation:

Siderosis Bulbi	
Definition	Degenerative eye condition caused by an iron-containing foreign body that is not promptly removed from the eye. (Option C)
Cause	Chisel and hammer injury (most common cause)
Mechanism	Oxidation and other chemical reactions with surrounding tissues, leading to toxic effects on cellular enzyme systems and subsequent cell death.
Clinical Manifestations	[Image not available] [Image not available] [Image not available] Lens deposits (radially distributed iron deposits on the anterior lens capsule leading to cataract) Heterochromia iridis (iris pigmentation): Reddish-brown staining of the iris. Retinal changes (pigmentary retinopathy), leading to atrophy of the retina and RPE (retinal pigment epithelium), potentially leading to profound visual loss. Glaucoma (trabecular damage) Electroretinography (ERG): Progressive attenuation of the b-wave over time.

- Chalcosis (Option A) is caused by copper, not iron, and shows distinct features like Kayser-Fleischer rings.

- Retinitis pigmentosa (Option B) is genetic, not due to trauma.
- Glaucoma (Option D) can be a complication but is not the primary diagnosis here.

Solution for Question 5:

Correct Answer: C) Immediate irrigation with water or saline for 20-30 minutes

Explanation:

Irrigation with water or saline is the priority to dilute and remove the chemical, minimizing damage.

Chemical Injuries	
Description	Chemical eye injuries are caused by acidic or alkaline substances entering the eye (accidental or due to assault), presenting a severe ocular emergency that demands immediate treatment.
Cause	Alkali burn (most common, more severe) Ammonia (most common), NaOH (sodium hydroxide), lime [CaO]. Acid (coagulates the surface proteins) Hydrofluoric acid (used in glass etching), sulphuric acid Others: Detergents, cosmetics, fertilizers, insecticides, etc.
Pathophysiology	

Management	Immediate management: The first and the most crucial step while treating a chemical injury is copious irrigation (water or saline for at least 20-30 minutes or until the eye's pH is restored) by double eversion of the lid. (Option C) Debridement: Carefully excise dead conjunctiva and corneal epithelium using a cotton swab.(Option A ruled out) Medical management: Antibiotics (Option A ruled out) Cycloplegics (relax the ciliary muscle), Steroids only for an initial 7 days (as they reduce collagen synthesis and impair healing). (Option B ruled out) Ascorbic acid (vitamin C) 10% E/D 2hrly. (promotes collagen synthesis and improves wound healing) Citrate 10% E/D 2 hourly for ten days (controls inflammation and inhibits collagenase) Tetracycline (collagenase inhibitor) Acetylcysteine 10% (anti-collagenase) Surgical management Early: Advancement of tenon capsule, limbal cell transplantation, amniotic membrane grafting ... (content truncated)
Complications	Secondary glaucoma Limbal stem cell deficiency Pseudopterygium Corneal opacity (keratoplasty or keratoprosthesis). (Option D ruled out)

Solution for Question 6:

Correct answer: B) Muscle entrapment due to an orbital floor fracture; surgical evaluation

Explanation:

The "teardrop sign" or "hanging drop sign" on the X-ray suggests an orbital floor fracture, often associated with muscle entrapment, which leads to restricted eye movement and diplopia. The appropriate next step in management would be to evaluate for surgical intervention to relieve the entrapment and repair the fracture.

Orbital floor fracture (also known as blow-out fracture):

A blow-out fracture usually occurs due to a sudden rise in orbital pressure from an impacting object, like a fist or tennis ball, causing the eyeball to be displaced and transmit, rather than absorb, the impact.

Classification:

- Pure blow-out fractures are isolated to the orbital walls (floor or medial wall) without involving the orbital rim.
- Impure blow-out fractures involve the orbital walls, the orbital rim, and surrounding facial bones.

Clinical features:

- Periorbital oedema and blood extravasation in and around the orbit (such as subconjunctival ecchymosis)
- Emphysema (subcutaneous) of the eyelids
- Paraesthesia and anaesthesia in the distribution of infraorbital nerve (lower lid, cheek, side of nose, upper lip, and upper teeth).
- Ipsilateral epistaxis (bleeding from maxillary sinus)
- Proptosis (orbital oedema and haemorrhage).
- Enophthalmos and mechanical ptosis
- Diplopia (oedema, haemorrhage, entrapment of muscle (inferior rectus/inferior oblique) at the fracture site.
- Hess chart testing (map eye movements for assessment and monitoring).

[Image not available]

Orbital imaging:

- X-rays: A nose-chin (Water's) view showing 'hanging drop' opacity of the superior maxillary antrum from orbital contents herniating through the floor.

[Image not available]

Plain X-ray orbit (AP view) showing herniated orbital contents (arrow) with blow-out fracture of the orbital floor

- Computerised tomography (CT) scanning and magnetic resonance imaging(MRI): Evaluate the extent of the fracture and herniation of contents into maxillary antrum (hanging drop or tear drop sign).

[Image not available]

CT scan of coronal section of orbit showing fracture floor with herniation of orbital contents

Management:

- Antibiotics and anti-inflammatory drugs
 - Ice packs and nasal decongestants
 - Surgery (dislodge the muscle and do microplating) Indications for surgery: No improvement in diplopia or enophthalmos even after 10 days (indicates muscle entrapment). Enophthalmos is > 2mm Half of the floor is fractured
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 - Half of the floor is fractured
-

Previous Year Questions

1. The patient presents with foreign body (chuna) in his eyes and complaints of irritability and lacrimation. What is the first step management in this patient?

- A. Antibiotics eye drops
- B. Ascorbic eye drops
- C. Irrigation of eye with normal saline
- D. Steroids

2. A patient has a history of a flying foreign body injury to the eye. Hemochromatosis is subsequently diagnosed. Which of the following substances is most likely involved in the deposition?

- A. Iron
- B. Aluminum
- C. Wood
- D. Glass

3. Which investigation is most suitable for localizing a foreign body in the eye of an industrial worker admitted to the hospital due to an eye injury caused by cutting stone with a hammer and iron chisel?

- A. MRI
- B. CT
- C. X-ray
- D. B scan

4. A patient presents to the emergency and his ocular examination reveals the following. Which of the following is not done initially?

[Image not available]

- A. Removal of the foreign body
 - B. Check for visual acuity
 - C. Slit lamp examination
 - D. Do primary survey
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	2
Question 4	1

Solution for Question 1:

Correct Option C - Irrigation of the eye with normal saline:

- Chemical injury to the conjunctiva and cornea is a severe ocular emergency that must be treated immediately.
- The first and the most crucial step in treating chemical injuries is copious eye irrigation by double eversion of the lid with normal saline.
- Deliver a minimum of 2 liters of water in 20-30 minutes or until pH is restored.

Incorrect Options:

Option A - Antibiotics drops in the eye:

- Antibiotics are used to prevent infection, but it is not the initial step.

Option B - Ascorbic eye drops:

- Ascorbic eye drops promote collagen synthesis and improve wound healing but it is not the initial step.

Option D - Steroids:

- Steroids reduce inflammation, neutrophil infiltration, and address anterior uveitis. Cautious use is recommended after 2 weeks, as steroids may interfere with collagen synthesis.

Solution for Question 2:

Hemochromatosis is a condition characterized by excessive iron accumulation in the body. In the context of a flying foreign body injury, if the foreign body were made of iron, it could lead to iron deposition in the affected tissues, contributing to the development of hemochromatosis.

Intraocular Foreign Bodies (IOFBs)	
Description	An IOFB can lodge in any part of the anterior or posterior segments, causing mechanical damage, infection, or toxic effects on intraocular structures.
Types	Inert: Glass, wood, plastic, rubber, gold, silver, platinum Reactive: Organic and Inorganic (iron causing siderosis, copper causing chalcosis).
Diagnosis	History. Examination (wound examination, foreign body location, associated signs such as lid laceration and anterior segment damage should be noted). CT (investigation of choice) with axial and coronal cuts detects and localizes a metallic IOFB. MRI is contraindicated (metallic IOFB).
Treatment	Magnetic removal (metallic IOFB) Forceps removal through pars plana vitrectomy Prophylaxis against infection (topical/systemic antibiotics) Conservative and supportive management (topical corticosteroids, analgesics, etc).

Modes of Damage:

- Depend on the size and velocity of the foreign body (FB).
- Metallic FB are usually sterile due to the heat generated on their emission.
- Pieces of wood or stone carry a greater chance of infection, often leading to endophthalmitis.
- Inert FB: No reaction. (Options C and D)
- Lead & Aluminium: Cause fibrosis (Option B).
- Toxic Non-Metallic FB (Zinc, Copper, Mercury, Aluminium): Incite suppuration.
- Siderosis (Iron FB - Degenerative Changes Due to Electrolytic Dissociation and Dissemination): Earliest sign: Deposition in the anterior capsule of the lens, appearing as a rusty ring (pathognomonic). Heterochromia iridis: Hyperchromic. Visual loss: Due to pigmentary retinal dystrophy. Secondary chronic glaucoma. Diagnosis: Prussian blue reaction.

- Earliest sign: Deposition in the anterior capsule of the lens, appearing as a rusty ring (pathognomonic).
- Heterochromia iridis: Hyperchromic.
- Visual loss: Due to pigmentary retinal dystrophy.
- Secondary chronic glaucoma.
- Diagnosis: Prussian blue reaction.
- Chalcosis (Copper from Alloy Deposited on Membranous Structures): Kayser-Fleischer Ring: Deposition in Descemet's membrane (seen in Wilson's disease). Sunflower cataract. No degenerative changes: Visual prognosis is good.
- Kayser-Fleischer Ring: Deposition in Descemet's membrane (seen in Wilson's disease).
- Sunflower cataract.
- No degenerative changes: Visual prognosis is good.
- Caterpillar Hair: Produces granulomatous iridocyclitis, known as Ophthalmia Nodosa.
- Earliest sign: Deposition in the anterior capsule of the lens, appearing as a rusty ring (pathognomonic).
- Heterochromia iridis: Hyperchromic.
- Visual loss: Due to pigmentary retinal dystrophy.
- Secondary chronic glaucoma.
- Diagnosis: Prussian blue reaction.
- Kayser-Fleischer Ring: Deposition in Descemet's membrane (seen in Wilson's disease).
- Sunflower cataract.
- No degenerative changes: Visual prognosis is good.

Solution for Question 3:

Correct Option B - CT:

- CT scan is the investigation of choice in case of an intraocular foreign body.

Incorrect Options:

Options A, C, and D are incorrect.

Solution for Question 4:

Correct Option A - Removal of the foreign body:

- Removal of the foreign body: This is not done initially, especially in cases of penetrating injuries. The foreign body should be left in place until the patient is taken to an appropriate medical facility with the necessary equipment and expertise to handle such cases.
- Removal of the penetrating foreign body can result in the drainage of the aqueous humor.

Incorrect Options:

Options B, C, and D are incorrect. These are the primary tests done initially for checking the degree of damage.

Lids and Disorders

1. A 6-month old baby was brought to the clinic with the following finding. Which among the following statements is incorrect regarding the likely diagnosis?

[Image not available]

- A. More common in boys
- B. Undergoes spontaneous regression in majority
- C. Oral propranolol can be used for treating this condition
- D. Most common orbital tumor of children.

2. Match the image with the condition: Image Condition a. 1. Styne b. 2. Chalazion c. 3. Internal Hordeolum

Image	Condition
[Image not available] a.	1. Styne
[Image not available] b.	2. Chalazion
[Image not available] c.	3. Internal Hordeolum

- A. a-2,b-3,c-1
- B. a-1,b-2,c-3
- C. a-3,b-2,c-1
- D. a-1,b-3,c-2

3. Which is the most common organism causing bacterial blepharitis?

- A. Staphylococci
 - B. Streptococci
 - C. Propionibacterium acnes
 - D. Moraxella
-

4. A 3-year-old girl has been having difficulty opening her left eye since birth. An image is given below. What is the condition?

[Image not available]

- A. Symblepharon
 - B. Ankyloblepharon
 - C. Blepharitis
 - D. Blepharophimosis
-

5. A 45-year-old woman with hypothyroidism is now complaining of loss of eyelashes. What is the diagnosis?

- A. Madarosis
 - B. Poliosis
 - C. Tylosis
 - D. Trichiasis
-

6. A neonatologist has noted the following finding in a newborn baby. What is the condition called?

[Image not available]

- A. Trichiasis
- B. Pseudotrichiasis
- C. Distichiasis
- D. None of the above

7. A 70-year-old patient presents with complaints of epiphora. On examination, there is an eversion of the punctum noted. Which among the following is incorrect regarding the type of ectropion seen in this patient?

- A. Most common type of ectropion
- B. Can cause lagophthalmos
- C. Age-related changes are implicated in etiopathogenesis
- D. Starts from lateral eyelid and progresses medially

8. Match the following types of entropion with their correct pair

1) Mechanical entropion	A) Trachoma
2) Cicatricial entropion	B) Phthisis bulbi
3) Congenital entropion	C) Senile entropion
4) Involutional entropion	D) Most commonly involves lower eyelid

1) Mechanical entropion	A) Trachoma
2) Cicatricial entropion	B) Phthisis bulbi
3) Congenital entropion	C) Senile entropion
4) Involutional entropion	D) Most commonly involves lower eyelid

- A. 1-A, 2-B, 3-C, 4-D
- B. 1-C, 2-A, 3-B, 4-D
- C. 1-B, 2-A, 3-D, 4-C
- D. 1-B, 2-A, 3-C, 4-D

9. A 75-year-old male patient presents with the following ocular findings and complaints of watering and photophobia. On examination, there are signs of minimal horizontal lid laxity. Which of the following surgical procedures would be the most appropriate for a long-term correction of his condition?

[Image not available]

- A. Transverse sutures
- B. Wies operation
- C. Jones operation
- D. Quickert procedure

10. Which among the following structures is part of the anterior lamella of the eyelid?

- A. Orbicularis oculi
- B. Orbital septum
- C. Tarsal plate
- D. Conjunctiva

11. All are causes of ipsilateral pseudoptosis except

- A. Enophthalmos
 - B. Lid retraction
 - C. Hypotropia
 - D. Phthisis bulbi
-

12. A 50 year male known smoker presented with hemoptysis and cough. A lung cancer was suspected. On examination, the patient had mild ptosis with good levator function. What would be the preferred management? (LPS-Levator palpebrae superioris)

- A. Mullerectomy
 - B. Minimal resection of LPS
 - C. Maximal resection of LPS
 - D. Frontalis sling surgery
-

13. Which of the following measurements is considered indicative of moderate ptosis?

- A. 1 mm droop
 - B. 2 mm droop
 - C. 3 mm droop
 - D. 4 mm droop
-

14. Which muscle is most commonly affected in congenital ptosis?

- A. Orbicularis oculi
 - B. Levator palpebrae superioris
 - C. Frontalis
 - D. Superior oblique
-

15. Which of the following is a distinguishing feature of aponeurotic ptosis compared to other forms of ptosis?

- A. Accompanied by ocular motor dysfunction
 - B. Presents with normal function of the levator muscle
 - C. Results from a neurological condition affecting the third cranial nerve
 - D. Usually involves a sudden onset of symptoms
-

16. A 45-year-old female presents with bilateral ptosis that worsens throughout the day, particularly in the evening, and improves with rest. She also complains of double vision and generalized fatigue. Which of the following is the most likely cause of her symptoms?

- A. Horner's Syndrome
 - B. Myasthenia Gravis
 - C. Oculopharyngeal Muscular Dystrophy
 - D. Botulism
-

17. A 60-year old male, heavy smoker, presents with persistent shoulder pain radiating to arm for several months. He presents with the following ocular findings. A chest x-ray reveals mass at the apex of the right lung. Which of the following symptoms is most likely to accompany this patient?

[Image not available]

- A. Mydriasis
 - B. Contralateral ptosis
 - C. Anhidrosis and miosis
 - D. Ophthalmoplegia
-

18. Identify the structure marked 'X' in the given image.

[Image not available]

- A. Orbicularis oculi muscle
 - B. Tarsal plate
 - C. Palpebral conjunctiva
 - D. Levator aponeurosis
-

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Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	1
Question 4	2
Question 5	1
Question 6	3
Question 7	4
Question 8	3
Question 9	2
Question 10	1
Question 11	2
Question 12	1
Question 13	3
Question 14	2
Question 15	2
Question 16	2
Question 17	3
Question 18	2

Solution for Question 1:

Correct Answer: A) More common in boys

Explanation:

The given clinical scenario is suggestive of Capillary Hemangioma.

Description	Most common orbital tumor in children (Option D) More common in girls than boys (3:2 ratio) (Option A)
Onset & Growth	Visible at birth in about one-third of cases Rapid growth phase in the first 6 months ~75% regress by 7 years old spontaneously (Option B)
Appearance	Red, raised nodular mass ("strawberry nevus") Deep orbital mass: proptosis and globe displacement
Histopathology	[Image not available] Lobules of plump endothelial cells separated by fibrous septa
Management	Only done for complicated cases (affecting vision loss, systemic, etc.) and for cosmetic indications. Oral Propranolol (Option C) Intralesional corticosteroid Radiotherapy/Cryotherapy Surgery

Eyelid tumors:

BENIGN TUMORS	
Papilloma:	Most common benign tumor arising from surface epithelium
Xanthelasma	Middle aged woman, lipid deposits present within histiocytes
Hemangioma:	Capillary & Cavernous hemangioma
Neurofibroma:	Variant called Plexiform neurofibroma: S-shaped appearance Bag of worms feel on palpation
Keratoacanthoma:	[Image not available] Non-pigmented protrusion with keratin filled central crater
Naevi:	Congenital naevi ; Acquired naevi
PRE-MALIGNANT TUMORS	
Actinic keratosis	Over sun-exposed areas
Xeroderma pigmentosum	Defective nucleotide excision repair
MALIGNANT TUMORS	

Basal cell cancer:	Most common malignant tumor of eyelid Most common site: lower lid > medial canthus
Squamous cell cancer:	2nd most common malignant tumor of eyelid
Sebaceous gland carcinoma:	Recurrence - common
Malignant melanoma:	Most common type: superficial spreading type Worst prognosis type: Nodular melanoma

Solution for Question 2:

Correct Answer: A) a-2,b-3,c-1

Explanation:

Condition	Definition & Features	Treatment
[Image not available] External Hordeolum (Stye)	Acute suppurative inflammation of a lash follicle and associated glands of Zeis or Moll. Most common causative organism - <i>Staphylococcus aureus</i> . Symptoms: Acute pain with swelling on the anterior lid margin, watering and photophobia.	Hot compresses 2-3 times daily. Evacuate pus by epilating involved cilia when pus point forms. Rarely, surgical incision for large abscess. Antibiotic eye drops and ointment. Systemic anti-inflammatory and analgesics. Systemic antibiotics for early infection control. Identify and treat predisposing conditions in recurrent cases.
[Image not available] Internal Hordeolum	Suppurative inflammation of the meibomian gland with duct blockage.	Similar to external hordeolum treatment. Drain pus by a vertical incision from the tarsal conjunctiva when pus forms.

[Image not available] Chalazion	Chronic non-infective lipogranulomatous inflammation of the meibomian gland. Painless swelling of the eyelid, mild heaviness, blurred vision. Nodule is present slightly away from the lid margin. Malignant change into meibomian gland adenocarcinoma may be seen in elderly.	Conservative treatment: Hot fomentation, topical antibiotics, oral anti-inflammatory drugs. Intralesional steroid injection (triamcinolone). Incision (vertical on the conjunctival side, and horizontal on skin side), and curettage for conventional treatment. Diathermy for marginal chalazion. Oral tetracycline as prophylaxis in recurrent cases, especially with acne rosacea or seborrhoeic dermatitis.
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Solution for Question 3:

Correct Answer: A) Staphylococci

Explanation:

Staphylococci is the most common organism causing bacterial blepharitis

Bacterial blepharitis:

Table rendering failed

Solution for Question 4:

Correct Answer: B) Ankyloblepharon

Explanation:

The scenario is of adhesion of upper and lower lid - ankyloblepharon

Condition	Definition	Etiology
-----------	------------	----------

[Image not available] Symblepharon	Palpebral and bulbar conjunctiva adhesions.	Healing of raw surfaces on the palpebral and bulbar conjunctiva. Causes: Thermal or chemical burns, membranous conjunctivitis, injuries, conjunctival ulcerations, ocular pemphigus, Stevens-Johnson syndrome.
[Image not available] Ankyloblepharon	Adhesions between the margins of the upper and lower eyelids.	Congenital anomaly. Acquired adhesions after healing of chemical burns, thermal burns, ulcers, and traumatic wounds of the lid margins. Often associated with symblepharon.

Solution for Question 5:

Correct Answer: A) Madarosis

Explanation:

The scenario is of loss of eyelashes (madarosis) secondary to hypothyroidism

Condition	Definition	Local Causes	Systemic Causes
[Image not available] Madarosis (Option A)	Partial or complete loss of eyelashes or eyebrows.	Chronic blepharitis Cicatrizing conjunctivitis Complications from cryotherapy, radiotherapy, or surgery for eyelid lesions	Alopecia Psoriasis Hypothyroidism Leprosy Systemic lupus erythematosus Acquired syphilis
[Image not available] Poliosis (Option B)	Greying of eyelashes and eyebrows.	Chronic anterior blepharitis Sympathetic ophthalmitis Idiopathic uveitis	Vogt-Koyanagi-Harada syndrome Waardenburg syndrome Vitiligo Marfan syndrome Tuberous sclerosis

Tylosis (Option C) - thickening and scarring of the lid margin, a complication of long-standing bacterial blepharitis

[Image not available]

Solution for Question 6:

Correct Answer: C) Distichiasis

Explanation:

The case is of distichiasis - extra eyelash row

Eyelash rubbing the cornea can happen in:

- Trichiasis - Inward misdirection of lashes with the normal position of the lid margin (Option A ruled out)

[Image not available]

- Pseudotrichiasis - Inward turning of lashes along with the lid margin (Option B ruled out)
- Distichiasis - Extra row of eyelash (Option C)

[Image not available]

Clinical features	Foreign body sensation Photophobia Irritation Pain Lacrimation Reflex blepharospasm and photophobia (if the cornea is abraded) Conjunctival congestion
Complications	Recurrent corneal abrasions Superficial corneal opacities Corneal vascularization Non-healing corneal ulcer
Treatment	Epilation Electrolysis Cryoeplilation Surgical correction

Solution for Question 7:

Correct Answer: D) Starts from lateral eyelid and progresses medially

Explanation:

The given clinical scenario is indicative of involutional ectropion.

Involutional ectropion:

- Most common form (Option A)
- Caused by age-related laxity of lower eyelid structures. (Option C)
- It usually begins medially and extends laterally. (Option D)
- Progression leads to: Conjunctival inflammation, Keratinization, Lagophthalmos (Option B)

Type of Ectropion	Description	Causes	Clinical Features	Treatment
Involutional Ectropion	Most common; age-related	Horizontal laxity Canthal tendon laxity	Epiphora Tarsal thinning Lagophthalmos	Medial conjunctivoplasty for mild cases. Horizontal lid shortening for moderate cases. Byron Smith's modified Kuhnt-Szymanowski operation for severe cases. Lateral tarsal strip technique for generalized ectropion.
Cicatricial Ectropion	Scarring of anterior lamella	Trauma/surgery Scarring conditions (e.g., blepharitis)	Scarring Skin fibrosis	Mild cases: V-Y operation. Mild to moderate cases: Z-plasty (Elschnig's operation). Severe cases: Excision of scar tissue and full-thickness skin grafting.
Paralytic Ectropion	Facial nerve palsy (Bell's)	Bell's palsy Trauma/surgery	Exposure keratopathy	Temporary measures: Topical lubricants, eyelid taping, suture tarsorrhaphy. Permanent measures (if needed): Horizontal lid tightening with/without middle lamellar buttress, palpebral sling operation.

Mechanical Ectropion	Mass or lesion pushing eyelid	Tumors Conjunctival growths	Eyelid displacement Conjunctival chemosis	Treat the underlying mechanical causes.
Congenital Ectropion	Present at birth	Tarsal plate atrophy Down syndrome	Associated eyelid malformation	Mild cases may not require treatment. Moderate or severe cases: Horizontal lid tightening and full-thickness skin graft.

Grading of Ectropion:

GRADE	FEATURE
0	Normal eyelid appearance and function
I	Only punctum is everted
II	Lid margin everted and palpebral conjunctiva visible
III	Fornix is also visible

Solution for Question 8:

Correct Answer: C) 1-B, 2-A, 3-D, 4-C

Explanation:

1) Mechanical entropion	B) Phthisis bulbi
2) Cicatricial entropion	A) Trachoma
3) Congenital entropion	D) Most commonly involves lower eyelid
4) Involutional entropion	C) Senile entropion

Types of Entropion:

Type	Description	Etiology
1) Congenital Entropion	Present since birth, more common in the lower eyelid	Lower Eyelid: Improper development of lower lid retractors. Upper Eyelid: Mechanical effects due to microphthalmos.

2) Cicatricial Entropion	Commonly involves the upper eyelid.	Common causes: Trachoma, chemical burns, pemphigus, Stevens-Johnson syndrome, symblepharon.
3) Senile (Involutional) Entropion	Affects the lower eyelid in elderly individuals.	Horizontal lid laxity Vertical lid instability Overriding of pretarsal orbicularis. Laxity of orbital septum
4) Mechanical Entropion	Lack of support from the globe to the lids.	Occurs in conditions like phthisis bulbi, enophthalmos and after enucleation or evisceration operations.

Solution for Question 9:

Correct Answer: B) Wies operation

Explanation: The patient's symptoms of watering and photophobia, along with the image of an inward-turned lower lid margin, suggest entropion. The most appropriate surgical procedure for correcting entropion with minimal horizontal laxity is the Weis operation.

Involutional Entropion:

Description and etiology	Usually seen in lower eyelid in elderly people Horizontal lid laxity due to weakening of orbicularis muscle Vertical lid instability due to weakening of capsulopalpebral fascia Overriding of pretarsal orbicularis/Laxity of orbital septum
Clinical features	Foreign body sensation Lacrimation Photophobia Horizontal lid laxity
Surgical Techniques	i. Transverse Suture and Everting Suture: (Option A) Temporary cure (up to 18 months) Suitable for very old patients
ii. Wies Operation: (Option B) Long-term cure with minimal horizontal lid laxity	

iii. Plication of Lower Lid Retractors (Jones Operation): (Option C) Used in severe cases or after recurrence Shortens lower lid retractors.	
iv. Quickert Procedure: (Option D) For cases with marked horizontal lid laxity Combines horizontal lid shortening with Wies procedure	

Grading of entropion:

Grade-I	Only the posterior lid border is in-rolled
Grade-II	includes in-rolling up to the intermarginal strip
Grade-III	Whole lid margin including the anterior border is in-rolled

Complications include recurrent corneal abrasions, superficial corneal opacities, corneal vascularisation and even corneal ulceration.

Solution for Question 10:

Correct Answer: A) Orbicularis oculi

Explanation:

- The lid is made up of mainly thick connective tissue, which is called tarsal plate.
- The eyelid is divided by a grey line into anterior and posterior lamella: Anterior lamella structures include skin, subcutaneous tissue and orbicularis oculi (Option A). Posterior lamella structures include tarsal plate (Option C) and conjunctiva (Option D).
- Anterior lamella structures include skin, subcutaneous tissue and orbicularis oculi (Option A).
- Posterior lamella structures include tarsal plate (Option C) and conjunctiva (Option D).
- The space between supraorbital margin and infraorbital margin is filled by a thin connective tissue known as orbital septum (Option B).
- Anterior lamella structures include skin, subcutaneous tissue and orbicularis oculi (Option A).
- Posterior lamella structures include tarsal plate (Option C) and conjunctiva (Option D).

[Image not available]

Solution for Question 11:

Correct Answer: B) Lid retraction

Explanation:

Pseudoptosis: Apparent drooping of eyelid, unrelated to neurological or muscular abnormalities.

Causes of pseudoptosis:

Ipsilateral pseudoptosis	Contralateral pseudoptosis
Microphthalmos Enophthalmos (Option A) Phthisis bulbi (Option D) Hypotropia (Option C) Dermatochalasis	Eyelid retraction (Option B) High myopia Proptosis

Solution for Question 12:

Correct Answer: A) Mullerectomy

Explanation:

The given clinical scenario is suggestive of Small cell carcinoma of lung, which is associated with Horner syndrome. Preferred management of horner syndrome is Mullerectomy (aka) Fasanella-Servat operation.

Management of ptosis:

Tarso-conjunctivo-Mullerectomy (Fasanella-Servat Operation)	Mild ptosis (1.5-2 mm) with good levator function Horner syndrome (Option A)
LPS resection Conjunctival approach (Blaskowic's operation) Skin approach (Everbusch's operation)	Mild ptosis - Minimal resection of LPS (Option B)
Moderate ptosis - Moderate resection of LPS	

Severe ptosis - Maximal resection of LPS (Option C)	
Frontalis Sling Operation (Brow Suspension)	Severe ptosis with zero LPS function (Option D) Marcus gunn jaw winking ptosis - LPS excision with frontalis sling surgery Neurogenic ptosis.
Timing of surgery in Congenital ptosis:	Mild-moderate ptosis: after 4 years of age
Severe ptosis: as early as possible	

Solution for Question 13:

Correct Answer: C) 3 mm droop

Explanation:

Measurement of the degree of Ptosis:

- In unilateral cases, the difference in the vertical height of the palpebral fissures on the two sides indicates the degree of ptosis.
- In bilateral cases, measure the amount of cornea covered by the upper lid and subtract 2mm.

[Image not available]

Ptosis grading

Mild Ptosis	2mm droop
Moderate Ptosis	3mm droop
Severe Ptosis	4mm droop

Margin reflex distance (MRD)

- MRD 1: Distance between the upper lid margins and corneal light reflex. The normal value is 4-4.5mm.
- MRD 2: Distance between the pupillary reflex in the centre to the lower lid margin. Normal values 5 - 5.5mm.

Solution for Question 14:

Correct Answer: B) Levator palpebrae superioris

Explanation:

Congenital Ptosis: Associated with congenital weakness or maldevelopment of the levator palpebrae superioris (LPS) muscle.

Congenital ptosis probably results from a failure of neuronal migration or development with muscular sequelae secondary to this. A minority of patients have a family history.

Characteristic features:

- Drooping of one or both upper lids.
- Lid crease is either diminished or absent.
- Lid lag on downgaze due to tethering effect of abnormal LPS muscle.

Types of congenital ptosis:

1. Simple congenital ptosis:

- Unilateral or bilateral
- Non-progressive
- No fluctuation: Unlike myogenic ptosis, congenital ptosis does not worsen throughout the day or improve with rest.

2. Congenital ptosis with associated weakness of superior rectus muscle.

3. Blepharophimosis syndrome:

- Congenital ptosis
- Blepharophimosis
- Telecanthus
- Epicanthus inversus.

4. Congenital synkinetic ptosis (Marcus Gunn jaw-winking ptosis)

- Retraction of the ptotic lid with jaw movements due to stimulation of ipsilateral pterygoid muscle.
- It occurs due to trigemino-oculomotor synkinesis. The pathophysiology of Marcus Gunn Jaw winking phenomenon is a congenital misdirection of mandibular branch of trigeminal nerve, i.e., this synkinetic movement results from a congenital aberrant connection between motor branches of the trigeminal nerve controlling muscles of mastication and superior division of the oculomotor nerve controlling the levator palpebrae superioris.

[Image not available]

Solution for Question 15:

Correct Answer: B) Presents with normal function of the levator muscle

Explanation:

Acquired Aponeurotic Ptosis: Develops due to defect of levator aponeurosis in the presence of a normal functioning muscle.

Also known as involutional ptosis.

It includes:

- Senile Ptosis
- Postoperative Ptosis
- Ptosis associated with blepharochalasis.
- Traumatic dehiscence or disinsertion of aponeurosis.

Accompanied by ocular motor dysfunction (Option A): Aponeurotic ptosis is not associated with ocular motor dysfunction, which is more common in neurological forms of ptosis.

Results from a neurological condition affecting the third cranial nerve (Option C): This describes ptosis due to cranial nerve III palsy, not aponeurotic ptosis.

Usually involves a sudden onset of symptoms (Option D): Aponeurotic ptosis generally has a gradual onset rather than sudden, distinguishing it from other types of ptosis that may present more abruptly.

Solution for Question 16:

Correct Answer: B) Myasthenia gravis

Explanation:

Myasthenia gravis is a neuromuscular disorder characterized by weakness in voluntary muscles, which worsens with activity and improves with rest.

- It often presents with ptosis, diplopia, and generalized muscle weakness.

Myogenic ptosis is caused by a myopathy of the levator muscle itself, or by impairment of transmission of impulses at the neuromuscular junction (neuromyopathic).

Acquired Myogenic Ptosis: Occurs due to acquired disorders of the LPS muscle or the myoneural junction.

Acquired myogenic ptosis occurs in myasthenia gravis, myotonic dystrophy and progressive external ophthalmoplegia

It is seen in conditions like:

- Myasthenia gravis (M/C)
- Dystrophia myotonica
- Ocular myopathy
- Oculopharyngeal muscular dystrophy
- Thyrotoxicosis
- Lambert Eaton myasthenic syndrome
- Trauma to LPS muscle

Horner's Syndrome (Option A): Mild, constant ptosis which does not worsen with activity or improve with rest and no generalized fatigue or double vision.

Oculopharyngeal Muscular Dystrophy (OPMD) (Option C): Progressive, constant ptosis but no fluctuation or improvement with rest and no daily variation in symptoms.

Botulism (Option D): Sudden onset of ptosis. Symptoms do not improve with rest and not associated with daily fluctuation

Solution for Question 17:

Correct Answer: C) Anhidrosis and miosis

Explanation:

The given clinical vignette suggests that the patient most likely has Horner's syndrome, which can result from a Pancoast tumor at the apex of the lung. This tumor can compress the sympathetic nerves, leading to the classic triad of symptoms: ptosis, miosis (constricted pupil), and ipsilateral facial anhidrosis (lack of sweating).

Acquired Neurogenic Ptosis: Acquired ptosis with drooping of the upper eyelid caused by a disruption in the nerve supply to the muscles responsible for elevating the eyelid.

Causes:

- **Third Cranial (Oculomotor) Nerve Palsy:** The third nerve can be damaged by trauma, ischemia, infection, tumor, or compression, causing ptosis and diplopia. Oculomotor nerve palsy causes ptosis as it supplies the levator palpebrae superioris muscle.

The ptotic eyelid can be elevated with a frontalis sling or levator aponeurosis advancement.

- **Horner's Syndrome:** Involves a lesion in the sympathetic pathway, affecting the Müller's muscle, leading to mild ptosis accompanied by miosis (pupil constriction) and anhidrosis (loss of sweating).
- Ophthalmoplegic migraine
- Multiple Sclerosis

Diagnosis:

- Topical apraclonidine test is now considered the test of choice. Apraclonidine is an alpha-2 adrenergic agonist that also acts as a very weak alpha 1-adrenergic agonist; when topically applied in Horner's syndrome, it causes dilation of the affected pupil and possibly lid elevation, without change to the normal pupil.

Dilated pupil (mydriasis) (Option A): Horner's syndrome is characterized by a constricted pupil (miosis) rather than a dilated pupil (mydriasis).

Contralateral ptosis (Option B): Horner's syndrome typically affects only one side of the face, so the ptosis is unilateral and not contralateral.

Ophthalmoplegia (Option D): Not typically associated with Horner's syndrome. The syndrome primarily affects the sympathetic pathway and does not directly involve the muscles that move the eye.

Solution for Question 18:

Correct Answer: B) Tarsal plate

Explanation: The structure 'X' is the tarsal plate which belongs to the posterior lamella of the eyelid.

The eyelid is divided by a gray line into anterior and posterior lamella:

Anterior lamella	Posterior lamella
Skin Subcutaneous tissue Orbicularis oculi	Tarsal plate Conjunctiva

[Image not available]

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Previous Year Questions

1. All of the following are functions of superior oblique muscle except?

- A. Adduction
- B. Abduction
- C. Depression
- D. Intorsion

2. All of the following are features of oculomotor nerve palsy except?

- A. Exotropia
- B. Dilated pupil
- C. Limited Abduction
- D. Ptosis

3. Which muscle is implicated in ptosis in Horner syndrome?

- A. Müller muscle
- B. Levator aponeurosis
- C. Horner muscle
- D. Orbicularis oculi

4. Internal hordeolum is due to inflammation of

(or)

Internal hordeolum is due to inflammation of

- A. Zeis glands
 - B. Meibomian glands
 - C. Lacrimal glands
 - D. Moll glands
-

5. Which of the following is characterized by chronic granulomatous inflammation in the upper eyelid, causing painless swelling?

- A. Internal hordeolum
 - B. External hordeolum
 - C. Chalazion
 - D. Trachoma
-

6. What is the recommended course of treatment for mild congenital ptosis?

- A. Blaskowicz's operation
 - B. Everbusch's operation
 - C. Fasanella-Servat operation
 - D. Frontalis Sling operation
-

7. What is the appropriate treatment for ptosis resulting from Horner's syndrome?

- A. Blaskowicz's operation
- B. Frontalis Sling operation

- C. Fasanella servat operation
 - D. Everbusch's operation
-

8. What is the most frequently observed eye manifestation in this disease in a 45-year-old patient who presents with progressive weakness throughout the day and is found to possess antibodies against nicotinic acetylcholine receptors?

- A. Miosis
 - B. Lagophthalmos
 - C. Exophthalmos
 - D. Ptosis
-

9. In a case of long-standing, uncontrolled diabetes, what is the most common characteristic of third nerve palsy?

- A. Limited abduction
 - B. Pseudoptosis
 - C. Ptosis
 - D. Pupillary dysfunction
-

10. Identify the instrument.

[Image not available]

- A. Evisceration spatula
- B. Chalazion clamp
- C. Enucleation spoon
- D. Bone punch

11. In a patient presenting with recurrent chalazion, which cancer should be suspected?

- A. Sebaceous cell carcinoma
 - B. Squamous cell carcinoma
 - C. Basal cell carcinoma
 - D. Kaposi's sarcoma
-

12. What is the order of insertion of muscles on Annulus of Zinn? A. Medial rectus B. Inferior rectus C. Lateral rectus D. Superior rectus

- A. A, B, C, D
 - B. B, C, A, D
 - C. D, C, A, B
 - D. C, A, B, D
-

13. What is the recommended treatment for a patient with Bell's palsy, who is experiencing symptoms of dry eye, redness, and irritation, along with the presence of multiple punctate epithelial erosions in the cornea as observed through a slit lamp examination?

- A. Tarsorrhaphy
 - B. Contact lens.
 - C. Corneal replacement
 - D. LASIK
-

14. What is the likely diagnosis for a patient who has a painless, firm nodule on the upper eyelid that is growing away from the lid margin?

- A. Styne
 - B. Entropion
 - C. Chalazion
 - D. Blepharitis
-

15. Please identify the medical condition depicted in the provided image.

[Image not available]

- A. Entropion
 - B. Ectropion
 - C. Trichiasis
 - D. Ptosis
-

16. Identify the condition given in the image.

[Image not available]

- A. Symblepharon
 - B. Ankyloblepharon
 - C. Congenital blepharophimosis
 - D. Congenital ptosis
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	1
Question 4	2
Question 5	3
Question 6	3
Question 7	3
Question 8	4
Question 9	3
Question 10	2
Question 11	1
Question 12	1
Question 13	1
Question 14	3
Question 15	1
Question 16	2

Solution for Question 1:

Correct Option A - Adduction:

- Adduction is not caused by superior oblique muscle.
- It is caused by inferior rectus muscle and superior rectus muscle.

Incorrect Options:

- Option B, C, and D: Superior oblique muscle causes Abduction, Depression, and Intorsion.

Solution for Question 2:

Correct Option C - Limited abduction:

- Limited abduction (inability to move the eye outward) is not a feature of oculomotor nerve palsy.
- This movement is controlled by the lateral rectus muscle, which is innervated by the abducens nerve (cranial nerve VI).
- Dysfunction of the abducens nerve leads to issues with abduction, not the oculomotor nerve.

Incorrect Options:

Option A, B, and D: .

- Oculomotor nerve palsy, or third nerve palsy, typically presents with several distinct features due to the dysfunction of the muscles and structures innervated by the oculomotor nerve (cranial nerve III). These features include: Ptosis: Drooping of the upper eyelid due to paralysis of the levator palpebrae superioris muscle. Exotropia: Outward deviation of the eye because the medial rectus muscle (which normally pulls the eye inward) is paralyzed. Dilated pupil: Also known as mydriasis, this occurs because the oculomotor nerve controls the constrictor pupillae muscle, and its dysfunction leads to unopposed action of the dilator pupillae muscle.
- Ptosis: Drooping of the upper eyelid due to paralysis of the levator palpebrae superioris muscle.
- Exotropia: Outward deviation of the eye because the medial rectus muscle (which normally pulls the eye inward) is paralyzed.
- Dilated pupil: Also known as mydriasis, this occurs because the oculomotor nerve controls the constrictor pupillae muscle, and its dysfunction leads to unopposed action of the dilator pupillae muscle.

Oculomotor nerve palsy, or third nerve palsy, typically presents with several distinct features due to the dysfunction of the muscles and structures innervated by the oculomotor nerve (cranial nerve III). These features include:

- Ptosis: Drooping of the upper eyelid due to paralysis of the levator palpebrae superioris muscle.
- Exotropia: Outward deviation of the eye because the medial rectus muscle (which normally pulls the eye inward) is paralyzed.
- Dilated pupil: Also known as mydriasis, this occurs because the oculomotor nerve controls the constrictor pupillae muscle, and its dysfunction leads to unopposed action of the dilator pupillae muscle.

Solution for Question 3:

Correct Option A - Müller muscle:

- Ptosis in Horner syndrome is due to the paralysis of the Müller's muscle, a smooth muscle responsible for a minor portion of upper eyelid retraction.

Incorrect Options:

Options B, C, and D are not responsible for causing ptosis in Horner's syndrome.

Solution for Question 4:

Correct Option - Meibomian glands:

- Internal hordeolum, is an inflammatory condition of the eyelid caused by the infection or blockage of the Meibomian glands. The Meibomian glands are specialized sebaceous glands located along the eyelid margin.
- When the Meibomian glands become obstructed or infected, typically due to bacterial colonization, it can lead to the development of an internal hordeolum. This causes localized inflammation and the formation of a painful, red lump on the inside of the eyelid.

[Image not available]

Incorrect Options:

Options A, C, and D are incorrect. Internal hordeolum is due to the inflammation of Meibomian gland.

Solution for Question 5:

Correct Option C - Chalazion:

- Chalazion is a chronic, non-infectious granulomatous (lipogranuloma) inflammation of the meibomian glands. It presents as a painless, firm swelling on the upper or lower eyelid. It is caused by blockage and subsequent inflammation of the oil glands.
- Histopathology shows a lipo granulomatous chronic inflammatory picture with extracellular fat deposits surrounded by lipid-laden epithelioid cells, multinucleated giant cells, and lymphocytes.

Incorrect options:

Option A - Internal hordeolum: This refers to an infection of the meibomian glands, which are located in the eyelid. It typically presents as a painful, red lump on the eyelid margin.

Option B - External hordeolum: Also known as a sty, it is an infection of the eyelash follicle or associated glands. It appears as a painful, red bump on the outside of the eyelid.

Option D - Trachoma is a contagious bacterial infection caused by Chlamydia trachomatis. It primarily affects the conjunctiva and can lead to scarring and distortion of the eyelid.

Solution for Question 6:

Correct Option C - Fasanella-Servat operation:

- Fasanella-Servat operation aka Tarso-conjunctivo mullerectomy.
- It is performed in cases having mild ptosis (1.5-2 mm) and good levator function. In it, the upper lid is everted and the upper tarsal border, along with its attached Muller's muscle and conjunctiva are resected.

Incorrect options

Options A and B - Blaskowicz's and Everbusch's operation is done for moderate ptosis (3 mm), it involves resection of LPS muscle.

Option D - Frontalis Sling operation: Frontalis Sling operation is performed for severe ptosis (4 mm), it involves resection of frontalis muscle.

Solution for Question 7:

Correct Option C - Fasanella-Servat operation:

- Horner syndrome is characterized by classic triad of: Mild ptosis due to paralysis of Muller's muscle. Miosis due to paralysis of dilator pupillae. Ipsilateral anhidrosis.
- Mild ptosis due to paralysis of Muller's muscle.
- Miosis due to paralysis of dilator pupillae.
- Ipsilateral anhidrosis.
- The treatment of choice for mild ptosis due to Muller's muscle paralysis is Fasanella-Servat operation, aka Tarso-conjunctiva muller ectomy, which involves resection of Muller's muscle, providing strength.
- Mild ptosis due to paralysis of Muller's muscle.
- Miosis due to paralysis of dilator pupillae.
- Ipsilateral anhidrosis.

Incorrect Options:

Options A and D - Blaskowicz's and Everbusch's operation is done for moderate ptosis (3 mm), it involves resection of LPS muscle.

Option B - Frontalis Sling operation: Frontalis Sling operation is performed for severe ptosis (4 mm), it involves resection of frontalis muscle.

Solution for Question 8:

Correct Option D - Ptosis:

- The clinical presentation of progressive fatigue worsening at the end of the day, along with the presence of antibodies against nicotinic acetylcholine receptors, suggests the diagnosis of myasthenia gravis.
- Myasthenia gravis is an autoimmune disorder characterized by muscle weakness and fatigue, particularly in the muscles controlled by voluntary movements.
- Ocular features of myasthenia gravis are ptosis (bilateral), diplopia, and nystagmus in extreme gazes.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 9:

Correct Option C - Ptosis:

- Neurogenic ptosis it is the most common feature of third nerve palsy in cases of long-standing, uncontrolled diabetes.
- Ocular manifestation seen in third nerve palsy are: A - Profound ptosis due to weakness of the levator muscle. B - Abduction and depression in the primary position due to unopposed action of the lateral rectus and superior oblique muscles. C - Normal abduction as the lateral rectus is intact D - Limited adduction due to medial rectus weakness. E - Limited elevation due to weakness of the superior rectus and inferior oblique. F - Limited depression due to weakness of the inferior rectus. G - Dilated pupil and defective accommodation due to parasympathetic palsy.
- A - Profound ptosis due to weakness of the levator muscle.
- B - Abduction and depression in the primary position due to unopposed action of the lateral rectus and superior oblique muscles.
- C - Normal abduction as the lateral rectus is intact
- D - Limited adduction due to medial rectus weakness.
- E - Limited elevation due to weakness of the superior rectus and inferior oblique.
- F - Limited depression due to weakness of the inferior rectus.

- G - Dilated pupil and defective accommodation due to parasympathetic palsy.
- A - Profound ptosis due to weakness of the levator muscle.
- B - Abduction and depression in the primary position due to unopposed action of the lateral rectus and superior oblique muscles.
- C - Normal abduction as the lateral rectus is intact
- D - Limited adduction due to medial rectus weakness.
- E - Limited elevation due to weakness of the superior rectus and inferior oblique.
- F - Limited depression due to weakness of the inferior rectus.
- G - Dilated pupil and defective accommodation due to parasympathetic palsy.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 10:

Correct Option B: Chalazion clamp

- The above instrument is a chalazion clamp because there is a solid blade (to be kept externally on the skin). There is a ring-shaped blade that needs to be placed on the inner conjunctival surface.
- The handle has a thumbscrew, which can be tightened and released to achieve the required pressure. There is a forceps-like handle. After aligning the position to enclose the chalazion in the ring, the clamp is tightened and fixed.

Incorrect Options:

Option A: Evisceration spatula: Used for removing only intraocular contents leaving the sclera and muscles of the eye intact.

[Image not available]

Option C: Enucleation spoon: Used to remove entire intraocular contents from the eye socket leaving the lining and muscle of the eye intact.

[Image not available]

Option D: Bone punch: This device helps remove tissues like polyps and tumors that obstruct the nose and cause infections.

[Image not available]

Solution for Question 11:

Correct Option A - Sebaceous cell carcinoma

- Recurrent chalazion can be a potential sign of sebaceous cell carcinoma, which is a rare but aggressive malignancy originating from the sebaceous glands. Sebaceous cell carcinoma can mimic chalazion-like lesions, making it important to consider this cancer in cases of persistent or recurrent chalazion.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for explanation.

Solution for Question 12:

Correct Option A - A, B, C, D:

Medial rectus > inferior rectus > lateral rectus > Spiral of tillaux

Recti muscles:

- It originates from the Annulus of Zinn
- Annulus of Zinn is a tendinous ring located at the apex of the orbit
- Insertion of the recti muscle: on the sclera, anterior to the equator: Spiral of tillaux

[Image not available]

- Distance from the limbus: Medial rectus muscle: 5.5 mm Inferior rectus muscle: 6.5 mm Lateral rectus muscle: 6.9 mm Superior rectus muscle: 7.7 mm
- Medial rectus muscle: 5.5 mm
- Inferior rectus muscle: 6.5 mm
- Lateral rectus muscle: 6.9 mm
- Superior rectus muscle: 7.7 mm

- Medial rectus muscle: 5.5 mm
- Inferior rectus muscle: 6.5 mm
- Lateral rectus muscle: 6.9 mm
- Superior rectus muscle: 7.7 mm

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 13:

Correct Option A - Tarsorrhaphy:

- In the case of an elderly patient with Bell's palsy who presents with dry eye, redness, and irritation, along with multiple punctate epithelial erosions in the cornea, the most appropriate management option is Tarsorrhaphy.
- It is a surgical procedure in which the eyelids are partially sewn together to reduce the exposure of the cornea.

Incorrect Options:

Option B - Contact lens: As the cornea is already eroded contact lenses should be avoided

Option C - Corneal replacement: Corneal replacement, such as corneal transplantation or corneal grafting is not recommended because bell's palsy will gain result in exposure keratopathy.

Option D - LASIK: LASIK (laser-assisted in situ keratomileusis) is a refractive surgery procedure used to correct myopia, hyperopia, and astigmatism.

Solution for Question 14:

Correct Option C - Chalazion:

- A hard, painless nodule that is growing outwards, away from the lid margin is a classic presentation of a chalazion.
- It is typically painless but can cause discomfort or cosmetic concerns if it becomes large.

[Image not available]

Incorrect Options:

Option A - Sty: A sty, also known as a hordeolum externum, is a painful, red, tender lump in the lid margin pointing anteriorly through the lid margin commonly caused by staphylococcus aureus...

Option B - Entropion: Entropion is a condition in which the eyelid turns inward, causing the lashes to rub against the eye's surface. It can cause irritation, redness, and tearing.

Option D - Blepharitis: Blepharitis is a chronic inflammation of the eyelids that can cause redness, itching, and flaking of the eyelid skin.

Solution for Question 15:

Correct Option A - Entropion:

- The above image is of Entropion. Entropion is the inward rolling of the eye margin towards the globe which is seen here.

Incorrect Options:

Option B - Ectropion is the outward rolling of the eyelid away from the globe.

Option C - Misdirection of the eyelids is known as trichiasis.

[Image not available]

Option D: Drooping of the eyelid of the eyelid is known as ptosis.

[Image not available]

Solution for Question 16:

Correct Option B - Ankyloblepharon:

- The condition of the given image represents Ankyloblepharon. Ankyloblepharon refers to a condition where the upper and lower eyelids are joined together by thin tags as seen in the image...
- The condition can be congenital or acquired caused after healing of chemical, thermal burns, and traumatic wound of the lid margin and is often associated with symblepharon.

Incorrect Options:

Option A - Symblepharon: Symblepharon is characterized by the adhesion of the palpebral and bulbar conjunctiva of the eyeball seen after burns, chemical injuries, ulcers, and surgery.

Option C - Congenital blepharophimosis: Congenital blepharophimosis is characterized by horizontal shortening of the eyelids, ptosis, congenital ectropion, and telecanthus.

Option D - Congenital ptosis: Congenital ptosis is characterized by drooping of the upper eyelid present at birth. This drooping can be mild or severe and can obstruct vision if severe.

Optics-II

1. A 6 y/o child presents with complaints of double vision. The ophthalmologist performs Worth's Four Dot Test, revealing that the child sees two red lights and three green lights as shown below in the image. Which test result most accurately describes the likely visual condition in this child?

[Image not available]

- A. Binocular Single Vision
- B. Left Eye Suppression
- C. Right Eye Suppression
- D. Diplopia

2. A 25-year-old patient presents with blurred vision and headaches. On examination, the patient is found to have regular astigmatism with the following prescription: Sphere: +2.00 D, Cylinder: -1.50 D, Axis: 90°. What is the refractive power of the eye along the 90° meridian?

- A. +2.00 D
- B. +0.50 D
- C. +3.50 D
- D. +1.25 D

3. A 45 y/o patient presents with difficulty in reading and blurred vision. An eye examination reveals astigmatism. Based on the classification of astigmatism, which of the following statements is accurate?

- A. In With-the-Rule (WTR) astigmatism, the horizontal meridian has greater curvature than the vertical meridian.
 - B. Compound myopic astigmatism is characterized by both principal meridians focusing light in front or behind the retina.
 - C. Irregular astigmatism involves a predictable pattern of refractive power changes along the meridians.
 - D. Bi-oblique astigmatism has principal meridians that are perpendicular to each other.
-

4. Which of the following statements about common treatments for presbyopia is correct?

- A. Bifocal lenses are designed to provide only distance vision and are not effective for near vision.
 - B. Multifocal IOLs (intraocular lenses) provide multiple focal points to help with both near and distant vision.
 - C. Conductive Keratoplasty (CK) offers permanent benefits for presbyopia.
 - D. Refractive Lens Exchange (RLE) involves using medication to improve lens flexibility and alleviate presbyopia.
-

5. A 45 y/o patient presents with significant astigmatism and is seeking surgical correction. The patient is not a candidate for LASIK. Which of the following procedures is most suitable for correcting high degrees of astigmatism in this case?

- A. Limbal Relaxing Incisions (LRI)
 - B. Photorefractive Keratectomy (PRK)
 - C. Toric Intraocular Lens (IOL) Implantation
 - D. Conductive Keratoplasty (CK)
-

6. Which of the following surgical procedures is most suitable for correcting hypermetropia up to +4D and allows for faster visual recovery?

- A. Photorefractive Keratectomy (PRK)
 - B. Conductive Keratoplasty (CK)
 - C. Laser Thermal Keratoplasty (LTK)
 - D. Laser in situ Keratomileusis (LASIK)
-

7. A 30-year-old patient with a history of myopia is considering LASIK surgery. Which of the following is NOT an indication of the procedure?

- A. The patient's refraction has been stable for the past 14 months
 - B. The patient is over 20 years old
 - C. Presence of corneal ectasia
 - D. Myopia of -7 D
-

8. A 32 y/o man with -14 D myopia presents for a consultation regarding refractive surgery. He is not a candidate for laser-based procedures due to thin corneas. Which of the following is the most appropriate surgical option for his condition?

- A. Phakic intraocular lens (IOL) implantation
 - B. Radial keratotomy (RK)
 - C. Clear lens extraction
 - D. Intercorneal ring (ICR) implantation
-

9. Match the following 1) Titmus fly test a) 2) TNO test b) 3) Frisby test c) 4) Random Dot E test d)

1) Titmus fly test	[Image not available] a)
2) TNO test	[Image not available] b)
3) Frisby test	[Image not available] c)
4) Random Dot E test	[Image not available] d)

- A. 1 - c, 2 - b, 3 - a, 4 - d
 B. 1 - b, 2 - c, 3 - a, 4 - d
 C. 1 - b, 2 - c, 3 - d, 4 - a
 D. 1 - c, 2 - b, 3 - d, 4 - a

10. Match the following images appropriately: 1. Streak Retinoscope a) 2. Priestly Smith Retinoscope b) 3. Trial Frame c) 4. Refraction Box d)

1. Streak Retinoscope	[Image not available] a)
2. Priestly Smith Retinoscope	[Image not available] b)
3. Trial Frame	[Image not available] c)
4. Refraction Box	[Image not available] d)

- A. 1- a, 2- d, 3- c, 4- b
 B. 1- b, 2- c, 3- a, 4- d
 C. 1- c, 2- a, 3- d, 4- b
 D. 1- d, 2- c, 3- a, 4- b

11. A 7-year-old child is evaluated for binocular vision using a synoptophore. The child successfully perceives two different images simultaneously and can superimpose incomplete but similar images into a single complete image. Which grade of binocular single vision is demonstrated by this child?

- A. Grade I - Simultaneous Perception
 - B. Grade II - Fusion
 - C. Grade III - Stereopsis
 - D. None of the above
-

12. A 7-year-old child with amblyopia in the right eye is prescribed occlusion therapy. The optometrist recommends patching the sound left eye for a certain number of hours each day. What is the typical recommended duration of patching for effective treatment in children with amblyopia?

- A. 1-2 hours per day
 - B. 3-4 hours per day
 - C. 4-6 hours per day
 - D. 12 hours per day
-

13. A 3-year-old girl presents with decreased vision in her right eye. Ophthalmic examination reveals a dense corneal opacity in the right eye, but no other significant structural abnormalities. Which type of amblyopia is most likely in this case?

- A. Strabismic amblyopia
- B. Toxic amblyopia

- C. Anisometropic amblyopia
 - D. Stimulus deprivation amblyopia
-

14. Which of the following is an absolute optical indication for contact lens use?

- A. Corneal ulcer
 - B. Bullous keratopathy
 - C. Unilateral aphakia
 - D. Aniridia
-

15. A 25-year-old female with myopia is looking for contact lenses that will be comfortable for long hours of wear and require minimal maintenance. She is involved in outdoor activities and is seeking a lens that provides comfort, ease of use, and reduced risk of irritation over extended periods. Which type of contact lens would be most suitable for her?

- A. Hard contact lenses
 - B. Rigid gas permeable lenses
 - C. Soft contact lenses
 - D. PMMA lenses
-

16. A 30-year-old patient undergoes a refraction assessment, and the optometrist performs a Jackson Cross Cylinder (JCC) test to fine-tune the cylindrical correction. During the test, the patient reports that one of the orientations of the cylinder appears clearer than the other. What is the purpose of using the Jackson Cross Cylinder in this context?

- A. To determine the spherical refractive error

- B. To measure the visual acuity at distance
 - C. To verify the axis and power of the cylindrical correction
 - D. To test for binocular vision and eye alignment
-

17. A 38-year-old patient presents for an eye examination. During the duochrome test, the patient reports that they can see the red letters more clearly than the green letters. What is the most likely refractive error for this patient?

- A. Hyperopia
 - B. Myopia
 - C. Astigmatism
 - D. Presbyopia
-

18. A patient presents with a spectacle prescription of -4.00 D sphere and +1.00 D cylinder at 90°. What is the transposed prescription?

- A. -3.00 D sphere -1.00 D cylinder at 180°
 - B. -5.00 D sphere -1.00 D cylinder at 90°
 - C. +4.00 D sphere -1.00 D cylinder at 90°
 - D. -1.00 D sphere +4.00 D cylinder at 180°
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	2
Question 4	2
Question 5	3
Question 6	4
Question 7	3
Question 8	1
Question 9	2
Question 10	4
Question 11	2
Question 12	3
Question 13	4
Question 14	3
Question 15	3
Question 16	3
Question 17	2
Question 18	1

Solution for Question 1:

Correct Answer: D) Diplopia

Explanation: Seeing two red lights and three green lights on Worth's Four Dot Test indicates the presence of diplopia (double vision), where the patient perceives two distinct images due to misalignment.

Binocular Single Vision Tests

Name of the test	Procedure	Results
[Image not available] Worth's Four Dot test	The patient wears a red lens on the right eye and a green lens on the left eye. They are shown a box with four lights: one red, two green, and one white. The patient is asked how many lights they see.	All four lights: BSV (binocular single vision). (Option A ruled out) All four lights with manifest squint: Harmonious ARC (anomalous retinal correspondence). Two red lights: Right eye suppression. (Option C ruled out) Three green lights: Left eye suppression. (Option B ruled out) Alternating red and green lights: Alternating suppression. Two red and three green lights: Diplopia (double vision). (Option D)
[Image not available] 4 Prism Base-out test	A 4-diopter base-out prism is placed in front of the eye suspected of having CSS or microtropia. The movement of both eyes is observed.	Bifoveal fixation (normal binocular vision): Both eyes move to maintain fixation and converge to correct for the prism-induced diplopia. Microtropia with suppression: When the prism is placed in front of the suppressed eye, no corrective eye movement is observed because the image falls within the suppression scotoma.
[Image not available] Bagolini striated glass test	Striated glasses are placed in front of each eye at 45° and 135°. The patient fixates on a single-point light source. Each eye sees an oblique streak of light perpendicular to the other	If the two streaks cross to form an 'X': BSV or harmonious ARC. If two lines are seen but don't intersect: Diplopia (double vision). If only one streak is seen: Suppression. If a gap is seen in one of the streaks: Central suppression scotoma (CSS).
[Image not available] Synoptophore	The patient views pictures inserted in two tubes, each with an optical system equivalent to a 6-meter distance. The images are moved to adjust for horizontal, vertical, and torsional misalignment.	Detects suppression and ARC. Assesses binocular function and misalignments in different positions of gaze. Helps plan surgery for squint by measuring and adjusting for misalignments in various gaze positions.

Solution for Question 2:

Correct Answer: A) +2.00 D

Explanation:

To calculate the refractive power along the 90° meridian:

- The sphere represents the base refractive power (+2.00 D).
- The cylinder (-1.50 D) represents the additional power applied at the axis perpendicular to the cylinder axis (in this case, at 180°).
- At 90° (the cylinder axis), the refractive power is equal to the sphere alone: +2.00 D.
- The cylinder power affects the meridian perpendicular to the cylinder axis, not the axis itself. For the given prescription: Sphere: +2.00 D, Cylinder: -1.50 D, Axis: 90°;
- The axis of the cylinder (90°) indicates the meridian where the cylinder has no effect. The cylinder power is applied at the meridian perpendicular to the axis, which is 180° in this case.
- Therefore, the power along the 90° meridian is +2.00 D.

Solution for Question 3:

Correct Answer: B) Compound myopic astigmatism is characterized by both principal meridians focusing light in front or behind the retina.

Explanation:

Compound myopic astigmatism occurs when both principal meridians focus light in front of the retina or behind it.

Astigmatism: A refractive error where different meridians of the eye have different refractive errors. Divided into 2 types: Regular and irregular

I. Regular Astigmatism: When refraction error changes uniformly from one meridian to the other. (2 principle meridians present)

Type	Description
With-the-Rule (WTR)	The vertical meridian has greater curvature than the horizontal meridian. (Option A ruled out)
Against-the-Rule (ATR)	The horizontal meridian is more curved than the vertical meridian.
Oblique Astigmatism	Principal meridians are at oblique angles (e.g., 45° and 135°), still perpendicular to each other.
Bi-oblique Astigmatism	Principal meridians are not perpendicular to each other (e.g., 30° and 100°). (Option D ruled out)

[Image not available]

[Image not available]

[Image not available]

Regular Astigmatism Based on Focal Lines:

Regular Astigmatism Based on Focal Lines:

Regular Astigmatism Based on Focal Lines:

Regular Astigmatism Based on Focal Lines:

[Image not available]

Type	Description
Simple Astigmatism	One meridian focuses on the retina; the other focuses in front (simple myopic) or behind (simple hypermetropic).
Compound Astigmatism (Option B)	Both meridians focus light either in front (compound myopic) or behind (compound hypermetropic) the retina.
Mixed Astigmatism	One meridian focuses in front of the retina (myopic), and the other behind (hypermetropic).

II. Irregular Astigmatism Involves multiple, changing meridians of refractive power, making it geometrically complex. (Option C ruled out). Usually due to Corneal scars.

IV. Sturm's Conoid:

[Image not available]

Sturm's conoid is a geometric concept used to visualize and understand the effects of astigmatism on light refraction in the eye. It provides a model for how light rays are focused in an astigmatic eye.

Concept	Description
Representation of Light Refraction	Light rays are refracted by the cornea/lens into a conoid shape, not a single point.

Two Focal Lines	Light focuses on two perpendicular lines (primary and secondary focal lines) in an astigmatic eye.
Circle of Least Confusion (CoLC)	The area between focal lines with the sharpest image.
Corrective Lenses	Spherical lenses align the CoLC with the retina; cylindrical lenses correct astigmatism by straightening the conoid.

Solution for Question 4:

Correct Answer: B) Multifocal IOLs (intraocular lenses) provide multiple focal points to help with both near and distant vision.

Explanation:

Multifocal IOLs are designed to offer multiple focal points, allowing for improved near and distance vision.

I. Surgical and Lens-Based Treatments for Presbyopia:

Treatment	Description	Advantages	Limitations
Lens Extraction and IOL Implantation	Removal of the natural lens and replacement with an artificial IOL. (Option D ruled out) Types: Clear Lens Exchange (CLE) Refractive Lens Exchange (RLE) Presbyopic Lens Exchange (PreLEx)	Permanent solution; maintains focus at varying distances.	Potential issues with glare and contrast sensitivity.
Multifocal IOLs (Option B)	Lenses with multiple focal points for different distances.	Reduces or eliminates the need for glasses.	Potential for glare and reduced contrast sensitivity; ~10% may need IOL exchange.
Bifocal IOLs	Lenses with two focal points for distance and near vision. (Option A ruled out)	Provides clear vision at two distances.	May not fully address all visual needs.

Conductive Keratoplasty (CK)	Uses radiofrequency energy to reshape the cornea, potentially providing multifocal functionality.	Can offer benefits for presbyopia.	Primarily used for hyperopia; results may not be permanent. (Option C ruled out)
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II. Emerging and Experimental Treatments for Presbyopia:

Treatment	Description	Advantages	Limitations
Laser-Induced monovision	Uses LASIK or PRK to reshape one eye for near vision while keeping the other for distance, reducing the need for glasses or contacts.	Less need for glasses or contacts for both near and distant vision. Vision correction tailored to individual needs. Rapid visual recovery post-procedure. Touch-ups can maintain optimal vision if needed.	Difficulty adjusting, leading to imbalance or discomfort. Impaired spatial judgment, affecting activities like driving. Less clarity in distance or near vision. Further near vision deterioration with presbyopia, requires additional correction.
Corneal Multifocality	Laser procedures to reshape the cornea to create a multifocal effect.	Potential to address presbyopia without lens replacement.	Experimental; still in development.
Scleral Expansion Surgery	Alters the sclera to restore accommodation.	Aims to restore accommodation.	Inconsistent results; less popular.
Intracorneal Inlays	Small devices are inserted into the cornea to correct presbyopia.	Can significantly benefit presbyopia.	Issues with biocompatibility; potential complications.
Laser Modification of the Natural Lens	Femtosecond lasers modify the natural lens's elasticity to restore accommodation.	An experimental approach aiming to restore natural focus.	Remains experimental; not widely available.

III. Low Vision Aids for Presbyopia:

Type	Description	Functional Benefits
Optical Devices	Magnifiers: Handheld, stand, and electronic magnifiers to enlarge text and images. Bifocal and Multifocal Lenses: Provide different optical powers for distance and near vision. (Option A ruled out)	Improves ability to see close-up objects; helps with reading and other tasks.

Non-Optical Devices	Lighting: Enhanced lighting solutions, such as task lamps. Large Print Materials: Books and labels in larger fonts.	Improves visibility; aids in reading and daily tasks.
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Solution for Question 5:

Correct Answer: C) Toric Intraocular Lens (IOL) Implantation

Explanation:

Toric Intraocular Lens (IOL) Implantation is appropriate for correcting high degrees of astigmatism, as it uses a specially designed lens with varying refractive powers to address significant astigmatism.

Astigmatism Correction:

I. Incisional Techniques:

Procedure	Description	Suitability	Limitations
Limbal Relaxing Incisions (LRI) / Arcuate Keratotomy (Option A ruled out)	Paired arcuate incisions were made on opposite sides of the steeper corneal meridian to flatten it, thereby reducing astigmatism. Compression sutures may be added for significant cases.	Suitable for reducing large corneal astigmatism, often post-keratoplasty. (unlike this patient)	Correction level depends on incision parameters.

Procedure	Description	Suitability	Limitations
LASIK	Involves creating a corneal flap and using an excimer laser to reshape the underlying stroma, correcting both spherical and cylindrical errors. Contraindicated in patients with corneal thickness less than 450µm.	Astigmatism up to 5D	Faster recovery, but flap complications are possible.
PRK / LASEK	Surface ablation techniques using an excimer laser to reshape	Low to moderate astigmatism	Longer healing time, higher discomfort.

(Option B ruled out)	the cornea after removing the epithelium. Corrects up to 3D of astigmatism. Contraindicated in patients unsuitable for LASIK.		
Procedure	Description	Suitability	Limitations
Toric Intraocular Lens (IOL) Implantation (Option C)	Replacement of the natural lens with a toric IOL designed to correct astigmatism by having different refractive powers in different meridians.	Suitable for astigmatism correction during cataract surgery or high degrees of astigmatism.	Implant rotation postoperatively can reduce effectiveness.

IV. Conductive Keratoplasty (CK)

Procedure	Description	Suitability	Limitations
Conductive Keratoplasty (CK) (Option D ruled out)	Radiofrequency energy heats and shrinks collagen fibres in the peripheral cornea, increasing corneal curvature.	Best suited for low to moderate hypermetropic astigmatism.	Results may not be permanent and could regress over time.

Solution for Question 6:

Correct Answer: D) Laser in situ Keratomileusis (LASIK)

Explanation:

LASIK is ideal for correcting hypermetropia up to +4D and offers faster visual recovery and greater postoperative comfort compared to surface ablation procedures like PRK.

Treatment Modalities for Hypermetropia:

I. Laser Refractive Procedures:

Procedure	Suitability	Limitations
LASIK	Moderate hypermetropia (up to +4D)	Faster recovery, more comfortable than PRK.

Surface Ablation (PRK) (Option A ruled out)	Low to moderate hypermetropia	Longer healing time, possible regression.
Laser Thermal Keratoplasty (LTK) (Option C ruled out)	Low degrees of hypermetropia	Effectiveness may diminish over time.

II. Conductive Keratoplasty (CK):

Procedure	Description	Suitability	Limitations
Conductive Keratoplasty (Option B ruled out)	Radiofrequency energy heats and shrinks collagen fibres in the peripheral cornea, increasing corneal curvature.	Low to moderate hyperopia	May require repeat procedures; non-permanent.

High hypermetropia may require:

- Clear Lens Extraction with IOL Implantation
- Phakic IOL Implantation

Solution for Question 7:

Correct Answer: C) Presence of corneal ectasia

Explanation:

- Refractive error correction surgeries like LASIK have gained widespread popularity.
- Such procedures are typically performed once the refractive error has stabilized, usually after the age of 20.
- This ensures that the vision has settled and is less likely to change significantly, making the surgery more effective and reliable.
- LASIK is contraindicated if there is a presence of ectasia or any other corneal pathology.
- Measuring central corneal thickness is crucial in preoperative evaluation. The residual stromal bed is calculated by subtracting the flap thickness and maximum ablation depth from the central corneal thickness. The minimal residual bed for LASIK is debated, but 250–300 μm is generally considered the minimum.

LASIK (Laser-Assisted In Situ Keratomileusis):

- Procedure: An anterior corneal flap of 130-160 microns is elevated using an automated microkeratome or femtosecond laser. Midstromal tissue is ablated with an excimer laser to flatten the cornea.
- An anterior corneal flap of 130-160 microns is elevated using an automated microkeratome or femtosecond laser.
- Midstromal tissue is ablated with an excimer laser to flatten the cornea.
- An anterior corneal flap of 130-160 microns is elevated using an automated microkeratome or femtosecond laser.
- Midstromal tissue is ablated with an excimer laser to flatten the cornea.

[Image not available]

- Indications: Myopia up to -8 D. (Option D ruled out) Patients must be over 20 years old, have stable refraction for at least 12 months, be motivated, and have no corneal pathology. (Options A & B ruled out) Contraindicated in cases of corneal ectasia or any other corneal pathology.
- Myopia up to -8 D. (Option D ruled out)
- Patients must be over 20 years old, have stable refraction for at least 12 months, be motivated, and have no corneal pathology. (Options A & B ruled out)
- Contraindicated in cases of corneal ectasia or any other corneal pathology.
- Advances: Custom LASIK (C-LASIK): Corrects spherical, cylindrical, and aberrations for enhanced vision (beyond 6/6). Femto-LASIK (All laser LASIK or No blade LASIK or iLASIK): Uses femtosecond laser for flap creation, offering greater precision. Custom Femto-LASIK (CF-LASIK): Combines femtosecond flap creation with corneal topography and wavefront-guided correction. Epi-LASIK (E-LASIK): This involves separating only the epithelial layer with a specialized device, avoiding complications related to the stromal flap.
- Custom LASIK (C-LASIK): Corrects spherical, cylindrical, and aberrations for enhanced vision (beyond 6/6).
- Femto-LASIK (All laser LASIK or No blade LASIK or iLASIK): Uses femtosecond laser for flap creation, offering greater precision.
- Custom Femto-LASIK (CF-LASIK): Combines femtosecond flap creation with corneal topography and wavefront-guided correction.
- Epi-LASIK (E-LASIK): This involves separating only the epithelial layer with a specialized device, avoiding complications related to the stromal flap.
- Advantages: Minimal or no postoperative pain. Rapid recovery of vision. Lower risk of perforation and residual haze compared to other methods.
- Minimal or no postoperative pain.
- Rapid recovery of vision.

- Lower risk of perforation and residual haze compared to other methods.
- Disadvantages: Very Expensive. Requires skilled surgery. Potential flap-related complications and other risks like infection, dry eye, and glare.
- Very Expensive.
- Requires skilled surgery.
- Potential flap-related complications and other risks like infection, dry eye, and glare.
- Myopia up to -8 D. (Option D ruled out)
- Patients must be over 20 years old, have stable refraction for at least 12 months, be motivated, and have no corneal pathology. (Options A & B ruled out)
- Contraindicated in cases of corneal ectasia or any other corneal pathology.
- Custom LASIK (C-LASIK): Corrects spherical, cylindrical, and aberrations for enhanced vision (beyond 6/6).
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- Custom Femto-LASIK (CF-LASIK): Combines femtosecond flap creation with corneal topography and wavefront-guided correction.
- Epi-LASIK (E-LASIK): This involves separating only the epithelial layer with a specialized device, avoiding complications related to the stromal flap.
- Minimal or no postoperative pain.
- Rapid recovery of vision.
- Lower risk of perforation and residual haze compared to other methods.
- Very Expensive.
- Requires skilled surgery.
- Potential flap-related complications and other risks like infection, dry eye, and glare.

Solution for Question 8:

Correct answer: A) Phakic intraocular lens (IOL) implantation

Explanation: Phakic IOL implantation is a suitable choice for patients with high myopia (generally greater than -8 D) who are not candidates for laser-based refractive surgery, such as LASIK, due to thin corneas. In this procedure, a lens is implanted in front of the natural lens to correct vision without removing the natural lens, preserving accommodation.

Refractive surgeries for myopia	
Incisional Method	
Radial Keratotomy (RK) (Option B ruled out)	Involves radial cuts in the cornea to flatten it, correcting myopia. Used for low to moderate myopia (2-6 D). No longer recommended due to the advent of safer procedures like LASIK.
Newer Methods	
Photorefractive Keratectomy (PRK)	It uses an excimer laser to reshape the cornea by removing a thin corneal layer. Suitable for correcting myopia up to -6 D.
Refractive Lenticule Extraction (SMILE):	A minimally invasive laser procedure where a small lenticule is removed to reshape the cornea. Effective for myopia up to -10 D, with or without astigmatism.
Intercorneal Rings (ICR): (Option D ruled out)	Rings are inserted into the cornea to flatten it and reduce refractive power. Results can be unpredictable, with risks such as corneal inflammation, so is not recommended
Orthokeratology	A non-surgical and reversible method involving rigid contact lenses worn overnight to reshape the cornea. Corrects myopia up to -5 D temporarily.
High Myopia Solutions	
Clear Lens Extraction (Option C ruled out)	Removal of the natural lens (without cloudiness) to correct very high myopia (-16 D to -18 D). Less common now due to risks like retinal detachment.
Phakic Intraocular Lens (IOL) Implantation (Option A)	Implantable lens placed without removing the natural lens. Best for myopia exceeding -8 D, especially in patients unsuitable for laser surgery.

Solution for Question 9:

Correct Answer: B) 1 - b, 2 - c, 3 - a, 4 - d

Explanation:

Tests for stereopsis

Titmus fly test	[Image not available]
Random Dot E test	[Image not available]
Lang test	[Image not available]
Random dot test	[Image not available]
Frisby test	[Image not available]
TNO Test	[Image not available]

Solution for Question 10:

Correct Answer: D) 1- d, 2- c, 3- a, 4- b

Explanation:

- Retinoscopy, also known as skiascopy or the shadow test is an objective method for determining the refractive error of the eye through neutralization.
- Principle: The procedure is based on observing the reflection of light as it passes through the pupil. The direction of this light reflection varies depending on the eye's refractive state. This variation is used to determine the eye's refractive error.

Aspect	Image
Streak Retinoscope Type: Self-illuminated retinoscope. Advantages: More sensitive than spot retinoscopy Spot retinoscope produces a spot of light that can be made larger or smaller whereas streak retinoscope produces a line or streak of light that can be rotated, made wider or narrower, or changed from convergent to divergent light. for detecting astigmatism. Popularity: More commonly used due to its enhanced ability to detect astigmatism.	[Image not available]
Priestly Smith Retinoscope Type: A specific type of mirror retinoscope. Design: Combines plane and concave mirrors. Function: Provides better light control and enhances the accuracy of retinoscopy. Cost: Generally inexpensive and commonly used.	[Image not available]

Trial Frame Pre-requisite for retinoscopy. Adjustable type, suitable for use in both children and adults.	[Image not available]
Refraction Box Contains a variety of spherical and cylindrical lenses with plus and minus powers. Includes a pinhole, occluder, and prisms.	[Image not available]

Solution for Question 11:

Correct Answer: B) Grade II - Fusion

Explanation: The ability to superimpose two incomplete but similar images into one complete image is characteristic of Grade II - Fusion.

Binocular Single Vision

- Coordinated use of both eyes to create a single mental image.
- Tested with a synoptophore.

Grades of Binocular Single Vision

Grade I - Simultaneous Perception	Grade II - Fusion	Grade III - Stereopsis
[Image not available] Ability to see two dissimilar objects simultaneously. (Option A ruled out) Tested by projecting different images to each eye (e.g., bird in one eye, cage in the other). Example: Seeing a bird inside a cage.	[Image not available] Power to superimpose two incomplete but similar images into one complete image. Tested by manipulating eye convergence/ divergence for positive and negative fusion ranges.	[Image not available] Ability to perceive depth (third dimension). (Option C ruled out) Tested with stereopsis slides using the synoptophore.

Solution for Question 12:

Correct Answer: C) 4-6 hours per day

Explanation:

For effective occlusion therapy in treating amblyopia, it is typically recommended to patch the stronger eye for 4-6 hours per day as per 'Amblyopia Treatment Study'. This duration helps stimulate the weaker eye (amblyopic eye) and encourages improvement in its visual acuity.

Treatment of amblyopia: Amblyopia treatment should be started as early as possible (younger the child, better the prognosis). Amblyopia therapy works best when initiated in young children under 3 years of age and comprise:

Occlusion therapy: Occlusion of the sound (normal) eye to force use of amblyopic eye is the main stay in the treatment of amblyopia

Methods of occlusion. Occlusion can be accomplished by an adhesive patch on skin, gauze pad and tape, use of Doyme's rubber occluder which can be stuck to spectacle lens, opaque contact lens, adhesive tape on glasses, and ground glass occluder.

- Full-time occlusion: Up to 2 years, the occlusion should be done in 2:1, i.e. 2 days in sound eye and one day in amblyopic eye; at the age of 3 years, 3:1; at the age of 4 years, 4:1; At the age of 5 years, 5:1; and after the age of 6 years 6:1.
- Part-time occlusion: 4-6 hours/day, as shown by Amblyopia Treatment Study (ATS), and is as effective as full-time occlusion.
- Maintenance occlusion: After vision equalization, part-time occlusion continues until at least 8 years of age.
- Ensure no media opacity or refractive errors before starting occlusion therapy.

Other Methods

- Penalization: Blurring vision in the sound eye with atropine or over-plus lenses.
- Computerized Home Vision Therapy: Uses specially designed software as an adjunct to occlusion therapy.
- Pharmacologic Manipulation Levodopa/Carbidopa: Studied as an adjunct to occlusion therapy. Other agents: Citicoline and omega fatty acids
- Levodopa/Carbidopa: Studied as an adjunct to occlusion therapy.
- Other agents: Citicoline and omega fatty acids
- Pleoptics: Pleoptic exercises are done to improve foveal fixation, though less commonly used now.
- Levodopa/Carbidopa: Studied as an adjunct to occlusion therapy.
- Other agents: Citicoline and omega fatty acids

Solution for Question 13:

Correct Answer:D) Stimulus deprivation amblyopia

Explanation:

Stimulus deprivation amblyopia occurs when there is a physical obstruction preventing adequate visual input from reaching the brain, such as a dense corneal opacity. In this case, the corneal opacity is blocking the vision in the right eye, leading to amblyopia, also known as "lazy eye."

Amblyopia: Definition, Types, and Clinical Characteristics:

Definition:

- Partial, reversible vision loss in one or both eyes.
- Absence of organic diseases of the ocular media, retina, visual pathway
- The eye appears normal, but vision is impaired.

Types of Amblyopia:

Type	Cause	Key Features
Strabismic Amblyopia (Option A ruled out)	The brain suppresses input from one eye due to constant squint (strabismus).	Common cause; underdevelopment of vision in the deviated eye.
Stimulus Deprivation Amblyopia (Option D)	Hindered vision due to obstruction from early life (e.g., congenital cataracts, ptosis, dense central corneal opacity).	Lack of visual stimulation affects normal development.
Anisometropic Amblyopia (Option C ruled out)	A significant difference in refractive error between eyes.	The brain favours the clearer eye, leading to amblyopia in the other.
Ametropic Amblyopia	High uncorrected refractive errors in both eyes.	Bilateral; is less severe compared to other types.
Meridional Amblyopia	Uncorrected astigmatism affecting specific visual meridians.	Affects one or both eyes depending on the astigmatism degree.

Clinical Characteristics:

- Reduced Visual Acuity: Affects recognition acuity more than resolution acuity. Graded by severity: Mild: 20/25 to 20/60 Moderate: 20/70 to 20/100 Severe: 20/200 or worse
- Affects recognition acuity more than resolution acuity.
- Graded by severity: Mild: 20/25 to 20/60 Moderate: 20/70 to 20/100 Severe: 20/200 or worse

- Mild: 20/25 to 20/60
- Moderate: 20/70 to 20/100
- Severe: 20/200 or worse
- Neutral Density Filter Test: Slight improvement in visual acuity through a filter (unlike organic eye diseases).
- Crowding Phenomenon: Difficulty identifying letters on crowded charts (e.g., Snellen chart).
- Fixation Pattern: Eccentric fixation (focusing away from the fovea) indicates severity.
- Color Vision: Usually normal but affected in severe cases (vision below 6/36).
- Affects recognition acuity more than resolution acuity.
- Graded by severity: Mild: 20/25 to 20/60 Moderate: 20/70 to 20/100 Severe: 20/200 or worse
- Mild: 20/25 to 20/60
- Moderate: 20/70 to 20/100
- Severe: 20/200 or worse
- Mild: 20/25 to 20/60
- Moderate: 20/70 to 20/100
- Severe: 20/200 or worse

Treatment of amblyopia: Amblyopia treatment should be started as early as possible (younger the child, better the prognosis). Amblyopia therapy works best when initiated in young children under 3 years of age.

Toxic amblyopia (Option B): Toxic amblyopia, also known as toxic/nutritional optic neuropathy, is caused by damage to the optic nerve fibres due to exposure to toxins like tobacco, alcohol, and certain medications.

Solution for Question 14:

Correct Answer: C) Unilateral aphakia

Explanation:

Unilateral aphakia (the absence of the lens in one eye) is an absolute optical indication for contact lens use. A contact lens can provide the necessary optical correction for the missing lens, which is crucial for restoring proper vision in the affected eye.

Contact Lens Indications and Contraindications:

Indications for Contact Lens Use:

- Cosmetic: For patients with refractive errors seeking aesthetic benefits.
- Absolute Indications: Anisometropia Unilateral aphakia (Option B) Keratoconus Irregular astigmatism
- Anisometropia
- Unilateral aphakia (Option B)
- Keratoconus
- Irregular astigmatism
- Anisometropia
- Unilateral aphakia (Option B)
- Keratoconus
- Irregular astigmatism
- Corneal diseases: Corneal ulcers, bullous keratopathy. (Options A & B)
- Iris diseases: Aniridia, coloboma, albinism. (Option D)
- Glaucoma
- Amblyopia
- Post-surgery: Bandage lenses promote healing after keratoplasty and micro corneal perforations.
- Chemical burns: Prevents symblepharon and preserves fornix depth.
- Exposure keratitis: Protects the cornea in cases of incomplete eyelid closure.
- Trichiasis: Shields cornea from irritation due to misdirected eyelashes.
- Used in procedures like gonioscopy, electroretinography, and fundus photography.
- Utilized during surgeries such as goniotomy, vitrectomy, and endocular photocoagulation.
- Masks corneal scars, corrects ptosis, and improves appearance in phthisis bulbi with cosmetic scleral lenses.
- Ideal for athletes, pilots, and actors for a wider field of view and stable vision.

Contraindications for Contact Lens Use:

- Mental incompetence, poor motivation.
- Chronic dacryocystitis, blepharitis, and recurrent styes.
- Chronic conjunctivitis.
- Dry eye syndromes.
- Corneal dystrophies and degenerations.

- Recurrent inflammatory conditions like episcleritis, scleritis, and iridocyclitis.

Solution for Question 15:

Correct Answer: C) Soft contact lenses

Explanation:

Soft contact lenses are the most suitable for long wear due to their comfort, flexibility, and lower risk of causing irritation to the eye.

Contact Lens Types and Properties:

Contact lenses are categorized based on their material and properties, mainly for correcting refractive errors:

Type of Contact Lens	Material	Key Properties	Common Disadvantages	Usage
Hard Lenses (Option A)	PMMA (polymethylmethacrylate) (Option D)	High optical quality, durable, cheap.	Poor oxygen permeability- limited wearing time, hydrophobic, risk of corneal abrasions.	Rarely used clinically.
RGP Lenses (Option B)	Silicone acrylate, Cellulose acetate butyrate (CAB)	Oxygen permeable, hard.	RGP lenses, unlike most soft contact lenses, must be fit to the individual patient's eye, making this a more laborious process.	Moderate use ("semi-soft" lenses).
Soft Lenses (Option C)	HEMA (hydroxyethyl-methacrylate)	Soft, pliable, hydrophilic, oxygen permeable, comfortable and well tolerated.	Prone to protein deposits, cracking, and infections.	Most frequently prescribed.

Solution for Question 16:

Correct Answer: C) To verify the axis and power of the cylindrical correction

Explanation:

The Jackson Cross Cylinder (JCC) is used during the refraction process to verify and fine-tune both the axis and power of the cylindrical correction for astigmatism.

[Image not available]

JCC consists of two cylinders with equal strength but opposite signs, placed at right angles to each other. It is mounted in a handle and commonly used with strengths of $\pm 0.25\text{D}$ and $\pm 0.50\text{D}$. The test helps to verify the power and axis of a cylindrical lens correction.

- The $\pm 0.25\text{D}$ cross cylinder is placed in the trial frame, aligned with the axis of the cylinder in the trial frame.
- In the first position (same sign), the cylindrical correction is enhanced by 0.25D .
- In the second position (opposite sign), the correction is diminished by 0.25D .
- If the patient's vision does not improve in either position, the power of the cylinder in the trial frame is correct.
- If the vision improves in either position, adjust the cylinder power accordingly and recheck until the optimal correction is achieved.
- The $\pm 0.50\text{D}$ cross cylinder is placed in the trial frame with its axis at 45° to the cylinder's axis.
- The patient is asked whether they notice any difference in visual clarity.
- If there is no change, the axis of the cylindrical correction is correct.
- If visual acuity improves in one position, rotate the cross cylinder (in the direction of the positive or negative cylinder components) and recheck the visual clarity.
- Repeat this process until the axis is properly aligned and the best correction is found.

Cylindrical Correction Tests:

Three key tests are used for verifying and refining cylindrical correction during subjective refraction:

Test	Purpose	Key Features
Jackson Cross Cylinder Test	Verifies cylindrical power and axis	Uses $\pm 0.25\text{D}$ or $\pm 0.50\text{D}$ cross cylinder to refine cylinder strength and axis. Adjusted based on vision improvement.

Astigmatic Fan Test	Confirms cylindrical correction	A dial with radiating lines at 10° intervals, known as an astigmatic fan is used. Some lines appear sharp in astigmatism. The concave cylinder is adjusted until all lines are equally sharp.
Stenopaic Slit Test	Identifies astigmatic axis and correction	A rotating slit identifies the axis. The clearest vision occurs when the slit aligns with the astigmatic axis. Not currently used.

Solution for Question 17:

Correct Answer: B) Myopia

Explanation:

In the duochrome test, individuals with myopia (nearsightedness) tend to see the red letters (longer wavelength) more clearly. Conversely, individuals with hyperopia (farsightedness) would generally see the green letters (shorter wavelength) more clearly. The longer wavelength (red) is refracted less than the shorter (green).

Spherical Correction Tests:

The following are three common tests used in subjective refraction to assess and refine spherical correction:

Test	Purpose	Procedure	Interpretation
Pinhole Test	Confirms the accuracy of optical correction in the trial frame	The patient views through a pinhole aperture.	An improvement in visual acuity while looking through a pin-hole indicates that optical correction in the trial frame is incorrect.
Duochrome Test	Refines the spherical component of the prescription	The patient views red and green letters to check for chromatic aberration.	Clear red letters: Myopia; Clear green letters: Hypermetropia; Both clear: Emmetropia.
Fogging Technique	Refines spherical correction (primarily for hypermetropia)	A strong convex lens is added (+2D for myopia, +4D for hypermetropia),	Helps the patient differentiate clarity, used to determine spherical

ique		and then gradually reduced until full vision is regained.	correction but not refractive errors.
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Solution for Question 18:

Correct Answer: A) -3.00 D sphere -1.00 D cylinder at 180°

Explanation:

-3.00 D sphere -1.00 D cylinder at 180° is the transposed spectacle prescription.

Transposition Process:

- Formula: New Spherical = Original Spherical + Cylinder
- Calculation: -4.00 D sphere + 1.00 D cylinder
- Result: $-4.00 + 1.00 = -3.00$ D sphere
- Original Cylinder Value: +1.00 D
- Transposed Cylinder Value: -1.00 D
- Original Axis: 90°
- Adjustment: Add 90° to the original axis
- Calculation: $90^\circ + 90^\circ = 180^\circ$

Transposed Prescription:

- New Spherical Value: -3.00 D
- New Cylinder Value: -1.00 D
- New Axis: 180°

Correct Transposed Prescription: -3.00 D sphere -1.00 D cylinder at 180°

Options B, C & D are numerically incorrect.

Optics-I

1. Which of the following are features of direct ophthalmoscopy? Erect, virtual image is formed 2.5x magnification Large field of view Peripheral retinal lesions can be examined Routine fundus examination Semi-dark room required

- A. 1,5,6
- B. 1,2,3,4,5,6
- C. 1,3,6
- D. 3,4

2. Match the following Powers with the type of astigmatism they belong to. Powers Types of Astigmatism i) +2D Cyl 1. Simple hypermetropic ii) +1D Sph & +2D Cyl 2. Compound hypermetropic iii) -2D Sph & +3D Cyl @180 degrees 3. Mixed against the rule iv) -2D Sph & +1D Cyl @90 degrees 4. Compound myopic with the rule v) -1D Sph and +2D Cyl @90 degrees 5. Mixed with the rule

Powers	Types of Astigmatism
i) +2D Cyl	1. Simple hypermetropic
ii) +1D Sph & +2D Cyl	2. Compound hypermetropic
iii) -2D Sph & +3D Cyl @180 degrees	3. Mixed against the rule
iv) -2D Sph & +1D Cyl @90 degrees	4. Compound myopic with the rule
v) -1D Sph and +2D Cyl @90 degrees	5. Mixed with the rule

- A. I-a, ii-b, iii-c, iv-d, v-e
- B. I-a, ii-b, iii-e, iv-d, v-c
- C. I-a, ii-b, iii-d, iv-c, v-e
- D. I-e, ii-c, iii-b, iv-d, v-a

3. All of the following are true regarding the procedure shown below except

[Image not available]

- A. It's an objective method for refraction testing
 - B. Red reflex moving with the movement of this instrument implies Hypermetropia
 - C. Results are unreliable if the patient's accommodation is active
 - D. Cycloplegics are only indicated for children below 5 years of age
-

4. A 50-year-old male presented to the ophthalmology clinic with complaints of difficulty in reading newspapers and books. Which among the following statements is inappropriate regarding the management of this patient?

- A. Roughly +1D convex lenses can be prescribed
 - B. Coexisting refractive error (myopia/hypermetropia) should be corrected first
 - C. 1.25% Pilocarpine can be beneficial in treatment
 - D. Conductive keratoplasty can be performed
-

5. A 55-year-old patient presents with difficulty reading up close and notices that their vision becomes blurry when trying to focus on objects at near distances. On examination, the near point is found to be farther away than normal. Which of the following best explains the patient's condition? A) The patient has a reduced amplitude of accommodation B) The patient's far point is at infinity C) The patient has a myopic refractive error D) The patient's near point has moved closer

- A. The patient has a reduced amplitude of accommodation B) The patient's far point is at infinity C) The patient has a myopic refractive error D) The patient's near point has moved closer
- B. The patient's far point is at infinity

- C. The patient has a myopic refractive error
 - D. The patient's near point has moved closer
-

6. All of the following are changes not related to accommodation except

- A. Anterior lens curvature almost remains the same
 - B. Posterior lens curvature is increased
 - C. Increased zonular tension
 - D. Ciliary muscle contraction
-

7. A patient post cataract lens extraction refused IOL placement and hence was prescribed spectacles. Later, he presented with complaints of visual discomfort. Choose the right combination of effects the patient would have likely felt. 1) Pincushion effect of distortion 2) Barrel effect of distortion 3) Transverse magnification increases as distance from the optical axis increases 4) Transverse magnification decreases as distance from the optical axis increases

- A. 1,3
 - B. 1,4
 - C. 2,3
 - D. 2,4
-

8. A 45-year-old patient presents with blurry vision and difficulty focusing after undergoing cataract surgery 3 weeks ago. On examination, the patient has no lens in the eye and no evidence of intraocular lens (IOL) implantation. Which of the following is the most appropriate initial management for this patient?

- A. Prescribe corrective glasses with a high refractive index

- B. Fit the patient with an intraocular lens
 - C. Perform a keratoplasty to correct vision
 - D. Initiate topical corticosteroids for inflammation
-

9. In the case of astigmatism, the vertical meridian is more curved than the horizontal. What kind of lens should be given?

- A. Concave cylindrical lens at 180°
 - B. Convex cylindrical lens at 180
 - C. Concave cylindrical lens at 90
 - D. None of the above
-

10. A few rays of light are focused on the retina in one meridian and a few others are focused in front of the retina in another meridian. What is the diagnosis?

- A. Simple myopic astigmatism
 - B. Simple hypermetropic astigmatism
 - C. Mixed astigmatism
 - D. Compound myopic astigmatism
-

11. A 70-year-old patient presents with difficulty focusing on both near and distant objects. Examination reveals high hypermetropia. What is the expected funduscopy finding

- A. Smaller optic disc
- B. Shotsilk retina
- C. Pseudopapillitis

D. All of the above

12. A 65-year-old patient presents with gradually worsening blurry vision over the past year. On examination, the lens shows yellowish discolouration with increased opacity, consistent with nuclear sclerosis. What kind of myopia has he developed secondary to this?

- A. Axial myopia
 - B. Curvatural myopia
 - C. Index myopia
 - D. Positional myopia
-

13. Identify X:

[Image not available]

- A. Angle alpha
 - B. Angle beta
 - C. Angle gamma
 - D. Angle kappa
-

14. Snellen chart is used to assess which of the following?

- A. Colour vision
 - B. Contrast sensitivity
 - C. Peripheral vision
 - D. Visual acuity
-

15. A 65-year-old patient with a mature cataract presents for an eye examination. During the assessment, the ophthalmologist observes the absence of a specific Purkinje image. Which Purkinje image is most likely to be absent in this case?

- A. First Purkinje image
 - B. Second Purkinje image
 - C. Third Purkinje image
 - D. Fourth Purkinje image
-

16. Which of the following is the most sensitive test for both congenital and acquired colour vision defects?

- A. Ishihara Test
 - B. Farnsworth-Munsell 100 Hue Test
 - C. Hardy-Rand-Rittler (HRR) Test
 - D. Holmgren Wool Test
-

17. A 40-year-old woman presents with a 3-day history of painful vision loss in her right eye. She has a history of multiple sclerosis. On examination, her visual acuity is 20/30 in the right eye, and a RAPD is noted. She reports difficulty distinguishing red and green colors in the right eye, but her left eye is normal on color vision testing. Which of the following is the most likely cause of her acquired color blindness?

- A. Retinal vein occlusion
- B. Optic neuritis
- C. Diabetic retinopathy
- D. Macular degeneration

18. Which of the following statements regarding congenital colour blindness is TRUE?

- A. It is more common in females than males.
 - B. It is an autosomal recessive condition.
 - C. It primarily affects the perception of red and green colours and is the most common type.
 - D. It always results in total loss of colour vision.
-

19. Which type of colour blindness is caused by the absence of green cones in the retina?

- A. Protanopia
 - B. Deutanopia
 - C. Tritanopia
 - D. Achromatopsia
-

20. At what age do infants typically begin to perceive and distinguish between different colors?

- A. At birth
 - B. 1-2 months
 - C. 3-4 months
 - D. 6-9 months
-

21. A 55-year-old male presents with complaints of difficulty driving at night and reading in dim light, despite having normal visual acuity during his eye exam. He underwent cataract surgery six months ago but still experiences these visual issues. Which of the following tests is most appropriate to assess his current visual problem?

- A. Snellen Chart
- B. Ishihara Test
- C. Letter Contrast Sensitivity Test
- D. Visual Field Test

22. A patient's visual acuity improves significantly when looking through a small, opaque disc with a tiny hole, as shown in the image below. What is the likely cause of this improvement in visual acuity?

[Image not available]

- A. Cataracts
- B. Refractive error
- C. Optic nerve damage
- D. Macular degeneration

23. Match the following visual acuity charts used to assess children of preschool age. 1) Allen test cards a) 2) HOTV test b) 3) Styicar Test c) 4) Cardiff Acuity test d)

1) Allen test cards	[Image not available] a)
2) HOTV test	[Image not available] b)

3) Stycar Test	[Image not available] c)
4) Cardiff Acuity test	[Image not available] d)

- A. 1-a; 2-b; 3-c; 4-d
- B. 1-a; 2-b; 4-c; 3-d
- C. 4-a; 2-b; 3-c; 1-d
- D. 1-a; 3-b; 2-c; 4-d

24. Which among the following is used to check visual acuity in children less than 1 year? 1) 2) 3) 4)

1)	[Image not available]
2)	[Image not available]
3)	[Image not available]
4)	[Image not available]

- A. 1 and 4
- B. 2 and 3
- C. All except 4
- D. All except 1

25. Which among the following is used to check visual acuity in illiterate adults? 1) 2) 3) 4)

1)	[Image not available]
2)	[Image not available]

3)	[Image not available]
4)	[Image not available]

- A. 1 and 3
 - B. 2 and 4
 - C. All except 1
 - D. None of the above
-

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Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	4
Question 4	1
Question 5	1
Question 6	4
Question 7	1
Question 8	2
Question 9	1
Question 10	1
Question 11	4
Question 12	3
Question 13	4
Question 14	4
Question 15	4
Question 16	2
Question 17	2
Question 18	3
Question 19	2
Question 20	3
Question 21	3
Question 22	2

Question 23	1
Question 24	4
Question 25	2

Solution for Question 1:

Characteristic	Direct Ophthalmoscopy	Indirect Ophthalmoscopy
Image of ophthalmoscope	[Image not available]	[Image not available]
Image Characteristics	Erect, virtual	Real, inverted
Magnification	About 15x (more in myopes, less in hypermetropes)	About 2.5x with a +20D lens, smaller with stronger lenses
Field of View	Limited, detailed central retina examination	Larger field of view
Uses	Routine fundus examination	Essential for assessing retinal detachment and peripheral retinal lesions
Prerequisites	Semi-darkroom	Darkroom, dilated pupils, convex lens (+20D), binocular ophthalmoscope

Solution for Question 2:

Powers	Types of Astigmatism
i) +2D Cyl	Simple hypermetropic
ii) +1D Sph & +2D Cyl	Compound hypermetropic
iii) -2D Sph & +3D Cyl @180 degrees	Mixed against the rule
iv) -2D Sph & +1D Cyl @90 degrees	Compound myopic with the rule
v) -1D Sph and +2D Cyl @90 degrees	Mixed with the rule

Steps to identify types of astigmatism:

- Astigmatism is based on the power of cylindrical lens. +1D Sph = No astigmatism.
- +1D Sph = No astigmatism.
- [+] power means Hypermetropic and [-] means Myopic. +2D Cyl = Hypermetropic -2D Cyl = Myopic
- +2D Cyl = Hypermetropic
- -2D Cyl = Myopic
- Simple astigmatism is when the power is present only for cylindrical lens. +2D Cyl = Simple Hypermetropic -2D Cyl = Simple Myopic
- +2D Cyl = Simple Hypermetropic
- -2D Cyl = Simple Myopic
- Compound astigmatism is when the same sign power is present for both Spherical and Cylindrical lenses. +1D Sph +2D Cyl = Compound Hypermetropic -1D Sph -2D Cyl = Compound Myopic
- +1D Sph +2D Cyl = Compound Hypermetropic
- -1D Sph -2D Cyl = Compound Myopic
- Mixed astigmatism is when the power of both lenses has opposite signs and the numerical value of the power of cylindrical lens is greater than spherical lens, irrespective of the sign. +1D Sph -2D Cyl = Mixed -1D Sph +2D Cyl = Mixed
- +1D Sph -2D Cyl = Mixed
- -1D Sph +2D Cyl = Mixed
- With the rule and against the rule can be calculated using this simple table.
- +1D Sph = No astigmatism.
- +2D Cyl = Hypermetropic
- -2D Cyl = Myopic
- +2D Cyl = Simple Hypermetropic
- -2D Cyl = Simple Myopic
- +1D Sph +2D Cyl = Compound Hypermetropic
- -1D Sph -2D Cyl = Compound Myopic
- +1D Sph -2D Cyl = Mixed
- -1D Sph +2D Cyl = Mixed

	180 degrees	90 Degrees
With the rule	[-]	[+]

Against the rule	[+]	[-]
------------------	-----	-----

- In the case of Spherical lens having greater power than cylindrical lens the concept of simple transposition is applied. Consider -xD Sph & +yD Cyl where $x > y$,
- Consider -xD Sph & +yD Cyl where $x > y$,
- Consider -xD Sph & +yD Cyl where $x > y$,
- New Sph = Old Sph - Old Cyl
- New Cyl = Opposite sign of Old Cyl
- New Axis = Perpendicular to Old Axis
- -2D Sph & +1D Cyl @90 degrees=? New Sph = -2 +1 = -1D New Cyl = -1D New Axis = 180 degrees -1D Sph -1D Cyl @180 degrees = Compound Myopic astigmatism with the rule
- New Sph = -2 +1 = -1D
- New Cyl = -1D
- New Axis = 180 degrees
- -1D Sph -1D Cyl @180 degrees = Compound Myopic astigmatism with the rule
- New Sph = -2 +1 = -1D
- New Cyl = -1D
- New Axis = 180 degrees
- -1D Sph -1D Cyl @180 degrees = Compound Myopic astigmatism with the rule

Note: This is based on the power of the cylindrical lens and axis only.

Solution for Question 3:

Retinoscopy:

Description	Also known as skiascopy or shadow test Objective method to assess refractive errors (Option A)
Principle	The direction of the reflected light traveling over the pupil depends upon the refractive state of the eye
Types of Retinoscopes	Mirror retinoscope: Plane (used commonly) Concave (in hazy media and high ametropia)

Self-illuminated retinoscope: Spot Streak retinoscope (more sensitive)	
Wet retinoscopy: used with cycloplegics Dry retinoscopy: without cycloplegics	
Static retinoscopy: No active accommodation Dynamic retinoscopy: Performed with active accommodation	
Procedure	The patient sits 1 meter from the examiner and light is reflected onto the eye. The examiner observes the red reflex and notes its movement in horizontal and vertical meridians.
Cycloplegics	Cycloplegics can be used in any age group if active accommodation is present. (Option D) Commonly used cycloplegics: Atropine: used in < 5 years age, Homatropine: 5-25 years Phenylephrine (Mydriatic) in elderly
Observations	[Image not available] With movement: Emmetropia or hypermetropia (Option B) Against movement: Myopia >1D No movement: Myopia 1D
Difficulties in Retinoscopy	Poor red reflex (small pupil, hazy media, high ametropia) Active accommodation (eliminated by cycloplegia) (Option C) Scissor or conflicting shadows

Solution for Question 4:

Given clinical scenario is suggestive of Presbyopia.

Definition	Physiological insufficiency of accommodation is typically associated with aging.
Etio - Pathogenesis	Age-related factors: Decreased elasticity of the lens capsule. Sclerosis of the lens. Possible decline in ciliary muscle power.

Premature causes: Uncorrected hypermetropia Chronic glaucoma	
Symptoms	Difficulty in near vision Asthenopic symptoms (eye fatigue) Intermittent diplopia
Treatment Options	Optical Treatment: Convex glasses based on age Correct refractive error for distance first (Option B) Types of Glasses: Unifocal, bifocal, or progressive (varifocal).
Pharmacological therapy: 1.25% Pilocarpine (FDA-approved) (Option C)	
Surgical options: Conductive keratoplasty (Option D) LASIK (Monovision/presbyopic multifocal-LASIK) Corneal inlays Multifocal IOL placement following lens extraction	

Approximate Convex glasses power by age (based on Donders table)

40 years	+1.00D
45 years	+1.50D
50 years	+2.00D (Option D)
55 years	+2.25D
60 years	+2.50D

Solution for Question 5:

- The amplitude of accommodation refers to the ability of the eye to focus on objects at different distances, specifically the difference between the near point and far point.
- As individuals age, the amplitude of accommodation naturally decreases, resulting in difficulty focusing on nearby objects (presbyopia). This is likely the cause of the patient's difficulty reading up close.

Changes related to accommodation:

Far Point (Punctum Remotum)	Emmetropic eye - the far point is at infinity (Option B) Hypermetropic eye - behind the eye Myopic eye - in front of the eye
Near Point (Punctum Proximum)	Increases with age (Option D)
Range of Accommodation	The distance between the far point and the near point
Amplitude of Accommodation	The difference in dioptric power needed to focus between the near point (P) and the far point (R). Expressed as $A = P - R$ The amplitude of accommodation decreases with age (Option A)

Myron Yanoff and Duker's textbook of ophthalmology, 6th edition, Page no: 26-27, 307.e3-e4, e8

Solution for Question 6:

Accommodation:

Definition	The mechanism by which the eye increases the power of the crystalline lens to focus on near objects, allowing diverging rays to be focused on the retina.
Changes	The curvature of the anterior surface of the lens increases (Option A) Posterior surface curvature almost remains the same (Option B) Contraction of the ciliary muscle (Option D) Relaxation of zonules (Option C) secondary to ciliary muscle contraction (Helmholtz's capsular theory)
Far Point (Punctum Remotum)	Emmetropic eye - the far point is at infinity Hypermetropic eye - behind the eye Myopic eye - in front of the eye
Near Point (Punctum Proximum)	Increases with age
Range of Accommodation	The distance between the far point and the near point.
Amplitude of Accommodation	The difference in dioptric power needed to focus between the near point (P) and the far point (R). Expressed as $A = P - R$ The amplitude of accommodation decreases with age.

Solution for Question 7:

Post-lens extraction, the patient will have hypermetropia. Convex spectacles are prescribed for the correction of hypermetropia, which could cause the Pincushion effect.

Pincushion distortion (Statement-1)	Barrel distortion (Statement-2)
[Image not available]	[Image not available]
Transverse magnification increases as distance from the optical axis increases. (Statement-3)	Transverse magnification decreases as distance from the optical axis increases. (Statement-4)
Seen with convex lenses.	Seen with concave lenses.

Solution for Question 8:

In this scenario, the patient is presenting with aphakia, which is the absence of a lens in the eye, a condition that can occur after cataract surgery if an intraocular lens (IOL) was not implanted or if it was removed. IOL implantation is the recommended corrective procedure.

Aphakia:

Definition	Absence of crystalline lens from the eye.
Aetiology	Congenital absence (rare) Surgical removal (common) Traumatic extrusion of lens (rare) Posterior lens dislocation (optical aphakia)
Optical Effects	High hypermetropia Reduced eye power to +44 D (from +60 D)
Symptoms	Defective vision (far and near) Erythropsia and cyanopsia (abnormal red and blue color defect)

Signs	Limbal/corneal scar (in surgical cases) Deep anterior chamber Iridodonesis & Jet-black pupil Only 2 Purkinje images (normal: 4) Hypermetropic small optic disc
Treatment	Correct refractive error with: Spectacles: the problem of spherical aberration and roving ring scotoma (jack-in-the-box phenomenon) Contact lenses Intraocular lenses (IOL) - best method Refractive LASIK surgery

Solution for Question 9:

This is a case of "with-the-rule astigmatism" in which a concave cylindrical lens at 180° or a convex cylindrical lens at 90° is used for correction

With-the-Rule Astigmatism (WTR)	Against-the-Rule Astigmatism (ATR)
The vertical meridian is more curved than the horizontal. Correction requires a concave cylindrical lens at 180° or a convex cylindrical lens at 90°. It is called "with-the-rule" because the vertical meridian is normally more convex than the horizontal due to eyelid pressure.	The horizontal meridian is more curved than the vertical meridian. Correction requires a convex cylindrical lens at 180° or a concave cylindrical lens at 90°.

Solution for Question 10:

- A refractive error where the eye's refraction varies across different meridians.
- As a result, light rays entering the eye do not converge to a single point, instead form focal lines.
- Regular Astigmatism Refractive power changes uniformly from one meridian to another. There are 2 principal meridians. Oblique Astigmatism- two principal meridians are not horizontal and vertical but are at right angles to each other (e.g., 45° and 135°). Bioblique Astigmatism- two principal meridians are not at right angles to each other (e.g., one at 30°

and the other at 100°). Irregular Astigmatism Irregular change in refractive power in different meridians, with multiple meridians that do not admit geometrical analysis

- Refractive power changes uniformly from one meridian to another. There are 2 principal meridians.
- Oblique Astigmatism- two principal meridians are not horizontal and vertical but are at right angles to each other (e.g., 45° and 135°).
- Bioblique Astigmatism- two principal meridians are not at right angles to each other (e.g., one at 30° and the other at 100°).
- Irregular change in refractive power in different meridians, with multiple meridians that do not admit geometrical analysis

[Image not available]

Regular Astigmatism	Refractive power changes uniformly from one meridian to another. There are 2 principal meridians. Oblique Astigmatism- two principal meridians are not horizontal and vertical but are at right angles to each other (e.g., 45° and 135°). Bioblique Astigmatism- two principal meridians are not at right angles to each other (e.g., one at 30° and the other at 100°).
Irregular Astigmatism	Irregular change in refractive power in different meridians, with multiple meridians that do not admit geometrical analysis

Solution for Question 11:

All are findings in a hypermetropic eye.

Hypermetropia, or long-sightedness, is a refractive error in which parallel rays of light from distant objects focus behind the retina when the eye's accommodation is relaxed.

Type of Hypermetropia	Cause
Axial Hypermetropia	Shortening of the eyeball's axial length. Commonest form.
Curvatural Hypermetropia	Flatter curvature of the cornea, lens, or both.
Index Hypermetropia	Decreased refractive index of the lens, usually due to cortical sclerosis in old age or in treated diabetics.

Positional Hypermetropia	Posterior displacement of the crystalline lens.
Aphakia	Absence of the crystalline lens, either congenital or acquired (e.g., after surgery), leads to high hypermetropia.
Consecutive Hypermetropia	Overcorrected myopia after refractive surgery (e.g., LASIK, ICL) or underpowered intraocular lens (IOL) after cataract surgery.

Features:

- Small eyeball in high hypermetropia
- Smaller cornea
- Shallow anterior chamber
- Fundus Examination: Optic disc - Small (Option A), ill-defined margins, possibly simulating papillitis (pseudopapillitis) (Option C). Retina - Shot silk appearance (shining retina) (Option B).
- Optic disc - Small (Option A), ill-defined margins, possibly simulating papillitis (pseudopapillitis) (Option C).
- Retina - Shot silk appearance (shining retina) (Option B).
- Optic disc - Small (Option A), ill-defined margins, possibly simulating papillitis (pseudopapillitis) (Option C).
- Retina - Shot silk appearance (shining retina) (Option B).

Solution for Question 12:

Myopia, or shortsightedness, is a refractive error where parallel rays of light from distant objects focus in front of the retina when the eye's accommodation is relaxed.

Type of Myopia	Cause
Axial Myopia (Option A)	Increase in the anteroposterior length of the eyeball (most common form).
Curvatural Myopia (Option B)	Increased curvature of the cornea, lens, or both.
Positional Myopia (Option D)	Anterior placement of the crystalline lens in the eye.

Index Myopia (Option C)	Increase in the refractive index of the crystalline lens, often due to nuclear sclerosis.
Myopia due to Excessive Accommodation	Caused by spasm of accommodation, leading to excessive focusing power.

Table rendering failed

Solution for Question 13:

Angle kappa is the angle between the visual axis and the anatomical (pupillary) axis.

[Image not available]

Axes of the Eye:

Optic Axis	Passes through the center of the cornea (P), center of the lens (N), and meets the retina (R) on the nasal side of the fovea.
Visual Axis	Line joining the fixation point (O), nodal point (N), and the fovea (F).
Fixation Axis	Line joining the fixation point (O) and the center of rotation of the eye (C).

Visual Angles:

Angle Alpha (Option A)	Angle ANO is formed between the optic axis (AR) and visual axis (OF) at the nodal point (N).
Angle Gamma (Option C)	Angle ACO is formed between the optic axis (AR) and fixation axis (OC) at the center of rotation (C).
Angle Kappa (Option D)	Angle OPA formed between the visual axis (OF) and pupillary line (AP). The center of the cornea (P) is considered equivalent to the center of the pupil. Clinical significance: Positive angle kappa causes pseudo-exotropia. Negative angle kappa causes pseudo-esotropia.

Solution for Question 14:

Correct Answer: D) Visual acuity

Explanation:

Snellen's chart is used to measure visual acuity by assessing the eye's ability to distinguish details or separate points.

Table rendering failed

Solution for Question 15:

Fourth Purkinje image (lens- vitreous humour):

- This image is formed by the posterior surface of the lens.
- In mature cataracts, this image is usually absent because the lens opacity obstructs the reflection from its posterior surface.

[Image not available]

First Purkinje image (Option A): (air-cornea) This image is formed by the anterior surface of the cornea. It is usually visible in cataract cases, as the cornea remains clear.

Second Purkinje image (Option B): (cornea-aqueous humour) This image is formed by the posterior surface of the cornea. It typically remains visible even with cataracts, as the cornea's posterior surface is not affected by lens opacity.

Third Purkinje image (Option C): (aqueous humour-lens) This image is formed by the anterior surface of the lens. In cases of mature cataract, this image might be visible but can be affected by increasing opacity.

Solution for Question 16:

Farnsworth-Munsell 100 Hue test is the most sensitive test for both congenital and acquired colour vision defects.

Tests for Colour Vision designed for:

- Screening defective colour vision from normal.
- Qualitative classification of colour blindness (protons, deutan, tritan).
- Quantitative analysis of deficiency degree (mild, moderate, marked).

Commonly employed tests:

- Pseudoisochromatic charts (most common, using Ishihara's plates)
- Hardy-Rand-Rittler plates (HRR)
- American Optical Colour Plate test
- Edridge-Green lantern test
- Farnsworth-Munsell 100 hue test
- Farnsworth D15 hue discrimination test
- City University colour vision test
- Nagel's anomaloscope
- Holmgren's wools test

Farnsworth-Munsell 100 Hue Test:

- Highly sensitive test that can detect both congenital and acquired colour vision defects across all types of red-green, blue-yellow, and total colour blindness.
- It evaluates a person's ability to discern small differences in hues.
- It consists of 85 hue caps and colour vision is judged by error score.
- Ishihara Test (Option A): Primarily used to detect red-green colour blindness. It is not very effective for diagnosing blue-yellow defects or for detecting acquired colour vision defects.
- Hardy-Rand-Rittler (HRR) Test (Option C): This test can detect both red-green and blue-yellow colour vision defects and is useful for both congenital and acquired conditions, but it is less sensitive than the Farnsworth-Munsell 100 Hue Test.
- Holmgren Wool Test (Option D): Older test using coloured wool yarns to assess colour vision. It is less sensitive than modern tests and is rarely used for clinical evaluation today.

[Image not available]

Ishihara Test (Option A): Primarily used to detect red-green colour blindness. It is not very effective for diagnosing blue-yellow defects or for detecting acquired colour vision defects.

[Image not available]

Hardy-Rand-Rittler (HRR) Test (Option C): This test can detect both red-green and blue-yellow colour vision defects and is useful for both congenital and acquired conditions, but it is less sensitive than the Farnsworth-Munsell 100 Hue Test.

[Image not available]

Holmgren Wool Test (Option D): Older test using coloured wool yarns to assess colour vision. It is less sensitive than modern tests and is rarely used for clinical evaluation today.

[Image not available]

Solution for Question 17:

Optic neuritis, often associated with multiple sclerosis, commonly affects the optic nerve, leading to disturbances in vision, including red-green colour vision defect.

Acquired colour blindness:

- Loss or alteration of colour vision due to external factors, typically involving damage to the eyes, optic nerve, or brain rather than a genetic cause.
- Unlike congenital colour blindness, which is present from birth, acquired colour blindness can develop at any point in life and maybe progressive or reversible, depending on the underlying cause.
- Associated with central scotoma or decreased visual acuity

Causes:

- Ocular Diseases: Conditions such as glaucoma, cataracts, age-related macular degeneration, diabetic retinopathy, and optic neuritis.
- Neurological Disorders: Multiple sclerosis, optic nerve damage, or brain lesions in areas responsible for colour perception.
- Drug Toxicity: Certain medications, such as deferoxamine, hydroxychloroquine, sildenafil, ethambutol or digoxin can induce colour vision defects.
- Chemical Exposure: Long-term exposure to industrial solvents, heavy metals, or other toxins.
- Trauma: Head or eye injuries that affect the visual pathways.

Types:

- Blue-yellow impairment: Seen in retinal lesions (CSR, macular oedema, shallow retinal detachment)
- Red-green deficiency: Seen in optic nerve lesions (optic neuritis, Leber's optic atrophy, compression of optic nerve)
- Acquired blue colour defect (blue blindness): Occur in old age due to increased sclerosis of the crystalline lens.

Retinal vein occlusion (Option A) typically causes sudden vision loss, often associated with retinal hemorrhages, but does not cause a specific color vision defect in isolation.

Diabetic retinopathy (Option C) causes gradual vision loss due to retinal changes (like microaneurysms, hemorrhages, and edema) but does not typically present with acute color vision defects like those seen in optic neuritis.

Macular degeneration (Option D) primarily affects central vision and may cause difficulty with reading or recognizing faces but does not typically cause red-green color vision defects.

Solution for Question 18:

- Group of inherited conditions that affect an individual's ability to perceive colours correctly.
- It typically occurs due to defects in the photoreceptor cells (cones) of the retina that are responsible for colour vision.
- This condition is usually present from birth and remains constant throughout life.
- Red-Green Color Blindness: The most common type (Option C).
- Typically inherited through an X-linked recessive pattern, which is why it is more common in males (Option A,B).
- Congenital colour blindness usually involves partial colour vision deficiency, not total colour blindness (Option D).

Congenital colour blindness is classified into:

Dyschromatopsia:

- Anomalous trichromacy
- Dichromacy
- Blue cone monochromatism

Achromatopsia/Rod monochromatism

Solution for Question 19:

Deuteranopia is a type of red-green colour blindness where green cone cells are absent or nonfunctional, leading to difficulty distinguishing between red and green hues.

Classification of colour blindness:

Trichromatic colour vision	Dichromatic colour vision	Monochromatic colour vision
The mechanism to appreciate all three primary colours is present but defective for one or two	One of the three primary colours is completely absent	Either have only one type of cone or total colour blindness.
Protanomalous: Defective red colour appreciation. Deuteranomalous: Defective green colour appreciation. Tritanomalous: Defective blue colour appreciation.	Protanopia: Complete red colour defect. (Option A) Deuteranopia: Complete green colour defect. Tritanopia: Absence of blue colour appreciation. (Option C)	Cone monochromatism: Only one type of cone presents with poor day vision and normal night vision. Rod monochromatism/ Achromatopsia: Total colour blindness with day blindness. (Option D)

Solution for Question 20:

While newborns see mostly in shades of grey, their colour perception begins to develop around 3-4 months of age, when they can start distinguishing between primary colours like red and blue.

- Fovea development begins at 6 weeks.
- Colour perception begins at 12 weeks(3 months).

Solution for Question 21:

The patient's difficulty with night driving and reading in dim light, despite normal visual acuity, suggests a problem with contrast sensitivity. The Letter Contrast Sensitivity Test is specifically designed to evaluate how well the patient can distinguish between varying contrasts, which is often impacted in conditions like post-cataract surgery or early-stage optic nerve diseases.

Contrast sensitivity tests assess the smallest contrast a person can distinguish, offering insight into visual quality beyond standard visual acuity. Two main methods are used:

- Letter Contrast Sensitivity Tests Modified visual acuity charts with decreasing contrast (from 50% to 4%). Commonly used charts include the Pelli-Robson and Regan charts. Patients, placed 1 meter away, identify the smallest visible letter. Piel Robson Chart Contrast Sensitivity Grating Tests Present sinusoidal pattern gratings (bars) where patients determine tilt direction. Contrast and spatial frequency (bar spacing) are key variables. Cambridge contrast sensitivity gratings involve identifying patterns until they blend with a blank card. Snellen Chart (Option A) measures visual acuity but does not assess the ability to see in low-contrast situations. The Ishihara Test (Option B) evaluates color vision, not contrast sensitivity. The Visual Field Test (Option D) checks for peripheral vision defects, which are unrelated to contrast sensitivity.
- Modified visual acuity charts with decreasing contrast (from 50% to 4%).
- Commonly used charts include the Pelli-Robson and Regan charts.
- Patients, placed 1 meter away, identify the smallest visible letter.
- Piel Robson Chart
- Contrast Sensitivity Grating Tests Present sinusoidal pattern gratings (bars) where patients determine tilt direction. Contrast and spatial frequency (bar spacing) are key variables. Cambridge contrast sensitivity gratings involve identifying patterns until they blend with a blank card. Snellen Chart (Option A) measures visual acuity but does not assess the ability to see in low-contrast situations. The Ishihara Test (Option B) evaluates color vision, not contrast sensitivity. The Visual Field Test (Option D) checks for peripheral vision defects, which are unrelated to contrast sensitivity.
- Present sinusoidal pattern gratings (bars) where patients determine tilt direction.
- Contrast and spatial frequency (bar spacing) are key variables.
- Cambridge contrast sensitivity gratings involve identifying patterns until they blend with a blank card.
- Snellen Chart (Option A) measures visual acuity but does not assess the ability to see in low-contrast situations. The Ishihara Test (Option B) evaluates color vision, not contrast sensitivity. The Visual Field Test (Option D) checks for peripheral vision defects, which are unrelated to contrast sensitivity.
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[Image not available]

- Present sinusoidal pattern gratings (bars) where patients determine tilt direction.
- Contrast and spatial frequency (bar spacing) are key variables.
- Cambridge contrast sensitivity gratings involve identifying patterns until they blend with a blank card.
- Snellen Chart (Option A) measures visual acuity but does not assess the ability to see in low-contrast situations. The Ishihara Test (Option B) evaluates color vision, not contrast sensitivity. The Visual Field Test (Option D) checks for peripheral vision defects, which are unrelated to contrast sensitivity.

[Image not available]

Snellen Chart (Option A) measures visual acuity but does not assess the ability to see in low-contrast situations.

The Ishihara Test (Option B) evaluates color vision, not contrast sensitivity.

The Visual Field Test (Option D) checks for peripheral vision defects, which are unrelated to contrast sensitivity.

Solution for Question 22:

The pinhole test reduces the amount of light scattering in the eye and sharpens vision by narrowing the beam of light entering the pupil. An improvement in vision suggests a refractive error, such as myopia or astigmatism, that can be corrected with lenses.

Table rendering failed

- Cataracts (Option A): They cloud the lens, but using a pinhole does not typically improve vision, as the issue is related to lens opacification.
- Optic nerve damage (Option C): This condition, such as optic neuropathy, does not improve with the pinhole, as the issue is with the nerve, not light entering the eye.
- Macular degeneration (Option D): The visual acuity is unlikely to improve with a pinhole and may even decrease since the underlying problem is retinal damage, not a refractive error.

Solution for Question 23:

Visual acuity assessment- preschool	
1) Allen test cards Designed for children aged 2 to 4 years. 4 flashcards are used to test containing 7 schematic figures. They are identified from different distances points	[Image not available] a)
2) HOTV test Designed for children older than 4 years. Test involves identifying the letters "H," "O," "T," and "V." The letters decrease in size from the top to the bottom of the chart.	[Image not available] b)
3) Stycar Test (Screening Test for Young Children and Retardates)	[Image not available] c)
4) Cardiff Acuity card test Designed for children aged 2 to 4 years. Uses cards with pictures of varying sizes to assess visual acuity. Children are shown the cards and asked to indicate or focus on the picture.	[Image not available] d)

Solution for Question 24:

[Image not available]

1) Cardiff Acuity test

Designed for children aged 2 to 4 years. (Option C)

Uses cards with pictures of varying sizes to assess visual acuity.

Children are shown the cards and asked to indicate or focus on the picture.

[Image not available]

2) Tellers chart

Utilises infants' natural tendency to look at patterns i.e., it is a preferential Looking Test.

Suitable for infants

[Image not available]

3) Optokinetic Drum Test (OKN)

A rotating striped drum elicits involuntary eye movements.

Adjusting the stripe width helps estimate visual acuity, especially for non-verbal children.

[Image not available]

4) Visually evoked response test (VER)

Measures brain activity in response to visual stimuli

Solution for Question 25:

1) Snellen's chart Measures visual acuity by assessing the eye's ability to distinguish details or separate points.	[Image not available]
2) Tumbling "E" Visual Acuity Chart Involves matching the orientation of the letter "E" in different sizes. Used to check visual acuity illiterate. (Option B)	[Image not available]
3) LOGMAR chart LOGMAR is a refined method for measuring and expressing visual acuity, offering advantages over the traditional Snellen chart: Precision: LOGMAR charts are more sensitive and allow finer measurements compared to Snellen charts. Tracking Changes: The consistent progression aids in detecting subtle improvements or declines in vision.	[Image not available]

Reliability: The standardized format and equally legible letters enhance measurement reliability.	
4) Landolt's broken ring test The "C" is presented in various directions (top, bottom, left, or right), and the size of the "C" decreases progressively, similar to Snellen charts. Procedure: The individual is shown on the chart from a standardized distance. For each "C" presented, they indicate the direction of the gap (either verbally or by pointing). The test continues until the individual can no longer accurately determine the gap's direction. Used to check visual acuity in illiterate. (Option B)	[Image not available]
5) Jaeger chart Gold standard for reading vision tests	[Image not available]

Previous Year Questions

1. During the Worth 4 Dot Test, a patient sees only three green dots. What is the likely diagnosis?

- A. Right eye suppression
- B. Left eye suppression
- C. Crossed diplopia with esodeviation
- D. Crossed diplopia with exodeviation

2. Which of the following do not represent the corrected refractive error?

[Image not available]

- A. Myopia
- B. Hyperopia
- C. Presbyopia
- D. Astigmatism

3. What diagnosis is suggested by having different refractive powers (+5D in one eye and +1D in the other eye)?

- A. Refractive errors
- B. Amblyopia
- C. Anisometropic amblyopia
- D. Malingering

4. What is Anisometropia ?

- A. Difference in visual acuity
 - B. Difference in refractive error between 2 eyes
 - C. Difference in the axial length of the eyeball
 - D. Difference in size of eyeball
-

5. Which of the following is not a suitable application for a stenopic slit?

- A. Fincham's test
 - B. Corneal tattooing
 - C. Optical iridectomy
 - D. To find out the axis in astigmatism
-

6. What is the recommended treatment for a 3-year-old child with amblyopia in the right eye?

- A. Observe for 3 months
 - B. Occlusion of the right eye
 - C. Occlusion of the left eye
 - D. Surgical correction
-

7. Maximum visual field is seen in?

- A. Temporal

- B. Nasal
 - C. Inferior
 - D. Superior
-

8. In myopia, image is formed at:

- A. Anterior to retina
 - B. Posterior to retina
 - C. On retina
 - D. Uvea
-

9. The test given below in the image is used for :

[Image not available]

- A. Ocular deviation
 - B. Stereopsis test
 - C. Refractive errors
 - D. Colour Blindness
-

10. A newborn is hypermetropic by

- A. + 2.5 D
 - B. + 5 D
 - C. + 10 D
 - D. + 1 D
-

11. Which of the following is not used to assess color vision?

- A. Holmgren wool test
 - B. Ishihara chart
 - C. FM 100 Hue test
 - D. Ames test
-

12. At a distance of 60 m, what is the measurement of the angle formed by the largest letter on Snellen's chart?

- A. 5
 - B. 30
 - C. 50
 - D. 10
-

13. What is the main underlying factor responsible for regular astigmatism when studying the optics of a patient?

- A. Abnormal curvature of the cornea
 - B. Excessive eye rubbing
 - C. Prolonged exposure to bright light
 - D. Vitamin A deficiency
-

14. What is the recommended treatment for a 3-year-old child with amblyopia in the right eye?

- A. Observation
 - B. Spectacles or refraction with a near addition
 - C. Occlusion Therapy
 - D. Radial keratotomy
-

15. What is considered the most reliable tonometer in the medical field?

- A. Schiottz
 - B. Goldmann applanation tonometer
 - C. Tonopen
 - D. Pascal dynamic contour
-

16. What should be the power of prescription glasses for a 50-year-old man who is presbyopic?

- A. +1D
 - B. +2D
 - C. +3D
 - D. +4D
-

17. Identify the instrument.

[Image not available]

- A. Maddox wing
- B. Stenopic slit
- C. Maddox rod
- D. Phoropter

18. What would be the recommended treatment for simple myopic astigmatism among the options provided?

- A. +1.00 DS
 - B. -1.00 DC × 180 Degree
 - C. -1.00 DS
 - D. -1.00DS – 1.00 DC × 180 Degree
-

19. Which of the following lenses would be recommended to correct myopia?

- A. Convex
 - B. Concave
 - C. Cylindrical
 - D. Abraham lens
-

20. What is the likely diagnosis for a 5-year-old girl presenting with decreased best corrected visual acuity in the right eye after a school vision screening revealed diminution of acuity in the right eye without any examination findings of the eye and visual pathway?

- A. Brain lesion
 - B. Retinoblastoma
 - C. Congenital cataract
 - D. Amblyopia
-

21. The probable cause for a patient with a corneal scar having a visual acuity of 6/36, which improves to 6/18 with a pinhole, is most likely due to...

- A. Glaucoma
 - B. Malingering
 - C. Irregular astigmatism
 - D. Cataract
-

22. What are the advantages of using contact lenses instead of eyeglasses?

- A. Less prismatic effect
 - B. Protection from UV rays
 - C. Decrease infection
 - D. Decrease inflammation
-

23. In myopia, an image is formed at?

- A. Anterior to retina
 - B. Posterior to retina
 - C. On retina
 - D. None
-

24. What is the term used to describe the refractive error seen in the ocular examination image of a 60-year-old patient who has complained of reduced distant vision, but no longer requires spectacles for near vision?

[Image not available]

- A. Index myopia
 - B. Axial myopia
 - C. Curvatural myopia
 - D. Positional myopia
-

25. Which of the following refractive errors is associated with the image given below?

[Image not available]

- A. Presbyopia
 - B. Astigmatism
 - C. Hypermetropia
 - D. Myopia
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	3
Question 4	2
Question 5	2
Question 6	3
Question 7	1
Question 8	1
Question 9	2
Question 10	1
Question 11	4
Question 12	1
Question 13	1
Question 14	3
Question 15	2
Question 16	2
Question 17	3
Question 18	2
Question 19	2
Question 20	4
Question 21	3
Question 22	1

Question 23	1
Question 24	1
Question 25	2

Solution for Question 1:

Correct Option A - Right eye suppression:

- The Worth 4 Dot Test is used to assess binocular vision. Right eye suppression is diagnosed when the patient reports seeing only three green dots.

Incorrect Options:

Option B: A patient with left eye suppression will only see two red dots.

Option C and D: A patient with diplopia will see five dots- two red and three green dots.

Solution for Question 2:

Correct Option C - Presbyopia:

- Presbyopia does not represent in the above given image
- Presbyopia happens around the age of 40 years for an emmetropic person, less than 40 years in hypermetropic patients, and more than 40 years for myopic patients.
- At 40 years, around +1 D spherical lens is used to correct it. With every year that passes, around 0.5 D increases happen.
- The first image is of Myopia correction -It is done by concave lens
- Second image is of Hyperopia correction -It is done by convex lens
- Third image is of Astigmatism correction -It is done by cylindrical lens

Incorrect Options:

- Options A, B & D: Myopia, Hyperopia and Astigmatism are represented by the first, second and third images respectively

Solution for Question 3:

Correct Option C - Anisometropic Amblyopia:

- In the scenario, the child has two different refractive powers in both eyes; +5D in one eye and +1D in the other eye
- Anisometropia is the difference between the refractive power of both eyes when > 2.5 D
- Anisometropia can lead to Amblyopia

Incorrect Options:

Options A & D - Refractive errors & Malingering:

- Refractive errors and Malingering cannot be the answer

Option B - Amblyopia:

- Amblyopia can also be the answer but the better option is Anisometric Amblyopia
-

Solution for Question 4:

Correct Option B - Difference in refractive error between 2 eyes:

- Anisometropia is when total refraction of the two eyes is unequal.
- The difference in power of two eyes is more than 2.5 diopters.
- Anisometropia up to 2.5D is well tolerated and that between 2.5 and 4D can be tolerated depending upon the individual sensitivity.
- However, if it is more than 4D, it is not tolerated and is a matter of concern.

Incorrect Options:

Option A - Difference in visual acuity:

- Differences in visual acuity could be caused by conditions like myopia, hypermetropia, astigmatism, and presbyopia.

Option C - Difference in axial length:

- The difference in the axial length of the eyeball is seen in axial myopia.

Option D - Difference in size of eyeball:

- The difference in the size of the eyeball is seen in conditions like Microphthalmia and Buphthalmos.

Solution for Question 5:

Correct Option B - Corneal tattooing:

- The stenopic slit is not typically used for corneal tattooing. Corneal tattooing is a procedure in which pigment is injected into the cornea to alter its appearance. It is usually performed for cosmetic purposes like pterygium and pinguecula.

Incorrect Option:

Option A - Fincham's test: A stenopeic slit can be used in Fincham's test to differentiate the halos of acute congestive glaucoma and immature cataract.

Option C - Optical iridectomy: A stenopaic slit is used to determine the site of optical iridectomy.

Option D - To find out the axis in astigmatism: A stenopaic slit can be used to detect the axis of astigmatism.

Solution for Question 6:

Correct Option C - Occlusion of the left eye:

- In the above case, the patient is a 3-year-old child, and the treatment of amblyopia in such a case is by occlusion of the normal eye, which is the left eye here. It is occluded for 4 to 6 hours in a day. It is reversible if it is treated before the age of 8 years.

Incorrect Options:

- Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 7:

Correct Option A - Temporal:

- The maximum visual field is seen in the temporal region, i.e., 90 to 100 degrees.

Incorrect Options:

Option B - Nasal: The extent of the nasal region visual field is 60 degrees.

Option C - Inferior: The extent of inferior visual fields is 70 degrees.

Option D - Superior: The extent of superior visual field is the least, i.e., 50 degrees.

Solution for Question 8:

Correct Option A - Anterior to retina:

- In myopia (nearsightedness), the eyeball is too long or the cornea is too steep, causing the focal point to fall in front of the retina rather than on it. This results in a blurred image for distant objects.

Incorrect Options:

Option B - Posterior to retina: This option is not correct for myopia. Myopia involves the focal point being anterior to the retina.

Option C - On retina: Myopia results in the focal point being anterior to the retina, not on it. This can be corrected with concave lenses.

Option D - Uvea: The uvea is not directly related to the location of the focal point in myopia.

Solution for Question 9:

Correct Option B - Stereopsis test:

- The test shown in the image is Titmus stereotest, used for assessing stereopsis, which is the ability to perceive depth and three-dimensional vision
- The Titmus test is a three-dimensional polarized vectograph that comprises two plates presented in a booklet. To take the test, one needs to wear polarized spectacles before viewing the plates.
- The right plate shows a large fly, which tests for gross stereopsis, while the left plate shows nine squares, each containing four circles and animals. The test should be taken at a distance of 40 cm.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 10:

Correct Option: A

- Hypermetropia refers to a refractive error where the eye has difficulty focusing on near objects. In newborns, a certain degree of hypermetropia is considered normal. A refractive error of +2.5 D to 3.0 D is within the expected range for a hypermetropic newborn.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 11:

Correct Option D - Ames test:

- The Ames test is not used for color vision testing. Instead, it is a widely used bacterial test to assess the mutagenic potential of various substances. The test measures the ability of a chemical to induce mutations in certain strains of bacteria, indicating its potential to cause genetic damage.

Incorrect Options:

Option A - Holmgren wool test: The Holmgren wool test is a color vision test that assesses an individual's ability to match and identify colors using different shades of wool. It is commonly used to evaluate color vision deficiencies, particularly red-green color deficiencies.

Option B - Ishihara chart: The Ishihara chart is a well-known color vision test that uses plates with colored dots or numbers embedded within backgrounds of different colors and shades. It is widely used to detect red-green color deficiencies and evaluate color vision abnormalities.

Option C - FM 100 Hue test: The FM 100 Hue test is a color vision test that assesses an individual's ability to order and arrange colored caps or discs in a specific hue sequence. It measures the degree of hue discrimination and can help diagnose color vision deficiencies.

Solution for Question 12:

Correct Option: A

- The angle subtended by every alphabet on Snellen's chart is 5 minutes of arc.

[Image not available]

Incorrect Options:

Options B, C, and D are incorrect. All the letters on Snellen's chart subtend an angle of 5 minutes.

Solution for Question 13:

Correct Option A - Abnormal curvature of the cornea:

- Astigmatism is primarily caused by an abnormal curvature of the cornea, which is the clear front surface of the eye. In a normal eye, the cornea has a smooth and evenly curved shape, allowing light to focus properly on the retina. However, in astigmatism, the cornea is irregularly shaped, resulting in change in the refractive power uniformly from one meridian to another.

Incorrect Options:

Option B - Excessive eye rubbing: Excessive eye rubbing can irritate the eyes and potentially lead to infections like keratoconjunctivitis.

Option C - Prolonged exposure to bright light: Prolonged exposure to bright light can cause eye strain and discomfort.

Option D - Vitamin A deficiency: Vitamin A deficiency can lead to xerophthalmia characterized by night blindness and dryness of the eyes.

Solution for Question 14:

Correct Option C - Occlusion Therapy:

- In the above case, the patient is a 3 year old child and the treatment of amblyopia in such a case is by occlusion of the normal eye, which is the left eye here. It is occluded for 4 to 6 hours in a day. It is reversible if it is treated before the age of 8 years.

Incorrect Options: Option A - Observation is not the preferred treatment for a 3-year-old child with amblyopia in the right eye because amblyopia is a condition that can worsen if left untreated. Option B - Spectacles or refraction with a near addition may be helpful but maturing of the fovea of the amblyopic eye does not occur. Option D - Radial keratotomy is not a treatment for amblyopia. Radial keratotomy is a type of refractive surgery used to correct myopia by making incisions in the cornea to flatten it and reduce its curvature.

Solution for Question 15:

Correct Option B - Goldmann applanation tonometer.

- The gold standard tonometers for measuring intraocular pressure (IOP) is the applanation tonometer like Goldmann applanation tonometry.

Incorrect Options:

Option A - Schiottz: The Schiottz tonometer is an indentation tonometer that measures IOP based on the amount of force required to flatten the cornea.

Option C - Tonopen: The Tonopen is a handheld electronic tonometer that measures IOP by using both applanation and indentation method..

Option D - Pascal dynamic contour: The Pascal dynamic contour tonometer is a newer technology that measures IOP by analyzing the contour of the cornea during gentle deformation.

Solution for Question 16:

Correct Option B: +2D

- In presbyopia, the near focusing ability of the eye diminishes, requiring additional positive or converging power to be added to the eye's optical system for clear near vision.
- It happens around the age of 40 years for an emmetropic person, less than 40 years in hypermetropic patients, and more than 40 years for myopic patients.
- At 40 years, around +1 D spherical lens is used to correct it. With every year that passes, around 0.5 D increases happen.
- So, +2D is the most appropriate power of prescription glasses for a 50-year-old man with presbyopia.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 17:

Correct Option C - Maddox rod:

- The instrument shown in the image is a Maddox rod. It is a diagnostic tool used in ophthalmology to assess eye alignment and evaluate for ocular deviations or muscle imbalances. The Maddox rod consists of a cylindrical lens with parallel lines running along its length.
- The Maddox rod is typically used in the evaluation of horizontal strabismus (ocular misalignment) and is particularly useful in detecting and measuring the presence of a phoria (latent misalignment) or atropia (manifest misalignment).

Incorrect Options:

Option A - Maddox wing: The Maddox wing consists of two slits, the right eye sees the vertical white and horizontal red arrows and the left eye sees a vertical and horizontal line of numbers used for detecting the amount of phoria.

[Image not available]

Option B - Stenopic slit: It is useful to determine the site of iridectomy and helps in detecting the axis of astigmatism.

[Image not available]

Option D - Phoropter: A phoropter is a refractive instrument used during an eye examination to determine a patient's refractive error and prescribe corrective lenses. It contains a variety of lenses that can be adjusted to assess and refine a person's vision.

[Image not available]

Solution for Question 18:

Correct option B: $-1.00 \text{ DC} \times 180 \text{ Degree}$:

- Simple myopic astigmatism refers to a condition where there is myopia (nearsightedness) along with astigmatism in one eye and normal other eye.
- This prescription that indicates a correction for myopic astigmatism is $-1.00 \text{ DC} \times 180 \text{ Degree}$. The -1.00 DC (D- dioptre and C - cylindrical) represents the cylindrical power, which addresses astigmatism. A negative cylindrical power indicates that the astigmatism is in the myopic (nearsighted) direction. The " $\times 180 \text{ Degree}$ " denotes the axis of astigmatism. In this case, the astigmatism is at 180 degrees, meaning that the irregular curvature of the cornea or lens causing the astigmatism is aligned vertically. By prescribing $-1.00 \text{ DC} \times 180 \text{ Degree}$,

the correction aims to address both the nearsightedness (myopia) and the astigmatism in the correct axis. This combination is appropriate for a patient with simple myopic astigmatism.

- The -1.00 DC (D- dioptre and C - cylindrical) represents the cylindrical power, which addresses astigmatism. A negative cylindrical power indicates that the astigmatism is in the myopic (nearsighted) direction.
- The "× 180 Degree" denotes the axis of astigmatism. In this case, the astigmatism is at 180 degrees, meaning that the irregular curvature of the cornea or lens causing the astigmatism is aligned vertically.
- By prescribing -1.00 DC × 180 Degree, the correction aims to address both the nearsightedness (myopia) and the astigmatism in the correct axis. This combination is appropriate for a patient with simple myopic astigmatism.
- The -1.00 DC (D- dioptre and C - cylindrical) represents the cylindrical power, which addresses astigmatism. A negative cylindrical power indicates that the astigmatism is in the myopic (nearsighted) direction.
- The "× 180 Degree" denotes the axis of astigmatism. In this case, the astigmatism is at 180 degrees, meaning that the irregular curvature of the cornea or lens causing the astigmatism is aligned vertically.
- By prescribing -1.00 DC × 180 Degree, the correction aims to address both the nearsightedness (myopia) and the astigmatism in the correct axis. This combination is appropriate for a patient with simple myopic astigmatism.

Incorrect options

Option A: +1.00 DS

- This option indicates a simple hyperopic (farsighted) correction, which is not suitable for myopic astigmatism.

Option C: -1.00 DS

- This option represents a simple myopic correction without addressing astigmatism. It only corrects for nearsightedness and does not address the irregular curvature of the cornea or lens that causes astigmatism.

Option D: -1.00 DS – 1.00 DC × 180 Degree

- This option includes both a myopic correction (-1.00 DS) and an astigmatic correction (-1.00 DC × 180 Degree).

Solution for Question 19:

Correct Option B - Concave:

- The above is a case of myopic refractive error and the lens prescribed in this case is a concave lens. The boy reads the book too closely and also narrows his eyes while watching the television which are all presenting features of myopia. Myopia is also known as short sightedness.

Incorrect Options:

Option A - convex lens is used in the case of presbyopia and in the case of hypermetropia.

Option C - cylindrical lens is used in the case of astigmatism.

Option D - Abraham lens is a 66 D used for peripheral iridotomy with Nd:Yag laser..

Solution for Question 20:

Correct Option D - Amblyopia:

- The above is a case of amblyopia. In amblyopia, there is a decrease in best corrected visual acuity in one or both eyes with no apparent pathology of the eye and visual pathway..

Incorrect Options:

Option A - A lesion in the visual pathway can result in loss of vision but no examination findings were observed..

Option B - Retinoblastoma shows leukocoria which is absent here.

Option C - Loss of red reflex is seen in congenital cataract.

Solution for Question 21:

Correct Option C - Irregular astigmatism:

- The presentation suggests that the patient is a case of irregular astigmatism. Here the refractive error of the patient improved to 6/18 using a pinhole camera is suggestive of refractive error.
- The history of corneal scar that makes the surface irregular may indicate that the above case is that of irregular astigmatism.

Incorrect Options:

Option A - Pinhole does not improve vision in the case of glaucoma.

Option B - Malingering diagnosis can only be established after excluding all other causes which is not the case here.

Option D - Pinhole does not improve vision in case of cataract.

Solution for Question 22:

Correct Option A - Less prismatic effect:

- Contact lenses are placed directly on the eye's surface, reducing the prismatic effect that can occur with spectacles.
- Prismatic effect refers to the displacement of light as it passes through a lens, causing objects to appear shifted or distorted. Thicker the lens greater the prismatic effect.
- Contact lenses, being in direct contact with the eye, minimize the prismatic effect.

Incorrect Options:

Options B, C, and D are incorrect. These are not the advantages of using contact lens over spectacles.

Solution for Question 23:

Correct Option:

Option A - Anterior to retina:

Myopia, also known as nearsightedness, is a refractive error of the eye where the eyeball is longer than normal or the cornea has excessive curvature. This causes light rays to focus in front of the retina instead of directly on it.

Incorrect Options:

Option B - Posterior to retina: In hypermetropia, image is found behind the retina.

Option C - On retina: In emmetropic individual image is formed at the retina.

Option D - None: In myopic patients the image is formed in front of the retina.

Solution for Question 24:

Correct Option A - Index myopia:

- The type of refractive error seen in the given ocular examination image of a 60-year-old patient who complains of decreased distant vision but no longer uses glasses for near vision is called index myopia.
- The given image is of a nuclear cataract causing index myopia.
- Index myopia is caused by an increase in the refractive power of the eye's lens due to changes in the lens's refractive index, leading to convergence of rays in front of the retina resulting in myopia.

Incorrect Options:

Option B - Axial myopia: Axial myopia is the most common type of myopia and occurs due to the elongation of the eyeball (buphthalmos), causing the image to be focused in front of the retina instead of on it.

Option C - Curvatural myopia: Curvatural myopia, also known as refractive myopia, is caused by a change in the curvature of the cornea (Keratoconus), which causes light entering the eye to converge in front of the retina instead of on it.

Option D - Positional myopia: Positional myopia is a temporary form of myopia that occurs when the eye's natural lens changes shape due to the position of the eye. Commonly seen in weill marchesani syndrome which lead to anterior dislocation of lens

Solution for Question 25:

Correct Option B - Astigmatism:

- Astigmatism is a refractive error that results from an irregularly shaped cornea or lens in the eye, which causes blurred or distorted vision. It can cause difficulties with both near and distance vision. In astigmatism, the cornea or lens has different curvatures in different meridians, which leads to a focus of light at two different points instead of a single point. This results in distorted images, which can be seen in the image provided.

[Image not available]

Incorrect Options:

Option A - Presbyopia: Presbyopia is an age-related condition in which the ability of the eye to focus on close objects gradually decreases over time. It is caused by the hardening of the lens, which makes it less flexible and thus unable to change shape easily.

Option C - Hypermetropia; Hypermetropia, also known as farsightedness, is a refractive error in which distant objects can be seen clearly, but nearby objects appear blurry. It is caused by the eyeball being too short, or the cornea being too flat, which causes light to focus behind the retina instead of on it.

Option D - Myopia: Myopia, also known as nearsightedness, is a refractive error in which nearby objects can be seen clearly, but distant objects appear blurry. It is caused by the eyeball being too long, or the cornea being too steep, which causes light to focus in front of the retina instead of on it.

Squint

1. A 3-year-old girl presents with esotropia, which increases on focusing near objects. This issue has progressively worsened over the past few months, and she has difficulty with close visual tasks. A refractive error of +6.50 D has been identified. What is the most likely diagnosis?

- A. Accommodative refractive esotropia
- B. Accommodative non-refractive esotropia
- C. Accommodative mixed esotropia
- D. Infantile esotropia

2. A 7-year old child is brought to the clinic by her parents due to concerns about her right eyes appearing misaligned. On examination, her squint remains constant in all directions of gaze. She does not report any diplopia, and there are no limitations in eye movement. What is the most likely diagnosis?

- A. Concomitant squint
- B. Incomitant squint
- C. Paralytic squint
- D. Pseudo-strabismus

3. A 65 year old woman presented to the OPD with her head tilted towards the right side. On examination, she was having left hypertropia which increases on looking right or medially. The muscle which is most likely paralyzed is:

- A. Right superior oblique
- B. Left superior oblique

- C. Right inferior oblique
 - D. Left inferior oblique
-

4. A 40-year-old patient presents for binocular vision assessment. The ophthalmologist performs the Worth's Four Dot Test. The patient wears red-green glasses and reports seeing three green dots. What does this finding most likely indicate?

- A. Diplopia
 - B. Suppression of right eye
 - C. Suppression of left eye
 - D. Normal binocular vision
-

5. During a Maddox rod test, the patient is asked to view a distant light source. With the Maddox rod placed in front of the right eye, the patient reports seeing a vertical red line to the right of the white light. What type of ocular misalignment does this result suggest?

- A. Esophoria
 - B. Exophoria
 - C. Right hyperphoria
 - D. Left hyperphoria
-

6. A 45-year-old male presents with eyelid retraction on downgaze. He has a history of ocular nerve injury. Which of the following best describes this condition?

- A. Von Graefe's Sign
- B. Hutchinson's Pupil

- C. Pseudo-Von Graefe Sign
 - D. Marcus Gunn Jaw-Winking Phenomenon
-

7. A 7-year-old girl has limited abduction of the left eye, with globe retraction and narrowing of the palpebral fissure on adduction and widening of the fissure on attempted abduction. What is the most likely diagnosis?

- A. Duane Retraction Syndrome Type I
 - B. Duane Retraction Syndrome Type II
 - C. Duane Retraction Syndrome Type III
 - D. Sixth Cranial Nerve Palsy
-

8. A 60-year-old patient presents with the following symptoms: right-sided facial weakness, right-sided lateral gaze impairment, and weakness on the left side of the body. Imaging reveals a ventral pontine lesion. Which of the following syndromes best describes this patient's condition?

- A. Gradenigo Syndrome
 - B. Millard-Gubler Syndrome
 - C. Moebius Syndrome
 - D. Foville Syndrome
-

9. A 45-year-old man has double vision and an inward deviation of his left eye. On examination it is found to be associated with 6th Cranial Nerve Palsy. What finding is typical for sixth cranial nerve palsy?

- A. Vertical diplopia
- B. Pupils are affected

- C. Head turn towards the same side, avoiding diplopia
 - D. Diplopia worsens on head turning towards opposite side
-

10. A 45-year-old patient reports diplopia and is found to have left trochlear nerve palsy. Which of the following findings is not typically associated with it?

- A. Diplopia relieves while going downstairs
 - B. No diplopia with head tilt away from the left side
 - C. Hypertropia of the affected eye
 - D. Increase in hypertropia on opposite gaze
-

11. A 65-year-old patient presents with the following findings. Which of the following is least likely to be associated with this condition?



- A. Weber's Syndrome
 - B. Benedict's Syndrome
 - C. Nothmangel's Syndrome
 - D. Brown's Syndrome
-

12. A 60-year-old man presents with sudden onset double vision and difficulty aligning his eyes, which began three weeks ago. He has a history of hypertension and recently experienced a mild stroke. Examination reveals that his eye is positioned downward and outward. Which of the following statements about this condition is incorrect?

- A. Ptosis is present
- B. Mydriasis is present
- C. Pupil is fixed
- D. Accommodation increases

13. How much is the difference in deviation between upward gaze and downward gaze in A-V pattern strabismus?

- A. A-pattern ≥ 10 PD, V-pattern ≥ 15 PD
- B. A-pattern ≥ 15 PD, V-pattern ≥ 10 PD
- C. A-pattern ≤ 10 PD, V-pattern ≤ 15 PD
- D. A-pattern ≤ 15 PD, V-pattern ≤ 10 PD

14. A 45-year-old male complains of double vision, describing seeing two overlapping images when looking in a specific direction. He reports that this problem started suddenly about two weeks ago and is most noticeable when looking towards the right side. There is no history of trauma. On examination, he has limited movement of the right eye towards the right. What is the most likely cause of this condition?

- A. Blowout fracture
- B. Thick pterygium
- C. Subluxation of lens
- D. Paralytic strabismus

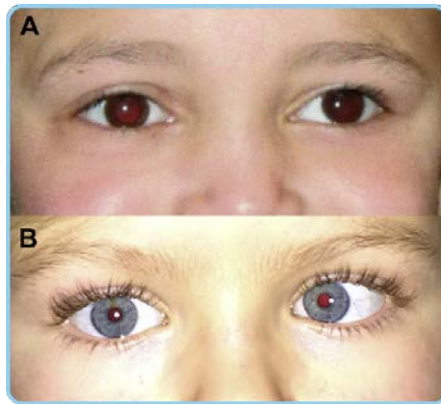
15. Which of the following is known for its specific pattern of muscle insertions around the sclera, essential for the alignment and movement of the extraocular muscles?

- A. Annulus of Zinn
 - B. Spiral of Tillaux
 - C. Scleral spur
 - D. Tenon's capsule
-

16. A 10-year-old child is having trouble catching a ball, climbing stairs, and performing fine motor tasks. She has a tendency to avoid activities requiring precise hand skills. No developmental delay was detected. Which of the following tests would be appropriate for diagnosing her condition?

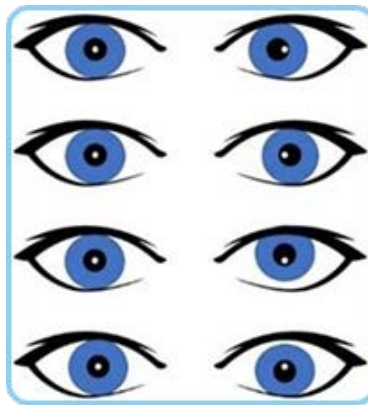
- A. HESS chart
 - B. Cover-uncover test
 - C. Titmus fly test
 - D. Maddox rod test
-

17. The parents of an 8-month-old girl have observed her eyes misaligning, especially when she focuses on objects up close, with one eye occasionally turning inward. They consulted an ophthalmologist, who conducted the test shown below. What is the name of this test?



- A. Hirschberg test
- B. Bruckner test
- C. Prism cover test
- D. Krimsky test

18. A 10-year-old boy's parents noticed his left eye turning inward, especially when tired or focusing on near objects. An ophthalmologist diagnosed strabismus and performed the test shown below. Which statements about this test are correct?



- A. It indicates the angle of squint
- B. In an aligned eye, the reflex is centred
- C. Reflex at limbus indicates 45° squint

D. All of the above

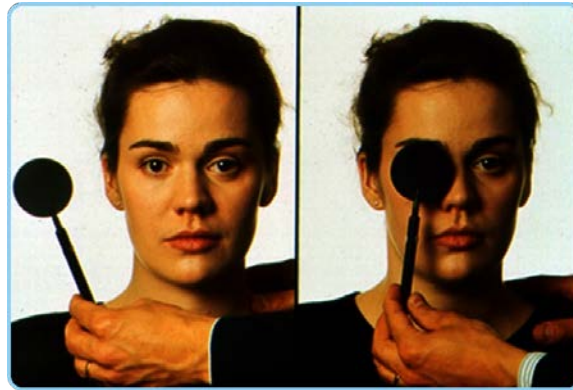
19. A 32-year-old patient presents with eye strain and headaches. The ophthalmologist plans to perform a series of tests to evaluate the patient's strabismus. All of the following are objective tests used to assess strabismus except?

- A. Maddox Rod test
 - B. Prism Bar Cover test
 - C. Cover Uncover test
 - D. Hirschberg test
-

20. A 35-year-old patient presents with increased diplopia on upward gaze and restricted movement of the left eye. The patient also has a history of recent orbital trauma. To determine whether the limitation is due to a mechanical restriction or a nerve palsy, which of the following tests is most appropriate?

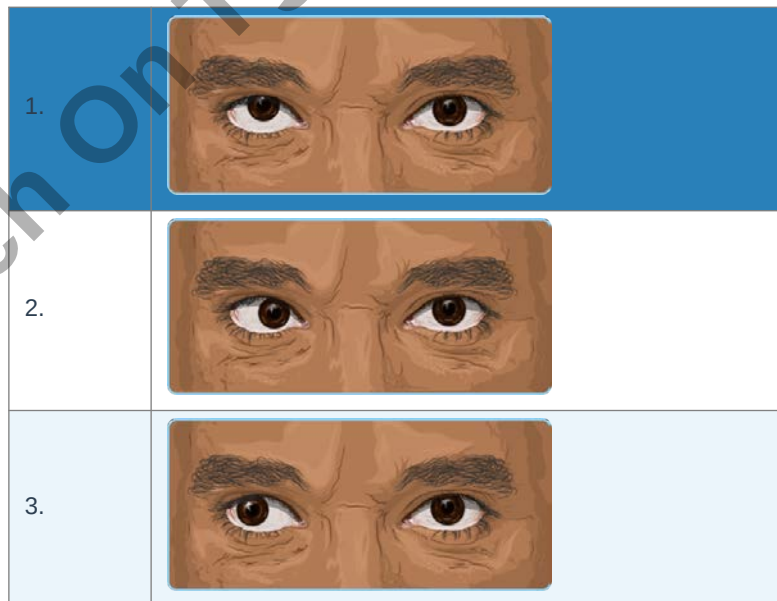
- A. Hess screen test
 - B. Forced duction test
 - C. Cover test
 - D. Worth's 4-dot test
-

21. A 10-year-old patient presents to the clinic with complaints of occasional double vision and eye strain during schoolwork. On examination, you suspect she may have a squint and perform the following test. What is this test known as?



- A. Maddox Rod Test
- B. Synoptophore
- C. Cover-Uncover Test
- D. Maddox Wing Test

22. Which of the following options correctly identifies the type of strabismus depicted in each image? 1. 2. 3.



- A. 1- Esotropia; 2-Exotropia; 3-Hypotropia

- B. 1- Esotropia; 2-Hypertropia; 3- Exotropia
 - C. 1- Hypertropia; 2-Exotropia; 3-Hypotropia
 - D. 1- Hypertropia; 2-Esotropia; 3-Exotropia
-

23. Which mechanism is responsible for maintaining coordinated movement of both eyes and achieving binocular vision?

- A. Conjugate and Dysconjugate movements
 - B. Synergistic and Antagonistic muscles
 - C. Yoke muscles
 - D. All of the above
-

24. A 32-year-old patient presents with diplopia and difficulty looking upwards and to the right, with limited elevation of the right eye. You suspect a lesion affecting a muscle responsible for vertical movement. Which axis describes the rotational movement, and which muscle is likely affected?

- A. Fick's x-axis; Right Inferior Rectus
 - B. Fick's y-axis; Right Medial Rectus
 - C. Fick's z-axis; Right Inferior Oblique
 - D. Fick's x-axis; Right Superior Rectus
-

25. Which physiological principle explains that during eye movement, an increase in neural stimulation to a contracting muscle is accompanied by a proportional decrease in stimulation to its opposing muscle?

- A. Hering's Law
- B. Sherrington's Law

- C. Henle's Law
 - D. Donder's Law
-

26. A 25-year-old patient presents with a condition in which, upon examination, the left eye fails to adduct during right gaze. The patient reports diplopia, which is more noticeable when looking to the right. Which extraocular muscle is most likely overacting in this patient's condition?

- A. Right Medial Rectus
 - B. Left Lateral Rectus
 - C. Right Lateral Rectus
 - D. Left Medial Rectus
-

27. A 45-year-old patient presents with vertical diplopia that worsens when looking down, especially while walking downstairs. Examination reveals weakness in intorsion, depression, and abduction of the affected eye. Which muscle and cranial nerve are most likely affected?

- A. Superior Rectus; CN III
 - B. Inferior Oblique; CN III
 - C. Superior Oblique; CN IV
 - D. Inferior Rectus; CN IV
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	2
Question 4	2
Question 5	1
Question 6	3
Question 7	1
Question 8	2
Question 9	3
Question 10	1,2
Question 11	4
Question 12	4
Question 13	1
Question 14	4
Question 15	2
Question 16	3
Question 17	2
Question 18	4
Question 19	1
Question 20	2
Question 21	3
Question 22	4

Question 23	4
Question 24	4
Question 25	2
Question 26	3
Question 27	3

Solution for Question 1:

Correct Answer: A) Accommodative refractive esotropia

Explanation:

The above scenario is a case of accommodative reflex esotropia, which has a convergent squint with high hypermetropia.

Esotropia

It is an inward deviation of one eye and is the most common type of squint in children.

It is of the following types:

- Infantile esotropia (Option D) - onset is before 6 months and mostly bilateral in presentation
- Accommodative esotropia Accommodative refractive esotropia Accommodative non-refractive esotropia Accommodative mixed esotropia
- Accommodative refractive esotropia
- Accommodative non-refractive esotropia
- Accommodative mixed esotropia
- Acquired Nonaccommodative esotropia
- Sensory esotropia
- Consecutive esotropia
- Accommodative refractive esotropia
- Accommodative non-refractive esotropia
- Accommodative mixed esotropia

Accommodative esotropia

	Accommodative Refractive Esotropia (Option A)	Accommodative Non-refractive Esotropia (Option B)	Accommodative Mixed Esotropia (Option C)
Cause	High hypermetropia (+4 to +7 D)	Abnormal large AC/A (accommodative convergence/accommodation) ratio	Combination of hypermetropia and high AC/A ratio
Treatment	Correction of hypermetropia (spectacles)	Corrected by bifocal glasses with +3 DS added for near vision Miotics (Echothiophate) Surgery: Bilateral medial rectus recession	Surgery: Bilateral medial rectus recession

Solution for Question 2:

Correct Answer: A) Concomitant squint

Explanation:

The squint remains constant in all directions of gaze with no associated diplopia, suggesting that it is a case of concomitant squint.

Squint/Strabismus: Misalignment of the visual axis. This condition is also called Heterotropia.

- Classification: It is divided into latent squint (phorias) and manifest (tropia).
- Latent squint - classification
- Manifest squint - classification

Concomitant squint:

- It's a type of manifest squint in which the angle of deviation remains equal in all the 9 gazes; there is no associated limitation of ocular movements (Option A).
- Primary deviation is equal to secondary deviation.
- Primary deviation is the amount of deviation in a squinted eye.
- Secondary deviation: It is the amount of deviation in the normal eye (behind the cover).
- Concomitant squint commonly occurs during early childhood or can be congenital (congenital squint).
- Eye movements are normal and no double vision (diplopia).

- Amblyopia develops later (negative adaptation).
- Abnormal retinal correspondence also plays a role. It is a positive adaptation.

Incomitant squint (Option B): This condition is just the opposite of concomitant squints, where the degree of squint varies in different aspects.

- Secondary deviation is higher than primary deviation in paralytic squint.
- In contrast to concomitant squints, there is a late presentation.
- There is palsy of other muscles, which causes limited ocular movements (ocular movements are restricted).
- Diplopia is also a very important feature of incomitant squints.

Paralytic squint (Option C): Paralytic squint is a type of incomitant squint in which ocular deviation results from complete or incomplete paralysis of one or more extraocular muscles.

- History of diplopia and confusion
- Secondary deviation is greater than primary deviation.
- Ocular movements are restricted towards the action of paralyzed muscle.
- Head is turned towards the action of paralyzed muscle.

Pseudo-strabismus (Option D): Pseudostrabismus is a condition when the eyes are well aligned but appear to be misaligned.

- Causes of Pseudo-esotropia: Epicanthal folds High myopia (high negative angle kappa)
- Epicanthal folds
- High myopia (high negative angle kappa)
- Causes of Pseudo-Exotropia: Telecanthus or hypertelorism High hypermetropia (high positive angle kappa)
- Telecanthus or hypertelorism
- High hypermetropia (high positive angle kappa)
- Epicanthal folds
- High myopia (high negative angle kappa)
- Telecanthus or hypertelorism
- High hypermetropia (high positive angle kappa)

Solution for Question 3:

Correct Answer: B) Left superior oblique

Explanation:

- The Park 3-step test is a diagnostic test used to identify which muscle is paretic in the case of an acquired hypertropia.
- It is designed for the diagnosis of a single paretic vertical muscle and is unreliable when there are multiple paretic muscles or in restrictive strabismus.

Applying the Park 3-step test:

- The left eye is hypertropic, meaning the affected muscle must be either: A depressor of the left eye (superior oblique or inferior rectus) An elevator of the right eye (superior rectus or inferior oblique)
- A depressor of the left eye (superior oblique or inferior rectus)
- An elevator of the right eye (superior rectus or inferior oblique)
- A depressor of the left eye (superior oblique or inferior rectus)
- An elevator of the right eye (superior rectus or inferior oblique)
- Yes, → This suggests a problem with the left superior oblique or the right superior rectus.
- The question doesn't explicitly state this, but given the patient's right head tilt, it suggests a compensatory posture.
- In superior oblique palsy, the head tilts opposite the affected muscle to minimize diplopia.
- This means the actual hypertropia worsens with a left tilt, confirming left superior oblique palsy.

Final Diagnosis: Left superior oblique palsy → Compensatory right head tilt.

Solution for Question 4:

Correct Answer: B) Suppression of right eye

Explanation:

- If the patient only sees three green dots, it suggests suppression of the right eye, as the red lens over the right eye would normally allow them to see the red dot.

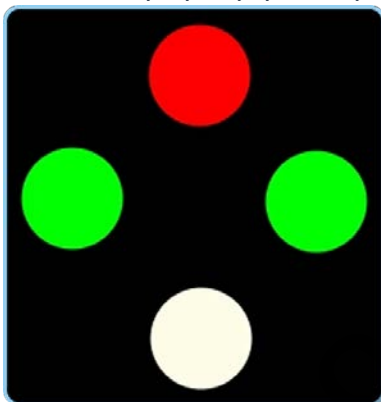
- The fact that only green dots are seen indicates that the right eye is not contributing to vision.

WORTH'S FOUR DOT TEST

Patient wears goggles with red lens in front of the right eye and green lens in front of the left eye, and views a box with 4 lights - one red, two green and one white.

Results are interpreted as below:

- Normal binocular single vision (Option D): Patient sees all the four lights in the absence of manifest squint (image A).
- Abnormal retinal correspondence: Patient sees all four lights even in the presence of a manifest squint.
- Left eye suppression (Option C): Patient sees only two red lights (Image B).
- Right eye suppression (Option B): Patient sees only three green lights (Image C).
- Alternating suppression: Patient sees three green lights and two red lights, alternately.
- Diplopia (Option A): Patient sees 5 lights (2 red and 3 green) (Image D & E).



Solution for Question 5:

Correct Answer: A) Esophoria

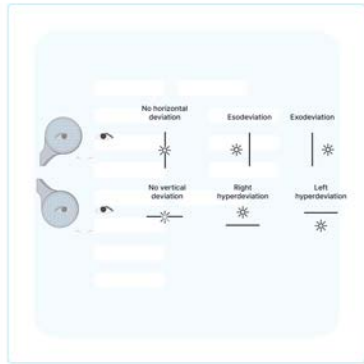
Explanation:

In the Maddox rod test, if the vertical red line (seen by the right eye) appears to the right of the white light, it suggests that the eyes are converging, which is indicative of esophoria.

MADDOX ROD TEST

- Patient is asked to fixate on a point light in the centre of a Maddox tangent scale at a distance of 6 metres.
- A Maddox rod is placed in front of one eye with the axis of rod parallel to the axis of deviation.
- The Maddox rod converts the point source into a line.
- Thus, the patient will see a point light with one eye and a red line with the other.
- Due to dissimilar images of the two eyes, fusion is broken and heterophoria becomes manifest.

Measure the Horizontal Deviation with Maddox rod test	
	The red lines of the Maddox rod are placed horizontally in front of the right eye so that a vertical line is formed. The patient is then asked whether the white light is superimposed on the horizontal red line or not. Patient might see the light source right or left to the red line. If superimposed - normal. If the patient is seeing the red line to the left of point light: Exotropia or crossed diplopia (as image moves inwards, eye moves outwards). (Option B) If the patient is seeing the red line to the right of point light: Esotropia or uncrossed diplopia. (Option A)
Measure the Vertical Deviation with Maddox rod test	
	The red lines of the Maddox rod are placed vertically in front of the right eye so that a horizontal line is formed. The patient is then asked whether the white light is superimposed on the horizontal red line or not. Patient might see the point light source up or down to the horizontal line. If superimposed - normal/orthophoria. If the point light source is above the red line: right hyperphoria or left hypophoria. (Option C) If the point light source is below the red line: right hypophoria or left hyperphoria. (Option D)



Solution for Question 6:

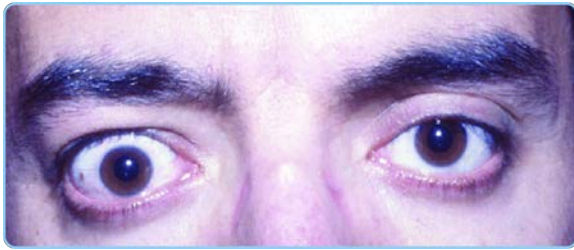
Correct Answer: C) Pseudo-Von Graefe Sign

Explanation:

- Pseudo-Von Graefe Sign is due to aberrant regeneration of the third cranial nerve following a palsy.
- The co-contraction of the levator palpebrae superioris and inferior rectus muscles leads to eyelid retraction on downgaze, distinguishing it from other conditions.
- This Pseudo-Von Graefe Sign is not associated with pupil abnormalities, and the pupil typically remains unaffected.



Von Graefe's Sign (Option A) is seen in Graves' disease, characterised by delayed descent of the upper eyelid during downgaze caused by restrictive orbitopathy affecting the eyelid.



Hutchinson's Pupil (Option B): Typically seen in uncal herniation, characterised by a unilateral dilated pupil that is unresponsive to light due to compression of the third cranial nerve, often associated with raised intracranial pressure.



Marcus Gunn Jaw-Winking Phenomenon (Option D) involves ptosis that improves with jaw movement due to aberrant innervation between the trigeminal nerve (motor division for the jaw) and the levator palpebrae superioris.

Reference:

Solution for Question 7:

Correct Answer: A) Duane Retraction Syndrome Type I

Explanation:

The clinical presentation described limited abduction, globe retraction on adduction, and widening of the palpebral fissure on attempted abduction, which is characteristic of Duane Retraction Syndrome Type I.

Duane's Retraction Syndrome:

- Neuropathology: Caused by absent sixth cranial nerve nucleus and innervation of the lateral rectus muscle by a branch of the inferior division of the third cranial nerve.
- Clinical Findings: Absent lateral rectus muscle firing during attempted abduction, simultaneous contraction of both horizontal recti during attempted adduction, leading to

Globe retraction Decreased palpebral fissure on adduction.

- Globe retraction
- Decreased palpebral fissure on adduction.
- Esotropia and Head Posture: Patients often develop an abnormal head posture to maintain single binocular vision or accept esotropia and abnormal retinal correspondence.
- Huber's Classification: Type I: Limited abduction with possible tether phenomenon. Type II: Limited adduction and tendency towards exotropia. Type III: Limited abduction and adduction, with possible horizontal strabismus.
- Type I: Limited abduction with possible tether phenomenon.
- Type II: Limited adduction and tendency towards exotropia.
- Type III: Limited abduction and adduction, with possible horizontal strabismus.
- Treatment: Indications for Surgery: Chronic abnormal head posture, manifest deviation, significant upshoots or downshoots, and disfiguring globe retraction.
- Indications for Surgery: Chronic abnormal head posture, manifest deviation, significant upshoots or downshoots, and disfiguring globe retraction.
- Globe retraction
- Decreased palpebral fissure on adduction.
- Type I: Limited abduction with possible tether phenomenon.
- Type II: Limited adduction and tendency towards exotropia.
- Type III: Limited abduction and adduction, with possible horizontal strabismus.
- Indications for Surgery: Chronic abnormal head posture, manifest deviation, significant upshoots or downshoots, and disfiguring globe retraction.

Solution for Question 8:

Correct Answer: B) Millard-Gubler Syndrome

Explanation:

The Given Clinical scenario with facial weakness, lateral gaze impairment & weakness on the left side of the body along with ventral pontine lesion points toward Millard-Gubler Syndrome

Syndromes Associated with Abducens Nerve:

Syndromes	Cause	Features
Foville Syndrome (Option D)	Damage to the dorsal pontine area affects the abducens nucleus, facial nucleus, and descending corticospinal fibres.	Sixth Nerve Palsy Seventh Nerve Palsy Contralateral Hemiparesis Central Horner Syndrome
Millard-Gubler Syndrome (Option B)	Ventral pontine injury affects the abducens nerve fascicle and facial nerve fibres.	Lateral Rectus Weakness Ipsilateral facial paralysis Contralateral Hemiparesis
Moebius Syndrome (Option C)	Congenital condition affecting cranial nerve development.	Facial Nerve Palsy Eye Movement Abnormalities: The eyes may exhibit esotropia (inward turning) or exotropia (outward turning). Orofacial Dysmorphism: Abnormalities in facial structure, which may include difficulty with suckling and phonation in infants.
Gradenigo Syndrome (Option A)	Mastoiditis leads to infection and inflammation near the petrous apex of the temporal bone.	Abducens Nerve Palsy: Difficulty moving the eye laterally, often associated with pain. Facial Pain: Pain may be present due to involvement of the trigeminal nerve. Deafness: Associated with mastoiditis, which results in pain and deafness.

Solution for Question 9:

Correct Answer: C) Head turn towards the same side, avoiding diplopia

Explanation:

- In sixth cranial nerve palsy, the affected eye cannot abduct properly, leading to diplopia that often improves with a head turn towards the affected side.
- This compensatory mechanism helps align the eyes to avoid double vision.

Sixth Cranial Nerve Palsy:

- Causes: Typically caused by infarction, demyelination, or compression from intrinsic pontine tumors.
- Features: Esotropia (inward eye deviation) due to Lateral rectus palsy - Restricted Abduction of the affected eye Ipsilateral Horizontal Gaze Palsy (Option A) Often compensated by turning their head toward the affected side. (Option C) Pupil function remains unaffected (Option B)
- Esotropia (inward eye deviation) due to Lateral rectus palsy - Restricted Abduction of the affected eye
- Ipsilateral Horizontal Gaze Palsy (Option A)
- Often compensated by turning their head toward the affected side. (Option C)
- Pupil function remains unaffected (Option B)
- Long-Term Effects: Muscle Changes: Prolonged palsy may cause tightening and contracture of the medial rectus muscle, leading to restrictive strabismus.
- Muscle Changes: Prolonged palsy may cause tightening and contracture of the medial rectus muscle, leading to restrictive strabismus.
- Associated Syndromes: One-and-a-Half Syndrome: I/L gaze palsy and internuclear ophthalmoplegia Foville Syndrome: Lateral Gaze Palsy, Ipsilateral Facial Palsy, Contralateral Hemiparesis Millard Gubler Syndrome: I/L 6,7 Cranial nerve palsy with C/L Hemiplegia
- One-and-a-Half Syndrome: I/L gaze palsy and internuclear ophthalmoplegia
- Foville Syndrome: Lateral Gaze Palsy, Ipsilateral Facial Palsy, Contralateral Hemiparesis
- Millard Gubler Syndrome: I/L 6,7 Cranial nerve palsy with C/L Hemiplegia
- Management: Acute Phase: Manage diplopia with eye patching or frosting a spectacle lens; prisms are often not well tolerated. Surgical Intervention: Indicated if the deviation stabilises for at least 6 months. Choices depend on lateral rectus muscle function: Good Function: Ipsilateral recess-resect procedure. Poor Function: Muscle transposition surgery may be needed.
- Acute Phase: Manage diplopia with eye patching or frosting a spectacle lens; prisms are often not well tolerated.
- Surgical Intervention: Indicated if the deviation stabilises for at least 6 months. Choices depend on lateral rectus muscle function: Good Function: Ipsilateral recess-resect procedure. Poor Function: Muscle transposition surgery may be needed.
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- Poor Function: Muscle transposition surgery may be needed.
- Esotropia (inward eye deviation) due to Lateral rectus palsy - Restricted Abduction of the affected eye
- Ipsilateral Horizontal Gaze Palsy (Option A)

- Often compensated by turning their head toward the affected side. (Option C)
- Pupil function remains unaffected (Option B)
- Muscle Changes: Prolonged palsy may cause tightening and contracture of the medial rectus muscle, leading to restrictive strabismus.
- One-and-a-Half Syndrome: I/L gaze palsy and internuclear ophthalmoplegia
- Foville Syndrome: Lateral Gaze Palsy, Ipsilateral Facial Palsy, Contralateral Hemiparesis
- Millard Gubler Syndrome: I/L 6,7 Cranial nerve palsy with C/L Hemiplegia
- Acute Phase: Manage diplopia with eye patching or frosting a spectacle lens; prisms are often not well tolerated.
- Surgical Intervention: Indicated if the deviation stabilises for at least 6 months. Choices depend on lateral rectus muscle function: Good Function: Ipsilateral recess-resect procedure. Poor Function: Muscle transposition surgery may be needed.
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- Poor Function: Muscle transposition surgery may be needed.
- Good Function: Ipsilateral recess-resect procedure.
- Poor Function: Muscle transposition surgery may be needed.

Solution for Question 10:

Correct Answer: A) Diplopia relieves while going downstairs

Explanation:

In trochlear nerve palsy, there is vertical diplopia which worsens while going downstairs

Trochlear Nerve (CN IV):

- Nucleus is located in the midbrain, near the inferior colliculus & the only cranial nerve that exits from the dorsal aspect of the brainstem.
- It is the smallest of the cranial nerves, while having the longest intracranial course.
- Innervates the contralateral superior oblique muscle
- Nuclear Lesion: Causes contralateral superior oblique muscle palsy.
- Cavernous / Orbital Lesion: Results in ipsilateral superior oblique muscle palsy.

Trochlear Nerve (CN IV) Palsy:

- Congenital: May become apparent at any age. Children often have abnormal head postures; adults may experience intermittent diplopia.
- Acquired: Most commonly due to trauma & presents with more noticeable symptoms.
- Clinical Features: Hypertropia (Option C) Increase in hypertropia on opposite gaze (Option D) Excyclotorsion (outward rotational movement of the eye) Vertical Diplopia - Increases on Downgaze (Option A) Depressed Chin with face towards the opposite side Head tilt towards the opposite side (Option B) Poor depression in adduction is typical. Crossed Diplopia
- Hypertropia (Option C)
- Increase in hypertropia on opposite gaze (Option D)
- Excyclotorsion (outward rotational movement of the eye)
- Vertical Diplopia - Increases on Downgaze (Option A)
- Depressed Chin with face towards the opposite side
- Head tilt towards the opposite side (Option B)
- Poor depression in adduction is typical.
- Crossed Diplopia
- Hypertropia (Option C)
- Increase in hypertropia on opposite gaze (Option D)
- Excyclotorsion (outward rotational movement of the eye)
- Vertical Diplopia - Increases on Downgaze (Option A)
- Depressed Chin with face towards the opposite side
- Head tilt towards the opposite side (Option B)
- Poor depression in adduction is typical.
- Crossed Diplopia



- Management: Resolution: Vascular issues/trauma often resolve spontaneously within 3–6 months. Compressive Lesions: Require surgical intervention. Surgical Treatment: (Knapp Management) Inferior Oblique Weakening Enhancing Superior Oblique Action
- Resolution: Vascular issues/trauma often resolve spontaneously within 3–6 months.
- Compressive Lesions: Require surgical intervention.
- Surgical Treatment: (Knapp Management) Inferior Oblique Weakening Enhancing Superior Oblique Action
- Inferior Oblique Weakening

- Enhancing Superior Oblique Action
- Resolution: Vascular issues/trauma often resolve spontaneously within 3–6 months.
- Compressive Lesions: Require surgical intervention.
- Surgical Treatment: (Knapp Management) Inferior Oblique Weakening Enhancing Superior Oblique Action
- Inferior Oblique Weakening
- Enhancing Superior Oblique Action
- Inferior Oblique Weakening
- Enhancing Superior Oblique Action

Solution for Question 11:

Correct Answer: D) Brown's Syndrome

Explanation:

Brown's Syndrome is a congenital ocular motility disorder caused by the fibrous tightening of the superior oblique tendon, causing a restriction in eye movement due to issues, and is not associated with third cranial nerve (CN III) palsy.

Syndromes associated with Third Cranial Nerve Palsy (Oculomotor Nerve):

Syndrome	Cause	Features
Weber's Syndrome	Injury to cerebral peduncle.	Third nerve palsy on the same side (ipsilateral) as the affected eye, combined with contralateral hemiplegia (paralysis on the opposite side of the body).
Benedict's Syndrome	Injury to the Red nucleus	Third nerve palsy with contralateral hemi-tremors, chorea & athetosis.
Nothnagel's Syndrome	Injury to the Superior Cerebellar Peduncle	Third nerve palsy with ipsilateral cerebellar ataxia (balance and coordination problems on the same side as the affected eye).
Claude's Syndrome	Lesion in the midbrain affects both the oculomotor nerve (CN III) and	Third nerve palsy, ipsilateral cerebellar ataxia, and contralateral

e	the red nucleus.	hemi-tremors.
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Solution for Question 12:

Correct Answer: D) Accommodation increases

Explanation:

The given scenario is diagnosed as paralytic strabismus, caused by 3rd nerve (oculomotor) palsy in which accommodation is completely lost due to paralysis of the ciliary muscles.

Features of third nerve palsy

- Ptosis: Due to paralysis of the LPS (levator palpebrae superioris) muscle



- The eyeball is deviated down and outwards (due to actions of the lateral rectus and superior oblique muscles).
- The pupil is fixed and dilated (due to paralysis of the sphincter pupillae muscle)
- Ocular movements are restricted
- Accommodation is completely lost (due to paralysis of the ciliary muscle)
- Crossed diplopia is elicited by raising the eyelid.
- Absence of Both direct and consensual reflexes in the affected eye
- These compensatory head postures are motor adaptations used to minimise the effects of double vision
 - Horizontal Diplopia: The patient often tilts the head downward
 - Vertical Diplopia: The head is positioned with the chin either raised or lowered
 - Torsional Diplopia: The head is tilted sideways
- Horizontal Diplopia: The patient often tilts the head downward

- Vertical Diplopia: The head is positioned with the chin either raised or lowered
- Torsional Diplopia: The head is tilted sideways
- Horizontal Diplopia: The patient often tilts the head downward
- Vertical Diplopia: The head is positioned with the chin either raised or lowered
- Torsional Diplopia: The head is tilted sideways

Solution for Question 13:

Correct Answer: A) A pattern ≥ 10 PD, V pattern ≥ 15 PD

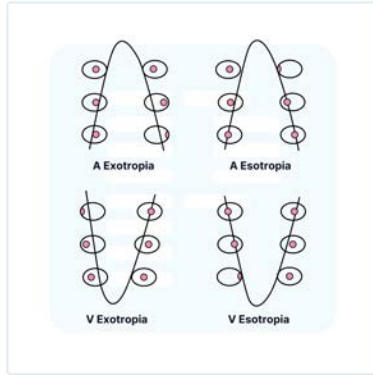
Explanation:

The difference between upgaze and downgaze in A-V pattern strabismus is A-pattern ≥ 10 PD, V-pattern ≥ 15 PD.

A-V Pattern Squint

- It consists of an A-pattern and a V-pattern.
- Causes: Oblique muscle dysfunction (most common) Rectus muscle dysfunction
- Oblique muscle dysfunction (most common)
- Rectus muscle dysfunction
- Oblique muscle dysfunction (most common)
- Rectus muscle dysfunction

Amount of deviation	'A' pattern	'V' pattern
Between upward and downward gaze	≥ 10 PD or prism dioptre	≥ 15 PD or prism dioptre
Esotropia (in upward gaze)	Increases	Decreases
Esotropia (in downward gaze)	Decreases	Increases
Exotropia (in upward gaze)	Decreases	Increases
Exotropia (in downward gaze)	Increases	Decreases



Solution for Question 14:

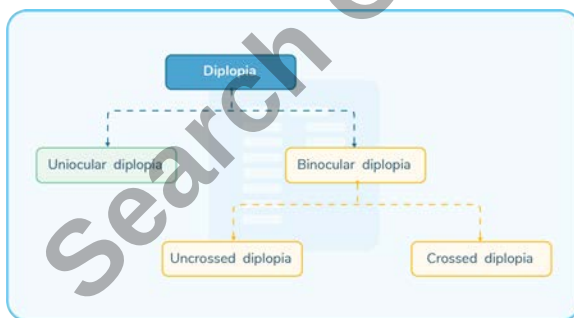
Correct Answer: D) Paralytic strabismus

Explanation:

The given scenario is diagnosed as binocular diplopia most commonly caused by paralysis of extra ocular muscles (paralytic strabismus).

Diplopia refers to the simultaneous perception of two images of a single object. It can be classified in the following ways,

1. It is classified as:



2. It can also be classified as,

- Horizontal diplopia
- Vertical diplopia
- Torsional diplopia

Causes of diplopia:

Unocular diplopia	Binocular diplopia
Subluxation of lens (Option C) Polycoria (multiple pupils) Incipient cataract Keratoconus	Paralysis of extraocular muscles (e.g. paralytic strabismus) (Option D) Orbital lesions Orbital fractures (e.g. blowout fracture) (Option A) Thick pterygium (Option B) Symblepharon Thyroid ophthalmopathy Decentred spectacles Anisometropia

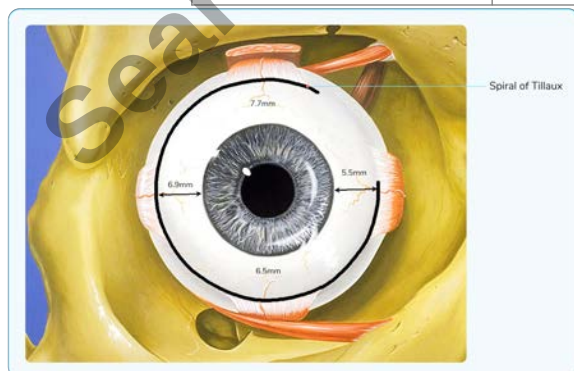
Solution for Question 15:

Correct Answer: B) Spiral of Tillaux

Explanation:

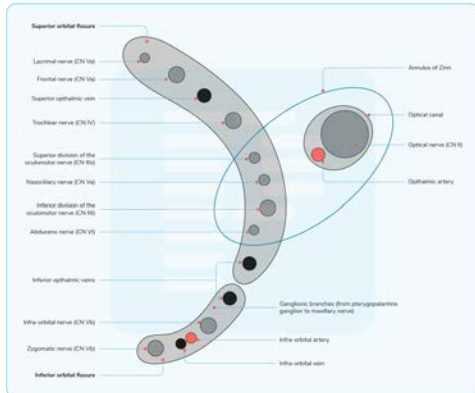
The Spiral of Tillaux refers to the spiral arrangement of the insertion points of the extraocular muscles on the sclera, which is essential for coordinating eye movements and maintaining proper ocular alignment.

Medial Rectus	Inserts closest to the limbus, about 5.5 mm from the limbus.
Inferior Rectus	Inserts 6.5 mm from the limbus.
Lateral Rectus	Inserts 6.9 mm from the limbus
Superior Rectus	Inserts the farthest, about 7.7 mm from the limbus.



Annulus of Zinn (Option A) is a fibrous ring at the back of the orbit around the optic nerve.

- It serves as the origin for the four rectus muscles (lateral, medial, superior, and inferior), providing a stable attachment point that allows these muscles to control eye movement.



Solution for Question 16:

Correct Answer: C) Titmus fly test

Explanation:

The scenario describes a case of stereopsis, and the appropriate test is the Titmus fly test.

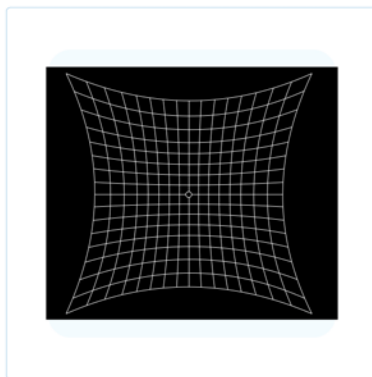


- Stereopsis is the ability to perceive the 3rd dimension (depth perception).
- Titmus fly test: The patient has to wear polarized glass, and the fly will be seen from different angles. Evaluate how well the child can perceive the relative depth of the images.
- The patient has to wear polarized glass, and the fly will be seen from different angles.

- Evaluate how well the child can perceive the relative depth of the images.
- The patient has to wear polarized glass, and the fly will be seen from different angles.
- Evaluate how well the child can perceive the relative depth of the images.

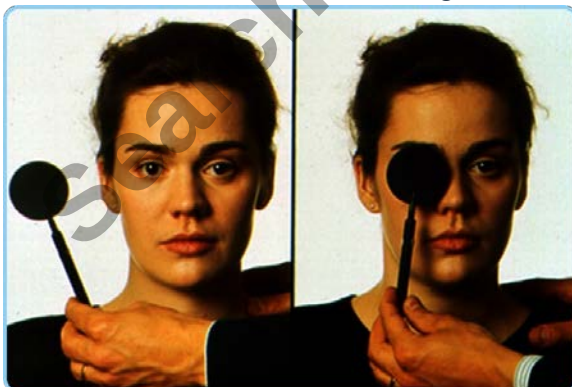
HESS chart (Option A) is used to measure the degree of deviation, especially if torsional, and particularly to measure any progressive increase or decrease by Hess screen test.

- The patient wears red goggles over the right eye and green goggles over the left eye.
- To test a specific eye, the green goggles are placed in front of it while the red goggles serve as a fixator.
- The patient aligns a green light with a red light projected on an electronic board.
- After the test, the goggles are switched so the green goggles are no longer in front of the right eye.



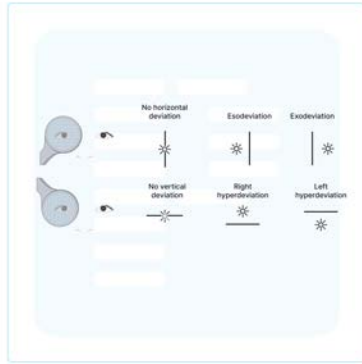
Cover-uncover test (Option B)

- This test is used to detect manifest (tropia) and latent (phoria) strabismus by covering one eye and then uncovering it to observe any corrective movement.
- It evaluates ocular misalignment, not depth perception.



Maddox rod test (Option D) is used for detection of latent squint.

- By convention, the Maddox rod is always placed over the right eye and the patient is asked to fixate on a bright white light either at distance or near.
- A binocular patient's right eye sees a red line, while the uncovered left eye sees the white light.



Solution for Question 17:

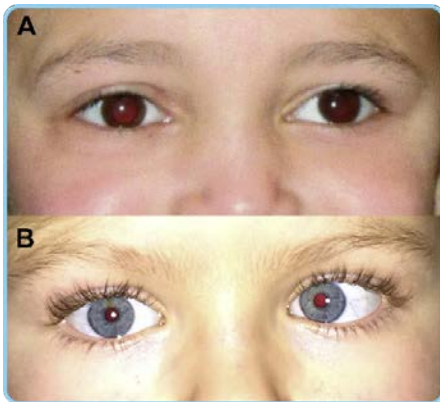
Correct Answer: B) Bruckner test

Explanation:

The above image depicts the Bruckner test, showing a brighter reflex in a squinted eye.

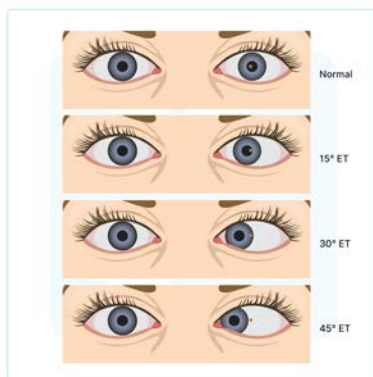
Bruckner test

- The infant is seated on the mother's lap while the lights are dimmed.
- The examiner sits at arm's length in front of the patient and shines the beam of a direct ophthalmoscope onto the patient's eyes, wide enough to illuminate both eyes simultaneously.
- Fundus reflex is depicted as, Squinted eye: Brighter reflex Fixing eye (normal eye): Dark or dull reflex
- Squinted eye: Brighter reflex
- Fixing eye (normal eye): Dark or dull reflex
- Squinted eye: Brighter reflex
- Fixing eye (normal eye): Dark or dull reflex



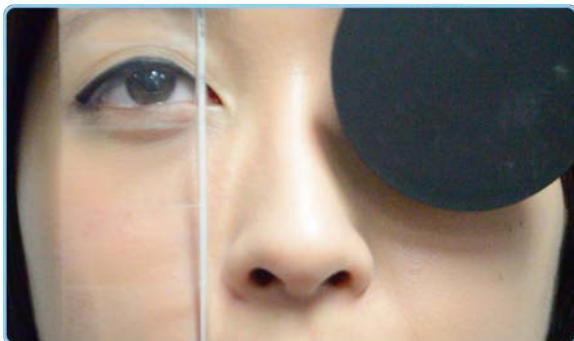
Hirschberg test (Option A) involves throwing light into the eye from a distance and the patient is asked to look at the light (an infant does this reflexively).

- A rough indication of the angle of the squint can be obtained from the position of the corneal reflex.



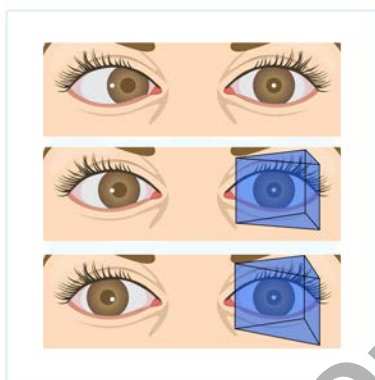
Prism cover test (Option C):

- Prisms of increasing strength with an apex towards the deviation are placed in front of one eye, and the patient is asked to fixate on an object with the other.
- The cover-uncover test is performed until there is no recovery movement of the eye under the cover.
- This will tell the amount of deviation in prism dioptres.



Krimsky test (Option D):

- The patient is asked to fixate on a point light and prisms of increasing power (with apex towards the direction of manifest squint) are placed in front of the normal fixating eye till the corneal light reflex is centered in the squinting eye.
- The power of the prism required to centre the light reflex in the squinting eye equals the amount of squint in prism dioptres.



Solution for Question 18:

Correct Answer: D) All of the above

Explanation:

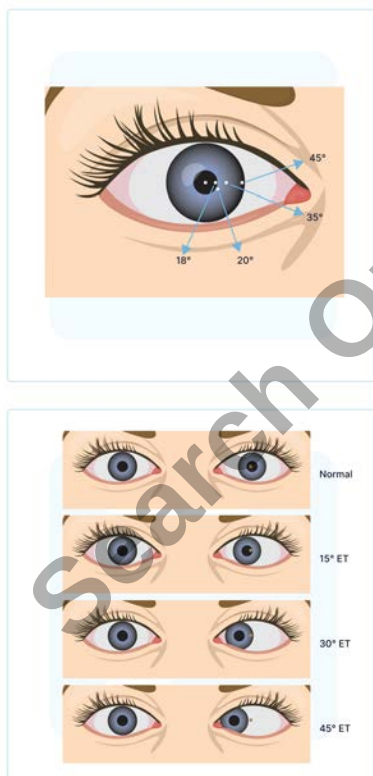
- The above image depicts Hirschberg's test, which is used to indicate the angle of squint by measuring the light reflex on the cornea.

Hirschberg's Test

- The angle of the squint can be roughly estimated by observing the position of the corneal reflex when light is directed into the eye. (Option A)

Procedure:

- Light is thrown into the eye with the ophthalmoscope or a focused light beam from a torch.
- The patient is asked to fixate at a point light held at a distance.
- In the fixing eye, the corneal reflex will be in the centre of the pupil. (Option B)
- The light is observed to be decentred in the squinting eye as, Pupillary border: 18-20° (outer: 20°, inner: 18°). Between the limbus and the papillary border: 35°. At the limbus: 45°. (Option C)
- Pupillary border: 18-20° (outer: 20°, inner: 18°).
- Between the limbus and the papillary border: 35°.
- At the limbus: 45°. (Option C)
- Pupillary border: 18-20° (outer: 20°, inner: 18°).
- Between the limbus and the papillary border: 35°.
- At the limbus: 45°. (Option C)



Solution for Question 19:

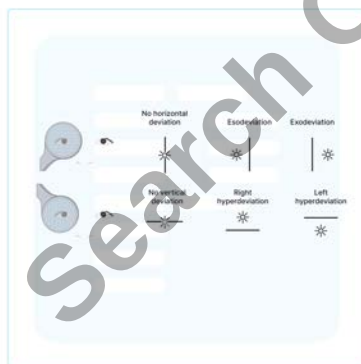
Correct Answer: A) Maddox Rod test

Explanation:

- The Maddox Rod Test is primarily subjective.
- It requires the patient to interpret the position of lines projected onto their retina, and their perception is used to measure the angle of deviation.
- This contrasts with purely objective tests, which rely on direct measurements or observations without relying on the patient's subjective interpretation of visual stimuli.

Maddox Rod Test:

- Procedure: The patient is positioned in front of a bright spot of light in a dark room. A Maddox rod, consisting of multiple red glass cylinders or a Maddox groove disc, is placed in a trial frame before one eye.
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- A Maddox rod, consisting of multiple red glass cylinders or a Maddox groove disc, is placed in a trial frame before one eye.



- Testing Horizontal Deviation: Cylinders Horizontal: The red line seen through the Maddox rod will be vertical. Observation: If there is no horizontal deviation (orthophoria), the bright spot of light will align with the vertical red line. Esophoria: The red line will appear to the right of the light (uncrossed diplopia). Exophoria: The red line will appear to the left of the light

(crossed diplopia).

- Cylinders Horizontal: The red line seen through the Maddox rod will be vertical.
- Observation: If there is no horizontal deviation (orthophoria), the bright spot of light will align with the vertical red line. Esophoria: The red line will appear to the right of the light (uncrossed diplopia). Exophoria: The red line will appear to the left of the light (crossed diplopia).
- If there is no horizontal deviation (orthophoria), the bright spot of light will align with the vertical red line.
- Esophoria: The red line will appear to the right of the light (uncrossed diplopia).
- Exophoria: The red line will appear to the left of the light (crossed diplopia).
- Testing Vertical Deviation: Cylinders Vertical: The red line will be horizontal. Observation: If there is no vertical deviation, the red line will pass through the light. Hyperphoria: If the red line appears above or below the light, it indicates hyperphoria. The direction of the deviation helps identify whether the right or left eye is affected.
- Cylinders Vertical: The red line will be horizontal.
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- If there is no horizontal deviation (orthophoria), the bright spot of light will align with the vertical red line.
- Esophoria: The red line will appear to the right of the light (uncrossed diplopia).
- Exophoria: The red line will appear to the left of the light (crossed diplopia).
- If there is no horizontal deviation (orthophoria), the bright spot of light will align with the vertical red line.
- Esophoria: The red line will appear to the right of the light (uncrossed diplopia).
- Exophoria: The red line will appear to the left of the light (crossed diplopia).

- Cylinders Vertical: The red line will be horizontal.
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List of Objective and Subjective Tests for Ocular Alignment and Function:

Objective Tests	Subjective Tests
Corneal Light Reflex Test (Hirschberg Test, Krimsky Test) Bruckner Test Cover-Uncover Test Prism Bar Test Synoptophore	Maddox Rod Test Maddox Wing Test Lancaster Red-Green Test Hess Screen Haploscopic Devices (Major Amblyoscope)

Solution for Question 20:

Correct Answer: B) Forced duction test

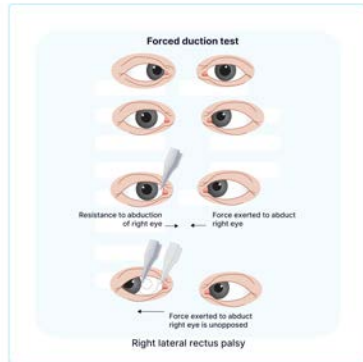
Explanation:

- The forced duction test is the primary method used to determine whether an eye movement limitation is due to mechanical restriction or a paralytic condition.

Mechanical Tests of Eye Movement Limitations: Mechanical tests help differentiate between eye movement limitations due to paralysis, paresis, or mechanical restriction.

1. Forced Duction Test:

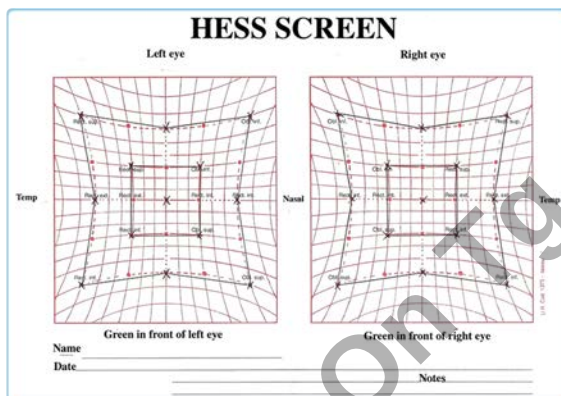
- Purpose: To determine if a limitation in eye movement is due to mechanical restriction or nerve palsy.
- Procedure: Apply topical anaesthetic to the limbal conjunctiva. The patient is instructed to look in the direction of the movement being tested, and the maximum eye range is observed. Using toothed forceps to grasp the conjunctiva, the examiner holds the eye at the opposite limbus and attempts to rotate it as far as possible in the same direction.
- Apply topical anaesthetic to the limbal conjunctiva.
- The patient is instructed to look in the direction of the movement being tested, and the maximum eye range is observed.
- Using toothed forceps to grasp the conjunctiva, the examiner holds the eye at the opposite limbus and attempts to rotate it as far as possible in the same direction.
- Apply topical anaesthetic to the limbal conjunctiva.
- The patient is instructed to look in the direction of the movement being tested, and the maximum eye range is observed.
- Using toothed forceps to grasp the conjunctiva, the examiner holds the eye at the opposite limbus and attempts to rotate it as far as possible in the same direction.
- Interpretation: Positive Test: Resistance to movement indicates a mechanical restriction (e.g., dysthyroid orbitopathy, muscle entrapment). Negative Test: No resistance suggests a nerve palsy, as the eye can be moved freely.
- Positive Test: Resistance to movement indicates a mechanical restriction (e.g., dysthyroid orbitopathy, muscle entrapment).
- Negative Test: No resistance suggests a nerve palsy, as the eye can be moved freely.
- Positive Test: Resistance to movement indicates a mechanical restriction (e.g., dysthyroid orbitopathy, muscle entrapment).
- Negative Test: No resistance suggests a nerve palsy, as the eye can be moved freely.



2. Testing Oblique Muscles: To test the superior and inferior obliques.

3. Active Force Generation Test: To assess the strength of muscle contraction against resistance.

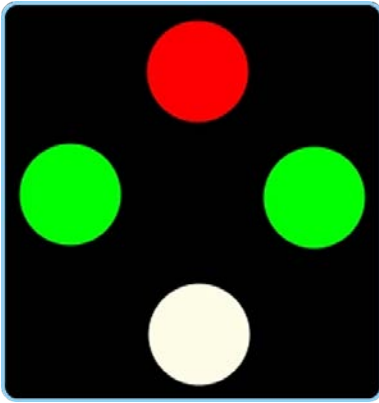
Hess Screen Test (Option A) measures the degree of misalignment but does not differentiate between mechanical restriction and nerve palsy directly.



Cover Test (Option C) is useful for detecting and measuring deviations but not for determining the cause of limited movement.



Worth's 4-Dot Test (Option D) assesses binocular vision and suppression but does not provide specific information about mechanical restrictions or nerve palsy.



Solution for Question 21:

Correct Answer: C) Cover-Uncover Test

Explanation:

The given image is of Cover-Uncover Test which is used to differentiate between latent squint (heterophoria) and manifest squint (heterotopia).

- Cover test: Detects Heterotropia (Manifest Squint)
- Uncover test: Detects Heterophoria (Latent Squint)

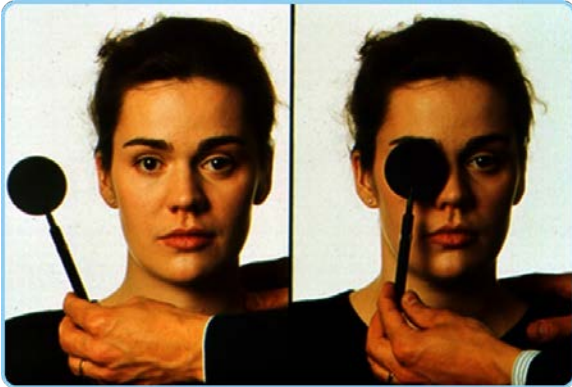
Cover-Uncover Test:

- The cover and uncover tests assess horizontal and vertical strabismus, but not torsional deviations. They also evaluate the ability of each eye to fixate on a target.

Cover Test:

- Detects Tropias: Measures constant deviations of the visual axis.
- Procedure: Cover one eye with a paddle, thumb, or occluder. Observe the uncovered eye for movement:
- Cover one eye with a paddle, thumb, or occluder.
- Observe the uncovered eye for movement:
- Cover one eye with a paddle, thumb, or occluder.

- Observe the uncovered eye for movement:



- Nasal movement: Exotropia
- Temporal movement: Esotropia
- Upward movement: Hypotropia
- Downward movement: Hypertropia
- For young children, use small toys instead of bright lights.
- Quantify the deviation with the simultaneous prism and cover test by introducing a prism before one eye until movement stops.
- Repeat with the prism before the other eye.

Uncover Test:

- Identifies latent deviations or tropias.
- Procedure: Cover one eye and then observe the covered eye when the cover is removed. If the eye reverts to fixation, it indicates a phoria or intermittent tropia.
- Cover one eye and then observe the covered eye when the cover is removed.
- If the eye reverts to fixation, it indicates a phoria or intermittent tropia.
- Cover one eye and then observe the covered eye when the cover is removed.
- If the eye reverts to fixation, it indicates a phoria or intermittent tropia.

Alternate Cover Test: Detects phorias by dissociating the visual axes.

- Alternate covering each eye to maximize visual axis dissociation.
- Care must be taken to permit time for each eye to reside behind the cover (the cover must not be "fanned" before the eyes).
- Use prisms to quantify phorias or tropias. If measurements are unclear, estimate the midpoint between defined movements.

Maddox Rod Test (Option A) is primarily used to assess the presence and amount of heterophoria by creating a red line that the patient tries to align with a point of light.

- It's helpful in diagnosing and quantifying latent squints, but not as direct for distinguishing between latent and manifest squint.

Synoptophore (Option B) is used for measuring the angle of squint and assessing binocular vision.

- It's also used for evaluating fusion and correspondence, but not specifically designed to differentiate between latent and manifest squints.

Maddox Wing Test (Option D) is similar to the Maddox Rod Test but is typically used to measure the amount of deviation in exophoria and esophoria.

- It is more focused on detecting and quantifying latent deviations rather than distinguishing between latent and manifest squints.

Solution for Question 22:

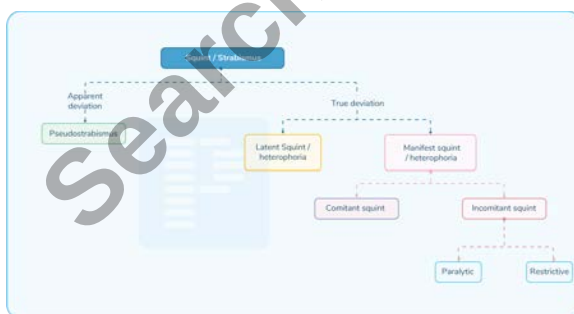
Correct Answer: D) 1- Hypertropia; 2-Esotropia; 3-Exotropia

Explanation:

Squint / Strabismus / Heterotropia:

- A condition where the visual axes of both eyes do not meet at a point (Misalignment of Visual axes)
- Pseudo-Strabismus: Eyes appear misaligned but are actually well-aligned.

Classification of Strabismus:



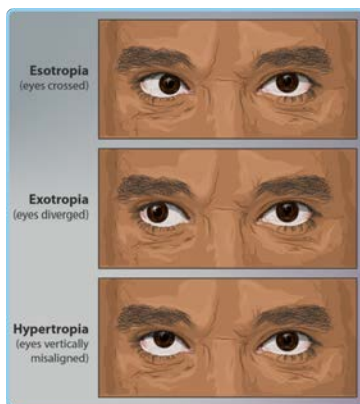
Heterophoria (Latent Strabismus): (Diagnosed by Cover-Uncover test)

- A tendency for the visual axes to misalign & is usually corrected by the eye's ability to fuse images.

- Types: Esophoria: Convergent tendency. Exophoria: Divergent tendency. Hyperphoria: Vertical deviation based on the upwardly deviating eye (e.g., right hyperphoria). Cyclophoria: Torsional deviation.
- Esophoria: Convergent tendency.
- Exophoria: Divergent tendency.
- Hyperphoria: Vertical deviation based on the upwardly deviating eye (e.g., right hyperphoria).
- Cyclophoria: Torsional deviation.
- Esophoria: Convergent tendency.
- Exophoria: Divergent tendency.
- Hyperphoria: Vertical deviation based on the upwardly deviating eye (e.g., right hyperphoria).
- Cyclophoria: Torsional deviation.

Manifest Strabismus (Heterotropia):

- Visible misalignment of the eyes, which is always present and not corrected by the eye's fusional capacity.
- Types: Esotropia: Inward misalignment of the eyes. (Common in Hypermetropes) Exotropia: Outward misalignment of the eyes. (Common in Myopes) Hypertropia: Upward misalignment of one eye relative to the other. Hypotropia: Downward misalignment of one eye relative to the other.
- Esotropia: Inward misalignment of the eyes. (Common in Hypermetropes)
- Exotropia: Outward misalignment of the eyes. (Common in Myopes)
- Hypertropia: Upward misalignment of one eye relative to the other.
- Hypotropia: Downward misalignment of one eye relative to the other.
- Esotropia: Inward misalignment of the eyes. (Common in Hypermetropes)
- Exotropia: Outward misalignment of the eyes. (Common in Myopes)
- Hypertropia: Upward misalignment of one eye relative to the other.
- Hypotropia: Downward misalignment of one eye relative to the other.



• Comitant vs. Incomitant Strabismus

Feature	Incomitant Strabismus	Comitant Strabismus
Magnitude of Squint	Varies with eye position	Same in all positions
Diplopia	Present	Absent
False Projection	Present	Absent
Abnormal Head Posture	Present	Absent
Secondary Deviation	Secondary > Primary	Secondary = Primary
Ocular Movements	Restricted	Full

- Incomitant Strabismus: Paralytic Strabismus: Resulting from afferent pathway damage, leading to muscle weakness or paralysis. Restrictive Strabismus: Caused by mechanical restrictions like muscle fibrosis or orbital lesions.

- Paralytic Strabismus: Resulting from afferent pathway damage, leading to muscle weakness or paralysis.

- Restrictive Strabismus: Caused by mechanical restrictions like muscle fibrosis or orbital lesions.

- Paralytic Strabismus: Resulting from afferent pathway damage, leading to muscle weakness or paralysis.

- Restrictive Strabismus: Caused by mechanical restrictions like muscle fibrosis or orbital lesions.

Solution for Question 23:

Correct Answer: D) All of the above

Explanation:

All of the above mechanisms are responsible for maintaining coordinated movement of both eyes and achieving binocular vision

Mechanism of Binocular Vision Formation: Binocular vision relies on the precise coordination of extraocular muscles in both eyes, ensuring single, clear vision through complex neural mechanisms.

Key components of this system include:

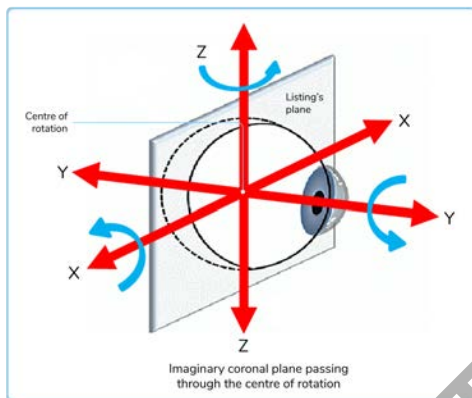
Conjugate & Dysconjugate Movements (Option A)	Synkinesis refers to the simultaneous movement of both eyes in the same direction, essential for maintaining binocular alignment. This is known as conjugate movement. Dysconjugate movements occur when the eyes move in opposite directions, such as during convergence (both eyes move inward) or divergence (both eyes move outward).
Synergistic & Antagonistic Muscles (Option B)	Synergistic muscles contract together to move both eyes in a specific direction. Eg: In Dextroversion (rightward gaze), the right lateral rectus and left medial rectus act as synergists. Antagonistic muscles are those that relax or are inhibited during these movements. Eg: In Dextroversion, the right medial rectus and left lateral rectus are antagonists.
Yoke Muscles (Option C)	Yoke muscles are pairs of synergistic muscles in each eye that work together to achieve coordinated eye movements. Eg: In the Rightward gaze, the right lateral rectus and left medial rectus are yoke muscles that contract together to move the eyes in the same direction.
Neurological Control	Hering's Law of Equal Innervation ensures that both eyes receive equal and simultaneous neural input to the synergistic (yoke) muscles during conjugate movements. Sherrington's Law of Reciprocal Innervation states that increased activation of a muscle is accompanied by decreased activation of its direct antagonist muscle.
Ocular Motor Nuclei and Pathways	The control of eye movements involves several brainstem nuclei, including the oculomotor (III), trochlear (IV), and abducens (VI) nuclei. These nuclei coordinate eye movements through neural pathways, such as the medial longitudinal fasciculus (MLF), which connects various eye movement centres.

Solution for Question 24:

Correct Answer: D) Fick's x-axis; Right Superior Rectus

Explanation:

- The vertical movements of the eye occur with the x axis as the centre of rotation in primary position.
- The right superior rectus elevates the eye, particularly in the direction of upward gaze.
- A lesion in the right superior rectus explains the patient's difficulty with upward movement.



Axis	Direction	Movements	Muscles Involved
X-Axis (Horizontal)	Runs horizontally from the outer corner to the inner corner.	Elevation (upward) and Depression (downward)	Superior Rectus, Inferior Rectus
Y-Axis (Anteroposterior)	Runs anteroposteriorly from the front (cornea) to the back.	Intorsion (inward rotation) and Extorsion (outward rotation)	Superior Oblique, Inferior Oblique
Z-Axis (Vertical)	Runs vertically from the top of the head to the chin.	Adduction (inward) and Abduction (outward)	Medial Rectus, Lateral Rectus

- Listing's Plane is a conceptual plane that passes through the eye's centre of rotation and is perpendicular to the visual axis when the eye is in a straight-ahead gaze.
- It describes how the eye rotates when moving from the primary position to any other direction and helps explain the torsional movements the eye makes to maintain visual stability.

Solution for Question 25:

Correct Answer: B) Sherrington's Law

Explanation:

Sherrington's Law of Reciprocal Innervation of the same eye, states that when one muscle contracts (agonist), its opposing muscle (antagonist) in the same eye must relax to allow coordinated movement.

Hering's Law (Option A) refers to the equal innervation of corresponding eye muscles to ensure simultaneous and symmetrical movements.

Henle's law of accommodation (Option C) refers to anatomical observations related to the length-tension relationship in muscles, particularly in the context of accommodation in the eye.

Donder's Law (Option D) pertains to the torsional movements of the eye.

Reference:

Solution for Question 26:

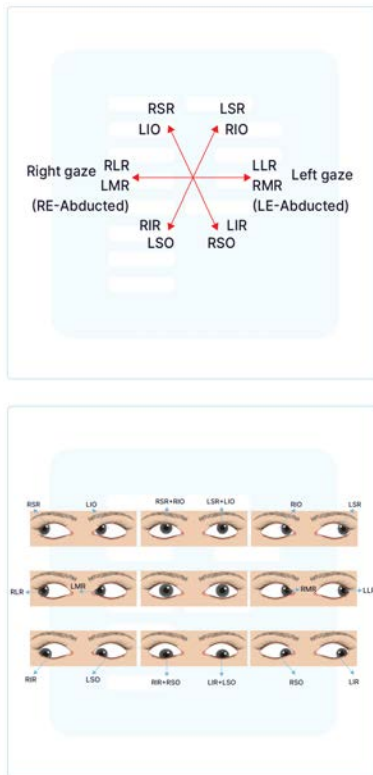
Correct Answer: C) Right Lateral Rectus

Explanation:

Given that the left eye fails to adduct and diplopia is more noticeable during right gaze, it suggests that the left medial rectus is not functioning properly.

- Hering's Law of Equal Innervation would mean that the right lateral rectus (yoke muscle) should be balanced with the left medial rectus (vs Sherrington's law: This law about reciprocal innervation states that increased innervation and contraction of a muscle is automatically associated with reciprocal decrease in innervation and contraction of its antagonist).
- If the left medial rectus is weak or underacting, the corresponding yoke muscle, the right lateral rectus, might be compensating and thus appearing to overact.
- Yoke muscles are pairs of extraocular muscles, one in each eye, that work together to ensure coordinated eye movements in the same direction.

- These muscles function according to Hering's Law of Equal Innervation, which ensures that corresponding muscles in both eyes receive equal neural impulses for synchronised movement.
- The yoke muscle pairs facilitate various eye movements:



- Leftward Gaze (Levoversion): Left Lateral Rectus (abducts the left eye) Right Medial Rectus (adducts the right eye)
- Left Lateral Rectus (abducts the left eye)
- Right Medial Rectus (adducts the right eye)
- Rightward Gaze (Dextroversion): Right Lateral Rectus (abducts the right eye) Left Medial Rectus (adducts the left eye)
- Right Lateral Rectus (abducts the right eye)
- Left Medial Rectus (adducts the left eye)
- Left Lateral Rectus (abducts the left eye)
- Right Medial Rectus (adducts the right eye)
- Right Lateral Rectus (abducts the right eye)
- Left Medial Rectus (adducts the left eye)
- Upward Gaze (Elevation): Right and Left Superior Rectus (elevate the eyes) Right and Left Inferior Oblique (assist in elevation)

- Right and Left Superior Rectus (elevate the eyes)
- Right and Left Inferior Oblique (assist in elevation)
- Downward Gaze (Depression): Right and Left Inferior Rectus (depress the eyes) Right and Left Superior Oblique (assist in depression)
- Right and Left Inferior Rectus (depress the eyes)
- Right and Left Superior Oblique (assist in depression)
- Right and Left Superior Rectus (elevate the eyes)
- Right and Left Inferior Oblique (assist in elevation)
- Right and Left Inferior Rectus (depress the eyes)
- Right and Left Superior Oblique (assist in depression)
- Both Medial Rectus muscles contract to bring the eyes inward (toward the nose) during near-vision tasks, such as reading.

Solution for Question 27:

Correct Answer: C) Superior Oblique; Trochlear nerve (CN IV)

Explanation:

- The superior oblique muscle, innervated by the trochlear nerve (CN IV), is responsible for intorsion, depression, and abduction of the eye.
- Dysfunction of this muscle causes vertical diplopia, particularly noticeable when looking down, such as when walking downstairs.

Extraocular muscles and actions			
Muscle	Primary Action	Secondary Actions	Innervation
Medial Rectus	Adduction (moves eye inward)	None	Oculomotor nerve (CN III)
Lateral Rectus	Abduction (moves eye outward)	None	Abducens nerve (CN VI)
Superior Rectus	Elevation (moves eye upward)	Intorsion (inward rotation), Adduction	Oculomotor nerve (CN III)
Inferior Rectus	Depression (moves eye downward)	Extorsion (outward rotation), Adduction	Oculomotor nerve (CN III)

Superior Oblique	Intorsion (inward rotation)	Depression, Abduction	Trochlear nerve (CN IV)
Inferior Oblique	Extorsion (outward rotation)	Elevation, Abduction	Oculomotor nerve (CN III)

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Previous Year Questions

1. The cover-uncover test is performed to diagnose various eye conditions. Which of the following is not detected by this test?

- A. Latent Squint
- B. Manifest Squint
- C. Lateral Nystagmus
- D. Amblyopia

2. The patient has an infection of the cavernous sinus, involving the nerve located on its medial side. To confirm the involvement of the abducens nerve, you will ask the patient to look in which direction?

- A. Superior
- B. Inferior
- C. Medial
- D. Lateral

3. Which nerve innervates the lateral rectus muscle?

- A. Sixth nerve
- B. Seventh nerve
- C. Third nerve
- D. Fourth nerve

4. Which drug is commonly used for refraction in a child with accommodative squint?

- A. 1% Atropine
 - B. Cyclopentolate
 - C. Tropicamide
 - D. Homatropine
-

5. What is the probable cause of uniocular diplopia in a young boy who has experienced trauma to his right eye?

- A. Paresis of extraocular muscle
 - B. Displacement of one eyeball
 - C. Symblepharon
 - D. Iridodialysis
-

6. What are the yoke muscles for Levodepression?

- A. LSO and RIR
 - B. RSO and LIR
 - C. RIO and LIR
 - D. LIO and RIR
-

7. What should be the initial course of action for a 2-year-old child displaying inward deviation of both eyes?

- A. Forced ductions test
 - B. Examination under anaesthesia
 - C. Refractive error testing
 - D. Examination of fundus
-

8. A 50-year-old female patient visits the ophthalmology clinic with complaints of experiencing vertical diplopia while walking downstairs. Upon further inquiry, it is noted that the patient tends to tilt her head to one side in order to alleviate the diplopia. Which cranial nerve palsy should be suspected in this patient?

- A. 3rd nerve
 - B. 6th nerve
 - C. 7th nerve
 - D. 4th nerve
-

9. The following image indicates:



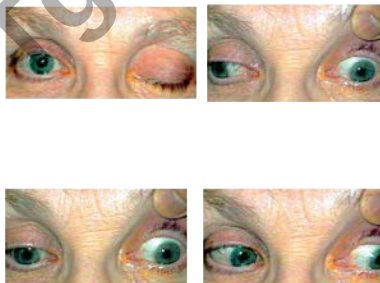
- A. Trochlear nerve palsy
- B. Abducens nerve palsy
- C. Internuclear ophthalmoplegia

D. Oculomotor palsy

10. Which of the following statements regarding oculomotor nerve palsy is false?

- A. Ptosis
 - B. Mid – dilated pupil
 - C. Loss of light reflex in the affected eye
 - D. Lateral and upward gaze in primary position
-

11. A 60-year-old male presents with left eyelid drooping and experiencing double vision. Upon examination, it was observed that the pupil in the affected eye was dilated. The following eye movement examination was conducted. Which nerve is most likely injured in this case?



- A. Optic nerve
 - B. Trochlear
 - C. Oculomotor
 - D. Abducens
-

12. Which muscle serves as the yoke muscle for the right lateral rectus?

- A. Right superior oblique
 - B. Left medial rectus
 - C. Left lateral rectus
 - D. Right medial rectus
-

13. A patient was prescribed with -1D spherical and -2D cylindrical at 180 degrees. What type of refractive error does this patient have?

- A. With the rule astigmatism
 - B. Against the rule astigmatism
 - C. Simple astigmatism
 - D. Mixed astigmatism
-

14. The center for vertical gaze is _____

- A. Pontine Paramedian Reticular formation
 - B. Raphe Nuclues
 - C. Rostral interstitial nucleus of the medial longitudinal fasciculus
 - D. Substantia nigra
-

15. Yoke muscle of the right lateral rectus in dextroversion is _____.

- A. Left superior oblique

- B. Left medial rectus
- C. Left inferior rectus
- D. Right superior rectus

16. Which of the following statements is not true in relation to accommodative esotropia?

- A. It is the most common type of squint
- B. Miotics are used as the treatment strategy in the high AC/A ratio
- C. It is associated with high myopia
- D. Refractive type can be fully corrected by use of spectacles

17. In a 3-year-old child who has recently developed 15-degree accommodative esotropia, what is the most effective treatment approach?

- A. Orthoptic exercise
- B. Surgery
- C. INTACS
- D. Refractive correction

18. The lesion in which of the following nerve is responsible for right eye squint, as depicted below.



- A. Oculomotor
 - B. Trochlear
 - C. Trigeminal
 - D. Abducent
-

19. The movement lost on left gaze as depicted in image is seen in?



- A. 3rd CN palsy
 - B. 4th CN palsy
 - C. 6th CN palsy
 - D. 7th CN palsy
-

20. Which of the following conditions is associated with the finding observed by a physician during an eye examination of a small child?



- A. Myopia
 - B. Astigmatism
 - C. Hyperopia
 - D. Emmetropia
-

21. In a 40-year-old patient with diabetes, a squint was observed. The secondary deviation was higher than the primary deviation during examination. The forced duction test yielded negative results. Which type of squint is expected to be predominant in this individual?

- A. Comitant squint
 - B. Restrictive squint
 - C. Paralytic squint
 - D. Pseudosquint
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	1
Question 4	1
Question 5	4
Question 6	2
Question 7	3
Question 8	4
Question 9	4
Question 10	4
Question 11	3
Question 12	2
Question 13	1
Question 14	3
Question 15	2
Question 16	3
Question 17	4
Question 18	4
Question 19	1
Question 20	3
Question 21	3

Solution for Question 1:

Amblyopia, often called "lazy eye," is a condition where the vision in one eye is reduced because the brain favors the other eye. It can be caused by strabismus, but the cover-uncover test itself does not diagnose amblyopia

Cover-Uncover Test is used to differentiate between latent squint (heterophoria) and manifest squint (heterotropia).

- Cover test: Detects Heterotropia (Manifest Squint) (Option B)
- Uncover test: Detects Heterophoria (Latent Squint) (Option A)

Cover-Uncover Test:

- The cover and uncover tests assess horizontal and vertical strabismus, but not torsional deviations. They also evaluate the ability of each eye to fixate on a target.

Cover Test:

- Detects Tropias: Measures constant deviations of the visual axis.
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- Observe the uncovered eye for movement:

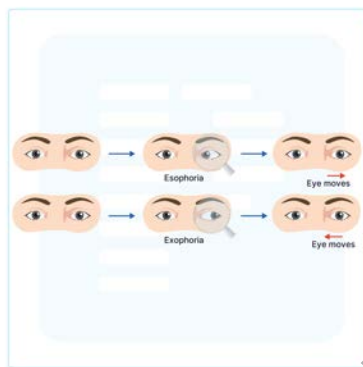


- Nasal movement: Exotropia
- Temporal movement: Esotropia
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- Downward movement: Hypertropia
- For young children, use small toys instead of bright lights.

- Quantify the deviation with the simultaneous prism and cover test by introducing a prism before one eye until movement stops. Repeat with the prism before the other eye.

Uncover Test:

- Identifies latent deviations or tropias.
- Procedure: Cover one eye and then observe the covered eye when the cover is removed. If the eye reverts to fixation, it indicates a phoria or intermittent tropia. If the eye remains deviated, it suggests a tropia with fixation preference for the fellow eye.
- Cover one eye and then observe the covered eye when the cover is removed.
- If the eye reverts to fixation, it indicates a phoria or intermittent tropia.
- If the eye remains deviated, it suggests a tropia with fixation preference for the fellow eye.
- Cover one eye and then observe the covered eye when the cover is removed.
- If the eye reverts to fixation, it indicates a phoria or intermittent tropia.
- If the eye remains deviated, it suggests a tropia with fixation preference for the fellow eye.



Alternate Cover Test: Detects phorias by dissociating the visual axes.

- Alternate covering each eye to maximize visual axis dissociation.
- Care must be taken to permit time for each eye to reside behind the cover (the cover must not be "fanned" before the eyes).
- Use prisms to quantify phorias or tropias. If measurements are unclear, estimate the midpoint between defined movements.

Lateral Nystagmus (Option C): Nystagmus is involuntary, rhythmic eye movements. The cover-uncover test might exacerbate nystagmus, though it is not the primary test for detecting it.

Solution for Question 2:

Correct Option D - Lateral:

- Abducens nerve supplied lateral rectus muscle which is involved in abduction movement
- Therefore patient is asked to look in lateral direction

Incorrect Options:

- Option A, B & C: The patient is not asked to look in superior, inferior and medial directions to test for Abducens nerve
-

Solution for Question 3:

Correct Option A - Sixth nerve:

- The sixth nerve innervates the lateral rectus muscle

Incorrect Options:

- Option B, C & D: Seventh, third and fourth nerves do not innervate the lateral rectus muscle
-

Solution for Question 4:

Correct Option A - 1% Atropine:

- High hypermetropia of around +4 to +7 D causes accommodative squint due to excessive accommodation. Correction of hypermetropia results in full correction of convergent squint.
- Atropine is often used in pediatric ophthalmology to measure hypermetropia. It works by dilating the pupil and temporarily paralyzing accommodation, or the eye's ability to adjust its focus to see nearby objects clearly.

Incorrect Options:

Options B, C, and D are incorrect. 1% Atropine is used to measure hypermetropia accurately.

Solution for Question 5:

Correct Option D - Iridodialysis:

- The most likely cause of uniocular diplopia (persistence of diplopia even when the normal eye is closed) in a young boy with trauma to the right eye is iridodialysis.
- Iridodialysis refers to a separation or tearing of the iris from its ciliary attachment, which can cause the pupil to become distorted and result in double vision.
- Other causes of uniocular diplopia may include Subluxated clear lens or intraocular lens, incipient cataract, and keratoconus.

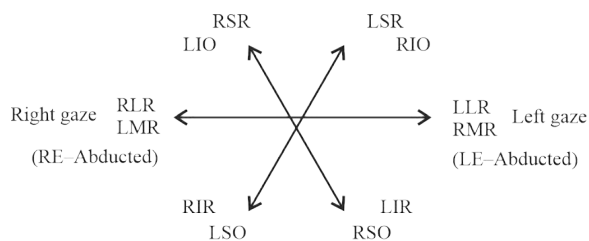
Incorrect Options:

Options A, B, and C are incorrect. These conditions causes binocular diplopia.

Solution for Question 6:

Correct Option B - RSO (Right Superior Oblique) and LIR (Left Inferior Rectus):

- The yoke muscle for levo depression are: Left inferior rectus responsible for the movement of left eye outward and down. Right superior oblique responsible for the movement of the right eye inward and down.
- Left inferior rectus responsible for the movement of left eye outward and down.
- Right superior oblique responsible for the movement of the right eye inward and down.
- Left inferior rectus responsible for the movement of left eye outward and down.
- Right superior oblique responsible for the movement of the right eye inward and down.



Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 7:

Correct Option C - Refractive error testing:

- The inward deviation of both the eyes in a 2-year-old child is suggestive of convergent squint due to accommodative esotropia.
- Refractive accommodative esotropia is the most common cause of convergent squint in the age of 2 to 3 years.
- Refractive error can cause strabismus (eye misalignment) and may contribute to the medial deviation of both eyes in this case. Therefore, the first step would be to evaluate the child's refractive error through appropriate testing.

Incorrect Options:

Options A, B, and D are incorrect. These tests will not be useful to diagnose the cause of esotropia.

Solution for Question 8:

Correct Option D - 4th nerve:

- The vertical diplopia in the absence of ptosis, combined with a characteristic head posture, suggests fourth nerve disease.

- The fourth cranial nerve controls the superior oblique muscle. The main actions of the superior oblique are intorsion and depression in adduction.
- A compensatory head posture avoids diplopia. To compensate for the weakness of intorsion, there is a contralateral head tilt. To alleviate the weakened depression of the eye, the chin is slightly depressed
- Peripheral lesions cause ipsilateral and nuclear lesions contralateral superior oblique weakness.

Incorrect Options:

Option A - Third nerve: Third nerve cranial palsy presents with ptosis, abduction and depression in the primary position, dilated pupil, and limited adduction, elevation and depression.

Option B - 6 CN Palsy (Abducens Nerve Palsy): Sixth nerve palsy presents with inability to move the eye laterally and could cause horizontal diplopia.

Option C - 7 CN Palsy (Facial Nerve Palsy): The seventh cranial nerve controls the muscles of facial expression and is responsible for taste sensation on the anterior two-thirds of the tongue.

Solution for Question 9:

Correct Option D - Oculomotor palsy:

- The image indicates the following findings which are suggestive of oculomotor nerve palsy/third nerve palsy. A - Profound ptosis due to weakness of the levator muscle. B - Abduction and depression in the primary position due to unopposed action of the lateral rectus and superior oblique muscles. C - Normal abduction as the lateral rectus is intact D - Limited adduction due to medial rectus weakness. E - Limited elevation due to weakness of the superior rectus and inferior oblique. F - Limited depression due to weakness of the inferior rectus.
- A - Profound ptosis due to weakness of the levator muscle.
- B - Abduction and depression in the primary position due to unopposed action of the lateral rectus and superior oblique muscles.
- C - Normal abduction as the lateral rectus is intact
- D - Limited adduction due to medial rectus weakness.
- E - Limited elevation due to weakness of the superior rectus and inferior oblique.

- F - Limited depression due to weakness of the inferior rectus.
- A - Profound ptosis due to weakness of the levator muscle.
- B - Abduction and depression in the primary position due to unopposed action of the lateral rectus and superior oblique muscles.
- C - Normal abduction as the lateral rectus is intact
- D - Limited adduction due to medial rectus weakness.
- E - Limited elevation due to weakness of the superior rectus and inferior oblique.
- F - Limited depression due to weakness of the inferior rectus.

Incorrect Options:

Option A - Trochlear nerve palsy: It presents with vertical diplopia and torsional diplopia with characteristic head posture to the contralateral side.

Option B - Abducens nerve palsy: It is characterized by esotropia in the affected eye with marked limitation of abduction (weakness of lateral rectus muscle) and normal adduction.

Option C - Internuclear ophthalmoplegia: It is characterized by defective ipsilateral adduction, and the contralateral abducting eye showing ataxic nystagmus.

Solution for Question 10:

Correct Option D - Lateral and upward gaze in primary position:

- In a patient with oculomotor nerve palsy the presentation is abduction and depression in the primary gaze, i.e., eye is down and out. It is due to the unopposed action of the lateral rectus and superior oblique muscles.

Incorrect Options:

Option A - Ptosis: Ptosis, or drooping of the eyelid, is a common symptom of oculomotor nerve palsy due to weakened levator palpebrae superioris muscle..

Option B - Mid-dilated pupil: This statement is true. Oculomotor nerve palsy can lead to a mid-dilated pupil due to parasympathetic palsy.

Option C - Loss of light reflex in the affected eye: Both the direct and consensual light reflex in the affected eye is absent.

Solution for Question 11:

Correct Option C - Oculomotor:

- The left eye ptosis, diplopia and the examination finding as shown in the image is suggestive of third nerve palsy i.e., oculomotor nerve.
- The diplopia occurs in all directions except downgaze and dilated pupil due to parasympathetic palsy.
- Examination findings seen in the image consistent with third nerve palsy are: Left eye ptosis can be seen in the image is suggestive of one sided lesion in the abducens nerve involving levator palpebrae superioris. It shows limited adduction due to medial rectus muscle weakness. It shows limited depression due to inferior rectus muscle weakness. Normal abduction as the lateral rectus is intact supplied by the 6th nerve.
- Left eye ptosis can be seen in the image is suggestive of one sided lesion in the abducens nerve involving levator palpebrae superioris.
- It shows limited adduction due to medial rectus muscle weakness.
- It shows limited depression due to inferior rectus muscle weakness.
- Normal abduction as the lateral rectus is intact supplied by the 6th nerve.
- Left eye ptosis can be seen in the image is suggestive of one sided lesion in the abducens nerve involving levator palpebrae superioris.
- It shows limited adduction due to medial rectus muscle weakness.
- It shows limited depression due to inferior rectus muscle weakness.
- Normal abduction as the lateral rectus is intact supplied by the 6th nerve.

Incorrect Options:

Option A - Optic nerve: Damage to an optic nerve can result in monocular vision loss.

Option B - Trochlear: Patients with trochlear nerve palsy present with vertical diplopia and torsional diplopia with characteristic head posture to the contralateral side.

Option D - Abducens: Abducens nerve palsy is characterized by esotropia in the affected eye with marked limitation of abduction (weakness of lateral rectus muscle) and normal adduction.

Solution for Question 12:

Correct Option B - Left medial rectus:

- Yoke muscles are contralaterally paired extraocular muscles that work synergistically to direct the gaze in a given direction. For example, in directing the gaze to the right, the right lateral and left medial rectus operate together as yoke muscles.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 13:

Correct Option A - With the rule astigmatism:

- The spectacle prescription should first be converted to a retinoscopic reading. The negative sign in the spherical component indicates myopia. If we solve the prescription, the values on the retinoscope will be -1 on the horizontal axis and -3 on the vertical axis because, in cylindrical lenses, the axis and power of the lens are perpendicular to each other.
- So, the patient has with-the-rule astigmatism as the vertical axis (63 D) is more curved than the horizontal

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 14:

Correct Option C - Rostral interstitial nucleus of the medial longitudinal fasciculus:

- Rostral interstitial nucleus of the medial longitudinal fasciculus is the center for vertical gaze.

- The Rostral Interstitial Nucleus of Cajal (riMLF) is a part of the midbrain, specifically located in the interstitial nucleus of Cajal.
- It is involved in the control and coordination of vertical eye movements.

Incorrect Option:

Option A. Pontine Paramedian Reticular formation:

- The Pontine Paramedian Reticular Formation (PPRF) is the center for horizontal gaze..
- The PPRF is located in the pons and coordinates the activity of the abducens nucleus and the contralateral oculomotor nucleus, facilitating horizontal gaze.

Option B. Raphe Nucleus:

- The raphe nuclei are a group of nuclei found in the brainstem. They are involved in regulating sleep, mood, and certain autonomic functions.

Option D.

- The substantia nigra is a critical brain region for the production of dopamine which affects many systems of the central nervous system ranging from movement control, cognitive executive functions, and emotional limbic activity

Solution for Question 15:

Correct Option B - Left medial rectus:

- Yoke muscles (contralateral synergist) are pairs of muscles, one in each eye, that produce conjugate ocular movements.
- In dextroversion, the muscle involved for the right eye movement is the right lateral rectus muscle and the contralateral synergist in the left eye is medial rectus muscle..

Incorrect Options:

Options A, C, and D are incorrect. Left medial rectus muscle is the yoke muscle of right lateral rectus in dextroversion.

Solution for Question 16:

Correct Option C - It is associated with high myopia:

- This statement is false. Accommodative esotropia is not associated with high myopia, but rather with high hypermetropia. In accommodative esotropia, the excessive effort to focus on near objects leads to an inward deviation of the eyes.

Incorrect Options:

Option A - "It is the most common type of squint": This statement is true. Accommodative esotropia is indeed the most common type of strabismus, particularly in children.

Option B - "Miotics are used as the treatment strategy in the high AC/A ratio": This statement is also true. Miotics, such as eye drops containing medications like pilocarpine or echothiophate iodide, can be used in cases where there is a high AC/A ratio (accommodative convergence/accommodation ratio).

Option D - "Refractive type can be fully corrected by the use of spectacles": This statement is true. The corrective lenses help to correct the underlying hypermetropia, allowing for improved alignment of the eyes and reducing or eliminating the esotropia.

Solution for Question 17:

Correct Option D - Refractive correction:

- The above case is that of accommodative esotropia and the choice of treatment is refractive correction.
- Accommodative esotropia is caused due to high uncorrected hypermetropia of around +4 to +7 D. So, to treat accommodative esotropia convex glasses are used for correction of hypermetropia..

Incorrect Options:

Option A - Orthoptic exercise is used for the improvement in the binocular vision to treat paralytic squint .

Option B - Surgery is used for the non-accommodative esotropia..

Option C - INTACS (intracorneal ring segments) are used for the management of keratoconus.

Solution for Question 18:

Correct Option D - Abducent:

- The nerve most likely lesioned is the abducent nerve (CN VI). The abducent nerve controls the movement of the lateral rectus muscle, which is responsible for moving the eye outward (toward the ear).
- When the abducent nerve is affected or lesioned, it can result in the affected eye being unable to move outward properly, causing the eye to deviate inward, leading to a squint or esotropia.

Incorrect Options:

Option A - Oculomotor (CN III) controls several eye muscles, including the superior rectus, inferior rectus, and medial rectus, and its lesion typically results in ptosis, limited adduction, limited elevation, and limited depression.

Option B - Trochlear (CN IV) results in ipsilateral hypertropia, limited left depression, and a compensatory head posture.

Option C - Trigeminal (CN V) is primarily a sensory nerve for the face and is not directly involved in controlling eye movements.

Solution for Question 19:

Correct Option A - 3rd CN palsy:

- The image depicts limited adduction of the right eye due to weakness of medial rectus muscle seen in 3rd nerve palsy.
- The oculomotor nerve is responsible for innervating several extraocular muscles, including the medial rectus, superior rectus, inferior rectus, and inferior oblique muscles.

Incorrect Options:

Options B, C, and D are incorrect. Adduction is normal in these nerve palsies.

Solution for Question 20:

Correct Option C - Hyperopia:

- The image is suggestive of convergent squint.
- High hyperopia/hypermotropia (+4 to +7D) results in increased exertion of accommodation and results in convergent squint.

Incorrect Options:

Options A, B, and D are incorrect. High hypermetropia results in accommodative esotropia.

Solution for Question 21:

Correct Option C - Paralytic squint:

- In the case of a 40-year-old patient with diabetes who presents with a squint, where the secondary deviation is higher than the primary deviation and the forced duction test yields negative results, is suggestive of Paralytic squint.
- Paralytic squint causes greater deviation in the yoke muscle of the sound eye compared to the eye with the paralyzed muscle. This is due to the strong impulse of innervation required to fix the paralyzed eye, which is transmitted to the yoke muscle of the sound eye. This is based on Hering's law of equal innervation.
- Forced duction test is done to differentiate the mechanical causes from extraocular muscle palsy causes of incomitant squint. It is negative in incomitant squints caused by extraocular muscle palsy (paralytic squint).

Incorrect Options:

Option A - Comitant squint: In comitant squint, the degree of deviation remains the same in all directions of gaze. This means that the primary and secondary deviations are equal.

Option B - Restrictive squint: Restrictive squint occurs when there is a limited range of movement of the affected eye due to restrictions in extraocular muscle function or mechanical factors. The forced duction test is positive in these cases.

Option D - Pseudosquint: Pseudosquint refers to an apparent squint where the eyes appear misaligned, but the alignment is actually normal. It is commonly seen in infants and young children due to the wide nasal bridge and an apparent deviation of the visual axes.

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Retinal Investigations and Vascular Disorders

1. A 29-year-old man presents with sudden, severe visual loss following a recent road traffic accident involving chest trauma. Fundus examination reveals multiple white retinal patches near the optic disc, cotton wool spots, and minimal intraretinal hemorrhage. Based on these findings, what is the most likely diagnosis?

- A. Diabetic Retinopathy
- B. Retinal Vein Occlusion
- C. Purtscher Retinopathy
- D. Central Serous Retinopathy

2. A 30-week premature infant weighing 1.4 kg is diagnosed with stage 3 retinopathy of prematurity (ROP) in zone II without plus disease. What is the most appropriate next step in management?

- A. Laser photocoagulation
- B. Intravitreal anti-VEGF injection
- C. Weekly observation and follow-up
- D. Cryotherapy of retina

3. A 65-year-old male presents to the emergency department with sudden, painless vision loss in his right eye. On examination, the right eye shows a pale retina with a cherry-red spot at the macula. The patient has a history of hypertension and smoking. Which of the following is the most appropriate initial treatment to improve retinal blood flow and potentially restore vision?

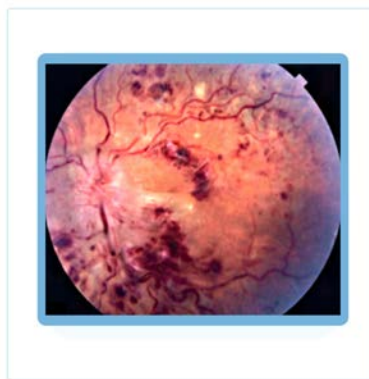
- A. Oral corticosteroids

- B. Intravitreal anti-VEGF injection
 - C. Ocular massage and anterior chamber paracentesis
 - D. Systemic anticoagulation therapy
-

4. A 64-year-old male with long-standing hypertension comes for a routine eye checkup. On fundoscopy, the ophthalmologist observes the following changes as shown in the image below. What is the most likely stage of hypertensive retinopathy?

- A. Grade I
 - B. Grade II
 - C. Grade III
 - D. Grade IV
-

5. A 45-year-old male presents with complaints of severe headache and blurred vision. His blood pressure is 220/120 mmHg. Fundoscopy image is given below. Which of the following fundus changes is most characteristic of this diagnosis?



- A. Arteriovenous nicking
- B. Copper-wiring

- C. Papilledema
 - D. Hard exudates
-

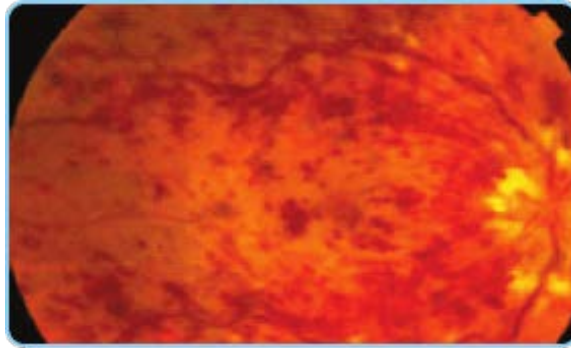
6. A 55-year-old patient with a history of uncontrolled hypertension presents for a routine eye examination. Fundoscopy reveals arteriolar narrowing and changes in the arteriovenous (A-V) ratio. What is the most likely A-V ratio finding in this patient with hypertensive retinopathy?

- A. 2:3
 - B. 1:2
 - C. 1:3
 - D. 3:2
-

7. All of the following are considered risk factors for the development of central retinal vein occlusion (CRVO), except?

- A. Age over 50 years
 - B. Open-angle glaucoma
 - C. Hyperthyroidism
 - D. Hypertension
-

8. A 65-year-old male presents with sudden, severe vision loss in his left eye. On examination, a relative afferent pupillary defect (RAPD) is present. Fundus examination reveals the following changes: Based on these findings, what is the most likely diagnosis?



- A. Non-ischemic CRVO
- B. Ischemic CRVO
- C. Branch Retinal Vein Occlusion (BRVO)
- D. Diabetic Retinopathy

9. Which of the following investigations is most useful in distinguishing ischemic CRVO from non-ischemic CRVO?

- A. Fundus Autofluorescence (FAF)
- B. Optical Coherence Tomography (OCT)
- C. Goldmann Perimetry and Electroretinogram (ERG)
- D. Slit-lamp Examination

10. A 60-year-old male presents with sudden, painless vision loss in his right eye. Fundus examination reveals retinal hemorrhages with massive venous congestion, cotton wool spots, and significant macular edema. Which of the following is least likely to be a complication of this patient's condition?

- A. Neovascular glaucoma
- B. Vitreous hemorrhage
- C. Rhegmatogenous retinal detachment

D. Neovascularization of the iris

11. A 72-year-old man with a history of hypertension presents with sudden, painless vision loss in his right eye. Which of the following findings is most consistent with the diagnosis of central retinal artery occlusion?

- A. Retinal hemorrhages and cotton wool spots
 - B. Optic disc edema and retinal hemorrhages
 - C. Dilated retinal veins with tortuosity
 - D. Pale retina with a cherry-red spot at the macula
-

12. A 58-year-old male with a 20-year history of poorly controlled type 2 diabetes and an HbA1c of 10.5% came for an ophthalmic examination. Fundoscopy reveals microaneurysms, hard exudates, and retinal thickening. Which of the following is the most likely cause of future vision loss in this patient?

- A. Vitreous hemorrhage
 - B. Retinal detachment
 - C. Macular edema
 - D. Neovascular glaucoma
-

13. A cherry-red spot is seen on the retina due to thickening and loss of transparency of the retina at the posterior pole. Which of the following diseases are least associated with this finding on fundoscopy?

- A. Tay-Sachs disease
- B. Farber disease
- C. Central retinal artery occlusion

D. Central retinal vein occlusion

14. A patient is being examined for loss of vision in the ophthalmology clinic. His peripheral visual field is lost, and only his central vision remains. Which of the following would not present with central (tunnel) vision?

- A. Advanced glaucoma
 - B. Retinitis pigmentosa
 - C. CRAO with cilioretinal artery
 - D. Tobacco amblyopia
-

15. A premature infant, born at 26 weeks of gestation, is receiving oxygen therapy in the neonatal intensive care unit. Retinal examination at 4 weeks post-birth reveals signs of retinal vascular abnormalities. Which of the following mechanisms is most likely responsible for the pathogenesis of retinopathy of prematurity in this patient?

- A. Increased oxygen supply leading to retinal vasodilation
 - B. Disruption of VEGF production
 - C. Excessive blood glucose levels
 - D. Retinal hypoxia leading to abnormal collagen formation
-

16. Which of the following is the most significant risk factor for Retinopathy of Prematurity (ROP)?

- A. High birth weight
- B. Prematurity and low birth weight
- C. Prolonged exposure to blue light

D. Deficiency of Vitamin A

17. A 32-week premature infant presents with dilated and tortuous retinal vessels in the posterior pole, poor pupil dilation, and vitreous haze, indicating a rapid progression of retinopathy of prematurity (ROP). What is the most likely diagnosis?

- A. Rush disease
 - B. Plus disease
 - C. Cicatricial ROP
 - D. Stage 4 ROP
-

18. A 23-year-old male presents with sudden, painless vision loss in his left eye. Fundus examination reveals peripheral retinal vasculitis, vessel sheathing and recurrent vitreous hemorrhages. Fluorescein angiography reveals vessel leakage. What is the most appropriate initial treatment?

- A. Intravitreal anti-VEGF therapy
 - B. Systemic corticosteroids
 - C. Laser photocoagulation
 - D. Vitrectomy
-

19. A 3-year-old child is brought to the clinic with a white reflex (leukocoria) noticed in the right eye. An ocular B-scan ultrasound performed to evaluate the child shows a well-defined, echogenic mass within the globe, with associated calcifications. Which is the most likely diagnosis?

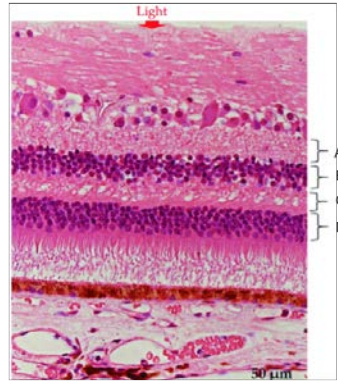


- A. Retinal detachment
 - B. Vitreous hemorrhage
 - C. Retinoblastoma
 - D. Choroidal malignant melanoma
-

20. What is the approximate distance of the fovea from the optic disc in a normal eye?

- A. 1 disc diameter temporal to the optic disc
 - B. 2 disc diameters nasal to the optic disc
 - C. 2 disc diameters temporal to the optic disc
 - D. 1 disc diameter nasal to the optic disc
-

21. In which layer of the retina, as labeled in the histology image, are the cell bodies of bipolar cells located?



- A. A
- B. B
- C. C
- D. D

22. A 65-year-old male with a history of hypertension and atherosclerosis presents with sudden, painless loss of vision in his left eye. On fundoscopy, you note a pale retina with a cherry-red spot at the macula. What is the most likely underlying cause of this condition?

- A. Diabetic Retinopathy
- B. Central Retinal Artery Occlusion
- C. Central Retinal Vein Occlusion
- D. Optic Neuritis

23. Which of the following arteries is primarily responsible for supplying the inner layers of the retina?

- A. Central retinal artery
- B. Short posterior ciliary arteries

- C. Long posterior ciliary arteries
 - D. Ophthalmic artery
-

24. A 55-year-old male with a history of poorly controlled diabetes and hypertension presents with blurred vision. Fundus examination shows bright yellow, well-defined deposits arranged in a star-like pattern around the macula and cloud-like white patches near the optic disc. What is the most likely cause of the yellow deposits?

- A. Soft exudates due to axoplasmic flow blockage
 - B. Hard exudates due to leakage of lipids and proteins
 - C. Retinal hemorrhages from ruptured microaneurysms
 - D. Cotton-wool spots from ischemic swelling of nerve fibers
-

25. The blood-retinal barrier separates the blood from the ocular cavity. Which of the following does not constitute the outer blood-retinal barrier?

- A. Bruch's membrane
 - B. Retinal pigmented epithelium
 - C. Zona occludens
 - D. Retinal vascular endothelial cells
-

26. A 60-year-old patient with a history of diabetes comes in for a routine check-up for diabetic retinopathy. The patient also has a cataract, causing a hazy view of the ocular media. Which method of study would be most appropriate for assessing the retina of this patient?

- A. Direct ophthalmoscopy

- B. Indirect ophthalmoscopy
 - C. Slit-lamp biomicroscopy
 - D. Amsler grid test
-

27. A 60-year-old patient with age-related macular degeneration (ARMD) undergoes Fundus Fluorescein Angiography (FFA), revealing pooling of dye under the retinal pigment epithelium (RPE) and leakage from newly formed vessels beneath the retina. What abnormality is most likely detected on FFA in this patient?

- A. Hyperfluorescence due to dye pooling under detached RPE and neovascularization
 - B. Hypofluorescence due to blockage of background fluorescence by retinal hemorrhage
 - C. Hypofluorescence due to occlusion of retinal vasculature
 - D. Hyperfluorescence due to staining of the optic disc
-

28. A 35-year-old male presents with difficulty in vision, especially at night, and he reports bumping into objects frequently. Upon examination, he is found to have tunnel vision, and the fundus reveals no abnormalities in the posterior pole. Which region of the retina is most likely responsible for his symptoms?

- A. Macula lutea
 - B. Fovea centralis
 - C. Optic disc
 - D. Peripheral retina
-

29. Which of the following electroretinogram (ERG) findings suggests gross disturbances in retinal circulation?

- A. Extinguished ERG response B) Subnormal ERG with a reduced b-wave C) Negative ERG response D) Normal ERG with a diminished c-wave
 - B. Subnormal ERG with a reduced b-wave
 - C. Negative ERG response
 - D. Normal ERG with a diminished c-wave
-

30. A 14-year-old patient presents with progressive visual impairment. Upon examination, there is evidence of vitelliform lesions in the macula. Electro-oculography (EOG) is performed to assess retinal function. What Arden ratio would be expected in a patient with Best disease on electro-oculography (EOG)?

- A. Greater than 2.0
 - B. Less than 1.5
 - C. Between 1.5 and 2.0
 - D. Exactly 1.85
-

31. A patient with chronic diabetes mellitus complains of diminished vision. Her fundoscopy showed clinically significant macular edema. Which of the following would not be seen as a progressive change leading to diabetic retinopathy?

- A. Macular ischemia
 - B. Neovascularization
 - C. Vitreous hemorrhage
 - D. Retinal vasoconstriction
-

32. Which of the following statements is false about laser treatment used for vascular diseases of the retina?

- A. Focal laser therapy for diabetic macular edema targets leaking microaneurysms within 300 μm of the fovea.
 - B. Grid laser acts on diffuse retinal thickening, at least 500 μm from the fovea and optic disc.
 - C. Subthreshold lasers use very short pulses with longer intervals minimizing collateral damage.
 - D. Limitations include central vision risks, visual field defects, and the need for multiple sessions.
-

33. A 60-year-old diabetic patient presents with worsening vision. Fundoscopy reveals retinal edema, microaneurysms, and hard exudates. Which of the following mechanisms is not responsible for the development of diabetic retinopathy?

- A. Protein kinase C activation
 - B. Reduced blood viscosity
 - C. Increased proangiogenic factors
 - D. Breakdown of the blood-retinal barrier
-

34. A 55-year-old patient with a long history of diabetes mellitus presents for a routine eye examination. The ophthalmologist noted several findings in the retinal examination. Which of the following features is least likely to be observed in early nonproliferative diabetic retinopathy (NPDR)?

- A. Microaneurysm
- B. Dot and blot hemorrhages
- C. Hard exudates
- D. Venous beading

35. A 60-year-old patient with chronic diabetes is diagnosed with diabetic retinopathy. What is the most specific finding for confirming proliferative diabetic retinopathy (PDR) in this patient?

- A. Cotton-wool spots
 - B. Retinal hemorrhage
 - C. Neovascularization of disc
 - D. Intraretinal microvascular abnormalities
-

36. A 58-year-old male with Type 2 diabetes undergoes a fundus examination, revealing extensive retinal hemorrhages, cotton wool spots, venous beading, and moderate intraretinal microvascular abnormalities (IRMA) in one quadrant. What is the most appropriate follow-up interval for this patient?

- A. 2 months
 - B. 4 months
 - C. 6 months
 - D. 12 months
-

Correct Answers

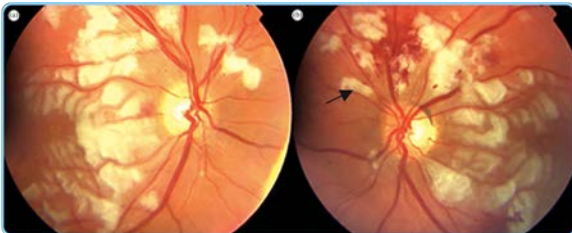
Question	Correct Answer
Question 1	3
Question 2	3
Question 3	3
Question 4	3
Question 5	3
Question 6	3
Question 7	3
Question 8	2
Question 9	3
Question 10	3
Question 11	4
Question 12	3
Question 13	4
Question 14	4
Question 15	2
Question 16	2
Question 17	2
Question 18	2
Question 19	3
Question 20	3
Question 21	2
Question 22	2

Question 23	1
Question 24	2
Question 25	4
Question 26	2
Question 27	1
Question 28	4
Question 29	3
Question 30	2
Question 31	4
Question 32	1
Question 33	3
Question 34	4
Question 35	3
Question 36	2

Solution for Question 1:

Purtscher Retinopathy is a condition typically associated with trauma (especially head or chest trauma) and is characterized by multiple retinal white patches around the optic nerve head and fovea, known as Purtscher flecken, cotton wool spots, and minimal intraretinal hemorrhage.

The occlusion of peripapillary arterioles and embolization is the key pathophysiological mechanism. In this case, the sudden severe visual loss after trauma, along with the described retinal findings, suggests Purtscher Retinopathy.



Feature	Description
Etiology	Trauma (head, chest), fractures, systemic diseases like SLE, pancreatitis, pre-eclampsia.
Fundoscopy features	Purtscher Flecken: White patches near the optic disc and fovea due to arteriolar occlusion. Cotton Wool Spots: Infarcts in the nerve fiber layer, typically restricted to the posterior pole. Retinal Hemorrhages: Small, low-to-moderate hemorrhages.
Diagnostic Criteria	Requires 3 out of 5: Purtscher flecken Hemorrhages Cotton wool spots Plausible cause Investigation compatible with the diagnosis
Ocular investigations	Fundus Fluorescein Angiography (FFA): Reveals macular ischemia, capillary nonperfusion, and optic nerve head leakage. Optical Coherence Tomography (OCT): Demonstrates retinal thickening, subretinal fluid, and photoreceptor loss in later stages. OCT Angiography: Can show retinal capillary ischemia.
Management	Supportive care, steroids, anti-VEGF for macular edema.

Diabetic Retinopathy (Option A) is associated with microvascular changes due to diabetes, including dot-blot hemorrhages, microaneurysms, and neovascularization, but not Purtscher flecken or trauma-related visual loss.

Retinal Vein Occlusion (Option B) presents with diffuse retinal hemorrhages, cotton wool spots, and optic disc swelling but is not typically linked to trauma.

Central Serous Retinopathy (Option D) involves serous retinal detachment usually due to stress, without the hemorrhages or cotton wool spots seen in Purtscher Retinopathy.

Solution for Question 2:

Stage 3 ROP without plus disease in zone II is considered to be Low risk Pre threshold. It typically warrants close observation with follow-up rather than immediate treatment unless there are signs of progression.

Anatomical classification of ROP by Zones:

Zone I	Bounded by a circle, the radius of which is twice the distance between the center of the optic disc and center of the macula.
Zone II	Extends from the edge of the zone 1 peripherally to a point tangential to the nasal ora serrata.
Zone III	Consists of residual temporal crescent of retina anterior to zone II.

- Threshold Disease: Stage 3+ ROP with plus disease. Located in zone I or II. Involves 5 continuous or 8 discontinuous clock hours. Requires laser therapy within 72 hours.
- Located in zone I or II.
- Involves 5 continuous or 8 discontinuous clock hours.
- Requires laser therapy within 72 hours.
- Prethreshold Disease: Classified by the Early Treatment for Retinopathy of Prematurity (ET-ROP) group into two types:
 - Located in zone I or II.
 - Involves 5 continuous or 8 discontinuous clock hours.
 - Requires laser therapy within 72 hours.

Type I	Type II
High-Risk Prethreshold Disease, requiring laser photocoagulation. Zone I: Plus disease, any stage. Zone I: Non-plus disease, stage 3. Zone II: Plus disease, stages 2 and 3. (Option A)	Low-Risk Prethreshold Disease, requiring weekly follow-up. Zone I: Non-plus disease, stages 1 and 2. Zone II: Non-plus disease, stage 3. (Option C)

Treatment of ROP:

- Done in the avascular peripheral retina.
- Use diode laser or frequency doubled Nd:YAG with LIO (laser indirect ophthalmoscope) delivery.
- Used for high-risk pre-threshold, threshold, and aggressive posterior ROP.
- Bevacizumab: Effective in patients with zone I, stage 3 ROP with plus disease.
- Zone I disease responds more than zone II.
- Recurrence can occur following cessation of anti-VEGF treatment.

Cryotherapy of retina (Option D) is not preferred for stage 3 ROP, especially in zone II without plus disease. It has been largely replaced by laser therapy due to its higher risk of causing inflammation and less favorable outcomes.

Solution for Question 3:

The goal of the initial treatment is to restore retinal blood flow. Ocular massage and anterior chamber paracentesis are commonly used methods to dislodge emboli and improve perfusion.

Treatment Goals	
Restore Blood Flow	Dislodge or break down the blockage in the central retinal artery as quickly as possible.
Minimize Retinal Damage	Limit the duration of ischemia to reduce permanent damage to retinal tissue.
Address Underlying Conditions	Identify and manage underlying conditions like atherosclerosis, hypertension, and cardiac diseases.
Treatment Options	
Ocular Massage (within 24 hrs)	Apply pressure to the globe to dislodge the embolus; performed using a three-mirror contact lens.
Anterior Chamber Paracentesis	Withdraw aqueous humor to decrease intraocular pressure, potentially dislodging the embolus.
Medications to Lower IOP	
Topical Apraclonidine & Timolol	Reduce aqueous humor production to lower IOP
Intravenous Acetazolamide (Option C)	Provides sustained IOP reduction
Vasodilators	
Sublingual Isosorbide Dinitrate (Option D)	Dilates blood vessels to improve blood flow past the blockage
Carbon Dioxide Inhalation	
"Rebreathing" into a Paper Bag	Increases carbon dioxide levels to induce vasodilation
Carbogen Breathing	Controlled inhalation of high oxygen (95%) and carbon dioxide (5%) mixtures to induce vasodilation and slow down ischemia.
Hyperosmotic agents	
Mannitol or glycerol (Option A&B)	Rapid IOP-lowering effect and raised intravascular volume

Transluminal Nd:YAG laser embolysis/embolectomy	For BRAO and CRAO where an occluding embolus is visible; Good final visual results achieved
Thrombolysis	
Local Intra-arterial or Intravenous	Directly dissolve clots with thrombolytic agents; efficacy in CRAO is inconclusive.
Local Intraocular Thrombolysis	Use of rtPA for thrombolysis; large trials show no significant benefit
Systemic Management	
Urgent Referral	Immediate evaluation by a stroke specialist due to high stroke risk.
Management of Risk Factors	
Smoking Cessation	Reduce overall vascular risk.
Cardiac Arrhythmia Management	Refer to a physician for conditions like atrial fibrillation.
Antiplatelet Therapy	To reduce further blood clot formation, unless contraindicated.
Oral Anticoagulation	Warfarin or other anticoagulants for conditions like atrial fibrillation
Carotid Endarterectomy	Considered for significant carotid artery stenosis.

Solution for Question 4:

The funduscopy image shows Grade III Hypertensive Retinopathy characterized by generalized arteriolar narrowing, hard exudates arranged in a macular star pattern, and cotton wool spots with no papilledema.

Hypertensive retinopathy refers to retinal changes seen in patients with systemic hypertension. The condition's development involves several pathological mechanisms, including vasospasm, arteriosclerosis, increased vascular permeability, and raised intracranial pressure.

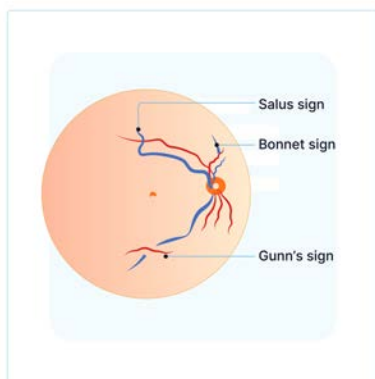
The pathophysiology of hypertensive retinopathy is thought to progress through overlapping, non-sequential stages: The vasoconstrictive phase, Exudative phase, Sclerotic phase, and complications arising from the sclerotic phase.

- Vasospasm: The initial reaction to elevated blood pressure is retinal arteriolar narrowing due to vasospasm. This is more noticeable in younger patients and leads to ischemia in the

retinal pigment epithelium (RPE), choroid, and optic nerve head.

- **Arteriosclerosis:** Arteriosclerotic changes reflect chronic hypertension. These changes are associated with the thickening of vessel walls, leading to arteriolar reflex alterations and arteriovenous (A-V) crossing abnormalities.
- **Increased Vascular Permeability:** Severe hypertension causes hypoxia and breakdown of the inner blood-retinal barrier, leading to retinal hemorrhages, exudates, and macular edema.
- **Raised Intracranial Pressure:** Seen in malignant hypertension, causing hypertensive optic neuropathy characterized by papilledema and optic nerve ischemia.
- **Clinical Features:** 1. Benign (Chronic) Hypertensive Retinopathy:
 - **Generalized arteriolar narrowing:** May be mild or marked, with vasoconstrictive and sclerotic phases.
 - **Focal arteriolar narrowing:** Localized vasoconstriction near the disc.
 - **A-V nicking:** Compression of veins at A-V crossings, Salus's sign (right-angle deflection of veins at arteriovenous crossings), Bonnet sign (banking of veins distal to arteriovenous crossings), and Gunn sign (tapering of veins on either side of the crossings)
 - .
 - **Arteriolar reflex changes:** Brightening of light reflex in younger individuals, with copper wiring (reddish-brown reflex of the arterioles occurs due to progressive sclerosis and hyalinization) and silver wiring (opaque- white reflex of the arterioles due to the continued sclerosis) seen in advanced disease.
 - **Retinal hemorrhages:** Flame-shaped hemorrhages at the posterior pole.
 - **Hard exudates:** Yellowish deposits in the retina, arranged in a macular-star pattern.
 - **Cotton wool spots:** Fluffy white areas representing nerve fiber infarctions.
- 2. Malignant Hypertensive Retinopathy:
 - **Acute retinal changes:** Marked arteriolar narrowing, flame-shaped hemorrhages, and cotton wool spots.
 - **Choroidal involvement:** Focal retinal pigment epithelium changes, Elschnig's spots, and Siegrist streaks due to choroidal ischemia.
 - **Optic neuropathy:** Disc edema and hemorrhages in the peripapillary area due to ischemia from vasoconstriction of choroidal vessels.

1. Benign (Chronic) Hypertensive Retinopathy:



2. Malignant Hypertensive Retinopathy:

Stage	Key Features
Grade I (Option A)	Mild generalized arteriolar narrowing, broadening of the light reflex.
Grade II (Option B)	Focal arteriolar narrowing, A-V nicking (Salu's sign)
Grade III (Option C)	Flame-shaped hemorrhages, cotton wool spots, and more pronounced A-V nicking (Bonnet and Gunn signs), hemorrhages, and exudates.
Grade IV (Option D)	All features of Grade III + Papilledema, Silver-wiring of arterioles

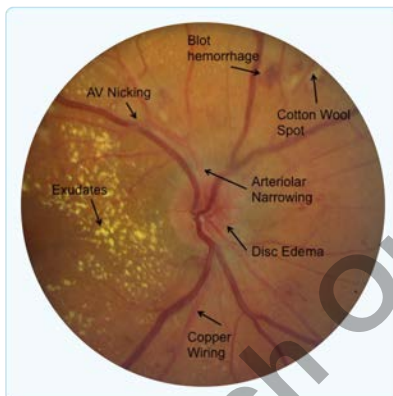
- Mild Retinopathy: Blood pressure control.
- Moderate Retinopathy: Blood pressure control and evaluation of vascular risk factors.
- Malignant Retinopathy: Careful and gradual reduction of blood pressure to prevent sudden perfusion loss.

Solution for Question 5:

Fundoscopy findings:

- Papilledema,
- exudate,
- retinal hemorrhages,
- venous dilation,
- narrowing of the arterioles,
- Elschnig's spots (small black spots surrounded by yellow halos formed due to choroidopathy),
- Siegrist's streaks (linear configuration of the pigment along the choroidal arterioles formed due to fibrinoid necrosis)

These findings indicate Malignant Hypertensive Retinopathy. Papilledema or Disc edema is a hallmark of malignant hypertensive retinopathy or acute hypertensive crisis, characterized by raised intracranial pressure and optic nerve head ischemia.



Feature	Chronic Hypertensive Retinopathy	Malignant Hypertensive Retinopathy
Arteriolar Narrowing	Generalized narrowing, focal narrowing, A-V nicking (Option A)	Marked narrowing due to spasm
Retinal Reflex Changes	Copper-wiring (Option B), silver-wiring	Not prominent
Retinal Hemorrhages	Flame-shaped at the posterior pole	More numerous flame-shaped hemorrhages
Exudates	Hard exudates in macular-star pattern (Option D)	May have focal intraretinal periarterial transudates (FIPTs)
Cotton Wool Spots	Present, indicating ischemia	More prominent

Choroidal Involvement	No significant choroidal involvement	Elschnig spots, Siegrist streaks
Optic Neuropathy	Absent or minimal	Disc edema and optic nerve ischemia

Solution for Question 6:

In hypertensive retinopathy, arteriolar narrowing occurs as a result of vasoconstriction and thickening of the vessel walls. This causes a change in the normal A-V ratio to 1:3. In normal fundoscopy, the ratio of the diameters of first- and second-order retinal arterioles to the diameters of venules is approximately 2:3.

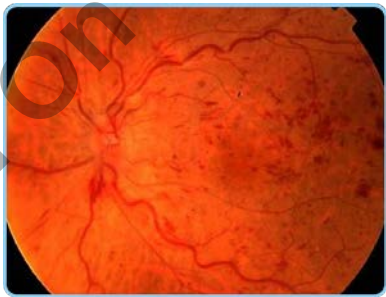
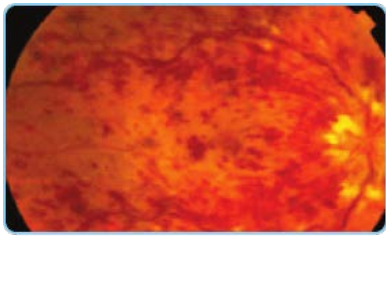
Solution for Question 7:

Hyperthyroidism is not associated with an increased risk of central retinal vein occlusion (CRVO). It primarily affects metabolism and does not directly influence the vascular system, predisposing to CRVO.

Factors	Details
Primary Risk Factors	Age > 50 years (primary risk factor) (Option A), Hypertension (Option D), Open-angle glaucoma (Option B), Diabetes, Hyperlipidemia.
Additional Risk Factors	Smoking, optic disc edema, oral contraceptives, hypercoagulable states (e.g., polycythemia, multiple myeloma, antiphospholipid syndrome, hyperhomocysteinemia, etc.), syphilis, HIV, sickle cell disease.
Pathophysiology	The exact etiology is unknown, but the occlusion of the Central retinal vein posterior to the lamina cribrosa of the optic nerve typically by thrombosis, is a key mechanism. This is contributed by venous stasis and

hypercoagulability. Atherosclerosis in the artery can lead to compression of the adjacent vein.

Solution for Question 8:

Feature	Non-ischemic CRVO (Venous Stasis Retinopathy)	Ischemic CRVO (Hemorrhagic Retinopathy)
Prevalence	75% of cases	25% of cases
Vision	Mild to moderate blurring	Severe, sudden vision loss
RAPD	Absent or mild	Present
Early Fundus Findings	Mild venous congestion, peripheral flame-shaped hemorrhages, mild/no macular edema	Massive venous engorgement, "splashed-tomato" hemorrhages, cotton wool spots, macular edema
Late Fundus Findings	Vein sheathing, cilioretinal collaterals, absorbed hemorrhages, chronic cystoid edema	Marked vein sheathing, collaterals, neovascularization, chronic cystoid edema
Complications	15-30% convert to ischemic CRVO	Neovascular glaucoma, vitreous hemorrhage, proliferative retinopathy
Image		

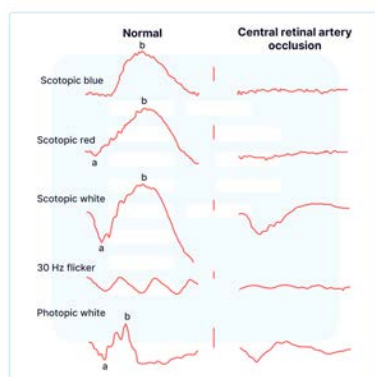
Non-ischemic CRVO (Option A) is more common (75% of cases), but it typically presents with milder symptoms, including gradual vision blurring and the absence or mild presence of RAPD. Fundus findings are less severe, with mild venous congestion and flame-shaped hemorrhages in the periphery.

Branch Retinal Vein Occlusion (BRVO) (Option C) typically affects a portion of the retina rather than the entire retina, and the hemorrhages are more localized to the area of the occluded vein. Vision loss is generally less severe compared to ischemic CRVO.

Diabetic Retinopathy (Option D) can present with retinal hemorrhages and macular edema, but it is not associated with sudden, severe vision loss and RAPD as in this case.

Solution for Question 9:

Goldmann Perimetry and ERG are crucial for differentiating between ischemic and non-ischemic CRVO. In ischemic CRVO, there is significant retinal ischemia leading to visual field defects and a reduction in the amplitude of the b-wave on ERG. These findings help to confirm the diagnosis of ischemic CRVO.



Investigations	Details
Ocular Examination	Visual acuity monitoring IOP measurement Slit-lamp examination for neovascularization of the iris (NVI) Gonioscopy to exclude neovascularization of the angle (NVA) Fundus examination (direct and indirect)
Ocular Investigations	Goldmann perimetry: Important for differentiating visual field defects in ischemic from non-ischemic CRVO. ERG: Reduced b-wave amplitude in ischemic CRVO. Fundus autofluorescence (FAF): Demonstrates fern-like perivenular hypo autofluorescence due to retinal edema. FFA: Assess retinal perfusion (Ischemic CRVO shows >10 disc areas of capillary non-perfusion). OCT: To evaluate macular edema, sub-retinal fluid, and epiretinal membrane (ERM) formation.
Systemic Investigations	Check for hypertension, diabetes, dyslipidemia, hypercoagulable states, homocysteinemia.

Medical Treatment	Intravitreal steroids (triamcinolone, dexamethasone) Intravitreal anti-VEGF (Bevacizumab, Ranibizumab)
Laser Treatment	Panretinal Photocoagulation (PRP): Indicated when neovascularization occurs (NVI, NVA, NVE, or NVD). Grid Laser: Not recommended for macular edema in CRVO.
Surgical Treatment	Pars plana vitrectomy (PPV) for complications (vitreous hemorrhage, retinal detachment, ERM) Glaucoma drainage devices for neovascular glaucoma.

Fundus Autofluorescence (FAF) (Option A) demonstrates fern-like perivenular hypoafluorescence due to retinal edema in both non-ischemic and ischemic CRVO.

Optical Coherence Tomography (OCT) (Option B) is valuable in evaluating macular edema, which may be present in both non-ischemic and ischemic CRVO. Goldmann perimetry and ERG are most useful in differentiating ischemic from non-ischemic CRVO.

Slit-lamp examination (Option D) is essential for detecting anterior segment changes such as neovascularization of the iris (NVI). It is mainly used to monitor for complications like neovascular glaucoma.

Solution for Question 10:

Rhegmatogenous retinal detachment is not commonly associated with ischemic CRVO. It is caused when fluid accumulates beneath the sensory retina, separating it from the underlying pigment epithelium. This separation typically results from a break in the retina, such as a hole or tear, allowing fluid to seep through as usually seen in Myopia.

Ischemic CRVO can lead to severe complications such as Neovascularization of the iris (Option D), Vitreous hemorrhage (Option B), Proliferative retinopathy, and Neovascular glaucoma (Option A), which is a frequent complication, occurring in about 20% of ischemic CRVO cases within 3 months, also referred to as "90-day glaucoma."

Learning Objective:

Feature	Details	
Risk Factor	Age > 50 years (primary risk factor), Hypertension, Diabetes Mellitus,	

tors	Hyperlipidemia, Glaucoma (increased IOP), Smoking, Hypercoagulable states	
Pathogenesis	Thrombosis driven by venous stasis, endothelial damage, and hypercoagulability	
Types	Non-ischemic CRVO (Venous stasis retinopathy) (75% of cases) Mild to moderate, unilateral blurring of vision Fundus: Mild venous congestion and tortuosity, flame-shaped hemorrhages, mild macular edema	Ischemic CRVO (Hemorrhagic retinopathy) Severe, sudden visual loss, massive venous engorgement, "splashed-tomato" hemorrhages, cotton wool spots, macular edema Marked vein sheathing, collaterals, neovascularization, chronic cystoid edema
Investigations	Ocular Exam: Visual acuity, IOP, Slit-lamp exam (NVI), Gonioscopy (NVA) Goldmann Perimetry and ERG: Differentiating ischemic vs non-ischemic CRVO Fundus Fluorescein Angiography (FFA): Assess retinal perfusion Optical Coherence Tomography (OCT): Assess macular edema, sub-retinal fluid	
Treatment	Systemic Management: Control of systemic conditions (hypertension, diabetes, hyperlipidemia, hyperhomocysteinemia) Medical Treatment: Intravitreal anti-VEGF (e.g., Bevacizumab), Intravitreal steroids (Triamcinolone, Dexamethasone implant) Laser Therapy: Panretinal photocoagulation (PRP) for neovascularization Surgical: Pars plana vitrectomy for complications	
Complications	Neovascular glaucoma, Neovascularization of the iris (NVI), Vitreous hemorrhage, Proliferative retinopathy, Chronic macular edema	

Solution for Question 11:

In central retinal artery occlusion, the cherry-red spot is seen at the center of the macula due to the vascular choroid shining through the thin foveal retina, contrasting with the surrounding pale retina caused by ischemia.

Central Retinal Artery Occlusion (CRAO)

Risk factors:

- Atherosclerosis
- Age
- Hypertension
- Cardiac Diseases Atrial fibrillation: promotes blood clot formation, increasing CRAO risk Mitral valve prolapse: this can lead to blood flow abnormalities and clot formation Bacterial endocarditis: causes vegetation that can travel and occlude the retinal artery
- Atrial fibrillation: promotes blood clot formation, increasing CRAO risk
- Mitral valve prolapse: this can lead to blood flow abnormalities and clot formation
- Bacterial endocarditis: causes vegetation that can travel and occlude the retinal artery
- Atrial fibrillation: promotes blood clot formation, increasing CRAO risk
- Mitral valve prolapse: this can lead to blood flow abnormalities and clot formation
- Bacterial endocarditis: causes vegetation that can travel and occlude the retinal artery
- Thrombophilic disorders Hyperhomocysteinemia: elevated homocysteine levels increase clot formation risk Antiphospholipid syndrome: an autoimmune disorder causing increased clot formation risk Inherited anticoagulant defects: genetic conditions disrupting natural blood clotting regulation
- Hyperhomocysteinemia: elevated homocysteine levels increase clot formation risk
- Antiphospholipid syndrome: an autoimmune disorder causing increased clot formation risk
- Inherited anticoagulant defects: genetic conditions disrupting natural blood clotting regulation
- Giant cell arteritis (GCA) Inflammatory condition affecting blood vessels, often with severe visual loss
- Inflammatory condition affecting blood vessels, often with severe visual loss
- Hyperhomocysteinemia: elevated homocysteine levels increase clot formation risk
- Antiphospholipid syndrome: an autoimmune disorder causing increased clot formation risk
- Inherited anticoagulant defects: genetic conditions disrupting natural blood clotting regulation

- Inflammatory condition affecting blood vessels, often with severe visual loss
- Other factors Smoking Diabetes mellitus Oral contraceptive pills
- Smoking
- Diabetes mellitus
- Oral contraceptive pills
- Smoking
- Diabetes mellitus
- Oral contraceptive pills

Clinical features:

Symptoms:

- Sudden, painless, profound vision loss
- Painless, except in GCA: GCA-associated CRAO may present with headache, jaw claudication, scalp tenderness, and/or pain around the temple

Clinical Signs:

- Visual Acuity (VA): severely reduced, often to counting fingers or worse
- Relative Afferent Pupillary Defect (RAPD): Profound and sometimes total (amaurotic pupil)

Fundus Examination:

- Retinal Whitening: Pale and opaque retina, particularly in the posterior pole
- Cherry-Red Spot: Bright red area at the fovea due to the contrast between pale retina and underlying choroidal circulation
- Cattle truck appearance: Due to segmentation of blood column in retinal veins
- Attenuated Arteries: Narrowed and segmented retinal arterioles
- Optic Disc Changes: Pale optic disc over time due to optic atrophy from prolonged ischemia is seen in chronic cases.
- Fluorescein Angiography (FA)
- Arterial Filling: Delayed
- Masking of Background Fluorescence: Edematous retina blocks normal fluorescence from the choroid
- Patent Cilioretinal Artery: Area of normal perfusion if a cilioretinal artery supplies the fovea, potentially preserving central vision
- Other Diagnostic Tests Reveals retinal edema, structural changes, and possibly an embolic plaque within the retinal artery and other systemic causes.
- Cranial MRI or CT

- Carotid duplex scanning
- Echocardiography
- Thrombophilia screening

Fluorescein Angiography (FA)

Other Diagnostic Tests

Reveals retinal edema, structural changes, and possibly an embolic plaque within the retinal artery and other systemic causes.

Solution for Question 12:

Complications of Diabetic Retinopathy:

1. Macular Edema: (Option C)

- Most common cause of vision loss in diabetic retinopathy is leakage from dilated capillaries.
 - Clinically significant macular edema (CSME): Retinal thickening at the center of the macula. Retinal thickening and/or adjacent hard exudates at or within 500 μ of the center of the macula. An area of retinal thickening greater than or equal to one disc area, any part of which is within 1 disc diameter of the center of the macula. Types: Focal or Diffuse macular edema Diagnosis: Slit-lamp Biomicroscopy: Loss of foveal reflex and macular thickening Fluorescein Angiography Identifies leakage and also diagnoses ischemic maculopathy. Treatment: Anti-VEGF Therapy: Bevacizumab, ranibizumab or aflibercept. Laser treatment should be deferred for 6 months after commencement of anti-VEGF injection. Intravitreal Triamcinolone acetonide: Reduce macular edema and improve visual acuity. Photocoagulation: Focal Photocoagulation: Treatment of choice for focal diabetic macular edema (DME) not involving the center of the fovea using diode or argon burns. Grid Photocoagulation: Considered in recalcitrant cases unresponsive to anti-VEGF agents or intravitreal steroids. Burns are applied to macular areas of diffuse retinal thickening, treating no closer than 500 μ from the foveola and 500 μ from the optic disc.
2. Vitreous Hemorrhage (VH): (Option A) Seen as blood floating in the vitreous, occluding the view of the retina. Secondary to proliferative diabetic retinopathy (PDR) is a cause of severe vision loss in diabetic patients. In PDR, fibrovascular tissue lies flat on the retina or attaches to the posterior vitreous, leading to vitreous traction, retinal separation, and vessel rupture. This is the most common cause of VH in adults.
3. Retinal Detachment: (Option B) PDR causes

scars to form in the back of the eye. When the scars pull the retina away from the back of the eye, it's called tractional retinal detachment. 4. Neovascular Glaucoma: (Option D) PDR causes abnormal blood vessels to grow out of the retina and block fluid from draining out of the eye, leading to neovascular glaucoma. Can cause vision loss and blindness. 5. Rubeosis Iridis: Neovascularization of iris, is when blood vessels develop on the anterior surface of the iris in response to retinal ischemia. Prone to spontaneous hyphema as the blood vessels are fragile. Treatment: Panretinal photocoagulation with or without intravitreal injection of anti-VEGF.

- Retinal thickening at the center of the macula.
- Retinal thickening and/or adjacent hard exudates at or within 500 μ of the center of the macula.
- An area of retinal thickening greater than or equal to one disc area, any part of which is within 1 disc diameter of the center of the macula.
- Types: Focal or Diffuse macular edema
- Diagnosis: Slit-lamp Biomicroscopy: Loss of foveal reflex and macular thickening
Fluorescein Angiography Identifies leakage and also diagnoses ischemic maculopathy.
Treatment: Anti-VEGF Therapy: Bevacizumab, ranibizumab or aflibercept. Laser treatment should be deferred for 6 months after commencement of anti-VEGF injection. Intravitreal Triamcinolone acetonide: Reduce macular edema and improve visual acuity.
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- 3. Retinal Detachment: (Option B)

- PDR causes scars to form in the back of the eye. When the scars pull the retina away from the back of the eye, it's called tractional retinal detachment.
- 4. Neovascular Glaucoma: (Option D)
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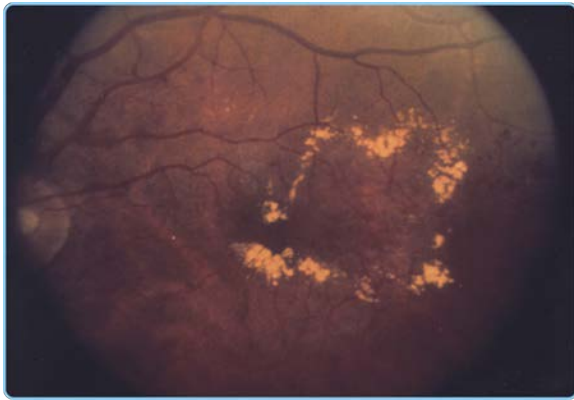
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- Treatment:
 - Anti-VEGF Therapy: Bevacizumab, ranibizumab or aflibercept. Laser treatment should be deferred for 6 months after commencement of anti-VEGF injection.
 - Laser treatment should be deferred for 6 months after commencement of anti-VEGF injection.
 - Intravitreal Triamcinolone acetonide: Reduce macular edema and improve visual acuity.
 - Photocoagulation: Focal Photocoagulation: Treatment of choice for focal diabetic macular edema (DME) not involving the center of the fovea using diode or argon burns. Grid Photocoagulation: Considered in recalcitrant cases unresponsive to anti-VEGF agents or intravitreal steroids. Burns are applied to macular areas of diffuse retinal thickening, treating no closer than 500 μ from the foveola and 500 μ from the optic disc. 2. Vitreous Hemorrhage (VH): (Option A) Seen as blood floating in the vitreous, occluding the view of the retina. Secondary to proliferative diabetic retinopathy (PDR) is a cause of severe vision loss in diabetic patients. In PDR, fibrovascular tissue lies flat on the retina or attaches to the posterior vitreous, leading to vitreous traction, retinal separation, and vessel rupture. This is the most common cause of VH in adults. 3. Retinal Detachment: (Option B) PDR causes scars to form in the back of the eye. When the scars pull the retina away from the back of the eye, it's called tractional retinal detachment. 4. Neovascular Glaucoma: (Option D) PDR causes abnormal blood vessels to grow out of the retina and block fluid from draining out of the eye, leading to neovascular glaucoma. Can cause vision loss and blindness. 5. Rubeosis Iridis: Neovascularization of iris, is when blood vessels develop on the anterior surface of the iris in response to retinal ischemia. Prone to spontaneous hyphema as the blood vessels are fragile. Treatment: Panretinal photocoagulation with or without intravitreal injection of anti-VEGF.
 - Focal Photocoagulation: Treatment of choice for focal diabetic macular edema (DME) not involving the center of the fovea using diode or argon burns.
 - Grid Photocoagulation: Considered in recalcitrant cases unresponsive to anti-VEGF agents or intravitreal steroids. Burns are applied to macular areas of diffuse retinal thickening, treating no closer than 500 μ from the foveola and 500 μ from the optic disc.
- 2. Vitreous Hemorrhage (VH): (Option A)
 - Seen as blood floating in the vitreous, occluding the view of the retina.
 - Secondary to proliferative diabetic retinopathy (PDR) is a cause of severe vision loss in diabetic patients.

- In PDR, fibrovascular tissue lies flat on the retina or attaches to the posterior vitreous, leading to vitreous traction, retinal separation, and vessel rupture. This is the most common cause of VH in adults.
 - 3. Retinal Detachment: (Option B)
 - PDR causes scars to form in the back of the eye. When the scars pull the retina away from the back of the eye, it's called tractional retinal detachment.
 - 4. Neovascular Glaucoma: (Option D)
 - PDR causes abnormal blood vessels to grow out of the retina and block fluid from draining out of the eye, leading to neovascular glaucoma.
 - Can cause vision loss and blindness.
 - 5. Rubeosis Iridis:
 - Neovascularization of iris, is when blood vessels develop on the anterior surface of the iris in response to retinal ischemia.
 - Prone to spontaneous hyphema as the blood vessels are fragile.
 - Treatment: Panretinal photocoagulation with or without intravitreal injection of anti-VEGF.
 - Slit-lamp Biomicroscopy: Loss of foveal reflex and macular thickening
 - Fluorescein Angiography Identifies leakage and also diagnoses ischemic maculopathy.
 - Treatment:
 - Anti-VEGF Therapy: Bevacizumab, ranibizumab or aflibercept. Laser treatment should be deferred for 6 months after commencement of anti-VEGF injection.
 - Laser treatment should be deferred for 6 months after commencement of anti-VEGF injection.
 - Intravitreal Triamcinolone acetonide: Reduce macular edema and improve visual acuity.
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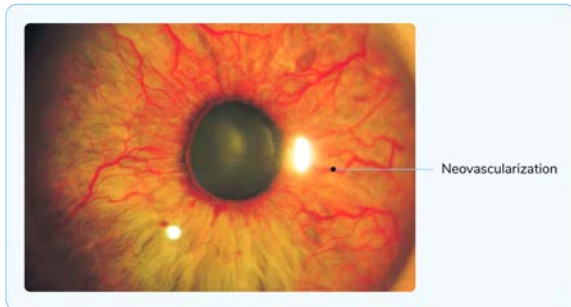
Iritis: Neovascularization of iris, is when blood vessels develop on the anterior surface of the iris in response to retinal ischemia. Prone to spontaneous hyphema as the blood vessels are fragile. Treatment: Panretinal photocoagulation with or without intravitreal injection of anti-VEGF.

- Focal Photocoagulation: Treatment of choice for focal diabetic macular edema (DME) not involving the center of the fovea using diode or argon burns.
- Grid Photocoagulation: Considered in recalcitrant cases unresponsive to anti-VEGF agents or intravitreal steroids. Burns are applied to macular areas of diffuse retinal thickening, treating no closer than 500 μ from the foveola and 500 μ from the optic disc.
- 2. Vitreous Hemorrhage (VH): (Option A)
 - Seen as blood floating in the vitreous, occluding the view of the retina.
 - Secondary to proliferative diabetic retinopathy (PDR) is a cause of severe vision loss in diabetic patients.
 - In PDR, fibrovascular tissue lies flat on the retina or attaches to the posterior vitreous, leading to vitreous traction, retinal separation, and vessel rupture. This is the most common cause of VH in adults.
- 3. Retinal Detachment: (Option B)
 - PDR causes scars to form in the back of the eye. When the scars pull the retina away from the back of the eye, it's called tractional retinal detachment.
- 4. Neovascular Glaucoma: (Option D)
 - PDR causes abnormal blood vessels to grow out of the retina and block fluid from draining out of the eye, leading to neovascular glaucoma.
 - Can cause vision loss and blindness.
- 5. Rubeosis Iridis:
 - Neovascularization of iris, is when blood vessels develop on the anterior surface of the iris in response to retinal ischemia.
 - Prone to spontaneous hyphema as the blood vessels are fragile.
 - Treatment: Panretinal photocoagulation with or without intravitreal injection of anti-VEGF.



- Laser treatment should be deferred for 6 months after commencement of anti-VEGF injection.
- Focal Photocoagulation: Treatment of choice for focal diabetic macular edema (DME) not involving the center of the fovea using diode or argon burns.
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- 2. Vitreous Hemorrhage (VH): (Option A)
- Seen as blood floating in the vitreous, occluding the view of the retina.
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- In PDR, fibrovascular tissue lies flat on the retina or attaches to the posterior vitreous, leading to vitreous traction, retinal separation, and vessel rupture. This is the most common cause of VH in adults.
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- PDR causes abnormal blood vessels to grow out of the retina and block fluid from draining out of the eye, leading to neovascular glaucoma.
- Can cause vision loss and blindness.
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 - Neovascularization of iris, is when blood vessels develop on the anterior surface of the iris in response to retinal ischemia.
 - Prone to spontaneous hyphema as the blood vessels are fragile.
 - Treatment: Panretinal photocoagulation with or without intravitreal injection of anti-VEGF.

2. Vitreous Hemorrhage (VH): (Option A)
3. Retinal Detachment: (Option B)
4. Neovascular Glaucoma: (Option D)
5. Rubeosis Iridis:



Solution for Question 13:

Central retinal vein occlusion causes mild blurring of vision and, on fundoscopy, shows superficial flame-shaped hemorrhages, papilloedema, mild or no macular edema in acute cases and sheathing of main veins, chronic cystoid edema in chronic cases. A cherry-red spot is not seen in central retinal vein occlusion.

Conditions with Cherry-red spot:



A) Metabolic storage diseases:

Causes pathological accumulation of lipid-based material in various tissues.

The spot becomes less evident with time and there is optic atrophy.

They are:

- GM1 gangliosidosis
- Mucopolipidosis type I (sialidosis)
- GM2 gangliosidosis: Tay-Sachs disease (Option A) and Sandhoff disease
- Niemann-Pick disease: three types A-C
- Type A and B present with a cherry-red spot
- Type C presents with ocular motility issues
- Farber disease (Option B)

B) Central retinal and cilioretinal artery occlusion (Option C)

- Retinal clouding from edema
- Acute sign

C) Retinotoxicity

- Quinine
- Dapsone
- Gentamicin
- Carbon monoxide
- Methanol

D) Commotio retinae

Edema due to blunt trauma

E) Miscellaneous

- Conditions giving a dark or reddish central macular appearance
- Ex: Macular hole

Solution for Question 14:

Tobacco amblyopia presents with loss of central vision and not with its preservation.

- It is caused by excessive use of tobacco, and the toxic agent is cyanide in tobacco which leads to degeneration of ganglion cells, especially in the macula.

- It characteristically presents with impairment of central vision, which is bilateral and progressive, with patients complaining of difficulty in doing near work and fogginess.

Central vision, tubular vision, or tunnel vision refers to the loss of peripheral vision while preserving central vision.

Causes of tubular vision:

- Terminal stage of advanced glaucomatous field defect (Option A): The visual field loss is gradual, both centrally and peripherally. Only a small island of central and temporal vision is left. Prolonged damage leads to loss of central vision as well. The temporal island is more resistant and hence is lost only in the end, leading to loss of even light perception.
- The visual field loss is gradual, both centrally and peripherally.
- Only a small island of central and temporal vision is left.
- Prolonged damage leads to loss of central vision as well.
- The temporal island is more resistant and hence is lost only in the end, leading to loss of even light perception.
- Advanced stage of retinitis pigmentosa (Option B): It is a disorder that, if sporadic, has a more favorable diagnosis with a retained central vision till the 6th decade of life. Central vision occurs in advanced cases and is lost ultimately after a prolonged duration.
- It is a disorder that, if sporadic, has a more favorable diagnosis with a retained central vision till the 6th decade of life.
- Central vision occurs in advanced cases and is lost ultimately after a prolonged duration.
- Central retinal artery occlusion (CRAO) with cilioretinal artery (Option C): The cilioretinal artery is found in one-third of the eyes. It provides a second arterial supply to the fovea and hence in the case of CRAO, the central vision is maintained whereas the peripheral is lost.
- The cilioretinal artery is found in one-third of the eyes.
- It provides a second arterial supply to the fovea and hence in the case of CRAO, the central vision is maintained whereas the peripheral is lost.
- Vigabatrin: It is a drug used in the treatment of epilepsy. It can cause vision loss in 30% of infants and causes bilateral tunnel vision in adults. Vigabatrin causes damage to the central retina and leads to visual acuity.
- It is a drug used in the treatment of epilepsy.
- It can cause vision loss in 30% of infants and causes bilateral tunnel vision in adults.
- Vigabatrin causes damage to the central retina and leads to visual acuity.
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- Only a small island of central and temporal vision is left.
- Prolonged damage leads to loss of central vision as well.

- The temporal island is more resistant and hence is lost only in the end, leading to loss of even light perception.
- It is a disorder that, if sporadic, has a more favorable diagnosis with a retained central vision till the 6th decade of life.
- Central vision occurs in advanced cases and is lost ultimately after a prolonged duration.
- The cilioretinal artery is found in one-third of the eyes.
- It provides a second arterial supply to the fovea and hence in the case of CRAO, the central vision is maintained whereas the peripheral is lost.
- It is a drug used in the treatment of epilepsy
- It can cause vision loss in 30% of infants and causes bilateral tunnel vision in adults
- Vigabatrin causes damage to the central retina and leads to visual acuity

Solution for Question 15:

In retinopathy of prematurity, the premature infant's retina is vulnerable to abnormal vascular growth. The disruption of VEGF production, often due to fluctuations in oxygen levels (e.g., excess oxygen therapy), impairs the normal retinal vascular development. This can lead to abnormal blood vessel formation and progression of ROP.

Retinopathy of prematurity (ROP) is a bilateral proliferative retinopathy in premature infants with low birth weight, commonly due to high oxygen exposure.

- Birth weight of <1.5 kg
- Gestational age <32 weeks
- Supplemental oxygen therapy
- Retinal vessels extend to the nasal edge by the eighth month of gestation but the temporal periphery vascularizes about a month after birth, by the effect of vascular endothelial growth factor (VEGF).
- Premature birth downregulates VEGF, halting vessel migration.
- Immature vessels are damaged by oxygen exposure, causing vasoconstriction and eventually obliteration of retinal arteries and veins.
- As the infant grows, metabolic demands increase leading to excess production of oxygen-regulated VEGF and non-oxygen-regulated insulin-like growth factor (IGF).
- This leads to neovascularization and fibrous proliferation.

- These changes cause progressive stages of retinopathy of prematurity (ROP).
- Retinal Changes: Early signs: Dilatation of retinal veins, hazy white patches in the peripheral retina. Proliferation of new vessels into the vitreous from the retina, forming fibrous tissue. Progression: Retinal detachment, microphthalmia, and varying degrees of vision loss.
- Early signs: Dilatation of retinal veins, hazy white patches in the peripheral retina.
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- Early signs: Dilatation of retinal veins, hazy white patches in the peripheral retina.
- Proliferation of new vessels into the vitreous from the retina, forming fibrous tissue.
- Progression: Retinal detachment, microphthalmia, and varying degrees of vision loss.

Increased oxygen supply leading to retinal vasodilation (Option A): Excessive oxygen therapy causes retinal vessel suppression, not vasodilation.

Excessive blood glucose levels (Option C): ROP is not directly caused by blood glucose levels, but by oxygen fluctuations.

Retinal hypoxia leading to abnormal collagen formation (Option D): Hypoxia triggers VEGF production, causing abnormal vessel growth, not collagen deposition.

Solution for Question 16:

Correct Answer: B) Prematurity and Low birth weight

Explanation:

Prematurity and Low birth weight is the most significant risk factor for ROP.

In retinopathy of prematurity, the premature infant's retina is vulnerable to abnormal vascular growth. The disruption of VEGF production, often due to fluctuations in oxygen levels (e.g., excess oxygen therapy), impairs the normal retinal vascular development. This can lead to abnormal blood vessel formation and progression of ROP.

Retinopathy of prematurity (ROP) is a bilateral proliferative retinopathy in premature infants with low birth weight, commonly due to high oxygen exposure.

Risk Factors:

- Birth weight of <1.5 kg
- Gestational age <32 weeks
- Supplemental oxygen therapy

Pathogenesis:

- Retinal vessels extend to the nasal edge by the eighth month of gestation but the temporal periphery vascularizes about a month after birth, by the effect of vascular endothelial growth factor (VEGF).
- Premature birth downregulates VEGF, halting vessel migration.
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- Early signs: Dilatation of retinal veins, hazy white patches in the peripheral retina.
- Proliferation of new vessels into the vitreous from the retina, forming fibrous tissue.
- Progression: Retinal detachment, microphthalmia, and varying degrees of vision loss.

Option A (High birth weight): Babies with higher birth weights are at a lower risk for ROP.

Option C (Prolonged exposure to blue light): No scientific evidence links blue light exposure to ROP.

Option D (Deficiency of Vitamin A): Vitamin A deficiency can cause xerophthalmia and night blindness, but it is not a primary cause of ROP.

Solution for Question 17:

Plus disease is indicated by severe retinal vessel changes, including dilatation and tortuosity and a rapid progression of retinopathy of prematurity (ROP). It is often associated with poor pupillary dilation and vitreous haze.

Classification of ROP:

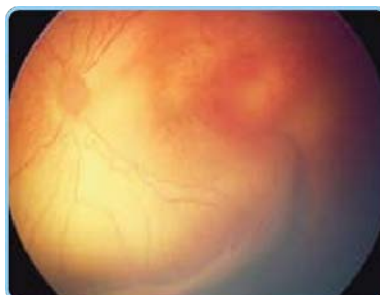
- About 20% of infants with active ROP develop cicatricial (scar tissue) complications.
- Findings include temporal vitreoretinal fibrosis and vascular arcade straightening with macula and disc dragging.
- Severe cases may show retrolental fibrovascular tissue, leading to retinal folds and retinal detachment (retrolental fibroplasia).
- Plus Disease (Option B) Rush Disease/ Aggressive Posterior ROP (Option A) Defined by dilatation and tortuosity of posterior pole vessels in at least two quadrants with any stage of ROP. Associated with engorgement and dilatation of the iris vessel, poor pupil dilation and vitreous haze. Indicates a tendency for disease progression. Located in zone I or posterior zone II with severe plus disease. Characterized by flat proliferation with vascular loops, hemorrhages and absence of any stage. Requires immediate treatment due to rapid progression. Can progress directly to stage 5 ROP without passing through earlier stages.
- Defined by dilatation and tortuosity of posterior pole vessels in at least two quadrants with any stage of ROP.
- Associated with engorgement and dilatation of the iris vessel, poor pupil dilation and vitreous haze.
- Indicates a tendency for disease progression.
- Located in zone I or posterior zone II with severe plus disease.
- Characterized by flat proliferation with vascular loops, hemorrhages and absence of any stage.
- Requires immediate treatment due to rapid progression.
- Can progress directly to stage 5 ROP without passing through earlier stages.
- Stage 4 ROP (Option A) involves tractional retinal detachment that extends from the periphery to the posterior pole, not characterized by vessel changes typical of plus disease.

Plus Disease
(Option B)

Rush Disease/ Aggressive Posterior ROP
(Option A)



Defined by dilatation and tortuosity of posterior pole vessels in at least two quadrants with any stage of ROP. Associated with engorgement and dilatation of the iris vessel, poor pupil dilation and vitreous haze. Indicates a tendency for disease progression.



Located in zone I or posterior zone II with severe plus disease. Characterized by flat proliferation with vascular loops, hemorrhages and absence of any stage. Requires immediate treatment due to rapid progression. ... (content truncated)

Stage 4 ROP (Option A) involves tractional retinal detachment that extends from the periphery to the posterior pole, not characterized by vessel changes typical of plus disease.

Solution for Question 18:

Eales disease is an idiopathic peripheral retinal vasculopathy that presents with recurrent vitreous hemorrhages. In the early stage of the disease, characterized by active retinal vasculitis, systemic corticosteroids are the treatment of choice. They reduce inflammation and help prevent further vascular damage and hemorrhage.



Venous congestion, perivascular exudates and sheets of hemorrhages present near the affected veins in Eales disease

Feature	Description
Population Affected	Young males
Etiology	Hypersensitivity reaction (Type IV or III) to tuberculoproteins, causing retinal vasculitis, resulting in retinal hypoxia and vasoproliferation
Vasculitis Manifestation	Vessel sheathing, extensive leakage on fluorescein angiography
Retinal Changes	Hemorrhages, soft exudates, retinal edema, neovascularization, and arteriovenous shunt vessels
Complications	Recurrent vitreous hemorrhage, tractional retinal detachment, secondary glaucoma
Initial Treatment	Systemic steroids for early vasculitis
Further Management	Photocoagulation for vessels under traction, vitreoretinal surgery for severe cases

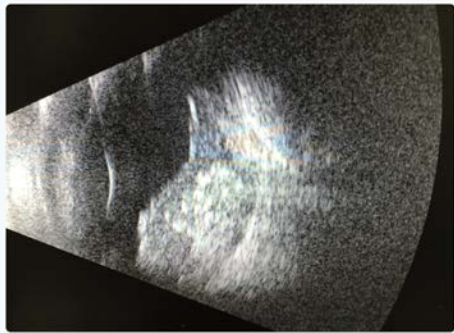
Intravitreal anti-VEGF therapy (Option A) is used in conditions involving neovascularization, such as diabetic retinopathy, they are not the primary treatment for Eales disease, which is mainly driven by inflammation rather than VEGF-mediated neovascularization.

Laser photocoagulation (Option C) is indicated for treating areas of retinal ischemia and neovascularization in the later stages of the disease, not in the early stages.

Vitrectomy (Option D) is usually done for cases where there is persistent vitreous hemorrhage that does not resolve on its own, or if there is tractional retinal detachment.

Solution for Question 19:

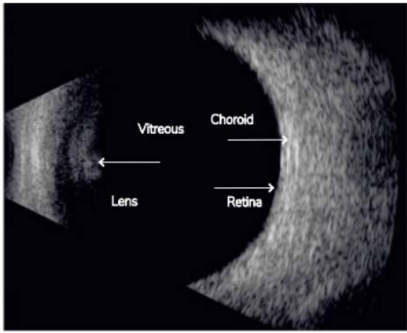
Retinoblastoma is characterised on B-scan ultrasound by a highly reflective intraocular mass, often accompanied by calcifications.



Retinoblastoma

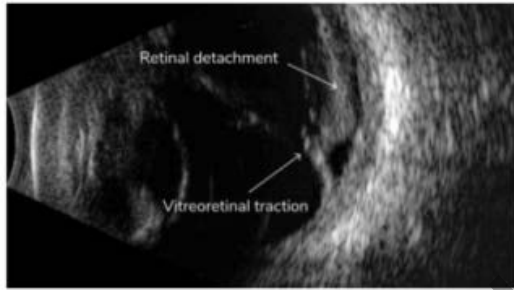
B-scan:

- Two-dimensional dotted section of the eye.
- Used to visualize eye and orbit structures in real-time with a gray-scale display.
- The frequency of the ultrasound used is 7.5-10 MHz.
- The vitreous cavity appears anechoic (black) with the posterior lens capsule visible anteriorly.
- The retina, choroid, and posterior lens capsule are hyperechoic structures.
- Dynamic B-scan: Differentiates between retinal detachments and vitreous detachments.
- Applications:
 - Useful for intraocular structure visualization in opaque media.
 - Identifies vitreoretinal and orbital mass lesions.
 - Detects vascular abnormalities and retinal detachments.
 - Localization of intraocular and intraorbital foreign bodies.
 - Option A: Opaque and corrugated with retinal folds.
 - Option B: Echogenic dotted and linear pattern.
 - Option C:
 -



Applications:

Option A: Opaque and corrugated with retinal folds.



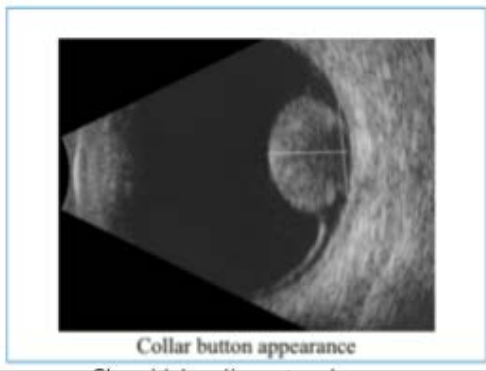
Retinal detachment

Echogenic dotted and linear pattern.



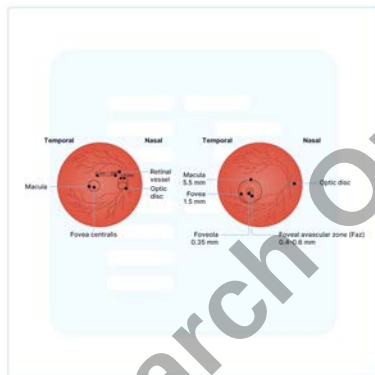
Vitreous Hemorrhage

Option C:



Solution for Question 20:

The fovea is situated approximately 2 disc diameters (DD) away from the optic disc on the temporal side. 1 disc diameter is equal to about 1.5 mm, so the fovea is located approximately 3 mm from the optic disc. (Options A, B, and C)



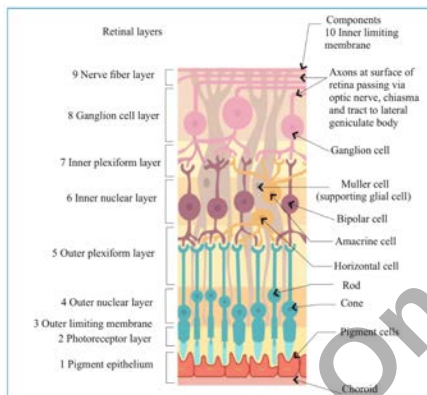
Structure	Description	Function
Macula lutea (clinical posterior pole)	A 5.5 mm diameter cone-rich region in the retina.	Responsible for central vision and color vision.
Fovea	A 1.5 mm diameter area at the center of the macula, 2 disc diameters temporal to the optic disc.	Provides sharp, detailed vision.
Foveola	A 0.35 mm diameter central depression within the fovea, often shows a bright foveal reflex.	Contains only cones, facilitating the highest visual acuity.

Foveal avascular zone	A 0.4–0.6 mm diameter zone surrounds the foveola, devoid of retinal blood vessels.	Enhances visual clarity by reducing interference from blood vessels.
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Solution for Question 21:

The Inner Nuclear Layer is made up of the cell bodies of bipolar cells, horizontal cells, and amacrine cells. It is labeled as B in the Histology image given.

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Layer	Description
1. Pigment Epithelium	Outermost layer with a single pigmented cell layer; adheres to Bruch's membrane; supports neurosensory retina, and reduces reflection.
Subretinal Space: This is the layer between the Pigment epithelium and the layer of Photoreceptors.	
2. Rods and Cones Layer	Contains the outer segments of photoreceptors (rods and cones, which are the first-order neurons); rods detect low-light vision, cones manage color and detailed vision.
3. External Limiting Membrane	A porous membrane allows processes from rods and cones to pass through.
4. Outer Nuclear Layer (Henle's layer) (Option D)	Contains the nuclei of the photoreceptor cells (rods and cones).

5. Outer Plexiform Layer (Option C)	Includes connections between rods, cones, bipolar cell dendrites, and horizontal cells.
6. Inner Nuclear Layer (Option B)	Made up of the cell bodies of bipolar cells (which are the second-order neurons), horizontal cells, amacrine cells, and Müller cells; and also contains retinal capillaries.
7. Inner Plexiform Layer (Option A)	Houses synaptic connections between bipolar cell axons, ganglion cell dendrites, and amacrine cell processes.
8. Ganglion Cell Layer	Contains ganglion cell bodies, which serve as third-order neurons; midget cells synapse with a single bipolar cell, and polysynaptic cells synapse with multiple.
9. Nerve Fiber Layer	Composed of ganglion cell axons that form the optic nerve after passing through the lamina cribrosa.
10. Internal Limiting Membrane	The innermost layer separating the retina from the vitreous; is formed by the terminal expansions of Müller cells, serving as a basement membrane.

The Neuroglial cells, particularly the vertical Müller cells, bind the retina's neural elements. These cells provide structural support and nourish the retinal tissue.

Solution for Question 22:

- The presentation of sudden, painless vision loss with a cherry-red spot at the macula is highly suggestive of central retinal artery occlusion (CRAO).
- In cases of central retinal artery occlusion (CRAO), the central retinal artery is blocked, causing widespread retinal ischemia.
- Obstruction at the level of the lamina cribrosa.
- Sudden, painless vision loss (occurs within seconds).
- History of transient visual loss (amaurosis fugax) in some patients.
- Marked narrowing of retinal arteries, mild narrowing of veins.
- Milky white retina due to ischemic edema.
- Cilioretinal artery may preserve part of the macula's normal color.
- Cherry-red spot at the center of the macula due to the vascular choroid shining through the thin retina.

- Cattle trucking: Segmentation of blood column in retinal veins.
- Atrophic changes: Grossly attenuated arteries and optic atrophy in long-standing cases.
- Delayed arterial filling.
- Masking of choroidal vasculature due to retinal edema.

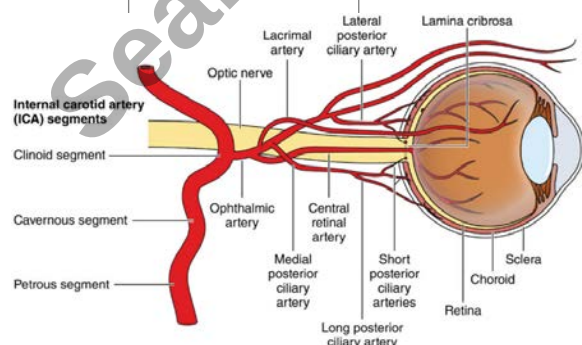
Solution for Question 23:

The central retinal artery is the main vessel responsible for supplying the inner layers of the retina, including the nerve fiber layer, ganglion cells, and inner retinal layers. The short posterior ciliary arteries primarily supply the choroid and the outer retinal layers.

The retina is supplied mainly by the central retinal artery, which enters the optic nerve on its lower surface. It emerges from the optic disc and divides into two branches to form a superior and an inferior trunk, and each trunk divides into two branches: Nasal and temporal.

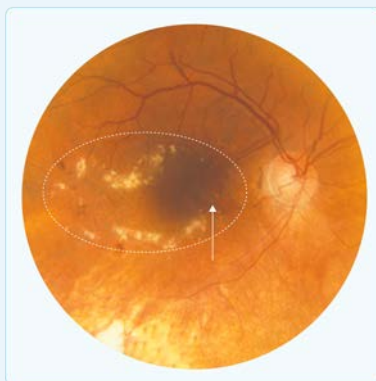
Venous drainage of the optic disc is mainly carried out by the central retinal vein. Retinal veins follow the arteries and drain into the cavernous sinus or superior ophthalmic vein. Retinal venous occlusion can cause hemorrhages - Splash tomato appearance.

Layer/Region	Blood Supply	Clinical Relevance
Outer four layers	Supplied by the choroidal vascular system (short and long posterior ciliary arteries).	Avascular layers; rely on choroidal vessels for nutrition
Inner six layers	Supplied by the central retinal artery (branch of ophthalmic artery).	Occlusion causes retinal ischemia.
Macula (in some individuals)	Supplied by the cilioretinal artery (from posterior ciliary arteries).	Preserves central vision during CRAO.



Solution for Question 24:

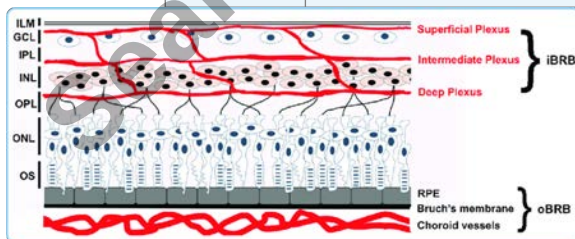
Hard and Soft Exudates

Feature	Soft Exudates (Cotton-Wool Spots)	Hard Exudates
Appearance	Cloud-like, small, ill-defined, whitish patches	Bright yellow, well-defined lipid deposits
Formation Mechanism	Caused by ischemia and axonal swelling; blockage of axoplasmic flow (Option A and D)	Formed by leakage of fluid rich in proteins and lipids
Location	Nerve fibre layer	Seen around leaking microaneurysms (Option C), especially in the macula
Common Shape	Indistinct and variable; may change shape quickly	Radial, star-shaped figure in the macula (macular fan/star)
Main Causes	Hypertensive retinopathy Diabetic retinopathy Toxemia of pregnancy Anemic retinopathy Lupus erythematosus (LE) Polyarteritis nodosa (PAN) Scleroderma Leukemic retinopathy Retinopathy associated with AIDS	Diabetic retinopathy Hypertensive retinopathy Coats' disease Circinate retinopathy
Image		

Solution for Question 25:

The blood-retinal barrier (BRB) consists of an inner and outer barrier that separates the blood from the ocular cavity, where the retinal vascular endothelial cells form the inner blood-retinal barrier.

Features	Outer blood-retinal barrier	Inner blood-retinal barrier
Components	Formed by the adjacent cells of the retinal pigment epithelium (RPE) (Option B). Maintained by tight junctions named "zonula occludens" (Option C). RPE rests on the Bruch's membrane (Option A), which separates the neural retina from the choroid capillaries.	Formed by the tight junctions between the endothelial cells of the retinal blood vessels. Depends on capillary integrity.
Function	It prevents the leakage of extracellular fluid and nutrients from the choroid to the outer retina.	Controls permeability of the retinal blood vessels.
Factors affecting the BRB	Prostaglandins Leukotrienes Protein kinase C Nitric oxide VEGF Other cytokines	
Associated pathology	Outer BRB damage is seen in conditions like: Age-related macular degeneration Central serous chorioretinopathy Accelerated hypertension Toxemia of pregnancy Breakdown of outer BRB leads to: Retinal detachment or RPE detachment	Retinal macroaneurysms cause a breakage in the inner blood-retinal barrier and accumulation of fluid within the outer neurosensory retina. Involved in conditions like: Cystoid macular edema. Diabetic retinopathy Retinal vein occlusion



Solution for Question 26:

Direct vs. Indirect Ophthalmoscopy:

Feature	Direct Ophthalmoscopy	Indirect Ophthalmoscopy
Instrument used		
Image	Erect, virtual	Inverted, real
Magnification	High magnification (~15x in emmetropes, more in myopes, less in hypermetropes)	Moderate magnification (~2.5x with a +20D lens)
Field of View	Small (~10° or 1-2 disc diameters)	Large (~35° or 8 disc diameters)
Visualization	Central retina only	Central and peripheral retina (up to the ora serrata)
Refractive Media	Poor visualization in hazy or cloudy media (Option A)	Can visualize through hazy media due to stronger illumination (option B)
Examination Distance	Close to the eye (~15 mm from the cornea)	Performed at a greater distance (~50 cm or an arm's length)
Patient Position	Performed in a seated/standing position in a semi-dark room	Performed with the patient in a supine or sitting position
Lens Requirement	No external lens is required; correcting lenses for refractive errors are built into the ophthalmoscope	Requires a convex lens (commonly +20D) to be placed in front of the patient's eye
Peripheral Lesion Detection	Limited ability to detect peripheral lesions	Excellent for detecting peripheral retinal lesions such as retinal detachment or tears.

Slit-Lamp Biomicroscopy (Option C): Although useful for anterior segment examination and some posterior pole assessments, it is ineffective in hazy media due to scattering of light.

Amsler grid test (Option D) is a widely used tool for monitoring vision changes, particularly in individuals at risk for age-related macular degeneration (AMD) and in other macular lesions. However, it is less effective in hazy media, such as cataracts or other opacities in the eye.

Solution for Question 27:

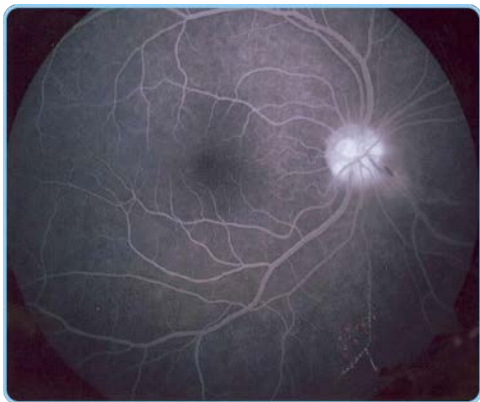
The patient has ARMD, a condition associated with the pooling of dye under the RPE and subretinal neovascularization, both of which lead to hyperfluorescence on FFA. Principle: Based on the detection of fluorescent light emitted by fluorescein dye in circulation.

- Absorption & Emission Wavelengths: Irradiating light: 420–490 nm. Emitted fluorescence: 510–530 nm.
- Irradiating light: 420–490 nm.
- Emitted fluorescence: 510–530 nm.
- Retinal pigments and red blood cells can absorb fluorescent light, masking deeper structures. Pigment migration allows visibility of deeper layers.
- Irradiating light: 420–490 nm.
- Emitted fluorescence: 510–530 nm.

Diagnostic uses:

- Diabetic Retinopathy: Reveals neovascularization, and assesses vascular tree integrity.
- Fundus Disorders: Identifies depth of pathological involvement.
- Optic Nerve Pathologies: Useful in detecting optic nerve issues such as papilloedema.
- Neovascularization of Iris: Detects leakage from iris vessels, an early sign of rubeosis.

Normal Study

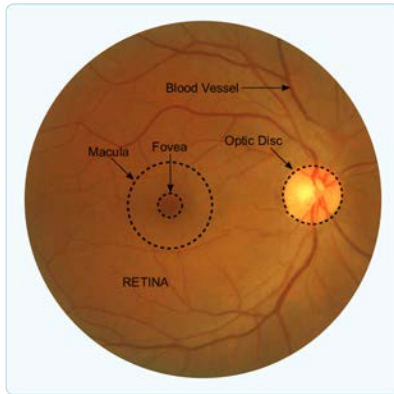


Abnormalities Detected by Fundus Fluorescein Angiography (FFA)

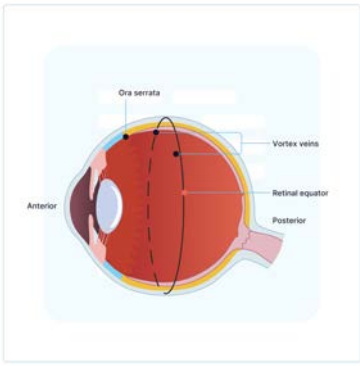
Fluorescence Type	Cause	Examples
Hyperfluorescence	Window defect in retinal pigment epithelium (RPE)	Atrophy showing background choroidal fluorescence
Pooling of dye under detached RPE	Age-related macular degeneration (ARMD) (Option A)	
Pooling of dye under sensory retina	Central serous retinopathy (CSR)	
Leakage into the neurosensory retina due to the breakdown of the inner blood-retinal barrier	Cystoid macular edema (CME)	
Leakage from choroidal or retinal neovascularization	Proliferative diabetic retinopathy, subretinal neovascular membrane in ARMD	
Staining (long retention of dye by tissues)	Drusen, normal optic disc (Option D)	
Leakage from optic nerve head	Papilloedema	
Hypofluorescence	Blockage of background fluorescence	Retinal hemorrhage (Option B), hard exudates, pigmented clumps
Occlusion of retinal/choroidal vasculature	Central retinal artery occlusion (Option C), diabetic retinopathy capillary occlusion	
Loss of vasculature	Choroideremia, myopic degeneration	

Solution for Question 28:

The presentation of diminished night vision and loss of peripheral vision is suggestive of a defect in the peripheral retina, which is crucial for peripheral and night vision. Additionally, no abnormality was found in the posterior pole, which includes the macula, fovea, and optic disc.

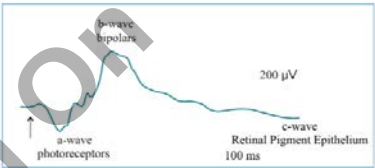


Part	Description
Posterior Pole	The region behind the retinal equator includes the optic disc and macula lutea; best examined using slit-lamp biomicroscopy or by direct ophthalmoscopy.
Optic Disc (Option C)	Known as the optic nerve head, it is a pink, vertically oval structure, marking the start of the optic nerve. The physiological Cup is a depression in the optic disc through which the central retinal artery and vein pass; the absence of photoreceptors results in the physiological blind spot.
Macula Lutea (Option A)	A dark, yellow spot at the posterior pole, measuring 5.5 mm in diameter; is responsible for central, high-acuity vision and color detection.
Fovea Centralis (Option B)	Central area of the macula (1.5 mm in diameter); contains only cones and is the most sensitive part of the retina for visual acuity.
Foveola	A 0.35 mm diameter shining pit within the fovea, represents the area of highest visual acuity, with the foveal avascular zone (FAZ) surrounding it.
Peripheral Retina (Option D)	Extends from the retinal equator to the ora serrata, examined via indirect ophthalmoscopy or Goldmann three-mirror lens. It contains rods, hence responsible for night vision.

Ora Serrata	 ... (content truncated)
-------------	---

Solution for Question 29:

Electroretinogram (ERG)

Definition	Electroretinography (ERG) records changes in the eye's resting potential. It is induced by a flash of light.	
Parts of ERG	 A-wave: Negative wave, likely from rods and cones (photoreceptors). B-wave: Large positive wave, generated by Müller cells (inner retinal layers) but reflects bipolar cell activity. C-wave: Positive wave, related to the metabolic activity of the pigment epithelium (RPE), observed in dark-adapted eyes. Foveal ERG provides information about the macula used in diagnosing cataracts.	
Interpretation	Response	Inference

Extin guish ed	Complete failure in rod and cone function (e.g., pigmentary retinal dystrophy, complete retinal artery occlusion, old retinal detachment, advanced siderosis).	
Subn ormal	Large areas of the retina not functioning (e.g., early retinitis pigmentosa, before ophthalmoscopic signs).	
Nega tive	Gross disturbances in the retinal circulation.	
Clinic al us es	Used in diagnosis and prognosis for retinal disorders like retinitis pigmentosa, Leber's congenital amaurosis, retinal ischemia, and chorioretinal degenerations. Assessment in Unclear Conditions: Evaluates retinal junction when fundus examination is not possible (e.g., dense cataract, corneal opacity). Infant Vision Assessment: Assesses retinal junction in infants with suspected impaired vision.	
Patte rns in ERG	Pattern ERG	Changes in PERG responses can occur in the early stages of glaucoma by measuring the activity of retinal ganglion cells (RGCs), often correlating with elevated intraocular pressure (IOP).
Multif ocal ERG	mfERG aids in diagnosing macular diseases like Stargardt dystrophy, cone dystrophy, and occult macular dystrophy. mfERG or similar objective tests can be used to screen chloroquine and hydroxychloroquine toxicity.	

Solution for Question 30:

In Best disease, the Arden ratio on EOG is typically less than 1.5, indicating abnormal retinal pigment epithelium function, which is characteristic of this condition.

Electro-oculography (EOG):

- Measures changes in resting ocular potential via electrodes placed at inner and outer canthi when eyes move side to side.
- Changes in potential reflect the activity of retinal pigment epithelium (regulated by the bestrophin gene) and outer segments of visual receptors.

Clinical Use:

- Useful in detecting retinal pigment epithelium dystrophies and degenerations before visual symptoms manifest.
- Helpful in diagnosing Best disease (Arden index <1.5) and differentiating it from adult-onset vitelliform macular dystrophy and pattern dystrophy. (Option B)
- EOG apparatus can also be used to record eye movements in nystagmus (electronystagmography (ENG)).

Arden Index: The result is calculated by dividing the light peak (maximal potential in light) by the dark trough (minimal potential in dark).

Arden Ratio	Interpretation
>1.85	Normal
$1.85-1.5$	Borderline
<1.5	Abnormal

Solution for Question 31:

Retinal vasoconstriction is seen in the pathogenesis of hypertensive retinopathy whereas microvascular adhesion is seen in diabetic retinopathy.

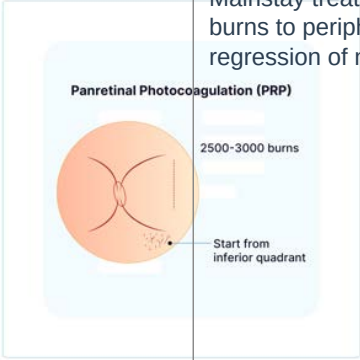
Retinal vascular disorders	Diabetic retinopathy Retinal venous occlusive disease Retinopathy of prematurity Sickle cell retinopathy Retinal artery occlusive disease, etc.
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General pathogenesis	
Diabetic retinopathy (DR): Capillary basement membrane thickening, endothelial cell damage, RBCs deformation, and rouleaux formation, increased stickiness of platelets, increased plasma viscosity, loss of capillary pericytes → microvascular occlusion (cotton wool spots), retinal ischemia → leads to the clinical appearances of DR. Progress of retinal changes in diabetic retinopathy: Diabetic macular edema → Macular ischemia (Option A) → Retinal new vessels (Option B) → Vitreous hemorrhage (Option C) → Tractional retinal detachment. Hypertensive retinopathy: Systemic hypertension causes vasoconstriction of retinal arterioles (Option D) Arteriosclerosis, increased vascular permeability, and retinal hemorrhages lead to retinopathy. Retinopathy of prematurity: In premature infants Linked to high-oxygen exposure Leads to abnormal neovascularization and retinal hypoxia Sickle cell retinopathy: Due to abnormal RBCs ... (content truncated)	

Solution for Question 32:

Focal laser therapy for diabetic macular edema targets leaking microaneurysms within 500 µm of the fovea.

Condition	Therapy	Details
Diabetic retinopathy	Diabetic Macular Edema (DMO)	
Focal laser	Targets leaking microaneurysms within 500 µm of the fovea	
Grid laser	Applied to areas of diffuse retinal thickening, at least 500 µm from the fovea and optic disc (Option B)	
Modified Grid Laser	Combines focal treatment with grid laser for diffuse thickening	

Subthreshold (micropulse) diode laser	Uses very short pulses with longer intervals, minimizing collateral damage (Option C) while stimulating RPE, and allows subfoveal treatment.	
Proliferative Diabetic Retinopathy (PDR)		
 ... (content truncated)	Mainstay treatment; applies laser burns to peripheral retina to induce regression of neovascularization.	
Targeted Retinal Photocoagulation	Utilizes FA to target peripheral capillary non-perfusion, aiming for neovascularization regression.	
Branch Retinal Vein Occlusion (BRVO)	Sector Photocoagulation	Targets on ischemic areas, often combined with anti-VEGF injections.
Macular Laser	Considered for persistent macular edema with good central perfusion.	
Subthreshold Laser	Alternative for macular edema treatment	
Central Retinal Vein Occlusion (CRVO)	Panretinal Photocoagulation (PRP)	Crucial for treating neovascularization, often supplemented with anti-VEGF injections.
Sickle Cell Retinopathy	Laser or Cryotherapy	Ablates peripheral non-perfused retina to minimize neovascularization stimulus.
Coats Disease	Laser Ablation	Targets leaking areas, often requiring multiple treatments.
Eales Disease	Scatter Photocoagulation or Cryotherapy	Treats non-perfused retina to reduce neovascularization.

Limitations and considerations (Option D)	Visual field defects Central vision risks Multiple sessions Informed consent	
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Solution for Question 33:

Reduced blood viscosity is not a contributing factor to diabetic retinopathy; instead, increased blood viscosity is observed.

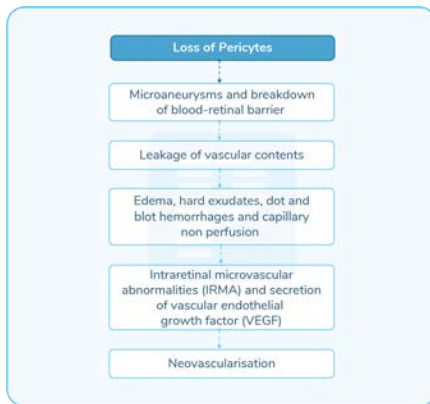
Risk Factors of Diabetic Retinopathy:

Pathogenesis of Diabetic Retinopathy:

- Cellular Damage: Produces damage to the cells of the retina, endothelial cells, loss of pericytes, and thickening of basement membrane by: Increased sorbitol accumulation Accumulation of advanced glycation end products (AGEs) Protein kinase C activation (Option A) Activation of eicosanoids, nitric oxides and intracellular adhesion molecules Excessive oxidative stress
- Increased sorbitol accumulation
- Accumulation of advanced glycation end products (AGEs)
- Protein kinase C activation (Option A)
- Activation of eicosanoids, nitric oxides and intracellular adhesion molecules
- Excessive oxidative stress
- Hematological and Biochemical Changes: Increased platelet adhesiveness Increased blood viscosity (Option B) Red blood cell deformation and rouleaux formation Altered serum lipids Increased leukostasis Increased fibrinolysis
- Increased platelet adhesiveness
- Increased blood viscosity (Option B)
- Red blood cell deformation and rouleaux formation
- Altered serum lipids
- Increased leukostasis
- Increased fibrinolysis
- Effects of Microangiopathy: Breakdown of the blood-retinal barrier (Option D) Weakened capillary walls Microvascular occlusion Neovascularization induced by: Proangiogenic

factors: Vascular endothelial growth factor (VEGF), platelet-derived growth factor (PDGF) and hepatocyte growth factor (HGF) (Option C)

- Breakdown of the blood-retinal barrier (Option D)
- Weakened capillary walls
- Microvascular occlusion
- Neovascularization induced by: Proangiogenic factors: Vascular endothelial growth factor (VEGF), platelet-derived growth factor (PDGF) and hepatocyte growth factor (HGF) (Option C)
- Proangiogenic factors: Vascular endothelial growth factor (VEGF), platelet-derived growth factor (PDGF) and hepatocyte growth factor (HGF) (Option C)
- Reduced anti-angiogenic factors: Decreased levels of endostatin, angiostatin, pigment-derived growth factor (PEDF), thrombospondin-1, and platelet factor 4
- Increased sorbitol accumulation
- Accumulation of advanced glycation end products (AGEs)
- Protein kinase C activation (Option A)
- Activation of eicosanoids, nitric oxides and intracellular adhesion molecules
- Excessive oxidative stress
- Increased platelet adhesiveness
- Increased blood viscosity (Option B)
- Red blood cell deformation and rouleaux formation
- Altered serum lipids
- Increased leukostasis
- Increased fibrinolysis
- Breakdown of the blood-retinal barrier (Option D)
- Weakened capillary walls
- Microvascular occlusion
- Neovascularization induced by: Proangiogenic factors: Vascular endothelial growth factor (VEGF), platelet-derived growth factor (PDGF) and hepatocyte growth factor (HGF) (Option C)
- Proangiogenic factors: Vascular endothelial growth factor (VEGF), platelet-derived growth factor (PDGF) and hepatocyte growth factor (HGF) (Option C)
- Proangiogenic factors: Vascular endothelial growth factor (VEGF), platelet-derived growth factor (PDGF) and hepatocyte growth factor (HGF) (Option C)



Solution for Question 34:

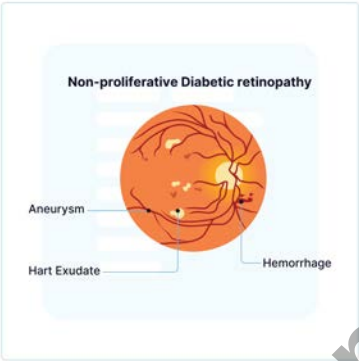
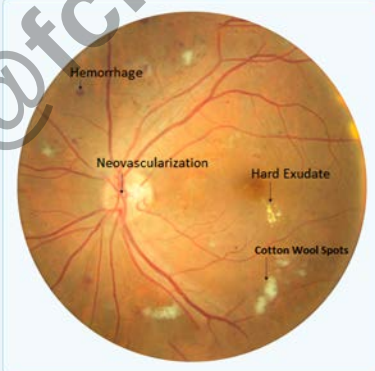
Venous beading is typically associated with severe non-progressive diabetic retinopathy (NPDR) or preproliferative diabetic retinopathy, and indicates sluggish retinal circulation.

Classification of Non-Progressive Diabetic Retinopathy (NPDR)

Severity	Definition
Very mild nonproliferative diabetic retinopathy	Microaneurysms only
Mild nonproliferative diabetic retinopathy	Microaneurysms plus hard exudates, cotton-wool spots, and/or mild retinal hemorrhages
Moderate nonproliferative diabetic retinopathy	Microaneurysms plus mild intraretinal microvascular abnormalities or moderate retinal hemorrhages
Severe nonproliferative diabetic retinopathy	Any one of the following (4-2-1 Rule) Severe retinal hemorrhages in four quadrants Venous beading in at least two Moderately severe intraretinal microvascular abnormalities (IRMA) in at least one quadrant
Very severe nonproliferative diabetic retinopathy	Any two of the following (4-2-1 Rule) Severe retinal hemorrhages in four quadrants Venous beading in at least two Moderately severe intraretinal microvascular abnormalities (IRMA) in at least one quadrant

Clinical Features:

Early NPDR or Background Diabetic Retinopathy	Severe NPDR or Preproliferative Diabetic Retinopathy
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Affects small retinal vessels. Earliest sign: Microaneurysm (Option A) Fluorescein angiography (FA): Tiny hyperfluorescent dots. Retinal hemorrhages: Both deep, i.e., in the inner nuclear or outer plexiform layer (dot and blot hemorrhage) and superficial, i.e., in nerve fiber layer (splinter or flame-shaped) hemorrhages (Option B) Hard exudates: White or yellow waxy patches with well-defined margins, often at the posterior pole, in the outer plexiform layer. (Option C) FA: Hypofluorescence with dense exudates. Edema is not marked.	Increased inner retinal hypoxia: Leads to multiple retinal hemorrhages, cotton-wool spots, venous beading, vascular loops, intraretinal microvascular abnormalities (IRMAs), and large areas of capillary nonperfusion. Cotton-Wool Spots: Also called soft exudates or nerve fiber infarcts. Result from ischemia, not exudation. Caused by obstruction of axoplasmic flow, leading to swelling and a white, fluffy appearance. FA: Focal hypofluorescence due to local ischemia. Venous Beading: Indicates sluggish retinal circulation. Typically adjacent to large areas of capillary nonperfusion. (Option D) Intraretinal Microvascular Abnormalities (IRMA): Dilated capillaries act as collateral channels. Bypass capillary bed and is often seen adjacent to capillary nonperfusion. FA: Focal hypofluorescence
 The diagram illustrates Non-proliferative Diabetic retinopathy. It shows a cross-section of the retina with a central optic disc. Labels point to an 'Aneurysm' (a small bulge in a retinal vessel), a 'Hard Exudate' (a small, pale, well-defined spot), and a 'Hemorrhage' (a small, dark, flame-shaped spot).	 The fundus photograph shows the retina with several pathological features labeled: 'Hemorrhage' (a dark, flame-shaped spot), 'Neovascularization' (new, fragile blood vessels growing from the optic disc), 'Hard Exudate' (a pale, well-defined spot), and 'Cotton Wool Spots' (pale, fluffy patches of retinal nerve fiber infarction).

Solution for Question 35:

Neovascularization of the Disc (NVD) is the most specific finding for confirming proliferative diabetic retinopathy (PDR).

Clinical Features of Proliferative Diabetic Retinopathy (PDR):

- Includes all features of preproliferative diabetic retinopathy or severe NPDR: Multiple retinal hemorrhages, cotton-wool spots, venous beading, vascular loops, intraretinal microvascular abnormalities (IRMAs), and large areas of capillary nonperfusion along with

neovascularization. (Options A, B, D)

- Neovascularization (2 types): Neovascularization of the Disc (NVD): Arises within 1 disc diameter of the optic nerve from the optic nerve head. (Option C) Neovascularization Elsewhere (NVE): Arises more than 1 disc diameter from the optic nerve, along the large vessels. Appears adjacent to areas of capillary closure and the accompanying fibrous tissue varies in extent. Fluorescein angiography (FA): Irregular expanding hyperfluorescence.
- Neovascularization of the Disc (NVD): Arises within 1 disc diameter of the optic nerve from the optic nerve head. (Option C)
- Neovascularization Elsewhere (NVE): Arises more than 1 disc diameter from the optic nerve, along the large vessels.
- Appears adjacent to areas of capillary closure and the accompanying fibrous tissue varies in extent.
- Fluorescein angiography (FA): Irregular expanding hyperfluorescence.
- Severe Vision Loss: Result in vision as poor as hand movements or worse due to severe vitreous hemorrhage or retinal detachment.
- Fibrovascular Tissue Formation: Formed due to condensation of connective tissue around the new vessels. Lie flat on the retina or attach to the posterior vitreous face. Can lead to vitreous traction, retinal separation, and blood vessel tearing. The most common cause of vitreous hemorrhage in adults.
- Formed due to condensation of connective tissue around the new vessels.
- Lie flat on the retina or attach to the posterior vitreous face.
- Can lead to vitreous traction, retinal separation, and blood vessel tearing.
- The most common cause of vitreous hemorrhage in adults.
- Rubeosis Iridis: Neovascularization of iris.
- Neovascular glaucoma: Occurs due neovascularization of anterior segment angle.
- Neovascularization of the Disc (NVD): Arises within 1 disc diameter of the optic nerve from the optic nerve head. (Option C)
- Neovascularization Elsewhere (NVE): Arises more than 1 disc diameter from the optic nerve, along the large vessels.
- Appears adjacent to areas of capillary closure and the accompanying fibrous tissue varies in extent.
- Fluorescein angiography (FA): Irregular expanding hyperfluorescence.
- Formed due to condensation of connective tissue around the new vessels.
- Lie flat on the retina or attach to the posterior vitreous face.
- Can lead to vitreous traction, retinal separation, and blood vessel tearing.
- The most common cause of vitreous hemorrhage in adults.

Solution for Question 36:

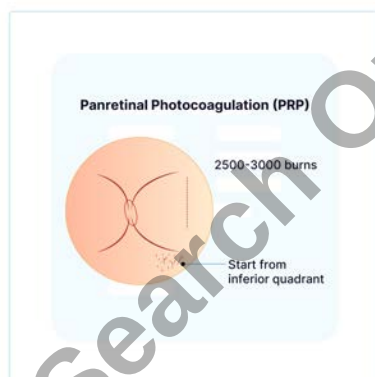
The presence of extensive retinal hemorrhages, venous beading, and moderate IRMA in one quadrant indicates severe non-proliferative diabetic retinopathy (NPDR). For severe NPDR, the follow-up interval is every 4 months.

Abbreviated Early Treatment Diabetic Retinopathy Study (ETDRS) classification of diabetic retinopathy

Category	Management
Non-Proliferative Diabetic Retinopathy (NPDR):	
No Diabetic Retinopathy	Review in 12 months
Very Mild NPDR Microaneurysms only	Review most patients in 12 months (Option D)
Mild NPDR Any or all of the: microaneurysms, retinal hemorrhages, exudates, and cotton-wool spots, up to the level of moderate NPDR. No intraretinal microvascular anomalies (IRMA) or significant beading.	Review range 6-12 months, depending on the severity of signs, stability, systemic factors, and patient's circumstances
Moderate NPDR Severe retinal hemorrhages in 1-3 quadrants or mild IRMA Significant venous beading can be present in no more than 1 quadrant Cotton wool spots commonly present	Review in approximately 6 months (Option C)
Severe NPDR The 4-2-1 rule; is one or more of the: Severe hemorrhages in all 4 quadrants Significant venous beading in 2 or more quadrants Moderate IRMA in 1 or more quadrants	Review in 4 months (Option B)
Very severe NPDR Two or more of the criteria for severe NPDR	Review in 2-3 months (Option A)
Proliferative Diabetic Retinopathy (PDR):	
Mild-moderate PDR New vessels on the disc (NVD) or new vessels elsewhere (NVE), but extent insufficient to meet the high-risk criteria	Treatment is considered according to the severity of signs, stability, systemic factors, and the patient's circumstances such as reliability of attendance for review. If not treated, review in up to 2 months

High-risk PDR NVD greater than ■ disc area Any NVD with vitreous hemorrhage NVE greater than ½ disc area with vitreous hemorrhage	Treatment should be performed immediately when possible and certainly on the same day if symptomatic presentation with a good retinal view
Advanced diabetic eye disease Is characterized by tractional retinal detachment. Significant persistent vitreous hemorrhage and neovascular glaucoma.	Vitreoretinal surgery with photocoagulation

- PRP of ischemic area reduces metabolic demand and prevents the release of vasoproliferative factors by conversion of hypoxic areas into anoxic areas and creates inert scars from leaking vascular anomalies.
- Laser used: Argon, frequency-doubled YAG or diode laser
- In PRP, the inferior quadrant of the retina is first coagulated.
- Untreated Areas: A circular area with a radius of 2.5 DD around the fovea is left untouched during treatment.
- Each treatment requires 2000-3000 burns, administered over two to four sessions.
- Anti-VEGF Therapy: Bevacizumab, ranibizumab or aflibercept are used to control neovascularization.
- Bevacizumab, ranibizumab or aflibercept are used to control neovascularization.
- Vitreoretinal Surgery: Advanced cases with tractional retinal detachment or unresolved vitreous hemorrhage require vitrectomy.
- Bevacizumab, ranibizumab or aflibercept are used to control neovascularization.



Macular and Fundus Disorders; Retinal Detachment and Retinoblastoma

1. Which of the following is a characteristic finding in the fundus examination of a patient with Stargardt disease?

- A. Beaten bronze appearance
- B. Cotton wool spots and Hard exudates
- C. Dark/silent choroid
- D. Honeycomb appearance

2. A 14-month-old child is diagnosed with retinoblastoma in the right eye. The tumor is confined to the retina, located posterior to equator, measures 2.5 mm in greatest dimension, and is located 2 mm from the fovea. There is no vitreous or subretinal seeding. According to the International Classification of Retinoblastoma (ICRB), how would this tumor be classified, and what will be the appropriate treatment?

- A. Group A; Systemic chemotherapy with focal consolidation
- B. Group B; Plaque radiotherapy
- C. Group B; Enucleation
- D. Group A; Laser photocoagulation

3. A 2-year-old child is brought to the ophthalmology clinic with a white pupillary reflex in the right eye, noticed by the parents in flash photography. On examination, the child has strabismus. Which of the following is NOT a condition to be considered in the differential diagnosis of this presentation?

- A. Coats' disease
 - B. Persistent hyperplastic primary vitreous
 - C. Retinopathy of prematurity
 - D. Optic nerve glioma
-

4. A 2-year-old child presents with leukocoria in the right eye. Fundus examination reveals a white, elevated mass in the retina. Histopathological examination of the tumor shows rosettes with central lumen. Which of the following is the most specific histological feature for this condition?

- A. Homer-Wright rosettes
 - B. Flexner-Wintersteiner rosettes
 - C. Pseudorosettes
 - D. Undifferentiated tumor cells
-

5. A 2-year-old child presents with the following findings as shown in the image. Genetic testing reveals a mutation in the RB1 gene. Which of the following statements best describes the genetic findings associated with this condition?



- A. Retinoblastoma can be inherited in an autosomal dominant manner, with affected individuals having a 100% chance of passing on the mutation
- B. Retinoblastoma typically arises from a germline mutation in the RB1 gene, followed by a second somatic mutation.

C. Affected children with retinoblastoma often have mutations in both the RB1 and TP53 genes.

D. A single mutation in the RB1 gene is usually sufficient for the development of retinoblastoma in both familial and sporadic cases.

6. A 55-year-old woman with a history of proliferative diabetic retinopathy presents with sudden vision changes in her right eye. She reports seeing floaters and flashes of light. On fundoscopy, there is evidence of fibrous tissue contraction and retinal elevation, but no retinal breaks are seen. What is the most appropriate treatment approach for this patient?

- A. Laser photocoagulation
- B. Cryotherapy with scleral buckling
- C. Pars plana vitrectomy
- D. Pneumatic retinopexy

7. A 60-year-old man presents with a sudden onset of flashes of light and a shadow resembling a curtain in the peripheral vision of his right eye. He denies any recent trauma. On examination, fundus evaluation reveals a retinal detachment with a retinal break in the upper temporal quadrant. What is the most appropriate next step in the management of this patient?

- A. Intravitreal anti-VEGF injection
- B. Pneumatic retinopexy with gas injection
- C. Pars plana vitrectomy with silicone oil tamponade
- D. Drainage of subretinal fluid with diathermy to seal the break

8. A 55-year-old man reports the sudden appearance of flashes of light, followed by a dark cloud or veil in the peripheral vision of his left eye. Examination reveals a greyish reflex in the affected quadrant of the retina. What type of retinal detachment is most likely in this patient?

- A. Rhegmatogenous Retinal Detachment
- B. Tractional Retinal Detachment
- C. Exudative Retinal Detachment
- D. Macular Hole

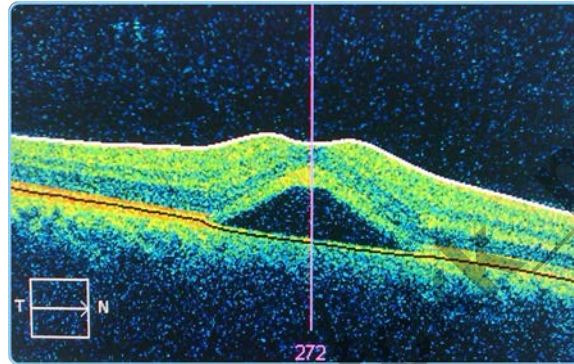
9. A 10-year-old child presents with a mild reduction in vision in both eyes. Fundus examination reveals a "sunny-side up egg" appearance in the macular region with yellowish subretinal material. The child's father has a similar history of visual impairment since childhood. What is the most likely diagnosis?

- A. Stargardt Disease
- B. Best Vitelliform Macular Dystrophy
- C. Retinitis Pigmentosa
- D. Leber Congenital Amaurosis

10. A 16-year-old patient presents with progressive vision loss. On examination, the ophthalmologist notes a central scotoma and yellowish-white flecks in the macular region of the retina. A diagnosis of Stargardt disease is made. All of the following are true statements about Stargardt disease, except:

- A. It is also known as juvenile macular dystrophy
- B. Fundus examination showing flame-shaped hemorrhages
- C. Vitamin A supplementation should be avoided
- D. Fundus autofluorescence

11. A 35-year-old male presents with sudden, painless vision loss in his right eye for two weeks. He reports distortion of images. He has been intermittently using corticosteroids for allergies. Visual acuity in the right eye is 6/18. FFA and OCT findings are given below. What is the most likely diagnosis?



- A. Central serous chorioretinopathy
- B. Age-related macular degeneration
- C. Cystoid macular edema
- D. Retinal vein occlusion

12. Which of the following is not related to retinitis pigmentosa?

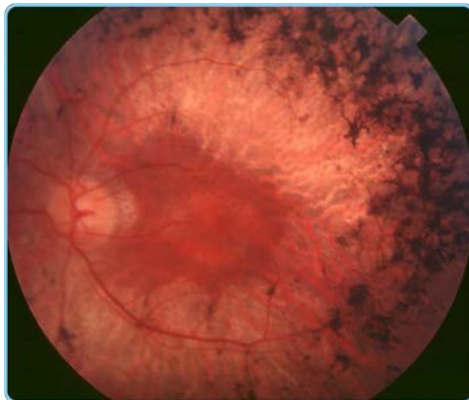
- A. Refsum's disease

- B. Laurence-Moon-Bardet-Biedl syndrome
- C. Kearns-Sayer syndrome
- D. Conductive deafness

13. A 30-year-old woman presents with progressive night blindness and gradual loss of peripheral vision over the past few years. Her family history reveals that her father had similar symptoms. On examination, you observe characteristic retinal changes, including bone spicule pigmentation. To evaluate the functional status of her retina more accurately, which diagnostic test would you choose to confirm the suspicion of retinitis pigmentosa (RP)?

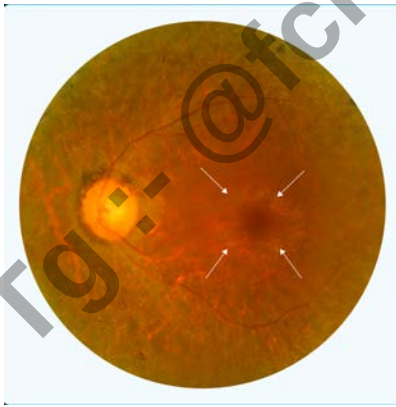
- A. Electroretinography (ERG)
- B. Optical Coherence Tomography (OCT)
- C. Visual Field Testing
- D. Electrooculogram (EOG)

14. A 32-year-old female presents with difficulty seeing in low light and peripheral vision loss that has gradually worsened over several years. On fundoscopy, the following findings are noted. Which of the following conditions is most likely responsible for these findings?



- A. Retinitis Pigmentosa
 - B. Chronic Central Serous Chorioretinopathy
 - C. Age-related Macular Degeneration
 - D. Toxoplasmic Retinochoroiditis
-

15. A 32-year-old woman presents with gradual central vision loss in both eyes. Fundus examination reveals concentric rings of retinal pigment epithelium (RPE) atrophy and hyperpigmentation centred on the fovea, creating a characteristic appearance. The image showing the appearance is given below: Which of the following is LEAST likely to be associated with this condition?



- A. Long-term hydroxychloroquine use
 - B. Cone dystrophy
 - C. Chloroquine maculopathy
 - D. Chronic central serous chorioretinopathy
-

16. A 68-year-old woman presents with gradual vision loss in her right eye over the past 6 months. Her ophthalmologist suspects age-related macular degeneration (AMD) and orders further tests. What is the most sensitive modality to detect exudative AMD?

- A. Optical Coherence Tomography (OCT)
 - B. Slit-lamp biomicroscopy using a +90 or +78 D lens
 - C. Indocyanine green angiography (ICGA)
 - D. Fluorescein angiography
-

17. A 72-year-old man presents with gradual vision loss in both eyes over the past year. His ophthalmologist diagnoses him with age-related macular degeneration (AMD). Which of the following statements about AMD is correct?

- A. AMD is a hereditary degeneration exclusively affecting the macula
 - B. Wet AMD accounts for 90% of all AMD cases but only 10% of severe vision loss
 - C. Dry AMD is characterised by choroidal neovascularisation
 - D. AMD is the most common cause of permanent central visual loss after 6th decade
-

18. Which of the following statements about angioid streaks is incorrect?

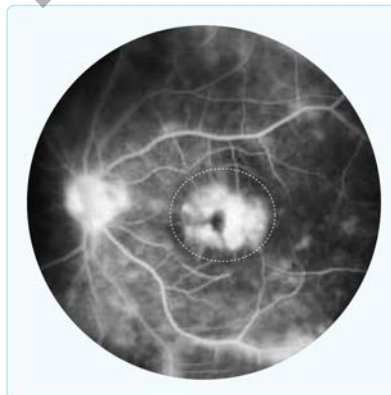
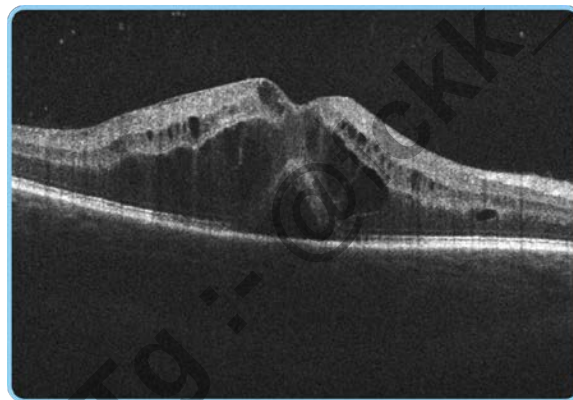
- A. They represent breaks in Bruch's membrane
 - B. They are commonly associated with sickle cell anemia
 - C. Choroidal neovascularization (CNV) frequently develops in these patients
 - D. Steroids are primarily used for the prevention of CNV
-

19. A 28-year-old woman with a history of high myopia since childhood presents for a routine eye examination. She reports occasional floaters and difficulty seeing at night. On fundus examination, the ophthalmologist notes a tilted optic disc, peripapillary atrophy, and a myopic crescent. Which of the following is NOT typically associated with the condition?

- A. Foster-Fuchs spots

- B. Lattice degeneration
- C. Posterior staphyloma
- D. Optic disc edema

20. A 65-year-old presents with visual blurring in the left eye, 20/40 vision, and a history of cataract surgery three months ago. Fundus examination shows retinal thickening and a dull foveal reflex. OCT and FFA findings are given below. What is the most likely diagnosis?



- A. Diabetic Retinopathy
- B. Central Retinal Vein Occlusion
- C. Cystoid Macular Edema
- D. Central Serous Retinopathy

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Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	4
Question 4	2
Question 5	2
Question 6	3
Question 7	2
Question 8	1
Question 9	2
Question 10	2
Question 11	1
Question 12	4
Question 13	1
Question 14	1
Question 15	4
Question 16	1
Question 17	4
Question 18	4
Question 19	4
Question 20	3

Solution for Question 1:

Answer: (A) Beaten bronze appearance

Explanation:

The following are the characteristic fundus findings seen in Stargardt Disease

- Beaten bronze appearance on fundus examination



Snail line appearance

Snail line appearance

- Snail line appearance



- On FFA (Fundus fluorescein angiography) : Dark choroid/ silent choroid sign.



Fig: of Stargardt Disease

Stargardt Disease

- Most common juvenile macular dystrophy.
- Autosomal recessive
- It is a dystrophy of the RPE.
- It is a variant of Fundus flavi-maculatus.
- It is the most common macular dystrophy.
- The gene involved is STD most common, ABCA.

Clinical features

Symptoms:

- It manifests in the initial decades.
- There will be gradual diminishing of central vision. The vision drops to less than 6/60.
- There will be decreased color vision or impaired dark adaptation.

Signs:

- Beaten bronze appearance
- Dark choroid/ silent choroid sign
- Normal FFA: background fluorescence d/t dye in choroidal vessels

(Option B) Cotton wool spots and Hard exudates are seen in fundus examination of diabetic retinopathy patients

(Option C) Dark choroid/ silent choroid sign is seen on Fundus fluorescein angiography (FFA)

(Option D) Honey-comb appearance is the characteristic finding in cystoid macular edema

Solution for Question 2:

Answer: D) Group A; Laser photocoagulation Explanation: According to the ICRB, this tumor would be classified as Group A. The criteria for Group A tumors are:

- All tumors are 3 mm or smaller in greatest dimension
- Confined to the retina
- Located >3 mm from the foveola and >1.5 mm from the optic disc

Laser photocoagulation is the primary treatment for Group A retinoblastoma, especially when the tumor is located posterior to the equator.

ICRB Stage	Features	Treatment
A	Small intraretinal tumors (<3mm) Located >3 mm from the foveola and >1.5 mm from the optic disc	Laser photocoagulation- Posterior to equator (Option D) Cryotherapy- Anterior to equator Thermotherapy
B	Large tumors >3 mm in dimension Any size tumors located <3 mm from fovea, and <1.5 mm from the optic disc margin	Chemoreduction + focal consolidation Plaque radiotherapy
C	Subretinal seeds ≤ 3 mm from tumor Vitreous seeds ≤ 3 mm from tumor Both subretinal and vitreous seeds ≤ 3 mm from tumor	Chemoreduction + focal consolidation Possible intravitreal chemotherapy for vitreous seeding
D	Subretinal seeds > 3 mm from tumor Vitreous seeds > 3 mm from tumor Both subretinal and vitreous seeds > 3 mm from retinoblastoma	Chemoreduction + possible intravitreal chemotherapy External beam radiotherapy in some cases Enucleation if poor response to other treatments
E	Extensive retinoblastoma occupying > 50% of the globe with or without: Neovascular glaucoma Hemorrhage Extension of tumor to optic nerve or anterior chamber	Enucleation Possible adjuvant chemotherapy if high-risk features

Option A: While systemic chemotherapy with focal consolidation can be used for retinoblastoma, it's typically reserved for more advanced tumors (Groups B-E).

Option B: Plaque radiotherapy is usually considered for Group B tumors.

Option C: Enucleation is reserved for advanced cases where the tumor is very large, has spread extensively within the eye, or when other treatments have failed.

Solution for Question 3:

Correct Answer: D) Optic nerve glioma

Explanation: The case presented describes a classic presentation of retinoblastoma, for which optic nerve glioma is not a differential diagnosis.

Optic nerve gliomas usually present with proptosis, visual loss, and optic nerve swelling, but not typically with leukocoria or an intraocular mass with calcification. Various conditions other than retinoblastoma, which present as leukocoria are collectively called pseudoglioma.

Differential diagnosis of Retinoblastoma

Leukocoria	Congenital cataract Inflammatory deposits in vitreous following a plastic cyclitis or choroiditis, Coloboma of the choroid, Retinopathy of prematurity (Option C) Persistent hyperplastic primary vitreous (Option B) Toxocara endophthalmitis Exudative retinopathy of Coats (Option A)
Endophytic retinoblastoma	Tuberous sclerosis Neurofibromatosis Astrocytoma
Exophytic Retinoblastoma	Exudative retinopathy of Coats

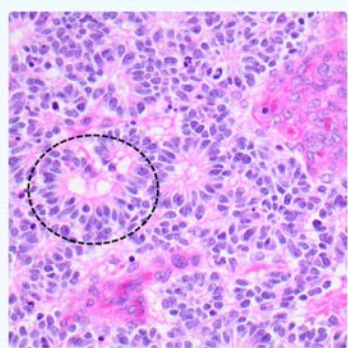
Solution for Question 4:

Correct Answer: B) Flexner-Wintersteiner rosettes

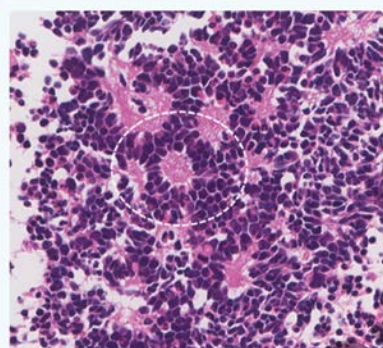
Explanation: The question describes a classic presentation of retinoblastoma, which is the most common primary intraocular malignancy in children.

The key features pointing to this diagnosis are:

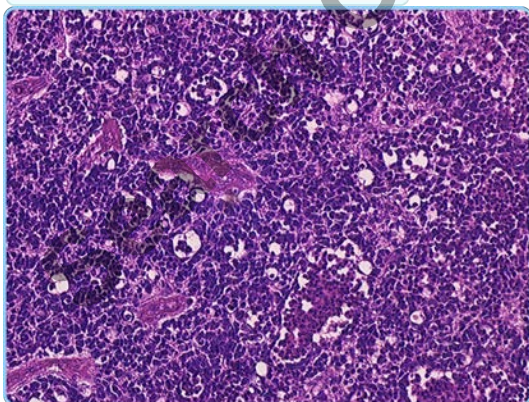
Most specific histopathological feature of retinoblastoma is Flexner-Wintersteiner rosettes, which are rosettes with a central lumen as shown in the diagram below.



Flexner-Wintersteiner rosettes



Homer Wright rosettes



Histopathology and Diagnostic modalities for Retinoblastoma

Histopathology

Flexner-Wintersteiner rosettes (highly specific)- These are composed of an 'empty' lumen surrounded by

	columnar cells Homer-Wright rosettes - These are pseudo-rosettes and consist of cells surrounding a central lumen made up of their processes Undifferentiated tumor cells Areas of necrosis and calcification
Clinical Presentation	Leukocoria (60%) Strabismus (20%) Painful red eye (7%) Poor vision (5%) Asymptomatic (3%)
Fundus Examination	White, elevated mass in the retina Can be endophytic, exophytic, or mixed growth
Imaging	CT scan: Calcification in 90% of cases MRI: Better for evaluating optic nerve and extraocular extension Ultrasound: Useful for initial screening
Genetic Testing	RB1 gene mutation on chromosome 13q14

Homer-Wright rosettes (Option A), Pseudorosettes (Option C), Undifferentiated tumor cells (Option D) are histological features seen in retinoblastoma, but are not highly specific.

Solution for Question 5:

Correct Answer: B) Retinoblastoma typically arises from a germline mutation in the RB1 gene, followed by a second somatic mutation.

Explanation:

Retinoblastoma is typically caused by a germline mutation in the RB1 gene (inherited or de novo) followed by a second somatic mutation or "second hit" in the retina, leading to the development of the tumor.

Retinoblastoma is caused by inactivation of both alleles of the RB1 tumor suppressor gene. There are two forms:

Heritable (germline):

- Seen in about 40% of cases, this form often results in bilateral retinoblastoma.

- It involves a germline mutation of the RB1 gene in all cells, followed by a somatic mutation in the retinal cells.
- They are prone to develop non-ocular tumors like osteosarcoma.

Non-heritable (sporadic):

- This form usually involves a unilateral tumor and arises from two somatic mutations that occur in the retinal cells after embryo formation, without a germline mutation.
- Cannot be passed to offspring and is not prone for developing non-ocular tumors.

Knudson's Two-Hit Hypothesis:

This hypothesis explains the development of retinoblastoma. According to this model:

- In heritable retinoblastoma, the child inherits a germline mutation in one allele of the RB1 gene (the first hit). A subsequent somatic mutation in the other allele in the retinal cells (the second hit) leads to tumor formation.
- In sporadic cases, both alleles of the RB1 gene are inactivated by somatic mutations (both hits occurring postnatally).

Option A: While retinoblastoma is inherited in an autosomal dominant pattern, this option does not explain the second-hit mechanism.

Option C: TP53 mutations are seen in Li-Fraumeni syndrome, not in retinoblastoma, which is primarily associated with RB1 mutations.

Option D: A single mutation in the RB1 gene is insufficient to cause retinoblastoma; a second somatic mutation is required.

Solution for Question 6:

Correct Answer: C) Pars plana vitrectomy

Explanation:

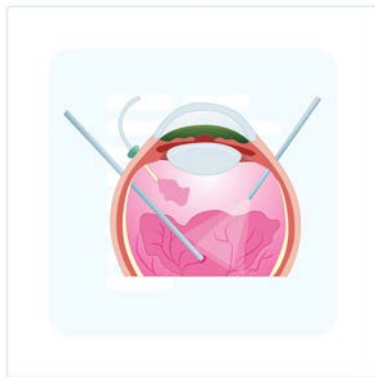
The patient's symptoms and funduscopy findings, including fibrous tissue contraction and retinal elevation without retinal breaks, are consistent with tractional retinal detachment. In this condition, the primary treatment approach is pars plana vitrectomy, which removes the vitreous and addresses the traction caused by the fibrous tissue, thereby relieving the retinal detachment.

Clinical features:

- Presence of vitreoretinal bands associated with the underlying disease.
- Absence of retinal breaks.
- A concave, localized detachment that usually does not extend to the ora serrata.
- Maximum retinal elevation at sites of vitreoretinal traction.
- Severely reduced retinal mobility with no shifting fluid.
- Focal traction from cellular membranes can occasionally cause retinal tears, leading to a combined tractional-rhegmatogenous detachment.

Treatment for tractional retinal detachment:

- Pars plana vitrectomy (PPV) is the standard approach for tractional retinal detachment.
- It involves removing cutting the vitreoretinal tractional bands and internal tamponade with either a long-acting gas or silicon oil.



Treatment for exudative retinal detachment:

- Treatment focuses on managing the primary disease responsible for the exudative retinal detachment.
- Since exudative retinal detachment can be caused by transudate, exudate, or hemorrhage, fluid absorption results in the detachment resolving on its own.
- If the detachment is due to intraocular tumors, enucleation may be necessary.

Laser photocoagulation (Option A) is used in proliferative retinopathies to prevent further neovascularization but does not address existing tractional detachment.

Cryotherapy with scleral buckling (Option B) is effective in rhegmatogenous retinal detachment, not for tractional detachment.

Pneumatic retinopexy (Option D) is not suitable for tractional or exudative detachments, as it primarily addresses rhegmatogenous detachment with retinal breaks.

Solution for Question 7:

Answer: B) Pneumatic retinopexy with gas injection

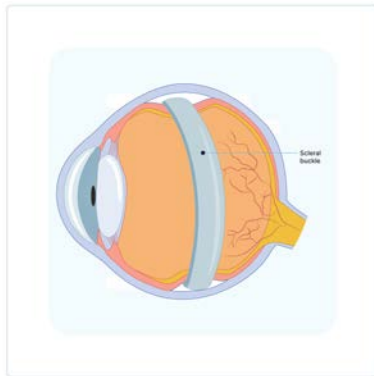
Explanation:

This patient has a fresh retinal detachment with a small break in the upper quadrant. Pneumatic retinopexy is an appropriate choice for treating retinal detachments with small breaks in the superior retina.

Treatment of Rhegmatogenous Retinal Detachment:

The primary goals of treating rhegmatogenous retinal detachment are sealing retinal breaks, reducing vitreous traction, and flattening the retina by draining subretinal fluid (SRF) and providing external or internal tamponade.

- All retinal breaks must be identified and sealed by inducing aseptic chorioretinitis.
- Techniques include cryocoagulation (for scleral buckling and pneumoretinopexy) or endo-laser photocoagulation (during vitrectomy).
- SRF is carefully drained by inserting a fine needle through the sclera and choroid to allow fluid to escape, promoting retinal reattachment.
- In some cases, SRF drainage may not be necessary.
- Scleral buckling: A surgical technique where material sutured onto the sclera (explant) creates an inward indentation (buckle). It is commonly used in simple cases by inserting a silicone band or sponge. Scleral buckling is an external retinal detachment surgery, which involves suturing an explant (material) onto the sclera to create an inward indentation (buckle).
- A surgical technique where material sutured onto the sclera (explant) creates an inward indentation (buckle).
- It is commonly used in simple cases by inserting a silicone band or sponge.
- Scleral buckling is an external retinal detachment surgery, which involves suturing an explant (material) onto the sclera to create an inward indentation (buckle).

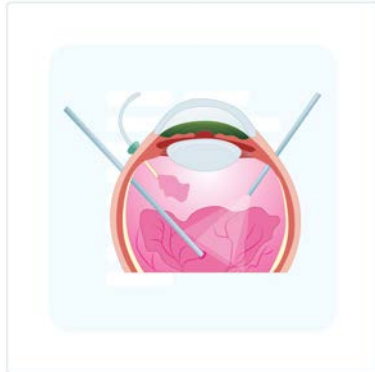


- **Pneumatic retinopexy:** (Option B) A minimally invasive quick and OP based procedure to reattach the retina. Uses an intravitreal gas bubble with cryotherapy or laser to seal retinal breaks. No scleral buckling is needed. Common gases: Sulfur hexafluoride (SF6) and perfluoropropane (C3F8). Used for uncomplicated retinal detachments. Ideal for small breaks or multiple breaks covering less than two clock hours in the upper peripheral retina.
- A minimally invasive quick and OP based procedure to reattach the retina.
- Uses an intravitreal gas bubble with cryotherapy or laser to seal retinal breaks.
- No scleral buckling is needed.
- Common gases: Sulfur hexafluoride (SF6) and perfluoropropane (C3F8).
- Used for uncomplicated retinal detachments.
- Ideal for small breaks or multiple breaks covering less than two clock hours in the upper peripheral retina.



- **Pars plana vitrectomy** is performed for complicated or recurrent cases of RD and tractional detachments. This involves removing the vitreous, draining SRF, and flattening the retina using silicone oil or perfluorocarbon liquids. Endolaser photocoagulation is applied around breaks and retinotomy sites to create chorioretinal adhesions, followed by internal tamponade with gas (SF6 or C3F8) or silicone oil.

- This involves removing the vitreous, draining SRF, and flattening the retina using silicone oil or perfluorocarbon liquids.
- Endolaser photocoagulation is applied around breaks and retinotomy sites to create chorioretinal adhesions, followed by internal tamponade with gas (SF6 or C3F8) or silicone oil.



Intravitreal anti-VEGF injection (Option A): Primarily used for neovascular diseases, such as wet AMD, but not for retinal detachment.

Pars plana vitrectomy with silicone oil tamponade (Option C): usually reserved for complicated cases like extensive vitreoretinal traction or multiple breaks. In this case, the detachment is relatively simple, and pneumatic retinopexy is more suitable.

Drainage of subretinal fluid with diathermy to seal the break (Option D): While it can be part of the management for retinal detachment, it is not the primary treatment for simple rhegmatogenous detachments with small breaks.

Solution for Question 8:

Answer: A) Rhegmatogenous Retinal Detachment

Explanation:

Retinal detachment refers to the separation of the neurosensory retina from the retinal pigment epithelium.

Etiology:

- Age: most common in 40-60 years.

- Sex. More common in males (M:F - 3:2).
- Myopia. About 40% cases of rhegmatogenous retinal detachment are myopic.
- Aphakia and pseudophakia.
- Trauma. It may also act as a predisposing factor.
- Senile posterior vitreous detachment (PVD): It is associated with retinal detachment in many cases.
- Retinal degenerations predisposed to retinal detachment: Lattice degeneration, Snail track degeneration, Acquired or degenerative retinoschisis, and Focal pigment clumps.

Retinal detachment can be classified into three types:

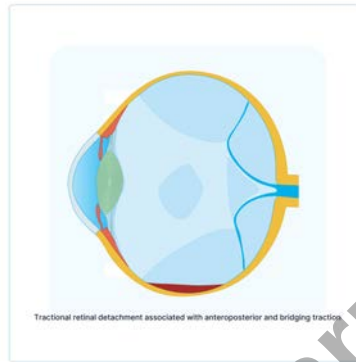
Rhegmatogenous Retinal Detachment: This is the most common type and occurs due to a retinal tear or break, allowing subretinal fluid to enter and separate the sensory retina from the pigmentary epithelium.

Tractional Retinal Detachment: Results from the pulling of the retina by fibrous or vascular tissue, commonly seen in fibrovascular proliferative conditions like diabetic retinopathy, without a retinal break.

Exudative Retinal Detachment: Caused due to choroidal pathologies or by tumors pushing the retina or fluid accumulation under the retina due to inflammation or vascular abnormalities, without any associated retinal tear or traction.

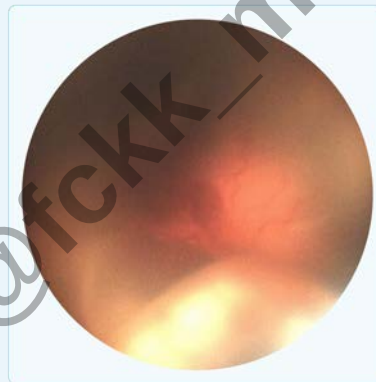
Type of Retinal Detachment	Clinical Features
Rhegmatogenous Retinal Detachment	 ... (content truncated)

Tractional Retinal Detachment



... (content truncated)

Exudative Retinal Detachment



... (content truncated)

Tractional Retinal Detachment (Option B): Often seen in proliferative diabetic retinopathy or trauma, presenting with gradual vision loss, not acute symptoms.

Exudative Retinal Detachment (Option C): Associated with systemic or ocular inflammatory diseases and presents without the classic flashes or floaters.

Macular Hole (Option D): Causes central vision loss but does not present with peripheral shadows or flashes.

Solution for Question 9:

Answer: B) Best Vitelliform Macular Dystrophy

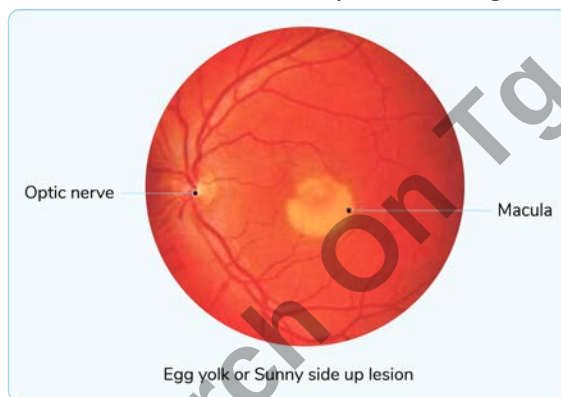
Explanation:

Best Vitelliform Macular Dystrophy (BVMD), also known as juvenile-onset vitelliform macular dystrophy, is the second most common macular dystrophy after Stargardt disease.

- Caused by mutations in the BEST1 gene located on chromosome 11q13.
- Bestrophin, the protein produced by this gene, functions as a transmembrane ion channel in the retinal pigment epithelium (RPE).
- Inherited in an autosomal dominant (AD) pattern.
- Usually begins in childhood

Clinical Features:

- Stages of the disease include: Pre-vitelliform: abnormal EOG, normal fundus in asymptomatic patients. Vitelliform: a characteristic "egg yolk or sunny-side up" macular lesion. The lesions can vary in size and developmental stage between the two eyes.
- Pre-vitelliform: abnormal EOG, normal fundus in asymptomatic patients.
- Vitelliform: a characteristic "egg yolk or sunny-side up" macular lesion. The lesions can vary in size and developmental stage between the two eyes.
- Pre-vitelliform: abnormal EOG, normal fundus in asymptomatic patients.
- Vitelliform: a characteristic "egg yolk or sunny-side up" macular lesion. The lesions can vary in size and developmental stage between the two eyes.



- Pseudohypopyon: partially regressed lesion.
- Vitelliruptive: Lesion fragments, visual acuity declines - "scrambled egg appearance".



- Atrophic: Pigment disappears, leading to RPE atrophy.
- Visual acuity - stable early on but declines in adulthood due to complications like choroidal neovascularization (CNV), scarring, or geographic atrophy.

Diagnosis:

- Fundus Autofluorescence (FAF): hyper autofluorescence yellowish material.
- Fluorescein Angiography (FA): hypofluorescence due to lesion masking.
- Optical Coherence Tomography (OCT): hyperreflective material beneath and within the RPE.
- Electro-oculography (EOG): Severely subnormal across all stages (Arden index <1.5).
- Hallmark- normal electroretinogram (ERG) but abnormal electrooculogram (EOG).

Stargardt Disease (Option A): Causes central vision loss with a characteristic "beaten-bronze" appearance and yellow flecks but is not associated with the vitelliform lesion.

Retinitis Pigmentosa (Option C): Characterized by night blindness and peripheral vision loss, not macular lesions.

Leber Congenital Amaurosis (Option D): A severe form of retinal dystrophy with early-onset blindness but lacks the specific macular changes seen in BVMD.

Solution for Question 10:

Answer: (B) Fundus examination showing flame-shaped hemorrhages

Explanation:

Fundus examination showing flame-shaped hemorrhages is seen in hypertensive retinopathy.

All other options are true regarding Stargardt disease.

Stargardt disease, also known as juvenile macular dystrophy, is the most common macular dystrophy. (Option A)

Investigations:

- FFA: A dye is injected, and there is a dark background choroid fluorescence due to dense deposition of lipofuscin on retinal pigment epithelium.
- Since there's no choroidal background fluorescence seen, it is called "dark choroid syndrome" or "silent choroid syndrome".
- Fundus autofluorescence: No dye is seen, and all lipofuscin deposits are shining. (Option D)
- EOG abnormality.
- ERG: Late stages may affect the ERG, mainly photopic ERG abnormality.

Treatment

- Avoid bright sunlight. Use sunglasses
- Avoid vitamin A, as it promotes lipofuscinosis. (Option C)

Solution for Question 11:

Correct Answer: A) Central serous chorioretinopathy

Explanation:

This patient's sudden onset of painless vision loss, stress, corticosteroid use, and characteristic FFA "smoke-stack (mushroom-shaped)" pattern with subretinal fluid seen on OCT strongly suggest central serous chorioretinopathy.

Central Serous Chorioretinopathy (CSCR)

CSCR is characterized by the spontaneous detachment of the neurosensory retina in the macular region, with or without retinal pigment epithelium (RPE) detachment.

Pathogenesis: Increased sympathetic drive and corticosteroids affect choroidal vascular permeability, raising hydrostatic pressure, and leading to RPE defects and serous detachment of the neurosensory retina.

Risk Factors:

- Males aged 20-50
- Type A personality
- Emotional stress
- Steroid use
- Systemic lupus erythematosus, Cushing's disease
- Pregnancy (3rd trimester)

Clinical Manifestations:

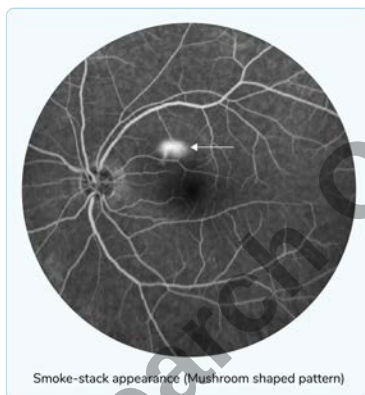
- Sudden painless vision loss (6/9 - 6/24)
- Relative positive scotoma, micropsia, and metamorphopsia
- Recurrence in 30-50% of cases

Examination Findings:

- Mild macular elevation with circular ring reflex
- RPE detachment (yellow-grey elevations)
- Absence or distortion of the foveal reflex
- Subretinal deposits

Investigations:

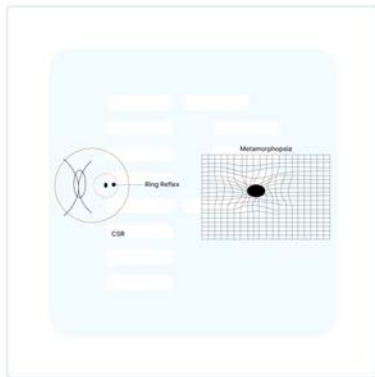
- Fundus fluorescein angiography (FFA): Shows "ink-blot" or "smoke-stack" leakage patterns.



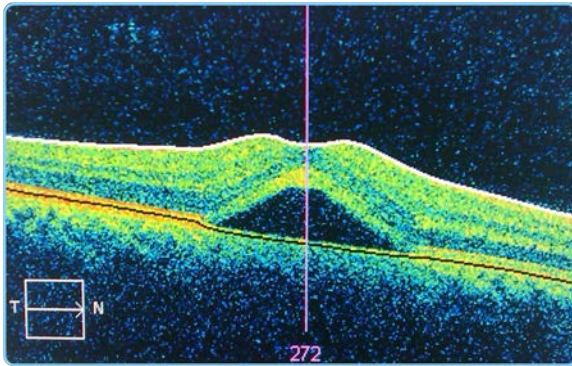


Enlarging ink-blot pattern

- Indocyanine green (ICG): Shows hyperfluorescence due to choroidal hyperpermeability.
- OCT: Shows neurosensory retinal and RPE detachment.



Circular ring reflex



Treatment:

Treatment:

Age-related macular degeneration (Option B): ARMD typically occurs in older patients (>50) and is associated with drusen and neovascularization, which are absent here.

Cystoid macular edema (Option C): CME usually follows ocular surgeries or inflammation and presents with cystic spaces on OCT, not the subretinal detachment seen here.

Retinal vein occlusion (Option D): RVO presents with retinal hemorrhages and cotton wool spots on examination, not the isolated macular elevation and leakage seen in CSCR.

Solution for Question 12:

Answer: (D) Conductive deafness

Explanation:

Retinitis pigmentosa associated with sensory neural deafness- Usher syndrome

Systemic associations in Retinitis Pigmentosa

- Refsum's disease: It is a defect in the phytanic acid metabolism. RP+ Cerebellar ataxia+ Polyneuropathy. It is managed with a low phytanic acid diet. (Option A)
- Laurence-Moon-Bardet-Biedl syndrome (m/c): involves RP, mental retardation, obesity, polydactyly, and paraplegia. (Option B)
- Usher's syndrome: It is retinitis pigmentosa (RP) with Sensory neural deafness.
- Kearns-Sayer syndrome: involves RP, heart block, and chronic progressive external ophthalmoplegia (CPEO). (Option C)

- Bassen-Kornzweig syndrome involves RP, acanthocytosis, abetalipoproteinemia, and ataxia. It is managed with a low-fat diet and vitamin supplements.

Ocular associations in Retinitis Pigmentosa

- Keratoconus
- Posterior subcapsular cataract (PSC)
- Open-angle glaucoma (OAG)
- Cystoid macular edema (CME)

Solution for Question 13:

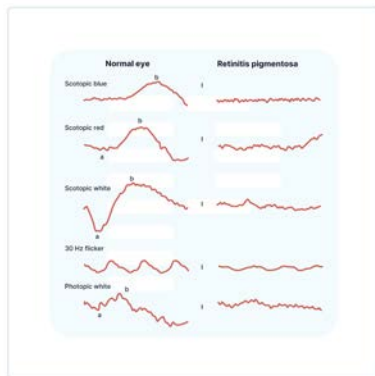
Answer: (A) Electroretinography (ERG)

Explanation:

ERG is a sensitive diagnostic test in retinitis pigmentosa that shows reduced scotopic, subnormal responses in early disease and becomes completely extinguished in advanced disease.

Investigation of Retinitis Pigmentosa

- Electroretinography (ERG) is the investigation of choice - reduced scotopic rod in early disease and reduced photopic with progression. (Option A)
- EOG - subnormal - absence of light rise. (Option D)
- Perimetry: initially small mid-peripheral scotomas which gradually merge and may progress, leaving only a small island of residual central vision.
- Electroretinography (ERG): There's a decreased amplitude of both "a" and "b" waves.
- OCT - to identify cystoid macular edema (CME). (Option B)
- Dark adaptometry: Dark adaptation time is prolonged.
- Decreased Arden ratio may be present.



Treatment of Retinitis Pigmentosa:

- The treatable RP includes Refsum's disease and Bassen-Kornzweig syndrome.

The treatment includes-

- Regular follow-ups: Annual check-ups help detect complications, provide support, and track new treatment options.
- Low-vision aids: Includes devices, rehabilitation, and access to social services. Avoid smoking.
- Light protection: Outdoor sunglasses blocking wavelengths up to 550 nm with side protection. Indoors, amber lenses blocking 511–527 nm may enhance contrast sensitivity.
- Outdoor sunglasses blocking wavelengths up to 550 nm with side protection.
- Indoors, amber lenses blocking 511–527 nm may enhance contrast sensitivity.
- Treatment for CMO: Oral acetazolamide or topical carbonic anhydrase inhibitors may reduce cystoid macular edema.
- Vitamin A: High-dose vitamin A (15,000 units/day) provides marginal benefits but should be used with caution due to risks like liver toxicity and increased lung cancer risk in smokers. Avoid during pregnancy. Lutein, possibly with zeaxanthin, is a safer alternative.
- A retinal implant (ARGUS-II) can also be performed.
- Low-vision aids (LVA) can also be used.
- It can also lead to cone dystrophy.
- Outdoor sunglasses blocking wavelengths up to 550 nm with side protection.
- Indoors, amber lenses blocking 511–527 nm may enhance contrast sensitivity.

Visual field testing (Option C) is important for evaluating the extent of vision loss, especially in RP where peripheral vision is often affected. However, it does not directly assess the functional status of the retinal photoreceptors.

Solution for Question 14:

Correct Answer: A) Retinitis Pigmentosa

Explanation:

The image given above is retinitis pigmentosa funduscopy findings showing a pale waxy disc, bony spicules, and attenuation of arteries.

The Triad of retinitis pigmentosa includes a pale waxy disc due to consecutive optic atrophy, attenuation of arteries, and pigmentation disturbance in the form of bony spicules.

- Retinitis pigmentosa is a dystrophy of rods and cones, initially, rods are affected.
- Most common hereditary fundus dystrophy.
- It can be sporadic (40-50%) or hereditary.
- If hereditary, autosomal dominant is the most common mode of inheritance followed by autosomal recessive (AR). X-linked recessive inheritance is the least severe form.

Clinical features:

Symptoms:

- Nyctalopia (night blindness)
- Peripheral visual field loss
- Mid-peripheral superior scotoma leads to ring/ doughnut scotoma resulting in tubular/ tunnel vision

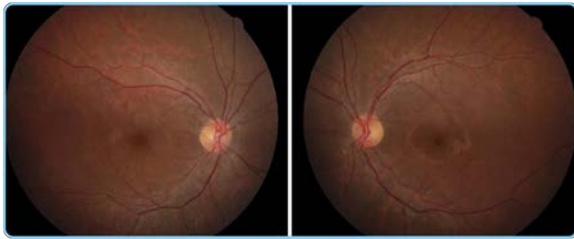
Signs:

- Triad of a pale waxy disc due to consecutive optic atrophy, attenuation of arteries, and pigmentation disturbance in the form of bony spicules.
- Tapetal reflex: shining reflex at macula - seen in female carriers of XLR inheritance.

Atypical Retinitis Pigmentosa

- Atypical retinitis pigmentosa includes heterogeneous disorders that share clinical features with typical pigmentary retinal dystrophy.
- Conditions included are:

RP sine pigmento - absence or paucity of pigment accumulation



- Retinitis punctata albescens - scattered whitish-yellow spots with arteriolar attenuation, usually sparing the macula.

Solution for Question 15:

Correct Answer: D) Chronic central serous chorioretinopathy

Explanation:

The image shows the characteristic appearance of Bull's eye maculopathy, a retinal disorder characterised by concentric rings of RPE atrophy and hyperpigmentation centred on the fovea, creating an appearance resembling a bull's eye target.

While chronic central serous chorioretinopathy can cause retinal pigment epithelium changes, it typically does not present with the characteristic bull's eye maculopathy appearance.

Etiological causes of bull's eye maculopathy include:

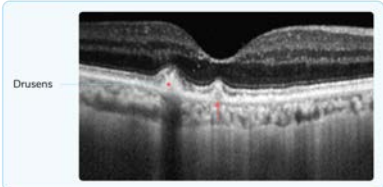
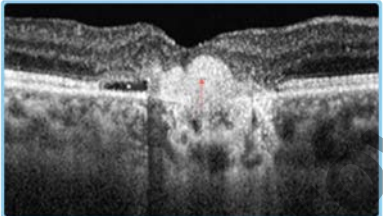
- Chloroquine maculopathy (Option C)
- Hydroxychloroquine (Option A)
- Cone dystrophy (Option B)
- Battens disease

Solution for Question 16:

Answer: A) Optical Coherence Tomography (OCT)

Explanation:

OCT is currently considered the most sensitive modality for detecting exudative (wet) AMD.

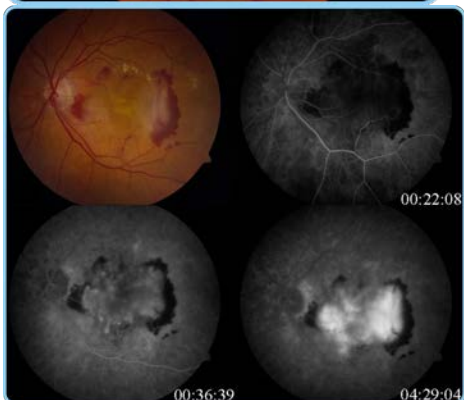
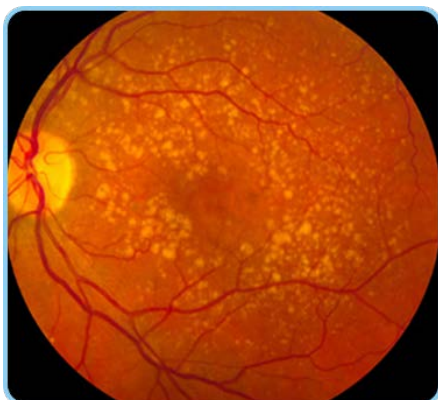
	Dry AMD	Wet AMD
Diagnostic Methods	 Slit-lamp biomicroscopy using a +90 or +78 D lens (Clinical diagnosis) Optical Coherence Tomography (OCT): outer retinal tubulation is seen in the outer nuclear layer. Fundus autofluorescence Amsler grid test	 Slit-lamp biomicroscopy using a +90 or +78 D lens Optical Coherence Tomography (OCT) (Most sensitive): RPE detachment is seen. Fluorescein angiography Indocyanine green angiography
Treatment Options	AREDS supplements (vitamins C, E, zinc, copper, lutein, zeaxanthin) Lifestyle modifications (smoking cessation, healthy diet) Regular monitoring to prevent progression to advanced disease	Intravitreal anti-VEGF injections (e.g., bevacizumab, ranibizumab, aflibercept) Photodynamic therapy (in select cases) Rarely, laser photocoagulation (for extrafoveal CNV)

Solution for Question 17:

Correct Answer: D) AMD is the most common cause of permanent central visual loss after 6th decade

Explanation:

Age-related Macular Degeneration (AMD) is a nonhereditary degeneration involving the choriocapillaris, Bruch's membrane, retinal pigment epithelium, and photoreceptors. It is indeed the most common cause of permanent central visual loss after 6th decade.

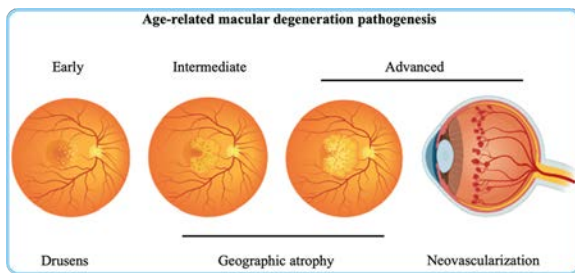


Based on the clinical findings of macular degeneration, It is of 2 types:

- Exudative (wet)
- Non-exudative (dry)

Category	Dry AMD	Wet AMD
Definition	Nonexudative macular degeneration	Exudative macular degeneration
Prevalence	90% of all AMD cases	10% of all AMD cases
Vision Loss	Accounts for about 10% of legal blindness in AMD	Major blinding form, accounts for 90% of severe vision loss in AMD
Key Features	Drusen, RPE abnormalities, geographic atrophy	Choroidal neovascularization (CNV), subretinal fluid
Pathophysiology	Accumulation of drusen, dysfunction of RPE	Growth of new blood vessels from choroid, leakage of fluid
Progression	Generally slow	Can be rapid

DRY AMD:



(Option A) AMD is a nonhereditary degeneration primarily affecting the macula. However, it also involves the choriocapillaris, Bruch's membrane, retinal pigment epithelium, and photoreceptors.

(Option B) Wet AMD accounts for only 10% of all AMD cases but is responsible for 90% of severe vision loss in AMD patients.

(Option C) Choroidal neovascularization is a characteristic of wet AMD, not dry AMD. Dry AMD is characterized by drusen, RPE abnormalities, and geographic atrophy.

Solution for Question 18:

Answer: D) Steroids are primarily used for the prevention of CNV

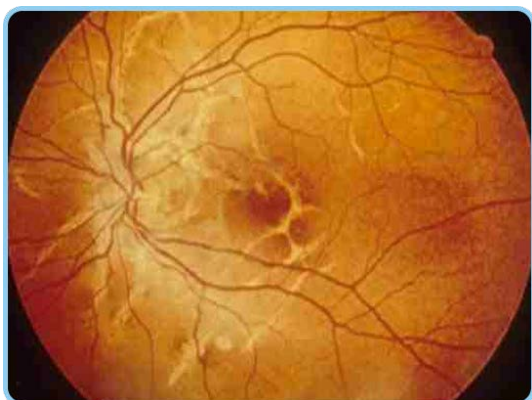
Explanation:

Angioid streaks are crack-like dehiscences in Bruch's membrane, appearing as irregular reddish-brown or gray lines radiating from the optic disc.

Bruch's membrane: The innermost layer of the choroid is the Bruch's membrane. It is a membrane between the choroid and the retinal pigment epithelium.

It lines the layer of choriocapillaris.

- Asymptomatic angioid streaks do not require treatment.
- However, anti-VEGF therapy can be used to prevent choroidal neovascularization.



Angioid Streaks:

Definition	Crack-like dehiscences in Bruch's membrane, appearing as reddish-brown or gray lines radiating from the optic disc (Option A)
Association with conditions	Pseudoxanthoma elasticum (PXE) - Most common Ehlers-Danlos syndrome Paget's disease Sickle cell disease (Option B)
Complication	Choroidal neovascularization (CNV) frequently develops in affected patients (Option C)
Treatment	Anti-VEGF therapy for choroidal neovascularization

Solution for Question 19:

Answer: D) Optic disc edema

Explanation:

The patient's history and fundus findings are consistent with pathological myopia.

- Pathological myopia is a progressive form of myopia characterized by excessive axial elongation of the eyeball and degenerative changes in the retina and choroid.
- The optic disc in pathological myopia often appears large and pale due to the stretching of the eye, but it is not edematous.

Pathological Myopia

Definition	Pathological/degenerative/progressive myopia is a rapidly progressive error that starts in childhood at 5-10 years of age and results in high myopia (>-6D) during early adult life.
Etiology	Rapid axial growth of the eyeball outside normal biological variations Genetic factors (major role) More common in women than men Autosomal dominant pathological myopia Heredity- Familial, more common in women and in certain races Autosomal Dominant - linked to genes 18p11.31 and 12q21-23 Sex-linked recessive - high myopia (>6D) General growth process (minor role) nutritional deficiency endocrine disturbances
Symptoms	Defective vision with considerable failure in visual function Muscae volitantes (floaters) Difficulty in night vision in very high myopes with degenerative changes
Signs	Prominent eyeballs, often appear elongated Exophthalmos, especially in unilateral cases Cornea is large The anterior chamber is deep Pupils are slightly large and react sluggishly to light Fundus examination reveals: The optic disc appears large and pale Characteristic myopic crescent Tigroid fundus Myopic crescent (temporal to disc margin)
Complications	 Diffuse chorioretinal atrophy appearing as an ill-defined yellowish lesion in the posterior fundus of a high myope with peripapillary atrophy ... (content truncated)

Solution for Question 20:

Correct Answer: C) Cystoid Macular Edema

Explanation:

Cystoid Macular Edema (CME) is associated with fluid accumulation in the outer plexiform and inner nuclear layers of the retina, leading to cystoid spaces.

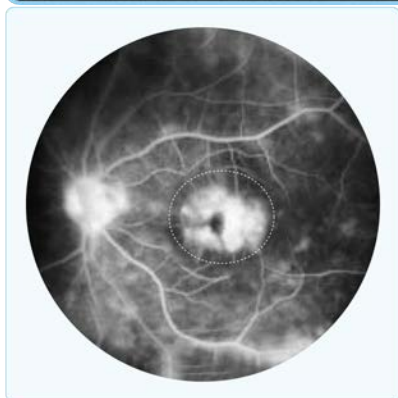
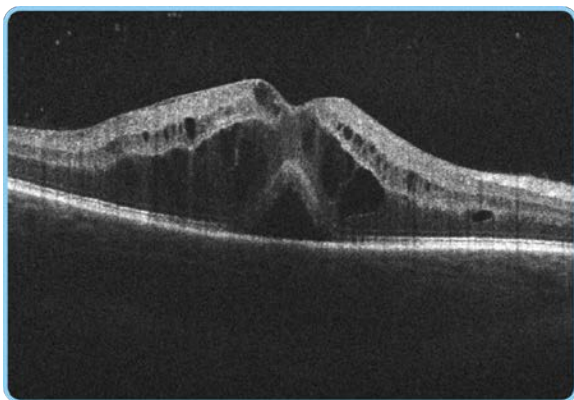
- The OCT findings of multiple cystic spaces centered around the foveola.
- Fundus Fluorescein Angiography shows leakage and dye accumulation in the macular region, with a "flower petal appearance" in late frames. The history of recent cataract surgery further supports this diagnosis, as CME is a known postoperative complication. (also known as Irvine gass syndrome)

Etiology:

- Ocular Treatment Complications: Post-surgery (e.g., cataract extraction, retinal detachment surgery), ocular laser therapy, and topical ocular medications.
- Retinal Vascular Disorders: Diabetic retinopathy, central and branch retinal vein occlusion, retinal telangiectasia, and hypertensive retinopathy.
- Intraocular Inflammations: Pars planitis, uveitis.
- Retinitis pigmentosa.
- Vitreomacular Traction: Associated with macular epiretinal membrane (ERM) and vitreomacular traction (VMT) syndrome.
- Systemic Diseases: Includes leukemia, chronic renal failure, and multiple myeloma.
- Miscellaneous Causes: niacin toxicity, PG analogues (risk factor for development of CME)

Clinical Features:

- Visual Loss: Initial minimal to moderate vision loss, potentially leading to permanent vision decrease if persistent.
- Ophthalmoscopy: Reveals loss of foveal contour, retinal thickening, and a "honeycomb appearance" of the macula.
- Fluorescein Angiography: Shows leakage and dye accumulation in the macular region, with a "flower petal appearance" in late frames.
- Optical Coherence Tomography (OCT): Demonstrates loss of foveal depression, cystoid spaces, and overall retinal thickening.



Complications:

- Long-standing CME may result in a lamellar macular hole.

Treatment:

- Addressing the Cause: Intravitreal anti-VEGF or photocoagulation for diabetic CME; discontinuing causative topical medications.
- Topical NSAIDs: Such as ketorolac, diclofenac, or profenac, are effective in preventing and treating CME.
- Topical/Systemic Steroids: Useful in established cases.
- Systemic Carbonic Anhydrase Inhibitors (CAIs): Oral acetazolamide may be beneficial, particularly in cases associated with retinitis pigmentosa.

Diabetic Retinopathy (Option A): Typically shows diffuse retinal thickening and other diabetic changes, not the cystoid spaces seen in CME.

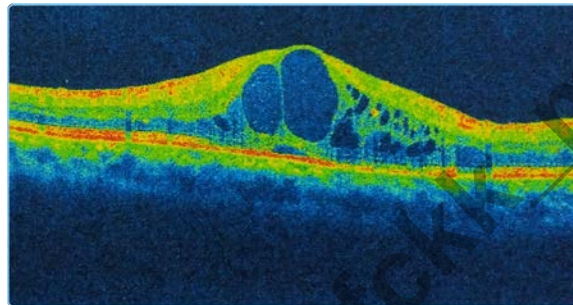
Central Retinal Vein Occlusion (Option B): Often associated with retinal hemorrhages and diffuse edema, but CME's cystoid spaces are more specific.

Central Serous Retinopathy (Option D) is characterized by "ink-blot" or "smoke-stack" leakage patterns on FFA, and neurosensory retinal and RPE detachment on OCT. CSR typically does not present with the cystoid pattern seen in CME.

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Previous Year Questions

1. A patient presents with vision problems and has a history of cataract surgery. OCT finding is shown below. What is the syndrome most likely associated with these findings?



- A. Posner-Schlossman Syndrome
- B. Irvine-Gass Syndrome
- C. Central Serous Retinopathy
- D. Elschnig Pearls

2. Which of the following is not seen in very severe NPDR?

- A. Venous beading
- B. Neovascularization of disc
- C. Retinal hemorrhage
- D. Microaneurysm

3. A 60-year-old male patient presents to the ophthalmology clinic with complaints of visual disturbances. He reports that he is unable to see on the left side from the

left eye and the left side from his right eye. However, he notices that his central vision remains intact and unaffected. The patient denies any recent trauma or other significant medical history. What is the most likely visual deficit that the patient is experiencing?

- A. Quadrantanopia
- B. Heteronymous hemianopia
- C. Homonymous hemianopia
- D. Homonymous hemianopia with macular sparing

4. A 55-year-old hypertensive patient presented with complaints of blurred vision and headaches. On retinal examination, the following findings were observed: multiple flame-shaped hemorrhages, cotton wool spots, and disc edema. Based on the Keith-Wagner classification, which grade of hypertensive retinopathy is this patient most likely classified as?

- A. GRADE 1
- B. GRADE 2
- C. GRADE 3
- D. GRADE 4

5. A 55-year-old male patient presents to the clinic with a complaint of sudden visual blurring that lasted for a short duration. He reports that during the episode, he completely lost vision in his left eye for a few minutes and then regained his vision. The patient denies any pain, headache, or other associated symptoms during the episode. What is the most likely diagnosis?

- A. Amaurosis fugax
- B. Central retinal artery occlusion
- C. Central retinal vein occlusion

D. Hypertensive retinopathy

6. Which of the following statements is incorrect about Retinitis pigmentosa?

- A. ERG show decreased amplitude of a and b wave
 - B. Rod dystrophy > Cone dystrophy
 - C. Late presentation is loss of peripheral vision
 - D. Night vision is lost late and rare
-

7. A Rheumatoid arthritis patient who is noncompliant with treatment presents with complaints of superior field loss of vision for the past 1 week. What is the cause of this condition?

- A. Branched retinal artery occlusion
 - B. Central retinal artery occlusion
 - C. Branched retinal vein occlusion
 - D. Central retinal vein occlusion
-

8. A 22-year-old patient with high myopia and thick glasses presents with floaters in his eyes. What is the likely cause of this condition?

- A. Subarachnoid hemorrhage
 - B. Retinal detachment
 - C. Vitreous hemorrhage
 - D. Branched retinal artery occlusion
-

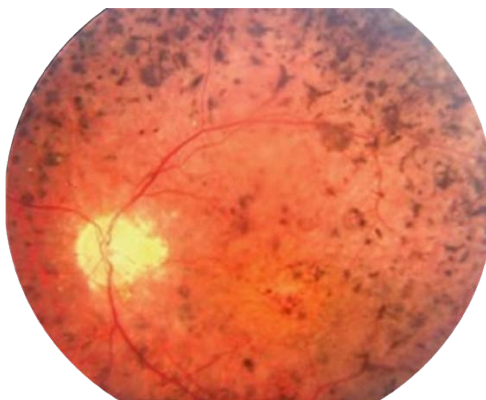
9. What could be the cause of sudden painless loss of vision?

- A. Chemical injuries to the eyeball
 - B. Acute congestive glaucoma
 - C. Acute iridocyclitis
 - D. Central Serous retinopathy
-

10. Which of the following conditions is associated with the Curtain falling sign?

- A. Retinal detachment
 - B. Macular degeneration
 - C. Optic atrophy
 - D. Acute anterior uveitis
-

11. A patient presents with a gradual loss of night vision and peripheral vision. What is the most likely diagnosis based on fundoscopy findings?



- A. Retinitis pigmentosa

- B. Retinal hemorrhage
 - C. Diabetic retinopathy
 - D. Hypertensive retinopathy
-

12. A child presents with leukocoria and an intraocular mass. Microscopic examination reveals rosettes with scanty cytoplasm, and the mass is positive for synaptophysin and chromogranin. What is the diagnosis?

- A. Retinoblastoma
 - B. Medulloblastoma
 - C. Optic glioma
 - D. Choroidal melanoma
-

13. What is the name of the first neuron in the optic pathway?

- A. Rods and cones
 - B. Ganglion cells
 - C. Cells in lateral geniculate body
 - D. Bipolar cells
-

14. Which of the following conditions is associated with retinitis pigmentosa?

- A. Arteriolar attenuation
- B. Neovascularization
- C. Retinal artery thrombosis
- D. Papilledema

15. A 72-year-old had undergone cataract surgery with a decrease in vision after one month of surgery. He is a known case of DM. On examination, Honeycomb appearance/ flower petal pattern of the retina is seen in?

- A. Cystoid macular edema
 - B. Central serous retinopathy
 - C. Retinal detachment
 - D. Diabetic retinopathy
-

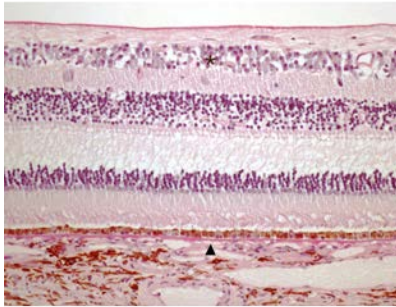
16. Which criteria is not included in the definition of severe NPDR?

- A. Microaneurysms in all quadrants
 - B. Retinal hemorrhage in all quadrants
 - C. Looping in all quadrants
 - D. Beading in two quadrants
-

17. At what point would you suggest conducting a dilated fundus examination for a person diagnosed with diabetes on their 45th birthday?

- A. Immediately
 - B. Before his 50th birthday
 - C. After his 50th birthday
 - D. When the complains of decrease in vision
-

18. Please identify the structure indicated by the arrow.

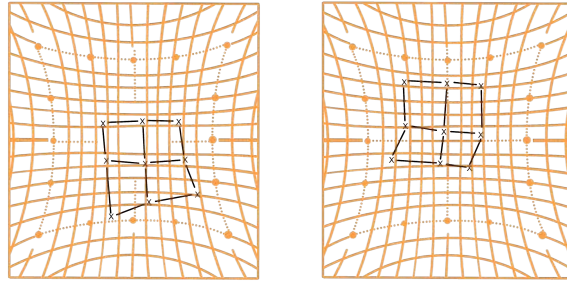


- A. Outer limiting membrane
- B. Inner plexiform layer
- C. Choroid
- D. Outer nuclear layer

19. ETDRS study was related to

- A. Central retinal artery occlusion
- B. Diabetic macular edema
- C. Primary open angle glaucoma
- D. Non – infectious uveitis

20. The chart depicted in the image is _____?



- A. Hess chart
- B. Snellen chart
- C. Pelli-Robson chart
- D. ETDRS chart

21. To confirm the diagnosis, what additional investigation should be conducted for a child exhibiting symptoms of night blindness and delayed dark adaptation?

- A. Dark adaptometry
- B. ERG
- C. EOG
- D. Retinoscopy

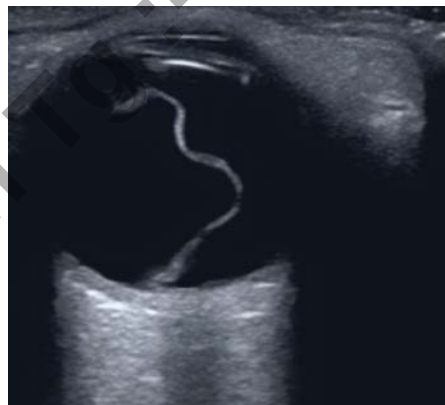
22. What is true regarding retinoscopy using a plane mirror?

- A. In hypermetropia the red glow moves in opposite direction
- B. In myopia the red glow moves in same direction
- C. In emmetropia the red glow moves in opposite direction
- D. Done at 1 meter away from patient

23. What is the most probable diagnosis for a 3-year-old boy who has displayed a growing clumsiness over the last half a year, has leukocoria and lacks red reflex in his left eye? After diagnosis, enucleation was performed, and the specimen displayed Flexner-Winter Steiner rosettes under microscopic examination?

- A. Congenital glaucoma
- B. Retinoblastoma
- C. Toxocariasis
- D. Neurofibroma

24. A 61-year-old university professor complains of vision loss in his left eye. He compares the sensation to a curtain being pulled over his affected eye. Additionally, he experienced flashing lights and floaters a week ago. The provided image shows a B-scan of his left eye. What could be the possible diagnosis for his condition?



- A. Glaucoma
- B. Posterior vitreous detachment
- C. Retinal detachment
- D. Corneal ulcer

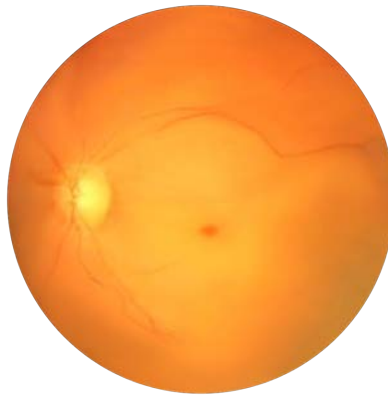
25. What is the most probable diagnosis for a 3-year-old boy who has been displaying increased clumsiness for the past 6 months, and upon examination, a cauliflower-like growth, leukocoria, and an absent red reflex in the left eye? Enucleation was performed, and microscopic examination of the specimen reveals the presence of Flexner-Wintersteiner rosettes.

- A. Retinoblastoma
- B. Neurofibroma
- C. Retinopathy of prematurity
- D. Toxocariasis

26. In a three-year-old boy, his parents have noticed a progressive decline in coordination over the last six months. During a physical examination, a growth resembling a cauliflower is observed in the left eye, along with leukocoria and the absence of a red reflex. Microscopic analysis of the sample reveals the presence of Flexner-Wintersteiner rosettes. Which gene is associated with this condition?

- A. Rb gene
- B. ATP 7B gene
- C. Rhodopsin gene
- D. FBN 1 gene

27. The image below displays the findings of a fundoscopy. What is the probable diagnosis?

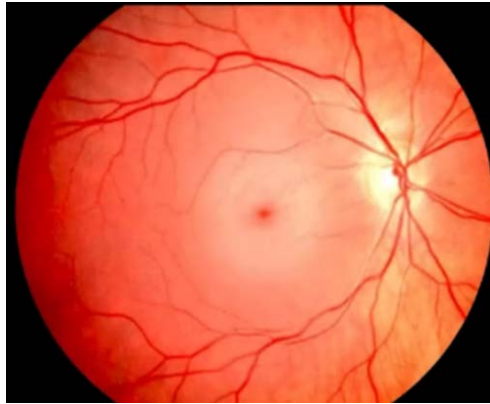


- A. CRAO
- B. CRVO
- C. Diabetic retinopathy
- D. Cystoid macular edema

28. What is the most probable diagnosis for a 7-year-old patient who presented with a white pupillary reflex and underwent enucleation, with histopathological examination revealing the presence of Flexner Winterstein rosettes?

- A. Retinoblastoma
- B. Medulloblastoma
- C. Rhabdomyosarcoma
- D. Astrocytoma

29. What would be the diagnosis for a diabetic patient experiencing an abrupt vision loss, having a visual acuity of 6/60, with a fundus picture provided?

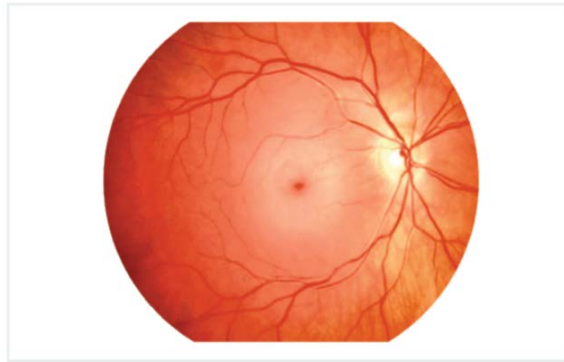


- A. CRVO
- B. CRAO
- C. Diabetic retinopathy
- D. Hypertensive retinopathy

30. Shifting fluid sign is seen in?

- A. Exudative retinal detachment
- B. Traction retinal detachment
- C. Rhegmatogenous retinal detachment
- D. Retinal dialysis

31. What is the likely cause of sudden loss of vision in the left eye for the past two hours in a 58-year-old man who experienced it after his retirement party? The fundoscopic examination reveals a similar image shown below.



- A. BRAO
- B. BRVO
- C. CRAO
- D. CRVO

32. Which of the following is not true about retinitis pigmentosa?

- A. Retinal pigmentation
- B. Pale waxy disc
- C. Narrowing of vessels
- D. ERG – normal

33. What is the commonly used chemotherapy regimen for the treatment of retinoblastoma?

- A. Vinblastine, etoposide and bleomycin
- B. Vinblastine, vincristine and etoposide
- C. Vinblastine, vincristine and cisplatin
- D. Vincristine, carboplatin and etoposide

34. What is the probable diagnosis for a 3-year-old boy who has presented with cognitive impairment and difficulty in mobility, along with the provided fundoscopy image?

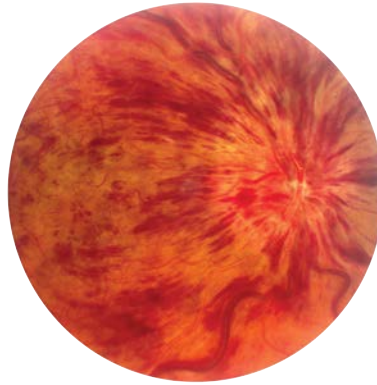


- A. Tay-Sachs disease
- B. Hunter disease
- C. Hurler syndrome
- D. Gaucher disease

35. In which diseases are microaneurysms observed?

- A. Retinal detachment
- B. Diabetic retinopathy
- C. Uveitis
- D. Glaucoma

36. What is the condition characterized by painless vision loss?



- A. Central retinal vein occlusion
- B. Branched retinal vein occlusion
- C. Central retinal arterial occlusion
- D. Branched retinal artery occlusion

37. What is the probable diagnosis for a middle-aged man with diabetes, who has reported a decline in vision and whose fluorescein angiography displays the depicted image?



- A. Mild-non proliferative diabetic retinopathy
- B. Severe non-proliferative diabetic retinopathy
- C. Advanced eye disease

D. Proliferative diabetic retinopathy

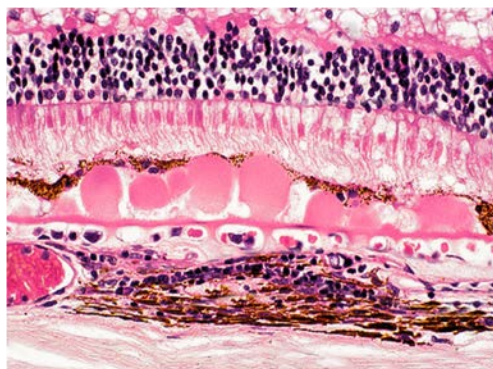
38. Which of the following is not a possible cause of the salt and pepper appearance of the fundus in a patient?

- A. Rubella
 - B. Syphilis
 - C. Refsum's disease
 - D. Tuberculosis
-

39. Roth spots are seen in _____.

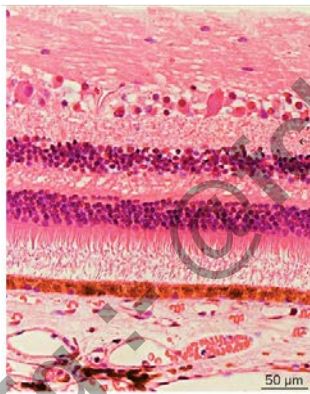
- A. Acute leukemia
 - B. Uveitis
 - C. Optic neuritis
 - D. Vogt-Koyanagi-Harada syndrome
-

40. Identify the lesion in the histopathological image of retina.



- A. Drusen
 - B. Hard exudates
 - C. Cotton wool spots
 - D. Microaneurysms
-

41. Identify the marked layer of the retina



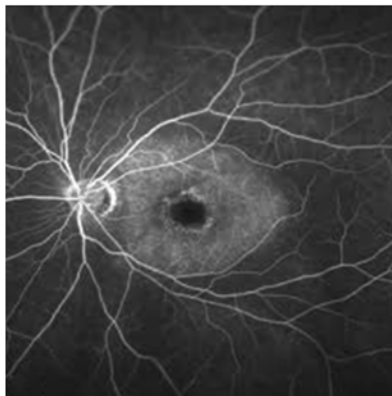
- A. Inner plexiform layer
 - B. Ganglion cell layer
 - C. Inner nuclear layer
 - D. Outer plexiform layer
-

42. The image is suggestive of?



- A. Macular hole
- B. Cystoid macular edema
- C. Central serous chorioretinopathy
- D. Age related Macular degeneration

43. What could be the likely diagnosis for a 65-year-old female patient with a prior diagnosis of chronic arthritis, undergoing treatment, who presents with complaints of impaired vision in both eyes and fundus angiography depicted the following finding?



- A. Choroiditis
- B. Chloroquine toxicity
- C. Steroid – induced glaucoma

D. Macular edema

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	4
Question 4	4
Question 5	1
Question 6	4
Question 7	3
Question 8	2
Question 9	4
Question 10	1
Question 11	1
Question 12	1
Question 13	4
Question 14	1
Question 15	1
Question 16	3
Question 17	1
Question 18	3
Question 19	2
Question 20	1
Question 21	2
Question 22	4

Question 23	2
Question 24	3
Question 25	1
Question 26	1
Question 27	1
Question 28	1
Question 29	2
Question 30	1
Question 31	3
Question 32	4
Question 33	4
Question 34	1
Question 35	2
Question 36	1
Question 37	4
Question 38	4
Question 39	1
Question 40	1
Question 41	1
Question 42	2
Question 43	2

Solution for Question 1:

- Irvine-Gass Syndrome, also known as cystoid macular edema (CME), is a condition commonly seen after cataract surgery.

- The OCT finding typically shows loss of foveal depression, cystoid spaces, and overall retinal thickening, which is indicative of this syndrome.

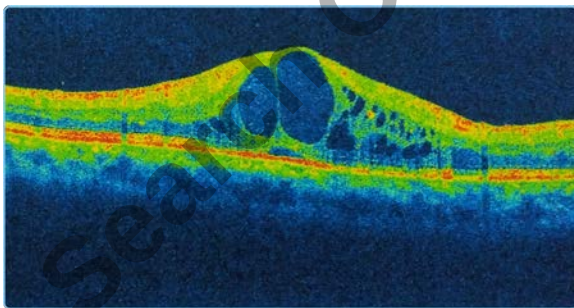
Etiology:

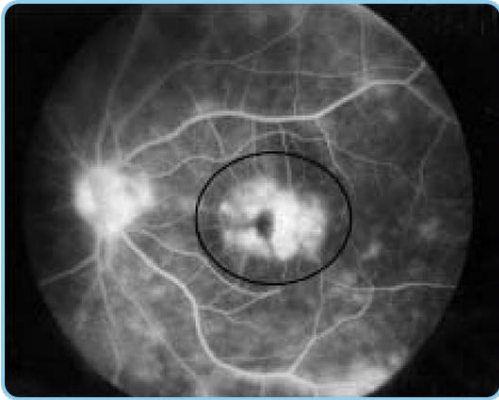
- Ocular Treatment Complications: Post-surgery (e.g., cataract extraction, retinal detachment surgery), ocular laser therapy, and topical ocular medications.
- Retinal Vascular Disorders: Diabetic retinopathy, central and branch retinal vein occlusion, retinal telangiectasia, and hypertensive retinopathy.
- Intraocular Inflammations: Pars planitis, uveitis.
- Retinitis pigmentosa.
- Vitreomacular Traction: Associated with macular epiretinal membrane (ERM) and vitreomacular traction (VMT) syndrome.
- Systemic Diseases: Includes leukemia, chronic renal failure, and multiple myeloma.
- Miscellaneous Causes: niacin toxicity , PG analogues (risk factor for development of CME)

Clinical Features:

- Visual Loss: Initial minimal to moderate vision loss, potentially leading to permanent vision decrease if persistent.
- Ophthalmoscopy: Reveals loss of foveal contour, retinal thickening, a yellow spot at the fovea, and a "honeycomb appearance" of the macula.
- Fluorescein Angiography: Shows leakage and dye accumulation in the macular region, with a "flower petal appearance" in late frames.

Optical Coherence Tomography (OCT): Demonstrates loss of foveal depression, cystoid spaces, and overall retinal thickening.





Complications:

- Long-standing CME may result in a lamellar macular hole.

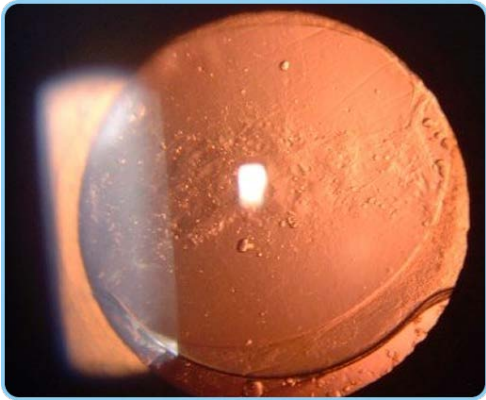
Treatment:

- Addressing the Cause: Intravitreal anti-VEGF or photocoagulation for diabetic CME; discontinuing causative topical medications.
- Topical NSAIDs: Such as ketorolac, diclofenac, or profenac, are effective in preventing and treating CME.
- Topical/Systemic Steroids: Useful in established cases.
- Systemic Carbonic Anhydrase Inhibitors (CAIs): Oral acetazolamide may be beneficial, particularly in cases associated with retinitis pigmentosa.

Posner-Schlossman Syndrome (Option A): This involves high IOP and mild uveitis, but is not associated with macular edema as seen in Irvine-Gass Syndrome.

Central Serous Retinopathy (Option C): is characterized by "ink-blot" or "smoke-stack" leakage patterns on FFA, and neurosensory retinal and RPE detachment on OCT. CSR typically does not present with the cystoid pattern seen in CME.

Elschnig's Pearls (Option D): Clustered, vacuolated subcapsular epithelial cells along the posterior capsule.



Solution for Question 2:

Correct Option B - Neovascularization of disc:

- NPDR stands for nonproliferative diabetic retinopathy
- In very severe NPDR there is no neovascularization of the disc seen.

Incorrect Options:

Option A - Venous beading:

- Venous beading is present in very severe NPDR

Option C - Retinal hemorrhage:

- Retinal hemorrhage is present in very severe NPDR

Option D - Microaneurysm:

- Microaneurysms are present in very severe NPDR
-

Solution for Question 3:

Correct Option D - Homonymous hemianopia with macular sparing

- This is the correct answer. This condition involves a loss of vision on the same side of both eyes, but with the central field of vision intact. This condition is usually associated with a stroke that affects the occipital lobe of the brain, which processes visual information. The patient's symptoms - inability to see on the left side from both eyes, but with central vision intact - align with this condition.

Incorrect Options:

Option A - Quadrantanopia: This is a type of visual field loss that happens in one of the four quarters of the visual field. But, the patient's symptoms indicate a loss of vision on the left side completely in both eyes, not just a quarter of the visual field.

Option B - Heteronymous hemianopia: This condition involves loss of half the field of view on different sides in each eye. For example, the right half in one eye and the left half in the other. But, the patient's symptoms suggest the same side (left) of both eyes is affected.

Option C - Homonymous hemianopia: This is a condition in which a patient loses half of the field of view on the same side in both eyes. This can occur because of stroke or brain injury. However, the patient's central vision remains intact, which suggests macular sparing.

Solution for Question 4:

Correct option D - GRADE 4

- Grade 4 hypertensive retinopathy represents severe hypertensive changes. This includes all the features mentioned in the patient's presentation: flame-shaped hemorrhages, cotton wool spots, and disc edema. Additionally, there can be signs of papilledema, hard exudates, and macular star formation. This grade indicates a more advanced stage of hypertensive retinopathy and is associated with significantly elevated blood pressure and potential risk of hypertensive emergency.

Incorrect Options

Option A - GRADE 1: Grade 1 hypertensive retinopathy typically involves arteriolar narrowing. There are no characteristic retinal findings such as hemorrhages, cotton wool spots, or disc edema. This grade is usually associated with mild hypertensive changes.

Option B - GRADE 2: Grade 2 hypertensive retinopathy is characterized by arteriolar narrowing along with mild retinal changes such as arteriovenous (AV) crossing changes (nickel-wire appearance). Hemorrhages and cotton wool spots are not prominent at this stage. There is

usually no disc edema.

Option C - GRADE 3: Grade 3 hypertensive retinopathy involves significant changes. It includes arteriolar narrowing, AV crossing changes, hemorrhages, and cotton wool spots, but usually no disc edema.

Solution for Question 5:

Correct Option A - Amaurosis Fugax

- Given the patient's clinical presentation of sudden, transient loss of vision in one eye, without pain, and the subsequent restoration of vision, along with the description of a curtain-like visual disturbance, is suggestive of amaurosis fugax

Incorrect Options:

Option B - Central Retinal Artery Occlusion: Central retinal artery occlusion involves the sudden blockage of the central retinal artery, leading to sudden and painless vision loss in one eye. However, in this condition, the vision loss is usually more severe and sustained, often leaving the patient with very poor vision or even complete blindness in the affected eye. The sudden and brief nature of the episode described by the patient makes central retinal artery occlusion less likely.

Option C - Central Retinal Vein Occlusion: Central retinal vein occlusion results from a blockage in the central retinal vein, causing a sudden painless loss of vision. However, the presentation of central retinal vein occlusion typically involves a more gradual and painless decrease in vision, without the sudden and transient nature of the episode described by the patient.

Option D - Hypertensive Retinopathy: Hypertensive retinopathy is associated with long-standing high blood pressure that leads to changes in the retinal blood vessels. It can cause symptoms such as blurred vision, but the sudden and transient nature of the episode, along with the complete loss and restoration of vision, is not consistent with hypertensive retinopathy.

Solution for Question 6:

Correct Option D - Night vision is lost late and rare:

- Night Blindness is the characteristic and earliest feature of retinitis pigmentosa and may present several years before the visible changes in the retina appear.
- It occurs due to degeneration of the rods.

Incorrect Options:

- Option A, B, and C: All these statements are true.

Solution for Question 7:

Correct Option C - Branched retinal vein occlusion:

- Branched retinal vein occlusion occurs as quadrantic occlusion due to occlusion at the level of AV crossing.
- It is mostly seen in the superotemporal retinal as AV crossings are more in this area, so the chances of blockage are higher.
- In this case, the patient is presenting with superior field loss of vision due to blockage in the inferior branch retinal vein.

Incorrect Options:

Option A - Branched retinal artery occlusion:

- In Branched retinal artery occlusion, a cherry red spot in the retina is noticed due to the surrounding edema.

Option B - Central retinal artery occlusion:

- In Central retinal artery occlusion, obstruction occurs at the level of the lamina cribrosa, causing sudden painless loss of vision within seconds.

Option D - Central retinal vein occlusion:

- In Central retinal vein occlusion, there is a mild blurring of vision, which improves over the day.
-

Solution for Question 8:

Correct Option B - Retinal detachment:

- Muscae volitantes, i.e. floating black opacities in front of the eyes is one of the symptoms in patients with pathological myopia.
- This is likely due to Rhegmatogenous retinal detachment due to the traction in posterior vitreous detachment.
- This causes the pigment to disperse in the vitreous cavity. This is called schaffer's sign or Tobacco dust.

Incorrect Options:

Option A - Subarachnoid hemorrhage:

- Subarachnoid hemorrhage causes subhyaloid hemorrhage.

Option C - Vitreous hemorrhage:

- Vitreous hemorrhage is caused by blunt force trauma.

Option D - Branched retinal artery occlusion:

- Branched retinal artery occlusion presents with cherry red spot in the retina.

Solution for Question 9:

Correct Option D - Central serous retinopathy:

- Central serous retinopathy is characterized by sudden painless loss of vision
- The other causes of sudden painless loss of vision are as follows: Optic neuritis Methyl alcohol amblyopia Nonischaemic central retinal vein occlusion.
- Optic neuritis
- Methyl alcohol amblyopia
- Nonischaemic central retinal vein occlusion.
- Optic neuritis
- Methyl alcohol amblyopia
- Nonischaemic central retinal vein occlusion.

Incorrect Options:

- Options A, B & C: Chemical injuries to eyeball, acute congestive glaucoma - and acute iridocyclitis cause sudden painful loss of vision

Solution for Question 10:

Correct Option A - Retinal detachment:

- Curtain falling in front of the eye is seen in case of rhegmatogenous retinal detachment.

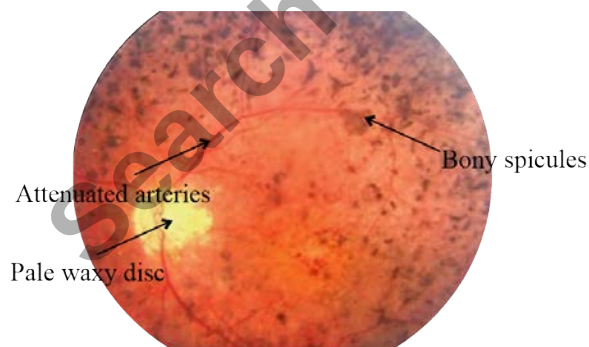
Incorrect Options:

- Options B, C & D: These are not associated with Curtain falling sign.

Solution for Question 11:

Correct Option A - Retinitis pigmentosa:

The patient's presentation and presence of bony spicules, pale waxy disc, and attenuated arteries are suggestive of retinitis pigmentosa.



Incorrect Options:

Options B, C, and D do not exhibit the funduscopy findings given in the question.

Solution for Question 12:

Correct Option A – Retinoblastoma:

- The presence of an intraocular mass in a child with leukocoria is suggestive of retinoblastoma. Flexner-Wintersteiner rosettes and Homer Wright rosettes are characteristic histopathological features of retinoblastoma. It is positive for synaptophysin, NSE, and chromogranin.

Incorrect Options:

Options B, C, and D do not align with the features described in the question.

Solution for Question 13:

Correct Option D - Bipolar cells:

- Bipolar cells are the first-order neurons. These cells receive signals from photoreceptor cells and transmit them to ganglion cells.

Bipolar cells are the first-order neurons. These cells receive signals from photoreceptor cells and transmit them to ganglion cells.

Incorrect Options:

Option B - Ganglion cells: Ganglion cells are the second-order neurons. They receive signals from bipolar cells and transmit these signals along their axons, forming the optic nerve to the lateral geniculate body.

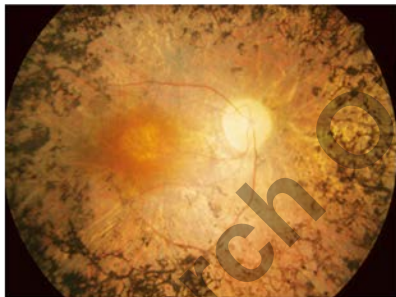
Option C - Cells in lateral geniculate body: Lateral Geniculate body is the third-order neuron. The optic nerve fibers synapse at the LGN in the thalamus.

Option D - Rods and Cones: These cells are photoreceptors and they are end receptors. They detect light and convert it into electrical signals.

Solution for Question 14:

Correct Option A - Arteriolar attenuation:

- Arteriolar attenuation is commonly observed in retinitis pigmentosa. Retinitis pigmentosa is a group of inherited retinal disorders characterized by progressive degeneration of the photoreceptor cells in the retina. This degeneration leads to various visual impairments, including night blindness, tunnel vision, and decreased peripheral vision.
- Classic examination findings seen in retinitis pigmentosa are: Pale, waxy disc. Attenuation of arteries. Pigmentation in the form of spicules.
- Pale, waxy disc.
- Attenuation of arteries.
- Pigmentation in the form of spicules.
- Pale, waxy disc.
- Attenuation of arteries.
- Pigmentation in the form of spicules.



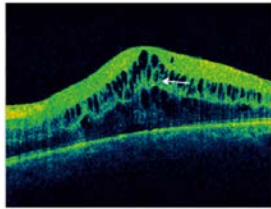
Incorrect options:

Option B, C, and D are incorrect. Refer to Option A for an explanation.

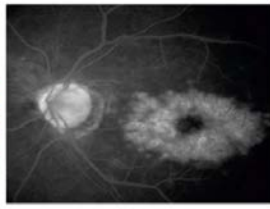
Solution for Question 15:

Correct Option A - Cystoid macular edema:

- The signs and symptoms given in the question stem leads to the diagnosis of CME (Cystoid macular edema)
- Cystic spaces are filled with fluid
- It is reversible
- Etiology - breaking of inner blood retinal barrier
- Clinical factors: diminishing of vision, metamorphopsia
- Investigation: optical coherence tomography
- Treatment: NSAIDs, steroids, acetazolamide, laser treatment.



Optical Coherence Tomography: CME



Angiography: CME

Incorrect Options:

Option B - Central serous retinopathy:

- Shallow, exudative retinal detachment is termed Central serous retinopathy
- After examining the patient, a ring reflex is seen because fluid is accumulated and this causes a shallow retinal detachment
- Fundus fluorescein angiography shows two patterns (Enlarging ink blot pattern and Mushroom shaped pattern).

Option C - Retinal detachment:

- Defined as the separation of the retinal pigment epithelium from neurosensory retina.

Option D - Diabetic retinopathy:

- Pathological changes in diabetic retinopathy are loss of pericyte, Thickening of basement membrane. Because of loss of pericyte and the microangiopathy there will be leakage, hypoxia and retinitis proliferans.
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- loss of pericyte, Thickening of basement membrane.
- Because of loss of pericyte and the microangiopathy there will be leakage ,hypoxia and retinitis proliferans.

Solution for Question 16:

Correct Option C - Looping in all quadrant:

- ETDRS classification
- Looping in all quadrants is not a criterion in severe NPDR.
- The NPDR can be divided into Very mild NPDR : Only microaneurysm Mild NPDR :Along with microaneurysm there are features of background diabetic retinopathy which means hemorrhagic exudates Moderate NPDR: All the features of preproliferative diabetic retinopathy Severe NPDR(described by 4:2:1 rule) Microaneurysms and hemorrhage are seen in all four quadrants (or) Looping and beading is seen in two quadrants (or) Intraretinal microvascular abnormalities may be present even in one quadrant. Very severe NPDR When the patient meets two or more than two criteria of the 4:2:1 rule, that is called very severe NPDR.
- Very mild NPDR : Only microaneurysm
- Mild NPDR :Along with microaneurysm there are features of background diabetic retinopathy which means hemorrhagic exudates
- Moderate NPDR: All the features of preproliferative diabetic retinopathy
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- Intraretinal microvascular abnormalities may be present even in one quadrant.
- When the patient meets two or more than two criteria of the 4:2:1 rule, that is called very severe NPDR.

Solution for Question 17:

Correct Option A - Immediately:

- A dilated fundus examination is recommended immediately as part of the initial evaluation for a person diagnosed with diabetes because of the delayed diagnosis commonly associated with type 2 diabetes.. This examination allows for the assessment of the retina and blood vessels in the eyes to detect any signs of diabetic retinopathy, a common complication of diabetes.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 18:

Correct Option C - Choroid

- The structure marked with arrow is the choroid
- The choroid is a critical structure in the eye, part of the uveal tract, which also includes the ciliary body and the iris.
- The main functions of the choroid are to provide oxygen and nutrients to the outer layers of the retina and to absorb scattered light, thereby reducing reflections within the eye.

Incorrect Options:

Option A - Outer limiting membrane: The outer limiting membrane is a structure in the retina that separates the outer nuclear layer from the inner segments of the photoreceptor cells. It is formed by the junctions between the Müller cells and the photoreceptor cells. The red arrow in the image does not point to the outer limiting membrane.

Option B - Inner plexiform layer: The inner plexiform layer is a layer of the retina located between the inner nuclear layer and the ganglion cell layer. It contains synaptic connections between bipolar cells, amacrine cells, and ganglion cells.

Option D - Outer nuclear layer: The outer nuclear layer is a layer of the retina located between the outer plexiform layer and the external limiting membrane. It contains the cell bodies of the photoreceptor cells (rods and cones).

Solution for Question 19:

Correct Option B - Diabetic macular edema:

- The ETDRS (Early Treatment of Diabetic Retinopathy Study) study was primarily conducted to study the treatment options for diabetic macular edema, a common complication of diabetic retinopathy characterized by fluid accumulation in the macula.
- Early Treatment of Diabetic Retinopathy It classifies the diabetic retinopathy into Non-proliferative diabetic retinopathy (NPDR) No DR Very mild NPDR - Microaneurysm only

Mild NPDR - microaneurysms, retinal haemorrhages, exudates, cotton-wool spots
 Moderate NPDR - Severe retinal haemorrhages in 1-3 quadrants or mild IRMA, and venous beading in 1 quadrant. Severe NPDR - One or more of severe haemorrhages in all 4 quadrants, significant venous beading in 2 or more quadrants Very severe NPDR - Two or more of the criteria for severe NPDR. PDR (Proliferative Diabetic Retinopathy) Mild-moderate PDR - New vessels on the disc (NVD) or new vessels elsewhere (NVE), but extent insufficient to meet the high-risk criteria. High-risk PDR - NVD greater than ETDRS standard photograph 10A, with vitreous haemorrhage, and NVE greater than 2 disc area with vitreous haemorrhage. Advanced diabetic eye disease - Vitreous hemorrhage, tractional retinal detachment, and neovascular glaucoma.

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- Advanced diabetic eye disease - Vitreous hemorrhage, tractional retinal detachment, and neovascular glaucoma.

Incorrect Options:

Options A, C, and D are incorrect. ETDRS study was conducted to study the treatment options for diabetic retinopathy.

Solution for Question 20:

Correct Option A - Hess chart:

- The Hess chart is a diagnostic tool used predominantly to evaluate binocular vision and ocular motility disorders, such as strabismus..
- In a Hess chart, two grids represent the two eyes – the right grid for the right eye and the left grid for the left eye. Each grid comprises horizontal and vertical lines that intersect to form a pattern, somewhat resembling a chessboard.

Incorrect Options:

Option B - Snellen Chart: This is the traditional eye examination tool used worldwide for assessing visual acuity. It contains lines of letters, numbers, or symbols in standardized sizes, with the topmost line usually being the largest.

Option C - Pelli-Robson Chart: This chart is a specific tool used to measure contrast sensitivity, a crucial aspect of vision that refers to the ability to discern differences in lightness and darkness in an image.

Option D - ETDRS Chart: Standing for Early Treatment Diabetic Retinopathy Study, this chart is a standardized tool used for testing visual acuity more precisely than the Snellen chart, particularly useful in research settings and monitoring progression of diseases like diabetic retinopathy.

Solution for Question 21:

Correct Option B - ERG:

- Night blindness and delayed dark adaptation are characteristic features of a condition called congenital stationary night blindness (CSNB). To confirm the diagnosis of CSNB, further investigation with electroretinography (ERG) is necessary.
- ERG measures the electrical responses generated by the retina in response to light stimuli and can help assess the functioning of retinal cells.

Incorrect Options:

Option A - Dark adaptometry: Dark adaptometry is a test that measures the time it takes for the eyes to adapt to darkness and regain visual sensitivity.

Option C - EOG (Electrooculography): EOG is a test that measures the electrical responses generated by the muscles surrounding the eye. It is primarily used to assess the functioning of the retinal pigment epithelium (RPE).

Option D -: Retinoscopy: Retinoscopy is a technique used to assess refractive errors of the eye and does not directly evaluate night blindness or dark adaptation.

Solution for Question 22:

Correct Option D - Done at 1 meter away from patient:

- Retinoscopy is a technique used to determine an individual's refractive error by analyzing the movement of light reflected from the retina. When performing retinoscopy with a plane mirror, the observer typically maintains a working distance of approximately 1 meter from the patient.
- Procedure: The examiner holds the retinoscope about 1 meter away from the patient. The examiner shines the light beam into the patient's eye while moving the retinoscope back and forth, horizontally and vertically. The reflected light forms a reflex that the examiner observes. Depending on the movement of reflex with respect to the light it can be interpreted as: With the movement: Patient is emmetropic/hypermotropic/myopia <1 . A movement against the light: Myopia <1 . No movement: Myopia -1
- The examiner holds the retinoscope about 1 meter away from the patient.
- The examiner shines the light beam into the patient's eye while moving the retinoscope back and forth, horizontally and vertically.
- The reflected light forms a reflex that the examiner observes.
- Depending on the movement of reflex with respect to the light it can be interpreted as: With the movement: Patient is emmetropic/hypermotropic/myopia <1 . A movement against the light: Myopia <1 . No movement: Myopia -1
- With the movement: Patient is emmetropic/hypermotropic/myopia <1 .
- A movement against the light: Myopia <1 .
- No movement: Myopia -1
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- The reflected light forms a reflex that the examiner observes.
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- A movement against the light: Myopia <1 .
- No movement: Myopia -1
- With the movement: Patient is emmetropic/hypermetropic/myopia <1 .
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- No movement: Myopia -1

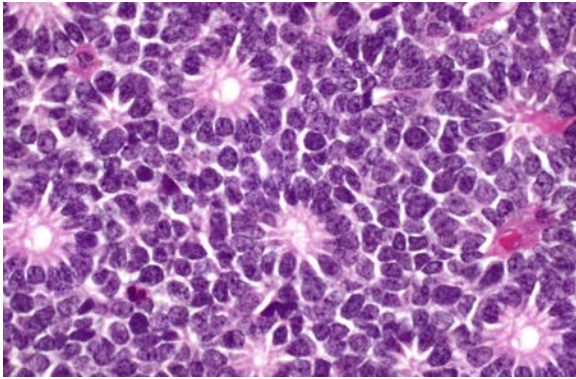
Incorrect options:

Options: A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 23:

Correct Option B - Retinoblastoma:

- Retinoblastoma is a malignant tumor that arises from immature retinal cells in young children, typically before the age of 5.
- The tumor can present with leukocoria (commonest presentation), strabismus, or an absent red reflex on physical examination.
- Flexner-Wintersteiner rosettes is a differentiated type of tumor characterized by columnar cells with a central lumen.



Incorrect Options:

Option A - Congenital glaucoma: Congenital glaucoma is characterized by epiphora, photophobia, blepharospasm, and buphthalmos.

Option C - Toxocariasis: It is a parasitic infection caused by roundworm *Toxocara canis* that can be acquired through ingestion of contaminated soil or food animals. It can present with leukocoria, strabismus, floaters, eosinophilia, and hypergammaglobulinemia.

Option D - Neurofibroma: Neurofibromas are tumors that arise from nerve sheath cells and are often associated with the genetic disorder neurofibromatosis. In plexiform neurofibroma S-shaped deformity of the leg is observed.

Solution for Question 24:

Correct Option C - Retinal detachment:

- The clinical presentation of flashes of light, floaters, and curtain like effect is suggestive of rhegmatogenous retinal detachment.
- B scan reveals the cyst representing vitreous fluid between the neurosensory layer and retinal pigmented epithelium.

Incorrect Options:

Option A - Glaucoma: It is a disease that damages the optic nerve, often caused by high pressure inside the eye. It typically causes a gradual loss or sudden loss of vision but curtain like effect is not seen..

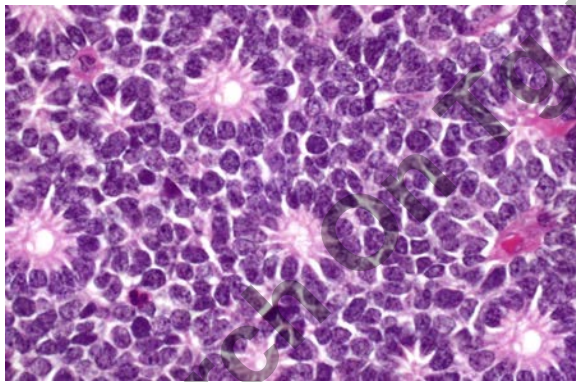
Option B - Posterior vitreous detachment Retinal : It is characterized by floaters, flashes of light, and blurred vision.

Option D - Corneal ulcers: They are painful sores on the surface of the eye that can cause redness, tearing, and sensitivity to light.

Solution for Question 25:

Correct Option A - Retinoblastoma:

- Retinoblastoma is a malignant tumor that arises from immature retinal cells in young children, typically before the age of 5.
- The tumor can present with leukocoria (most commonest presentation), strabismus , or an absent red reflex on examination.
- Flexner-Wintersteiner rosettes is a differentiated type of tumor characterized by columnar cells with a central lumen.



Incorrect Options:

Option B - Neurofibroma: Neurofibromas are tumors that arise from nerve sheath cells and are often associated with the genetic disorder neurofibromatosis. In plexiform neurofibroma S-shaped deformity of the lid is observed.

Option C - Retinopathy of prematurity: It may cause retinal detachment and leukocoria but there is no history of prematurity and low birth weight.

Option D - Toxocariasis: It is a parasitic infection caused by roundworm *Toxocara canis* that can be acquired through ingestion of contaminated soil or food. It can present with leukocoria, strabismus, floaters, eosinophilia, and hypergammaglobulinemia.

Solution for Question 26:

Correct Option A - Rb gene:

- The clinical presentation of leukocoria (commonest manifestation), strabismus, secondary glaucoma, absent red reflex on direct ophthalmoscope, and Flexner-Wintersteiner rosettes is suggestive of retinoblastoma.
- Mutation in the Rb gene located in the q14 band of the 13th chromosome can result in retinoblastoma.

Incorrect Options: Option B - ATP7B gene: Mutation in this gene results in Wilson's Disease.

Option C - Rhodopsin gene: Mutation in this gene results in autosomal dominant Retinitis Pigmentosa.

Option D - FBN-1 gene: Mutation in this gene results in Marfan's Syndrome.

Solution for Question 27:

Correct Option A - CRAO:

- The given image shows a cherry-red spot. The orange reflex from the intact choroid stands out at the thin foveola, in contrast to the surrounding pale retina, giving rise to a 'cherry-red spot' appearance.
- Cherry red spot is caused due to the following conditions Central retinal and cilioretinal artery occlusion Retinotoxicity Commotio retinae Metabolic storage diseases GM1 gangliosidosis GM2 gangliosidosis Tay Sachs disease Sandhoff disease Mucopolysaccharidosis type I (sialidosis) Niemann–Pick disease Farber disease
- Central retinal and cilioretinal artery occlusion
- Retinotoxicity
- Commotio retinae
- Metabolic storage diseases GM1 gangliosidosis GM2 gangliosidosis Tay Sachs disease Sandhoff disease Mucopolysaccharidosis type I (sialidosis) Niemann–Pick disease Farber disease

- GM1 gangliosidosis
- GM2 gangliosidosis Tay Sachs disease Sandhoff disease
- Tay Sachs disease
- Sandhoff disease
- Mucopolidosis type I (sialidosis)
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- Farber disease
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- Farber disease
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- GM2 gangliosidosis Tay Sachs disease Sandhoff disease
- Tay Sachs disease
- Sandhoff disease
- Mucopolidosis type I (sialidosis)
- Niemann–Pick disease
- Farber disease
- Tay Sachs disease
- Sandhoff disease

Incorrect Options:

Options B, C, and D are incorrect. Cherry red spot is not caused by these conditions.

Solution for Question 28:

Correct Option A - Retinoblastoma:

- The most likely diagnosis based on the histopathological examination showing Flexner Winterstein rosettes is Retinoblastoma.

Microscopic pathology of Retinoblastoma

- It is divided into differentiated and undifferentiated retinoblastoma.
- The differentiated type has three forms Flexner-Wintersteiner Rosette are the columnar cells with a central lumen The Pseudo rosette is called homer wright rosette Fleurettes
- Flexner-Wintersteiner Rosette are the columnar cells with a central lumen
- The Pseudo rosette is called homer wright rosette
- Fleurettes
- Flexner-Wintersteiner Rosette are the columnar cells with a central lumen
- The Pseudo rosette is called homer wright rosette
- Fleurettes

Incorrect Options:

Options B, C, and D are incorrect. These conditions are not characterized by Flexner Winterstein rosettes.

Solution for Question 29:

Correct Option B - CRAO:

- The given image shows a cherry-red spot. The orange reflex from the intact choroid stands out at the thin foveola, in contrast to the surrounding pale retina, giving rise to a 'cherry-red spot' appearance.
- Sudden loss of vision and cherry red spot are characteristic of Central Retinal Artery Occlusion.
- CRAO (Central Retinal Artery Occlusion): CRAO occurs when there is a sudden blockage in the central retinal artery, leading to a severe and immediate loss of vision.

Incorrect Options:

Options A, C, and D are incorrect. Cherry red spot is not seen in these conditions.

Solution for Question 30:

Correct Option A - Exudative retinal detachment:

- The shifting fluid sign is typically seen in exudative retinal detachment (RD).
- Exudative retinal detachment is a condition in which fluid accumulates beneath the retina, leading to its separation from the underlying layers. The shifting fluid sign refers to the movement or shifting of subretinal fluid with changes in body or head position.



Incorrect Options:

Option B - Tractional retinal detachment (TRD) is a type of retinal detachment characterized by the pulling or traction on the retina due to fibrous tissue or membranes. These fibrous tissues can exert traction on the retina, leading to its detachment from the underlying layers.

Option C - Rhegmatogenous retinal detachment (RD) is the most common type of retinal detachment and occurs due to a break or tear in the retina.

Option D - Retinal dialysis, also known as retinal breaks or retinal tears at the ora serrata, is a specific type of retinal pathology.

Solution for Question 31:

Correct Option C - CRAO:

- The given image shows a cherry-red spot. The orange reflex from the intact choroid stands out at the thin foveola, in contrast to the surrounding pale retina, giving rise to a 'cherry-red spot' appearance.
- Sudden loss of vision and cherry red spots are characteristic of Central Retinal Artery Occlusion.
- CRAO (Central Retinal Artery Occlusion): CRAO occurs when there is a sudden blockage in the central retinal artery, leading to a severe and immediate loss of vision. It is usually through embolism and the risk factors include the patient being a heart patient or having carotid artery disease, or being associated with mucormycosis of the orbit.

Incorrect Options:

Options A, B, and D are incorrect. Cherry red spots are not associated with these conditions.

Solution for Question 32:

Correct Option D - ERG - normal:

- The electroretinogram (ERG) is a sensitive diagnostic test for retinitis pigmentosa, it is characterized by decreased amplitude of both 'a' and 'b' waves.
- Classic examination findings associated with retinitis pigmentosa are: Pale waxy disc. Attenuation of arteries, and Retinal pigmentation disturbance in the form of bony spicules
- Pale waxy disc.
- Attenuation of arteries, and
- Retinal pigmentation disturbance in the form of bony spicules
- Pale waxy disc.
- Attenuation of arteries, and
- Retinal pigmentation disturbance in the form of bony spicules

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 33:

Correct Option D - Vincristine, carboplatin, and etoposide:

- Chemotherapy is the mainstay treatment in most of the cases of retinoblastoma. It involves carboplatin, etoposide and vin-cristine (CEV) are given in three–six cycles according to the grade of retinoblastoma.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 34:

Correct Option A - Tay-Sachs disease:

- The clinical presentation of inability to move, functional decline, and fundoscopic image of retina showing cherry red spot (develops between 3 months to 12 months of age) is suggestive of neurodegenerative disorder Tay Sachs disease (TSD) caused due to deficiency of the enzyme hexosaminidase A, which leads to GM2 gangliosides build up in the body.
- It is characterized by incoordination, exaggerated startle response, loss of muscle tone, progressive neurologic deterioration, functional decline, and loss of previously attained developmental milestones.

Incorrect Options:

Option B - Hunter's disease is characterized by short attention spans, do not respond to instructions, and have no sense of danger, aggressive behavior, and developmental delay, which progressively declines.

Option C - Hurler syndrome is the severe form of MPS I and is characterized by a broad spectrum of clinical problems, including skeletal abnormalities, hepatosplenomegaly, and severe intellectual disability.

Option D - Gaucher disease is characterized by hepatosplenomegaly, bleeding, thrombocytopenia, diffuse bone pain, osteonecrosis, peripheral neuropathy, and delayed puberty.

Solution for Question 35:

Correct Option B - Diabetic retinopathy:

- Microaneurysms are localized outpouchings of the capillary wall that may form by focal dilatation of the capillary wall where pericytes are absent or by fusion of two arms of a capillary loop.
- Microaneurysms are the earliest sign of Diabetic retinopathy.

Incorrect Options:

Options A, B, and C are incorrect. Microaneurysms are not seen in these diseases.

Solution for Question 36:

Correct Option A - Central retinal vein occlusion:

- The above image given is that of splashed sauce appearance and sudden painless loss of vision are classical features of ischaemic central retinal vein occlusion or CRVO.
- Splashed sauce appearance is characterized by severe tortuosity and engorgement of all branches of the central retinal vein involving the peripheral and posterior retina.

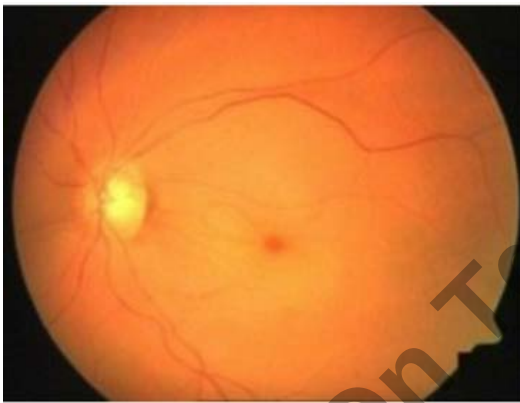
Incorrect Options:

Option B:



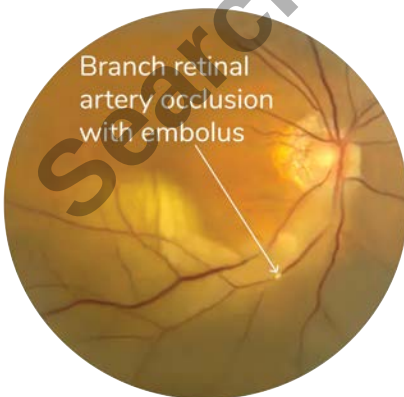
Early fundus features of BRVO on ophthalmoscopy include sectorial superficial retinal hemorrhages that rarely cross the horizontal raphe, cotton-wool spots, retinal edema, and dilated and tortuous retinal vein.

Option C:



CRAO presents with cherry red spot.

Option D:



Branch retinal artery occlusion describes decreased arterial blood flow to the retina leading to ischemic damage. The severity of visual loss depends upon the area of retinal tissue affected by the vascular occlusion.

Solution for Question 37:

Correct Option D - Proliferative diabetic retinopathy:

- In the above given image of FFA or fundus fluorescein angiography, neovascularization is noted which is classical of proliferative diabetic retinopathy. It is of two types: neovascularization of the disc or NVD and neovascularization elsewhere or NVE.
- According to ETDRS classification it is classified into: Non-proliferative diabetic retinopathy (NPDR) No DR Very mild NPDR: Microaneurysms only Mild NPDR: Any or all of: microaneurysms, retinal hemorrhages, exudates, cotton-wool spots, up to the level of moderate NPDR. No intraretinal microvascular anomalies (IRMA) or significant beading Moderate NPDR Severe retinal hemorrhages (more than ETDRS standard photograph 2A: about 20 medium-large per quadrant) in 1-3 quadrants or mild IRMA Significant venous beading can be present in no more than 1 quadrant Cotton-wool spots commonly present Severe NPDR: The 4-2-1 rule is one or more of the: Severe hemorrhages in all 4 quadrants Significant venous beading in 2 or more quadrants Moderate IRMA in 1 or more quadrants Very severe NPDR: Two or more of the criteria for severe NPDR PDR PDR (Proliferative Diabetic Retinopathy) Mild-moderate PDR: New vessels on the disc (NVD) or new vessels elsewhere (NVE), but extent insufficient to meet the high-risk criteria High-risk PDR NVD greater than ETDRS standard photograph 10A (about disc area). Any NVD with vitreous hemorrhage. NVE greater than 2 disc area with vitreous hemorrhage. Advanced diabetic eye disease: Vitreous hemorrhage, tractional retinal detachment, and neovascular glaucoma.
- Non-proliferative diabetic retinopathy (NPDR) No DR Very mild NPDR: Microaneurysms only Mild NPDR: Any or all of: microaneurysms, retinal hemorrhages, exudates, cotton-wool spots, up to the level of moderate NPDR. No intraretinal microvascular anomalies (IRMA) or significant beading Moderate NPDR Severe retinal hemorrhages (more than ETDRS standard photograph 2A: about 20 medium-large per quadrant) in 1-3 quadrants or mild IRMA Significant venous beading can be present in no more than 1 quadrant Cotton-wool spots commonly present Severe NPDR: The 4-2-1 rule is one or more of the: Severe hemorrhages in all 4 quadrants Significant venous beading in 2 or more quadrants Moderate IRMA in 1 or more quadrants Very severe NPDR: Two or more of the criteria for severe

NPDR PDR

- No DR
- Very mild NPDR: Microaneurysms only
- Mild NPDR: Any or all of: microaneurysms, retinal hemorrhages, exudates, cotton-wool spots, up to the level of moderate NPDR. No intraretinal microvascular anomalies (IRMA) or significant beading
- Moderate NPDR Severe retinal hemorrhages (more than ETDRS standard photograph 2A: about 20 medium-large per quadrant) in 1-3 quadrants or mild IRMA Significant venous beading can be present in no more than 1 quadrant Cotton-wool spots commonly present
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- Any NVD with vitreous hemorrhage.
- NVE greater than 2 disc area with vitreous hemorrhage.
- Advanced diabetic eye disease: Vitreous hemorrhage, tractional retinal detachment, and neovascular glaucoma.
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- Any NVD with vitreous hemorrhage.
- NVE greater than 2 disc area with vitreous hemorrhage.
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- NVD greater than ETDRS standard photograph 10A (about disc area).
- Any NVD with vitreous hemorrhage.

- NVE greater than 2 disc area with vitreous hemorrhage.
- Advanced diabetic eye disease: Vitreous hemorrhage, tractional retinal detachment, and neovascular glaucoma.
- NVD greater than ETDRS standard photograph 10A (about disc area).
- Any NVD with vitreous hemorrhage.
- NVE greater than 2 disc area with vitreous hemorrhage.

Incorrect Options:

Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 38:

Correct Option D - Tuberculosis:

- Amongst the above given options, salt and pepper fundus is not seen in tuberculosis.

Incorrect Options:

Options A, B, and C are incorrect. Salt and pepper fundus is seen in these conditions.

Solution for Question 39:

Correct Option A - Acute leukemia:

- Roth spots are retinal findings characterized by round or oval-shaped retinal hemorrhages with a pale or white center. The whitish element is thought to be composed of coagulated fibrin in most cases.
- Conditions associated with roth spots are: Acute leukemia. Bacteremia (subacute endocarditis, endogenous endophthalmitis). Diabetes. Hypertension. Anemia.
- Acute leukemia.
- Bacteremia (subacute endocarditis, endogenous endophthalmitis).

- Diabetes.
- Hypertension.
- Anemia.
- Acute leukemia.
- Bacteremia (subacute endocarditis, endogenous endophthalmitis).
- Diabetes.
- Hypertension.
- Anemia.

Incorrect Options:

Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 40:

Correct Option A - Drusen:

- Drusen are yellowish-white deposits that accumulate between the retinal pigment epithelium (RPE) and Bruch's membrane in the retina. They can be seen as small, round or oval-shaped lesions in the histopathological image of the retina.
- Drusen are a characteristic finding in age-related macular degeneration (AMD), particularly in the early stages known as dry AMD. They are thought to be accumulations of extracellular material, including lipids and proteins, which can disrupt the normal functioning of the retina and its blood supply.

Incorrect Options:

Option B - Hard exudates: Hard exudates are lipid deposits that can be seen as yellow or white lesions in the retina. They typically appear as small, round or elongated deposits with well-defined borders associated with diabetic retinopathy.

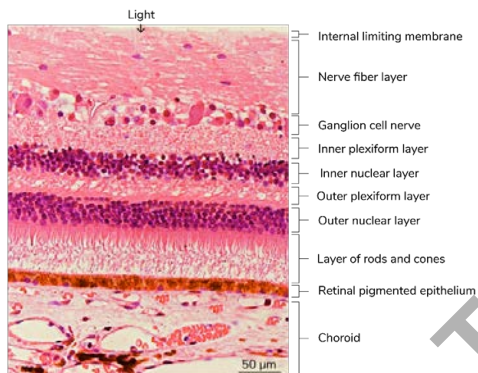
Option C - Cotton wool spots: Cotton-wool spots are composed of accumulations of neuronal debris within the nerve fiber layer characterized by small fluffy whitish superficial lesions that obscure underlying blood vessels associated with diabetic and hypertensive retinopathy.

Option D - Microaneurysms: Microaneurysms are localized saccular outpouchings of the capillary wall that may form by focal dilatation of the capillary wall where pericytes are absent or by fusion of two arms of a capillary loop seen in diabetic retinopathy.

Solution for Question 41:

Correct Option A - Inner plexiform layer:

- Inner plexiform layer- is marked in the given image
- The inner plexiform layer is one of the distinct layers within the retina, and it is located between the inner nuclear layer and the ganglion cell layer. It plays a crucial role in the transmission of visual information from the photoreceptor cells to the ganglion cells as it contains the synapse of ganglion.



Incorrect Options:

Option B - Ganglion cell layer : Ganglion cell layer is just present between the inner plexiform layer and nerve fiber layer.

Option C - Inner nuclear layer: Inner nuclear layer is present between the inner and outer plexiform layer and contains the bipolar cells, amacrine cells, and horizontal cells.

Option D - Outer plexiform layer: The outer plexiform layer is located between the outer nuclear layer and the inner segments of the photoreceptor cells and has the synapses of axons of rods and cones with the dendrites of bipolar cells/amacrine cells/horizontal cells.

Solution for Question 42:

Correct Option B - Cystoid macular edema:

- The given image depicts flower petal appearance of the retina around the macula suggestive of cystoid macular edema.
- The leakage is always radial at macula because accumulation of fluid takes place in the outer plexiform and inner nuclear layers, and the outer plexiform layer at macula is radially placed and is called Henle's layer.

Incorrect Options:

Options A, C, and D are incorrect. Flower petal pattern is characteristic of cystoid macular edema.

Solution for Question 43:

Correct Option B - Chloroquine toxicity:

- Chloroquine is an antimalarial medication that is sometimes used in the treatment of chronic arthritis. However, prolonged use of chloroquine can lead to retinal toxicity, specifically affecting the macula, which is responsible for central vision.
- The characteristic findings of chloroquine toxicity include a bull's eye pattern on fundoscopy and increased fluorescence in the macular area on fundus fluorescein angiography (FFA) as in the given image.

Incorrect Options:

Options A, C, and D are incorrect. Bull's eye maculopathy is not associated with these conditions.

Previous Year Questions

1. How does the World Health Organization (WHO) define blindness?

- A. $VA < 3/60$
- B. $VA < 6/60$
- C. $VA > 3/60$
- D. $VA \geq 6/60$

2. According to the National Blindness and Visual Impairment Survey conducted from 2015 to 2019, what is the leading cause of moderate to severe visual impairment?

- A. Refractive error
- B. Untreated cataract
- C. Glaucoma
- D. Diabetic retinopathy

3. A person comes with the visual acuity in the right eye of 6/60 and left eye of 3/60. He should be categorized into which type of blindness?

- A. Mild
- B. Moderate
- C. Severe
- D. No blindness

4. Which is the most common cause of preventable childhood blindness in India?

- A. Refractive error
 - B. Vitamin A deficiency
 - C. Retinopathy of prematurity
 - D. Cataract
-

5. Most common cause of blindness in India is?

- A. Cataract
 - B. Refractive error
 - C. Trachoma
 - D. Glaucoma
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	3
Question 4	2
Question 5	1

Solution for Question 1:

Correct Option A - VA $<3/60$:

- According to WHO, Blindness is defined as VA $<3/60$ in better eye with available correction

Incorrect Options:

- Options B, C & D: These are not the correct definitions of Blindness

Solution for Question 2:

Correct Option B - Untreated cataract:

- According to the National Blindness and Visual Impairment Survey conducted in India from 2015-19, untreated cataracts were found to be the most common cause of moderate to severe visual impairment.

Incorrect Options:

Option A, C and D are incorrect.

Solution for Question 3:

Correct Option C - Severe:

- Based on the provided visual acuity measurements the patient belongs to the category 2 of visual impairment classification i.e., severe visual impairment.

WHO's Classification of Visual Impairment based on severity		
CATEGORY OF VI	CORRECTED VA (BETTER Eye)	DEGREE OF VI
0	6/6 – 6/18	Normal
1	Less than 6/18 – 6/60	Moderate Visual Impairment
2	Less than 6/60 – 3/60	Severe Visual Impairment
3	Less than 3/60 – 1/60	Blind
4	Less than 1/60 – PL	Blind
5	NPL	Blind

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 4:

Correct Option B - Vitamin A deficiency:

- The most common cause of preventable childhood blindness is vitamin A deficiency leading to Xerophthalmia. It can be prevented by timely vitamin A prophylaxis.

Incorrect Options:

Options A, C, and D are incorrect.

Solution for Question 5:

Correct Option A - Cataract:

- Cataract is the most common cause of blindness in India. It is a condition characterized by clouding of the lens in the eye, which leads to impaired vision and eventual blindness if left untreated.

Incorrect Options:

Options B, C, and D are incorrect.
